

1. 写出下列复数的实部, 虚部, 模和幅角:

$$(1) 1+i\sqrt{3}; (2) 1-\cos\alpha+i\sin\alpha, 0\leq\alpha<2\pi; (3) e^{i\sin x}, x \text{ 为实数}; (4) e^{iz};$$

$$(5) e^z; (6) \sqrt[4]{-1}; (7) \sqrt{1+i}; (8) \sqrt{\frac{1+i}{1-i}}; (9) e^{1+i}; (10) e^{i\varphi(x)}, \varphi(x) \text{ 是实变数 } x$$

的实函数。

$$(1) \operatorname{Re}=1, \operatorname{Im}=\sqrt{3}, \operatorname{Am}=\sqrt{\operatorname{Re}^2+\operatorname{Im}^2}=2, \operatorname{Arg}=\arctan\left(\frac{\operatorname{Im}}{\operatorname{Re}}\right)+2k\pi=\frac{\pi}{3}+2k\pi;$$

$$(2) \operatorname{Re}=1-\cos\alpha, \operatorname{Im}=\sin\alpha, \operatorname{Am}=\sqrt{(1-\cos\alpha)^2+\sin^2\alpha}=\sqrt{2-2\cos\alpha}=2\sin\frac{\alpha}{2},$$

$$\tan(\operatorname{Arg})=\frac{\sin\alpha}{1-\cos\alpha}=\frac{2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}{2\sin^2\frac{\alpha}{2}}=\cot\frac{\alpha}{2}, \text{ 所以 } \operatorname{Arg}=\frac{\pi-\alpha}{2}+2k\pi;$$

$$(3) \operatorname{Am}=1, \operatorname{Arg}=\sin x+2k\pi, \operatorname{Re}=\cos(\sin x), \operatorname{Im}=\sin(\sin x);$$

$$(4) z=x+iy, e^{iz}=e^{-y+ix}, \operatorname{Am}=e^{-y}, \operatorname{Arg}=x+2k\pi, \operatorname{Re}=e^{-y}\cos x, \operatorname{Im}=e^{-y}\sin x;$$

$$(5) \operatorname{Am}=e^x, \operatorname{Arg}=y+2k\pi, \operatorname{Re}=e^x\cos y, \operatorname{Im}=e^x\sin y;$$

$$(6) \sqrt[4]{-1}=\left[e^{i(\pi+2n\pi)}\right]^{\frac{1}{4}}=e^{i\frac{2n+1}{4}\pi}, (n=0, 1, 2, 3), \operatorname{Am}=1, \operatorname{Arg}=\frac{2n+1}{4}\pi+2k\pi,$$

$$\operatorname{Re}=\cos\left(\frac{2n+1}{4}\pi\right), \operatorname{Im}=\sin\left(\frac{2n+1}{4}\pi\right);$$

$$(7) \sqrt{1+i}=\sqrt[4]{2}e^{i\left(\frac{\pi}{4}+2n\pi\right)/2}=\sqrt[4]{2}e^{i\left(\frac{\pi}{8}+n\pi\right)}, (n=0, 1), \operatorname{Am}=\sqrt[4]{2}, \operatorname{Arg}=\frac{\pi}{8}+n\pi+2k\pi,$$

$$\operatorname{Re}=\sqrt[4]{2}\cos\left(\frac{\pi}{8}+n\pi\right)=(-1)^n\sqrt[4]{2}\cos\frac{\pi}{8}, \operatorname{Im}=(-1)^n\sqrt[4]{2}\sin\frac{\pi}{8};$$

$$(8) \sqrt{\frac{1+i}{1-i}}=\left[\frac{\sqrt{2}e^{i(\pi/4+2n\pi)}}{\sqrt{2}e^{-i\pi/4}}\right]^{\frac{1}{2}}=e^{i\frac{\pi/2+2n\pi}{2}}=e^{i\left(\frac{\pi}{4}+n\pi\right)}, (n=0, 1), \operatorname{Am}=1, \operatorname{Arg}=\frac{\pi}{4}+n\pi+2k\pi,$$

$$\operatorname{Re}=\frac{(-1)^n}{\sqrt{2}}, \operatorname{Im}=\frac{(-1)^n}{\sqrt{2}};$$

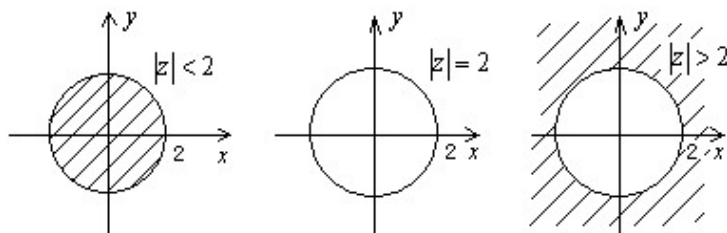
$$(9) \operatorname{Am}=e, \operatorname{Arg}=1+2k\pi, \operatorname{Re}=e\cos 1, \operatorname{Im}=e\sin 1;$$

$$(10) \operatorname{Am}=1, \operatorname{Arg}=\varphi(x)+2k\pi, \operatorname{Re}=\cos[\varphi(x)], \operatorname{Im}=\sin[\varphi(x)];$$

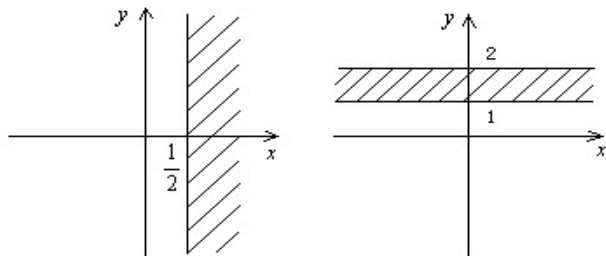
2. 把下列关系用几何图形表示出来:

(1) $|z| < 2$, $|z| = 2$, $|z| > 2$; (2) $\operatorname{Re} z > \frac{1}{2}$, $1 < \operatorname{Im} z < 2$; (3) $\arg(1-z) = 0$, $\arg(1+z) = \frac{\pi}{3}$, $\arg(z+1-i) = \frac{\pi}{2}$; (4) $0 < \arg(1-z) < \frac{\pi}{4}$, $0 < \arg(1+z) < \frac{\pi}{4}$, $\frac{\pi}{4} < \arg(z-1-2i) < \frac{\pi}{3}$; (5) $\alpha < \arg z < \beta$ 与 $\gamma < \operatorname{Re} z < \delta$ 的公共区域, $\alpha, \beta, \gamma, \delta$ 均为常数; (6) $|z-i| < 1$, $1 < |z-i| < \sqrt{2}$; (7) $|z-a| = |z-b|$, a, b 为常数; (8) $|z-a| + |z-b| = c$, 其中 a, b, c , 为常数, 且 $c > |a-b|$; (9) $|z| + \operatorname{Re} z < 1$; (10) $0 < \arg\left(\frac{z-i}{z+i}\right) < \frac{\pi}{4}$ 。

(1)



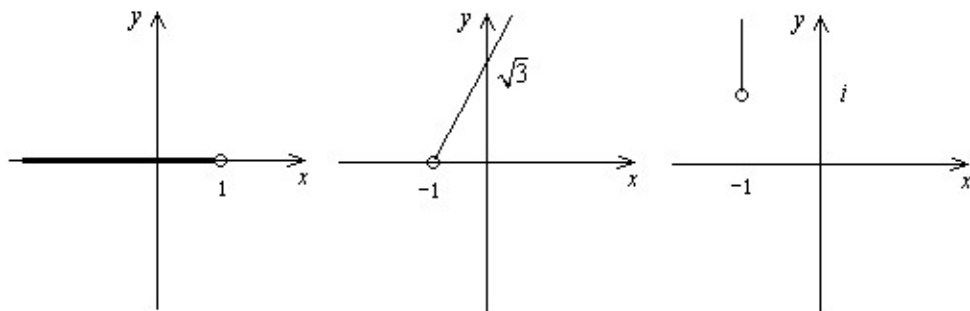
(2)



(3) $\arg(1-z) = \arg(1-x-iy) = 0 \Leftrightarrow 1-x > 0$ 且 $y=0$, 即 $x < 1$, $y=0$;

$\arg(1+z) = \arg(1+x+iy) = \frac{\pi}{3} \Leftrightarrow 1+x > 0$ 且 $y = \sqrt{3}(1+x)$;

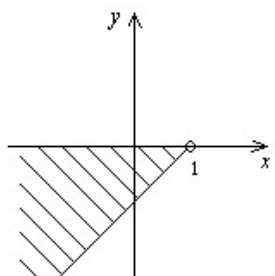
$\arg(z+1-i) = \arg[x+1+i(y-1)] = \frac{\pi}{2} \Leftrightarrow x+1=0$ 且 $y-1 > 0$ 。



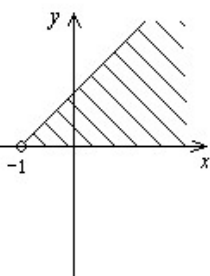
$$(4) \quad 0 < \arg(1-z) = \arg[(1-x)-iy] < \frac{\pi}{4} \Leftrightarrow 0 < -y < 1-x;$$

$$0 < \arg(1+z) = \arg[(1+x)+iy] < \frac{\pi}{4} \Leftrightarrow 0 < y < 1+x;$$

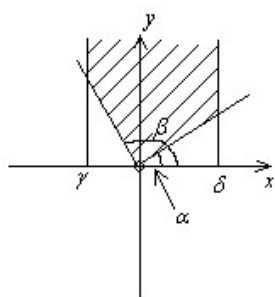
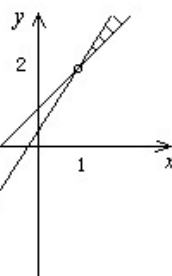
$$\frac{\pi}{4} < \arg(z-1-2i) = \arg[(x-1)+i(y-2)] < \frac{\pi}{3} \Leftrightarrow 0 < x-1 < y-2 < \sqrt{3}(x-1);$$



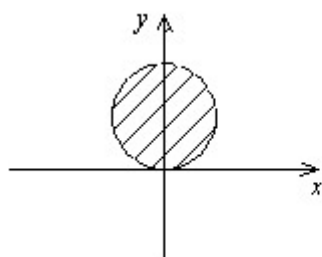
(5)



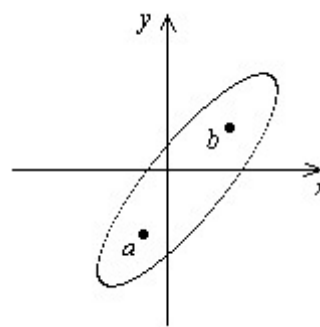
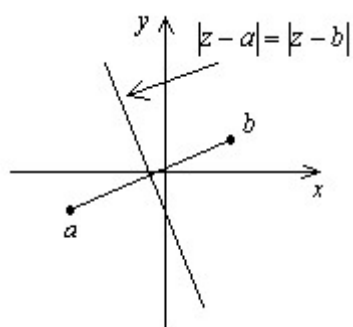
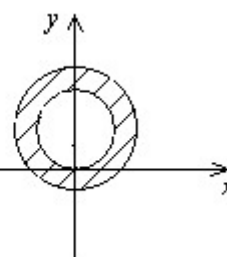
(6)



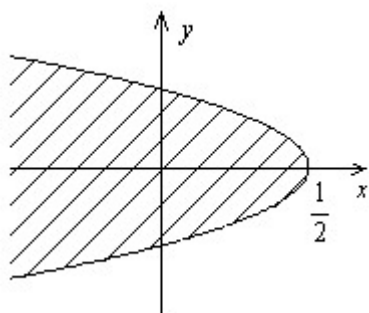
(7)



(8)

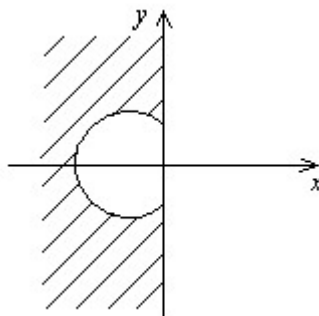


$$(9) \quad |z| + \operatorname{Re} z = \sqrt{x^2 + y^2} + x < 1, \text{ 化简得 } x < \frac{1}{2}(1 - y^2).$$



$$(10) \frac{z-i}{z+i} = \frac{x+i(y-1)}{x+i(y+1)} = \frac{x^2+y^2-1-2ix}{x^2+(y+1)^2}, \text{ 所以 } 0 < \arg\left(\frac{z-i}{z+i}\right) < \frac{\pi}{4} \Leftrightarrow$$

$$0 < -2x < x^2 + y^2 - 1, \text{ 即 } x < 0 \text{ 且 } (x+1)^2 + y^2 > 2。$$

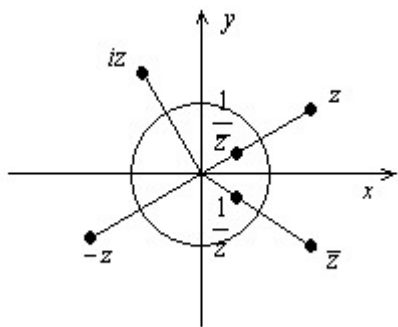


3. 已知一复数 z ，画出 iz ， $-z$ ， $\bar{z}$ ， $\frac{1}{\bar{z}}$ ， $\frac{1}{z}$ ，并指出它们之间的几何关系。

把 z 写成 $\rho e^{i\varphi}$ ，则 $iz = \rho e^{i(\varphi+\pi/2)}$ ，即把 z 逆时针旋转 90 度。 $-z = \rho e^{i(\varphi+\pi)}$ ，即把 z 逆时针

旋转 180 度。 $\bar{z} = \rho e^{-i\varphi}$ ，即 z 关于实轴的对称点。 $\frac{1}{\bar{z}} = \frac{1}{\rho} e^{i\varphi}$ ，即 z 关于单位圆的对称点。

$\frac{1}{z} = \frac{1}{\rho} e^{-i\varphi}$ ，即 $\bar{z}$ 关于单位圆的对称点。



4. 若 $|z|=1$ ，试证明 $\left| \frac{az+b}{bz+\bar{a}} \right| = 1$ ， a, b 为任意复数。

$$\left| \frac{az+b}{bz+\bar{a}} \right|^2 = \frac{(az+b)(\bar{a}\bar{z}+\bar{b})}{(\bar{b}z+\bar{a})(b\bar{z}+a)} = \frac{|a|^2 + a\bar{b}z + \bar{a}b\bar{z} + |b|^2}{|b|^2 + a\bar{b}z + \bar{a}b\bar{z} + |a|^2} = 1, \text{ 所以 } \left| \frac{az+b}{bz+\bar{a}} \right| = 1。$$

5. 证明下列各式:

$$(1) |z-1| \leq ||z|-1| + |z| |\arg z|;$$

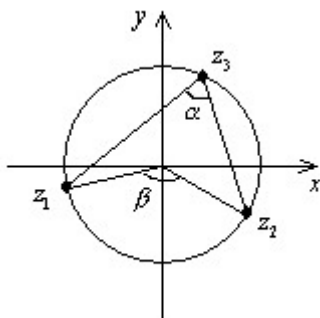
$$(2) \text{ 若 } |z_1| = |z_2| = |z_3|, \text{ 则 } \arg \frac{z_3 - z_2}{z_3 - z_1} = \frac{1}{2} \arg \frac{z_2}{z_1}.$$

$$(1) \text{ 先证 } \left| \frac{z}{|z|} - 1 \right| \leq |\arg z|.$$

$$\text{记 } z = \rho e^{i\varphi}, \quad \left| \frac{z}{|z|} - 1 \right| = |e^{i\varphi} - 1| = \sqrt{2 - 2\cos\varphi} = 2 \left| \sin \frac{\varphi}{2} \right| \leq |\varphi| = |\arg z|.$$

$$|z-1| = |z-|z|+|z|-1| \leq |z-|z|| + ||z|-1| = ||z|-1| + |z| \left| \frac{z}{|z|} - 1 \right| \leq ||z|-1| + |z| |\arg z|.$$

(2)



如图, z_1, z_2, z_3 在同一圆周上, $\alpha = \arg \frac{z_3 - z_2}{z_3 - z_1}$, $\beta = \arg \frac{z_2}{z_1}$ 。由于同弧所对圆周角是

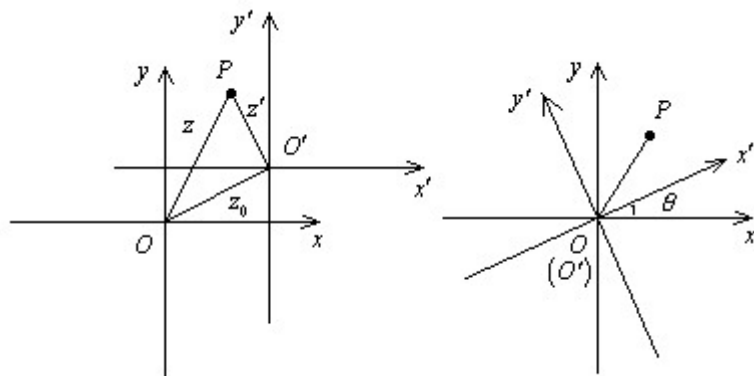
圆心角的一半, 所以 $\alpha = \frac{1}{2} \beta$, 即 $\arg \frac{z_3 - z_2}{z_3 - z_1} = \frac{1}{2} \arg \frac{z_2}{z_1}$ 。

6. 用复数 z 表示曲线上的变点。(1) 写出经过点 a 且与复数 b 所代表的矢量平行的直线方程; (2) 写出以 d 和 $-d$ 为焦点, 长轴长 $2a$ 的椭圆方程 ($a > |d|$)。

(1) 矢量 $z-a$ 与矢量 b 平行, 所以 $z-a = kb$, k 为实数;

(2) 由椭圆定义得 $|z-d| + |z+d| = 2a$ 。

7. 用复数运算法则推出: (1) 平面直角坐标平移公式; (2) 平面直角坐标旋转公式。



(1) 设坐标系 $x'O'y'$ 的原点 O' 在坐标系 xOy 中的坐标是 (x_0, y_0) 。 P 点在 xOy 系中的坐标是 (x, y) ，在 $x'O'y'$ 系中坐标 (x', y') 。如上面左图，令 $\overrightarrow{OP} = z$ ， $\overrightarrow{O'P} = z'$ ， $\overrightarrow{OO'} = z_0$ 。

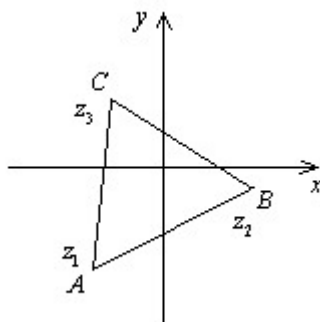
则 $z' = z - z_0$ ，即 $x' + iy' = x - x_0 + i(y - y_0)$ ，由此得 $x' = x - x_0$ ， $y' = y - y_0$ 。

(2) 将坐标系 xOy 绕原点逆时针旋转 θ 角得到坐标系 $x'O'y'$ 。如上面右图， $x'O'y'$ 系中 z'

只是比 xOy 系中 z 的幅角小 θ ，即 $z' = ze^{-i\theta}$ ，由此得 $x' = x \cos \theta + y \sin \theta$ ，

$y' = -x \sin \theta + y \cos \theta$ 。

8. 设复数 z_1 ， z_2 ， z_3 满足 $\frac{z_2 - z_1}{z_3 - z_1} = \frac{z_1 - z_3}{z_2 - z_3}$ 。证明： $|z_2 - z_1| = |z_3 - z_2| = |z_1 - z_3|$ 。



如图， $\frac{z_2 - z_1}{z_3 - z_1} = \frac{|AB|}{|AC|} e^{i\angle A}$ ， $\frac{z_1 - z_3}{z_2 - z_3} = \frac{|AC|}{|BC|} e^{i\angle C}$ 。所以 $\frac{|AB|}{|AC|} = \frac{|AC|}{|BC|}$ ， $\angle A = \angle C$ 。

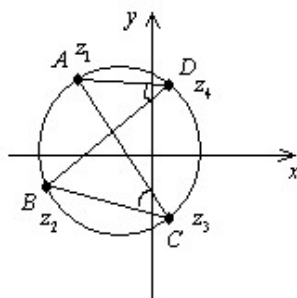
由 $\angle A = \angle C$ 可得 $|AB| = |BC|$ ，代入 $\frac{|AB|}{|AC|} = \frac{|AC|}{|BC|}$ 可得 $|AB| = |BC| = |AC|$ ，即

$$|z_2 - z_1| = |z_3 - z_2| = |z_1 - z_3|。$$

9. (1) 给出 z_1, z_2, z_3 三点共线的充要条件; (2) 给出 z_1, z_2, z_3, z_4 四点共圆的充要条件。

(1) 若三点共线, 则矢量 $z_1 - z_3$ 与矢量 $z_2 - z_3$ 平行, 反之也成立。所以三点共线的充要条件是 $\frac{z_1 - z_3}{z_2 - z_3} = \text{实数}$ 。

(2)

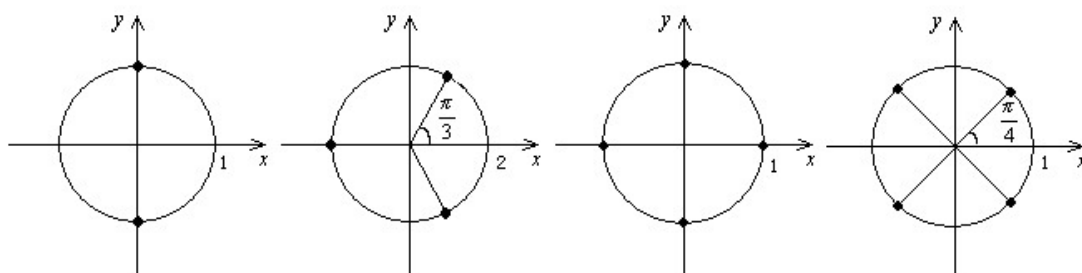


如图若四点共圆, 则有 $\angle ACB = \angle ADB$ (同弧所对圆周角相等)。反之也成立。写成复数形式即为 $\frac{z_1 - z_3}{z_2 - z_3} \bigg/ \frac{z_1 - z_4}{z_2 - z_4} = \text{实数}$ 。

10. 求下列方程的根, 并在复平面上画出它们的位置。

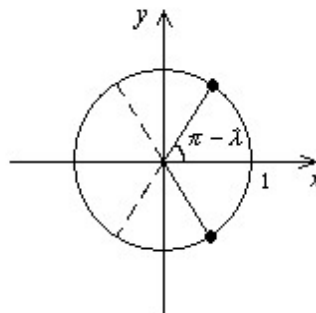
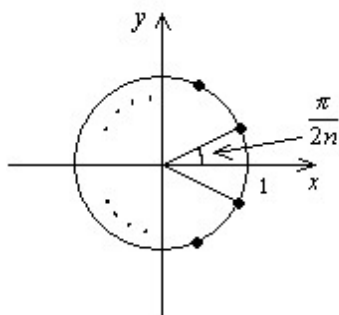
(1) $z^2 + 1 = 0$; (2) $z^3 + 8 = 0$; (3) $z^4 - 1 = 0$; (4) $z^4 + 1 = 0$; (5) $z^{2n} + 1 = 0$, n 为正整数; (6) $z^2 + 2z \cos \lambda + 1 = 0$, $0 < \lambda < \pi$ 。

(1) $z = \pm i$; (2) $z = 2e^{\pm i\frac{\pi}{3}}, -2$; (3) $z = \pm 1, \pm i$; (4) $z = e^{\pm i\frac{\pi}{4}}, e^{\pm i\frac{3\pi}{4}}$;



(5) $z = e^{i(\pi+2k\pi)/2n}$, $k = 0, 1, \dots, 2n-1$;

(6) $z = -e^{\pm i\lambda}$ 。



11. 设 $z = p + iq$ 是实系数方程 $a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n = 0$ 的根, 证明 $\bar{z} = p - iq$ 也是此方程的根。

对方程两边取共轭得 $a_0 + a_1 \bar{z} + a_2 \bar{z}^2 + \cdots + a_n \bar{z}^n = 0$, 即 $\bar{z}$ 也满足此方程。

12. 证明: $\sin^4 \varphi = \frac{1}{8}(\cos 4\varphi - 4\cos 2\varphi + 3)$ 。

$$\begin{aligned} e^{4i\varphi} - 4e^{2i\varphi} + 3 &= e^{4i\varphi} - 1 + 4(1 - e^{2i\varphi}) = e^{2i\varphi}(e^{2i\varphi} - e^{-2i\varphi}) - 4e^{i\varphi}(e^{i\varphi} - e^{-i\varphi}) \\ &= 2ie^{2i\varphi} \sin 2\varphi - 8ie^{i\varphi} \sin \varphi = 2i(\cos 2\varphi + i \sin 2\varphi) \sin 2\varphi - 8i(\cos \varphi + i \sin \varphi) \sin \varphi \\ &= 8\sin^4 \varphi + i(\sin 4\varphi - 4\sin 2\varphi) \end{aligned}$$

取等式两边实部即得证。

13. 把 $\sin n\varphi$ 和 $\cos n\varphi$ 用 $\sin \varphi$ 和 $\cos \varphi$ 表示出来。

$$\begin{aligned} \cos n\varphi + i \sin n\varphi &= (\cos \varphi + i \sin \varphi)^n = \sum_{k=0}^n \frac{n! i^k}{k!(n-k)!} \cos^{n-k} \varphi \sin^k \varphi \\ &= \sum_{k=0}^{[n/2]} (-1)^k \frac{n!}{(2k)!(n-2k)!} \cos^{n-2k} \varphi \sin^{2k} \varphi \\ &\quad + i \sum_{k=0}^{[(n-1)/2]} (-1)^k \frac{n!}{(2k+1)!(n-2k-1)!} \cos^{n-2k-1} \varphi \sin^{2k+1} \varphi \end{aligned}$$

比较两边实部和虚部得:

$$\cos n\varphi = \sum_{k=0}^{[n/2]} (-1)^k \frac{n!}{(2k)!(n-2k)!} \cos^{n-2k} \varphi \sin^{2k} \varphi ;$$

$$\sin n\varphi = \sum_{k=0}^{[(n-1)/2]} (-1)^k \frac{n!}{(2k+1)!(n-2k-1)!} \cos^{n-2k-1} \varphi \sin^{2k+1} \varphi .$$

14. 将下列和式表示成有限形式: (1) $\sum_{k=1}^n \cos k\varphi$; (2) $\sum_{k=1}^n \sin k\varphi$ 。

$$\sum_{k=1}^n e^{ik\varphi} = e^{i\varphi} \frac{1 - e^{in\varphi}}{1 - e^{i\varphi}} = e^{i\varphi} \frac{e^{i\frac{n\varphi}{2}} \left(e^{i\frac{n\varphi}{2}} - e^{-i\frac{n\varphi}{2}} \right)}{e^{i\frac{\varphi}{2}} \left(e^{i\frac{\varphi}{2}} - e^{-i\frac{\varphi}{2}} \right)} = e^{i\frac{n+1}{2}\varphi} \frac{\sin \frac{n\varphi}{2}}{\sin \frac{\varphi}{2}}$$

比较两边实部和虚部得:

$$\sum_{k=1}^n \cos k\varphi = \frac{\sin \frac{n\varphi}{2} \cos \frac{(n+1)\varphi}{2}}{\sin \frac{\varphi}{2}}, \quad \sum_{k=1}^n \sin k\varphi = \frac{\sin \frac{n\varphi}{2} \sin \frac{(n+1)\varphi}{2}}{\sin \frac{\varphi}{2}}。$$

15. 证明: $\sin \frac{\pi}{n} \cdot \sin \frac{2\pi}{n} \cdots \sin \frac{(n-1)\pi}{n} = \frac{n}{2^{n-1}}$ 。

记 $1, z_1, z_2, \dots, z_{n-1}$ 为方程 $z^n = 1$ 的 n 个根, 即 $z_k = e^{i\frac{2k}{n}\pi}$, $k=1, 2, \dots, n-1$ 。则有

$$z^n - 1 = (z-1)(z-z_1)(z-z_2) \cdots (z-z_{n-1}),$$

$$\text{所以 } (z-z_1)(z-z_2) \cdots (z-z_{n-1}) = \frac{z^n - 1}{z-1} = z^{n-1} + z^{n-2} + \cdots + z + 1。$$

令上式两边 $z=1$, 则有 $\prod_{k=1}^{n-1} \left(1 - e^{i\frac{2k}{n}\pi} \right) = n$ 。

$$1 - e^{i\frac{2k}{n}\pi} = -e^{i\frac{k\pi}{n}} \left(e^{i\frac{k\pi}{n}} - e^{-i\frac{k\pi}{n}} \right) = -2ie^{i\frac{k\pi}{n}} \sin \frac{k\pi}{n} = 2e^{-i\frac{\pi}{2}} e^{i\frac{k\pi}{n}} \sin \frac{k\pi}{n},$$

$$\prod_{k=1}^{n-1} \left(1 - e^{i\frac{2k}{n}\pi} \right) = 2^{n-1} e^{-i\frac{n-1}{2}\pi + i \sum_{k=1}^{n-1} \frac{k\pi}{n}} \prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = 2^{n-1} \prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = n, \quad \text{即 } \prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}。$$

16. 求下列序列 $\{a_n\}$ 的聚点和极限, 如果是实数序列, 则同时求出上下极限。

$$(1) a_n = (-1)^n \frac{n}{2n+1}; \quad (2) a_n = (-1)^n \frac{1}{2n+1}; \quad (3) a_n = n + (-1)^n (2n+1)i;$$

$$(4) a_n = 2n+1 + (-1)^n ni; \quad (5) a_n = \left(1 + \frac{i}{n} \right) \sin \frac{n\pi}{6}; \quad (6) a_n = \left(1 + \frac{1}{2n} \right) \cos \frac{n\pi}{3}。$$

- (1) 聚点 $\pm 1/2$, 极限无, 上极限 $1/2$, 下极限 $-1/2$;
 (2) 聚点 0 , 极限 0 , 上下极限 0 ;
 (3) 聚点 ∞ , 极限 ∞ ;
 (4) 聚点 ∞ , 极限 ∞ ;
 (5) 聚点 $0, \pm 1/2, \pm \sqrt{3}/2, \pm 1$, 极限无;
 (6) 聚点 $\pm 1/2, \pm 1$, 极限无, 上极限 1 , 下极限 -1 。

17. 证明序列 $a_n = 1 + \frac{1}{2} + \frac{1}{3} \cdots + \frac{1}{n} - \ln n$ 极限存在。

先证 $\frac{x}{x+1} \leq \ln(1+x) \leq x$, 其中 $x \geq 0$ 。

令 $f(x) = \ln(1+x) - x$, 则 $f'(x) = \frac{1}{1+x} - 1 = -\frac{x}{1+x} \leq 0$, 所以 $f(x) \leq f(0) = 0$, 不等式右半部分得证, 同样可证左半部分。

由此可得 $\frac{1}{n+1} < \ln\left(1 + \frac{1}{n}\right) < \frac{1}{n}$ 。

$a_{n+1} - a_n = \frac{1}{n+1} - \ln\left(1 + \frac{1}{n}\right) < 0$, 即 a_n 是递减序列。

$$\begin{aligned} \text{由 } \frac{1}{n} > \ln\left(1 + \frac{1}{n}\right) \text{ 得 } a_n &> \ln(1+1) + \ln\left(1 + \frac{1}{2}\right) + \ln\left(1 + \frac{1}{3}\right) + \cdots + \ln\left(1 + \frac{1}{n}\right) - \ln n \\ &= \ln\left(2 \cdot \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n}\right) - \ln n = \ln\left(1 + \frac{1}{n}\right) > 0 \end{aligned}$$

即 a_n 是递减有下界序列, 所以极限存在。

18. 证明 Lagrange 恒等式: $\left| \sum_{k=1}^n z_k w_k \right|^2 = \left(\sum_{k=1}^n |z_k|^2 \right) \left(\sum_{k=1}^n |w_k|^2 \right) - \sum_{k < j} |z_k \bar{w}_j - z_j \bar{w}_k|^2$ 。

$$\begin{aligned} \text{右边} &= \sum_{k,j} |z_k|^2 |w_j|^2 - \sum_{k < j} (z_k \bar{w}_j - z_j \bar{w}_k) (\bar{z}_k w_j - \bar{z}_j w_k) \\ &= \sum_{k,j} |z_k|^2 |w_j|^2 - \sum_{k < j} |z_k|^2 |w_j|^2 - \sum_{k < j} |z_j|^2 |w_k|^2 + \sum_{k < j} z_k \bar{z}_j w_k \bar{w}_j + \sum_{k < j} z_j \bar{z}_k w_k \bar{w}_j \\ &= \sum_{k,j} |z_k|^2 |w_j|^2 - \sum_{k \neq j} |z_k|^2 |w_j|^2 + \sum_{k \neq j} z_k \bar{z}_j w_k \bar{w}_j \\ &= \sum_k |z_k|^2 |w_k|^2 + \sum_{k \neq j} z_k \bar{z}_j w_k \bar{w}_j = \sum_k z_k w_k \sum_j \overline{z_j w_j} \\ &= \sum_k z_k w_k \overline{\sum_k z_k w_k} = \left| \sum_{k=1}^n z_k w_k \right|^2 = \text{左边}。 \end{aligned}$$

19. 试证明：从条件 $\lim_{n \rightarrow \infty} z_n = A$ 可以导出 $\lim_{n \rightarrow \infty} \frac{z_1 + z_2 + \cdots + z_n}{n} = A$ 。又当 $A = \infty$ 时上述结论还正确吗？

由 $\lim_{n \rightarrow \infty} z_n = A$ 知对于任意的 $\varepsilon > 0$ ，存在整数 N_1 ，使得当 $n > N_1$ 时有 $|z_n - A| < \frac{\varepsilon}{2}$ 。

对于给定的 N_1 ，存在 N_2 ，使得 $\frac{|z_1 - A| + |z_2 - A| + \cdots + |z_{N_1} - A|}{N_2} < \frac{\varepsilon}{2}$ 。当 $n > N = \max(N_1, N_2)$

$$\begin{aligned} \text{时, } \left| \frac{z_1 + z_2 + \cdots + z_n}{n} - A \right| &= \frac{1}{n} \left| (z_1 - A) + (z_2 - A) + \cdots + (z_n - A) \right| \\ &\leq \frac{1}{n} \left(|z_1 - A| + |z_2 - A| + \cdots + |z_{N_1} - A| \right) + \frac{1}{n} \left(|z_{N_1+1} - A| + |z_{N_1+2} - A| + \cdots + |z_n - A| \right) \\ &< \frac{|z_1 - A| + |z_2 - A| + \cdots + |z_{N_1} - A|}{N_2} + \frac{1}{n - N_1} \left(|z_{N_1+1} - A| + |z_{N_1+2} - A| + \cdots + |z_n - A| \right) \\ &< \frac{\varepsilon}{2} + \frac{1}{n - N_1} (n - N_1) \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

即 $\lim_{n \rightarrow \infty} \frac{z_1 + z_2 + \cdots + z_n}{n} = A$ 。

20. 设 $z = x + iy$, $z_0 = x_0 + iy_0$, $c = a + ib$, 并且已知 $f(z) = u(x, y) + iv(x, y)$, 证明

$$\lim_{z \rightarrow z_0} f(z) = c \text{ 与 } \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} u(x, y) = a, \quad \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} v(x, y) = b \text{ 等价。}$$

由于 $|f - c| = |(u - a) + i(v - b)|$, 所以 $|u - a|$ (或 $|v - b|$) $\leq |f - c| \leq |u - a| + |v - b|$ 。

若 $\lim_{z \rightarrow z_0} f(z) = c$, 对于任意的 $\varepsilon > 0$, 则存在 δ , 当 $|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$,

就有 $|u - a|$ (或 $|v - b|$) $\leq |f - c| < \varepsilon$, 即 $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} u(x, y) = a$, $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} v(x, y) = b$ 。

同样的, 若 $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} u(x, y) = a$, $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} v(x, y) = b$, 就有 $\lim_{z \rightarrow z_0} f(z) = c$ 。

21. 证明: $f(z) = \frac{1}{1 - z^2}$ 在单位圆 $|z| < 1$ 内连续但不一致连续。

易证 $f(z)$ 连续 (初等函数)。下面证 $f(z)$ 在单位圆内不一致连续。

定义在 D 上的函数 $f(z)$ 在 D 上一致连续的充要条件: 任意的 $\{x_n\} \subset D$, $\{y_n\} \subset D$, 只

要 $\lim_{n \rightarrow \infty} (x_n - y_n) = 0$, 就有 $\lim_{n \rightarrow \infty} [f(x_n) - f(y_n)] = 0$

$$\text{令 } x_n = \sqrt{1 - \frac{1}{n}}, \quad y_n = \sqrt{1 - \frac{2}{n}}, \quad \text{则 } x_n - y_n = \frac{1}{n} \frac{1}{\sqrt{1 - \frac{1}{n}} + \sqrt{1 - \frac{2}{n}}} \rightarrow 0,$$

$f(x_n) - f(y_n) = \frac{n}{2} \rightarrow \infty$ 。所以 $f(z)$ 在单位圆 $|z| < 1$ 内不一致连续。

22. 证明下列函数在 $z = 0$ 点连续:

$$(1) \quad f(z) = \begin{cases} \frac{[\operatorname{Re}(z^2)]^2}{z^2}, & z \neq 0, \\ 0, & z = 0 \end{cases}, \quad (2) \quad f(z) = |z|.$$

$$(1) \text{ 在 } z \neq 0 \text{ 处, } |f(z)| = \frac{(x^2 - y^2)^2}{|x^2 - y^2 + 2ixy|} \leq \frac{(x^2 - y^2)^2}{|x^2 - y^2|} = |x^2 - y^2|,$$

$\lim_{z \rightarrow 0} |f(z)| = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} |x^2 - y^2| = 0$, 即 $\lim_{z \rightarrow 0} f(z) = f(0)$, 所以 $f(z)$ 在 $z = 0$ 点连续。

$$(2) \lim_{\substack{z \rightarrow 0 \\ x \rightarrow 0 \\ y \rightarrow 0}} f(z) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \sqrt{x^2 + y^2} = 0 = f(0)。$$

23. 判断下列函数在何处可导（并求出导数），在何处解析：

$$(1) |z|; (2) \bar{z}; (3) z^m, m=0,1,2,\dots; (4) e^z; (5) (x^2+2y)+i(x^2+y^2);$$

$$(6) (x-y)^2+2i(x+y); (7) z \operatorname{Re} z; (8) 1/z; (9) \cos z; (10) \operatorname{sh} z。$$

由可导充分条件（25 题）判别：

(1) 全平面不可导，不解析；

(2) 全平面不可导，不解析；

(3) 全平面可导，解析， $(z^m)' = mz^{m-1}$ ；

(4) 全平面可导，解析， $(e^z)' = e^z$ ；

(5) 除 $(-1, -1)$ 点可导外，全平面其余处处不可导，全平面不解析；

(6) 除 $y=x-1$ 的线上处处可导外，其余点不可导，全平面不解析；

(7) $z=0$ 点可导， $(z \operatorname{Re} z)' \Big|_{z=0} = 0$ ，其余处处不可导，全平面不解析；

(8) 除 $z=0$ 点外在扩充全平面上可导，解析， $(1/z)' = -1/z^2$ ；

(9) 全平面可导，解析， $(\cos z)' = -\sin z$ ；

(10) 全平面可导，解析， $(\operatorname{sh} z)' = \operatorname{ch} z$ 。

24. 证明极坐标下的 Cauchy-Riemann 条件： $\frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi}$, $\frac{\partial v}{\partial \rho} = -\frac{1}{\rho} \frac{\partial u}{\partial \varphi}$ 。

由变换关系 $x = \rho \cos \varphi$, $y = \rho \sin \varphi$ 可得

$$\frac{\partial u}{\partial \rho} = \frac{\partial u}{\partial x} \cos \varphi + \frac{\partial u}{\partial y} \sin \varphi, \quad \frac{\partial u}{\partial \varphi} = -\rho \frac{\partial u}{\partial x} \sin \varphi + \rho \frac{\partial u}{\partial y} \cos \varphi,$$

$$\frac{\partial v}{\partial \rho} = \frac{\partial v}{\partial x} \cos \varphi + \frac{\partial v}{\partial y} \sin \varphi, \quad \frac{\partial v}{\partial \varphi} = -\rho \frac{\partial v}{\partial x} \sin \varphi + \rho \frac{\partial v}{\partial y} \cos \varphi。$$

$$\text{变换得 } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial \rho} \cos \varphi - \frac{1}{\rho} \frac{\partial u}{\partial \varphi} \sin \varphi, \quad (1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \rho} \sin \varphi + \frac{1}{\rho} \frac{\partial u}{\partial \varphi} \cos \varphi, \quad (2)$$

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial \rho} \cos \varphi - \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \sin \varphi, \quad (3)$$

$$\frac{\partial v}{\partial y} = \frac{\partial v}{\partial \rho} \sin \varphi + \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \cos \varphi. \quad (4)$$

代入直角坐标的 C-R 方程 $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ 有

$$\frac{\partial u}{\partial \rho} \cos \varphi - \frac{1}{\rho} \frac{\partial u}{\partial \varphi} \sin \varphi = \frac{\partial v}{\partial \rho} \sin \varphi + \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \cos \varphi, \quad (5)$$

$$\frac{\partial u}{\partial \rho} \sin \varphi + \frac{1}{\rho} \frac{\partial u}{\partial \varphi} \cos \varphi = -\frac{\partial v}{\partial \rho} \cos \varphi + \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \sin \varphi. \quad (6)$$

$$(5) \times \cos \varphi + (6) \times \sin \varphi \text{ 得 } \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi},$$

$$(5) \times \sin \varphi - (6) \times \cos \varphi \text{ 得 } \frac{\partial v}{\partial \rho} = -\frac{1}{\rho} \frac{\partial u}{\partial \varphi}.$$

25. 证明：若函数 $f(z)$ 的偏导数在 $z = z_0$ 点连续，且满足 C-R 方程，则 $f(z)$ 在 $z = z_0$ 点可导。

由 $f(z)$ 的偏导数在 $z = z_0$ 点连续可知 $u(x, y), v(x, y)$ 在 $z = z_0$ 点可微，所以有

$$u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0) = \frac{\partial u}{\partial x} \Big|_{(x_0, y_0)} \Delta x + \frac{\partial u}{\partial y} \Big|_{(x_0, y_0)} \Delta y + \varepsilon_1,$$

$$v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0) = \frac{\partial v}{\partial x} \Big|_{(x_0, y_0)} \Delta x + \frac{\partial v}{\partial y} \Big|_{(x_0, y_0)} \Delta y + \varepsilon_2.$$

上面的 $\varepsilon_1, \varepsilon_2$ 是 $|\Delta z| = \sqrt{\Delta x^2 + \Delta y^2}$ 的高阶无穷小。记 $a = \frac{\partial u}{\partial x} \Big|_{(x_0, y_0)} = \frac{\partial v}{\partial y} \Big|_{(x_0, y_0)},$

$$b = -\frac{\partial u}{\partial y} \Big|_{(x_0, y_0)} = \frac{\partial v}{\partial x} \Big|_{(x_0, y_0)}, \text{ 则以上两式写成}$$

$$u(z_0 + \Delta z) - u(z_0) - a\Delta x + b\Delta y = \varepsilon_1,$$

$$v(z_0 + \Delta z) - v(z_0) - b\Delta x - a\Delta y = \varepsilon_2.$$

$$\begin{aligned} \left| \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} - (a + ib) \right| &= \left| \frac{f(z_0 + \Delta z) - f(z_0) - a\Delta z - ib\Delta z}{\Delta z} \right| \\ &= \left| \frac{u(z_0 + \Delta z) - u(z_0) - a\Delta x + b\Delta y + i[v(z_0 + \Delta z) - v(z_0) - b\Delta x - a\Delta y]}{\Delta z} \right| \\ &= \left| \frac{\varepsilon_1 + i\varepsilon_2}{\Delta z} \right| \leq \frac{|\varepsilon_1|}{|\Delta z|} + \frac{|\varepsilon_2|}{|\Delta z|} \end{aligned}$$

当 $\Delta z \rightarrow 0$ 时, $\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \rightarrow a + ib$, 即 $f(z)$ 在 $z = z_0$ 点可导。

26. 设 $f(z) = \begin{cases} \frac{z^5}{|z|^4}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ 。(1) 证明: 当 $z \rightarrow 0$ 时, $\frac{f(z)}{z}$ 的极限不存在; (2) 若

$$u = \operatorname{Re} f(z), \quad v = \operatorname{Im} f(z), \quad \text{证明: } u(x, 0) = x, \quad v(0, y) = y, \quad u(0, y) = v(x, 0) = 0;$$

(3) 证明: u, v 的偏导数存在, 且 C-R 方程成立, 但 (1) 中已证明 $f'(0)$ 不存在, 这个结论和 25 题矛盾吗?

(1) $z \neq 0$ 时, $\frac{f(z)}{z} = \frac{z^4}{|z|^4} = \frac{(x^2 - y^2)^2 - 4x^2y^2 + 4ixy(x^2 - y^2)}{(x^2 + y^2)^2}$ 。在直线 $y = kx$ 上

$$\frac{f(z)}{z} = \frac{(1 - k^2)^2 - 4k^2 + 4ik(1 - k^2)}{(1 + k^2)^2}, \quad \text{可见 } z \text{ 沿不同直线趋于 } 0 \text{ 将有不同极限值, 所以}$$

$$\frac{f(z)}{z} \text{ 的极限不存在。}$$

(2) $z \neq 0$ 时, $f(z) = \frac{x^5 - 10x^3y^2 + 5xy^4 + i(5x^4y - 10x^2y^3 + y^5)}{(x^2 + y^2)^2}$ 。

$$\text{所以 } u = \begin{cases} \frac{x^5 - 10x^3y^2 + 5xy^4}{(x^2 + y^2)^2}, & (x, y) \neq 0 \\ 0, & (x, y) = 0 \end{cases}, \quad v = \begin{cases} \frac{5x^4y - 10x^2y^3 + y^5}{(x^2 + y^2)^2}, & (x, y) \neq 0 \\ 0, & (x, y) = 0 \end{cases}.$$

容易看出 $u(x, 0) = x, \quad v(0, y) = y, \quad u(0, y) = v(x, 0) = 0$ 。

(3) 仿 (1) 的方法, u , v 在 $z=0$ 处的偏导数不存在。

27. 利用极坐标下的 C-R 方程 (24 题) 证明: $f'(z) = \frac{\rho}{z} \left(\frac{\partial u}{\partial \rho} + i \frac{\partial v}{\partial \rho} \right) = \frac{1}{z} \left(\frac{\partial v}{\partial \varphi} - i \frac{\partial u}{\partial \varphi} \right)$ 。

利用 24 题 (1) (3) 式,

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial \rho} \cos \varphi - \frac{1}{\rho} \frac{\partial u}{\partial \varphi} \sin \varphi + i \frac{\partial v}{\partial \rho} \cos \varphi - i \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \sin \varphi$$

代入极坐标 C-R 方程,

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial \rho} \cos \varphi + \frac{\partial v}{\partial \rho} \sin \varphi + i \frac{\partial v}{\partial \rho} \cos \varphi - i \frac{\partial u}{\partial \rho} \sin \varphi \\ &= \frac{\rho}{z} (\cos \varphi + i \sin \varphi) \left(\frac{\partial u}{\partial \rho} \cos \varphi + \frac{\partial v}{\partial \rho} \sin \varphi + i \frac{\partial v}{\partial \rho} \cos \varphi - i \frac{\partial u}{\partial \rho} \sin \varphi \right) \\ &= \frac{\rho}{z} \left(\frac{\partial u}{\partial \rho} + i \frac{\partial v}{\partial \rho} \right) \end{aligned}$$

再利用极坐标 C-R 方程有 $f'(z) = \frac{\rho}{z} \left(\frac{\partial u}{\partial \rho} + i \frac{\partial v}{\partial \rho} \right) = \frac{1}{z} \left(\frac{\partial v}{\partial \varphi} - i \frac{\partial u}{\partial \varphi} \right)$ 。

28. 设 $\rho = \rho(x, y)$, $\varphi = \varphi(x, y)$ 是实变量 x, y 的实函数。若 $f(z) = \rho(\cos \varphi + i \sin \varphi)$ 是

$z = x + iy$ 的解析函数, 证明: $\frac{\partial \rho}{\partial x} = \rho \frac{\partial \varphi}{\partial y}$, $\frac{\partial \rho}{\partial y} = -\rho \frac{\partial \varphi}{\partial x}$ 。

$$\frac{\partial u}{\partial x} = \frac{\partial \rho}{\partial x} \cos \varphi - \rho \sin \varphi \frac{\partial \varphi}{\partial x}, \quad \frac{\partial u}{\partial y} = \frac{\partial \rho}{\partial y} \cos \varphi - \rho \sin \varphi \frac{\partial \varphi}{\partial y},$$

$$\frac{\partial v}{\partial x} = \frac{\partial \rho}{\partial x} \sin \varphi + \rho \cos \varphi \frac{\partial \varphi}{\partial x}, \quad \frac{\partial v}{\partial y} = \frac{\partial \rho}{\partial y} \sin \varphi + \rho \cos \varphi \frac{\partial \varphi}{\partial y}.$$

由 C-R 方程可得:

$$\frac{\partial \rho}{\partial x} \cos \varphi - \rho \sin \varphi \frac{\partial \varphi}{\partial x} = \frac{\partial \rho}{\partial y} \sin \varphi + \rho \cos \varphi \frac{\partial \varphi}{\partial y}, \quad (1)$$

$$\frac{\partial \rho}{\partial x} \sin \varphi + \rho \cos \varphi \frac{\partial \varphi}{\partial x} = -\frac{\partial \rho}{\partial y} \cos \varphi + \rho \sin \varphi \frac{\partial \varphi}{\partial y}. \quad (2)$$

$$(1) \times \cos \varphi + (2) \times \sin \varphi \text{ 得 } \frac{\partial \rho}{\partial x} = \rho \frac{\partial \varphi}{\partial y},$$

$$(1) \times \sin \varphi - (2) \times \cos \varphi \text{ 得 } \frac{\partial \rho}{\partial y} = -\rho \frac{\partial \varphi}{\partial x}.$$

29. 设 $r = r(\rho, \varphi)$, $\theta = \theta(\rho, \varphi)$ 是实变数 ρ, φ 的实函数。若 $f(z) = r(\cos \theta + i \sin \theta)$ 解

$$\text{析, 其中 } z = \rho e^{i\varphi}, \text{ 试证: } \frac{\partial r}{\partial \rho} = \frac{r}{\rho} \frac{\partial \theta}{\partial \varphi}, \quad \frac{\partial r}{\partial \varphi} = -\rho r \frac{\partial \theta}{\partial \rho}.$$

$$\frac{\partial u}{\partial \rho} = \frac{\partial r}{\partial \rho} \cos \theta - r \sin \theta \frac{\partial \theta}{\partial \rho}, \quad \frac{\partial u}{\partial \varphi} = \frac{\partial r}{\partial \varphi} \cos \theta - r \sin \theta \frac{\partial \theta}{\partial \varphi},$$

$$\frac{\partial v}{\partial \rho} = \frac{\partial r}{\partial \rho} \sin \theta + r \cos \theta \frac{\partial \theta}{\partial \rho}, \quad \frac{\partial v}{\partial \varphi} = \frac{\partial r}{\partial \varphi} \sin \theta + r \cos \theta \frac{\partial \theta}{\partial \varphi}.$$

由极坐标 C-R 方程 (24 题) 得:

$$\frac{\partial r}{\partial \rho} \cos \theta - r \sin \theta \frac{\partial \theta}{\partial \rho} = \frac{1}{\rho} \frac{\partial r}{\partial \varphi} \sin \theta + \frac{r}{\rho} \cos \theta \frac{\partial \theta}{\partial \varphi}, \quad (1)$$

$$\frac{\partial r}{\partial \rho} \sin \theta + r \cos \theta \frac{\partial \theta}{\partial \rho} = -\frac{1}{\rho} \frac{\partial r}{\partial \varphi} \cos \theta + \frac{r}{\rho} \sin \theta \frac{\partial \theta}{\partial \varphi}. \quad (2)$$

$$(1) \times \cos \varphi + (2) \times \sin \varphi \text{ 得 } \frac{\partial r}{\partial \rho} = \frac{r}{\rho} \frac{\partial \theta}{\partial \varphi},$$

$$(2) \times \sin \varphi - (1) \times \cos \varphi \text{ 得 } \frac{\partial r}{\partial \varphi} = -\rho r \frac{\partial \theta}{\partial \rho}.$$

30. 若函数 $f(z) = u + iv$ 在 G 内解析, 且 $f(z) \neq$ 常数, 试讨论下列函数是否也是 G 内的

解析函数: (1) $u - iv$; (2) $-u - iv$; (3) $-v + iu$; (4) $v + iu$ 。

由 C-R 方程判断, (2) (3) 解析, (1) (4) 不解析。

31. 设 $z = x + iy$, 已知解析函数 $f(z) = u(x, y) + iv(x, y)$ 的实部或虚部如下, 试求其导

数 $f'(z)$: (1) $u = e^{-y} \cos x$; (2) $u = \operatorname{ch} x \cos y$; (3) $v = \sin x \operatorname{sh} y$; (4) $v = \frac{x}{x^2 + y^2}$;

(5) $u = \ln(x^2 + y^2)$; (6) $v = x^3 + 6x^2y - 3xy^2 - 2y^3$ 。

$$(1) \quad \frac{\partial u}{\partial x} = -e^{-y} \sin x, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = e^{-y} \cos x,$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = e^{-y}(-\sin x + i \cos x) = i e^{-y+ix} = i e^{iz};$$

$$(2) f'(z) = \operatorname{sh} z; (3) f'(z) = \sin z; (4) f'(z) = -\frac{i}{z^2}; (5) f'(z) = \frac{2}{z};$$

$$(6) f'(z) = 3(2+i)z^2.$$

32. 根据下列条件确定解析函数 $f(z) = u + iv$ 。

$$(1) u = x + y; (2) u = \sin x \operatorname{ch} y; (3) v = \frac{x}{x^2 + y^2}; (4) v = \arctan \frac{y}{x}.$$

$$(1) dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy = -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy = d(y - x), \text{ 所以 } v = y - x + C \text{ (} C \text{ 为实常数),}$$

$$f = u + iv = (1-i)x + (1+i)y + iC = (1-i)z + iC;$$

$$(2) f = \sin z + iC; (3) f = \frac{i}{z} + C; (4) f = \ln z + C.$$

33. 若 $f(z) = u(x, y) + iv(x, y)$ 解析, 且 $u(x, y) - v(x, y) = (x - y)(x^2 + 4xy + y^2)$, 求 $f(z)$ 。

$$\text{由已知可得 } \frac{\partial u}{\partial x} - \frac{\partial v}{\partial x} = x^2 + 4xy + y^2 + (x - y)(2x + 4y),$$

$$\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} = -(x^2 + 4xy + y^2) + (x - y)(4x + 2y),$$

$$\text{再由 C-R 方程 } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \text{ 由以上四式可解出:}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 6xy, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 3(x^2 - y^2).$$

$$\text{解出 } u = 3x^2y - y^3 + C_1, v = 3xy^2 - x^3 + C_2, \text{ 由已知条件可确定 } C_1 = C_2 = C.$$

$$f(z) = u + iv = 3x^2y - y^3 + i(3xy^2 - x^3) + (1+i)C = iz^3 + (1+i)C$$

34. 若 $u(x, y)$ 具有连续三阶偏导数, 且 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, 证明函数 $\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$ 解析。

$$\text{令 } U = \frac{\partial u}{\partial x}, \quad V = -\frac{\partial u}{\partial y}, \quad \text{则 } \frac{\partial U}{\partial x} - \frac{\partial V}{\partial y} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial U}{\partial y} + \frac{\partial V}{\partial x} = \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial y \partial x} = 0,$$

即该函数满足 C-R 方程, 所以解析。

35. 如果 $u(x, y)$ 和 $v(x, y)$ 都是调和函数, 讨论下列函数是否也是调和函数:

$$(1) \quad U = u[v(x, y), 0]; \quad (2) \quad U = u[0, v(x, y)]; \quad (3) \quad U = u(x, y)v(x, y);$$

$$(4) \quad U = u(x, y) + v(x, y).$$

$$(1) \quad \frac{\partial U}{\partial x} = \frac{\partial u}{\partial x} \bigg|_{(v,0)} \frac{\partial v}{\partial x}, \quad \frac{\partial^2 U}{\partial x^2} = \frac{\partial^2 u}{\partial x^2} \bigg|_{(v,0)} \left(\frac{\partial v}{\partial x} \right)^2 + \frac{\partial u}{\partial x} \bigg|_{(v,0)} \frac{\partial^2 v}{\partial x^2},$$

$$\frac{\partial U}{\partial y} = \frac{\partial u}{\partial x} \bigg|_{(v,0)} \frac{\partial v}{\partial y}, \quad \frac{\partial^2 U}{\partial y^2} = \frac{\partial^2 u}{\partial x^2} \bigg|_{(v,0)} \left(\frac{\partial v}{\partial y} \right)^2 + \frac{\partial u}{\partial x} \bigg|_{(v,0)} \frac{\partial^2 v}{\partial y^2}.$$

$$\begin{aligned} \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} &= \frac{\partial^2 u}{\partial x^2} \bigg|_{(v,0)} \left[\left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right] + \frac{\partial u}{\partial x} \bigg|_{(v,0)} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \\ &= \frac{\partial^2 u}{\partial x^2} \bigg|_{(v,0)} \left[\left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right] \end{aligned}$$

上式右边一般不等于 0, 所以不是调和函数。

$$(2) \quad \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = \frac{\partial^2 u}{\partial y^2} \bigg|_{(0,v)} \left[\left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right], \quad \text{不是调和函数。}$$

$$(3) \quad \frac{\partial^2 U}{\partial x^2} = \frac{\partial^2 u}{\partial x^2} v + 2 \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + u \frac{\partial^2 v}{\partial x^2}, \quad \frac{\partial^2 U}{\partial y^2} = \frac{\partial^2 u}{\partial y^2} v + 2 \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} + u \frac{\partial^2 v}{\partial y^2},$$

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = 2 \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} \right), \quad \text{不是调和函数。}$$

$$(4) \quad \frac{\partial^2 U}{\partial x^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial x^2}, \quad \frac{\partial^2 U}{\partial y^2} = \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 v}{\partial y^2}, \quad \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = 0, \quad \text{是调和函数。}$$

36. 假设函数 $f(z)$ 在区域 G 内任意一点都满足 $f'(z) = 0$, 证明 $f(z)$ 在 G 内为常数。

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 0, \quad \text{所以 } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = 0, \quad \text{即 } u, v \text{ 都是常数, } f(z) \text{ 为常数。}$$

37. 若 $f(z)$ 在区域 G 内解析, 且 $\operatorname{Im} f(z) = 0$, 证明 $f(z)$ 在 G 内为常数。

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 0$, $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 0$, 所以 u 为常数, 又 $v = 0$, 所以 $f(z)$ 在 G 内为常数。

38. 若 $f(z) = u(x, y) + iv(x, y)$ 在区域 G 内解析, 且 $au + bv = c$, 其中 a, b, c 是不为 0 的实常数, 证明 $f(z)$ 在 G 内为常数。如果 a, b, c 是不为 0 的复常数, 结论还成立吗?

由已知可得 $a \frac{\partial u}{\partial x} + b \frac{\partial v}{\partial x} = 0$, $a \frac{\partial u}{\partial y} + b \frac{\partial v}{\partial y} = 0$, 代入 C-R 方程,

$a \frac{\partial u}{\partial x} - b \frac{\partial u}{\partial y} = 0$, $a \frac{\partial u}{\partial y} + b \frac{\partial u}{\partial x} = 0$, 两式消去 $\frac{\partial u}{\partial y}$ 得 $(a^2 + b^2) \frac{\partial u}{\partial x} = 0$, 由于 a, b 为

实数, 所以 $\frac{\partial u}{\partial x} = 0$ 。同样可得 $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = 0$, 所以 $f(z)$ 为常数。如果 a, b, c 是复数,

结论不成立。

39. 若 $f(z)$ 和 $g(z)$ 在 $z = a$ 点解析, 且 $f(a) = g(a) = 0$, 而 $g'(a) \neq 0$, 试证:

$$\lim_{z \rightarrow a} \frac{f(z)}{g(z)} = \frac{f'(a)}{g'(a)}.$$

$$\lim_{z \rightarrow a} \frac{f(z)}{g(z)} = \lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a} \bigg/ \frac{g(z) - g(a)}{z - a} = \frac{f'(a)}{g'(a)}.$$

40. 设 z 沿着从原点出发的射线运动, 其模无限增大, 试讨论函数 e^z 的变化趋势。

$e^z = e^x e^{iy}$, 若 $x > 0$ (即 $-\frac{\pi}{2} < \arg z < \frac{\pi}{2}$), $e^z \rightarrow \infty$ 。

若 $x < 0$ (即 $\frac{\pi}{2} < \arg z < \frac{3\pi}{2}$), $e^z \rightarrow 0$ 。

若 $x = 0$ (即 $\arg z = \pm \frac{\pi}{2}$), e^z 的实部虚部在 $[-1, 1]$ 之间振荡。

41. 证明下列公式:

$$(1) \sin(z_1 \pm z_2) = \sin z_1 \cos z_2 \pm \cos z_1 \sin z_2;$$

$$(2) \cos(z_1 \pm z_2) = \cos z_1 \cos z_2 \mp \sin z_1 \sin z_2;$$

$$(3) \operatorname{sh} z = -i \sin iz;$$

$$(4) \operatorname{ch} z = \cos iz;$$

$$(5) \cos^{-1} z = -i \ln(z + \sqrt{z^2 - 1});$$

$$(6) \tan^{-1} z = \frac{1}{2i} \ln \frac{1+iz}{1-iz};$$

$$(7) \operatorname{ch}^2 z - \operatorname{sh}^2 z = 1;$$

$$(8) 1 - \operatorname{th}^2 z = \operatorname{sech}^2 z.$$

$$\begin{aligned} (1) \sin z_1 \cos z_2 + \cos z_1 \sin z_2 &= \frac{(e^{iz_1} - e^{-iz_1})(e^{iz_2} + e^{-iz_2}) + (e^{iz_1} + e^{-iz_1})(e^{iz_2} - e^{-iz_2})}{4i} \\ &= \frac{e^{i(z_1+z_2)} - e^{-i(z_1+z_2)}}{2i} = \sin(z_1 + z_2) \end{aligned}$$

同样得, $\sin z_1 \cos z_2 - \cos z_1 \sin z_2 = \sin(z_1 - z_2)$

$$\begin{aligned} (2) \cos z_1 \cos z_2 - \sin z_1 \sin z_2 &= \frac{(e^{iz_1} + e^{-iz_1})(e^{iz_2} + e^{-iz_2}) + (e^{iz_1} - e^{-iz_1})(e^{iz_2} - e^{-iz_2})}{4} \\ &= \frac{e^{i(z_1+z_2)} + e^{-i(z_1+z_2)}}{2} = \cos(z_1 + z_2) \end{aligned}$$

同样得, $\cos z_1 \cos z_2 + \sin z_1 \sin z_2 = \cos(z_1 - z_2)$

$$(3) \operatorname{sh} z = \frac{e^z - e^{-z}}{2} = \frac{e^{-i(iz)} - e^{i(iz)}}{2} = -i \frac{e^{i(iz)} - e^{-i(iz)}}{2i} = -i \sin iz$$

$$(4) \operatorname{ch} z = \frac{e^z + e^{-z}}{2} = \frac{e^{-i(iz)} + e^{i(iz)}}{2} = \cos iz$$

$$(5) \text{ 令 } z = \cos w = \frac{e^{iw} + e^{-iw}}{2}, \text{ 则 } e^{2iw} - 2ze^{iw} + 1 = 0, \text{ 解出 } e^{iw} = z + \sqrt{z^2 - 1},$$

所以 $\cos^{-1} z = w = -i \ln(z + \sqrt{z^2 - 1})$

$$(6) \text{ 令 } z = \tan w = -i \frac{e^{iw} - e^{-iw}}{e^{iw} + e^{-iw}}, \text{ 解出 } e^{2iw} = \frac{1+iz}{1-iz}. \text{ 所以 } \tan^{-1} z = w = \frac{1}{2i} \ln \frac{1+iz}{1-iz}$$

$$(7) \operatorname{ch}^2 z - \operatorname{sh}^2 z = \frac{(e^z + e^{-z})^2 - (e^z - e^{-z})^2}{4} = 1$$

$$(8) 1 - \operatorname{th}^2 z = 1 - \left(\frac{e^z - e^{-z}}{e^z + e^{-z}} \right)^2 = \frac{4}{(e^z + e^{-z})^2} = \operatorname{sech}^2 z$$

42. 证明下列公式: (1) $(\operatorname{sh} z)' = \operatorname{ch} z$; (2) $(\operatorname{ch} z)' = \operatorname{sh} z$;

$$(3) (\operatorname{th} z)' = \operatorname{sech}^2 z; (4) (\operatorname{cth} z)' = -\operatorname{csch}^2 z$$

$$(1) (\operatorname{sh} z)' = \left(\frac{e^z - e^{-z}}{2} \right)' = \frac{e^z + e^{-z}}{2} = \operatorname{ch} z$$

$$(2) (\operatorname{ch} z)' = \left(\frac{e^z + e^{-z}}{2} \right)' = \frac{e^z - e^{-z}}{2} = \operatorname{sh} z$$

$$(3) (\operatorname{th} z)' = \left(\frac{e^z - e^{-z}}{e^z + e^{-z}} \right)' = \frac{(e^z + e^{-z})^2 - (e^z - e^{-z})^2}{(e^z + e^{-z})^2} = \left(\frac{2}{e^z + e^{-z}} \right)^2 = \operatorname{sech}^2 z$$

$$(4) (\operatorname{cth} z)' = \left(\frac{e^z + e^{-z}}{e^z - e^{-z}} \right)' = \frac{(e^z - e^{-z})^2 - (e^z + e^{-z})^2}{(e^z - e^{-z})^2} = - \left(\frac{2}{e^z - e^{-z}} \right)^2 = -\operatorname{csch}^2 z$$

43. 证明下列不等式: (1) $|\operatorname{sh} y| \leq |\sin(x + iy)| \leq \operatorname{ch} y$;

$$(2) |\operatorname{sh} y| \leq |\cos(x + iy)| \leq \operatorname{ch} y.$$

$$(1) \sin(x + iy) = \sin x \cos iy + \cos x \sin iy = \sin x \operatorname{ch} y + i \cos x \operatorname{sh} y,$$

$$\text{所以 } |\sin(x + iy)| = \sqrt{\sin^2 x \operatorname{ch}^2 y + \cos^2 x \operatorname{sh}^2 y}.$$

$$\text{代入 } \operatorname{ch}^2 y = 1 + \operatorname{sh}^2 y \text{ 得 } |\sin(x + iy)| = \sqrt{\sin^2 x + \operatorname{sh}^2 y} \geq \operatorname{sh} y,$$

$$\text{代入 } \operatorname{sh}^2 y = \operatorname{ch}^2 y - 1 \text{ 得 } |\sin(x + iy)| = \sqrt{\operatorname{ch}^2 y - \cos^2 x} \leq \operatorname{ch} y. \text{ 不等式得证。}$$

(2) 同 (1)。

44. 解下列方程: (1) $\operatorname{sh} z = 0$; (2) $2\operatorname{ch}^2 z - 3\operatorname{ch} z + 1 = 0$; (3) $\sin^2 z - \frac{5}{2}\sin z + 1 = 0$;

(4) $\tan z = i$ 。

(1) $\operatorname{sh} z = \frac{e^z - e^{-z}}{2} = 0$, 即 $e^{2z} = 1 = e^{i2k\pi}$, 所以 $z = ik\pi$, ($k = 0, \pm 1, \pm 2, \dots$);

(2) 解得 $\operatorname{ch} z = 1$ 或 $1/2$, 即 $e^{2z} = 1 = e^{i2k\pi}$ 或 $e^z = \frac{1}{2} \pm i \frac{\sqrt{3}}{2} = e^{i(\pm\pi/3+2k\pi)}$, 所以 $z = ik\pi$, $i(\pm\pi/3+2k\pi)$, ($k = 0, \pm 1, \pm 2, \dots$);

(3) $z = \pi/6 + 2k\pi$, $5\pi/6 + 2k\pi$, $\pi/2 - i \ln(2 \pm \sqrt{3}) + 2k\pi$, ($k = 0, \pm 1, \pm 2, \dots$);

(4) 无解。

46. 扇形区域 $0 < \arg z < \frac{\pi}{3}$ 经变换 $w = z^3$ 后边成什么区域? (上半平面)

47. 试证: 圆 $A(x^2 + y^2) + Bx + Cy + D = 0$ 经变换 $w = \frac{1}{z}$ 后仍为圆, 并讨论 $A = 0$ 及 $D = 0$ 的情况。

由于 $x^2 + y^2 = |z|^2 = z\bar{z}$, $x = \frac{1}{2}(z + \bar{z})$, $y = \frac{1}{2i}(z - \bar{z})$, 圆方程可写为

$Az\bar{z} + \frac{1}{2}(B - iC)z + \frac{1}{2}(B + iC)\bar{z} + D = 0$ 。令 $E = \frac{1}{2}(B + iC)$, 则方程写成

$Az\bar{z} + \bar{E}z + Ez + D = 0$, 这就是圆的标准方程。代入 $z = 1/w$, 得到

$Dw\bar{w} + Ew + \bar{E}\bar{w} + A = 0$, 仍是圆方程。

$A = 0$ 时, 将直线变换为圆, $D = 0$ 时, 将圆变换成直线。

48. $w = e^{iz}$ 把实轴上线段 $0 \leq x < 2\pi$ 变为什么图形?

由 $y = 0$ 得 $w = e^{ix}$, 所以 $|w| = 1$ 。 $0 \leq x < 2\pi$ 即是 $0 \leq \arg w < 2\pi$, 所以变为单位圆。

49. 双纽线 $\rho^2 = 2a^2 \cos 2\varphi$ 经变换 $w = z^2$ 后变为什么图形?

令 $z = \rho e^{i\varphi}$, 则 $w = \rho^2 e^{2i\varphi}$ 。令 $w = re^{i\theta}$, 则 $\theta = 2\varphi$, $r = \rho^2 = 2a^2 \cos \theta$, 即变换为圆。

50. 证明: $w = -i \frac{z-1}{z+1}$ 将直线 $y = ax$ 变为圆。

直线方程写为 $\bar{A}z + A\bar{z} = 0$, 其中 $A = \frac{1}{2}(a-i)$ 。代入 $z = \frac{1+iw}{1-iw}$ 得 $aw\bar{w} + w + \bar{w} - a = 0$, 即为圆方程。

51. 证明: 在变换 $w = \frac{1}{2}\left(z - \frac{1}{z}\right)$ 下, z 平面上以原点为圆心, e^β ($\beta > 0$) 为半径的圆变

为 w 平面上的椭圆, 焦点为 $\pm i$, 长短半轴分别为 $\text{ch } \beta$ 及 $\text{sh } \beta$ 。

令 $z = \rho e^{i\varphi}$, 则圆方程为 $\rho = e^\beta$ 。

$$\begin{aligned} w &= \frac{1}{2}(e^\beta e^{i\varphi} - e^{-\beta} e^{-i\varphi}) = \frac{1}{2}[e^\beta \cos \varphi + ie^\beta \sin \varphi - e^{-\beta} \cos \varphi + ie^{-\beta} \sin \varphi] \\ &= \text{sh } \beta \cos \varphi + i \text{ch } \beta \sin \varphi \end{aligned}$$

令 $w = x + iy$, 则 $x = \text{sh } \beta \cos \varphi$, $y = \text{ch } \beta \sin \varphi$, 消去 φ 得 $\frac{x^2}{\text{sh}^2 \beta} + \frac{y^2}{\text{ch}^2 \beta} = 1$, 即以 $\pm i$ 为

焦点, $\text{ch } \beta$ 及 $\text{sh } \beta$ 为长短半轴的椭圆。

52. 设 $w = u(x, y) + iv(x, y)$ 解析, 且 $\frac{dw}{dz} \neq 0$, 试证曲线族 $u(x, y) = C_1$, $v(x, y) = C_2$ (C_1 , C_2 为任意实常数) 互相正交。

设 $\mathbf{n}_1$, $\mathbf{n}_2$ 为过点 (x, y) 的两曲线在该点的法向量, 即 $\mathbf{n}_1 = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right)$, $\mathbf{n}_2 = \left(\frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}\right)$ 。

则 $\mathbf{n}_1 \cdot \mathbf{n}_2 = \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} = -\frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial x} = 0$, 即两曲线正交。

53. 判断下列函数是单值的还是多值的:

$$(1) z + \sqrt{z-1}; (2) \frac{1}{1+\ln z}; (3) \sqrt{\cos z}; (4) \ln \sin z; (5) \frac{\cos \sqrt{z}}{\sqrt{z}}; (6) \frac{\sin \sqrt{z}}{\sqrt{z}}.$$

明显 (1) ~ (4) 都是多值函数。

用 $\pm w$ 表示 z 的两个平方根, 即 $\sqrt{z} = w$ 或 $-w$ 。取 $\sqrt{z} = w$, 则 $\frac{\cos \sqrt{z}}{\sqrt{z}} = \frac{\cos w}{w}$, 取

$$\sqrt{z} = -w, \text{ 则 } \frac{\cos \sqrt{z}}{\sqrt{z}} = \frac{\cos(-w)}{-w} = -\frac{\cos w}{w}, \text{ 即 } \frac{\cos \sqrt{z}}{\sqrt{z}} \text{ 为多值函数, 同样可得, } \frac{\sin \sqrt{z}}{\sqrt{z}}$$

为单值函数。

54. 找出下列函数的枝点, 并讨论 z 绕各个枝点移动一周回到原处函数值的变化。若同时绕两个, 三个枝点, 又会出现怎样的情况?

$$(1) \sqrt{1-z^3}; (2) z + \sqrt{z^2-1}; (3) \sqrt{\frac{z-a}{z-b}}; (4) \frac{1}{1+\ln z}; (5) \frac{\cos \sqrt{z}}{\sqrt{z}};$$

$$(6) \sqrt[3]{z^2-4}; (7) \sqrt[3]{z^2(z+1)}; (8) \ln(z^2+1).$$

(1) $w = \sqrt{1-z^3} = \sqrt{(1-z)\left(z-e^{\frac{2\pi}{3}}\right)\left(z-e^{-\frac{2\pi}{3}}\right)}$, 当 z 逆时针绕 1 点一圈 (不包围 $e^{\frac{2\pi}{3}}$ 和 $e^{-\frac{2\pi}{3}}$) 回到原处, 因子 $\sqrt{1-z}$ 顺时针绕 0 点旋转 π , 另外两个因子 $\sqrt{z-e^{\frac{2\pi}{3}}}$ 和 $\sqrt{z-e^{-\frac{2\pi}{3}}}$

不变, 故 w 顺时针绕 0 点旋转 π 。当 z 逆时针绕 $e^{\frac{2\pi}{3}}$ (或 $e^{-\frac{2\pi}{3}}$) 点一圈 (不包围另外两点)

回到原处, w 逆时针绕 0 点旋转 π 。所以 $1, e^{\pm \frac{2\pi}{3}}$ 为枝点。若 z 逆时针绕 1 和 $e^{\frac{2\pi}{3}}$ 两点一

圈 (不包围 $e^{-\frac{2\pi}{3}}$) 回到原处, w 不变, 同样的, z 逆时针绕任意两个枝点一圈 (不包围另一个枝点) 回到原处, w 都不变。若 z 逆时针绕这三个枝点一圈回到原处, w 顺时针绕 0 点旋转 π , 所以 ∞ 也是枝点。

(2) 枝点是 ± 1 ;

(3) 枝点是 a, b (z 绕 b 逆时针一圈回到原处, 因子 $\frac{1}{\sqrt{z-b}}$ 顺时针绕 0 点旋转 π);

(4) 枝点是 $0, \infty$;

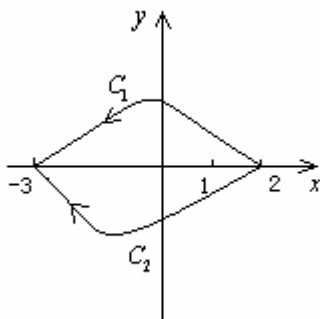
(5) 枝点是 $0, \infty$;

(6) 枝点是 $\pm 2, \infty$;

(7) 枝点是 $0, -1$;

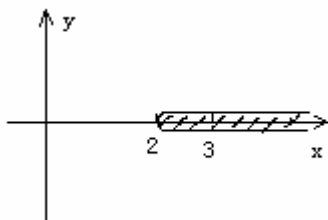
(8) 枝点是 $\pm i, \infty$;

55. 函数 $w = z + \sqrt{z-1}$, 规定 $w(2)=1$, 是分别求当 z 沿着图中的 C_1 和 C_2 连续变化时 $w(-3)$ 之值。



若规定 $z=2$ 处 $\arg(z-1) = 2\pi$, 则有 $w(2)=1$ 。 z 沿 C_1 连续变化到 -3 时, $\arg(z-1) = 3\pi$, 所以 $w(-3) = -3 + \sqrt{4}e^{i\frac{3\pi}{2}} = -3 - 2i$ 。沿 C_2 有 $w(-3) = -3 + 2i$ 。

56. 规定函数 $w = z\sqrt[3]{z-2}$ 在下图割线上岸的幅角为 0 , 试求该函数在割线下岸 $z=3$ 处的数值, 又问, 这个函数有几个单值分枝: 求出在其他分枝中割线下岸 $z=3$ 处的函数值。



$z-2$ 在割线上岸幅角为 0 , 下岸为 2π , 所以 $w(3) = 3e^{i\frac{2\pi}{3}}$ 。

有三个单值分枝。规定割线上岸幅角为 2π , 则下岸为 4π , $w(3) = 3e^{i\frac{4\pi}{3}}$,

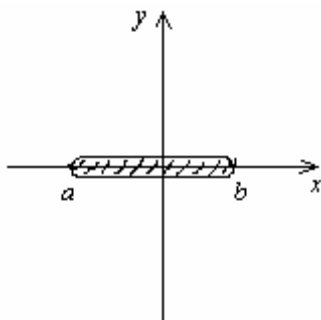
规定割线上岸幅角为 4π , 则下岸为 6π , $w(3) = 3$ 。

57. 函数 $w = \sqrt{(z-a)(z-b)}$ 的割线有多少种可能的做法? 试在两种不同做法下讨论单值分枝的规定。设 a, b 为实数, 且 $a \neq b$ 。

a, b 为枝点, 连接 a, b 的任意线段都可作为割线, 所以有无穷种做法。

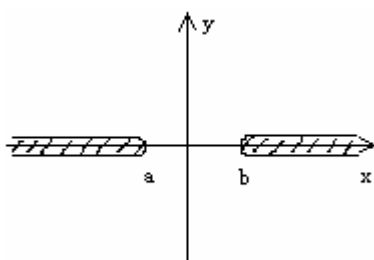
其中两种做法:

(1)



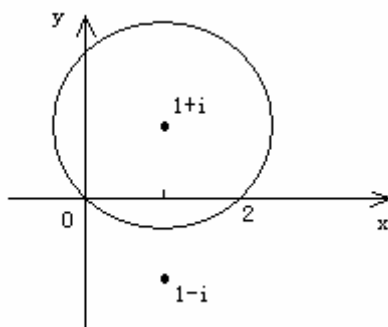
规定割线上岸 $\arg(z-a) + \arg(z-b) = \pi$ 和 3π 可得两个单枝分枝。

(2)



可规定正实轴割线上岸 $\arg(z-a) + \arg(z-b)$ 分别为 $0, 2\pi$ 。

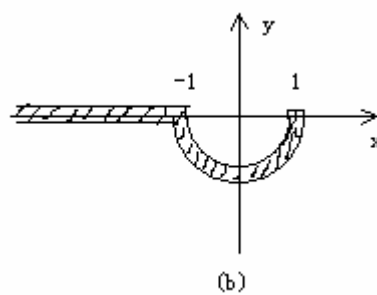
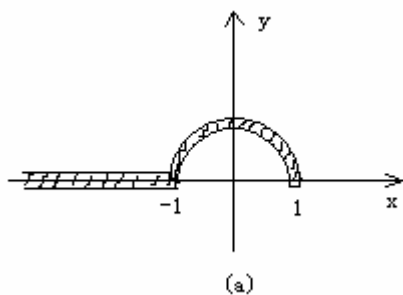
58. 规定函数 $w = \sqrt{z^2 - 2z + 2}$, $w(0) = \sqrt{2}$ 。求当 z 由原点出发沿圆 $|z - (1+i)| = \sqrt{2}$ 逆时针方向通过 x 轴时的函数值。又当 z 回到原点时函数之值如何?



$w = \sqrt{(z - \sqrt{2}e^{i\pi/4})(z - \sqrt{2}e^{-i\pi/4})}$ 。只要规定 $z=0$ 时 $\arg(z - \sqrt{2}e^{i\pi/4}) = -\frac{3\pi}{4}$, $\arg(z - \sqrt{2}e^{-i\pi/4}) = \frac{3\pi}{4}$ 就有 $w(0) = \sqrt{2}$ 。当 z 沿圆逆时针到达 $z=2$ 时, $\arg(z - \sqrt{2}e^{i\pi/4}) = -\frac{\pi}{4}$, $\arg(z - \sqrt{2}e^{-i\pi/4}) = \frac{\pi}{4}$, 所以 $w(2) = \sqrt{\sqrt{2}e^{-i\pi/4} \cdot \sqrt{2}e^{i\pi/4}} = \sqrt{2}$ 。当 z 回到原点时, $\arg(z - \sqrt{2}e^{i\pi/4}) = \frac{5\pi}{4}$, $\arg(z - \sqrt{2}e^{-i\pi/4}) = \frac{3\pi}{4}$, 所以

$$w(0) = \sqrt{\sqrt{2}e^{i5\pi/4}} \cdot \sqrt{2}e^{i3\pi/4} = \sqrt{2}e^{i\pi} = -\sqrt{2}。$$

59. 函数 $w = \ln(1-z^2)$ ，规定 $w(0)=0$ ，试讨论当 z 分别限制在以下两图中变化时， $w(3)$ 之值。

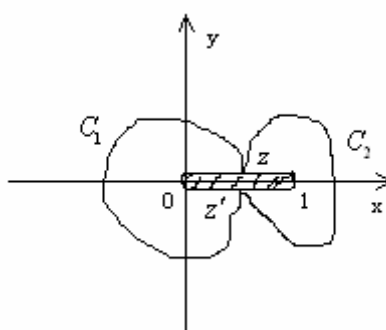


(a) $w = \ln[(1-z)(z+1)]$ 。规定 $z=0$ 时 $\arg(1-z)=0$ ， $\arg(z+1)=0$ 就有 $w(0)=0$ 。

z 从下半平面到达 $z=3$ 时有 $\arg(1-z)=\pi$ ， $\arg(z+1)=0$ ，所以 $w(3) = \ln(2e^{i\pi} \cdot 4e^{i0}) = 3\ln 2 + i\pi$ 。

(b) z 从上半平面到达 $z=3$ 时有 $\arg(1-z)=-\pi$ ， $\arg(z+1)=0$ ，所以 $w(3) = \ln(2e^{-i\pi} \cdot 4e^{i0}) = 3\ln 2 - i\pi$ 。

60. 函数 $w = \sqrt[4]{z(1-z)^3}$ 在割线上岸函数值与下岸函数值有何不同？割线如下图。



若割线上岸上一点 z 由左边(曲线 C_1)绕到割线下岸同一处(记为 z')，则 z 的辐角增加 2π ，

即 $z' = ze^{i2\pi}$ ， $1-z$ 的辐角不变，即 $(1-z)' = 1-z$ 。所以

$$w' = \sqrt[4]{z'(1-z)'^3} = \sqrt[4]{ze^{i2\pi}(1-z)^3} = we^{i\pi/2}。$$

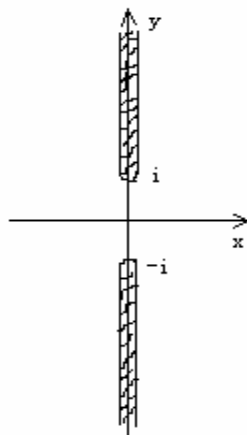
若 z 由右边 (曲线 C_2) 绕到割线下岸同一处, 则 z 的辐角不变, $1-z$ 的辐角减小 2π ,

$$w' = \sqrt[4]{z \left[(1-z)e^{-i2\pi} \right]^3} = we^{-i3\pi/2}。$$

61. 规定 $0 \leq \arg z < 2\pi$, 求 $w = \sqrt{z}$ 在 $z = i$ 处的导数值。

$$w'(z) = \frac{1}{2\sqrt{z}}, \quad w'(e^{i\pi/2}) = \frac{1}{2}e^{-i\pi/4} = \frac{1}{2\sqrt{2}}(1-i)。$$

62. 规定 $z = 0$ 处 $\arctan z = \pi$, 求在 $z = 2$ 处的导数值。割线做法如图。



$$\arctan z = \frac{1}{2i} \ln \frac{i-z}{z+i}, \quad (\arctan z)' = \frac{1}{z^2+1}, \quad (\arctan z)' \Big|_{z=2} = \frac{1}{5}。$$

虽然导函数 $f'(z)$ 是单值函数, 但它是在 $f(z)$ 的单值分枝中定义的, 否则极限值

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \text{ 不定。}$$

63. 证明: 若函数 $f(z)$ 在区域 G 内解析, 其模为一常数, 则函数 $f(z)$ 本身也必为一常数。

证: 令 $f(z) = Ae^{i\varphi(x,y)} = A\cos\varphi + iA\sin\varphi$, 其中 $\varphi(x,y)$ 为实函数。由于 $f(z)$ 解析, C-R

方程为: $-A\sin\varphi \frac{\partial\varphi}{\partial x} = A\cos\varphi \frac{\partial\varphi}{\partial y}, \quad A\sin\varphi \frac{\partial\varphi}{\partial y} = A\cos\varphi \frac{\partial\varphi}{\partial x}$ 。可由此解出 $\frac{\partial\varphi}{\partial x} = 0$,

$\frac{\partial\varphi}{\partial y} = 0$, 即 $\varphi(x,y)$ 为常数, 所以 $f(z)$ 为常数。

64. $f(z) = \frac{z^{1-p}(1-z)^p}{2z}$, $-1 < p < 2$ 。在实轴上沿 0 到 1 做割线, 规定沿割线上岸

$\arg z = \arg(1-z) = 0$, 试计算 $f(\pm i)$ 。

$$z = i \text{ 时, } \arg z = \frac{\pi}{2}, \arg(1-z) = -\frac{\pi}{4}, f(i) = \frac{(e^{i\pi/2})^{1-p} (2^{1/2} e^{-i\pi/4})^p}{2i} = 2^{\frac{p-1}{2}} e^{i\frac{3}{4}p\pi}。$$

$$z \text{ 从左边由割线上岸绕到 } z = -i, \text{ 则 } \arg z = \frac{3\pi}{2}, \arg(1-z) = \frac{\pi}{4}, f(-i) = 2^{\frac{p-1}{2}} e^{-i\frac{5}{4}p\pi}。$$

$$z \text{ 从右边由割线上岸绕到 } z = -i, \text{ 则 } \arg z = -\frac{\pi}{2}, \arg(1-z) = -\frac{7\pi}{4}, f(-i) = 2^{\frac{p-1}{2}} e^{-i\frac{5}{4}p\pi}。$$

65. 试按给定的路径计算下列积分:

(1) $\int_{-1}^1 \frac{dz}{z}$, (i) 沿路径 $C_1: |z|=1$ 的上半圆周, (ii) 沿路径 $C_2: |z|=1$ 的下半圆周;

(2) $\int_0^{2+i} \operatorname{Re} z dz$, (i) C_1 : 直线段 $[0, 2]$ 和 $[2, 2+i]$ 组成的折线, (ii) C_2 : 直线段 $z = (2+i)t$, $0 \leq t \leq 1$ 。

(1) (i) $\int_{C_1} \frac{dz}{z} = \int_{\pi}^0 \frac{de^{i\varphi}}{e^{i\varphi}} = \int_{\pi}^0 i d\varphi = -\pi i$,

(ii) $\int_{C_2} \frac{dz}{z} = \int_{-\pi}^0 \frac{de^{i\varphi}}{e^{i\varphi}} = \int_{-\pi}^0 i d\varphi = \pi i$;

(2) (i) $\int_{C_1} \operatorname{Re} z dz = \int_0^2 x dx + i \int_0^1 2 dy = 2 + 2i$,

(ii) $\int_{C_2} \operatorname{Re} z dz = \int_0^1 2t d(2+i)t = 2(2+i) \int_0^1 t dt = 2+i$ 。

66. 计算: (1) $\int_{|z|=1} \frac{dz}{z}$; (2) $\int_{|z|=1} \frac{dz}{|z|}$; (3) $\int_{|z|=1} \frac{|dz|}{z}$; (4) $\int_{|z|=1} \left| \frac{dz}{z} \right|$ 。

(1) 令 $z = e^{i\varphi}$, $\int_{|z|=1} \frac{dz}{z} = \int_0^{2\pi} \frac{de^{i\varphi}}{e^{i\varphi}} = \int_0^{2\pi} i d\varphi = 2\pi i$;

(2) $\int_{|z|=1} \frac{dz}{|z|} = \int_0^{2\pi} de^{i\varphi} = e^{i\varphi} \Big|_0^{2\pi} = 0$;

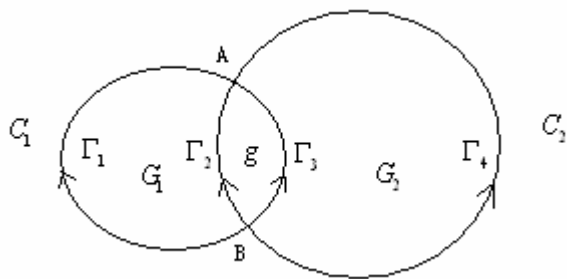
(3) $\int_{|z|=1} \frac{|dz|}{z} = \int_0^{2\pi} \frac{|ie^{i\varphi} d\varphi|}{e^{i\varphi}} = \int_0^{2\pi} e^{-i\varphi} d\varphi = ie^{-i\varphi} \Big|_0^{2\pi} = 0$;

(4) $\int_{|z|=1} \left| \frac{dz}{z} \right| = \int_0^{2\pi} d\varphi = 2\pi$ 。

67. 考虑两简单闭合曲线 C_1 , C_2 , 彼此相交于 A, B 两点。设 C_1 与 C_2 所包围的内部区域

分别是 G_1 与 G_2 , 其公共区域为 g 。若 $f(z)$ 在曲线 C_1 , C_2 上解析, 且在区域 $G_1 - g$ 及

$G_2 - g$ 内解析, 试证明: $\oint_{C_1} f(z) dz = \oint_{C_2} f(z) dz$ 。



如图, $\Gamma_1 \sim \Gamma_4$ 表示四条边界线, C_1 是 Γ_1 的负向加上 Γ_3 的正向, C_2 是 Γ_2 的负向加上 Γ_4 的正向。

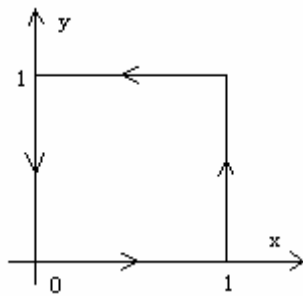
$$\oint_{C_1} f(z) dz - \oint_{C_2} f(z) dz = -\int_{\Gamma_1} f dz + \int_{\Gamma_3} f dz + \int_{\Gamma_2} f dz - \int_{\Gamma_4} f dz$$

Γ_1 的负向加上 Γ_2 的正向就是 $G_1 - g$ 的边界, 所以 $-\int_{\Gamma_1} f dz + \int_{\Gamma_2} f dz = 0$,

同样的, $\int_{\Gamma_3} f dz - \int_{\Gamma_4} f dz = 0$, 所以有 $\oint_{C_1} f(z) dz = \oint_{C_2} f(z) dz$ 。

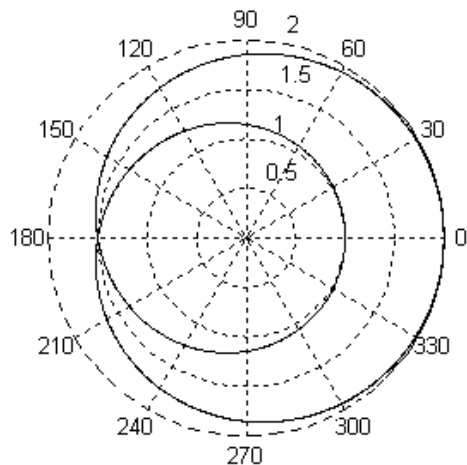
68. 对于任一解析函数的实部或虚部, Cauchy 定理仍成立吗? 如果成立, 试证明之, 如果不成立, 试说明理由, 并举一例。

不成立。取 $f(z) = z$, 则实部 $u(x, y) = x$ 。取如下积分路径:



$$\oint_C u dz = \int_0^1 x dx + i \int_0^1 dy + \int_1^0 x dx = i.$$

69. 证明: $\oint_C \frac{dz}{z} = 4\pi i$, 其中积分路径 C 为闭合曲线 $\rho = 2 - \sin^2 \frac{\varphi}{4}$ 。这个结果和围绕原点一圈 $\oint \frac{dz}{z} = 2\pi i$ 的结论有矛盾吗? 为什么?



$$\oint_C \frac{dz}{z} = \int_0^{4\pi} \frac{d \left[\left(2 - \sin^2 \frac{\varphi}{4} \right) e^{i\varphi} \right]}{\left(2 - \sin^2 \frac{\varphi}{4} \right) e^{i\varphi}} = \int_0^{4\pi} \frac{-\frac{1}{4} \sin \frac{\varphi}{2} e^{i\varphi} + i \left(2 - \sin^2 \frac{\varphi}{4} \right) e^{i\varphi}}{\left(2 - \sin^2 \frac{\varphi}{4} \right) e^{i\varphi}} d\varphi$$

$$= \int_0^{4\pi} i d\varphi - \frac{1}{2} \int_0^{4\pi} \frac{\sin \frac{\varphi}{2}}{3 + \cos \frac{\varphi}{2}} d\varphi$$

上式右边第二项被积函数以 4π 为周期，所以积分限可换为 $-2\pi \sim 2\pi$ ，被积函数又是奇函数，故积分为 0，所以 $\oint_C \frac{dz}{z} = 4\pi i$ 。由上图可看出， C 绕原点两圈，并不与围绕原点一圈

$\oint \frac{dz}{z} = 2\pi i$ 的结论矛盾。

70. 计算 $\oint_{|z|=3} \frac{2z^2 - 15z + 30}{z^3 - 10z^2 + 32z - 32} dz$ 。

$$\text{原式} = \oint_{|z|=3} \frac{2z^2 - 15z + 30}{(z-2)(z-4)^2} dz = 2\pi i \cdot \left. \frac{2z^2 - 15z + 30}{(z-4)^2} \right|_{z=2} = 4\pi i$$

71. 计算：(1) $\oint_C \frac{\sin \frac{\pi z}{4}}{z^2 - 1} dz$ ， C 分别为：(i) $|z| = \frac{1}{2}$ ，(ii) $|z-1| = 1$ ，(iii) $|z| = 3$ ；

(2) $\oint_C \frac{e^{iz}}{z^2 + 1} dz$ ， C 分别为：(i) $|z-i| = 1$ ，(ii) $|z| = 2$ ，(iii) $|z+i| + |z-i| = 2\sqrt{2}$ 。

(1) (i) 积分路径不包围任何奇点，故积分值为 0，

$$(ii) \text{ 积分路径包围奇点 } z=1, \oint_C \frac{\sin \frac{\pi z}{4}}{z^2-1} dz = 2\pi i \cdot \frac{\sin \frac{\pi z}{4}}{z+1} \bigg|_{z=1} = \frac{\pi}{\sqrt{2}} i,$$

(iii) 积分路径包围奇点 $z = \pm 1$, C_1, C_2 为单独包围 $z = \pm 1$ 的闭路径,

$$\oint_C \frac{\sin \frac{\pi z}{4}}{z^2-1} dz = \oint_{C_1} \frac{\sin \frac{\pi z}{4}}{z^2-1} dz + \oint_{C_2} \frac{\sin \frac{\pi z}{4}}{z^2-1} dz = 2\pi i \left(\frac{\sin \frac{\pi z}{4}}{z+1} \bigg|_{z=1} + \frac{\sin \frac{\pi z}{4}}{z-1} \bigg|_{z=-1} \right) = \sqrt{2}\pi i.$$

$$(2) (i) \oint_C \frac{e^{iz}}{z^2+1} dz = 2\pi i \cdot \frac{e^{iz}}{z+i} \bigg|_{z=i} = \frac{\pi}{e},$$

$$(ii) \oint_C \frac{e^{iz}}{z^2+1} dz = 2\pi i \left(\frac{e^{iz}}{z+i} \bigg|_{z=i} + \frac{e^{iz}}{z-i} \bigg|_{z=-i} \right) = -2\pi \operatorname{sh} 1,$$

(iii) 同 (ii)。

72. 计算: (1) $\oint_{|z|=2} \frac{\cos z}{z} dz$; (2) $\oint_{|z|=2} \frac{\sin z}{z^2} dz$; (3) $\oint_{|z|=2} \frac{z^2}{z-1} dz$; (4) $\oint_{|z|=2} \frac{z^2-1}{z^2+1} dz$;

(5) $\oint_{|z|=2} \frac{dz}{z^2}$; (6) $\oint_{|z|=2} \frac{dz}{z^2+z+1}$; (7) $\oint_{|z|=2} \frac{dz}{z^2-8}$; (8) $\oint_{|z|=2} \frac{dz}{z^2-2z+3}$;

(9) $\oint_{|z|=2} \frac{|z|e^z dz}{z^2}$; (10) $\oint_{|z|=2} \frac{dz}{z^2(z^2+16)}$ 。

(1) $\oint_{|z|=2} \frac{\cos z}{z} dz = 2\pi i \cdot \cos z \big|_{z=0} = 2\pi i$;

(2) $\oint_{|z|=2} \frac{\sin z}{z^2} dz = 2\pi i (\sin z)' \big|_{z=0} = 2\pi i$;

(3) $\oint_{|z|=2} \frac{z^2}{z-1} dz = 2\pi i \cdot z^2 \big|_{z=1} = 2\pi i$;

(4) $\oint_{|z|=2} \frac{z^2-1}{z^2+1} dz = 2\pi i \left(\frac{z^2-1}{z+i} \bigg|_{z=i} + \frac{z^2-1}{z-i} \bigg|_{z=-i} \right) = 0$;

(5) $\oint_{|z|=2} \frac{dz}{z^2} = 0$;

(6) $\oint_{|z|=2} \frac{dz}{z^2+z+1} = 2\pi i \left(\frac{1}{z+1/2-i\sqrt{3}/2} \bigg|_{z=-1/2-i\sqrt{3}/2} + \frac{1}{z+1/2+i\sqrt{3}/2} \bigg|_{z=-1/2+i\sqrt{3}/2} \right) = 0$;

$$(7) \oint_{|z|=2} \frac{dz}{z^2-8} = 0; \text{ (不包围奇点)}$$

$$(8) \oint_{|z|=2} \frac{dz}{z^2-2z+3} = 2\pi i \left(\frac{1}{z-1-i\sqrt{2}} \Big|_{z=1-i\sqrt{2}} + \frac{1}{z-1+i\sqrt{2}} \Big|_{z=1+i\sqrt{2}} \right) = 0;$$

$$(9) \oint_{|z|=2} \frac{|z|e^z dz}{z^2} = 2 \oint_{|z|=2} \frac{e^z dz}{z^2} = 4\pi i (e^z)' \Big|_{z=0};$$

$$(10) \oint_{|z|=2} \frac{dz}{z^2(z^2+16)} = 2\pi i \left(\frac{1}{z^2+16} \right)' \Big|_{z=0} = 0。$$

73. (1) 计算 $\oint_{|z|=1} \frac{e^z}{z^3} dz$; (2) 对于什么样的 a 值, 函数 $F(z) = \int_{z_0}^z e^t \left(\frac{1}{t} + \frac{a}{t^3} \right) dt$ 是单值的?

$$(1) \oint_{|z|=1} \frac{e^z}{z^3} dz = \frac{2\pi i}{2!} (e^z)'' \Big|_{z=0} = \pi i;$$

$$(2) \oint_C e^t \left(\frac{1}{t} + \frac{a}{t^3} \right) dt = \begin{cases} 0, C \text{ 不包围原点} \\ (2+a)\pi i, C \text{ 包围原点} \end{cases}。当 $a = -2$ 时, 对任意的闭曲线 (不过$$

原点) 该积分都是 0, 则 $F(z)$ 为单值函数。

74. 证明: 在挖去 $z=0$ 点的全平面上不存在一个解析函数 $f(z)$, 使其满足 $f'(z) = \frac{1}{z}$ 。

这个结论和 $\frac{d}{dz} \ln z = \frac{1}{z}$ 矛盾吗?

因为 $z=0$ 点是 $\frac{1}{z}$ 的奇点, $\frac{1}{z}$ 在绕原点路径上的积分不为 0, 所以无法定义变上限函数

$\int_{z_0}^z \frac{1}{t} dt$, 即找不到在除去 $z=0$ 点的全平面上解析的原函数。 $\ln z$ 在划定割线, 在单值分枝

内才有 $\frac{d}{dz} \ln z = \frac{1}{z}$, 他是在分割的平面上成立, 而不是全平面。

75. 设 G 是单连通区域, C 是它的边界, $z_1, z_2, \dots, z_n$ 是 G 内的 n 个不同的点。

$$P(z) = (z-z_1)(z-z_2) \cdots (z-z_n), f(z) \text{ 在 } G \text{ 中解析, 证明: } Q(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta) P(\zeta) - P(z)}{\zeta - z} d\zeta$$

是一个 $n-1$ 次多项式, 且 $Q(z_k) = f(z_k)$, $k=1, 2, \dots, n$ 。如果 G 是复连通区域, 上述结果还正确吗?

$$\begin{aligned} \text{证: } z \neq z_k \text{ 时, } Q(z) &= \sum_{i=1}^n \left[(\zeta - z_i) \frac{f(\zeta) P(\zeta) - P(z)}{P(\zeta) \zeta - z} \right]_{\zeta=z_i} \\ &= \sum_{i=1}^n \left[\frac{f(z_i)(\zeta - z_i)}{\prod_{j=1}^n (\zeta - z_j)} \right]_{\zeta=z_i} \cdot \frac{\prod_{j=1}^n (z - z_j)}{z - z_i} = \sum_{i=1}^n \left[\frac{f(z_i)}{\prod_{j \neq i} (z_i - z_j)} \prod_{j \neq i} (z - z_j) \right] \end{aligned}$$

即它是 $n-1$ 次多项式。

$$z = z_k \text{ 时, } Q(z_k) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z_k} d\zeta = f(z_k)。$$

若 G 是复连通区域, 上面的计算不成立。

76. 设 $f(z)$ 在 $|z| \leq R$ 的区域内解析, 且 $\zeta = \rho e^{i\varphi}$ ($0 \leq \rho < R$) 为圆内一点, 证明圆内的

$$\text{Poisson 公式: } f(\zeta) = \frac{R^2 - \rho^2}{2\pi} \int_0^{2\pi} \frac{f(Re^{i\theta})}{R^2 - 2R\rho \cos(\theta - \varphi) + \rho^2} d\theta。$$

$$\begin{aligned} \text{证: } f(\zeta) &= \frac{1}{2\pi i} \oint_{|z|=R} \frac{f(z)}{z - \zeta} dz = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(Re^{i\theta})}{Re^{i\theta} - \rho e^{i\varphi}} dRe^{i\theta} = \frac{1}{2\pi} \int_0^{2\pi} \frac{Rf(Re^{i\theta})}{R - \rho e^{i(\varphi - \theta)}} d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{R[R - \rho e^{-i(\varphi - \theta)}]}{[R - \rho e^{i(\varphi - \theta)}][R - \rho e^{-i(\varphi - \theta)}]} f(Re^{i\theta}) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{R^2 - R\rho e^{-i(\varphi - \theta)}}{R^2 - 2R\rho \cos(\theta - \varphi) + \rho^2} f(Re^{i\theta}) d\theta \quad (1) \end{aligned}$$

$$\text{令 } \zeta' = \frac{R^2}{\rho} e^{i\varphi}, \text{ 它在圆外, 所以有 } \frac{1}{2\pi i} \oint_{|z|=R} \frac{f(z)}{z - \zeta'} dz = 0 \quad (\text{函数 } \frac{f(z)}{z - \zeta'} \text{ 在圆内解析})。$$

$$0 = \oint_{|z|=R} \frac{f(z)}{z - \zeta'} dz = \frac{1}{2\pi} \int_0^{2\pi} \frac{\rho^2 - R\rho e^{-i(\varphi - \theta)}}{R^2 - 2R\rho \cos(\theta - \varphi) + \rho^2} f(Re^{i\theta}) d\theta \quad (2)$$

$$(1) - (2) \text{ 即得 } f(\zeta) = \frac{R^2 - \rho^2}{2\pi} \int_0^{2\pi} \frac{f(Re^{i\theta})}{R^2 - 2R\rho \cos(\theta - \varphi) + \rho^2} d\theta。$$

77. 若 $f(z)$ 在区域 G 内单值连续, 且沿 G 内任一闭合路径 C 均有 $\oint_C f(z) dz = 0$, 试证 $f(z)$ 在区域 G 内解析 (这是 Cauchy 定理的逆定理, 即 Morera 定理)。

因为 $f(z)$ 沿 G 内任一闭合路径积分都是 0, 则 $\int_{z_0}^z f(t) dt$ 与积分路径无关, 它定义了一个单值函数 $F(z) = \int_{z_0}^z f(t) dt$ 。

$$\left| \frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) \right| = \left| \frac{1}{\Delta z} \int_z^{z+\Delta z} [f(t) - f(z)] dt \right| \leq \frac{1}{|\Delta z|} \int_z^{z+\Delta z} |f(t) - f(z)| |dt|$$

由于 $f(z)$ 在 z 点连续, 对于任意 $\varepsilon > 0$, 存在 $\delta > 0$, 使当 $|t - z| < \delta$ 时, $|f(t) - f(z)| < \varepsilon$,

所以只要 $|\Delta z| < \delta$, 对于 $t \in [z, z + \Delta z]$, 有 $|t - z| \leq |\Delta z| < \delta$,

$$\left| \frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) \right| < \frac{1}{|\Delta z|} \cdot \varepsilon \cdot |\Delta z| = \varepsilon. \text{ 所以 } F'(z) = f(z), F(z) \text{ 为解析函数,}$$

而解析函数的导函数仍解析, 即 $f(z)$ 在区域 G 内解析。

78. 考虑函数 $f(z) = \frac{1}{z^2}$ 。(1) 它对于所有不通过原点的闭合围道 C 都有积分

$\oint_C f(z) dz = 0$, 但 $f(z)$ 在 $z = 0$ 点不解析。这个情况和 Morera 定理 (上题) 矛盾吗?

(2) 当 $z \rightarrow \infty$ 时, 此函数有界, 但并不是一个常数。这和 Liouville 定理矛盾吗?

(1) 对于过原点路径上的积分, 由于 $f(0) \rightarrow \infty$, 积分 $\rightarrow \infty$, 并不满足 Morera 定理条件;

(2) Liouville 定理要求全平面解析。

79. 设 G 为单连通区域, 其边界为简单闭合曲线 C 。若函数 $f(z)$ 在 $\bar{G} = G + C$ 中解析,

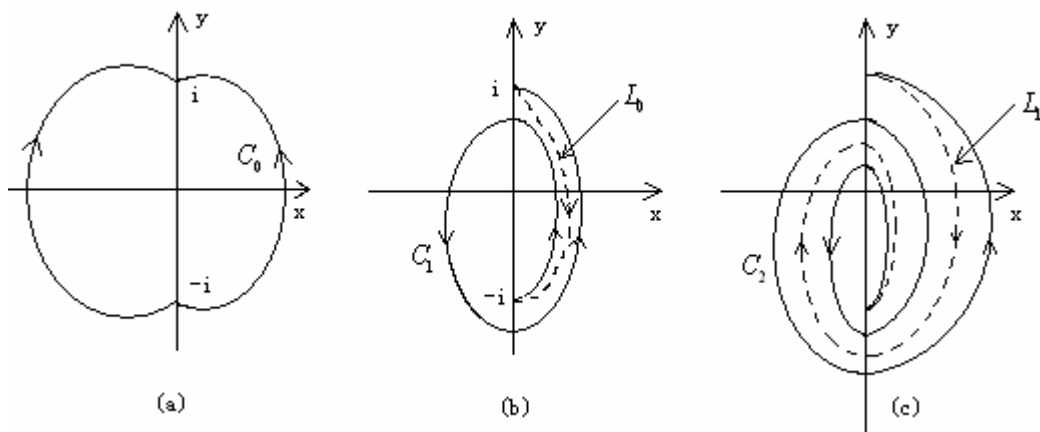
且在 C 上, $f(z) = 0$ 。证明: 在区域 G 内恒有 $f(z) = 0$ 。

由 Cauchy 积分公式 $f(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta$ 可证。

80. 计算 $\int_C \frac{dz}{z}$, 积分路径 C 为: (1) 没有割线的 z 平面上, 由 $-i$ 到 i 的各种可能路径;

(2) 沿负实轴割开的 z 平面上, 由 $-i$ 到 i 的各种可能路径。

(1) 可计算出沿右半圆从 $-i$ 到 i 的积分值为 πi ，沿左半圆从 $-i$ 到 i 的积分值为 $-\pi i$ 。



如上图 (a)，对于右半平面从 $-i$ 到 i 的任意路径 C_0 ，与 i 到 $-i$ 的右半圆构成闭合路径，该

闭路径积分值为 0，所以 $I_0 = \int_{C_0} \frac{dz}{z} = \pi i$ 。

如上图 (b)， C_1 为从 $-i$ 逆时针绕原点一圈后从右半平面到达 i 的曲线，记 C_1 上的积分为 I_1 。

L_0 为右半平面从 i 到 $-i$ 的曲线， L_0 上的积分即为 $-I_0$ 。 C_1 与 L_0 构成的闭曲线绕原点一圈，

所以 C_1 与 L_0 上的积分之和为 $2\pi i$ ，即 $I_1 - I_0 = 2\pi i$ ，所以 $I_1 = 3\pi i$ 。

如上图 (c)， C_2 为从 $-i$ 逆时针绕原点两圈后从右半平面到达 i 的曲线，记 C_2 上的积分为 I_2 。

L_1 为从 $-i$ 逆时针绕原点一圈后从右半平面到达 i 的反向曲线， L_1 上的积分即为 $-I_1$ 。 C_2 与

L_1 构成的闭曲线绕原点一圈，所以 $I_2 - I_1 = 2\pi i$ ，即 $I_2 = 5\pi i$ 。

依此类推，从 $-i$ 逆时针绕原点 n 圈后从右半平面到达 i 的曲线上的积分为 $I_n = (2n+1)\pi i$ ，

$n = 0, 1, 2, \dots$ 。

同样可得，对于从 $-i$ 顺时针绕原点 n 圈后从左半平面到达 i 的曲线上的积分为

$I'_n = -(2n+1)\pi i$ ， $n = 0, 1, 2, \dots$ 。

(2) 只能由右半平面直接从 $-i$ 到 i 的路径积分，积分值为 πi 。

81. 证明： $\oint_C \frac{dz}{(z-a)^n} = \begin{cases} 2\pi i, n=1 \\ 0, n \neq 1 \end{cases}$ ，其中 C 为包围 a 点的任一简单闭合围道， n 为整数。

证: 设 ε 为任意小的正数。
$$\int_{|z-a|=\varepsilon} \frac{dz}{(z-a)^n} = \int_0^{2\pi} \frac{d(a + \varepsilon e^{i\varphi})}{\varepsilon^n e^{in\varphi}} = i\varepsilon^{1-n} \int_0^{2\pi} e^{i(1-n)\varphi} d\varphi,$$

当 $n \neq 1$ 时, $\int_0^{2\pi} e^{i(1-n)\varphi} d\varphi = 0$, 即 $\int_{|z-a|=\varepsilon} \frac{dz}{(z-a)^n} = 0$,

当 $n=1$ 时, $\int_{|z-a|=\varepsilon} \frac{dz}{(z-a)^n} = i \int_0^{2\pi} d\varphi = 2\pi i$ 。

对于包围 a 点的任一简单闭合围道 C , 存在 ε , 使 $|z-a|=\varepsilon$ 在 C 包围的区域内, 则

$$\oint_C \frac{dz}{(z-a)^n} = \oint_{|z-a|=\varepsilon} \frac{dz}{(z-a)^n}, \text{ 得证。}$$

82. 计算 $\int_C \frac{dz}{\sqrt{z}}$ 。规定 $z=1$ 时 $\sqrt{z}=1$, 沿路径: (1) 单位圆的上半周从 1 到 -1; (2) 单位圆的下半周从 1 到 -1。

$z=1$ 时, $\arg z = 0$ 。(1) $z=-1$ 时, $\arg z = \pi$, $\int_C \frac{dz}{\sqrt{z}} = 2\sqrt{z} \Big|_1^{-1} = 2(\sqrt{e^{i\pi}} - 1) = 2(-1+i)$;

(2) $z=-1$ 时, $\arg z = -\pi$, $\int_C \frac{dz}{\sqrt{z}} = 2\sqrt{z} \Big|_1^{-1} = 2(\sqrt{e^{-i\pi}} - 1) = -2(1+i)$ 。

83. 设 $f(z)$ 在区域 G 内解析, C 为 G 内任一简单闭曲线, 证明对于 G 内, 但不在 C 上的

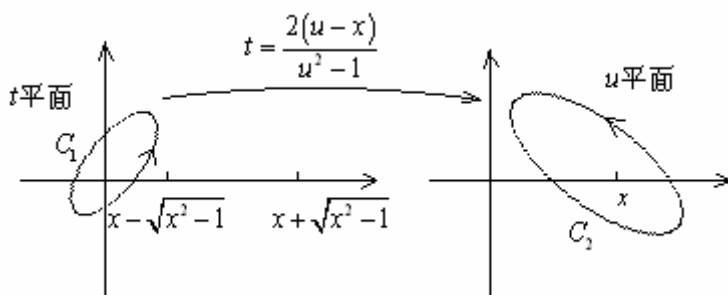
$$\oint_C \frac{f'(\zeta)}{\zeta-z} d\zeta = \oint_C \frac{f(\zeta)}{(\zeta-z)^2} d\zeta。$$

由 Cauchy 积分公式, $\oint_C \frac{f'(\zeta)}{\zeta-z} d\zeta = 2\pi i f'(z)$ 。由解析函数高阶导数公式,

$$\oint_C \frac{f(\zeta)}{(\zeta-z)^2} d\zeta = 2\pi i f'(z), \text{ 得证。}$$

84. 设 $\Psi(t, x) = \frac{1}{\sqrt{1-2xt+t^2}}$, t 是复变数。试证: $\frac{\partial^n \Psi(t, x)}{\partial t^n} \Big|_{t=0} = \frac{1}{2^n} \frac{d^n}{dx^n} (x^2-1)^n$ 。

证:



根据高阶微商公式, $\left. \frac{\partial^n \Psi(t, x)}{\partial t^n} \right|_{t=0} = \frac{n!}{2\pi i} \oint_{C_1} \frac{\Psi(t, x)}{t^{n+1}} dt = \frac{n!}{2\pi i} \oint_{C_1} \frac{1}{t^{n+1} \sqrt{1-2xt+t^2}} dt$ 。

上式中 C_1 是绕原点的围线, 且不包围 $\Psi(t, x)$ 的两个奇点 $x \pm \sqrt{x^2-1}$ 。

作变换 $\sqrt{1-2xt+t^2} = 1-ut$, 即 $t = \frac{2(u-x)}{u^2-1}$, 则 C_1 映射为绕 x 的 C_2 (方向不变),

上面的积分化为:

$$\begin{aligned} \frac{n!}{2\pi i} \oint_{C_1} \frac{1}{t^{n+1} \sqrt{1-2xt+t^2}} dt &= \frac{n!}{2\pi i} \oint_{C_2} \left[\frac{(u^2-1)^{n+1}}{2^{n+1}(u-x)^{n+1}} \frac{u^2-1}{-u^2+2ux-1} \frac{-2u^2+4ux-2}{(u^2-1)^2} \right] du \\ &= \frac{1}{2^n} \cdot \frac{n!}{2\pi i} \oint_{C_2} \frac{(u^2-1)^n}{(u-x)^{n+1}} du = \frac{1}{2^n} \frac{d^n}{du^n} (u^2-1)^n \Big|_{u=x} = \frac{1}{2^n} \frac{d^n}{dx^n} (x^2-1)^n. \end{aligned}$$

85. 设 $\Psi(t, x) = \exp(2tx - t^2)$, t 是复变数, 试证: $\left. \frac{\partial^n \Psi(t, x)}{\partial t^n} \right|_{t=0} = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$ 。

证: 同上题。 C_1 是 t 平面上逆时针绕原点的围线, 通过变换 $t = x-u$ 映射为 u 平面上逆时针绕 x 的围线 C_2 。

$$\begin{aligned} \left. \frac{\partial^n \Psi(t, x)}{\partial t^n} \right|_{t=0} &= \frac{n!}{2\pi i} \oint_{C_1} \frac{e^{2tx-t^2}}{t^{n+1}} dt = -\frac{n!}{2\pi i} \oint_{C_2} \frac{e^{2(x-u)x-(x-u)^2}}{(x-u)^{n+1}} du \\ &= (-1)^n e^{x^2} \frac{n!}{2\pi i} \oint_{C_2} \frac{e^{-u^2}}{(u-x)^{n+1}} du = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \end{aligned}$$

86. $f(z)$ 在 a 点的邻域内解析, 当 $\theta_1 \leq \arg(z-a) \leq \theta_2$, $z \rightarrow a$ 时, $(z-a)f(z)$ 一致地

趋于 k ，试证： $\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = ik(\theta_2 - \theta_1)$ 。其中 C_δ 为： $|z-a| = \delta$ ， $\theta_1 \leq \arg(z-a) \leq \theta_2$ （逆时针）。

证：可计算出 $\int_{C_\delta} \frac{dz}{z-a} = i(\theta_2 - \theta_1)$ ，

$$\int_{C_\delta} f(z) dz - ik(\theta_2 - \theta_1) = \int_{C_\delta} f(z) dz - \int_{C_\delta} \frac{k}{z-a} dz = \int_{C_\delta} \left[(z-a)f(z) - k \right] \frac{dz}{z-a}。$$

$(z-a)f(z)$ 一致地趋于 k ，即任意 $\frac{\varepsilon}{\theta_2 - \theta_1} > 0$ ，存在 $\delta_1 > 0$ （与 $\arg(z-a)$ 无关），使

$|z-a| < \delta_1$ 时， $|(z-a)f(z) - k| < \frac{\varepsilon}{\theta_2 - \theta_1}$ 。当 $\delta < \delta_1$ 时，对于 C_δ 上的点有 $|z-a| = \delta < \delta_1$ ，

$$\text{则 } \left| \int_{C_\delta} f(z) dz - ik(\theta_2 - \theta_1) \right| \leq \int_{C_\delta} |(z-a)f(z) - k| \left| \frac{dz}{z-a} \right| < \frac{\varepsilon}{\theta_2 - \theta_1} \cdot (\theta_2 - \theta_1) = \varepsilon$$

87. $f(z)$ 在 ∞ 点邻域内解析，当 $\theta_1 \leq \arg(z-a) \leq \theta_2$ ， $z \rightarrow \infty$ 时， $zf(z)$ 一致地趋于 K 。

试证： $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = iK(\theta_2 - \theta_1)$ 。其中 C_R 为： $|z| = R$ ， $\theta_1 \leq \arg z \leq \theta_2$ （逆时针）。

$$\text{证：同上题，有 } \int_{C_R} f(z) dz - iK(\theta_2 - \theta_1) = \int_{C_R} f(z) dz - \int_{C_R} \frac{K}{z} dz = \int_{C_R} \left[zf(z) - K \right] \frac{dz}{z}。$$

任意 $\frac{\varepsilon}{\theta_2 - \theta_1} > 0$ ，存在 $M > 0$ （与 $\arg z$ 无关），当 $|z| > M$ 时， $|zf(z) - K| < \frac{\varepsilon}{\theta_2 - \theta_1}$ 。

只要 $R > M$ ，在 C_R 上有 $|z| = R > M$ ，所以

$$\left| \int_{C_R} f(z) dz - iK(\theta_2 - \theta_1) \right| \leq \int_{C_R} |zf(z) - K| \left| \frac{dz}{z} \right| < \frac{\varepsilon}{\theta_2 - \theta_1} \cdot (\theta_2 - \theta_1) = \varepsilon。$$

88. 证明： $\frac{1}{2\pi i} \oint_{|z|=1} \frac{e^z}{z} dz = \frac{1}{\pi} \int_0^\pi e^{\cos \theta} \cos(\sin \theta) d\theta$ 。从而计算出 $\int_0^\pi e^{\cos \theta} \cos(\sin \theta) d\theta$ 。

$$\frac{1}{2\pi i} \oint_{|z|=1} \frac{e^z}{z} dz = \frac{1}{2\pi i} \int_{-\pi}^\pi \frac{e^{(\cos \theta + i \sin \theta)}}{e^{i\theta}} de^{i\theta} = \frac{1}{2\pi} \int_{-\pi}^\pi e^{\cos \theta} \cos(\sin \theta) d\theta + \frac{i}{2\pi} \int_{-\pi}^\pi e^{\cos \theta} \sin(\sin \theta) d\theta$$

上式右边第二项被积函数是奇函数，积分为 0，**第一项被积函数为偶函数**，所以

$$\frac{1}{2\pi i} \oint_{|z|=1} \frac{e^z}{z} dz = \frac{1}{\pi} \int_0^\pi e^{\cos\theta} \cos(\sin\theta) d\theta.$$

计算上式左边的积分得 $\int_0^\pi e^{\cos\theta} \cos(\sin\theta) d\theta = \pi e^z \Big|_{z=0} = \pi$

89. $f(z)$ 在全平面解析, 且 $\lim_{z \rightarrow \infty} \frac{f(z)}{z} = 0$, 证明 $f(z)$ 为常数。

证: 令 $F(z) = \begin{cases} \frac{f(z) - f(0)}{z}, & z \neq 0 \\ f'(0), & z = 0 \end{cases}$ 。 $z \neq 0$ 时 $F(z)$ 显然是可导的, $z = 0$ 时,

$$\frac{F(z) - F(0)}{z} = \frac{f(z) - f(0) - zf'(0)}{z^2}, \text{ 利用洛比达法则,}$$

$$\lim_{z \rightarrow 0} \frac{F(z) - F(0)}{z} = \lim_{z \rightarrow 0} \frac{f'(z) - f'(0)}{2z} = \frac{1}{2} f''(0). \text{ 即 } F(z) \text{ 在 } z = 0 \text{ 处也是可导的, 所以}$$

$F(z)$ 在全平面解析。因为 $\lim_{z \rightarrow \infty} \frac{f(z)}{z} = 0$, 则对于 $\varepsilon = 1$, 存在 $M > 1$, 当 $|z| > M > 1$ 时,

$$\text{有 } \left| \frac{f(z)}{z} \right| < \varepsilon = 1, \quad |F(z)| \leq \left| \frac{f(z)}{z} \right| + \left| \frac{f(0)}{z} \right| < \varepsilon + \frac{|f(0)|}{M} < 1 + |f(0)|, \text{ 即 } |F(z)| \text{ 有界, 根}$$

据 Liouville 定理, $F(z)$ 为常数。由于 $\lim_{z \rightarrow \infty} F(z) = \lim_{z \rightarrow \infty} \frac{f(z)}{z} - \lim_{z \rightarrow \infty} \frac{f(0)}{z} = 0$, 所以

$$F(z) = 0. \text{ 由此得, } z \neq 0 \text{ 时, } F(z) = \frac{f(z) - f(0)}{z} = 0, \quad f(z) = f(0), \text{ 即 } f(z) \text{ 为常}$$

数。

90. $f(z)$ 在全平面解析, 且 $|f(z)| \geq 1$, 证明 $f(z)$ 为常数。

证: 令 $F(z) = \frac{1}{f(z)}$, 因为 $|f(z)| \geq 1$, 所以 $f(z)$ 没有零点, 则 $F(z)$ 没有奇点, 即 $F(z)$

在全平面解析。 $|F(z)| = \frac{1}{|f(z)|} \leq 1$, 即 $F(z)$ 有界, 根据 Liouville 定理, $F(z)$ 为常数,

则 $f(z)$ 为常数。

91. 求 $|\sin z|$ 在闭区域 $0 \leq \operatorname{Re} z \leq 2\pi$, $0 \leq \operatorname{Im} z \leq 2\pi$ 中的最大值。

由最大模原理, 在边界上寻找最大值。

在 $y = 0$, $0 \leq x \leq 2\pi$ 上, $\sin z = \sin x$, 最大值为 1;

在 $x = 2\pi$, $0 \leq y \leq 2\pi$ 上, $|\sin z| = |\sin(2\pi + iy)| = |\sin(iy)| = |\operatorname{sh} y|$, 最大值为 $\operatorname{sh} 2\pi$;

在 $y = 2\pi$, $0 \leq x \leq 2\pi$ 上,

$$|\sin z| = |\sin(x + 2\pi i)| = |\operatorname{ch} 2\pi \sin x + i \operatorname{sh} 2\pi \cos x| = \sqrt{(\operatorname{ch} 2\pi \sin x)^2 + (\operatorname{sh} 2\pi \cos x)^2},$$

可求出最大值为 $\operatorname{ch} 2\pi$;

在 $x = 0$, $0 \leq y \leq 2\pi$ 上, $|\sin z| = |\sin(iy)| = |\operatorname{sh} y|$, 最大值为 $\operatorname{sh} 2\pi$;

所以最大值为 $\operatorname{ch} 2\pi$ 。

92. 函数 $f(z)$ 在 G 内解析, 且 z_0 为 G 内一点, 有 $f'(z_0) \neq 0$, 试证明:

$$\frac{2\pi i}{f'(z_0)} = \oint_C \frac{dz}{f(z) - f(z_0)}. \text{ 其中 } C \text{ 是以 } z_0 \text{ 为圆心的一个足够小的圆。}$$

$$\text{证: 令 } F(z) = \begin{cases} \frac{z - z_0}{f(z) - f(z_0)}, & z \neq z_0 \\ \frac{1}{f'(z_0)}, & z = z_0 \end{cases}, \quad z \neq z_0 \text{ 时 } F(z) \text{ 显然是可导的, 对于 } z = z_0 \text{ 点,}$$

$$\begin{aligned} \lim_{z \rightarrow z_0} \frac{F(z) - F(z_0)}{z - z_0} &= \lim_{z \rightarrow z_0} \frac{(z - z_0) f'(z_0) - f(z) + f(z_0)}{f'(z_0)(z - z_0)[f(z) - f(z_0)]}, \\ &= \lim_{z \rightarrow z_0} \frac{f'(z_0) - f'(z)}{f'(z_0)[f(z) - f(z_0)] + f'(z_0) f'(z)(z - z_0)} \\ &= \lim_{z \rightarrow z_0} \frac{-\frac{f'(z) - f'(z_0)}{z - z_0}}{f'(z_0) \frac{f(z) - f(z_0)}{z - z_0} + f'(z_0) f'(z)} = -\frac{f''(z_0)}{2[f'(z_0)]^2}. \end{aligned}$$

即 $F(z)$ 在 $z = z_0$ 点也是可导的, 所以 $F(z)$ 在 G 内解析, 因此有:

$$\frac{1}{2\pi i} \oint_C \frac{dz}{f(z) - f(z_0)} = \frac{1}{2\pi i} \oint_C \frac{F(z)}{z - z_0} dz = F(z_0) = \frac{1}{f'(z_0)}.$$

93. 函数 $f(z)$, $g(z)$ 及 $g(z)$ 的反函数均在 G 内单值解析, 且 $g'(z)$ 恒不为 0, 试计算

$$\frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{g(\zeta) - g(z)} d\zeta, \text{ 其中 } C \text{ 是 } G \text{ 内的简单闭曲线, } z \text{ 不在 } C \text{ 上.}$$

由于 $g(z)$ 的反函数在 G 内单值, 所以当且仅当 $\zeta = z$ 时 $g(\zeta) = g(z)$, 即 $\frac{f(\zeta)}{g(\zeta) - g(z)}$ 在

G 内只有一个奇点 $\zeta = z$ 。

若 C 不包围 z , 则 $\frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{g(\zeta) - g(z)} d\zeta = 0$ 。

若 C 包围 z , 同上题作法, 令 $F(\zeta) = \begin{cases} \frac{\zeta - z}{g(\zeta) - g(z)} f(\zeta), & \zeta \neq z \\ \frac{f(z)}{g'(z)}, & \zeta = z \end{cases}$, 它在 G 内解析, 所以

$$\text{有: } \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{g(\zeta) - g(z)} d\zeta = \frac{1}{2\pi i} \oint_C \frac{F(\zeta)}{\zeta - z} d\zeta = F(z) = \frac{f(z)}{g'(z)}.$$

94. 设 $\sum a_n$ 与 $\sum b_n$ 皆为正项级数, 试举反例, 说明下列说法不对:

(1) 若 $\lim_{n \rightarrow \infty} na_n = 0$, 则 $\sum a_n$ 收敛; (2) 若 $a_{2n} < a_{2n+1}$, 则 $\sum a_n$ 发散;

(3) 若 $\lim_{n \rightarrow \infty} \frac{a_{2n+1}}{a_n} = \infty$, 则 $\sum a_n$ 发散; (4) 若 $\sum a_n$ 与 $\sum b_n$ 发散, 则 $\sum \sqrt{a_n b_n}$ 发散。

(1) 取 $a_n = \frac{1}{n \ln n}$; (可用积分判别法断定级数 $\sum \frac{1}{n(\ln n)^p}$ 当 $p > 1$ 时收敛, $p \leq 1$ 时发散)

(2) 取 $a_n = \frac{2 - (-1)^n}{n^2}$, 则 $n \geq 1$ 时 $a_{2n} - a_{2n+1} = \frac{-8n^2 + 4n + 1}{(2n)^2 (2n+1)^2} < 0$, 而 $a_n \leq \frac{3}{n^3}$, $\sum \frac{3}{n^3}$

收敛, 所以 $\sum a_n$ 收敛;

(3) 取 $a_n = n^{-2 - \frac{(-1)^n}{2}}$, 则 $\lim_{n \rightarrow \infty} \frac{a_{2n+1}}{a_n} = \lim_{n \rightarrow \infty} n = \infty$, 而 $a_n \leq n^{-\frac{3}{2}}$, $\sum \frac{1}{n^{3/2}}$ 是收敛的, 所以 $\sum a_n$

收敛;

(4) 取 $a_n = \frac{1}{n^{2+(-1)^n}}$, $b_n = \frac{1}{n^{2-(-1)^n}}$, 因为 $\sum_n a_n = \sum_k \frac{1}{(2k)^3} + \sum_k \frac{1}{2k+1}$, 右边第一个级数

收敛, 第二个发散, 所以 $\sum a_n$ 发散, 同样的, $\sum b_n$ 也发散, 而 $\sum \sqrt{a_n b_n} = \sum \frac{1}{n^2}$ 收敛。

95. 指出下列谬误:

$$\begin{aligned}
 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots &= 1 + \left(\frac{1}{2} - 2 \times \frac{1}{2} \right) + \frac{1}{3} + \left(\frac{1}{4} - 2 \times \frac{1}{4} \right) + \frac{1}{5} + \left(\frac{1}{6} - 2 \times \frac{1}{6} \right) + \cdots \\
 &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \cdots \\
 &\quad - 2 \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \cdots \right) \\
 &= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \cdots \right) \\
 &\quad - \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \cdots \right) = 0
 \end{aligned}$$

不能随意改变求和顺序。

96. 判断下列级数的收敛性及绝对收敛性：(1) $\sum \frac{i^n}{\ln n}$ ；(2) $\sum \frac{i^n}{n}$ 。

$$(1) \sum_n \frac{i^n}{\ln n} = \sum_k \frac{(-1)^k}{\ln(2k)} + i \sum_k \frac{(-1)^k}{\ln(2k+1)}, \text{ 右边两个级数都收敛 (用 Leibnitz 判别法),}$$

所以 $\sum \frac{i^n}{\ln n}$ 收敛, 因为 $\left| \frac{i^n}{\ln n} \right| = \frac{1}{\ln n} > \frac{1}{n}$, 而 $\sum \frac{1}{n}$ 发散, 所以 $\sum \frac{i^n}{\ln n}$ 不绝对收敛

$$(2) \text{ 同上, } \sum \frac{i^n}{n} = \sum \frac{(-1)^k}{2k} + i \sum \frac{(-1)^k}{2k+1}, \text{ 所以 } \sum \frac{i^n}{n} \text{ 收敛。} \left| \frac{i^n}{n} \right| = \frac{1}{n}, \sum \frac{1}{n} \text{ 发散, 所以}$$

$\sum \frac{i^n}{n}$ 不绝对收敛。

97. 证明级数 $\sum_{n=1}^{\infty} \frac{z^{n-1}}{(1-z^n)(1-z^{n+1})}$, $|z| \neq 1$ 收敛, 并求其和。

$$\begin{aligned} S_N(z) &= \sum_{n=1}^N \frac{z^{n-1}}{(1-z^n)(1-z^{n+1})} = \frac{1}{z(1-z)} \sum_{n=1}^N \left(\frac{1}{1-z^n} - \frac{1}{1-z^{n+1}} \right) \\ &= \frac{1}{z(1-z)} \left(\frac{1}{1-z} - \frac{1}{1-z^2} + \frac{1}{1-z^2} - \frac{1}{1-z^3} + \cdots + \frac{1}{1-z^N} - \frac{1}{1-z^{N+1}} \right) \\ &= \frac{1}{z(1-z)} \left(\frac{1}{1-z} - \frac{1}{1-z^{N+1}} \right) \end{aligned}$$

若 $|z| < 1$, $N \rightarrow \infty$ 时 $S_N(z) \rightarrow \frac{1}{(1-z)^2}$, 若 $|z| > 1$, $S_N(z) \rightarrow \frac{1}{z(1-z)^2}$, 即

$$S(z) = \lim_{N \rightarrow \infty} S_N(z) = \begin{cases} \frac{1}{(1-z)^2}, & |z| < 1 \\ \frac{1}{z(1-z)^2}, & |z| > 1 \end{cases}.$$

98. 证明无穷乘积 $\prod_{n=0}^{\infty} (1+z^{2^n}) = (1+z)(1+z^2)(1+z^4)(1+z^8) \cdots$, ($|z| < 1$) 收敛,

并求其积。

记其前 N 项部分积为 $P_N(z)$ 。

$$\begin{aligned}
P_N(z) &= (1+z)(1+z^2)(1+z^4)\cdots(1+z^{2^{N-1}}) \\
&= \frac{1}{1-z}(1-z)(1+z)(1+z^2)(1+z^4)\cdots(1+z^{2^{N-1}}) \\
&= \frac{1}{1-z}(1-z^2)(1+z^2)(1+z^4)\cdots(1+z^{2^{N-1}}) \\
&= \frac{1}{1-z}(1-z^4)(1+z^4)\cdots(1+z^{2^{N-1}}) = \cdots = \frac{1-z^{2^N}}{1-z}
\end{aligned}$$

因为 $|z| < 1$, 所以 $P(z) = \lim_{N \rightarrow \infty} P_N(z) = \frac{1}{1-z}$ 。

99. 证明: (1) $\prod_{n=1}^{\infty} \cos \frac{z}{2^n} = \frac{\sin z}{z}$; (2) $\sum_{n=1}^{\infty} \frac{1}{2^n} \tan \frac{z}{2^n} = \frac{1}{z} - \cot z$ 。

$$\begin{aligned}
(1) \quad P_N(z) &= \cos \frac{z}{2} \cdot \cos \frac{z}{2^2} \cdots \cos \frac{z}{2^{N-1}} \cdot \cos \frac{z}{2^N} \\
&= \cos \frac{z}{2} \cdot \cos \frac{z}{2^2} \cdots \cos \frac{z}{2^{N-1}} \cdot \cos \frac{z}{2^N} \cdot \sin \frac{z}{2^N} \cdot \frac{1}{\sin \frac{z}{2^N}} \\
&= \cos \frac{z}{2} \cdot \cos \frac{z}{2^2} \cdots \cos \frac{z}{2^{N-1}} \cdot \sin \frac{z}{2^{N-1}} \cdot \frac{1}{2 \sin \frac{z}{2^N}} \\
&= \cos \frac{z}{2} \cdot \cos \frac{z}{2^2} \cdots \sin \frac{z}{2^{N-2}} \cdot \frac{1}{2^2 \sin \frac{z}{2^N}} = \cdots = \frac{\sin z}{2^N \sin \frac{z}{2^N}}
\end{aligned}$$

$$\prod_{n=1}^{\infty} \cos \frac{z}{2^n} = \lim_{N \rightarrow \infty} \frac{\sin z}{2^N \sin \frac{z}{2^N}} = \frac{\sin z}{z};$$

$$\begin{aligned}
(2) \quad \cot z + \sum_{n=1}^N \frac{1}{2^n} \tan \frac{z}{2^n} &= \frac{1 - \tan^2 \frac{z}{2}}{2 \tan \frac{z}{2}} + \frac{1}{2} \tan \frac{z}{2} + \frac{1}{2^2} \tan \frac{z}{2^2} + \cdots + \frac{1}{2^N} \tan \frac{z}{2^N} \\
&= \frac{1}{2 \tan \frac{z}{2}} + \frac{1}{2^2} \tan \frac{z}{2^2} + \cdots + \frac{1}{2^N} \tan \frac{z}{2^N}
\end{aligned}$$

$$\begin{aligned}
&= \frac{1 - \tan^2 \frac{z}{2^2}}{2^2 \tan \frac{z}{2^2}} + \frac{1}{2^2} \tan \frac{z}{2^2} + \frac{1}{2^3} \tan \frac{z}{2^3} \cdots + \frac{1}{2^N} \tan \frac{z}{2^N} \\
&= \frac{1}{2^2 \tan \frac{z}{2^2}} + \frac{1}{2^3} \tan \frac{z}{2^3} + \cdots + \frac{1}{2^N} \tan \frac{z}{2^N} \\
&= \cdots = \frac{1}{2^N \tan \frac{z}{2^N}}
\end{aligned}$$

令 $N \rightarrow \infty$, 则 $\frac{1}{2^N \tan \frac{z}{2^N}} \rightarrow \frac{1}{z}$, 所以 $\sum_{n=1}^{\infty} \frac{1}{2^n} \tan \frac{z}{2^n} = \frac{1}{z} - \cot z$ 。

100. 证明级数 $\sum e^{-n} \sin nz$ 在区域 $|\operatorname{Im} z| < 1$ 内解析。

证: 对于任意 $p \in (0, 1)$, 当 $|\operatorname{Im} z| \leq p$ 时, (参考习题 02 的 43 题)

$$|e^{-n} \sin nz| \leq e^{-n} \operatorname{ch} ny \leq e^{-n} \operatorname{ch} pn = \frac{1}{2} [e^{-(1-p)n} + e^{-(1+p)n}].$$

因为级数 $\sum [e^{-(1-p)n} + e^{-(1+p)n}]$

收敛, 所以级数 $\sum e^{-n} \sin nz$ 在区域 $|\operatorname{Im} z| \leq p$ 内一致收敛, 由 Weierstrass 定理, 级数

$\sum e^{-n} \sin nz$ 在区域 $|\operatorname{Im} z| < p$ 内解析。这里的 p 具有任意性。

任取区域 $|\operatorname{Im} z| < 1$ 内一点 z_0 , 存在 p_0 使 $|\operatorname{Im} z_0| < p_0 < 1$ 。由于级数 $\sum e^{-n} \sin nz$ 在区域

$|\operatorname{Im} z| < p_0$ 内解析, 故在 z_0 点解析。由 z_0 的任意性, $\sum e^{-n} \sin nz$ 在区域 $|\operatorname{Im} z| < 1$ 内解析。

101. x 为实数, 证明: (1) 级数 $\sum_{n=1}^{\infty} \frac{x^2}{(1+x^2)^n}$ 绝对收敛, 但不一致收敛;

(2) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n+x^2}$ 一致收敛, 但不绝对收敛。

(1) 这是正项等比级数, 显然绝对收敛。记和函数为 $S(x)$, 则 $S(x) = \begin{cases} 0, & x = 0 \\ -\frac{1}{1+x^2}, & x \neq 0 \end{cases}$,

$x=0$ 处为间断点, 而 $\frac{x^2}{(1+x^2)^n}$ 在整个实轴上是连续的, 所以 $\sum_{n=1}^{\infty} \frac{x^2}{(1+x^2)^n}$ 不一致收敛。

(2) 由于 $\frac{1}{n+x^2} \leq \frac{1}{n}$, 所以 $\frac{1}{n+x^2}$ 单调一致趋于 0, 由 Leibnitz 判敛法可知 $\sum_{n=1}^{\infty} \frac{(-1)^n}{n+x^2}$ 一致收敛。

致收敛。 $\left| \frac{(-1)^n}{n+x^2} \right| = \frac{1}{n+x^2}$, 它不绝对收敛。

102. 确定下列级数的收敛半径 (或收敛区域): (1) $\sum \frac{1}{n^n} z^n$; (2) $\sum \frac{1}{2^n n^n} z^n$;

(3) $\sum \frac{n!}{n^n} z^n$; (4) $\sum \frac{(-1)^n}{2^{2n} (n!)^2} z^n$; (5) $\sum n^{\ln n} z^n$; (6) $\sum z^n$; (7) $\sum \frac{1}{2^{2n}} z^{2n}$;

(8) $\sum \left(\frac{z}{1+z} \right)^n$; (9) $\sum (-1)^n (z^2 + 2z + 2)^n$; (10) $\sum 2^n \sin \frac{z}{3^n}$; (11) $\sum \frac{\ln(n^n)}{n!} z^n$;

(12) $\sum \left(1 - \frac{1}{n} \right)^n z^n$ 。

(1) $\lim_{n \rightarrow \infty} \left| n^n \right|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} n = \infty$, 所以收敛半径 $R = \infty$;

(2) $\lim_{n \rightarrow \infty} \left| 2^n n^n \right|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} 2n = \infty$, 所以收敛半径 $R = \infty$;

(3) $\lim_{n \rightarrow \infty} \left| \frac{n!}{n^n} \right| / \left| \frac{(n+1)!}{(n+1)^{n+1}} \right| = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$, 所以收敛半径 $R = e$;

(4) $\lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{2^{2n} (n!)^2} \right| / \left| \frac{(-1)^{n+1}}{2^{2(n+1)} [(n+1)!]^2} \right| = \lim_{n \rightarrow \infty} 4(n+1)^2 = \infty$, 所以收敛半径 $R = \infty$;

(5) $\lim_{n \rightarrow \infty} \left| n^{\ln n} \right|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} n^{\frac{\ln n}{n}} = \lim_{n \rightarrow \infty} e^{\frac{(\ln n)^2}{n}} = 1$, 所以收敛半径 $R = 1$;

(6) 显然收敛半径 $R = 1$;

(7) $\frac{1}{2^{2n}} z^{2n} = \left(\frac{z}{2} \right)^{2n}$, 收敛半径 $R = 2$;

(8) 收敛域为 $\left| \frac{z}{1+z} \right| < 1$, 化简得 $\operatorname{Re} z > -\frac{1}{2}$;

(9) 收敛域为 $|z^2 + 2z + 2| < 1$;

(10) $\sum 2^n \sin \frac{z}{3^n}$ 在全平面收敛, 即收敛域为全平面;

(11) $\lim_{n \rightarrow \infty} \left| \frac{n \ln n}{n!} \bigg/ \frac{(n+1) \ln(n+1)}{(n+1)!} \right| = \lim_{n \rightarrow \infty} \frac{n \ln n}{\ln(n+1)} = \infty$, 所以收敛半径 $R = \infty$;

(12) $\lim_{n \rightarrow \infty} \left| \left(1 - \frac{1}{n} \right)^n \right|^{1/n} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n} \right) = 1$, 所以收敛半径 $R = 1$ 。

103. 已知幂级数 $\sum a_n z^n$ 和 $\sum b_n z^n$ 的收敛半径分别为 R_1, R_2 , 试讨论下列幂级数的收敛半

径: (1) $\sum (a_n + b_n) z^n$; (2) $\sum a_n b_n z^n$; (3) $\sum \frac{1}{a_n} z^n$; (4) $\sum \frac{b_n}{a_n} z^n$ 。

(1) 设 $R_1 < R_2$ 。当 $|z| < R_1$ 时, 级数收敛, $R_1 < |z| < R_2$ 时, 级数发散; 由阿贝尔第一定理, 当 $|z| > R_2$ 时级数也发散, 所以 $R = R_1$ 。

若 $R_1 = R_2$, 当 $|z| < R_1 = R_2$ 时, 级数收敛, 当 $|z| > R_1 = R_2$ 时, 级数有可能收敛。

综上, $R \geq \min\{R_1, R_2\}$ 。

(2) 参考数学分析中关于上下极限的等式和不等式。

$$R = \lim_{n \rightarrow \infty} \frac{1}{|a_n b_n|^{1/n}} \geq \lim_{n \rightarrow \infty} \frac{1}{|a_n|^{1/n}} \cdot \lim_{n \rightarrow \infty} \frac{1}{|b_n|^{1/n}} = R_1 R_2;$$

$$(3) R = \frac{1}{\lim_{n \rightarrow \infty} \frac{1}{|a_n|^{1/n}}} \leq \frac{1}{\lim_{n \rightarrow \infty} \frac{1}{|a_n|^{1/n}}} = \frac{1}{R_1};$$

$$(4) R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{b_n} \right|^{1/n} \leq \lim_{n \rightarrow \infty} |a_n|^{1/n} \cdot \lim_{n \rightarrow \infty} \frac{1}{|b_n|^{1/n}} = \frac{1}{\lim_{n \rightarrow \infty} \frac{1}{|a_n|^{1/n}}} \cdot \lim_{n \rightarrow \infty} \frac{1}{|b_n|^{1/n}} = \frac{R_2}{R_1}。$$

104. 如果 $\sum a_n z^n$ 的收敛半径为 R , 试证明 $\sum (\operatorname{Re} a_n) z^n$ 的收敛半径 $R' \geq R$ 。

$$R' = \lim_{n \rightarrow \infty} \left| \frac{1}{\operatorname{Re} a_n} \right|^{1/n} = \lim_{n \rightarrow \infty} \left| \frac{2}{a_n + \bar{a}_n} \right|^{1/n} \geq \lim_{n \rightarrow \infty} \left(\frac{2}{|a_n| + |\bar{a}_n|} \right)^{1/n} = \lim_{n \rightarrow \infty} \left| \frac{1}{a_n} \right|^{1/n} = R$$

105. 设级数 $\sum a_n$ 收敛, 而 $\sum |a_n|$ 发散, 证明级数 $\sum a_n z^n$ 的收敛半径为 1。

$|z| < 1$ 时, 存在 δ 使 $|z| < \delta < 1$, 则 $|a_n z^n| < |a_n| \delta^n$ 。

因为 $\sum a_n$ 收敛, 有 $\lim_{n \rightarrow \infty} a_n = 0$, 对于 $\varepsilon = 1$, 当 n 充分大时 $|a_n| < \varepsilon = 1$, 则 $|a_n z^n| < \delta^n$,

因为 $\sum \delta^n$ 收敛, 所以 $\sum a_n z^n$ 绝对收敛。

$|z| > 1$ 时, $|a_n z^n| > |a_n|$, 因为 $\sum |a_n|$ 发散, 所以 $\sum a_n z^n$ 发散。

106. 若级数 $\sum a_n$ 收敛, 而 $\sum n|a_n|$ 发散, 证明级数 $\sum a_n z^n$ 的收敛半径为 1。

$|z| < 1$ 时, 同上题方法可证 $\sum a_n z^n$ 收敛。

$|z| > 1$ 时, 存在 ζ 使 $|z| > \zeta > 1$, 则 $|a_n z^n| > |a_n| \zeta^n$ 。因为 $\zeta > 1$, 所以当 n 充分大时有 $\zeta^n > n$,

则 $|a_n z^n| > n|a_n|$, 因为 $\sum n|a_n|$ 发散, 所以 $\sum a_n z^n$ 发散。

107. 证明: 级数 $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ 在 $|z| \leq 1$ 中一致收敛, 但由它逐项微商求得的级数在 $|z| < 1$ 内却不一致收敛。这个结果和 Weierstrass 定理矛盾吗?

$|z| \leq 1$ 时 $\left| \frac{z^n}{n^2} \right| \leq \frac{1}{n^2}$, 由于 $\sum \frac{1}{n^2}$ 收敛, 所以 $\sum \frac{z^n}{n^2}$ 在 $|z| \leq 1$ 中一致收敛。

假设 $\sum \frac{z^{n-1}}{n}$ 在 $|z| < 1$ 内一致收敛, 则对任意 $\varepsilon > 0$, 存在 N , 当 $n > N$ 时, 对任意整数 p

有 $\left| \frac{z^n}{n+1} + \frac{z^{n+1}}{n+2} + \cdots + \frac{z^{n+p-1}}{n+p} \right| < \frac{\varepsilon}{2}$, ($\forall |z| < 1$) 由于 $\frac{z^n}{n+1}$ 在全平面连续, 可令 z 从单位圆

内趋于 1 得 $\lim_{z \rightarrow 1} \left| \frac{z^n}{n+1} + \frac{z^{n+1}}{n+2} + \cdots + \frac{z^{n+p-1}}{n+p} \right| = \left| \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+p} \right| \leq \frac{\varepsilon}{2} < \varepsilon$, 即级数

$\sum \frac{1}{n}$ 收敛, 所以假设不成立, 即级数 $\sum \frac{z^{n-1}}{n}$ 在 $|z| < 1$ 内不一致收敛。这并不与 Weierstrass

定理矛盾, 该定理结论是逐项微商求得的级数在收敛域内的闭区域上一致收敛, $|z| < 1$ 非闭区域。

108. 证明 Riemann- ζ 函数 $\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}$ 在区域 $\operatorname{Re} z > 1$ 内解析, 并计算 $\zeta'(z)$ 。

证: 对任意 $p > 1$, 当 $\operatorname{Re} z \geq p$ 时, $\left| \frac{1}{n^z} \right| = \frac{1}{e^{x \ln n}} \leq \frac{1}{e^{p \ln n}} = \frac{1}{n^p}$ 。由于级数 $\sum \frac{1}{n^p}$ 收敛, 所以

$\sum \frac{1}{n^z}$ 在 $\operatorname{Re} z \geq p$ 内一致收敛, 所以 $\zeta(z)$ 在 $\operatorname{Re} z > p$ 内解析, 逐项可导。

任取区域 $\operatorname{Re} z > 1$ 内一点 z_0 , 存在 p_0 使 $\operatorname{Re} z_0 > p_0 > 1$ 。由于 $\zeta(z)$ 在 $\operatorname{Re} z > p_0$ 内解析,

逐项可导, 所以 $\zeta(z)$ 在 z_0 点解析, 逐项可导, 由 z_0 的任意性, $\zeta(z)$ 在 $\operatorname{Re} z > 1$ 内解析,

逐项可导。 $\zeta'(z) = \sum \left(\frac{1}{n^z} \right)' = -\sum \frac{\ln n}{n^z}$ 。

109. 将下列函数在指定点展成 Taylor 级数, 并给出其收敛半径:

(1) $\sin z$, 在 $z = n\pi$ 展开; (2) $1 - z^2$, 在 $z = 1$ 展开; (3) $\frac{1}{1+z+z^2}$, 在 $z = 0$ 展开;

(4) $\ln z$, 在 $z = i$ 展开, 规定: (i) $0 \leq \arg z < 2\pi$, (ii) $-\pi \leq \arg z < \pi$, (iii) $(\ln z)_{z=i} = -\frac{3}{2}\pi i$;

(5) $\arctan z$ 的主值, 在 $z = 0$ 展开; (6) $\frac{\sin z}{1-z}$, 在 $z = 0$ 展开;

(7) $\exp\left(\frac{1}{1-z}\right)$, 在 $z = 0$ 展开 (可只求前四项系数); (8) $\ln\left(\frac{1+z}{1-z}\right)$, 在 $z = \infty$ 展开。

$$(1) (\sin z)_{z=n\pi}^{(k)} = \sin\left(n\pi + \frac{k}{2}\pi\right) = \begin{cases} 0, k=2m \\ (-1)^{n+m}, k=2m+1 \end{cases}$$

$$\sin z = \sum_{k=0}^{\infty} \frac{1}{k!} (\sin z)_{z=n\pi}^{(k)} (z - n\pi)^k = \sum_{m=0}^{\infty} \frac{(-1)^{n+m}}{(2m+1)!} (z - n\pi)^{2m+1};$$

$$(2) 1 - z^2 = -(z-1)(z+1) = -(z-1)[2 + (z-1)] = -2(z-1) - (z-1)^2;$$

$$\begin{aligned} (3) \frac{1}{1+z+z^2} &= \frac{1}{(z-e^{i2\pi/3})(z-e^{-i2\pi/3})} = \frac{1}{\sqrt{3}i} \left(\frac{1}{z-e^{i2\pi/3}} - \frac{1}{z-e^{-i2\pi/3}} \right) \\ &= \frac{1}{\sqrt{3}i} \left(\frac{e^{i2\pi/3}}{1-e^{i2\pi/3}z} - \frac{e^{-i2\pi/3}}{1-e^{-i2\pi/3}z} \right) \\ &= \frac{1}{\sqrt{3}i} \sum_{n=0}^{\infty} \left[e^{i2\pi/3} (e^{i2\pi/3}z)^n - e^{-i2\pi/3} (e^{-i2\pi/3}z)^n \right] \end{aligned}$$

$$= \frac{1}{\sqrt{3}i} \sum_{n=0}^{\infty} [e^{i2(n+1)\pi/3} - e^{-i2(n+1)\pi/3}] z^n = \frac{2}{\sqrt{3}} \sum_{n=0}^{\infty} \sin \left[\frac{2}{3}(n+1)\pi \right] z^n \quad (|z| < 1);$$

$$(4) (\ln z)_{z=i}^{(k)} = \left(\frac{1}{z} \right)_{z=i}^{(k-1)} = \frac{(-1)^{k-1} (k-1)!}{z^k} \Big|_{z=i} = -\frac{(-1)^k (k-1)!}{i^k} = -\frac{i^{2k} (k-1)!}{i^k} = -i^k (k-1)!,$$

$$(k=1, 2, 3, \dots) \quad \ln z = (\ln z)_{z=i} + \sum_{k=1}^{\infty} \frac{1}{k!} (\ln z)_{z=i}^{(k)} (z-i)^k = (\ln z)_{z=i} - \sum_{k=1}^{\infty} \frac{i^k}{k} (z-i)^k,$$

$$(|z-i| < 1); \quad (i)(ii) \quad (\ln z)_{z=i} = \frac{1}{2}\pi i; \quad (iii) \quad (\ln z)_{z=i} = -\frac{3}{2}\pi i \text{ 代入上式即可。}$$

$$(5) \frac{1}{1+t^2} = \sum_{n=0}^{\infty} (-t^2)^n = \sum_{n=0}^{\infty} (-1)^n t^{2n}, \quad (|t| < 1)$$

对两边积分, 由于幂级数在收敛域内可逐项积分, 故有

$$\arctan z = \int_0^z \frac{1}{1+t^2} dt = \sum_{n=0}^{\infty} (-1)^n \int_0^z t^{2n} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} z^{2n+1}, \quad (|z| < 1)$$

$$(6) \left(\frac{\sin z}{1-z} \right)_{z=0}^{(n)} = \sum_{k=0}^n \frac{n!}{k!(n-k)!} \left(\frac{1}{1-z} \right)_{z=0}^{(n-k)} (\sin z)_{z=0}^{(k)} = \sum_{k=0}^n \frac{n!}{k!} \sin \frac{k}{2} \pi = \sum_{m=0}^{[(n-1)/2]} \frac{(-1)^m n!}{(2m+1)!}$$

$$\frac{\sin z}{1-z} = \left(\frac{\sin z}{1-z} \right)_{z=0} + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\frac{\sin z}{1-z} \right)_{z=0}^{(n)} z^n = \sum_{n=1}^{\infty} \sum_{m=0}^{[(n-1)/2]} \frac{(-1)^m}{(2m+1)!} z^n, \quad (|z| < 1)$$

$$(7) \text{ 记 } f(z) = e^{\frac{1}{1-z}}, \text{ 则 } f(0) = e. \quad f'(z) = \frac{1}{(1-z)^2} e^{\frac{1}{1-z}}, \quad f'(0) = e.$$

$$f''(z) = \left[\frac{2}{(1-z)^3} + \frac{1}{(1-z)^4} \right] e^{\frac{1}{1-z}}, \quad f''(0) = 3e.$$

$$f'''(z) = \left[\frac{6}{(1-z)^4} + \frac{6}{(1-z)^5} + \frac{1}{(1-z)^6} \right] e^{\frac{1}{1-z}}, \quad f'''(0) = 13e.$$

$$f^{(4)}(z) = \left[\frac{24}{(1-z)^5} + \frac{36}{(1-z)^6} + \frac{12}{(1-z)^7} + \frac{1}{(1-z)^8} \right] e^{\frac{1}{1-z}}, \quad f^{(4)}(0) = 73e.$$

$$\begin{aligned} f(z) &= f(0) + f'(0)z + \frac{1}{2} f''(0)z^2 + \frac{1}{6} f'''(0)z^3 + \frac{1}{24} f^{(4)}(0)z^4 + \dots \\ &= e \left(1 + z + \frac{3}{2}z^2 + \frac{13}{6}z^3 + \frac{73}{24}z^4 + \dots \right) \quad (|z| < 1) \end{aligned}$$

$$(8) \text{ 记 } f(z) = \ln \frac{1+z}{1-z}, \text{ 则 } f\left(\frac{1}{t}\right) = \ln \frac{t+1}{t-1} = \ln(t+1) - \ln(t-1).$$

$$\ln(t+1) = \ln(t+1)_{t=0} + \int_0^t \frac{1}{1+u} du = \ln(t+1)_{t=0} + \int_0^t \sum_{n=0}^{\infty} (-1)^n u^n du$$

$$= \ln(t+1)_{t=0} + \sum_{n=0}^{\infty} (-1)^n \int_0^t u^n du = \ln(t+1)_{t=0} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} t^{n+1}, \quad (|t| < 1)$$

$$-\ln(t-1) = -\ln(t-1)_{t=0} + \int_0^t \frac{1}{1-u} du = -\ln(t-1)_{t=0} + \sum_{n=0}^{\infty} \int_0^t u^n du$$

$$= -\ln(t-1)_{t=0} + \sum_{n=0}^{\infty} \frac{1}{n+1} t^{n+1}, \quad (|t| < 1)$$

± 1 为 $\ln \frac{t+1}{t-1}$ 的两个枝点，以连接两点的线段为割线，规定割线上岸

$\arg(t+1) - \arg(t-1) = (2k+1)\pi$, ($k = 0, \pm 1, \pm 2, \dots$), 则

$$f\left(\frac{1}{t}\right) = \ln \frac{t+1}{t-1} \Big|_{t=0} + \sum_{n=0}^{\infty} \frac{(-1)^n + 1}{n+1} t^{n+1} = (2k+1)\pi i + \sum_{n=0}^{\infty} \frac{2}{2n+1} t^{2n+1},$$

$$f(z) = (2k+1)\pi i + \sum_{n=0}^{\infty} \frac{2}{2n+1} z^{-(2n+1)}.$$

110. 求下列级数之和: (1) $\sum_{n=0}^{\infty} \frac{1}{2n+1} z^{2n+1}$, $|z| < 1$; (2) $\sum_{n=0}^{\infty} \frac{1}{(2n)!} z^{2n}$, $|z| < \infty$.

$$(1) \text{ 记 } f(z) = \sum_{n=0}^{\infty} \frac{1}{2n+1} z^{2n+1}, \text{ 则 } f'(z) = \sum_{n=0}^{\infty} z^{2n} = \frac{1}{1-z^2}, \quad (|z| < 1)$$

$$f(z) = f(0) + \int_0^z \frac{1}{1-t^2} dt = \frac{1}{2} \int_0^z \left(\frac{1}{1+t} + \frac{1}{1-t} \right) dt = \frac{1}{2} \ln \frac{1+z}{1-z}, \quad \text{为使 } f(0) = 0, \text{ 规定}$$

$$\ln \frac{1+z}{1-z} \Big|_{z=0} = 0.$$

级数 $\rightarrow$ 单值化、前后根群级以确定单值分支。

$$(2) \text{ 记 } f(z) = \sum_{n=0}^{\infty} \frac{1}{(2n)!} z^{2n}, \text{ 则 } f'(z) = \sum_{n=1}^{\infty} \frac{1}{(2n-1)!} z^{2n-1}, f''(z) = \sum_{n=1}^{\infty} \frac{1}{(2n-2)!} z^{2n-2} = \sum_{k=0}^{\infty} \frac{1}{(2k)!} z^{2k},$$

所以有 $f''(z) - f(z) = 0$, 由初始条件 $f(0) = 1$, $f'(0) = 0$ 解此方程得 $f(z) = \operatorname{ch} z$ 。

级数 $\rightarrow$ 递推关系

$\downarrow$ 微分方程。

111. 验证等式 $\frac{1}{a} - \frac{1}{a+b} + \frac{1}{a+2b} - \frac{1}{a+3b} + \dots = \int_0^1 \frac{t^{a-1}}{1+t^b} dt$, ($a > 0, b > 0$)。因此, 此

类无穷级数求和就化为求定积分。利用这个办法求下列级数之和:

$$(1) 1 - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \dots; (2) \frac{1}{2} - \frac{1}{5} + \frac{1}{8} - \frac{1}{11} + \dots; (3) 1 - \frac{1}{5} + \frac{1}{9} - \frac{1}{13} + \dots.$$

$$\text{证: } \frac{t^{a-1}}{1+t^b} = \sum_{n=0}^{\infty} (-1)^n t^{a+bn-1}, (|t| < 1)$$

$$\int_0^x \frac{t^{a-1}}{1+t^b} dt = \sum_{n=0}^{\infty} (-1)^n \int_0^x t^{a+bn-1} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{a+bn} x^{a+bn}, (|x| < 1)$$

由于级数 $\sum_{n=0}^{\infty} (-1)^n t^{a+bn-1}$ 在区间 $|t| \leq |x|$ ($|x| < 1$) 上一致收敛, 所以上式中求和与积分可交

换顺序。上式右边的幂级数在 $x=1$ 点是收敛的, 所以它在 $x=1$ 点左连续 (Abel 第二定理),

$$\text{即 } \lim_{x \rightarrow 1-0} \sum_{n=0}^{\infty} \frac{(-1)^n}{a+bn} x^{a+bn} = \sum_{n=0}^{\infty} \frac{(-1)^n}{a+bn}. \text{ 左边的积分是关于 } x \text{ 的连续函数, 所以}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{a+bn} = \lim_{x \rightarrow 1-0} \sum_{n=0}^{\infty} \frac{(-1)^n}{a+bn} x^{a+bn} = \lim_{x \rightarrow 1-0} \int_0^x \frac{t^{a-1}}{1+t^b} dt = \int_0^1 \frac{t^{a-1}}{1+t^b} dt.$$

(1) 这里 $a=1, b=3$, 所以和为

$$\begin{aligned} \int_0^1 \frac{1}{1+t^3} dt &= \frac{1}{3} \int_0^1 \left(\frac{1}{t+1} - \frac{t-2}{t^2-t+1} \right) dt = \frac{1}{3} \int_0^1 \frac{dt}{t+1} - \frac{1}{6} \int_0^1 \frac{2t-1}{t^2-t+1} dt + \frac{1}{2} \int_0^1 \frac{1}{\left(t-\frac{1}{2}\right)^2 + \frac{3}{4}} dt \\ &= \frac{1}{3} \ln(t+1) \Big|_{t=0}^{t=1} - \frac{1}{6} \int_0^1 \frac{d(t^2-t+1)}{t^2-t+1} + \frac{1}{\sqrt{3}} \int_0^1 \frac{d\left[\frac{2}{\sqrt{3}}\left(t-\frac{1}{2}\right)\right]}{1+\left[\frac{2}{\sqrt{3}}\left(t-\frac{1}{2}\right)\right]^2} \\ &= \frac{1}{3} \ln 2 - \frac{1}{6} \ln(t^2-t+1) \Big|_{t=0}^{t=1} + \frac{1}{\sqrt{3}} \arctan \left[\frac{2}{\sqrt{3}} \left(t - \frac{1}{2} \right) \right] \Big|_{t=0}^{t=1} \\ &= \frac{1}{3} \left(\frac{\pi}{\sqrt{3}} + \ln 2 \right) \end{aligned}$$

(2) $a=2, b=3$, 和为 $\int_0^1 \frac{t}{1+t^3} dt$ 。由于

$$\int_0^1 \frac{t}{1+t^3} dt + \int_0^1 \frac{1}{1+t^3} dt = \int_0^1 \frac{1}{t^2-t+1} dt = \frac{2}{\sqrt{3}} \arctan \left[\frac{2}{\sqrt{3}} \left(t - \frac{1}{2} \right) \right] \Big|_{t=0}^{t=1} = \frac{2\pi}{3\sqrt{3}}$$

$$\text{所以 } \int_0^1 \frac{t}{1+t^3} dt = \frac{2\pi}{3\sqrt{3}} - \int_0^1 \frac{1}{1+t^3} dt = \frac{1}{3} \left(\frac{\pi}{\sqrt{3}} - \ln 2 \right).$$

$$(3) \quad a=1, b=4, \text{ 和为 } \int_0^1 \frac{1}{1+t^4} dt = \frac{1}{2\sqrt{2}} \int_0^1 \left(\frac{t+\sqrt{2}}{t^2+\sqrt{2}t+1} - \frac{t-\sqrt{2}}{t^2-\sqrt{2}t+1} \right) dt$$

$$\begin{aligned} &= \frac{1}{4\sqrt{2}} \int_0^1 \frac{2t+\sqrt{2}}{t^2+\sqrt{2}t+1} dt + \frac{1}{2\sqrt{2}} \int_0^1 \frac{d\left[\sqrt{2}\left(t+\frac{1}{\sqrt{2}}\right)\right]}{1+\left[\sqrt{2}\left(t+\frac{1}{\sqrt{2}}\right)\right]^2} - \frac{1}{4\sqrt{2}} \int_0^1 \frac{2t-\sqrt{2}}{t^2-\sqrt{2}t+1} dt + \frac{1}{2\sqrt{2}} \int_0^1 \frac{d\left[\sqrt{2}\left(t-\frac{1}{\sqrt{2}}\right)\right]}{1+\left[\sqrt{2}\left(t-\frac{1}{\sqrt{2}}\right)\right]^2} \\ &= \frac{1}{4\sqrt{2}} \ln(2+\sqrt{2}) + \frac{1}{2\sqrt{2}} \arctan(\sqrt{2}+1) - \frac{\pi}{8\sqrt{2}} - \frac{1}{4\sqrt{2}} \ln(2-\sqrt{2}) + \frac{1}{2\sqrt{2}} \arctan(\sqrt{2}-1) + \frac{\pi}{8\sqrt{2}} \\ &= \frac{1}{4\sqrt{2}} \ln \frac{2+\sqrt{2}}{2-\sqrt{2}} + \frac{1}{2\sqrt{2}} \left[\arctan(\sqrt{2}+1) + \operatorname{arccot}(\sqrt{2}+1) \right] \\ &= \frac{1}{4\sqrt{2}} \ln(3+2\sqrt{2}) + \frac{1}{2\sqrt{2}} \cdot \frac{\pi}{2} = \frac{1}{4\sqrt{2}} \left[\ln(3+2\sqrt{2}) + \pi \right] \end{aligned}$$

112. 如果 k 和 n 是自然数, $a > 0, b > 0$, 证明:

$$\frac{k!}{(a+nb)(a+nb+1)\cdots(a+nb+k)} = \int_0^1 t^{a+nb-1} (1-t)^k dt, \text{ 并求下列级数之和:}$$

$$(1) \quad \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{3 \cdot 4 \cdot 5} + \frac{1}{5 \cdot 6 \cdot 7} + \cdots; \quad (2) \quad \frac{1}{1 \cdot 2 \cdot 3} - \frac{1}{3 \cdot 4 \cdot 5} + \frac{1}{5 \cdot 6 \cdot 7} - \cdots;$$

$$(3) \quad \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{4 \cdot 5 \cdot 6} + \frac{1}{7 \cdot 8 \cdot 9} + \cdots; \quad (4) \quad \frac{1}{2 \cdot 3 \cdot 4} - \frac{1}{4 \cdot 5 \cdot 6} + \frac{1}{6 \cdot 7 \cdot 8} - \cdots.$$

$$\text{证: } \int_0^x t^{p-1} (x-t)^k dt = \frac{1}{p} \int_0^x (t^p)' (x-t)^k dt = \frac{k}{p} \int_0^x t^p (x-t)^{k-1} dt$$

$$= \frac{k(k-1)}{p(p+1)} \int_0^x t^{p+1} (x-t)^{k-2} dt$$

=

$$= \frac{k! x^k}{p(p+1)\cdots(p+k)} x^p$$

$$\text{令 } p = a + nb, \quad x = 1 \text{ 即可得 } \frac{k!}{(a+nb)(a+nb+1)\cdots(a+nb+k)} = \int_0^1 t^{a+nb-1} (1-t)^k dt.$$

级数 $\sum_{n=0}^{\infty} \int_0^x t^{a+nb-1} (x-t)^k dt = x^{k+a} \sum_{n=0}^{\infty} \frac{k!}{(a+bn)(a+bn+1)\cdots(a+bn+k)} x^{bn}$ 是幂级数。

当 $k \geq 1$ 时, $\sum_n \int_0^x t^{a+nb-1} (x-t)^k dt \Big|_{x=1} = \sum_n \frac{k!}{(a+nb)(a+nb+1)\cdots(a+nb+k)}$ 显然是收敛

的, 由 Abel 第二定理, 幂级数 $\sum_n \int_0^x t^{a+nb-1} (x-t)^k dt$ 在 $x=1$ 处左连续, 即下式成立:

$$\lim_{x \rightarrow 1-0} \sum_n \int_0^x t^{a+nb-1} (x-t)^k dt = \sum_n \int_0^1 t^{a+nb-1} (1-t)^k dt \quad (|x| < 1)。$$

$$\text{级数 } \sum_{n=0}^{\infty} \frac{1}{(a+nb)(a+nb+1)\cdots(a+nb+k)} = \frac{1}{k!} \sum_{n=0}^{\infty} \int_0^1 t^{a+nb-1} (1-t)^k dt$$

$$= \frac{1}{k!} \lim_{x \rightarrow 1-0} \sum_{n=0}^{\infty} \int_0^x t^{a+nb-1} (x-t)^k dt$$

由于级数 $\sum_n t^{bn}$ 收敛半径为 1, 所以对于 $|x| < 1$, 求和与积分可交换顺序, 即

$$\text{上式} = \frac{1}{k!} \lim_{x \rightarrow 1-0} \int_0^x t^{a-1} (x-t)^k \sum_{n=0}^{\infty} t^{bn} dt = \frac{1}{k!} \int_0^1 \frac{t^{a-1} (1-t)^k}{1-t^b} dt$$

$$\text{同样可得 } \sum_{n=0}^{\infty} \frac{(-1)^n}{(a+nb)(a+nb+1)\cdots(a+nb+k)} = \frac{1}{k!} \int_0^1 \frac{t^{a-1} (1-t)^k}{1+t^b} dt。$$

$$(1) \text{ 这里 } a=1, b=2, k=2。 \text{ 和为 } \frac{1}{2} \int_0^1 \frac{(1-t)^2}{1-t^2} dt。$$

$$\frac{1}{2} \int_0^1 \frac{(1-t)^2}{1-t^2} dt = \frac{1}{2} \int_0^1 \left(\frac{2}{1+t} - 1 \right) dt = \ln 2 - \frac{1}{2};$$

$$(2) \text{ 和为 } \frac{1}{2} \int_0^1 \frac{(1-t)^2}{1+t^2} dt = \frac{1}{2} \int_0^1 \left(1 - \frac{2t}{1+t^2} \right) dt = \frac{1}{2} (1 - \ln 2)。$$

$$(3) \text{ } a=1, b=3, k=2。 \text{ 和为 } \frac{1}{2} \int_0^1 \frac{(1-t)^2}{1-t^3} dt$$

$$= \frac{\sqrt{3}}{2} \int_0^1 \frac{d \left[\frac{2}{\sqrt{3}} \left(t + \frac{1}{2} \right) \right]}{1 + \left[\frac{2}{\sqrt{3}} \left(t + \frac{1}{2} \right) \right]^2} - \frac{1}{4} \int_0^1 \frac{2t+1}{t^2+t+1} dt = \frac{1}{4} \left(\frac{\pi}{\sqrt{3}} - \ln 3 \right)$$

$$(4) \text{ } a=2, b=2, k=2。 \text{ 和为 } \frac{1}{2} \int_0^1 \frac{t(1-t)^2}{1+t^2} dt = \int_0^1 \left(\frac{1}{2} t - 1 + \frac{1}{1+t^2} \right) dt = \frac{1}{4} (\pi - 3)。$$

113. 求下列函数的 Laurent 展开: (1) $\frac{1}{z^2(z-1)}$, 在 $z=1$ 附近展开;

(2) $\frac{1}{z^2-3z+2}$, 展开区域为: (i) $1 < |z| < 2$, (ii) $2 < |z| < \infty$;

(3) $\frac{1}{z(z+1)}$, 展开区域为: (i) $1 < |z-i| < \sqrt{2}$, (ii) $0 < |z| < 1$;

(4) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, 展开区域为: (i) $3 < |z| < 4$, (ii) $4 < |z| < \infty$;

(5) $\frac{e^z}{z+2}$, 在 $|z| > 2$ 处展开; (6) $\frac{1}{1-\cos z}$, 在 $z=2n\pi$ 附近展开 (可只求出不为 0 的前四项系数)。

$$\begin{aligned} (1) \quad \frac{1}{z^2} &= [1+(z-1)]^{-2} = \sum_{n=0}^{\infty} \frac{(-2)(-2-1)(-2-2)\cdots(-2-n+1)}{n!} (z-1)^n \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n (n+1)!}{n!} (z-1)^n = \sum_{n=0}^{\infty} (-1)^n (n+1) (z-1)^n \quad (|z-1| < 1) \end{aligned}$$

$$\frac{1}{z^2(z-1)} = (z-1)^{-1} \sum_{n=0}^{\infty} (-1)^n (n+1) (z-1)^n = \sum_{k=-1}^{\infty} (-1)^{k+1} (k+2) (z-1)^k, \quad (0 < |z-1| < 1)$$

$$\begin{aligned} (2) \quad (i) \quad \frac{1}{z^2-3z+2} &= \frac{1}{z-2} - \frac{1}{z-1} = -\frac{1}{2} \frac{1}{1-z/2} - z^{-1} \frac{1}{1-z^{-1}} \quad \text{能拆不乘} \\ &= -\frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2^n} z^n - z^{-1} \sum_{n=0}^{\infty} z^{-n} = -\sum_{n=-1}^{\infty} z^n - \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} z^n \quad (1 < |z| < 2) \end{aligned}$$

$$(ii) \quad \frac{1}{z^2-3z+2} = z^{-1} \frac{1}{1-2z^{-1}} - z^{-1} \frac{1}{1-z^{-1}} = \sum_{k=-2}^{\infty} (2^{-k-1} - 1) z^k \quad (2 < |z| < \infty)$$

$$\begin{aligned} (3) \quad (i) \quad \frac{1}{z(z+1)} &= \frac{1}{z} - \frac{1}{z+1} = (z-i)^{-1} \frac{1}{1+\frac{i}{z-i}} - \frac{1}{1+i} \frac{1}{1+\frac{z-i}{1+i}} \\ &= \sum_{n=-1}^{\infty} i^{n+1} (z-i)^n - \sum_{n=0}^{\infty} \frac{(-1)^n}{(1+i)^{n+1}} (z-i)^n \quad (1 < |z-i| < \sqrt{2}) \end{aligned}$$

$$(ii) \quad \frac{1}{z(z+1)} = z^{-1} - \frac{1}{1+z} = z^{-1} + \sum_{n=0}^{\infty} (-1)^{n+1} z^n = \sum_{n=-1}^{\infty} (-1)^{n+1} z^n \quad (0 < |z| < 1)$$

$$(4) \text{ (i) } \frac{(z-1)(z-2)}{(z-3)(z-4)} = 1 + \frac{6}{z-4} - \frac{2}{z-3} = 1 - \frac{3}{2} \frac{1}{1-z/4} - 2z^{-1} \frac{1}{1-3z^{-1}}$$

$$= 1 - \frac{3}{2} \sum_{n=0}^{\infty} 4^{-n} z^n - 2 \sum_{n=-1}^{-\infty} 3^{-n-1} z^n \quad (3 < |z| < 4)$$

$$(ii) \frac{(z-1)(z-2)}{(z-3)(z-4)} = 1 + 6z^{-1} \frac{1}{1-4z^{-1}} - 2z^{-1} \frac{1}{1-3z^{-1}}$$

$$= 1 + \sum_{n=-1}^{-\infty} \left(\frac{3}{2} \cdot 4^{-n} - \frac{2}{3} \cdot 3^{-n} \right) z^n \quad (4 < |z| < \infty)$$

$$(5) \frac{1}{z+2} = z^{-1} \frac{1}{1+2/z} = \sum_{n=0}^{\infty} (-2)^n z^{-n-1} \quad (|z| > 2), \quad e^z = \sum_{n=0}^{\infty} \frac{1}{n!} z^n \quad (|z| < \infty), \text{ 两级数都}$$

是绝对收敛的, 所以可用 Cauchy 乘积计算 $\frac{e^z}{z+2}$ 。

$$\begin{aligned} \frac{e^z}{z+2} &= \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{z^k}{k!} \cdot (-2)^{n-k} z^{k-n-1} = \sum_{k=0}^{\infty} \sum_{n=k}^{\infty} \frac{(-2)^{n-k}}{k!} z^{2k-n-1} \\ &= \sum_{k=0}^{\infty} \sum_{n=2k}^{\infty} \frac{(-2)^{n-k}}{k!} z^{2k-n-1} + \sum_{k=0}^{\infty} \sum_{n=k}^{2k-1} \frac{(-2)^{n-k}}{k!} z^{2k-n-1} \\ &= \sum_{k=0}^{\infty} \sum_{m=-1}^{-\infty} \frac{(-2)^{k-m-1}}{k!} z^m + \sum_{k=0}^{\infty} \sum_{m=0}^{k-1} \frac{(-2)^{k-m-1}}{k!} z^m \\ &= \sum_{m=-1}^{-\infty} \sum_{k=0}^{\infty} \frac{(-2)^{k-m-1}}{k!} z^m + \sum_{m=0}^{\infty} \sum_{k=m+1}^{\infty} \frac{(-2)^{k-m-1}}{k!} z^m \quad (2 < |z| < \infty) \end{aligned}$$

$$(6) \frac{1}{1-\cos z} = \frac{1}{1-\cos(z-2n\pi)} = \frac{1}{2\sin^2\left(\frac{z-2n\pi}{2}\right)}, \text{ 可看出 } z=2n\pi \text{ 是二阶极点, 且它}$$

是关于 $z-2n\pi$ 的偶函数, 故可设 $\frac{1}{1-\cos z} = \sum_{n=-1}^{\infty} a_n (z-2n\pi)^{2n}$ 。

$$1 = (1-\cos z) \sum_{n=-1}^{\infty} a_n (z-2n\pi)^{2n} = \sum_{n=1}^{\infty} b_n (z-2n\pi)^{2n} \cdot \sum_{n=-1}^{\infty} a_n (z-2n\pi)^{2n}, \text{ 其中 } b_n = \frac{(-1)^{n+1}}{(2n)!}。$$

假设上式右边的 Cauchy 乘积收敛, 则 $1 = \sum_{n=0}^{\infty} \sum_{k=-1}^{n-1} a_k b_{n-k} (z-2n\pi)^{2n}$ 。比较系数可得

$$1 = a_{-1} b_1, \quad 0 = a_{-1} b_2 + a_0 b_1, \quad 0 = a_{-1} b_3 + a_0 b_2 + a_1 b_1, \quad 0 = a_{-1} b_4 + a_0 b_3 + a_1 b_2 + a_2 b_1, \quad \dots$$

解得 $a_{-1} = 2$, $a_0 = \frac{1}{6}$, $a_1 = \frac{1}{120}$, $a_2 = \frac{1}{3024}$, ...

即 $\frac{1}{1-\cos z} = 2(z-2n\pi)^{-2} + \frac{1}{6} + \frac{1}{120}(z-2n\pi)^2 + \frac{1}{3024}(z-2n\pi)^4 + \dots$ ($0 < |z-2n\pi| < 2\pi$)

114. 用级数相乘的方法求下列函数在 $z=0$ 附近的级数展开: (1) $-\ln(1-z)\ln(1+z)$;

(2) $\ln(1+z^2)\arctan z$; (3) $\exp\frac{1}{2}\left(z-\frac{a^2}{z}\right)$; (4) $e^z \sin \frac{1}{z}$.

由于幂级数在 $|z| < R$ (R 是收敛半径) 内都是绝对收敛的, 故可计算 Cauchy 乘积.

$$(1) \ln(1+z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} z^n, \quad \ln(1-z) = -\sum_{n=1}^{\infty} \frac{1}{n} z^n. \quad (|z| < 1)$$

$$-\ln(1-z)\ln(1+z) = \sum_{n=2}^{\infty} a_n z^n = \sum_{n=2}^{\infty} \sum_{k=1}^{n-1} \frac{(-1)^{n-k+1}}{k(n-k)} z^n$$

$$\text{当 } n \text{ 为奇数时, } a_n = \sum_{k=1}^{n-1} \frac{(-1)^{n-k+1}}{k(n-k)} = \sum_{k=1}^{n-1} \frac{(-1)^k}{k(n-k)}. \text{ 又有 } a_n = \sum_{k=1}^{n-1} \frac{(-1)^{n-k+1}}{k(n-k)} = -\sum_{j=1}^{n-1} \frac{(-1)^j}{j(n-j)},$$

(作变换 $n-k=j$) 所以 $a_n = -a_n$, 即 $a_n = 0$. 令上面的级数表达式中 n 只取偶数, 则

$$-\ln(1-z)\ln(1+z) = \sum_{n=1}^{\infty} \sum_{k=1}^{2n-1} \frac{(-1)^{k+1}}{k(2n-k)} z^{2n} = \sum_{n=1}^{\infty} \frac{1}{2n} \sum_{k=1}^{2n-1} \left[\frac{(-1)^{k+1}}{k} + \frac{(-1)^{k+1}}{2n-k} \right] z^{2n}$$

$$\text{作变换 } 2n-k=m, \text{ 则 } \sum_{k=1}^{2n-1} \frac{(-1)^{k+1}}{2n-k} = \sum_{m=1}^{2n-1} \frac{(-1)^{m+1}}{m}, \text{ 所以}$$

$$-\ln(1-z)\ln(1+z) = \sum_{n=1}^{\infty} \frac{1}{n} \sum_{k=1}^{2n-1} \frac{(-1)^{k+1}}{k} z^{2n} \quad (|z| < 1)$$

$$\begin{aligned} (2) \ln(1+z^2)\arctan z &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} z^{2n} \cdot \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} z^{2n+1} = \sum_{n=1}^{\infty} \sum_{k=0}^{n-1} \frac{(-1)^{n+1}}{(2k+1)(n-k)} z^{2n+1} \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n+1} \sum_{k=0}^{n-1} \left(\frac{2}{2k+1} + \frac{1}{n-k} \right) z^{2n+1} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n+1} \left(\sum_{k=0}^{n-1} \frac{2}{2k+1} + \sum_{k=1}^n \frac{1}{k} \right) z^{2n+1} \\ &= 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n+1} \left(\sum_{k=0}^{n-1} \frac{1}{2k+1} + \sum_{k=1}^n \frac{1}{2k} \right) z^{2n+1} = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n+1} \sum_{k=1}^{2n} \frac{1}{k} z^{2n+1} \quad (|z| < 1) \end{aligned}$$

(3)

$$\exp \frac{1}{2} \left(z - \frac{a^2}{z} \right) = \sum_{n=0}^{\infty} \frac{z^n}{2^n n!} \cdot \sum_{n=0}^{\infty} \frac{(-1)^n a^{2n}}{2^n n!} z^{-n} = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{(-1)^k a^{2k}}{2^n k! (n-k)!} z^{n-2k} = \sum_{k=0}^{\infty} \sum_{n=k}^{\infty} \frac{(-1)^k a^{2k}}{2^n k! (n-k)!} z^{n-2k}$$

$$\begin{aligned} \text{令 } n-2k=m, \text{ 则上式} &= \sum_{k=0}^{\infty} \sum_{m=-k}^{\infty} \frac{(-1)^k}{k! (k+m)!} \left(\frac{a}{2} \right)^{2k+m} \left(\frac{z}{a} \right)^m \\ &= \sum_{k=0}^{\infty} \sum_{m=-k}^{-1} \frac{(-1)^k}{k! (k+m)!} \left(\frac{a}{2} \right)^{2k+m} \left(\frac{z}{a} \right)^m + \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} \frac{(-1)^k}{k! (k+m)!} \left(\frac{a}{2} \right)^{2k+m} \left(\frac{z}{a} \right)^m \\ &= \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (k+m)!} \left(\frac{a}{2} \right)^{2k+m} \left(\frac{z}{a} \right)^m + \sum_{m=-1}^{-\infty} \sum_{k=-m}^{\infty} \frac{(-1)^k}{k! (k+m)!} \left(\frac{a}{2} \right)^{2k+m} \left(\frac{z}{a} \right)^m \\ &= \sum_{m=-\infty}^{\infty} J_m(a) \left(\frac{z}{a} \right)^m \quad (0 < |z| < \infty) \end{aligned}$$

$$\text{其中 } J_m(a) = \begin{cases} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (k+m)!} \left(\frac{a}{2} \right)^{2k+m}, & m=0, 1, 2, \dots \\ \sum_{k=-m}^{\infty} \frac{(-1)^k}{k! (k+m)!} \left(\frac{a}{2} \right)^{2k+m}, & m=-1, -2, \dots \end{cases}.$$

$$\begin{aligned} (4) \quad e^z \sin \frac{1}{z} &= \sum_{n=0}^{\infty} \frac{z^n}{n!} \cdot \sum_{n=0}^{\infty} \frac{(-1)^n z^{-2n-1}}{(2n+1)!} = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{(-1)^k}{(2k+1)! (n-k)!} z^{n-3k-1} \\ &= \sum_{k=0}^{\infty} \sum_{n=k}^{\infty} \frac{(-1)^k}{(2k+1)! (n-k)!} z^{n-3k-1} \end{aligned}$$

$$\begin{aligned} \text{令 } n-3k-1=m, \text{ 则上式} &= \sum_{k=0}^{\infty} \sum_{m=-(2k+1)}^{\infty} \frac{(-1)^k}{(2k+1)! (2k+m+1)!} z^m \\ &= \sum_{k=0}^{\infty} \sum_{m=-(2k+1)}^{-1} \frac{(-1)^k}{(2k+1)! (2k+m+1)!} z^m + \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} \frac{(-1)^k}{(2k+1)! (2k+m+1)!} z^m \\ &= \sum_{m=-1}^{-\infty} \sum_{k=\lceil \frac{n}{2} \rceil}^{\infty} \frac{(-1)^k}{(2k+1)! (2k+m+1)!} z^m + \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)! (2k+m+1)!} z^m \quad (0 < |z| < \infty) \end{aligned}$$

115. 将下列函数在 $z=0$ 点展开 (其中的多值函数均取主值分枝):

(1) $\sqrt{1+z^2} \ln(z+\sqrt{1+z^2})$; (2) $\sqrt{1-z^2} \sin^{-1} z$; (3) $(1+z)^{-n} \ln(1+z)$; (4) $\exp(\tan^{-1} z)$ 。

(1) 令 $f(z) = \sqrt{1+z^2} \ln(z+\sqrt{1+z^2})$, 可得微分方程 $f'(z) - \frac{z}{1+z^2} f(z) = 1$ 。

可看出方程在 $|z| < 1$ 上有解析解, 可设 $f(z) = \sum_{n=0}^{\infty} a_n z^n$, 由于在 $|z| < 1$ 上可逐项求导, 所以

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}, \text{ 则 } \sum_{n=1}^{\infty} n a_n z^{n-1} + \sum_{n=1}^{\infty} n a_n z^{n+1} - \sum_{n=0}^{\infty} a_n z^{n+1} = 1 + z^2, \text{ 即}$$

$$a_1 + (2a_2 - a_0)z + 3a_3 z^2 + \sum_{n=3}^{\infty} [(n+1)a_{n+1} + (n-2)a_{n-1}] z^n = 1 + z^2$$

比较系数可得 $a_1 = 1$, $2a_2 - a_0 = 0$, $3a_3 = 1$, $a_k = -\frac{k-3}{k} a_{k-2}$ ($k \geq 4$)。

又有 $a_0 = f(0) = 0$, 所以 $a_2 = 0$, $a_3 = 1/3$,

$$a_{2k} = -\frac{2k-3}{2k} a_{2(k-1)} = (-1)^2 \frac{(2k-3)(2k-5)}{(2k)(2k-2)} a_{2(k-2)} = \cdots = (-1)^{k-1} \frac{(2k-3)!!}{(2k)!!} a_2 = 0,$$

($k = 2, 3, 4, \cdots$)

$$a_{2k+1} = -\frac{2k-2}{2k+1} a_{2(k-1)+1} = (-1)^2 \frac{(2k-2)(2k-4)}{(2k+1)(2k-1)} a_{2(k-2)+1} = \cdots = (-1)^{k-1} \frac{(2k-2)!!}{(2k+1)!!}$$

($k = 2, 3, 4, \cdots$)。

$$\text{所以 } \sqrt{1+z^2} \ln(z + \sqrt{1+z^2}) = z + \sum_{k=1}^{\infty} (-1)^{k-1} \frac{(2k-2)!!}{(2k+1)!!} z^{2k+1} \quad (|z| < 1)$$

(2) 可得微分方程 $f'(z) + \frac{z}{1-z^2} f(z) = 1$, 在 $|z| < 1$ 上设 $f(z) = \sum_{n=0}^{\infty} a_n z^n$, 则

$$a_1 + (2a_2 + a_0)z + 3a_3 z^2 + \sum_{n=2}^{\infty} [(n+2)a_{n+2} - (n-1)a_n] z^{n+1} = 1 - z^2. \text{ 所以}$$

$$a_0 = f(0) = 0, \quad a_1 = 1, \quad a_2 = 0, \quad a_3 = -1/3, \quad a_{2k} = \frac{2k-3}{2k} a_{2k-2} = \cdots = \frac{(2k-3)!!}{(2k)!!} a_0 = 0$$

$$(k = 2, 3, 4, \cdots), \quad a_{2k+1} = \frac{2k-2}{2k+1} a_{2(k-1)+1} = \cdots = -\frac{(2k-2)!!}{(2k+1)!!} \quad (k = 2, 3, 4, \cdots).$$

$$\text{所以 } \sqrt{1-z^2} \sin^{-1} z = z - \sum_{k=1}^{\infty} \frac{(2k-2)!!}{(2k+1)!!} z^{2n+1} \quad (|z| < 1)$$

$$(3) \text{ 令 } f(z) = (1+z)^{-n} \ln(1+z), \text{ 则 } f^{(k)}(z) = \sum_{l=0}^k \frac{k!}{l!(k-l)!} [(1+z)^{-n}]^{(l)} [\ln(1+z)]^{(k-l)},$$

$$\left[(1+z)^{-n}\right]^{(l)} = \frac{(-1)^l (n+l-1)!}{(n-1)!(1+z)^{n+l}}, \quad [\ln(1+z)]^{(m)} = \left(\frac{1}{1+z}\right)^{(m-1)} = \frac{(-1)^{m-1} (m-1)!}{(1+z)^m} \quad (m \geq 1),$$

$$\text{代入 } f^{(k)}(z) \text{ 的表达式得 } f^{(k)}(0) = \frac{(-1)^{k-1} k!}{(n-1)!} \sum_{l=0}^{k-1} \frac{(n+l-1)!}{l!(k-l)}.$$

$$\text{所以 } f(z) = \sum_{k=1}^{\infty} \frac{f^{(k)}(0)}{k!} z^k = \frac{1}{(n-1)!} \sum_{k=1}^{\infty} (-1)^{k-1} \sum_{l=0}^{k-1} \frac{(n+l-1)!}{l!(k-l)} z^k \quad (|z| < 1)$$

$$(4) \text{ 可得微分方程 } f'(z) - \frac{1}{1+z^2} f(z) = 0. \text{ 在 } |z| < 1 \text{ 上设 } f(z) = \sum_{n=0}^{\infty} a_n z^n, \text{ 则}$$

$$a_1 - a_0 + \sum_{n=1}^{\infty} [(n+1)a_{n+1} - a_n + (n-1)a_{n-1}] z^n = 0.$$

$$a_0 = f(0) = 1, \quad a_1 = a_0 = 1, \quad a_n = \frac{1}{n} a_{n-1} - \frac{n-2}{n} a_{n-2}, \quad n = 2, 3, 4, \dots$$

$$\text{若令 } a_n = \frac{b_n}{n!}, \text{ 则有 } b_0 = 1, \quad b_1 = 1, \quad b_n = b_{n-1} - (n-1)(n-2)b_{n-2}, \quad n = 2, 3, 4, \dots.$$

116. 证明: 如果级数 $\sum_{k=1}^{\infty} u_k(z)$ 在区域 G 的边界 C 上一致收敛, $u_k(z)$ ($k=1, 2, \dots$) 在 $\bar{G}$

中解析, 则此级数在 $\bar{G}$ 中一致收敛。

证: 因为级数在 C 上一致收敛, 故对任意 $\varepsilon > 0$, n 充分大时对任意整数 p 及任意 $z \in C$ 有

$$|u_{n+1} + u_{n+2} + \dots + u_{n+p}| < \varepsilon. \text{ 由最大模原理, 该不等式对于任意 } z \in \bar{G} \text{ 都成立, 即此级数在}$$

$\bar{G}$ 中一致收敛。

117. 利用 Abel 第二定理证明: 如果 $\sum_{k=0}^{\infty} u_k$, $\sum_{k=0}^{\infty} v_k$ 与 $\sum_{k=0}^{\infty} w_k = \sum_{k=0}^{\infty} \sum_{l=0}^k u_l v_{k-l}$ 分别收敛于 A, B

和 C , 则 $C = AB$ 。

$$\text{证: 令 } U(x) = \sum_{k=0}^{\infty} u_k x^k, \quad V(x) = \sum_{k=0}^{\infty} v_k x^k, \quad W(x) = \sum_{k=0}^{\infty} w_k x^k = \sum_{k=0}^{\infty} \sum_{l=0}^k u_l v_{k-l} x^k.$$

由于 $U(x)$, $V(x)$ 在 $x=1$ 点收敛, 所以 $U(x)$, $V(x)$ 在区间 $|x| < 1$ 上绝对收敛 (阿贝尔

第一定理), 因此可用 Cauchy 乘积计算 $U(x) \cdot V(x)$, 即

$$U(x) \cdot V(x) = \sum_{k=0}^{\infty} \sum_{l=0}^k u_l x^l \cdot v_{k-l} x^{k-l} = \sum_{k=0}^{\infty} w_k x^k = W(x)$$

由于 $U(x)$, $V(x)$ 和 $W(x)$ 都在 $x=1$ 点收敛, 所以他们都在 $x=1$ 点左连续 (阿贝尔第二

定理), 即 $\lim_{x \rightarrow 1-0} U(x) \cdot V(x) = U(1) \cdot V(1) = \lim_{x \rightarrow 1-0} W(x) = W(1)$, 也就是 $C = AB$ 。

$$118. \text{ 定义 } \sin z = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} z^{2k+1}, \cos z = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k}, \quad (|z| < \infty)$$

试利用级数乘法证明 $\sin(a+b) = \sin a \cos b + \cos a \sin b$ 。

$$\begin{aligned} \text{证: 右边} &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} a^{2k+1} \cdot \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} b^{2k} + \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} a^{2k} \cdot \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} b^{2k+1} \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n \left[\frac{(-1)^k}{(2k+1)!} a^{2k+1} \frac{(-1)^{n-k}}{(2n-2k)!} b^{2n-2k} + \frac{(-1)^k}{(2k)!} a^{2k} \frac{(-1)^{n-k}}{(2n-2k+1)!} b^{2n-2k+1} \right] \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n \left[\frac{(-1)^n b^{2n+1}}{(2k+1)!(2n+1-2k-1)!} \left(\frac{a}{b}\right)^{2k+1} + \frac{(-1)^n b^{2n+1}}{(2k)!(2n+1-2k)!} \left(\frac{a}{b}\right)^{2k} \right] \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n b^{2n+1}}{(2n+1)!} \left[\sum_{k=0}^n \binom{2n+1}{2k+1} \left(\frac{a}{b}\right)^{2k+1} + \sum_{k=0}^n \binom{2n+1}{2k} \left(\frac{a}{b}\right)^{2k} \right] \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n b^{2n+1}}{(2n+1)!} \sum_{k=0}^{2n+1} \binom{2n+1}{k} \left(\frac{a}{b}\right)^k = \sum_{n=0}^{\infty} \frac{(-1)^n b^{2n+1}}{(2n+1)!} \left(1 + \frac{a}{b}\right)^{2n+1} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n (a+b)^{2n+1}}{(2n+1)!} = \sin(a+b) = \text{左边} \end{aligned}$$

119. 计算积分 $\oint_{\gamma} \left(\sum_{n=-2}^{\infty} z^n \right) dz$, 其中 γ 是单位圆内任一不经过原点的简单闭合曲线。

$$\oint_{\gamma} \left(\sum_{n=-2}^{\infty} z^n \right) dz = \oint_{\gamma} \frac{dz}{z^2} + \oint_{\gamma} \frac{dz}{z} + \oint_{\gamma} \left(\sum_{n=0}^{\infty} z^n \right) dz$$

由于在单位圆内, $\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$, 这是解析函数, 故有 $\oint_{\gamma} \left(\sum_{n=0}^{\infty} z^n \right) dz = 0$ 。

若 γ 包围原点, 则 $\oint_{\gamma} \frac{dz}{z} = 2\pi i$, $\oint_{\gamma} \frac{dz}{z^2} = 0$, 所以 $\oint_{\gamma} \left(\sum_{n=-2}^{\infty} z^n \right) dz = 2\pi i$;

若 γ 不包围原点, 则 $\oint_{\gamma} \frac{dz}{z} = 0$, $\oint_{\gamma} \frac{dz}{z^2} = 0$, 所以 $\oint_{\gamma} \left(\sum_{n=-2}^{\infty} z^n \right) dz = 0$ 。

附:

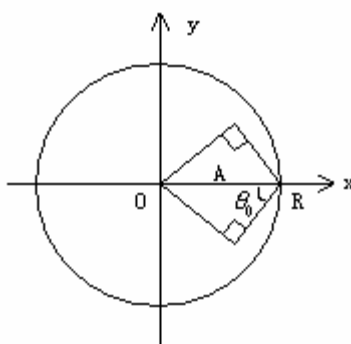
Abel 第一定理: 若幂级数 $\sum_{n=0}^{\infty} a_n z^n$ 在 $z_0 \neq 0$ 处收敛, 则它在圆 $|z| < |z_0|$ 内绝对收敛, 并且在

任意闭圆 $|z| \leq k|z_0|$ ($0 < k < 1$) 内一致收敛。

Abel 第二定理: 若复幂级数 $\sum_{n=0}^{\infty} a_n z^n$ 的收敛半径为 R , 且在 $z = R$ 处收敛, 则 (1) 该级数

级数在以 $z = R$ 为顶点, 以 $[0, R]$ 为角平分线, 开度为 $2\theta_0$ ($< \pi$) 的四边形角域 A (下图

所示) 上一致收敛; (2) $\lim_{z \in A, z \rightarrow R} \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} a_n R^n$ 。



若实幂级数 $\sum_{n=0}^{\infty} a_n x^n$ 的收敛半径为 R , 并且它在 $x = R$ 处收敛, 则该级数在闭区间 $[0, R]$

上一致收敛, 且在 $x = R$ 处左连续。

120. 判断下列函数奇点的性质, 如果是极点, 确定其阶数: (1) $\frac{1}{z^2+a^2}$; (2) $\frac{\cos az}{z^2}$;

(3) $\frac{\cos az - \cos bz}{z^2}$; (4) $\frac{\sin z}{z^2} - \frac{1}{z}$; (5) $\frac{1}{e^z-1} - \frac{1}{z}$; (6) $\sin \frac{1}{z}$; (7) $\frac{\sqrt{z}}{\sin \sqrt{z}}$;

(8) $\int_0^z \frac{e^{\sqrt{\zeta}} - e^{-\sqrt{\zeta}}}{\sqrt{\zeta}} d\zeta$.

(1) $\pm ai$ 是一阶极点. $f\left(\frac{1}{t}\right) = \frac{t^2}{1+a^2t^2}$, 0 不是 $f(1/t)$ 的奇点, 故 ∞ 不是 $f(z)$ 的奇点.

(2) 0 是二阶极点; $f\left(\frac{1}{t}\right) = t^2 \left[1 - \frac{1}{2!} \left(\frac{a}{t}\right)^2 + \frac{1}{4!} \left(\frac{a}{t}\right)^4 - + \dots \right] = t^2 - \frac{a^2}{2!} + \frac{a^4}{4!} t^{-2} - + \dots$

0 是 $f(1/t)$ 的本性奇点, 所以 ∞ 是 $f(z)$ 的本性奇点.

$$\begin{aligned} (3) \quad f(z) &= z^{-2} \left[\left(1 - \frac{1}{2!} a^2 z^2 + \frac{1}{4!} a^4 z^4 - + \dots \right) - \left(1 - \frac{1}{2!} b^2 z^2 + \frac{1}{4!} b^4 z^4 - + \dots \right) \right] \\ &= -\frac{1}{2!} (a^2 - b^2) + \frac{1}{4!} (a^4 - b^4) z^2 - + \dots \end{aligned}$$

所以 0 是 $f(z)$ 的可去奇点. $f(1/t) = -\frac{1}{2!} (a^2 - b^2) + \frac{1}{4!} (a^4 - b^4) t^{-2} - + \dots$, ∞ 是 $f(z)$ 的本性奇点.

(4) $f(z) = z^{-2} \left(z - \frac{1}{3!} z^3 + \frac{1}{5!} z^5 - + \dots \right) - z^{-1} = -\frac{1}{3!} z + \frac{1}{5!} z^3 - + \dots$, 0 是可去奇点.

$f\left(\frac{1}{t}\right) = -\frac{1}{3!} t^{-1} + \frac{1}{5!} t^{-3} - + \dots$, ∞ 是 $f(z)$ 的本性奇点.

(5) $\lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \frac{z - e^z + 1}{z(e^z - 1)} = \lim_{z \rightarrow 0} \frac{z + 1 - (1 + z + z^2/2 + \dots)}{z^2} = -\frac{1}{2}$, 所以 0 是可去奇点.

$$\lim_{z \rightarrow 2n\pi i} (z - 2n\pi i) f(z) = \lim_{z \rightarrow 2n\pi i} \frac{(z - 2n\pi i)(z - e^z + 1)}{z(e^z - 1)} = \lim_{z \rightarrow 2n\pi i} \frac{z - e^z + 1 + (z - 2n\pi i)(1 - e^z)}{e^z - 1 + ze^z} = 1$$

所以 $2n\pi i$ ($n = \pm 1, \pm 2, \dots$) 是一阶奇点. 由于 $2n\pi i \rightarrow \infty$, 所以在 ∞ 点的任一邻域内有无穷个奇点, 即 ∞ 是非孤立奇点.

(6) $f(z) = z^{-1} - \frac{1}{3!} z^{-3} + \frac{1}{5!} z^{-5} - + \dots$, 所以 0 是本性奇点.

$f(1/t) = t - \frac{1}{3!} t^3 + \frac{1}{5!} t^5 - + \dots$, 所以 ∞ 点是可去奇点.

(7) $\lim_{z \rightarrow 0} f(z) = 1$, 所以 0 是可去奇点.

$$\lim_{z \rightarrow n^2 \pi^2} (z - n^2 \pi^2) f(z) = \lim_{z \rightarrow n^2 \pi^2} \frac{(z - n^2 \pi^2) \sqrt{z}}{\sin \sqrt{z}} = \lim_{z \rightarrow n^2 \pi^2} \frac{\sqrt{z} + \frac{(z - n^2 \pi^2)}{2\sqrt{z}}}{\frac{\cos \sqrt{z}}{2\sqrt{z}}} = (-1)^n 2n^2 \pi^2$$

所以 $n^2 \pi^2$ ($n=1, 2, 3, \dots$) 是一阶极点。由于 $n^2 \pi^2 \rightarrow \infty$, 所以 ∞ 点的任意邻域包含无穷个奇点, 即 ∞ 点为非孤立奇点。

$$\begin{aligned} (8) \quad f(z) &= \int_0^z \frac{e^{\sqrt{\zeta}} - e^{-\sqrt{\zeta}}}{\sqrt{\zeta}} d\zeta = \int_0^z \zeta^{-\frac{1}{2}} \left(\sum_{n=0}^{\infty} \frac{\zeta^{n/2}}{n!} - \sum_{n=0}^{\infty} \frac{(-1)^n \zeta^{n/2}}{n!} \right) d\zeta \\ &= 2 \int_0^z \sum_{k=0}^{\infty} \frac{\zeta^k}{(2k+1)!} d\zeta = 2 \sum_{k=0}^{\infty} \frac{\zeta^{k+1}}{(k+1)(2k+1)!} \end{aligned}$$

$$f(1/t) = 2 \sum_{k=0}^{\infty} \frac{t^{-(k+1)}}{(k+1)(2k+1)!}, \text{ 即 } \infty \text{ 点为本性奇点。}$$

$$f(z) = 2 \int_0^z (e^{\sqrt{\zeta}} - e^{-\sqrt{\zeta}}) d\sqrt{\zeta} = 2(e^{\sqrt{z}} + e^{-\sqrt{z}} - 2), \text{ 即除 } \infty \text{ 点外无其他奇点。}$$

121. 求下列函数在指定点 z_0 的留数: (1) $\frac{e^{z^2}}{z-1}$, $z_0=1$; (2) $\frac{e^{z^2}}{(z-1)^2}$, $z_0=1$;

$$(3) \left(\frac{z}{1-\cos z} \right)^2, z_0=0; (4) \frac{z^2}{z^4-1}, z_0=i; (5) \frac{1}{z^2 \sin z}, z_0=0; (6) \frac{1+e^z}{z^4}, z_0=0;$$

$$(7) \frac{e^z}{(z^2-1)^2}, z_0=1; (8) \frac{1}{\operatorname{ch} \sqrt{z}}, z_0 = -\left(\frac{2n+1}{2} \pi \right)^2, n=0, 1, 2, \dots$$

$$(1) \operatorname{res} f(1) = \lim_{z \rightarrow 1} (z-1) f(z) = e.$$

$$(2) \operatorname{res} f(1) = \lim_{z \rightarrow 1} \frac{d}{dz} [(z-1)^2 f(z)] = 2e.$$

$$(3) 1 - \cos z = 1 - \left(1 - \frac{1}{2!} z^2 + \frac{1}{4!} z^4 - \frac{1}{6!} z^6 + \dots \right) = \frac{1}{2} z^2 \left(1 - \frac{1}{12} z^2 + \frac{1}{360} z^4 - \dots \right),$$

$$\begin{aligned} \left(\frac{z}{1-\cos z} \right)^2 &= 4z^{-2} \left(1 - \frac{1}{12} z^2 + \frac{1}{360} z^4 - \dots \right)^{-2} = 4z^{-2} \left[1 + 2 \left(\frac{1}{12} z^2 - \frac{1}{360} z^4 + \dots \right) + O(z^4) \right] \\ &= 4z^{-2} + \frac{2}{3} - \frac{1}{45} z^2 + \dots \end{aligned}$$

所以 $\operatorname{res} f(0) = 0$ 。

$$(4) f(z) = \frac{z^2}{(z+i)(z-i)(z+1)(z-1)}, i \text{ 是一阶极点。} \operatorname{res} f(i) = \lim_{z \rightarrow i} (z-i) f(z) = -\frac{1}{4}i。$$

$$(5) \text{ 显然 } 0 \text{ 是三阶极点。} \operatorname{res} f(0) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} z^3 f(z) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{z(1+\cos^2 z) - \sin 2z}{\sin^3 z}$$

$$= \frac{1}{2} \lim_{z \rightarrow 0} \frac{3\sin^2 z - 2z \sin z \cos z}{3\sin^2 z \cos z} = \frac{1}{2} \lim_{z \rightarrow 0} \left(\frac{1}{\cos z} - \frac{2z}{3\sin z} \right) = \frac{1}{6}$$

$$(6) \operatorname{res} f(0) = \frac{1}{3!} \lim_{z \rightarrow 0} \frac{d^3}{dz^3} z^4 f(z) = \frac{1}{6} \lim_{z \rightarrow 0} e^z = \frac{1}{6}。$$

$$(7) \operatorname{res} f(1) = \lim_{z \rightarrow 1} \frac{d}{dz} (z-1)^2 f(z) = 0。$$

$$(8) \operatorname{res} f \left[-\left(\frac{2n+1}{2} \pi \right)^2 \right] = \lim_{z \rightarrow -\left(\frac{2n+1}{2} \pi \right)^2} \left[z + \left(\frac{2n+1}{2} \pi \right)^2 \right] f(z)$$

$$= \lim_{z \rightarrow -\left(\frac{2n+1}{2} \pi \right)^2} \frac{2 \left[z + \left(\frac{2n+1}{2} \pi \right)^2 \right]}{e^{\sqrt{z}} + e^{-\sqrt{z}}} = \lim_{z \rightarrow -\left(\frac{2n+1}{2} \pi \right)^2} \frac{4\sqrt{z}}{e^{\sqrt{z}} - e^{-\sqrt{z}}} = (-1)^n (2n+1)\pi$$

122. 求下列函数在奇点处的留数: (1) $\frac{1}{z^3 - z^5}$; (2) $\frac{1}{(1+z^2)^{m+1}}$; (3) $\frac{z}{1 - \cos z}$;

(4) $\frac{\sqrt{z}}{\operatorname{sh} \sqrt{z}}$; (5) $e^{\frac{1}{1-z}}$; (6) $\cos \sqrt{\frac{1}{z}}$; (7) $\frac{1}{(z-1) \ln z}$;

(8) $\frac{1}{z} \left[1 + \frac{1}{z+1} + \frac{1}{(z+1)^2} + \cdots + \frac{1}{(z+1)^n} \right]$ 。

(1) $f(z) = \frac{1}{z^3(z+1)(z-1)}$, 0 是三阶极点, ± 1 是一阶极点。 $f(1/t) = \frac{t^5}{t^2-1}$, ∞ 点

是可去奇点。 $\operatorname{res} f(0) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} z^3 f(z) = 1$, $\operatorname{res} f(1) = \lim_{z \rightarrow 1} (z-1) f(z) = \frac{1}{2}$,

$\operatorname{res} f(-1) = \lim_{z \rightarrow -1} (z+1) f(z) = \frac{1}{2}。$

(2) $\pm i$ 是 $m+1$ 阶极点, ∞ 点是可去奇点。

$$\begin{aligned}\operatorname{res} f(i) &= \frac{1}{m!} \lim_{z \rightarrow i} \frac{d^m}{dz^m} \frac{1}{(z+i)^{m+1}} = \frac{1}{m!} \lim_{z \rightarrow i} \frac{(-1)^m (m+1)(m+2) \cdots (2m)}{(z+i)^{2m+1}} \\ &= \frac{1}{(m!)^2} \frac{(-1)^m (2m)!}{(2i)^{2m+1}} = -i \frac{(2m)!}{(m!)^2 2^{2m+1}} \\ \operatorname{res} f(-i) &= i \frac{(2m)!}{(m!)^2 2^{2m+1}}\end{aligned}$$

(3) 可看出 0 是一阶极点, $2n\pi$ ($n = \pm 1, \pm 2, \dots$) 是二阶极点, ∞ 点是非孤立奇点。

$$\operatorname{res} f(0) = \lim_{z \rightarrow 0} \frac{z^2}{1 - \cos z} = \lim_{z \rightarrow 0} \frac{z^2}{2 \sin^2 \frac{z}{2}} = 2。$$

$$1 - \cos z = 1 - \cos(z - 2n\pi) = \frac{1}{2}(z - 2n\pi)^2 \left[1 - \frac{1}{12}(z - 2n\pi)^2 + \frac{1}{360}(z - 2n\pi)^4 - + \dots \right]$$

$$\frac{z}{1 - \cos z} = (z - 2n\pi) [1 - \cos(z - 2n\pi)]^{-1} + 2n\pi [1 - \cos(z - 2n\pi)]^{-1}$$

$$= 2(z - 2n\pi)^{-1} \left[1 + \frac{1}{12}(z - 2n\pi)^2 + O(z - 2n\pi)^4 \right]$$

$$+ 4n\pi(z - 2n\pi)^{-2} \left[1 + \frac{1}{12}(z - 2n\pi)^2 + O(z - 2n\pi)^4 \right]$$

$$= 4n\pi(z - 2n\pi)^{-2} + 2(z - 2n\pi)^{-1} + \frac{n\pi}{3} + \frac{1}{6}(z - 2n\pi) + \dots$$

即 $\operatorname{res} f(2n\pi) = 2$ 。

$$(4) \quad f(z) = \frac{2\sqrt{z}}{e^{\sqrt{z}} - e^{-\sqrt{z}}}, \quad \text{奇点是 } -k^2\pi^2 \quad (k = 0, \pm 1, \pm 2, \dots), \quad \infty \text{ 点是非孤立奇点。}$$

$$\lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \frac{2}{e^{\sqrt{z}} + e^{-\sqrt{z}}} = 1, \quad \text{所以 } 0 \text{ 是可去奇点, } \operatorname{res} f(0) = 0。k \neq 0 \text{ 时,}$$

$$\operatorname{res} f(-k^2\pi^2) = \lim_{z \rightarrow -k^2\pi^2} \frac{2(z + k^2\pi^2)\sqrt{z}}{e^{\sqrt{z}} - e^{-\sqrt{z}}} = \lim_{z \rightarrow -k^2\pi^2} \frac{4z + 2(z + k^2\pi^2)}{e^{\sqrt{z}} + e^{-\sqrt{z}}} = (-1)^{k+1} 2k^2\pi^2。$$

$$(5) \quad f(z) = 1 + \frac{1}{1-z} + \frac{1}{2!} \left(\frac{1}{1-z} \right)^2 + \frac{1}{3!} \left(\frac{1}{1-z} \right)^3 + \dots = 1 - (z-1)^{-1} + \frac{1}{2}(z-1)^{-2} - \frac{1}{3!}(z-1)^{-3} + \dots$$

即 $\operatorname{res} f(1) = -1$ 。由于 $f(z)$ 只有两个孤立奇点 $1, \infty$, 所以 $\operatorname{res} f(\infty) = -\operatorname{res} f(1) = 1$ 。

$$(6) \quad f(z) = 1 - \frac{1}{2!} \left(\sqrt{\frac{1}{z}} \right)^2 + \frac{1}{4!} \left(\sqrt{\frac{1}{z}} \right)^4 - + \cdots = 1 - \frac{1}{2} z^{-1} + \frac{1}{24} z^{-2} - + \cdots$$

$$\text{即 } \operatorname{res} f(0) = -\frac{1}{2}。 \operatorname{res} f(\infty) = -\operatorname{res} f(0) = \frac{1}{2}。$$

(7) 若规定 $\ln z|_{z=1} = 0$ ，则 1 是二阶极点。此时

$$\operatorname{res} f(1) = \lim_{z \rightarrow 1} \frac{d}{dz} (z-1)^2 f(z) = \lim_{z \rightarrow 1} \frac{z \ln z - z + 1}{z(\ln z)^2} = \lim_{z \rightarrow 1} \frac{1}{\ln z + 2} = \frac{1}{2}。$$

若规定 $\ln z|_{z=1} = 2k\pi i$ ($k = \pm 1, \pm 2, \cdots$)，则 1 是一阶极点。此时

$$\operatorname{res} f(1) = \lim_{z \rightarrow 1} (z-1) f(z) = \lim_{z \rightarrow 1} \frac{1}{\ln z} = \frac{1}{2k\pi i}。$$

$$(8) \quad \operatorname{res} f(0) = \lim_{z \rightarrow 0} z f(z) = n+1。$$

$$\begin{aligned} \operatorname{res} f(-1) &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} \left[(z+1)^n f(z) \right] \\ &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} \left\{ \frac{1}{z} \left[(z+1)^n + (z+1)^{n-1} + \cdots + (z+1) + 1 \right] \right\} \\ &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} \left[\frac{1}{z} (z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + n+1) \right] \\ &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} \left(z^{n-1} + a_{n-1} z^{n-2} + \cdots + a_1 + \frac{n+1}{z} \right) \\ &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \left[(n-1)! + 0 + \cdots + 0 + (n+1) \frac{(-1)^{n-1} (n-1)!}{z^n} \right] \\ &= -n \end{aligned}$$

$$\operatorname{res} f(\infty) = -\operatorname{res} f(0) - \operatorname{res} f(-1) = -1。$$

123. 指出下列函数在 ∞ 点的性质，并求其留数：(1) $\frac{1}{z}$ ；(2) $\frac{\cos z}{z}$ ；(3) $\frac{z}{\cos z}$ ；

$$(4) \quad \frac{z^2+1}{e^z}；(5) \quad e^{\frac{1}{z^2}}；(6) \quad \sqrt{(z-1)(z-2)}。$$

(1) $f(1/t) = t$ ，所以 ∞ 是可去奇点。 ∞ 点的留数等于 $-f(1/t)$ 在 $t=0$ 点邻域内幂级数

展开中 t 的系数，即 $\operatorname{res} f(\infty) = -1$ 。

$$(2) \quad f(1/t) = t \cos \frac{1}{t} = t \left(1 - \frac{1}{2!} \frac{1}{t^2} + \frac{1}{4!} \frac{1}{t^4} - + \dots \right) = t - \frac{1}{2} t^{-1} + \frac{1}{24} t^{-3} - + \dots,$$

即 ∞ 是本性奇点, $\operatorname{res} f(\infty) = -1$ 。

$$(3) \quad \text{易知 } z = \left(k + \frac{1}{2}\right)\pi \text{ 是奇点, 由于 } \left(k + \frac{1}{2}\right)\pi \rightarrow \infty, \text{ 所以 } \infty \text{ 是非孤立奇点。}$$

$$(4) \quad f(1/t) = \left(\frac{1}{t^2} + 1\right) e^{-t^{-1}} = t^{-2} \left(1 - t^{-1} + \frac{1}{2} t^{-2} - + \dots\right) + \left(1 - t^{-1} + \frac{1}{2} t^{-2} - + \dots\right) \\ = 1 - t^{-1} + \frac{3}{2} t^{-2} + \dots$$

即 ∞ 是本性奇点, $\operatorname{res} f(\infty) = 0$ 。

$$(5) \quad f(1/t) = e^{-t^2}, \quad \infty \text{ 点为可去奇点, } \operatorname{res} f(\infty) = 0。$$

$$(6) \quad f(1/t) = \frac{\sqrt{(1-t)(1-2t)}}{t}, \quad 0 \text{ 为 } f(1/t) \text{ 的一阶极点, 故 } \infty \text{ 点为 } f(z) \text{ 的一阶极点。}$$

1 和 1/2 是 $f(1/t)$ 的分枝点。规定 $\arg(1-t)_{t=0} = 2n\pi, \arg(1/2-t)_{t=0} = 2m\pi$, 则

$$\left(\sqrt{1-t}\right)_{t=0}^{(k)} = (-1)^k \frac{1}{2} \left(\frac{1}{2}-1\right) \cdots \left(\frac{1}{2}-k+1\right) \frac{\sqrt{1-t}}{(1-t)^k} \bigg|_{t=0} = (-1)^n (-1)^k \frac{1}{2} \left(\frac{1}{2}-1\right) \cdots \left(\frac{1}{2}-k+1\right)$$

$$\left(\sqrt{1-2t}\right)_{t=0}^{(k)} = (-2)^k \frac{1}{2} \left(\frac{1}{2}-1\right) \cdots \left(\frac{1}{2}-k+1\right) \frac{\sqrt{1-2t}}{(1-2t)^k} \bigg|_{t=0} = (-1)^m (-2)^k \frac{1}{2} \left(\frac{1}{2}-1\right) \cdots \left(\frac{1}{2}-k+1\right)$$

$$f(1/t) = (-1)^{n+m} t^{-1} \left(1 - \frac{1}{2} t - \frac{1}{8} t^2 - \frac{1}{16} t^3 + \dots\right) \left(1 - t - \frac{1}{2} t^2 - \frac{1}{2} t^3 + \dots\right)$$

$$= (-1)^{n+m} \left(t^{-1} - \frac{3}{2} - \frac{1}{8} t - \frac{7}{16} t^2 + \dots\right)$$

$$\text{即 } \operatorname{res} f(\infty) = (-1)^{n+m} \frac{1}{8}。$$

124. 设 $f(z)$ 在 $z = \infty$ 的邻域内展开为 $f(z) = C_0 + \frac{C_1}{z} + \frac{C_2}{z^2} + \dots$, 试求 $f^2(z)$ 在 $z = \infty$ 处的留数。

$$f(z) = C_0 \left(1 + \frac{C_1}{C_0} z^{-1} + \frac{C_2}{C_0} z^{-2} + \dots\right)$$

$$f^2(z) = C_0^2 \left[1 + \left(\frac{C_1}{C_0} z^{-1} + \frac{C_2}{C_0} z^{-2} + \dots \right) \right]^2 = C_0^2 \left[1 + 2 \left(\frac{C_1}{C_0} z^{-1} + \frac{C_2}{C_0} z^{-2} + \dots \right) + O(z^{-2}) \right]$$

$$= C_0^2 + 2C_0 C_1 z^{-1} + O(z^{-2})$$

$f(z)$ 在 ∞ 点的留数等于 $-f(z)$ 在 $z = \infty$ 点邻域内幂级数展开中 z^{-1} 的系数, 即 $-2C_0 C_1$ 。

125. 证明: 若除有限个奇点外, $f(z)$ 在扩充 z 平面上解析, 则函数 $f(z)$ 的留数和为 0。

证: 设 z_k ($k=1, 2, \dots, n$) 是 $f(z)$ 的有限奇点, C_k 是包围 z_k 的逆时针闭曲线 (不包围其它奇点), C 为顺时针包围所有有限奇点的闭曲线。

$$\operatorname{res} f(\infty) = \frac{1}{2\pi i} \oint_C f(z) dz = - \sum_{k=1}^n \frac{1}{2\pi i} \oint_{C_k} f(z) dz = - \sum_{k=1}^n \operatorname{res} f(z_k)$$

$$\text{所以 } \operatorname{res} f(\infty) + \sum_{k=1}^n \operatorname{res} f(z_k) = 0。$$

126. $f(z)$ 为偶函数, $z=0$ 是他的孤立奇点, 证明 $\operatorname{res} f(0) = 0$ 。

因为 $f(z)$ 为偶函数, 所以 $f(z)$ 在 $z=0$ 附近的幂级数展开式中无奇数项, 当然也没有 z^{-1} 项, 因此 $\operatorname{res} f(0) = 0$ 。

127. $f(z)$ 和 $g(z)$ 分别以 $z=0$ 为其 m 阶和 n 阶零点, 问下列函数在 $z=0$ 处的性质如何?

$$(1) f(z) + g(z); (2) f(z) \cdot g(z); (3) \frac{f(z)}{g(z)}; (4) f[g(z)].$$

设 $f(z) = z^m \varphi(z)$, $g(z) = z^n \Psi(z)$ 。其中 $\varphi(z)$ 和 $\Psi(z)$ 都在 $z=0$ 的某邻域内解析, 且 $\varphi(0) \neq 0$, $\Psi(0) \neq 0$ 。

(1) 设 $m < n$, 则 $f(z) + g(z) = z^m [\varphi(z) + z^{n-m} \Psi(z)]$, 由于 $\varphi(z) + z^{n-m} \Psi(z)$ 是不以 $z=0$ 为零点, 在 $z=0$ 的某邻域内的解析函数, 所以 $z=0$ 是 $f(z) + g(z)$ 的 m 阶零点。

当 $m \neq n$ 时, $z=0$ 是 $f(z) + g(z)$ 的 $\min\{m, n\}$ 阶零点。

若 $m=n$, 则 $f(z)+g(z)=z^m[\varphi(z)+\Psi(z)]$, $z=0$ 有可能是 $\varphi(z)+\Psi(z)$ 的零点, 所以 $z=0$ 是 $f(z)+g(z)$ 的 k 阶零点 ($k \geq m=n$)。

(2) $f(z) \cdot g(z) = z^{m+n} \varphi(z) \cdot \Psi(z)$, 所以 $z=0$ 是 $f(z) \cdot g(z)$ 的 $m+n$ 阶零点。

(3) 若 $m > n$, 则 $\frac{f(z)}{g(z)} = z^{m-n} \frac{\varphi(z)}{\Psi(z)}$, 由于 0 不是 $\Psi(z)$ 的零点, 所以 $\frac{\varphi(z)}{\Psi(z)}$ 在 $z=0$ 的

某邻域内解析, 即 $z=0$ 是 $\frac{f(z)}{g(z)}$ 的 $m-n$ 阶零点。

若 $m < n$, 则 $\frac{f(z)}{g(z)} = \frac{1}{z^{n-m}} \frac{\varphi(z)}{\Psi(z)}$, 即 $z=0$ 是 $\frac{f(z)}{g(z)}$ 的 $n-m$ 阶极点。

若 $m=n$, 则 $\frac{f(z)}{g(z)} = \frac{\varphi(z)}{\Psi(z)}$, $z=0$ 是 $\frac{f(z)}{g(z)}$ 的可去奇点。

(4) $f[g(z)] = z^{mn} \Psi^m(z) \varphi[z^n \Psi(z)]$, $z=0$ 不是 $\Psi^m(z) \varphi[z^n \Psi(z)]$ 的零点, 且他在

$z=0$ 的某邻域内解析, 所以 $z=0$ 是 $f[g(z)]$ 的 mn 阶零点。

128. $f(z)$ 和 $g(z)$ 分别以 $z=0$ 为其 m 阶和 n 阶极点, 问下列函数在 $z=0$ 处的性质如何?

(1) $f(z)+g(z)$; (2) $f(z) \cdot g(z)$; (3) $\frac{f(z)}{g(z)}$; (4) $f\left[\frac{1}{g(z)}\right]$ 。

设 $f(z) = \frac{\varphi(z)}{z^m}$, $g(z) = \frac{\Psi(z)}{z^n}$ 。其中 $\varphi(z)$ 和 $\Psi(z)$ 都在 $z=0$ 的某邻域内解析, 且

$\varphi(0) \neq 0$, $\Psi(0) \neq 0$ 。

(1) 若 $m < n$, 则 $f(z)+g(z) = \frac{z^{n-m}\varphi(z)+\Psi(z)}{z^n}$, $z^{n-m}\varphi(z)+\Psi(z)$ 是不以 $z=0$ 为

零点, 在 $z=0$ 的某邻域内的解析函数, 所以 $z=0$ 是 $f(z)+g(z)$ 的 n 阶极点。

若 $m \neq n$, $z=0$ 是 $f(z)+g(z)$ 的 $\max\{m, n\}$ 阶极点。

若 $m=n$, $f(z)+g(z) = \frac{\varphi(z)+\Psi(z)}{z^n}$, $z=0$ 有可能是 $\varphi(z)+\Psi(z)$ 的零点, 所以 $z=0$

是 $f(z)+g(z)$ 的 k 阶极点 ($k \leq m=n$)。

$$(2) \quad f(z) \cdot g(z) = \frac{\varphi(z)\Psi(z)}{z^{m+n}}, \quad z=0 \text{ 是 } f(z) \cdot g(z) \text{ 的 } m+n \text{ 阶极点。}$$

$$(3) \quad \text{若 } m < n, \quad \frac{f(z)}{g(z)} = z^{n-m} \frac{\varphi(z)}{\Psi(z)}, \quad z=0 \text{ 是 } \frac{f(z)}{g(z)} \text{ 的 } n-m \text{ 阶零点;}$$

$$\text{若 } m > n, \quad \frac{f(z)}{g(z)} = \frac{1}{z^{m-n}} \frac{\varphi(z)}{\Psi(z)}, \quad z=0 \text{ 是 } \frac{f(z)}{g(z)} \text{ 的 } m-n \text{ 阶极点;}$$

$$\text{若 } m = n, \quad \frac{f(z)}{g(z)} = \frac{\varphi(z)}{\Psi(z)}, \quad z=0 \text{ 是 } \frac{f(z)}{g(z)} \text{ 的可去奇点。}$$

$$(4) \quad f\left[\frac{1}{g(z)}\right] = \frac{1}{z^{mn}} \varphi\left[\frac{z^n}{\Psi(z)}\right] \Psi^m(z), \quad z=0 \text{ 是 } f\left[\frac{1}{g(z)}\right] \text{ 的 } mn \text{ 阶极点。}$$

129. 讨论 $F(z) = \frac{f'(z)}{f(z)} = \frac{d}{dz} \ln f(z)$ 在 $z=a$ 点的性质, 若 a 点是 $f(z)$ 的: (1) m 阶零点;

(2) m 阶极点。如果 $z=a$ 是 $F(z)$ 的孤立奇点的话, 则求出函数 $F(z)$ 在该点的留数。

(1) 设 $f(z) = (z-a)^m \varphi(z)$, 其中 $\varphi(z)$ 在 $z=a$ 的某邻域内解析, 且 $\varphi(a) \neq 0$ 。

$$F(z) = \frac{d}{dz} [m \ln(z-a) + \ln \varphi(z)] = \frac{1}{z-a} \left[m + (z-a) \frac{\varphi'(z)}{\varphi(z)} \right].$$

由于 $z=a$ 不是 $\varphi(z)$ 的零点, 所以 $m + (z-a) \frac{\varphi'(z)}{\varphi(z)}$ 在 $z=a$ 的某邻域内解析, 且 $z=a$ 不是它的零点, 所以 $z=a$ 是 $F(z)$ 的一阶极点, $\operatorname{res} F(a) = \lim_{z \rightarrow a} (z-a) F(z) = m$ 。

(2) 设 $f(z) = \frac{\varphi(z)}{(z-a)^m}$, 则 $F(z) = \frac{1}{z-a} \left[-m + (z-a) \frac{\varphi'(z)}{\varphi(z)} \right]$, $z=a$ 是 $F(z)$ 的一阶

极点, $\operatorname{res} F(a) = \lim_{z \rightarrow a} (z-a) F(z) = -m$ 。

130. 设 $\varphi(z)$ 在 $z=a$ 点解析, 且 $\varphi(a) \neq 0$ 。若 (1) a 是 $f(z)$ 的 n 阶零点, (2) a 是 $f(z)$

的 n 阶极点, 试求函数 $F(z) = \varphi(z) \frac{f'(z)}{f(z)}$ 在 $z = a$ 点的留数。

(1) 由上题结论, $z = a$ 是 $\frac{f'(z)}{f(z)}$ 的一阶极点, 留数为 n 。因为 $\varphi(z)$ 在 $z = a$ 点解析, 且

$\varphi(a) \neq 0$, 所以 $z = a$ 是 $F(z)$ 的一阶极点,

$$\operatorname{res} F(a) = \lim_{z \rightarrow a} (z - a) \varphi(z) \frac{f'(z)}{f(z)} = \lim_{z \rightarrow a} \varphi(z) \cdot \lim_{z \rightarrow a} (z - a) \frac{f'(z)}{f(z)} = n \varphi(a)。$$

(2) $\operatorname{res} F(a) = -n \varphi(a)$ 。

131. 设 C 是区域 G 内的任意一条简单闭曲线, a 为 C 包围区域内一点。若函数 $f(z)$ 在 G 内解析, 且 $f(a) = 0$, $f'(a) \neq 0$, 此外, $f(z)$ 在 $\bar{G}$ 中无其它零点。试证:

$$a = \frac{1}{2\pi i} \oint_C \frac{zf'(z)}{f(z)} dz。$$

证: 可知 a 为 $f(z)$ 的一阶零点。若 $a \neq 0$, 由上题结论, $\frac{zf'(z)}{f(z)}$ 在 $z = a$ 点留数为 a , 因

为 $f(z)$ 在 $\bar{G}$ 中无其它零点, 则 $\frac{zf'(z)}{f(z)}$ 无其他奇点, 所以 $\frac{1}{2\pi i} \oint_C \frac{zf'(z)}{f(z)} dz = a$,

若 $a = 0$, 则 $z = a = 0$ 是 $\frac{zf'(z)}{f(z)}$ 的可去奇点, 即 $\frac{zf'(z)}{f(z)}$ 在 C 包围区域内解析, 故有

$$\frac{1}{2\pi i} \oint_C \frac{zf'(z)}{f(z)} dz = 0 = a。$$

132. 若 $z = 0$ 是 $f(z)$ 的 n 阶零点, 试求下列函数在 $z = 0$ 处的留数: (1) $\frac{f''(z)}{f'(z)}$;

(2) $\frac{f''(z)}{f(z)}$ 。

设 $f(z) = z^n \varphi(z)$, 其中 $\varphi(z)$ 在 $z = 0$ 的某邻域内解析, 且 $\varphi(0) \neq 0$ 。

$$\begin{aligned}
 (1) \quad F(z) &= \frac{f''(z)}{f'(z)} = \frac{n(n-1)z^{n-2}\varphi(z) + 2nz^{n-1}\varphi'(z) + z^n\varphi''(z)}{nz^{n-1}\varphi(z) + z^n\varphi'(z)} \\
 &= \frac{1}{z} \cdot \frac{n(n-1)\varphi(z) + 2nz\varphi'(z) + z^2\varphi''(z)}{n\varphi(z) + z\varphi'(z)}
 \end{aligned}$$

可判断 $z=0$ 是其一阶极点, 有 $\operatorname{res} F(0) = \lim_{z \rightarrow 0} zF(z) = n-1$ 。

$$\begin{aligned}
 (2) \quad F(z) &= \frac{f''(z)}{f(z)} = \frac{n(n-1)z^{n-2}\varphi(z) + 2nz^{n-1}\varphi'(z) + z^n\varphi''(z)}{z^n\varphi(z)} \\
 &= \frac{1}{z^2} \cdot \frac{n(n-1)\varphi(z) + 2nz\varphi'(z) + z^2\varphi''(z)}{\varphi(z)}
 \end{aligned}$$

$z=0$ 是其二阶极点。 $\operatorname{res} F(0) = \lim_{z \rightarrow 0} \frac{d}{dz} [z^2 F(z)] = 2n \frac{\varphi'(0)}{\varphi(0)}$ 。

可设 $\varphi(z) = a_0 + a_1 z + a_2 z^2 + \dots$, 则 $f(z) = a_0 z^n + a_1 z^{n+1} + a_2 z^{n+2} + \dots$

$$\varphi(0) = a_0 = \frac{1}{n!} f^{(n)}(0), \quad \varphi'(0) = a_1 = \frac{1}{(n+1)!} f^{(n+1)}(0), \quad \operatorname{res} F(0) = \frac{2n}{n+1} \frac{f^{(n+1)}(0)}{f^{(n)}(0)}。$$

133. 求下列各种条件下函数 $F(z) = \frac{f(z)}{g(z)}$ 在奇点 z_0 处的留数:

(1) z_0 是 $f(z)$ 的 m 阶零点, 是 $g(z)$ 的 $m+1$ 阶零点;

(2) z_0 是 $g(z)$ 的二阶零点, 但 $f(z_0) \neq 0$;

(3) z_0 是 $f(z)$ 的一阶零点, 是 $g(z)$ 的三阶零点;

(4) z_0 是 $f(z)$ 的一阶极点, 是 $g(z)$ 的一阶零点;

(1) 设 $f(z) = a_0(z-z_0)^m + a_1(z-z_0)^{m+1} + \dots$, 其中 $a_0 = \frac{f^{(m)}(z_0)}{m!}, a_1 = \frac{f^{(m+1)}(z_0)}{(m+1)!}, \dots$

$g(z) = b_0(z-z_0)^{m+1} + b_1(z-z_0)^{m+2} + \dots$, 其中 $b_0 = \frac{g^{(m+1)}(z_0)}{(m+1)!}, b_1 = \frac{g^{(m+2)}(z_0)}{(m+2)!}, \dots$

$$\operatorname{res} F(z_0) = \lim_{z \rightarrow z_0} (z-z_0) F(z) = \lim_{z \rightarrow z_0} \frac{a_0(z-z_0)^m + a_1(z-z_0)^{m+1} + \dots}{b_0(z-z_0)^{m+1} + b_1(z-z_0)^{m+2} + \dots}$$

$$= \lim_{z \rightarrow z_0} \frac{a_0 + a_1(z - z_0) + \cdots}{b_0 + b_1(z - z_0) + \cdots} = \frac{a_0}{b_0} = (m+1) \frac{f^{(m)}(z_0)}{g^{(m+1)}(z_0)}.$$

(2) 设 $g(z) = (z - z_0)^2 \Psi(z)$, 则 $\Psi(z_0) = \frac{1}{2} g''(z_0)$, $\Psi'(z_0) = \frac{1}{6} g'''(z_0)$ 。

$$\operatorname{res} F(z_0) = \lim_{z \rightarrow z_0} \frac{d}{dz} [(z - z_0)^2 F(z)] = \frac{f'(z_0)}{\Psi(z_0)} - \frac{f(z_0) \Psi'(z_0)}{\Psi^2(z_0)} = 2 \frac{f'(z_0)}{g''(z_0)} - \frac{2}{3} \frac{f(z_0) g'''(z_0)}{[g''(z_0)]^2}.$$

(3) 设 $f(z) = (z - z_0) \varphi(z)$, 则 $\varphi(z_0) = f'(z_0)$, $\varphi'(z_0) = \frac{1}{2} f''(z_0)$ 。

$$g(z) = (z - z_0)^3 \Psi(z), \text{ 则 } \Psi(z_0) = \frac{1}{6} g'''(z_0), \Psi'(z_0) = \frac{1}{24} g^{(4)}(z_0).$$

$$\operatorname{res} F(z_0) = \lim_{z \rightarrow z_0} \frac{d}{dz} [(z - z_0)^2 F(z)] = \frac{\varphi'(z_0)}{\Psi(z_0)} - \frac{\varphi(z_0) \Psi'(z_0)}{\Psi^2(z_0)} = 3 \frac{f''(z_0)}{g'''(z_0)} - \frac{3}{2} \frac{f'(z_0) g^{(4)}(z_0)}{[g'''(z_0)]^2}.$$

(4) 设 $f(z) = \frac{\varphi(z)}{z - z_0}$, 则 $\varphi(z_0) = \lim_{z \rightarrow z_0} [(z - z_0) f(z)]$, $\varphi'(z_0) = \lim_{z \rightarrow z_0} \frac{d}{dz} [(z - z_0) f(z)]$ 。

$$g(z) = (z - z_0) \Psi(z), \text{ 则 } \Psi(z_0) = g'(z_0), \Psi'(z_0) = \frac{1}{2} g''(z_0).$$

$$\operatorname{res} F(z_0) = \lim_{z \rightarrow z_0} \frac{d}{dz} [(z - z_0)^2 F(z)] = \frac{\varphi'(z_0)}{\Psi(z_0)} - \frac{\varphi(z_0) \Psi'(z_0)}{\Psi^2(z_0)} = \frac{\varphi'(z_0)}{g'(z_0)} - \frac{1}{2} \frac{\varphi(z_0) g''(z_0)}{[g'(z_0)]^2}.$$

134. 若函数 $f(z)$ 与 $g(z)$ 在闭区域 $\bar{G}$ 中解析, $g(z)$ 在 G 内有有限个一阶零点

$a_1, a_2, \dots, a_n$, 而 $g(0) \neq 0$, 试计算积分 $\frac{1}{2\pi i} \oint_C \frac{f(z)}{zg(z)} dz$, 其中 C 是 G 的边界, 且 $z=0$ 在

G 内。

$$\text{令 } F(z) = \frac{f(z)}{zg(z)}, \text{ 则 } \operatorname{res} F(0) = \lim_{z \rightarrow 0} \frac{f(z)}{g(z)} = \frac{f(0)}{g(0)},$$

$$\operatorname{res} F(a_k) = \lim_{z \rightarrow a_k} \frac{(z - a_k) f(z)}{zg(z)} = \lim_{z \rightarrow a_k} \frac{f(z)}{z \frac{g(z) - g(a_k)}{(z - a_k)}} = \frac{f(a_k)}{a_k g'(a_k)}, \quad (k = 1, 2, \dots, n)$$

$$\text{所以 } \frac{1}{2\pi i} \oint_C \frac{f(z)}{zg(z)} dz = \frac{f(0)}{g(0)} + \frac{f(a_1)}{a_1 g'(a_1)} + \frac{f(a_2)}{a_2 g'(a_2)} + \cdots + \frac{f(a_n)}{a_n g'(a_n)}.$$

135. 计算下列积分值:

- (1) $\oint_C \frac{dz}{1+z^4}$, C 为 $|z-1|=2$ 或 $|z-1|=1$;
- (2) $\oint_C \frac{\sin \frac{\pi z}{4}}{z^2-1} dz$, C 为: (i) $|z|=\frac{1}{2}$, (ii) $|z-1|=1$, (iii) $|z|=3$;
- (3) $\oint_{|z|=R} \frac{z^2}{e^{2\pi iz^3}-1} dz$, $n < R^3 < n+1$, n 为正整数;
- (4) $\oint_{|z|=n} \tan \pi z dz$, n 为正整数;
- (5) $\oint_{|z|=2} \frac{1}{z^3(z^{10}-2)} dz$; (6) $\oint_{|z|=1} \frac{e^z}{z^3} dz$; (7) $\oint_{|z|=2} e^{\frac{1}{z^2}} dz$;
- (8) $\oint_{|z|=R} \frac{e^z}{\operatorname{sh} mz} dz$, $\frac{n}{m}\pi < R < \frac{n+1}{m}\pi$, m, n 均为正整数。

(1) 被积函数有四个一阶极点: $e^{\pm i\frac{\pi}{4}}$, $e^{\pm i\frac{3\pi}{4}}$ 。

$$\operatorname{res} f(e^{i\pi/4}) = \lim_{z \rightarrow e^{i\pi/4}} (z - e^{i\pi/4}) f(z) = -\frac{1}{4\sqrt{2}}(1+i),$$

$$\operatorname{res} f(e^{-i\pi/4}) = \lim_{z \rightarrow e^{-i\pi/4}} (z - e^{-i\pi/4}) f(z) = \frac{1}{4\sqrt{2}}(-1+i),$$

$$\operatorname{res} f(e^{i3\pi/4}) = \lim_{z \rightarrow e^{i3\pi/4}} (z - e^{i3\pi/4}) f(z) = \frac{1}{4\sqrt{2}}(1-i),$$

$$\operatorname{res} f(e^{-i3\pi/4}) = \lim_{z \rightarrow e^{-i3\pi/4}} (z - e^{-i3\pi/4}) f(z) = \frac{1}{4\sqrt{2}}(1+i)。$$

若 C 为 $|z-1|=2$, 则 C 包围四个极点,

$$\text{原积分} = 2\pi i [\operatorname{res} f(e^{i\pi/4}) + \operatorname{res} f(e^{-i\pi/4}) + \operatorname{res} f(e^{i3\pi/4}) + \operatorname{res} f(e^{-i3\pi/4})] = 0。$$

若 C 为 $|z-1|=1$, 则 C 包围两个极点 $e^{\pm i\frac{\pi}{4}}$,

$$\text{原积分} = 2\pi i [\operatorname{res} f(e^{i\pi/4}) + \operatorname{res} f(e^{-i\pi/4})] = -\frac{\pi i}{\sqrt{2}}。$$

$$(2) \operatorname{res} f(1) = \lim_{z \rightarrow 1} \frac{\sin \frac{\pi z}{4}}{z+1} = \frac{1}{2\sqrt{2}}, \quad \operatorname{res} f(-1) = \lim_{z \rightarrow -1} \frac{\sin \frac{\pi z}{4}}{z-1} = \frac{1}{2\sqrt{2}}。$$

(i) C 不包围两个极点 ± 1 , 所以原积分=0。

(ii) C 包围 1, 原积分 $= 2\pi i \cdot \operatorname{res} f(1) = \frac{\pi}{\sqrt{2}} i$ 。

(iii) C 包围 ± 1 , 原积分 $= 2\pi i [\operatorname{res} f(1) + \operatorname{res} f(-1)] = \sqrt{2}\pi i$ 。

(3) 被积函数具有一阶极点 $0, \pm\sqrt[3]{k}, \sqrt[3]{k}e^{\pm i\frac{\pi}{3}}, \sqrt[3]{k}e^{\pm i\frac{2\pi}{3}}$ 。($k=1, 2, \dots$)

$$\operatorname{res} f(0) = \lim_{z \rightarrow 0} \frac{z^3}{e^{2\pi iz^3} - 1} = \frac{1}{2\pi i},$$

$$\operatorname{res} f(\pm\sqrt[3]{k}) = \lim_{z \rightarrow \pm\sqrt[3]{k}} \frac{(z \mp \sqrt[3]{k})z^2}{e^{2\pi iz^3} - 1} = \lim_{z \rightarrow \pm\sqrt[3]{k}} \frac{z^2 + 2z(z \mp \sqrt[3]{k})}{6\pi iz^2 e^{2\pi iz^3}} = \frac{1}{6\pi i},$$

$$\operatorname{res} f\left(\sqrt[3]{k}e^{\pm i\frac{\pi}{3}}\right) = \frac{1}{6\pi i}, \quad \operatorname{res} f\left(\sqrt[3]{k}e^{\pm i\frac{2\pi}{3}}\right) = \frac{1}{6\pi i}。$$

$|z|=R$ ($\sqrt[3]{n} < R < \sqrt[3]{n+1}$) 包围 $0, \pm\sqrt[3]{k}, \sqrt[3]{k}e^{\pm i\frac{\pi}{3}}, \sqrt[3]{k}e^{\pm i\frac{2\pi}{3}}$ 。($k=1, 2, \dots, n-1, n$)

共 $6n+1$ 个奇点。原积分 $= 2\pi i \left(6n \cdot \frac{1}{6\pi i} + \frac{1}{2\pi i} \right) = 2n+1$ 。

(4) 被积函数有一阶极点 $k + \frac{1}{2}$ ($k=0, \pm 1, \pm 2, \dots$)。

$$\operatorname{res} f\left(k + \frac{1}{2}\right) = \lim_{z \rightarrow k + \frac{1}{2}} \frac{\left(z - k - \frac{1}{2}\right) \sin \pi z}{\cos \pi z} = \lim_{z \rightarrow k + \frac{1}{2}} \frac{\sin \pi z + \pi \left(z - k - \frac{1}{2}\right) \cos \pi}{-\pi \sin \pi z} = -\frac{1}{\pi}。$$

$|z|=n$ 包围 $k + \frac{1}{2}$ ($k=-n, -(n-1), \dots, 0, 1, \dots, n-1$) 共 $2n$ 个奇点。

$$\text{原积分} = 2\pi i \left[2n \cdot \left(-\frac{1}{\pi} \right) \right] = -4ni。$$

(5) 被积函数 $f(z)$ 有三阶极点 0 , 一阶极点 $\pm\sqrt[10]{2}, \sqrt[10]{2}e^{\pm i\frac{k\pi}{5}}$, ($k=1, 2, 3, 4$)

$f(1/t) = \frac{t^{13}}{1-2t^{10}}$, 可看出 ∞ 点为解析点, 即 ∞ 点留数为 0 , 所以所有有限奇点留数之和为

0 , 而 $|z|=2$ 包围所有有限奇点, 所以原积分 $= 0$ 。

(6) $\operatorname{res} f(0) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} e^z = \frac{1}{2}$, 原积分 $= 2\pi i \cdot \frac{1}{2} = \pi i$ 。

(7) $f(z) = 1 + z^{-2} + \frac{1}{2}z^{-4} + \dots$, 所以 $\operatorname{res} f(0) = 0$, 原积分 $= 0$ 。

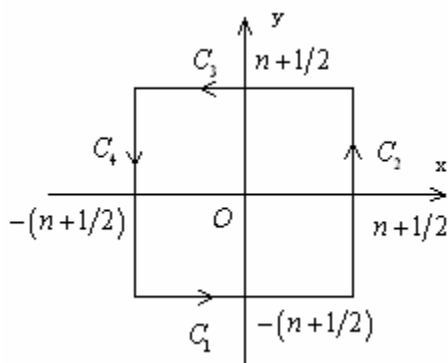
(8) 被积函数有一阶极点 $i\frac{k}{m}\pi$ ($k=0, \pm 1, \pm 2, \dots$)。

$$\operatorname{res} f\left(i\frac{k}{m}\pi\right) = \lim_{z \rightarrow ik\pi/m} \frac{(z - ik\pi/m)e^z}{\operatorname{sh} mz} = \frac{(-1)^k}{m} e^{i\frac{k}{m}\pi}。$$

$$\text{原积分} = 2\pi i \sum_{k=-n}^n \operatorname{res} f\left(i\frac{k}{m}\pi\right) = \frac{2\pi i}{m} \sum_{k=-n}^n \left(-e^{i\frac{\pi}{m}}\right)^k = (-1)^n \frac{2\pi i}{m} \frac{\cos \frac{2n+1}{2m}\pi}{\cos \frac{\pi}{2m}}。$$

136. 计算积分 $\oint_{C_n} \frac{\cot \pi z}{z^2} dz$, 其中 C_n 是以 $\left(\pm \frac{2n+1}{2}, \pm \frac{2n+1}{2}\right)$ 为顶点的正方形。令

$n \rightarrow \infty$, 就得到级数 $\sum_{m=1}^{\infty} \frac{1}{m^2}$ 之和。若把被积函数换成 $\frac{\csc \pi z}{z^2}$, 又能得到什么结果?



被积函数 $f(z) = \frac{\cot \pi z}{z^2}$ 三阶极点 0 和一阶极点 k ($k = \pm 1, \pm 2, \dots$)。

$$\operatorname{res} f(0) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} (z \cot \pi z) = \pi \lim_{z \rightarrow 0} \frac{\pi z \cos \pi z - \sin \pi z}{\sin^3 \pi z} = -\frac{\pi}{3},$$

$$\operatorname{res} f(k) = \lim_{z \rightarrow k} \frac{(z-k) \cos \pi z}{z^2 \sin \pi z} = \frac{1}{\pi k^2}。$$

$$\text{所以 } \oint_{C_n} \frac{\cot \pi z}{z^2} dz = 2\pi i \left[\operatorname{res} f(0) + \sum_{\substack{k=-n \\ k \neq 0}}^n \operatorname{res} f(k) \right] = 2\pi i \left(-\frac{\pi}{3} + \frac{2}{\pi} \sum_{k=1}^n \frac{1}{k^2} \right)。$$

在 C_1 上有 $|\cot \pi z| = \left| \frac{\cos \pi [x - i(n+1/2)]}{\sin \pi [x - i(n+1/2)]} \right| \leq \frac{\operatorname{ch} \pi(n+1/2)}{\operatorname{sh} \pi(n+1/2)} = \coth \pi(n+1/2)$, (参考习题

02 的第 43 题) 易证 $\coth x$ 为单调减函数, 所以有 $|\cot \pi z| \leq \coth \frac{3\pi}{2}$. ($n \geq 1$)

上式在 C_3 上同样成立。

$$\begin{aligned} \text{在 } C_2 \text{ 上有 } |\cot \pi z| &= \left| \frac{\cos \pi \left[(n+1/2) + iy \right]}{\sin \pi \left[(n+1/2) + iy \right]} \right| \\ &\leq \frac{\operatorname{ch} \pi y}{\sqrt{\sin^2 \pi (n+1/2) \operatorname{ch}^2 \pi y + \cos^2 \pi (n+1/2) \operatorname{sh}^2 \pi y}} \leq \frac{\operatorname{ch} \pi y}{|\sin \pi (n+1/2) \operatorname{ch} \pi y|} = 1 \end{aligned}$$

上式在 C_4 上同样成立。所以在 C_n 上有 $|\cot \pi z| \leq M = \max \left\{ \coth \frac{3\pi}{2}, 1 \right\}$ 。

$$\text{在 } C_n \text{ 上有 } |z| \geq n + \frac{1}{2}, \text{ 由此得 } \left| \oint_{C_n} \frac{\cot \pi z}{z^2} dz \right| \leq \oint_{C_n} \frac{|\cot \pi z|}{|z|^2} |dz| \leq M \cdot \frac{1}{(n+1/2)^2} \cdot 4(2n+1)。$$

所以当 $n \rightarrow \infty$ 时有 $\oint_{C_n} \frac{\cot \pi z}{z^2} dz \rightarrow 0$ 。由此可得 $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$ 。

若令 $f(z) = \frac{\csc \pi z}{z^2}$ ，则有 $\operatorname{res} f(0) = \frac{\pi}{6}$ ， $\operatorname{res} f(k) = \frac{(-1)^k}{\pi k^2}$ 。

$$\oint_{C_n} \frac{\csc \pi z}{z^2} dz = 2\pi i \left(\frac{\pi}{6} + \frac{2}{\pi} \sum_{k=1}^n \frac{(-1)^k}{k^2} \right)。$$

同样可得当 $n \rightarrow \infty$ 时， $\oint_{C_n} \frac{\csc \pi z}{z^2} dz \rightarrow 0$ ，所以 $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2} = \frac{\pi^2}{12}$ 。

137. 如果 $R(\sin \theta, \cos \theta)$ 在 $[0, 2\pi]$ 中有奇点, 通过变换 $z = e^{i\theta}$, $R(\sin \theta, \cos \theta)$ 变为

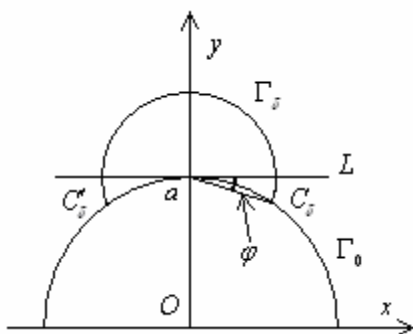
$f(z) = R\left(\frac{z^2 - 1}{2iz}, \frac{z^2 + 1}{2z}\right)$, 则 $f(z)$ 在单位圆周 $|z| = 1$ 上有奇点。设这些奇点 β_k

($k = 1, 2, \dots, m$) 均为一阶奇点, 证明:

$$\int_0^{2\pi} R(\sin \theta, \cos \theta) d\theta = 2\pi \sum_{|z| < 1} \operatorname{res} \left\{ \frac{f(z)}{z} \right\} + \pi \sum_{k=1}^m \operatorname{res} \left\{ \frac{f(z)}{z} \right\}_{z=\beta_k}.$$

其中 $R(\sin \theta, \cos \theta)$ 表示 $\sin \theta$ 和 $\cos \theta$ 的有理函数。

证:



如上图, Γ_0 是以原点为圆心, a 为半径的上半圆, Γ_δ 是以 $(0, a)$ 为圆心, δ 为半径的圆被 Γ_0 截断的圆外部分。 L 是 Γ_0 过点 $(0, a)$ 的切线, C_δ 和 C'_δ 是 Γ_δ 夹在 Γ_0 和 L 之间的部分, Γ'_δ 的上半圆部分 $\Gamma'_\delta = \Gamma_\delta - C_\delta - C'_\delta$ 。

参考第 86 题 (习题 04) 的证明, 设 $(z-a)f(z)$ 在 Γ_δ 上一致趋于 k , 则对任意 $\varepsilon > 0$, 存在 δ' 满足 $0 < \delta' < \varepsilon$, 使得当 $\delta < \delta' < \varepsilon$ 时有

$$\left| \int_{C_\delta} f(z) dz \right| = \left| \int_{C_\delta} f(z) dz - ik\varphi + ik\varphi \right| \leq \left| \int_{C_\delta} f(z) dz - ik\varphi \right| + k\varphi < (\varepsilon + k)\varphi.$$

由上图可解出 $\varphi = \arctan \frac{\delta}{2a\sqrt{1 - \left(\frac{\delta}{2a}\right)^2}}$, 则 $\varphi \leq \frac{\delta}{2a\sqrt{1 - \left(\frac{\delta}{2a}\right)^2}} < \frac{\delta'}{2a\sqrt{1 - \left(\frac{\delta'}{2a}\right)^2}}$ (这是

关于 δ 的单调增函数)。只要上面的 δ' 足够小就有

$$\varphi < \frac{\delta'}{2a\sqrt{1 - \left(\frac{\delta'}{2a}\right)^2}} < \frac{\delta'}{2a} \left[1 + \frac{1}{2} \left(\frac{\delta'}{2a} \right)^2 \right] < \frac{\delta'}{2a} \left(1 + \frac{1}{8a^2} \right) < \frac{1}{2a} \left(1 + \frac{1}{8a^2} \right) \varepsilon,$$

所以 $\left| \int_{C_\delta} f(z) dz \right| < \varepsilon \varphi + k\varphi < \frac{1}{2a} \left(1 + \frac{1}{8a^2} \right) (k\varepsilon + \varepsilon^2)$, 这就证明了 $\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = 0$ 。

$$\begin{aligned} \text{所以 } \lim_{\delta \rightarrow 0} \int_{\Gamma_\delta} f(z) dz &= \lim_{\delta \rightarrow 0} \left[\int_{\Gamma'_\delta} f(z) dz + \int_{C_\delta} f(z) dz + \int_{C'_\delta} f(z) dz \right] \\ &= \lim_{\delta \rightarrow 0} \int_{\Gamma'_\delta} f(z) dz = k\pi i = \pi i \lim_{z \rightarrow a} [(z-a)f(z)]. \end{aligned}$$

在单位圆上挖去奇点 β_k , 代之以以 β_k 为圆心, δ 为半径, 被单位圆截断的圆弧 C_k ($k=1, 2, \dots, m$), 用 C_0 表示单位圆剩下的部分。这样构成一个包围单位圆内奇点和 β_k

($k=1, 2, \dots, m$) 的围线 $C = C_0 + C_1 + C_2 + \dots + C_m$ 。则

$$\oint_C \frac{f(z)}{iz} dz = 2\pi \sum_{|z|<1} \operatorname{res} \left\{ \frac{f(z)}{z} \right\} + 2\pi \sum_{k=1}^m \operatorname{res} \left\{ \frac{f(z)}{z} \right\}_{z=\beta_k}$$

$$\begin{aligned} \text{又有 } \lim_{\delta \rightarrow 0} \oint_C \frac{f(z)}{iz} dz &= \lim_{\delta \rightarrow 0} \oint_{C_0} \frac{f(z)}{iz} dz + \lim_{\delta \rightarrow 0} \sum_{k=1}^m \oint_{C_k} \frac{f(z)}{iz} dz \\ &= \int_0^{2\pi} R(\sin \theta, \cos \theta) d\theta + \pi i \sum_{k=1}^m \lim_{z \rightarrow \beta_k} \frac{(z-\beta_k)f(z)}{iz} \\ &= \int_0^{2\pi} R(\sin \theta, \cos \theta) d\theta + \pi \sum_{k=1}^m \operatorname{res} \left\{ \frac{f(z)}{z} \right\}_{z=\beta_k} \end{aligned}$$

$$\text{所以有 } \int_0^{2\pi} R(\sin \theta, \cos \theta) d\theta = 2\pi \sum_{|z|<1} \operatorname{res} \left\{ \frac{f(z)}{z} \right\} + \pi \sum_{k=1}^m \operatorname{res} \left\{ \frac{f(z)}{z} \right\}_{z=\beta_k}。$$

138. 计算下列积分: (1) $\int_0^{2\pi} \frac{dx}{1-2p \cos x + p^2}$, $0 < p < 1$; (2) $\int_0^{2\pi} \frac{dx}{2 + \cos x}$;

(3) $\int_0^{2\pi} \frac{dx}{(a+b \cos x)^2}$, $a > b > 0$; (4) $\int_0^{2\pi} \cos^{2n} x dx$; (5) $\int_0^{2\pi} \exp(e^{i\theta}) d\theta$;

(6) $\int_0^\pi \frac{d\theta}{1 + \sin^2 \theta}$; (7) $\int_0^\pi \frac{d\theta}{(1 + \sin^2 \theta)^2}$; (8) $\int_0^\pi \cot(x-\alpha) dx$, $\operatorname{Im} \alpha \neq 0$ 。

$$(1) \text{ 令 } f(z) = \frac{1}{z} \frac{1}{1-2p\frac{z^2+1}{2z}+p^2} = -\frac{1}{p} \frac{1}{(z-p)\left(z-\frac{1}{p}\right)}, \text{ 则 } \operatorname{res} f(p) = \frac{1}{1-p^2}.$$

$$\text{原积分} = 2\pi \operatorname{res} f(p) = \frac{2\pi}{1-p^2}.$$

$$(2) \text{ 令 } f(z) = \frac{1}{z} \frac{1}{2+\frac{z^2+1}{2z}} = \frac{2}{(z+2-\sqrt{3})(z+2+\sqrt{3})}, \text{ 则 } \operatorname{res} f(-2+\sqrt{3}) = \frac{1}{\sqrt{3}}.$$

$$\text{原积分} = \frac{2\pi}{\sqrt{3}}.$$

$$(3) \text{ 令 } f(z) = \frac{1}{z} \frac{1}{\left(a+b\frac{z^2+1}{2z}\right)^2} = \frac{4}{b^2} \frac{z}{(z-z_1)^2(z-z_2)^2}, \text{ 其中 } z_1 = -\frac{a}{b} - \sqrt{\left(\frac{a}{b}\right)^2 - 1},$$

$$z_2 = -\frac{a}{b} + \sqrt{\left(\frac{a}{b}\right)^2 - 1}. \quad -\frac{a}{b} - \sqrt{\left(\frac{a}{b}\right)^2 - 1} < -\frac{a}{b} < -1, \text{ 所以 } |z_1| > 1, \text{ 在单位圆外,}$$

$$|z_2| = \left| -\frac{a}{b} + \sqrt{\left(\frac{a}{b}\right)^2 - 1} \right| = 1 / \left| -\frac{a}{b} - \sqrt{\left(\frac{a}{b}\right)^2 - 1} \right| < 1, \text{ 在单位圆内.}$$

$$\operatorname{res} f(z_2) = \lim_{z \rightarrow z_2} \frac{d}{dz} \left[(z-z_2)^2 f(z) \right] = \frac{a}{(a^2-b^2)^{3/2}},$$

$$\text{所以原积分} = \frac{2\pi a}{(a^2-b^2)^{3/2}}.$$

$$(4) \text{ 令 } f(z) = \frac{1}{z} \left(\frac{z^2+1}{2z} \right)^{2n} = \frac{1}{2^{2n}} \frac{(1+z^2)^{2n}}{z^{2n+1}} = \frac{1}{2^{2n}} \sum_{k=0}^{2n} \frac{(2n)!}{k!(2n-k)!} z^{2k},$$

$$\begin{aligned} \operatorname{res} f(0) &= \lim_{z \rightarrow 0} \left[\frac{1}{2^{2n}} \frac{d^{2n}}{dz^{2n}} \sum_{k=0}^{2n} \frac{(2n)!}{k!(2n-k)!} z^{2k} \right] \\ &= \frac{1}{2^{2n}} \frac{d^{2n}}{dz^{2n}} \left[1 + a_1 z + \cdots + \frac{(2n)!}{(n!)^2} z^{2n} + \cdots + z^{4n} \right] \Big|_{z=0} \\ &= \frac{1}{2^{2n}} \frac{d^{2n}}{dz^{2n}} \left[0 + 0 + \cdots + \frac{(2n)!}{(n!)^2} (2n)! + b_{2n+1} z + \cdots + b_{4n} z^{2n} \right] \Big|_{z=0} = \frac{(2n)!}{2^{2n} (n!)^2} \end{aligned}$$

$$\begin{aligned}
\text{所以原积分} &= 2\pi \frac{(2n)!}{2^{2n} (n!)^2} = 2\pi \frac{2n(2n-1)(2n-2)(2n-3)\cdots 2\cdot 1}{2^{2n} n^2 (n-1)^2 (n-2)^2 \cdots 2^2 \cdot 1^2} \\
&= 2\pi \frac{2n(2n-1)(2n-2)(2n-3)\cdots 2\cdot 1}{(2n)^2 (2n-2)^2 (2n-4)^2 \cdots 2^2} = 2\pi \frac{(2n-1)(2n-3)\cdots 3\cdot 1}{(2n)(2n-2)(2n-4)\cdots 2} \\
&= 2\pi \frac{(2n-1)!!}{(2n)!!}.
\end{aligned}$$

(5) 令 $f(z) = \frac{1}{z} e^z$, 则 $\operatorname{res} f(0) = 1$, 原积分 $= 2\pi$ 。

(6) 原积分 $= \int_0^\pi \frac{2d\theta}{3 - \cos 2\theta} = \int_0^{2\pi} \frac{d\varphi}{3 - \cos \varphi}$ 。

令 $f(z) = \frac{1}{z} \frac{1}{3 - \frac{z^2+1}{2z}} = -\frac{2}{(z-3-2\sqrt{2})(z-3+2\sqrt{2})}$, 则 $\operatorname{res} f(3-2\sqrt{2}) = \frac{1}{2\sqrt{2}}$ 。

原积分 $= \frac{\pi}{\sqrt{2}}$ 。

(7) 原积分 $= \int_0^\pi \frac{4d\theta}{(3 - \cos 2\theta)^2} = \int_0^{2\pi} \frac{2d\varphi}{(3 - \cos \varphi)^2}$ 。

令 $f(z) = \frac{1}{z} \frac{2}{\left(3 - \frac{z^2+1}{2z}\right)^2} = \frac{8z}{(z-3-2\sqrt{2})^2 (z-3+2\sqrt{2})^2}$, 则

$\operatorname{res} f(3-2\sqrt{2}) = \lim_{z \rightarrow 3-2\sqrt{2}} \frac{d}{dz} \frac{8z}{(z-3-2\sqrt{2})^2} = \frac{3}{8\sqrt{2}}$, 原积分 $= \frac{3}{4\sqrt{2}} \pi$ 。

(8) 令 $\alpha = a + ib$, 原积分 $= i \int_0^\pi \frac{e^{i(x-a-ib)} + e^{-i(x-a-ib)}}{e^{i(x-a-ib)} - e^{-i(x-a-ib)}} dx = i \int_0^\pi \frac{e^{i2(x-a-ib)} + 1}{e^{i2(x-a-ib)} - 1} dx$

$= \frac{i}{2} \int_{-2a}^{2\pi-2a} \frac{e^{2b} e^{i\theta} + 1}{e^{2b} e^{i\theta} - 1} d\theta$ (作代换 $2(x-a) = \theta$), 因为被积函数以 2π 为周期, 所以

原积分 $= \frac{i}{2} \int_0^{2\pi} \frac{e^{2b} e^{i\theta} + 1}{e^{2b} e^{i\theta} - 1} d\theta = \frac{1}{2} \oint_{|z|=1} \frac{e^{2b} z + 1}{z(e^{2b} z - 1)} dz$ 。

记 $f(z) = \frac{z + e^{-2b}}{2z(z - e^{-2b})}$, 则 $\operatorname{res} f(0) = -\frac{1}{2}$, $\operatorname{res} f(e^{-2b}) = 1$ 。

若 $b > 0$, 原积分 $= 2\pi i \cdot [\operatorname{res} f(0) + \operatorname{res} f(e^{-2b})] = \pi i$,

若 $b < 0$, 原积分 $= 2\pi i \cdot \operatorname{res} f(0) = -\pi i$ 。

139. 计算下列积分: (1) $\int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx$; (2) $\int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)(x^2+b^2)}$, $a > 0, b > 0$;

(3) $\int_{-\infty}^{\infty} \frac{x^{2m}}{x^{2n}+1} dx$, m, n 均为正整数, 且 $n > m$; (4) $\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{n+1}}$, n 为正整数;

(5) $\int_0^{\infty} \frac{x^2 dx}{(x^2+a^2)^2}$, $a > 0$; (6) $\int_{-\infty}^{\infty} \frac{dx}{x^2-2x+4}$;

(7) $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+1)(x^2-2x \cos \theta + 1)}$, θ 为实数, 且 $\sin \theta \neq 0$;

(8) $\int_{-\infty}^{\infty} \frac{dx}{(1+x^2) \operatorname{ch} \frac{\pi x}{2}}$ 。

(1) 令 $f(z) = \frac{z^2}{1+z^4} = \frac{z^2}{(z-e^{i\pi/4})(z-e^{-i\pi/4})(z-e^{i3\pi/4})(z-e^{-i3\pi/4})}$ 。

$\operatorname{res} f(e^{i\pi/4}) = \frac{1}{4\sqrt{2}}(1-i)$, $\operatorname{res} f(e^{i3\pi/4}) = -\frac{1}{4\sqrt{2}}(1+i)$ 。

所以 $\text{v.p.} \int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx = 2\pi i [\operatorname{res} f(e^{i\pi/4}) + \operatorname{res} f(e^{i3\pi/4})] = \frac{1}{\sqrt{2}}\pi$ 。

被积函数是偶函数, 所以 $\int_0^{\infty} f(x) dx = \frac{1}{2} \lim_{b \rightarrow \infty} 2 \int_0^b f(x) dx = \frac{1}{2} \lim_{b \rightarrow \infty} \left[\int_{-b}^0 f(x) dx + \int_0^b f(x) dx \right]$
 $= \frac{1}{2} \text{v.p.} \int_{-\infty}^{\infty} f(x) dx$, 即 $\int_0^{\infty} f(x) dx$ 收敛, 同样的, $\int_{-\infty}^0 f(x) dx$ 也收敛, 所以 $\int_{-\infty}^{\infty} f(x) dx$ 收

敛, 且等于 $\text{v.p.} \int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx$ 。后面类似的讨论省略。

(2) 令 $f(z) = \frac{1}{(z^2+a^2)(z^2+b^2)} = \frac{1}{(z+ai)(z-ai)(z+bi)(z-bi)}$, 则

$\operatorname{res} f(ai) = \frac{i}{2a(a^2-b^2)}$, $\operatorname{res} f(bi) = -\frac{i}{2b(a^2-b^2)}$ 。

$$\text{原积分} = \text{v.p.} \int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)(x^2 + b^2)} = 2\pi i [\text{res } f(ai) + \text{res } f(bi)] = \frac{\pi}{ab(a+b)}。$$

$$(3) \text{ 令 } f(z) = \frac{z^{2m}}{z^{2n} + 1}, \text{ 它在上半平面的奇点是 } e^{i\left(k+\frac{1}{2}\right)\frac{\pi}{n}} \quad (k=0, 1, 2, \dots, n-1)。$$

$$\begin{aligned} \text{res } f \left[e^{i\left(k+\frac{1}{2}\right)\frac{\pi}{n}} \right] &= \lim_{z \rightarrow e^{i\left(k+\frac{1}{2}\right)\frac{\pi}{n}}} \frac{z^{2m}}{(z^{2n} + 1)'} = \lim_{z \rightarrow e^{i\left(k+\frac{1}{2}\right)\frac{\pi}{n}}} \frac{z^{2m-2n+1}}{2n} \\ &= \frac{1}{2n} e^{ik(2m-2n+1)\frac{\pi}{n}} e^{i(2m-2n+1)\frac{\pi}{2n}} \end{aligned}$$

$$\begin{aligned} \text{原积分} &= \text{v.p.} \int_{-\infty}^{\infty} \frac{x^{2m}}{x^{2n} + 1} dx \\ &= 2\pi i \sum_{k=0}^{n-1} \text{res } f \left[e^{i\left(k+\frac{1}{2}\right)\frac{\pi}{n}} \right] = \frac{\pi i}{n} e^{i(2m-2n+1)\frac{\pi}{2n}} \sum_{k=0}^{n-1} e^{ik(2m-2n+1)\frac{\pi}{n}} \\ &= \frac{\pi i}{n} e^{i(2m-2n+1)\frac{\pi}{2n}} \frac{e^{i(2m-2n+1)\pi} - 1}{e^{i(2m-2n+1)\frac{\pi}{n}} - 1} = -\frac{\pi i}{n} \frac{2}{2i \sin \frac{(2m-2n+1)\pi}{2n}} \\ &= \frac{\pi}{n \sin \frac{(2m+1)\pi}{2n}}。 \end{aligned}$$

$$(4) \text{ 令 } f(z) = \frac{1}{(1+z^2)^{n+1}} = \frac{1}{(z+i)^{n+1}(z-i)^{n+1}}, \text{ 则}$$

$$\text{res } f(i) = \frac{1}{n!} \left[\frac{1}{(z+i)^{n+1}} \right]_{z=i}^{(n)} = \frac{1}{n!} \frac{(-1)^n (n+1)(n+2) \cdots (2n)}{(2i)^{2n+1}} = \frac{1}{2i} \frac{(2n)!}{2^{2n} (n!)^2} = \frac{1}{2i} \frac{(2n-1)!!}{(2n)!!}$$

$$\text{所以原积分} = \text{v.p.} \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{n+1}} = 2\pi i \text{res } f(i) = \frac{(2n-1)!!}{(2n)!!} \pi。$$

$$(5) \text{ 令 } f(z) = \frac{z^2}{(z^2 + a^2)^2} = \frac{z^2}{(z+ai)^2(z-ai)^2}, \text{ 则 } \text{res } f(ai) = \lim_{z \rightarrow ai} \frac{d}{dz} \frac{z^2}{(z+ai)^2} = \frac{1}{4ai}。$$

$$\text{同第(1)小题的讨论, 有原积分} = \frac{1}{2} \text{v.p.} \int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2 + a^2)^2} = \frac{1}{2} \cdot 2\pi i \text{res } f(ai) = \frac{\pi}{4a}。 \text{ (后面}$$

类似讨论省略)

(6) $\frac{1}{x^2-2x+4} = O\left(\frac{1}{x^2}\right)$, 所以 $\int_0^\infty \frac{dx}{x^2-2x+4}$ 收敛, $\int_{-\infty}^0 \frac{dx}{x^2-2x+4}$ 也收敛, 所以

$\int_{-\infty}^\infty \frac{dx}{x^2-2x+4}$ 收敛, 可直接计算其主值。(7) 可同样讨论)。

$$\text{令 } f(z) = \frac{1}{z^2-2z+4} = \frac{1}{(z-1-i\sqrt{3})(z-1+i\sqrt{3})}, \quad \text{res } f(1+i\sqrt{3}) = \frac{1}{2\sqrt{3}i},$$

$$\text{原积分} = \frac{\pi}{\sqrt{3}}.$$

$$(7) \text{ 令 } f(z) = \frac{z^2}{(z^2+1)(z^2-2z\cos\theta+1)} = \frac{z^2}{(z+i)(z-i)(z-e^{i\theta})(z-e^{-i\theta})},$$

$$\text{res } f(i) = -\frac{1}{4\cos\theta}, \quad \text{res } f(e^{i\theta}) = \frac{1}{2i\sin\theta(1+e^{-2i\theta})} = \frac{1}{4\cos\theta} - \frac{i}{4\sin\theta},$$

$$\text{res } f(e^{-i\theta}) = -\frac{e^{-2i\theta}}{2i\sin\theta(1+e^{-2i\theta})} = \frac{1}{4\cos\theta} + \frac{i}{4\sin\theta}.$$

$$\text{若 } \sin\theta > 0, \text{ 则 } e^{i\theta} \text{ 在上半平面, 原积分} = 2\pi i [\text{res } f(i) + \text{res } f(e^{i\theta})] = \frac{\pi}{2\sin\theta},$$

$$\text{若 } \sin\theta < 0, \text{ 则 } e^{-i\theta} \text{ 在上半平面, 原积分} = 2\pi i [\text{res } f(i) + \text{res } f(e^{-i\theta})] = -\frac{\pi}{2\sin\theta},$$

$$\text{所以原积分} = \frac{\pi}{2|\sin\theta|}.$$

$$(8) \text{ 令 } f(z) = \frac{1}{(1+z^2)\text{ch}\frac{\pi z}{2}}, \text{ 则 } z=i \text{ 为二阶极点, } z=(2k+1)i \text{ (} k=1,2,\dots \text{)} \text{ 为一阶}$$

$$\text{极点. } \text{res } f(i) = \lim_{z \rightarrow i} \frac{d}{dz} \frac{z-i}{(z+i)\text{ch}\frac{\pi z}{2}} = \lim_{z \rightarrow i} \frac{2i\text{ch}\frac{\pi z}{2} - \frac{\pi}{2}(z^2+1)\text{sh}\frac{\pi z}{2}}{(z+i)^2\text{ch}^2\frac{\pi z}{2}}$$

$$= \lim_{z \rightarrow i} \frac{\pi i \text{sh}\frac{\pi z}{2} - \pi z \text{sh}\frac{\pi z}{2} - \left(\frac{\pi}{2}\right)^2 (z^2+1)\text{ch}\frac{\pi z}{2}}{2(z+i)\text{ch}^2\frac{\pi z}{2} + \pi(z+i)^2\text{ch}\frac{\pi z}{2}\text{sh}\frac{\pi z}{2}}$$

$$= \lim_{z \rightarrow i} \frac{-\pi \text{sh}\frac{\pi z}{2} - \left(\frac{\pi}{2}\right)^2 (z+i)\text{ch}\frac{\pi z}{2}}{2(z+i)\text{ch}\frac{\pi z}{2}\frac{\text{ch}\frac{\pi z}{2}}{z-i} + \pi(z+i)^2\text{sh}\frac{\pi z}{2}\frac{\text{ch}\frac{\pi z}{2}}{z-i}}$$

$$= \frac{-\pi \operatorname{sh} \frac{\pi z}{2} - \left(\frac{\pi}{2}\right)^2 (z+i) \operatorname{ch} \frac{\pi z}{2}}{\pi (z+i) \operatorname{ch} \frac{\pi z}{2} \operatorname{sh} \frac{\pi z}{2} + \frac{\pi^2}{2} (z+i)^2 \operatorname{sh}^2 \frac{\pi z}{2}} \Bigg|_{z=i} = \frac{1}{2\pi i}。$$

$$\operatorname{res} f[(2k+1)i] = \lim_{z \rightarrow (2k+1)i} \frac{2}{\pi(z^2+1) \operatorname{sh} \frac{\pi z}{2}} = \frac{(-1)^{k-1}}{2\pi i} \frac{1}{k(k+1)}。 (k=1, 2, \dots)$$

取这样的积分路径：实轴上 $-R_n$ 到 R_n ，以原点为圆心， R_n 为半径的上半圆 C_n ，这里

$R_n = 2n$ 。该闭合路径包围奇点 $(2k+1)i$ ($k=0, 1, \dots, n-1$)，且路径上没有奇点。有

$$\int_{-2n}^{2n} f(x) dx + \int_{C_n} f(z) dz = 1 + \sum_{k=1}^{n-1} \frac{(-1)^{k-1}}{k(k+1)} = 1 + \sum_{k=1}^{n-1} (-1)^{k-1} \left(\frac{1}{k} - \frac{1}{k+1} \right)。 (*)$$

在 C_n 上有 $z_n = 2n(\cos \theta + i \sin \theta)$ ， $\operatorname{ch} \frac{\pi z_n}{2} = \frac{1}{2} [e^{n\pi(\cos \theta + i \sin \theta)} + e^{-n\pi(\cos \theta + i \sin \theta)}]$ ($0 \leq \theta \leq \pi$)，

当 $0 \leq \theta < \frac{\pi}{2}$ 时， $\cos \theta > 0$ ， $e^{n\pi(\cos \theta + i \sin \theta)} \rightarrow \infty, e^{-n\pi(\cos \theta + i \sin \theta)} \rightarrow 0$ ， $\operatorname{ch} \frac{\pi z_n}{2} \rightarrow \infty$ ，

当 $\frac{\pi}{2} < \theta \leq \pi$ 时， $\cos \theta < 0$ ， $e^{n\pi(\cos \theta + i \sin \theta)} \rightarrow 0, e^{-n\pi(\cos \theta + i \sin \theta)} \rightarrow \infty$ ， $\operatorname{ch} \frac{\pi z_n}{2} \rightarrow \infty$ ，

当 $\theta = \frac{\pi}{2}$ 时， $\operatorname{ch} \frac{\pi z_n}{2} = \cos n\pi$ ， $\left| \operatorname{ch} \frac{\pi z_n}{2} \right| = 1$ 。

综上， $0 \leq \theta \leq \pi$ 时存在 N ，当 $n > N$ 时有 $\left| \operatorname{ch} \frac{\pi z_n}{2} \right| \geq 1$ 。在 C_n 上

$$\left| z_n \cdot \frac{1}{(1+z_n^2) \operatorname{ch} \frac{\pi z_n}{2}} \right| \leq \frac{z_n}{1+z_n^2} (n > N)，所以 \lim_{n \rightarrow \infty} z_n f(z_n) = 0，由此得 \lim_{n \rightarrow \infty} \int_{C_n} f(z) dz = 0。$$

令 (*) 式 $n \rightarrow \infty$ 可得

$$\text{v.p.} \int_{-\infty}^{\infty} \frac{dx}{(1+x^2) \operatorname{ch} \frac{\pi x}{2}} = 1 + \sum_{k=1}^{\infty} (-1)^{k-1} \left(\frac{1}{k} - \frac{1}{k+1} \right) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} + \sum_{k=0}^{\infty} \frac{(-1)^k}{k+1} = 2 \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}。$$

易知 $\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} x^k$ ，($|x| < 1$) 由于右边级数在 $x=1$ 收敛，根据 Abel 第二定理，

该级数在 $x=1$ 点左连续，对上式两边取 $x \rightarrow 1-0$ 可得 $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} = \ln 2$ 。

所以 $\text{v.p.} \int_{-\infty}^{\infty} \frac{dx}{(1+x^2) \operatorname{ch} \frac{\pi x}{2}} = 2 \ln 2$ 。

由于 $\operatorname{ch} \frac{\pi x}{2} \geq 1$, 所以 $\left| \frac{1}{(1+x^2) \operatorname{ch} \frac{\pi x}{2}} \right| \leq \frac{1}{1+x^2}$, 因为 $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$ 收敛, 所以原积分收敛,

且等于其主值。

140. 计算下面积分: (1) $\int_0^{\infty} \frac{\cos x}{1+x^4} dx$; (2) $\int_0^{\infty} \frac{x \sin mx}{x^2+a^2} dx$, $a > 0, m > 0$;

(3) $\int_{-\infty}^{\infty} \frac{x \sin x}{x^2-2x+2} dx$; (4) $\int_0^{\infty} \frac{x^3 \sin mx}{x^4+4a^4} dx$, $a > 0, m > 0$; (5) $\int_{-\infty}^{\infty} \frac{\cos mx}{(x+b)^2+a^2} dx$,

$\int_{-\infty}^{\infty} \frac{\sin mx}{(x+b)^2+a^2} dx$, $a > 0, m > 0$; (6) $\int_{-\infty}^{\infty} \frac{a \cos x + x \sin x}{x^2+a^2} dx$, $a > 0$;

(7) $\int_{-\infty}^{\infty} \frac{\sin^2 ax}{(x^2+b^2)(x^2+c^2)} dx$, $a > 0, b > 0, c > 0$; (8) $\int_0^{\infty} \frac{\cos x}{\operatorname{ch} x} dx$ 。

(1) 令 $f(z) = \frac{e^{iz}}{1+z^4} = \frac{e^{iz}}{(z-e^{i\pi/4})(z-e^{-i\pi/4})(z-e^{i3\pi/4})(z-e^{-i3\pi/4})}$ 。

$\operatorname{res} f(e^{i\pi/4}) = \frac{e^{-1/\sqrt{2}}}{4i} e^{i\left(\frac{1}{\sqrt{2}} - \frac{\pi}{4}\right)}$, $\operatorname{res} f(e^{i3\pi/4}) = \frac{e^{-1/\sqrt{2}}}{4i} e^{-i\left(\frac{1}{\sqrt{2}} - \frac{\pi}{4}\right)}$ 。

原积分 = $\frac{1}{2} \text{v.p.} \int_{-\infty}^{\infty} \frac{\cos x}{1+x^4} dx$

$= \frac{1}{2} \operatorname{Re} \left\{ 2\pi i \left[\operatorname{res} f(e^{i\pi/4}) + \operatorname{res} f(e^{i3\pi/4}) \right] \right\} = \frac{\pi}{2} e^{-\frac{1}{\sqrt{2}}} \cos \left(\frac{1}{\sqrt{2}} - \frac{\pi}{4} \right)$ 。

(2) 令 $f(z) = \frac{ze^{imz}}{z^2+a^2} = \frac{ze^{imz}}{(z+ai)(z-ai)}$, 则 $\operatorname{res} f(ai) = \frac{1}{2} e^{-ma}$ 。

原积分 = $\frac{1}{2} \text{v.p.} \int_{-\infty}^{\infty} \frac{x \sin mx}{x^2+a^2} dx = \frac{1}{2} \operatorname{Im} [2\pi i \operatorname{res} f(ai)] = \frac{\pi}{2} e^{-ma}$ 。

(3) 令 $f(z) = \frac{ze^{iz}}{z^2-2z+2} = \frac{ze^{iz}}{(z-1-i)(z-1+i)}$, 则 $\operatorname{res} f(1+i) = \frac{1}{\sqrt{2}ei} e^{i\left(1+\frac{\pi}{4}\right)}$ 。

$\text{v.p.} \int_{-\infty}^{\infty} \frac{x \sin x}{x^2-2x+2} dx = \operatorname{Im} [2\pi i \cdot \operatorname{res} f(1+i)] = \frac{\sqrt{2}\pi}{e} \sin \left(1 + \frac{\pi}{4} \right)$ 。

由于 $\frac{x}{x^2-2x+2}$ 单调趋于 0, $\left| \int_0^b \sin x dx \right| = |1 - \cos b| \leq 2$, 所以原积分收敛 (狄里克莱判敛法), 等于其主值。

$$(4) \quad f(z) = \frac{z^3 e^{imz}}{z^4 + 4a^4} = \frac{z^3 e^{imz}}{\left(z - \sqrt{2}ae^{i\frac{\pi}{4}}\right)\left(z - \sqrt{2}ae^{-i\frac{\pi}{4}}\right)\left(z - \sqrt{2}ae^{i\frac{3\pi}{4}}\right)\left(z - \sqrt{2}ae^{-i\frac{3\pi}{4}}\right)}.$$

$$\text{则 } \operatorname{res} f\left(\sqrt{2}ae^{i\pi/4}\right) = \frac{1}{4}e^{-ma}e^{ima}, \quad \operatorname{res} f\left(\sqrt{2}ae^{i3\pi/4}\right) = \frac{1}{4}e^{-ma}e^{-ima}.$$

$$\begin{aligned} \text{原积分} &= \frac{1}{2} \text{v.p.} \int_0^\infty \frac{x^3 \sin mx}{x^4 + 4a^4} dx \\ &= \frac{1}{2} \operatorname{Im} \left\{ 2\pi i \left[\operatorname{res} f\left(\sqrt{2}ae^{i\pi/4}\right) + \operatorname{res} f\left(\sqrt{2}ae^{i3\pi/4}\right) \right] \right\} = \frac{\pi}{2} e^{-ma} \cos ma. \end{aligned}$$

$$(5) \quad \text{令 } f(z) = \frac{e^{imz}}{(z+b)^2 + a^2} = \frac{e^{imz}}{(z+b+ai)(z+b-ai)}, \quad \operatorname{res} f(-b+ai) = \frac{e^{-ma}}{2ai} e^{-imb}.$$

$$\text{所以 } \text{v.p.} \int_{-\infty}^\infty \frac{\cos mx}{(x+b)^2 + a^2} dx = \operatorname{Re} [2\pi i \operatorname{res} f(-b+ai)] = \frac{\pi}{a} e^{-ma} \cos mb,$$

$$\text{v.p.} \int_{-\infty}^\infty \frac{\sin mx}{(x+b)^2 + a^2} dx = \operatorname{Im} [2\pi i \operatorname{res} f(-b+ai)] = -\frac{\pi}{a} e^{-ma} \sin mb.$$

由于 $\left| \frac{\cos x}{(x+b)^2 + a^2} \right| \leq \frac{1}{(x+b)^2 + a^2}, \quad \left| \frac{\sin x}{(x+b)^2 + a^2} \right| \leq \frac{1}{(x+b)^2 + a^2}$ 所以原积分收敛, 等于其主值。

$$(6) \quad \text{令上小题第一个积分中 } b=0, m=1 \text{ 可得 } \int_{-\infty}^\infty \frac{\cos x}{x^2 + a^2} dx = \frac{\pi}{a} e^{-a},$$

$$\text{令第 (2) 小题中 } m=1 \text{ 可得 } \text{v.p.} \int_{-\infty}^\infty \frac{x \sin x}{x^2 + a^2} dx = \pi e^{-a}.$$

所以原积分 = $2\pi e^{-a}$ 。

$$(7) \quad \text{原积分} = \frac{1}{2} \int_{-\infty}^\infty \frac{1}{(x^2 + b^2)(x^2 + c^2)} dx - \frac{1}{2} \int_{-\infty}^\infty \frac{\cos 2ax}{(x^2 + b^2)(x^2 + c^2)} dx.$$

$$\text{令 } f(z) = \frac{1}{(z^2 + b^2)(z^2 + c^2)} = \frac{1}{(z+bi)(z-bi)(z+ci)(z-ci)},$$

$$\operatorname{res} f(bi) = -\frac{1}{2b(b^2 - c^2)i}, \quad \operatorname{res} f(ci) = \frac{1}{2c(b^2 - c^2)i}.$$

$$\text{则 } \int_{-\infty}^\infty \frac{1}{(x^2 + b^2)(x^2 + c^2)} dx = 2\pi i [\operatorname{res} f(bi) + \operatorname{res} f(ci)] = \frac{\pi}{bc(b+c)}.$$

$$\text{令 } f(z) = \frac{e^{2iaz}}{(z^2 + b^2)(z^2 + c^2)} = \frac{e^{2iaz}}{(z + bi)(z - bi)(z + ci)(z - ci)},$$

$$\text{res } f(bi) = -\frac{e^{-2ab}}{2b(b^2 - c^2)i}, \quad \text{res } f(ci) = \frac{e^{-2ac}}{2c(b^2 - c^2)i}。 \text{ 则}$$

$$\text{v.p.} \int_{-\infty}^{\infty} \frac{\cos 2ax}{(x^2 + b^2)(x^2 + c^2)} dx = \text{Re} \{ 2\pi i [\text{res } f(bi) + \text{res } f(ci)] \} = \frac{\pi}{b^2 - c^2} \left(\frac{e^{-2ac}}{c} - \frac{e^{-2ab}}{b} \right)$$

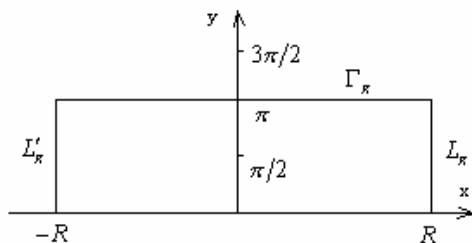
$$\text{由于 } \left| \frac{\sin^2 ax}{(x^2 + b^2)(x^2 + c^2)} \right| \leq \frac{1}{(x^2 + b^2)(x^2 + c^2)}, \text{ 所以原积分收敛且}$$

$$\text{原积分} = \frac{\pi(b - c + ce^{-2ab} - be^{-2ac})}{2bc(b^2 - c^2)}。$$

$$(8) \text{ 令 } f(z) = \frac{e^{iz}}{\text{ch } z}, \text{ 它有一阶极点 } \left(k + \frac{1}{2}\right)\pi i \quad (k = 0, 1, \dots)。$$

$$\text{res } f \left[\left(k + \frac{1}{2}\right)\pi i \right] = \lim_{z \rightarrow \left(k + \frac{1}{2}\right)\pi i} \frac{e^{iz}}{\text{sh } z} = \frac{(-1)^k}{i} e^{-\left(k + \frac{1}{2}\right)\pi}, \quad \text{res } f \left(\frac{\pi}{2} i \right) = \frac{1}{i} e^{-\frac{\pi}{2}}。$$

选取如下积分路径:



$$\int_{-R}^R f(x) dx + \int_{L_R} f(z) dz + \int_{\Gamma_R} f(z) dz + \int_{L_R'} f(z) dz = 2\pi i \text{res } f \left(\frac{\pi}{2} i \right) = 2\pi e^{-\frac{\pi}{2}}。 \quad (*)$$

$$\text{因为 } |\text{ch}(R + yi)| = |\text{ch } R \cos y + i \text{sh } R \sin y| = \sqrt{\text{ch}^2 R \cos^2 y + \text{sh}^2 R \sin^2 y}$$

$$= \sqrt{\cos^2 y + \text{sh}^2 R} \geq \text{sh } R, \text{ 所以 } \left| \int_{L_R} f(z) dz \right| = \left| \int_0^\pi \frac{e^{i(R+yi)}}{\text{ch}(R+yi)} dy \right| \leq \int_0^\pi \frac{e^{-y}}{\text{sh } R} dy = \frac{2(1 - e^{-\pi})}{e^R - e^{-R}}。$$

$$\text{因此 } \lim_{R \rightarrow \infty} \int_{L_R} f(z) dz = 0, \text{ 同样有 } \lim_{R \rightarrow \infty} \int_{L_R'} f(z) dz = 0。$$

$$\int_{\Gamma_R} f(z) dz = \int_R^{-R} \frac{e^{i(x+\pi i)}}{\text{ch}(x+\pi i)} dx = -e^{-\pi} \int_R^{-R} \frac{e^{ix}}{\text{ch } x} dx = e^{-\pi} \int_{-R}^R f(x) dx, \text{ 代入 } (*) \text{ 式并令}$$

$$R \rightarrow \infty \text{ 得 } \text{v.p.} \int_{-\infty}^{\infty} \frac{e^{ix}}{\operatorname{ch} x} dx = \frac{\pi e^{-\pi/2}}{1+e^{-\pi}} = \frac{\pi}{\operatorname{ch} \frac{\pi}{2}}。 \text{ 所以原积分} = \frac{\pi}{2 \operatorname{ch} \frac{\pi}{2}}。$$

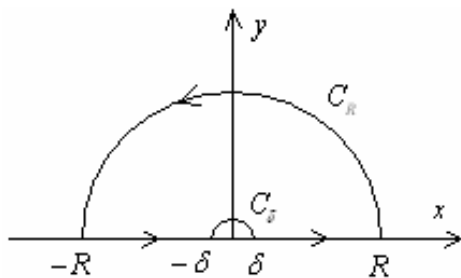
141. 计算下列积分: (1) $\int_0^{\infty} \frac{\sin mx}{x(x^2+a^2)} dx, a>0, m>0$; (2) $\text{v.p.} \int_{-\infty}^{\infty} \frac{dx}{x(x-1)(x-2)}$;

(3) $\int_0^{\infty} \frac{\cos ax - \cos bx}{x^2} dx, a>0, b>0$; (4) $\int_0^{\infty} \frac{\sin(x+a)\sin(x-a)}{x^2-a^2} dx, a>0$;

(5) $\int_0^{\infty} \frac{\sin^3 x}{x^3} dx$; (6) $\int_{-\infty}^{\infty} \frac{e^{px} - e^{qx}}{1-e^x} dx, 0<p<1, 0<q<1$;

(7) $\text{v.p.} \int_{-\infty}^{\infty} \frac{x \cos x}{x^2-5x-6} dx$; (8) $\text{v.p.} \int_{-\infty}^{\infty} \frac{\sin x}{(x^2+4)(x-1)} dx。$

(1) 取下图积分路径:



令 $f(z) = \frac{e^{imz}}{z(z^2+a^2)}$, 则 $\operatorname{res} f(ai) = -\frac{e^{-ma}}{2a^2}$, 所以

$$\begin{aligned} & \int_{-R}^{-\delta} \frac{e^{imx}}{x(x^2+a^2)} dx + \int_{C_\delta} \frac{e^{imz}}{z(z^2+a^2)} dz + \int_{\delta}^R \frac{e^{imx}}{x(x^2+a^2)} dx + \int_{C_R} \frac{e^{imz}}{z(z^2+a^2)} dz \\ &= 2\pi i \cdot \operatorname{res} f(ai) = -\frac{\pi i e^{-ma}}{a^2}。 \end{aligned} \quad (*)$$

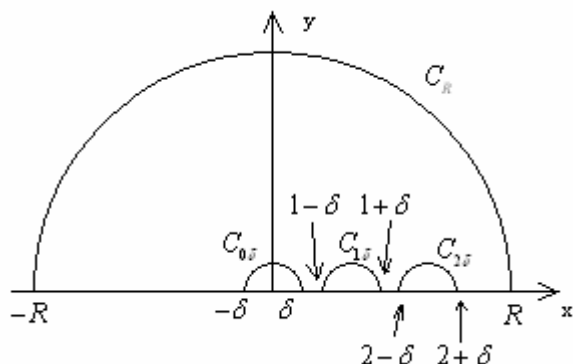
$$\lim_{\delta \rightarrow 0} \int_{C_\delta} \frac{e^{imz}}{z(z^2+a^2)} dz = -\pi i \lim_{z \rightarrow 0} z \cdot \frac{e^{imz}}{z(z^2+a^2)} = -\frac{\pi i}{a^2},$$

因为 $\lim_{z \rightarrow \infty} \frac{e^{imz}}{z(z^2+a^2)} = 0$, 所以 $\lim_{R \rightarrow \infty} \int_{C_R} \frac{e^{imz}}{z(z^2+a^2)} dz = 0$,

令 (*) 式 $\delta \rightarrow 0, R \rightarrow \infty$ 得到 $\text{v.p.} \int_{-\infty}^{\infty} \frac{e^{mx}}{x(x^2+a^2)} dx = \frac{\pi i}{a^2} (1 - e^{-ma})$,

$$\text{所以 } \int_0^{\infty} \frac{\sin mx}{x(x^2 + a^2)} dx = \frac{\pi}{2a^2} (1 - e^{-ma}) = \frac{\pi}{a^2} e^{-\frac{1}{2}ma} \operatorname{sh} \frac{ma}{2}.$$

(2) 取下图积分路径:



$$\text{令 } f(z) = \frac{1}{z(z-1)(z-2)}, \text{ 则 } \lim_{\delta \rightarrow 0} \int_{C_{0\delta}} f(z) dz = -\pi i \lim_{z \rightarrow 0} \frac{1}{(z-1)(z-2)} = -\frac{\pi}{2} i,$$

$$\lim_{\delta \rightarrow 0} \int_{C_{1\delta}} f(z) dz = -\pi i \lim_{z \rightarrow 1} \frac{1}{z(z-2)} = \pi i,$$

$$\lim_{\delta \rightarrow 0} \int_{C_{2\delta}} f(z) dz = -\pi i \lim_{z \rightarrow 2} \frac{1}{z(z-1)} = -\frac{\pi}{2} i$$

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = \pi i \lim_{z \rightarrow \infty} \frac{1}{(z-1)(z-2)} = 0.$$

$$\begin{aligned} & \int_{-R}^{-\delta} f(x) dx + \int_{C_{0\delta}} f(z) dz + \int_{\delta}^{1-\delta} f(x) dx + \int_{C_{1\delta}} f(z) dz + \int_{1+\delta}^{2-\delta} f(x) dx + \int_{C_{2\delta}} f(z) dz \\ & + \int_{2+\delta}^R f(x) dx + \int_{C_R} f(z) dz = 0 \end{aligned}$$

$$\text{令上式 } \delta \rightarrow 0, R \rightarrow \infty \text{ 得到 } \text{v.p.} \int_{-\infty}^{\infty} \frac{dx}{x(x-1)(x-2)} = 0.$$

(3) 令 $f(z) = \frac{e^{iaz} - e^{ibz}}{z^2}$, 取与第(1)小题相同的积分路径,

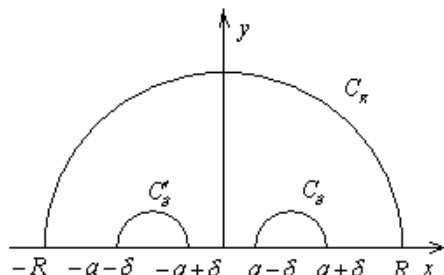
$$\lim_{\delta \rightarrow 0} \int_{C_{\delta}} f(z) dz = -\pi i \lim_{z \rightarrow 0} \frac{e^{iaz} - e^{ibz}}{z} = \pi(a-b),$$

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = \lim_{R \rightarrow \infty} \int_{C_R} \frac{e^{iaz}}{z^2} dz - \lim_{R \rightarrow \infty} \int_{C_R} \frac{e^{ibz}}{z^2} dz = 0,$$

对该路径的积分取 $\delta \rightarrow 0, R \rightarrow \infty$ 的极限可得 $\int_0^\infty \frac{\cos ax - \cos bx}{x^2} dx = \frac{\pi}{2}(b-a)$ 。

$$(4) \text{ 原积分} = \frac{1}{2} \int_0^\infty \frac{\cos 2a - \cos 2x}{x^2 - a^2} dx. \text{ 令 } f(z) = \frac{\cos 2a - e^{2iz}}{z^2 - a^2},$$

取如下积分路径:



$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -\pi i \lim_{z \rightarrow a} (z-a) f(z) = \frac{\pi(\cos 2a - e^{2ia})}{2ai},$$

$$\lim_{\delta \rightarrow 0} \int_{C'_\delta} f(z) dz = -\pi i \lim_{z \rightarrow -a} (z+a) f(z) = -\frac{\pi(\cos 2a - e^{-2ia})}{2ai}.$$

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0.$$

$$\int_{-R}^{-a-\delta} f(x) dx + \int_{C'_\delta} f(z) dz + \int_{-a+\delta}^{a-\delta} f(x) dx + \int_{C_\delta} f(z) dz + \int_{a+\delta}^R f(x) dx + \int_{C_R} f(z) dz = 0$$

$$\text{令 } \delta \rightarrow 0, R \rightarrow \infty \text{ 得 v.p.} \int_{-\infty}^\infty \frac{\cos 2a - e^{2ix}}{x^2 - a^2} dx = \frac{\pi}{a}(\sin 2a - i \cos 2a).$$

$$\text{所以原积分} = \frac{1}{4} \operatorname{Re} \left(\text{v.p.} \int_{-\infty}^\infty \frac{\cos 2a - e^{2ix}}{x^2 - a^2} dx \right) = \frac{\pi}{4a} \sin 2a.$$

$$(5) \sin^3 x = \left(\frac{1}{2} - \frac{1}{2} \cos 2x \right) \sin x = \frac{1}{2} \sin x - \frac{1}{4} (\sin 3x - \sin x) = \frac{1}{4} (3 \sin x - \sin 3x).$$

$$\text{原积分} = \int_0^\infty \frac{3 \sin x - \sin 3x}{4x^3} dx. \text{ 令 } f(z) = \frac{3e^{iz} - e^{3iz} - 2}{4z^3}, \text{ 取与第(1)小题相同的积分}$$

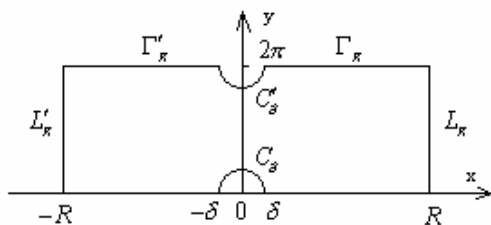
路径, 有

$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -\pi i \lim_{z \rightarrow 0} \frac{3e^{iz} - e^{3iz} - 2}{4z^2} = -\frac{3}{4} \pi i, \quad \lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0.$$

$$\text{令围线积分 } \delta \rightarrow 0, R \rightarrow \infty \text{ 得 v.p.} \int_{-\infty}^\infty \frac{3e^{ix} - e^{3ix} - 2}{4x^3} dx = \frac{3}{4} \pi i,$$

$$\text{原积分} = \frac{1}{2} \operatorname{Im} \left(\text{v.p.} \int_{-\infty}^{\infty} \frac{3e^{ix} - e^{3ix} - 2}{4x^3} dx \right) = \frac{3}{8} \pi .$$

(6) 令 $f(z) = \frac{e^{pz}}{1-e^z}$, $2k\pi i$ ($k=0, \pm 1, \pm 2, \dots$) 是其一阶极点。取如下积分路径:



$$\begin{aligned} & \int_{-R}^{-\delta} f(x) dx + \int_{C_\delta} f(z) dz + \int_{\delta}^R f(x) dx + \int_{L_R} f(z) dz + \int_{\Gamma_R} f(z) dz + \int_{C'_\delta} f(z) dz \\ & + \int_{\Gamma'_R} f(z) dz + \int_{L'_R} f(z) dz = 0 . \end{aligned} \quad (*)$$

$$\int_{\Gamma_R} f(z) dz = \int_R^\delta \frac{e^{p(x+2\pi i)}}{1-e^{(x+2\pi i)}} dx = -e^{2p\pi i} \int_\delta^R f(x) dx, \quad \int_{\Gamma'_R} f(z) dz = -e^{2p\pi i} \int_{-R}^{-\delta} f(x) dx,$$

代入 (*) 式得

$$\begin{aligned} & (1-e^{2p\pi i}) \left[\int_{-R}^{-\delta} f(x) dx + \int_{\delta}^R f(x) dx \right] + \int_{C_\delta} f(z) dz + \int_{L_R} f(z) dz + \int_{C'_\delta} f(z) dz \\ & + \int_{L'_R} f(z) dz = 0 . \end{aligned} \quad (**)$$

$$\begin{aligned} \left| \int_{\Gamma_R} f(z) dz \right| & \leq \int_0^{2\pi} \left| \frac{e^{p(R+iy)}}{1-e^{(R+iy)}} \right| dy = \int_0^{2\pi} \frac{e^{pR}}{\sqrt{(1-e^R \cos y)^2 + e^{2R} \sin^2 y}} dy \\ & = e^{pR} \int_0^{2\pi} \frac{1}{\sqrt{1-2e^R \cos y + e^{2R}}} dy \leq e^{pR} \int_0^{2\pi} \frac{1}{e^R - 1} dy = 2\pi \frac{e^{-(1-p)R}}{1-e^{-R}}, \end{aligned}$$

所以 $\lim_{R \rightarrow \infty} \int_{\Gamma_R} f(z) dz = 0$, 同样的, $\lim_{R \rightarrow \infty} \int_{\Gamma'_R} f(z) dz = 0$ 。

$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -\pi i \lim_{z \rightarrow 0} \frac{ze^{pz}}{1-e^z} = \pi i ,$$

$$\lim_{\delta \rightarrow 0} \int_{C'_\delta} f(z) dz = -\pi i \lim_{z \rightarrow 2\pi i} \frac{(z-2\pi i)e^{pz}}{1-e^z} = \pi i e^{2p\pi i} ,$$

令 (**) 式 $\delta \rightarrow 0, R \rightarrow \infty$ 可得 $\text{v.p.} \int_{-\infty}^{\infty} \frac{e^{px}}{1-e^x} dx = \pi i \frac{e^{2p\pi i} + 1}{e^{2p\pi i} - 1} = \pi \cot(p\pi) ,$

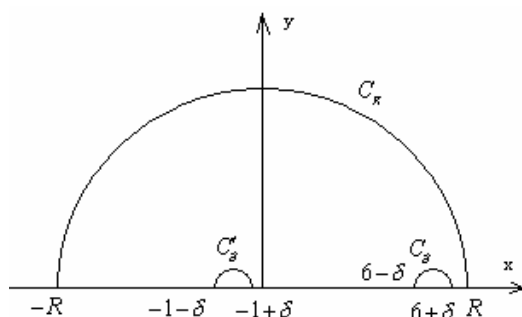
$$\text{所以 } \text{v.p.} \int_{-\infty}^{\infty} \frac{e^{px} - e^{qx}}{1 - e^x} dx = \text{v.p.} \int_{-\infty}^{\infty} \frac{e^{px}}{1 - e^x} dx - \text{v.p.} \int_{-\infty}^{\infty} \frac{e^{qx}}{1 - e^x} dx = \pi [\cot(p\pi) - \cot(q\pi)]。$$

当 x 充分大时, 有 $e^x - 1 \geq \frac{1}{2}e^x$ (即 $e^x \geq 2$)。不妨假设 $p > q$,

$$\left| \frac{e^{px} - e^{qx}}{1 - e^x} \right| \leq 2 \frac{e^{px} - e^{qx}}{e^x} = 2 [e^{-(1-p)x} - e^{-(1-q)x}]。 \text{由于 } \int_0^{\infty} [e^{-(1-p)x} - e^{-(1-q)x}] dx \text{ 收敛, 所以}$$

$$\int_0^{\infty} \frac{e^{px} - e^{qx}}{1 - e^x} dx \text{ 收敛, 类似地, } \int_{-\infty}^0 \frac{e^{px} - e^{qx}}{1 - e^x} dx \text{ 收敛, 所以原积分收敛, 等于其主值。}$$

$$(7) \text{ 令 } f(z) = \frac{ze^{iz}}{z^2 - 5z - 6} = \frac{ze^{iz}}{(z-6)(z+1)}。 \text{取如下积分路径:}$$



$$\left(\int_{-R}^{-1-\delta} + \int_{-1+\delta}^{6-\delta} + \int_{6+\delta}^R \right) f(x) dx + \left(\int_{C_\delta} + \int_{C'_\delta} + \int_{C_R} \right) f(z) dz = 0。 \quad (*)$$

$$\text{因为 } \lim_{z \rightarrow \infty} \frac{z}{(z-6)(z+1)} = 0, \text{ 所以 } \lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0。$$

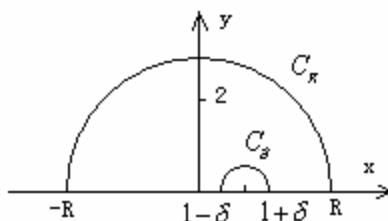
$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -\pi i \lim_{z \rightarrow 6} \frac{ze^{iz}}{z+1} = \frac{6\pi}{7i} e^{6i}, \quad \lim_{\delta \rightarrow 0} \int_{C'_\delta} f(z) dz = -\pi i \lim_{z \rightarrow -1} \frac{ze^{iz}}{z-6} = \frac{\pi}{7i} e^{-i}。$$

$$\text{令 } (*) \text{ 式 } \delta \rightarrow 0, R \rightarrow \infty \text{ 得 } \text{v.p.} \int_{-\infty}^{\infty} f(x) dx = \frac{\pi i}{7} (6e^{6i} + e^{-i}),$$

$$\text{原积分} = \text{Re} \left[\text{v.p.} \int_{-\infty}^{\infty} f(x) dx \right] = \frac{\pi}{7} (\sin 1 - 6 \sin 6)。$$

$$(8) \text{ 令 } f(z) = \frac{e^{iz}}{(z^2 + 4)(z-1)}, \quad \text{res } f(2i) = \lim_{z \rightarrow 2i} \frac{e^{iz}}{(z+2i)(z-1)} = -\frac{e^{-2}}{20i} (1+2i)。$$

取如下积分路径:



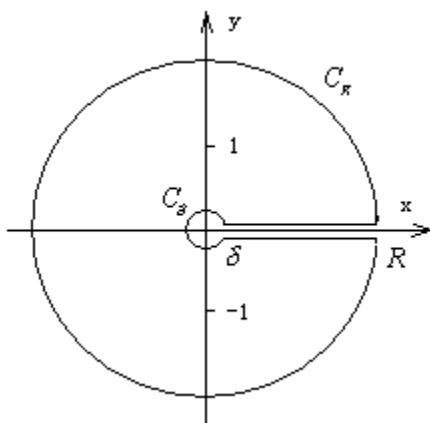
$$\left(\int_{-R}^{1-\delta} + \int_{1+\delta}^R \right) f(x) dx + \left(\int_{C_\delta} + \int_{C_R} \right) f(z) dz = 2\pi i \operatorname{res} f(2i) = -\frac{\pi e^{-2}}{10} (1+2i). \quad (*)$$

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0, \quad \lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -\pi i \lim_{z \rightarrow 1} \frac{e^{iz}}{z^2 + 4} = -\frac{\pi i}{5} e^i.$$

$$\text{令 } (*) \delta \rightarrow 0, R \rightarrow \infty \text{ 得原积分} = \operatorname{Im} \left[\text{v.p.} \int_{-\infty}^{\infty} f(x) dx \right] = \frac{\pi}{5} (\cos 1 - e^{-2}).$$

142. 计算下列积分: (1) $\int_0^\infty \frac{x^s}{(1+x^2)^2} dx$, $-1 < s < 3$; (2) $\int_0^\infty \frac{x^{-p}}{1+2x \cos \lambda + x^2} dx$, $-1 < p < 1$, $0 < \lambda < \pi$; (3) $\text{v.p.} \int_0^\infty \frac{x^{a-1}}{1-x} dx$, $0 < a < 1$; (4) $\int_0^\infty \frac{\ln x}{\sqrt{x}(x^2+a^2)^2} dx$, $a > 0$.

(1) 令 $f(z) = \frac{z^s}{(1+z^2)^2}$, 取如下积分路径, 规定 $0 \leq \arg \leq 2\pi$,



$$\text{则 } \operatorname{res} f(i) = \lim_{z \rightarrow i} \frac{d}{dz} \frac{z^s}{(z+i)^2} = -\frac{s-1}{4i} e^{i\frac{s\pi}{2}}, \quad \operatorname{res} f(-i) = \lim_{z \rightarrow -i} \frac{d}{dz} \frac{z^s}{(z-i)^2} = \frac{s-1}{4i} e^{i\frac{3s\pi}{2}}$$

(这里 $i = e^{i\frac{\pi}{2}}$, $-i = e^{i\frac{3\pi}{2}}$). 围道积分为:

$$\begin{aligned} & \int_\delta^R \frac{x^s}{(1+x^2)^2} dx + \int_R^\delta \frac{x^s e^{2is\pi}}{(1+x^2)^2} dx + \left(\int_{C_\delta} + \int_{C_R} \right) f(z) dz \\ &= 2\pi i [\operatorname{res} f(i) + \operatorname{res} f(-i)] = i\pi (s-1) e^{is\pi} \sin \frac{s\pi}{2}. \end{aligned}$$

$$\text{即 } (1 - e^{2is\pi}) \int_\delta^R \frac{x^s}{(1+x^2)^2} dx + \left(\int_{C_\delta} + \int_{C_R} \right) f(z) dz = i\pi (s-1) e^{is\pi} \sin \frac{s\pi}{2}.$$

令上式 $\delta \rightarrow 0, R \rightarrow \infty$, 由于 $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 2\pi i \lim_{z \rightarrow \infty} z \cdot \frac{z^s}{(1+z^2)^2} = 0$,

$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -2\pi i \lim_{z \rightarrow 0} z \cdot \frac{z^s}{(1+z^2)^2} = 0$, 所以

$$\int_0^\infty \frac{x^s}{(1+x^2)^2} dx = \frac{i\pi(s-1)e^{is\pi} \sin \frac{s\pi}{2}}{1-e^{2is\pi}} = \frac{\pi}{4}(1-s) \sec \frac{s\pi}{2}.$$

(2) 令 $f(z) = \frac{z^{-p}}{z^2 + 2z \cos \lambda + 1} = \frac{z^{-p}}{(z+e^{i\lambda})(z+e^{-i\lambda})}$, 取与上小题同样的积分路径, 规定

$0 \leq \arg z \leq 2\pi$, 则 $\operatorname{res} f \left[e^{i(\pi+\lambda)} \right] = \lim_{z \rightarrow e^{i(\pi+\lambda)}} \frac{z^{-p}}{z+e^{-i\lambda}} = -\frac{e^{-ip(\pi+\lambda)}}{2i \sin \lambda}$,

$\operatorname{res} f \left[e^{i(\pi-\lambda)} \right] = \lim_{z \rightarrow e^{i(\pi-\lambda)}} \frac{z^{-p}}{z+e^{i\lambda}} = \frac{e^{-ip(\pi-\lambda)}}{2i \sin \lambda}$ 。所以

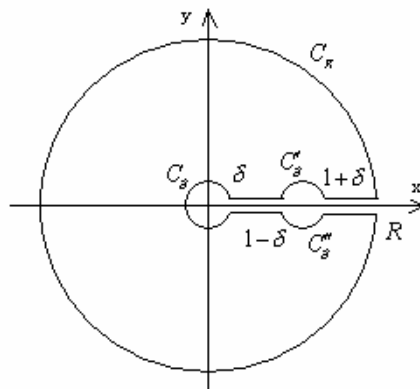
$$(1-e^{-2ip\pi}) \int_\delta^R \frac{x^{-p}}{1+2x \cos \lambda + x^2} dx + \left(\int_{C_\delta} + \int_{C_R} \right) f(z) dz = 2\pi i e^{-ip\pi} \frac{\sin p\lambda}{\sin \lambda}.$$

令上式 $\delta \rightarrow 0, R \rightarrow \infty$, 由于 $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 2\pi i \lim_{z \rightarrow \infty} z \cdot \frac{z^{-p}}{z^2 + 2z \cos \lambda + 1} = 0$,

$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -2\pi i \lim_{z \rightarrow 0} z \cdot \frac{z^{-p}}{z^2 + 2z \cos \lambda + 1} = 0$, 所以

$$\int_0^\infty \frac{x^{-p}}{1+2x \cos \lambda + x^2} dx = 2\pi i e^{-ip\pi} \frac{\sin p\lambda}{\sin \lambda (1-e^{-2ip\pi})} = \frac{\pi \sin p\lambda}{\sin p\pi \sin \lambda}.$$

(3) 令 $f(z) = \frac{z^{a-1}}{1-z}$, 取如下积分路径, 规定 $0 \leq \arg z \leq 2\pi$ 。



$$(1 - e^{2ia\pi}) \left(\int_{\delta}^{1-\delta} + \int_{1+\delta}^R \right) f(x) dx + \left(\int_{C_R} + \int_{C_{\delta}} + \int_{C'_{\delta}} + \int_{C''_{\delta}} \right) f(z) dz = 0,$$

令上式 $\delta \rightarrow 0, R \rightarrow \infty$, 因为 $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$, $\lim_{\delta \rightarrow 0} \int_{C_{\delta}} f(z) dz = 0$,

$$\lim_{\delta \rightarrow 0} \int_{C'_{\delta}} f(z) dz = -\pi i \lim_{z \rightarrow 1} (z-1) \frac{z^{a-1}}{1-z} = \pi i,$$

$$\lim_{\delta \rightarrow 0} \int_{C''_{\delta}} f(z) dz = -\pi i \lim_{z \rightarrow e^{2i\pi}} (z-1) \frac{z^{a-1}}{1-z} = \pi i e^{2ia\pi},$$

$$\text{所以原积分} = \pi i \frac{e^{2ia\pi} + 1}{e^{2ia\pi} - 1} = \pi \cot(a\pi).$$

(4) 令 $f(z) = \frac{(\ln z)^2}{\sqrt{z}(z^2 + a^2)^2}$, 取与第(1)小题相同的积分路径, 规定 $0 \leq \arg \leq 2\pi$,

$$\text{则 } \operatorname{res} f(ai) = \lim_{z \rightarrow ai} \frac{d}{dz} \frac{(\ln z)^2}{\sqrt{z}(z+ai)^2} = \lim_{z \rightarrow ae^{i\pi/2}} \frac{4(z+ai) \ln z - (5z+ai)(\ln z)^2}{2z^{3/2}(z+ai)^3}$$

$$= \frac{4 \ln a - 3(\ln a)^2 + \frac{3}{4}\pi^2 - 2\pi + 3\pi \ln a + i \left[4 \ln a - 3(\ln a)^2 + \frac{3}{4}\pi^2 + 2\pi - 3\pi \ln a \right]}{8\sqrt{2}a^{7/2}},$$

$$\operatorname{res} f(-ai) = \lim_{z \rightarrow -ai} \frac{d}{dz} \frac{(\ln z)^2}{\sqrt{z}(z-ai)^2} = \lim_{z \rightarrow ae^{i3\pi/2}} \frac{4(z-ai) \ln z - (5z-ai)(\ln z)^2}{2z^{3/2}(z-ai)^3}$$

$$= \frac{-4 \ln a + 3(\ln a)^2 - \frac{27}{4}\pi^2 - 6\pi + 9\pi \ln a + i \left[4 \ln a - 3(\ln a)^2 + \frac{27}{4}\pi^2 - 6\pi + 9\pi \ln a \right]}{8\sqrt{2}a^{7/2}}.$$

围道积分为:

$$\int_{\delta}^R \frac{(\ln x)^2}{\sqrt{x}(x^2 + a^2)^2} dx + \int_R^{\delta} \frac{(\ln x + 2\pi i)^2}{-\sqrt{x}(x^2 + a^2)^2} dx + \left(\int_{C_R} + \int_{C_{\delta}} \right) f(z) dz = 2\pi i [\operatorname{res} f(ai) + \operatorname{res} f(-ai)]$$

令 $\delta \rightarrow 0, R \rightarrow \infty$, 由于 $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$, $\lim_{\delta \rightarrow 0} \int_{C_{\delta}} f(z) dz = 0$, 所以

$$\int_0^{\infty} \frac{2(\ln x)^2 - 4\pi^2}{\sqrt{x}(x^2 + a^2)^2} dx + 4\pi i \int_0^{\infty} \frac{\ln x}{\sqrt{x}(x^2 + a^2)^2} dx = 2\pi i [\operatorname{res} f(ai) + \operatorname{res} f(-ai)],$$

$$\text{因此原积分} = \frac{1}{4\pi} \operatorname{Im} \{ 2\pi i [\operatorname{res} f(ai) + \operatorname{res} f(-ai)] \} = \frac{1}{2} \operatorname{Re} [\operatorname{res} f(ai) + \operatorname{res} f(-ai)]$$

$$= \frac{\pi}{2\sqrt{2}a^{7/2}} \left(\frac{3}{2} \ln a - 1 - \frac{3\pi}{4} \right).$$

143. 设 $P(z)$ 及 $Q(z)$ 分别为 m 阶及 n 阶多项式, 并且 $m \leq n-2$, 且 $Q(z)$ 无非负实根。

考虑函数 $\frac{P(z)}{Q(z)} \ln z$ 的积分, 证明 $\int_0^\infty \frac{P(x)}{Q(x)} dx = - \sum_{\text{全平面}} \operatorname{res} \left\{ \frac{P(z)}{Q(z)} \ln z \right\}, \quad 0 \leq \arg z \leq 2\pi$ 。

证: 令 $f(z) = \frac{P(z)}{Q(z)} \ln z$, 取与上题第 (1) 小题相同的积分路径, 规定 $0 \leq \arg z \leq 2\pi$,

$$\text{有 } \int_\delta^R \frac{P(x)}{Q(x)} \ln x dx + \int_R^\delta \frac{P(x)}{Q(x)} (\ln x + 2\pi i) dx + \left(\int_{C_R} + \int_{C_\delta} \right) f(z) dz = 2\pi i \sum_{\text{全平面}} \operatorname{res} \left\{ \frac{P(z)}{Q(z)} \ln z \right\},$$

$$\text{即 } \int_\delta^R \frac{P(x)}{Q(x)} dx + \left(\int_{C_R} + \int_{C_\delta} \right) f(z) dz = - \sum_{\text{全平面}} \operatorname{res} \left\{ \frac{P(z)}{Q(z)} \ln z \right\}. \quad (*)$$

由于 0 不是 $Q(z)$ 的零点, 则 $\lim_{z \rightarrow 0} z \cdot \frac{P(z)}{Q(z)} \ln z = \frac{P(0)}{Q(0)} \lim_{z \rightarrow 0} z \ln z = 0$, 所以

$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = 0$ 。因为 $m \leq n-2$, 所以 $\lim_{z \rightarrow \infty} z \cdot \frac{P(z)}{Q(z)} \ln z = \lim_{z \rightarrow \infty} \frac{\ln z}{z^{n-m-1}} = 0$, 则

$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$ 。令 (*) 式 $\delta \rightarrow 0, R \rightarrow \infty$ 即得 $\int_0^\infty \frac{P(x)}{Q(x)} dx = - \sum_{\text{全平面}} \operatorname{res} \left\{ \frac{P(z)}{Q(z)} \ln z \right\}$ 。

144. 利用上题结果计算下列积分: (1) $\int_0^\infty \frac{x}{(1+x+x^2)^2} dx$; (2) $\int_0^\infty \frac{1}{x^3+a^3} dx$;

(3) $\int_0^\infty \frac{1}{(x+a)(x^2+b^2)} dx, \quad a > 0, \quad b > 0$; (4) $\int_0^\infty \frac{1}{(x^2+a^2)(x^2+b^2)} dx$ 。

$$(1) \text{ 令 } f(z) = \frac{z \ln z}{(1+z+z^2)^2}, \quad \operatorname{res} f(e^{i2\pi/3}) = \lim_{z \rightarrow e^{i2\pi/3}} \frac{d}{dz} \frac{z \ln z}{(z - e^{i4\pi/3})^2} = -\frac{1}{3} - \frac{2}{9\sqrt{3}}\pi,$$

$$\operatorname{res} f(e^{i4\pi/3}) = \lim_{z \rightarrow e^{i4\pi/3}} \frac{d}{dz} \frac{z \ln z}{(z - e^{i2\pi/3})^2} = \frac{4}{9\sqrt{3}}\pi - \frac{1}{3}.$$

$$\text{原积分} = -\operatorname{res} f(e^{i2\pi/3}) - \operatorname{res} f(e^{i4\pi/3}) = \frac{2}{3} \left(1 - \frac{\sqrt{3}}{9} \pi \right).$$

$$(2) \text{ 令 } f(z) = \frac{\ln z}{z^3 + a^3},$$

$$\operatorname{res} f(ae^{i\pi/3}) = \lim_{z \rightarrow ae^{i\pi/3}} \frac{\ln z}{(z+a)(z-ae^{i5\pi/3})} = \frac{\sqrt{3}\pi - 3\ln a - i(\pi + 3\sqrt{3}\ln a)}{18a^2},$$

$$\operatorname{res} f(-a) = \lim_{z \rightarrow ae^{i\pi}} \frac{\ln z}{z^2 - az + a^2} = \frac{\ln a + \pi i}{3a^2},$$

$$\operatorname{res} f(ae^{i5\pi/3}) = \lim_{z \rightarrow ae^{i5\pi/3}} \frac{\ln z}{(z+a)(z-ae^{i2\pi/3})} = \frac{-5\sqrt{3}\pi - 3\ln a + i(3\sqrt{3}\ln a - 5\pi)}{18a^2}.$$

$$\text{原积分} = -\operatorname{res} f(ae^{i\pi/3}) - \operatorname{res} f(-a) - \operatorname{res} f(ae^{i5\pi/3}) = \frac{2\sqrt{3}\pi}{9a^2}.$$

$$(3) \text{ 令 } f(z) = \frac{\ln z}{(z+a)(z^2+b^2)}, \text{ 则 } \operatorname{res} f(-a) = \lim_{z \rightarrow ae^{i\pi}} \frac{\ln z}{z^2+b^2} = \frac{\ln a + \pi i}{a^2+b^2},$$

$$\operatorname{res} f(bi) = \lim_{z \rightarrow be^{i\pi/2}} \frac{\ln z}{(z+a)(z+bi)} = \frac{a\pi - 2b\ln b - i(2a\ln b + b\pi)}{4b(a^2+b^2)},$$

$$\operatorname{res} f(-bi) = \lim_{z \rightarrow be^{i3\pi/2}} \frac{\ln z}{(z+a)(z-bi)} = \frac{-3a\pi - 2b\ln b + i(2a\ln b - 3b\pi)}{4b(a^2+b^2)}.$$

$$\text{原积分} = -\operatorname{res} f(-a) - \operatorname{res} f(-bi) - \operatorname{res} f(bi) = \frac{1}{a^2+b^2} \left(\ln \frac{b}{a} + \frac{a\pi}{2b} \right).$$

$$(4) f(z) = \frac{\ln z}{(z^2+a^2)(z^2+b^2)}, \text{ 则 } \operatorname{res} f(ai) = \lim_{z \rightarrow ae^{i\pi/2}} \frac{\ln z}{(z+ai)(z^2+b^2)} = \frac{-\pi + 2i\ln a}{4a(a^2-b^2)},$$

$$\operatorname{res} f(-ai) = \lim_{z \rightarrow ae^{i3\pi/2}} \frac{\ln z}{(z-ai)(z^2+b^2)} = \frac{3\pi - 2i\ln a}{4a(a^2-b^2)},$$

$$\operatorname{res} f(bi) = \lim_{z \rightarrow be^{i\pi/2}} \frac{\ln z}{(z^2+a^2)(z+bi)} = \frac{\pi - 2i\ln b}{4b(a^2-b^2)},$$

$$\operatorname{res} f(-bi) = \lim_{z \rightarrow be^{i3\pi/2}} \frac{\ln z}{(z^2+a^2)(z-bi)} = \frac{-3\pi + 2i\ln b}{4b(a^2-b^2)}.$$

$$\text{原积分} = -\operatorname{res} f(ai) - \operatorname{res} f(-ai) - \operatorname{res} f(bi) - \operatorname{res} f(-bi) = \frac{\pi}{2ab(a+b)}.$$

145. 用类似于 143 题的方法证明 $\int_0^\infty f(x) \ln x dx = -\frac{1}{2} \operatorname{Re} \sum_{\text{全平面}} \operatorname{res} \left\{ f(z) (\ln z)^2 \right\}$,

$\int_0^\infty f(x) dx = -\frac{1}{2\pi} \operatorname{Im} \sum_{\text{全平面}} \operatorname{res} \left\{ f(z) (\ln z)^2 \right\}$, $0 \leq \arg z \leq 2\pi$ 。其中 $f(x)$ 满足和第 143

题中 $\frac{P(z)}{Q(z)}$ 同样的要求。

证: 令 $F(z) = f(z) (\ln z)^2$, 取与 143 题相同的积分路径, 规定 $0 \leq \arg z < 2\pi$, 同样有

$\lim_{\delta \rightarrow 0} \int_{C_\delta} F(z) dz = 0$, $\lim_{R \rightarrow \infty} \int_{C_R} F(z) dz = 0$ 。围线积分为

$$\int_\delta^R f(x) (\ln x)^2 dx + \int_R^\delta f(x) (\ln x + 2\pi i)^2 dx + \left(\int_{C_\delta} + \int_{C_R} \right) F(z) dz = 2\pi i \sum_{\text{全平面}} \operatorname{res} \{F(z)\},$$

$$\text{即 } 4\pi^2 \int_\delta^R f(x) dx - 4\pi i \int_\delta^R f(x) \ln x dx + \left(\int_{C_\delta} + \int_{C_R} \right) F(z) dz = 2\pi i \sum_{\text{全平面}} \operatorname{res} \{F(z)\}。$$

$$\text{令上式 } \delta \rightarrow 0, R \rightarrow \infty \text{ 得 } \pi \int_0^\infty f(x) dx - i \int_0^\infty f(x) \ln x dx = \frac{1}{2} i \sum_{\text{全平面}} \operatorname{res} \left\{ f(z) (\ln z)^2 \right\},$$

比较两边实部和虚部即得证。

146. 利用上题结论计算下列积分: (1) $\int_0^\infty \frac{\ln x}{x^2 + a^2} dx$, $a > 0$; (2) $\int_0^\infty \frac{\ln x}{(x+a)(x+b)} dx$,

$b > a > 0$; (3) $\int_0^\infty \frac{\ln x}{(x+a)^2} dx$, $a > 0$; (4) $\int_0^\infty \frac{\ln x}{(x+a)^2 + b^2} dx$, a, b 均为正数。

$$(1) \text{ 令 } f(z) = \frac{(\ln z)^2}{z^2 + a^2}, \text{ 则 } \operatorname{res} f(ai) = \lim_{z \rightarrow ae^{i\pi/2}} \frac{(\ln z)^2}{z + ai} = \frac{\pi \ln a}{2a} + i \frac{\pi^2 - 4 \ln^2 a}{8a},$$

$$\operatorname{res} f(-ai) = \lim_{z \rightarrow ae^{i3\pi/2}} \frac{(\ln z)^2}{z - ai} = -\frac{3\pi \ln a}{2a} + i \frac{4 \ln^2 a - 9\pi^2}{8a}。$$

$$\text{原积分} = -\frac{1}{2} \left\{ \operatorname{Re} [\operatorname{res} f(ai)] + \operatorname{Re} [\operatorname{res} f(-ai)] \right\} = \frac{\pi \ln a}{2a}。$$

$$(2) \text{ 令 } f(z) = \frac{(\ln z)^2}{(z+a)(z+b)}, \text{ 则 } f(-a) = \lim_{z \rightarrow ae^{i\pi}} \frac{(\ln z)^2}{z+b} = \frac{\ln^2 a - \pi^2}{b-a} + i \frac{2\pi \ln a}{b-a},$$

$$f(-b) = \lim_{z \rightarrow be^{i\pi}} \frac{(\ln z)^2}{z+a} = \frac{\pi^2 - \ln^2 b}{b-a} - i \frac{2\pi \ln b}{b-a}。$$

$$\text{原积分} = -\frac{1}{2} \{ \operatorname{Re} [\operatorname{res} f(-a)] + \operatorname{Re} [\operatorname{res} f(-b)] \} = \frac{\ln^2 b - \ln^2 a}{2(b-a)} = \frac{\ln ab \ln \frac{b}{a}}{2(b-a)}.$$

$$(3) \text{ 令 } f(z) = \frac{(\ln z)^2}{(z+a)^2}, \text{ 则 } \operatorname{res} f(-a) = \lim_{z \rightarrow ae^{i\pi}} \frac{d}{dz} (\ln z)^2 = -\frac{2 \ln a}{a} - i \frac{2\pi}{a}.$$

$$\text{原积分} = -\frac{1}{2} \operatorname{Re} [\operatorname{res} f(-a)] = \frac{\ln a}{a}.$$

$$(4) \text{ 令 } f(z) = \frac{(\ln z)^2}{(z+a)^2 + b^2}, \text{ 则 } \operatorname{res} f(-a+bi) = \lim_{z \rightarrow \sqrt{a^2+b^2}e^{i(\pi-\tan^{-1}\frac{b}{a})}} \frac{(\ln z)^2}{z+a+bi}$$

$$= \frac{1}{2b} \left(\pi - \tan^{-1} \frac{b}{a} \right) \ln(a^2+b^2) + i \frac{1}{2b} \left[\left(\pi - \tan^{-1} \frac{b}{a} \right)^2 - \frac{1}{4} \ln^2(a^2+b^2) \right],$$

$$\operatorname{res} f(-a-bi) = \lim_{z \rightarrow \sqrt{a^2+b^2}e^{i(\pi+\tan^{-1}\frac{b}{a})}} \frac{(\ln z)^2}{z+a-bi}$$

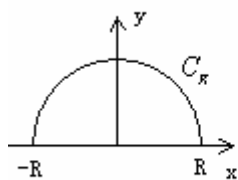
$$= -\frac{1}{2b} \left(\pi + \tan^{-1} \frac{b}{a} \right) \ln(a^2+b^2) + i \frac{1}{2b} \left[\frac{1}{4} \ln^2(a^2+b^2) - \left(\pi + \tan^{-1} \frac{b}{a} \right)^2 \right].$$

$$\text{原积分} = -\frac{1}{2} \{ \operatorname{Re} [\operatorname{res} f(-a+bi)] + \operatorname{Re} [\operatorname{res} f(-a-bi)] \} = \frac{1}{2b} \tan^{-1} \frac{b}{a} \ln(a^2+b^2).$$

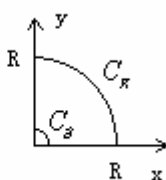
147. 按照指定的积分围道, 考虑适当的复变积分, 计算下列定积分:

$$(1) \int_0^\infty \frac{(1+x^2)\cos ax}{1+x^2+x^4} dx, \int_0^\infty \frac{x \sin ax}{1+x^2+x^4} dx, a > 0; (2) \int_0^\infty \frac{\cos x - e^x}{x} dx; (3) \int_{-\infty}^\infty \frac{e^{ax}}{1+e^x} dx,$$

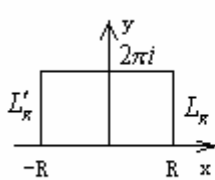
$$0 < a < 1; (4) \int_0^1 \frac{\sqrt[4]{x(1-x)^3}}{(1+x)^3} dx.$$



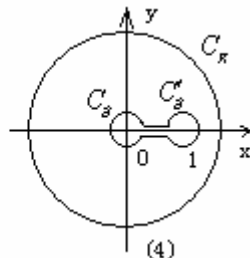
(1)



(2)



(3)



(4)

$$(1) \text{ 令 } f(z) = \frac{(1+z^2)e^{iaz}}{1+z^2+z^4}, \text{ 则}$$

$$\operatorname{res} f(e^{i\pi/3}) = \lim_{z \rightarrow e^{i\pi/3}} \frac{(1+z^2)e^{iaz}}{(z-e^{-i\pi/3})(z^2+z+1)} = -\frac{1}{2\sqrt{3}} e^{-\frac{\sqrt{3}}{2}a} \left(-\sin \frac{a}{2} + i \cos \frac{a}{2} \right),$$

$$\operatorname{res} f(e^{i2\pi/3}) = \lim_{z \rightarrow e^{i2\pi/3}} \frac{(1+z^2)e^{iaz}}{(z^2-z+1)(z-e^{-i2\pi/3})} = -\frac{1}{2\sqrt{3}} e^{-\frac{\sqrt{3}}{2}a} \left(\sin \frac{a}{2} + i \cos \frac{a}{2} \right),$$

对围道积分取 $R \rightarrow \infty$ 得

$$\begin{aligned} \int_0^\infty \frac{(1+x^2)\cos ax}{1+x^2+x^4} dx &= \frac{1}{2} \operatorname{Re} \left\{ 2\pi i \left[\operatorname{res} f(e^{i\pi/3}) + \operatorname{res} f(e^{i2\pi/3}) \right] \right\} \\ &= -\pi \operatorname{Im} \left[\operatorname{res} f(e^{i\pi/3}) + \operatorname{res} f(e^{i2\pi/3}) \right] = \frac{\pi}{\sqrt{3}} e^{-\frac{\sqrt{3}}{2}a} \cos \frac{a}{2}. \end{aligned}$$

令 $f(z) = \frac{ze^{iaz}}{1+z^2+z^4}$, 则

$$\operatorname{res} f(e^{i\pi/3}) = \lim_{z \rightarrow e^{i\pi/3}} \frac{ze^{iaz}}{(z-e^{-i\pi/3})(z^2+z+1)} = \frac{1}{2\sqrt{3}} e^{-\frac{\sqrt{3}}{2}a} \left(\sin \frac{a}{2} - i \cos \frac{a}{2} \right),$$

$$\operatorname{res} f(e^{i2\pi/3}) = \lim_{z \rightarrow e^{i2\pi/3}} \frac{ze^{iaz}}{(z^2-z+1)(z-e^{-i2\pi/3})} = \frac{1}{2\sqrt{3}} e^{-\frac{\sqrt{3}}{2}a} \left(\sin \frac{a}{2} + i \cos \frac{a}{2} \right)$$

对围道积分取 $R \rightarrow \infty$ 得

$$\begin{aligned} \int_0^\infty \frac{x \sin ax}{1+x^2+x^4} dx &= \frac{1}{2} \operatorname{Im} \left\{ 2\pi i \left[\operatorname{res} f(e^{i\pi/3}) + \operatorname{res} f(e^{i2\pi/3}) \right] \right\} \\ &= \pi \operatorname{Re} \left[\operatorname{res} f(e^{i\pi/3}) + \operatorname{res} f(e^{i2\pi/3}) \right] = \frac{\pi}{\sqrt{3}} e^{-\frac{\sqrt{3}}{2}a} \sin \frac{a}{2}. \end{aligned}$$

(2) 令 $f(z) = \frac{e^{iz}}{z}$, 则 $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$ (由 Jordan 引理的证明过程可看出对于上半

平面张角小于 π 的弧该定理仍成立), $\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -\frac{\pi i}{2} \lim_{z \rightarrow 0} e^{iz} = -\frac{\pi i}{2}$, 围道积分为:

$$\int_\delta^R \frac{e^{ix}}{x} dx + \int_R^\delta \frac{e^{-y}}{y} dy + \left(\int_{C_R} + \int_{C_\delta} \right) f(z) dz = 0, \text{ 令 } \delta \rightarrow 0, R \rightarrow \infty \text{ 得 } \int_0^\infty \frac{e^{ix} - e^{-x}}{x} dx = \frac{\pi i}{2},$$

两边取实部即可得 $\int_0^\infty \frac{\cos x - e^{-x}}{x} dx = 0$ 。

(3) 令 $f(z) = \frac{e^{az}}{1+e^z}$, 则 $\operatorname{res} f(\pi i) = \lim_{z \rightarrow \pi i} \frac{e^{az}}{(1+e^z)'} = -e^{ia\pi}$ 。类似于 141 题第 (6)

小题作法可证 $\lim_{R \rightarrow \infty} \int_{L_R} f(z) dz = 0$ 和 $\lim_{R \rightarrow \infty} \int_{L'_R} f(z) dz = 0$ (也可证 $\int_{-\infty}^\infty \frac{e^{ax}}{1+e^x} dx$ 收敛)。

围道积分为 $\int_{-R}^R \frac{e^{ax}}{1+e^x} dx + e^{2ia\pi} \int_R^{-R} \frac{e^{ax}}{1+e^x} dx + \left(\int_{L_R} + \int_{L'_R} \right) f(z) dz = -2\pi i e^{ia\pi}$ 。

令 $R \rightarrow \infty$ 可得 $\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx = \frac{-2\pi i e^{ia\pi}}{1-e^{2ia\pi}} = \frac{\pi}{\sin a\pi}$ 。

(4) 令 $f(z) = \frac{\sqrt[4]{z(1-z)^3}}{(1+z)^3}$ ，规定割线上岸 $\arg z = 0, \arg(1-z) = 0$ ，则割线上岸的积分

为 $\int_{\delta}^{1-\delta} \frac{\sqrt[4]{x(1-x)^3}}{(1+x)^3} dx$ ，割线下岸积分为 $\int_{1-\delta}^{\delta} \frac{\sqrt[4]{x[(1-x)e^{-2i\pi}]^3}}{(1+x)^3} dx = -i \int_{\delta}^{1-\delta} \frac{\sqrt[4]{x(1-x)^3}}{(1+x)^3} dx$ 。

由于 $\lim_{z \rightarrow \infty} z f(z) = \lim_{z \rightarrow \infty} \frac{z^4 \sqrt[4]{z(1-z)^3}}{(1+z)^3} = 0$ 与幅角无关，所以 $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$ 。

$\lim_{\delta \rightarrow 0} \int_{C_{\delta}'} f(z) dz = -2\pi i \lim_{z \rightarrow 0} \frac{z^4 \sqrt[4]{z(1-z)^3}}{(1+z)^3} = 0$ （与幅角无关），

$\lim_{\delta \rightarrow 0} \int_{C_{\delta}} f(z) dz = -2\pi i \lim_{z \rightarrow 1} \frac{(z-1)^4 \sqrt[4]{z(1-z)^3}}{(1+z)^3} = 0$ （与幅角无关），

$\text{res } f(-1) = \frac{1}{2} \lim_{\substack{z \rightarrow e^{i\pi} \\ 1-z \rightarrow 2}} \frac{d^2}{dz^2} [z^{1/4} (1-z)^{3/4}]$
 $= -\frac{3}{32} \lim_{\substack{z \rightarrow e^{i\pi} \\ 1-z \rightarrow 2}} [z^{-7/4} (1-z)^{3/4} + 2z^{-3/4} (1-z)^{-1/4} + z^{1/4} (1-z)^{-5/4}] = -\frac{3}{64} 2^{-1/4} e^{i\pi/4}$ 。

围道积分为

$(1-i) \int_{\delta}^{1-\delta} \frac{\sqrt[4]{x(1-x)^3}}{(1+x)^3} dx + \left(\int_{C_R} + \int_{C_{\delta}} + \int_{C_{\delta}'} \right) f(z) dz = 2\pi i \text{res } f(-1) = -i \frac{3}{64} 2^{3/4} \pi e^{i\pi/4}$ ，

令 $\delta \rightarrow 0, R \rightarrow \infty$ 得 $\int_0^1 \frac{\sqrt[4]{x(1-x)^3}}{(1+x)^3} dx = -i \frac{3}{64} 2^{3/4} \pi \frac{e^{i\pi/4}}{1-i} = \frac{3}{64} 2^{1/4} \pi$ 。

148. 按照指定的被积函数，选择适当的积分围道，计算下列积分：

(1) $\int_0^{\pi} \frac{\cos n\varphi}{a - ib \cos \varphi} d\varphi$ ， $a > 0, b > 0$ ，被积函数为 $\frac{z^n}{bz^2 + 2iaz + b}$ ；

(2) $\int_0^{\infty} \frac{x^b}{1+x^2} \cos\left(ax - \frac{1}{2}bx\right) dx$ ， $a \geq 0, -1 < b < 1$ ，被积函数为 $\frac{e^{iaz} z^b}{1+z^2}$ ；

$$(3) \int_0^{\infty} \frac{dx}{x[(\ln x)^2 + \pi^2]}, \text{ 被积函数为 } \frac{1}{z \ln z};$$

$$(4) \int_0^{\infty} \frac{x \tan^{-1} x}{(1+2x^2)^2} dx, \text{ 被积函数为 } \frac{z \ln(1-iz)}{(1+2z^2)^2};$$

$$(5) \int_0^{\infty} \frac{\cos(\ln x)}{1+x^2} dx, \text{ 被积函数为 } \frac{z^i}{z^2-1};$$

$$(6) \int_0^{\pi/2} \frac{r \sin 2\theta}{1-2r \cos 2\theta + r^2} \theta d\theta, \text{ 被积函数为 } \frac{2zr}{z^2(1+r)^2 + (1-r)^2} \frac{\ln(1-iz)}{1+z^2}.$$

$$\begin{aligned} (1) \int_{|z|=1} \frac{z^n}{bz^2+2iaz+b} dz &= i \int_0^{2\pi} \frac{e^{in\varphi}}{be^{2i\varphi}+2iae^{i\varphi}+b} e^{i\varphi} d\varphi = i \int_0^{2\pi} \frac{e^{in\varphi} d\varphi}{be^{i\varphi}+2ia+be^{-i\varphi}} \\ &= \frac{1}{2} \int_0^{2\pi} \frac{e^{in\varphi} d\varphi}{a-ib \cos \varphi} = \frac{1}{2} \int_0^{\pi} \frac{e^{in\varphi} d\varphi}{a-ib \cos \varphi} + \frac{1}{2} \int_{\pi}^{2\pi} \frac{e^{in\varphi} d\varphi}{a-ib \cos \varphi} \\ &= \frac{1}{2} \int_0^{\pi} \frac{e^{in\varphi} d\varphi}{a-ib \cos \varphi} + \frac{1}{2} \int_0^{\pi} \frac{e^{-in\theta} d\theta}{a-ib \cos \theta} \quad (\text{作代换 } \theta = 2\pi - \varphi) \\ &= \int_0^{\pi} \frac{\cos n\varphi}{a-ib \cos \varphi} d\varphi \end{aligned}$$

$$\text{记 } f(z) = \frac{z^n}{bz^2+2iaz+b} = \frac{z^n}{b(z-z_1)(z-z_2)}, \text{ 其中 } z_1 = \left(\sqrt{\frac{a^2}{b^2}+1} - \frac{a}{b} \right) i,$$

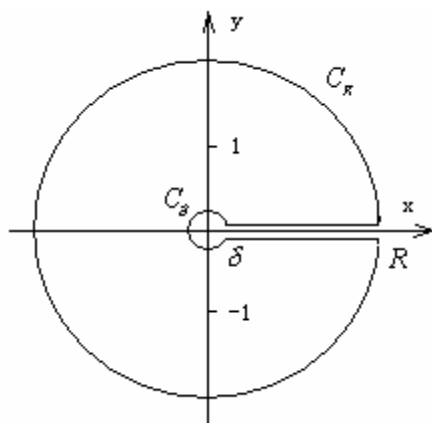
$$z_2 = -\left(\sqrt{\frac{a^2}{b^2}+1} + \frac{a}{b} \right) i, \text{ 显然 } z_1 \text{ 在单位圆内, } z_2 \text{ 在单位圆外.}$$

$$\operatorname{res} f(z_1) = \lim_{z \rightarrow z_1} \frac{z^n}{b(z-z_2)} = \frac{i^n \left(\sqrt{\frac{a^2}{b^2}+1} - \frac{a}{b} \right)^n}{2bi \sqrt{\frac{a^2}{b^2}+1}} = \frac{1}{2i \sqrt{a^2+b^2}} \left(\frac{i}{b} \right)^n \left(\sqrt{a^2+b^2} - a \right)^n,$$

$$\text{原积分} = 2\pi i \operatorname{res} f(z_1) = \frac{\pi}{\sqrt{a^2+b^2}} \left(\frac{i}{b} \right)^n \left(\sqrt{a^2+b^2} - a \right)^n.$$

$$(2) \text{ 令 } f(z) = \frac{e^{iaz} z^b}{1+z^2}, \text{ 规定 } 0 \leq \arg z \leq 2\pi, \text{ 则 } \operatorname{res} f(i) = \lim_{z \rightarrow e^{i\pi/2}} \frac{e^{iaz} z^b}{z+i} = \frac{e^{-a}}{2i} e^{i\frac{b}{2}\pi},$$

$$\operatorname{res} f(-i) = \lim_{z \rightarrow e^{i3\pi/2}} \frac{e^{iaz} z^b}{z-i} = -\frac{e^a}{2i} e^{i\frac{3b}{2}\pi}. \text{ 取如下积分路径:}$$



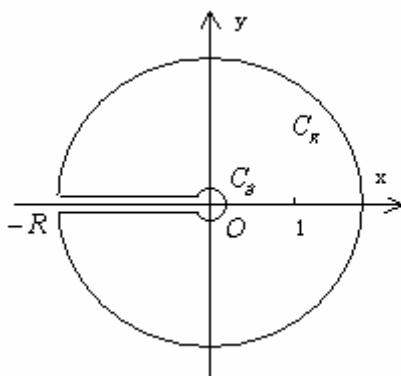
$$\begin{aligned} \text{易得 } \int_0^\infty \frac{x^b e^{iax}}{1+x^2} dx &= \frac{2\pi i}{(1-e^{2ib\pi})} [\operatorname{res} f(i) + \operatorname{res} f(-i)] = -i\pi \frac{\operatorname{sh}\left(a+i\frac{b}{2}\pi\right)}{\sin b\pi} \\ &= \pi \left(\frac{\operatorname{ch} a}{2\cos\frac{b\pi}{2}} - i \frac{\operatorname{sh} a}{2\sin\frac{b\pi}{2}} \right). \end{aligned}$$

$$\text{所以 } \int_0^\infty \frac{x^b \cos ax}{1+x^2} dx = \pi \frac{\operatorname{ch} a}{2\cos\frac{b\pi}{2}}, \quad \int_0^\infty \frac{x^b \sin ax}{1+x^2} dx = -\pi \frac{\operatorname{sh} a}{2\sin\frac{b\pi}{2}}.$$

$$\text{原积分} = \cos\frac{b\pi}{2} \int_0^\infty \frac{x^b \cos ax}{1+x^2} dx + \sin\frac{b\pi}{2} \int_0^\infty \frac{x^b \sin ax}{1+x^2} dx = \frac{\pi}{2} (\operatorname{ch} a - \operatorname{sh} a) = \frac{\pi}{2} e^{-a}.$$

(3) 令 $f(z) = \frac{1}{z \ln z}$, 规定 $-\pi \leq \arg z \leq \pi$, $\operatorname{res} f(1) = \lim_{z \rightarrow e^{i0}} \frac{1}{z(\ln z)'} = 1$. 积分路径如

下:



$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 2\pi i \lim_{z \rightarrow \infty} z f(z) = 2\pi i \lim_{z \rightarrow \infty} \frac{1}{\ln z} = 0,$$

$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -2\pi i \lim_{z \rightarrow 0} z f(z) = -2\pi i \lim_{z \rightarrow 0} \frac{1}{\ln z} = 0,$$

割线上岸有 $z = re^{i\pi}$, 积分为 $\int_R^\delta \frac{1}{re^{i\pi}(\ln r + \pi i)} d(re^{i\pi}) = -\int_\delta^R \frac{1}{x(\ln x + \pi i)} dx$, 同样可得割

线下岸积分为 $\int_\delta^R \frac{1}{x(\ln x - \pi i)} dx$ 。围道积分为:

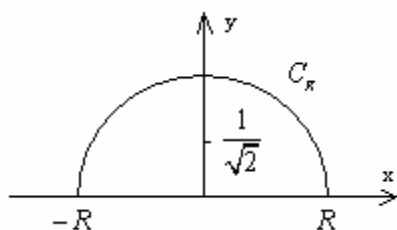
$$\int_\delta^R \frac{1}{x} \left(\frac{1}{\ln x - \pi i} - \frac{1}{\ln x + \pi i} \right) dx + \left(\int_{C_R} + \int_{C_\delta} \right) f(z) dz = 2\pi i。$$

令 $\delta \rightarrow 0, R \rightarrow \infty$ 得 $2\pi i \int_0^\infty \frac{dx}{x[(\ln x)^2 + \pi^2]} = 2\pi i$, 即 $\int_0^\infty \frac{dx}{x[(\ln x)^2 + \pi^2]} = 1$ 。

$$(4) \text{ 令 } f(z) = \frac{z \ln(1-iz)}{4\left(z^2 + \frac{1}{2}\right)^2} = \frac{z \ln(1-iz)}{4\left(z + \frac{1}{\sqrt{2}}i\right)^2 \left(z - \frac{1}{\sqrt{2}}i\right)^2}。$$

$$\begin{aligned} \operatorname{res} f\left(\frac{1}{\sqrt{2}}i\right) &= \lim_{z \rightarrow \frac{1}{\sqrt{2}}i} \frac{d}{dz} \frac{z \ln(1-iz)}{4\left(z + \frac{1}{\sqrt{2}}i\right)^2} = \lim_{z \rightarrow \frac{1}{\sqrt{2}}i} \frac{\left[\ln(1-iz) - \frac{iz}{1-iz}\right]\left(z + \frac{1}{\sqrt{2}}i\right) - 2z \ln(1-iz)}{4\left(z + \frac{1}{\sqrt{2}}i\right)^3} \\ &= -\frac{1}{8}(\sqrt{2}-1)。 \end{aligned}$$

$-i$ 和 ∞ 是 $f(z)$ 的枝点, 可沿虚轴从 $-i$ 向下延伸到 ∞ 作为割线, 取如下积分路径 (在单值分枝内):



$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = \pi i \lim_{z \rightarrow \infty} \frac{z^2 \ln(1-iz)}{(1+2z^2)^2} = 0$ 。围道积分为:

$$\int_{-R}^0 \frac{x \ln(1-ix)}{(1+2x^2)^2} dx + \int_0^R \frac{x \ln(1-ix)}{(1+2x^2)^2} dx + \int_{C_R} f(z) dz = 2\pi i \operatorname{res} f\left(\frac{1}{\sqrt{2}}i\right) = -i \frac{\pi}{4}(\sqrt{2}-1)。$$

$$\text{由于 } \int_{-R}^0 \frac{x \ln(1-ix)}{(1+2x^2)^2} dx = \int_R^0 \frac{y \ln(1+iy)}{(1+2y^2)^2} dy = -\int_0^R \frac{x \ln(1+ix)}{(1+2x^2)^2} dx,$$

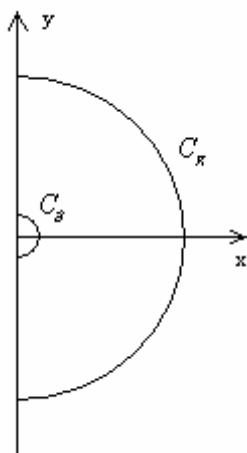
$$\text{所以 } \int_{-R}^0 \frac{x \ln(1-ix)}{(1+2x^2)^2} dx + \int_0^R \frac{x \ln(1-ix)}{(1+2x^2)^2} dx = \int_0^R \frac{x \ln\left(\frac{1-ix}{1+ix}\right)}{(1+2x^2)^2} dx = -2i \int_0^R \frac{x \tan^{-1} x}{(1+2x^2)^2} dx。$$

$$\text{令围道积分 } R \rightarrow \infty \text{ 得 } -2i \int_0^\infty \frac{x \tan^{-1} x}{(1+2x^2)^2} dx = -i \frac{\pi}{4} (\sqrt{2}-1),$$

$$\text{即 } \int_0^\infty \frac{x \tan^{-1} x}{(1+2x^2)^2} dx = \frac{\pi}{8} (\sqrt{2}-1)。$$

(5) 令 $f(z) = \frac{z^i}{z^2-1} = \frac{e^{i \ln z}}{z^2-1}$, 0 和 ∞ 是其枝点, 以负实轴作为割线, 规定

$-\pi < \arg z \leq \pi$, 则 $\text{res } f(1) = \lim_{z \rightarrow e^{i0}} \frac{e^{i \ln z}}{z+1} = \frac{1}{2}$ 。可取如下积分路径:



正虚轴上有 $z = re^{i\pi/2}$, 积分为 $\int_R^\delta \frac{e^{i(\ln r + \frac{\pi}{2}i)}}{-r^2-1} d(re^{i\pi/2}) = ie^{-\pi/2} \int_\delta^R \frac{e^{i \ln x}}{x^2+1} dx,$

负虚轴上 $z = re^{-i\pi/2}$ 积分为 $ie^{\pi/2} \int_\delta^R \frac{e^{i \ln x}}{x^2+1} dx$ 。围道积分为

$$2i \text{ch} \frac{\pi}{2} \int_\delta^R \frac{e^{i \ln x}}{x^2+1} dx + \left(\int_{C_R} + \int_{C_\delta} \right) f(z) dz = 2\pi i \text{res } f(1) = \pi i。$$

由于 $zf(z) = \frac{e^{(1+i) \ln z}}{z^2-1} = \frac{e^{(1+i)(\ln|z|+i \arg z)}}{z^2-1} = \frac{e^{(\ln|z|-\arg z)}}{z^2-1} e^{i(\ln|z|+\arg z)},$ $|z| \rightarrow 0$ 时 $\ln|z| \rightarrow -\infty,$

$e^{(\ln|z|-\arg z)} \rightarrow 0$ (与 $\arg z$ 无关), 所以 $\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = 0。$

$$\lim_{z \rightarrow \infty} zf(z) = \lim_{z \rightarrow \infty} \frac{z^{1+i}}{z^2-1} = \lim_{z \rightarrow \infty} \frac{(1+i)z^i}{2z} = \frac{1+i}{2} \lim_{z \rightarrow \infty} z^{-1+i} = \frac{1+i}{2} \lim_{z \rightarrow \infty} e^{(-1+i)(\ln|z|+i \arg z)}$$

$$= \frac{1+i}{2} \lim_{z \rightarrow \infty} e^{(-\ln|z| - \arg z)} e^{i(\ln|z| - \arg z)} = 0 \quad (\text{与 } \arg z \text{ 无关}),$$

所以 $\lim_{R \rightarrow 0} \int_{C_R} f(z) dz = 0$ 。令围道积分 $\delta \rightarrow 0, R \rightarrow \infty$ 得 $\int_0^\infty \frac{e^{i \ln x}}{x^2 + 1} dx = \frac{\pi}{2 \operatorname{ch} \frac{\pi}{2}}$ 。

(6) 见附录。

149. 变换 $t = \frac{bx+a}{x+1}$, 即 $x = \frac{t-a}{b-t}$, 试利用此类变换证明 $\int_{-1}^1 \left(\frac{1+t}{1-t} \right)^{m-1} g(t) dt = \int_0^\infty x^{m-1} f(x) dx$,

其中 $f(x) = \frac{2}{(x+1)^2} g\left(\frac{x-1}{x+1}\right)$ 。假定有关的积分均存在。

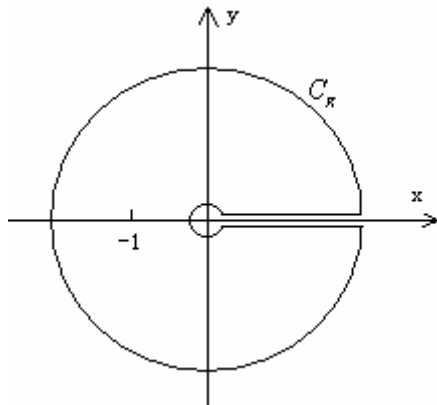
证: 令 $x = \frac{1+t}{1-t}$, 则 $t = \frac{x-1}{x+1}$, $dt = \frac{2}{(x+1)^2} dx$ 。所以

$$\int_{-1}^1 \left(\frac{1+t}{1-t} \right)^{m-1} g(t) dt = \int_0^\infty x^{m-1} g\left(\frac{x-1}{x+1}\right) \frac{2}{(x+1)^2} dx = \int_0^\infty x^{m-1} f(x) dx。$$

150. 利用上题结果, 计算下列积分: (1) $\int_{-1}^1 \left(\frac{1+t}{1-t} \right)^{m-1} dt$, $0 < m < 2$;

$$(2) \int_{-1}^1 \left(\frac{1+t}{1-t} \right)^{m-1} \frac{dt}{t^2 + 1}, \quad 0 < m < 2。$$

(1) 原积分 $= 2 \int_0^\infty \frac{x^{m-1}}{(x+1)^2} dx$ 。令 $f(z) = \frac{z^{m-1}}{(z+1)^2}$, 选取如下积分路径:



可得原积分 $= \frac{2(1-m)\pi}{\sin m\pi}$ 。

(2) 原积分 $= \int_0^\infty \frac{x^{m-1}}{x^2+1} dx = \frac{\pi}{2 \sin \frac{m\pi}{2}}$ (取与上小题相同的积分路径)。

151. 证明: $\int_{-1}^1 (1-t^2)^{m-1} h(t) dt = \int_0^\infty x^{m-1} f(x) dx$, 其中 $f(x) = \frac{1}{2} \left(\frac{2}{x+1} \right)^{2m} h\left(\frac{x-1}{x+1}\right)$ 。

并由此计算积分 $\int_{-1}^1 \frac{\sqrt{1-t^2}}{1+t^2} dt$ 。

令 $x = \frac{1+t}{1-t}$, 则 $t = \frac{x-1}{x+1}$,

$$\int_{-1}^1 (1-t^2)^{m-1} h(t) dt = \int_0^\infty \left[\frac{4x}{(x+1)^2} \right]^{m-1} h\left(\frac{x-1}{x+1}\right) \frac{2}{(x+1)^2} dx = \int_0^\infty x^{m-1} f(x) dx。$$

$$\int_{-1}^1 \frac{\sqrt{1-t^2}}{1+t^2} dt = 2 \int_0^\infty \frac{x^{1/2}}{(x+1)(x^2+1)} dx = (\sqrt{2}-1)\pi \quad (\text{取与上小题相同的积分路径})。$$

152. (1) 证明: $\int_{-1}^1 \ln\left(\frac{1+t}{1-t}\right) g(t) dt = \int_0^\infty f(x) \ln x dx$, 其中 $f(x) = \frac{2}{(x+1)^2} g\left(\frac{x-1}{x+1}\right)$ 。

(2) 计算积分 $\int_{-1}^1 \ln\left(\frac{1+t}{1-t}\right) \frac{dt}{1-ct}$, $|c| < 1$ 。

(1) 令 $x = \frac{1+t}{1-t}$, 则 $t = \frac{x-1}{x+1}$,

$$\int_{-1}^1 \ln\left(\frac{1+t}{1-t}\right) g(t) dt = \int_0^\infty \ln x g\left(\frac{x-1}{x+1}\right) \frac{2}{(x+1)^2} dx = \int_0^\infty f(x) \ln x dx。$$

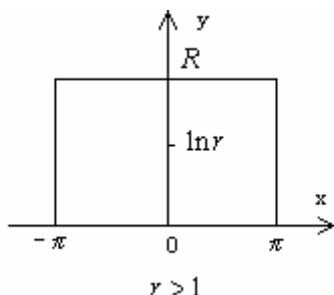
(2) $\int_{-1}^1 \ln\left(\frac{1+t}{1-t}\right) \frac{dt}{1-ct} = \frac{2}{1-c} \int_0^\infty \frac{\ln x}{(x+1)\left(x+\frac{1+c}{1-c}\right)} dx$ 。令 $f(z) = \frac{(\ln z)^2}{(z+1)\left(z+\frac{1+c}{1-c}\right)}$, 取

与上小题相同的积分路径可得原积分 $= \frac{1}{2c} \left(\ln \frac{1+c}{1-c} \right)^2$ 。

附录:

148. (6) 用原题给的被积函数没做出来: (

取被积函数为 $f(z) = \frac{z}{r - e^{-iz}}$, 积分路径如下图:



$r > 1$ 时, $i \ln r$ 是 $f(z)$ 的一阶极点, $\operatorname{res} f(i \ln r) = \lim_{z \rightarrow i \ln r} \frac{z}{(r - e^{-iz})'} = \frac{\ln r}{r}$ 。

$$\begin{aligned} \text{如上图, 底边积分} &= \int_{-\pi}^0 \frac{x}{r - e^{-ix}} dx + \int_0^{\pi} \frac{x}{r - e^{-ix}} dx = \int_{\pi}^0 \frac{x}{r - e^{ix}} dx + \int_0^{\pi} \frac{x}{r - e^{-ix}} dx \\ &= -2i \int_0^{\pi} \frac{x \sin x}{1 - 2r \cos x + r^2} dx。 \end{aligned}$$

$$\text{顶边积分} = \int_{\pi}^{-\pi} \frac{x + iR}{r - e^{-i(x+iR)}} dx = \int_{\pi}^{-\pi} \frac{x}{r - e^R e^{-ix}} dx + \int_{\pi}^{-\pi} \frac{iR}{r - e^R e^{-ix}} dx。$$

$$\text{由于 } \left| \int_{\pi}^{-\pi} \frac{x}{r - e^R e^{-ix}} dx \right| \leq \int_{-\pi}^{\pi} \frac{|x|}{\sqrt{r^2 - 2re^R \cos \theta + e^{2R}}} dx \leq \frac{2}{e^R - r} \int_0^{\pi} x dx = \frac{\pi^2}{e^R - r},$$

$$\left| \int_{\pi}^{-\pi} \frac{iR}{r - e^R e^{-ix}} dx \right| \leq \int_{-\pi}^{\pi} \frac{R}{e^R - r} dx = \frac{2\pi R}{e^R - r}, \text{ 所以顶边积分} \rightarrow 0 \quad (R \rightarrow \infty)。$$

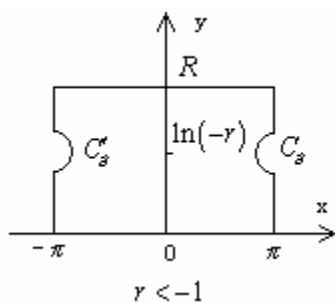
$$\begin{aligned} \text{左边积分} + \text{右边积分} &= i \int_R^0 \frac{-\pi + iy}{r - e^{-i(-\pi+iy)}} dy + i \int_0^R \frac{\pi + iy}{r - e^{-i(\pi+iy)}} dy = 2\pi i \int_0^R \frac{1}{r + e^y} dy \\ &= 2\pi i \int_0^R \frac{e^{-y}}{1 + re^{-y}} dy = -\frac{2\pi i}{r} \ln |1 + re^{-y}|_0^R = \frac{2\pi i}{r} [\ln(1+r) - \ln(1+re^{-R})]。 \end{aligned}$$

当 $R \rightarrow \infty$ 时, 该积分 $\rightarrow \frac{2\pi i}{r} \ln(1+r)$ 。

$$\begin{aligned} \text{所以 } R \rightarrow \infty \text{ 时围道积分} &= -2i \int_0^{\pi} \frac{x \sin x}{1 - 2r \cos x + r^2} dx + \frac{2\pi i}{r} \ln(1+r) \\ &= 2\pi i \operatorname{res} f(i \ln r) = 2\pi i \frac{\ln r}{r}, \end{aligned}$$

$$\text{所以 } \int_0^{\pi} \frac{x \sin x}{1 - 2r \cos x + r^2} dx = \frac{\pi}{r} \ln \left(1 + \frac{1}{r} \right), \text{ 即}$$

$$\int_0^{\pi/2} \frac{r \sin 2\theta}{1 - 2r \cos 2\theta + r^2} \theta d\theta = \frac{r}{4} \int_0^{\pi} \frac{x \sin x}{1 - 2r \cos x + r^2} dx = \frac{\pi}{4} \ln \left(1 + \frac{1}{r} \right)。$$



$r < -1$ 时, $\pm\pi + i\ln(-r)$ 是 $f(z)$ 的一阶极点, 如上图,

$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -\pi i \lim_{z \rightarrow \pi + i\ln(-r)} [z - \pi - i\ln(-r)] f(z) = -\pi i \frac{\ln(-r)}{r} - \frac{\pi^2}{r},$$

$$\lim_{\delta \rightarrow 0} \int_{C'_\delta} f(z) dz = -\pi i \lim_{z \rightarrow -\pi + i\ln(-r)} [z + \pi - i\ln(-r)] f(z) = -\pi i \frac{\ln(-r)}{r} + \frac{\pi^2}{r}.$$

左边积分+右边积分

$$= -\frac{2\pi i}{r} \ln|1 + re^{-y}|_0^R = \frac{2\pi i}{r} [\ln(-1-r) - \ln|1 + re^{-R}|] \rightarrow \frac{2\pi i}{r} \ln(-1-r) \quad (R \rightarrow \infty).$$

$$\text{围道积分} = -2i \int_0^\pi \frac{x \sin x}{1 - 2r \cos x + r^2} dx + \frac{2\pi i}{r} \ln(-1-r) - 2\pi i \frac{\ln(-r)}{r} = 0, \text{ 即}$$

$$\int_0^{\pi/2} \frac{r \sin 2\theta}{1 - 2r \cos 2\theta + r^2} \theta d\theta = \frac{r}{4} \int_0^\pi \frac{x \sin x}{1 - 2r \cos x + r^2} dx = \frac{\pi}{4} \ln\left(1 + \frac{1}{r}\right).$$

$-1 < r < 1$ 时, 极点位于下半平面, 所以围道积分为

$$-2i \int_0^\pi \frac{x \sin x}{1 - 2r \cos x + r^2} dx + \frac{2\pi i}{r} \ln(1+r) = 0, \text{ 即 } \int_0^{\pi/2} \frac{r \sin 2\theta}{1 - 2r \cos 2\theta + r^2} \theta d\theta = \frac{\pi}{4} (1+r).$$

153. 若函数 $f(z)$ 在右半平面 $\operatorname{Re} z > 0$ 内解析, 且满足 $f(z+1) = zf(z)$, $f(1) \neq 0$, 证明 $f(z)$ 能够解析延拓到全平面, $z = 0, -1, -2, \dots$ 除外。

令 $f_1(z) = \frac{1}{z} f(z+1)$, 他在 $\operatorname{Re} z > -1$ 内解析 (除 $z = 0$ 外), 由于在 $\operatorname{Re} z > 0$ 内有 $f_1(z) = f(z)$, 所以 $f_1(z)$ 就是 $f(z)$ 在 $\operatorname{Re} z > -1$ 内的解析延拓。同样的, 令 $f_2(z) = \frac{1}{z(z+1)} f(z+2)$, 它在 $\operatorname{Re} z > -2$ 内解析 (除 $z = 0, -1$ 外), 由于在 $\operatorname{Re} z > -1$ 内有 $f_2(z) = \frac{1}{z} f(z+1) = f_1(z)$, 即它是 $f_1(z)$ 在 $\operatorname{Re} z > -2$ 内的解析延拓。同样的, 可得到 $f(z)$ 在全平面上的解析延拓 ($z = 0, -1, -2, \dots$ 除外)。

154. 证明 $f_1(z) = 1 + az + a^2 z^2 + \dots$ 与 $f_2(z) = \frac{1}{1-z} - \frac{(1-a)z}{(1-z)^2} + \frac{(1-a)^2 z^2}{(1-z)^3} - \dots$ 互为解析延拓。

当 $|az| < 1$, 即 $|z| < \frac{1}{|a|}$ 时, $f_1(z) = \frac{1}{1-az}$ 。

当 $\left| \frac{(1-a)z}{1-z} \right| < 1$ 时, $f_2(z) = \frac{1}{1-z} \frac{1}{1 + \frac{(1-a)z}{1-z}} = \frac{1}{1-az}$ 。

将 $\left| \frac{(1-a)z}{1-z} \right| < 1$ 化成 $(|1-a|^2 - 1)z\bar{z} + z + \bar{z} - 1 < 0$, 这是圆内 (外) 方程 (见习题 02 第 47

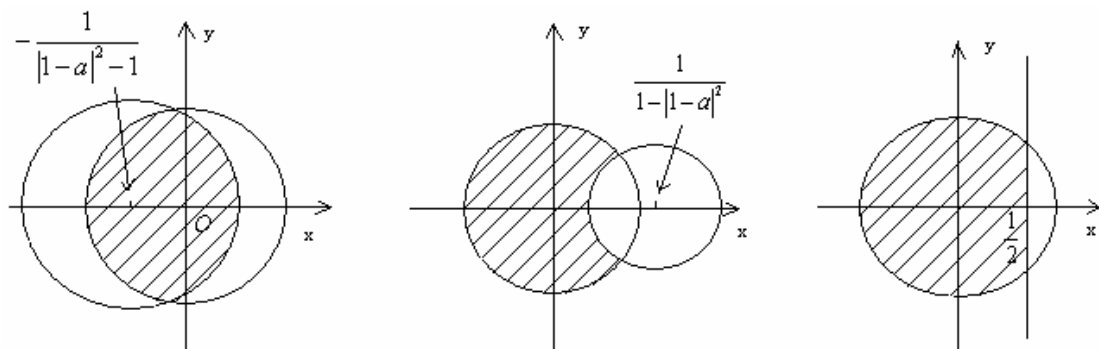
题), 圆心为 $c = -\frac{1}{|1-a|^2 - 1}$, 半径为 $R = \frac{|1-a|}{|1-a|^2 - 1}$ 。

若 $|1-a| > 1$, 这是圆内方程, 圆心在负实轴, 且 $R > |c|$; 若 $|1-a| < 1$, 这是圆外方程, 圆心在正实轴, 且 $R < |c|$ 。

当 $|1-a| = 1$, $(|1-a|^2 - 1)z\bar{z} + z + \bar{z} - 1 < 0$ 化为 $x < \frac{1}{2}$ 。

这三种情况 $(|1-a|^2 - 1)z\bar{z} + z + \bar{z} - 1 < 0$ 都与 $|z| < \frac{1}{|a|}$ 有交集, 如下图所示。

在交集上有 $f_1(z) = f_2(z)$, 所以两者互为解析延拓。



155. 证明级数 $\sum_{n=1}^{\infty} \left(\frac{1}{1-z^{n+1}} - \frac{1}{1-z^n} \right)$ 在区域 $|z| < 1$ 与 $|z| > 1$ 内分别代表两个解析函数，但不互为解析延拓。

$$S_N(z) = \sum_{n=1}^N \left(\frac{1}{1-z^{n+1}} - \frac{1}{1-z^n} \right) = -\frac{1}{1-z} + \frac{1}{1-z^2} - \frac{1}{1-z^2} + \frac{1}{1-z^3} - \cdots - \frac{1}{1-z^N} + \frac{1}{1-z^{N+1}}$$

$$= \frac{1}{1-z^{N+1}} - \frac{1}{1-z}.$$

$|z| < 1$ 时, $S(z) = \lim_{N \rightarrow \infty} S_N(z) = 1 - \frac{1}{1-z} = -\frac{z}{1-z}$, $|z| > 1$ 时, $S(z) = -\frac{1}{1-z}$ 。显然两者不互为解析延拓。

156. 已知: $f(z) = \sum_{n=0}^{\infty} z^{2^n} = z + z^2 + z^4 + z^8 + \cdots$, $|z| < 1$ 。(1) 证明: $z=1$ 是 $f(z)$ 的奇点; (2) 证明: $f(z) = z + f(z^2)$, 因此 $z^2=1$ 的根也都是 $f(z)$ 的奇点; (3) 类似的证明: $z^{2^k}=1$ 的 2^k 个根也是 $f(z)$ 的奇点, k 为任意正整数; (4) 由此证明: 不可能将 $f(z)$ 解析延拓到单位圆外。

(1) $f(1) = \sum_{n=0}^{\infty} 1 \rightarrow \infty$, 即 $z=1$ 是 $f(z)$ 的奇点。

(2) $f(z^2) = \sum_{n=0}^{\infty} z^{2^{n+1}} = \sum_{n=1}^{\infty} z^{2^n} = f(z) - z$ 。

(3) $f(z) = z + f(z^2) = z + z^2 + f(z^4)$

$$= \cdots = z + z^2 + z^4 + \cdots + z^{2^{k-1}} + f(z^{2^k})$$

(4) 对于任意包含单位圆上一段弧的区域, 总存在 N , 使得 $z^{2^N} = 1$ 的某个根落入该区域, 所以不可能将 $f(z)$ 解析延拓到单位圆外。

157. 求下列各积分的一致收敛区域: (1) $\int_0^1 \frac{t^{z-1}}{\sqrt{1-t}} dt$; (2) $\int_0^\infty \frac{\sin t}{t^z} dt$; (3) $\int_0^\infty e^{-zt^2} dt$;

(4) $\int_0^\infty \frac{e^{-zt}}{1+t} dt$ 。

(1) $\operatorname{Re} z \geq \delta > 0$ 时, 在 $t \in (0, 1)$ 上有 $\left| \frac{t^{z-1}}{\sqrt{1-t}} \right| \leq \frac{1}{t^{1-\delta} (1-t)^{1/2}}$, 由于 $1-\delta < 1$, $\frac{1}{2} < 1$, 所

以 $\int_0^1 \frac{dt}{t^{1-\delta} (1-t)^{1/2}}$ 收敛, 所以 $\int_0^1 \frac{t^{z-1}}{\sqrt{1-t}} dt$ 在 $\operatorname{Re} z \geq \delta$ 上一致收敛。

(2) 原积分 $= \int_0^1 \frac{\sin t}{t^z} dt + \int_1^\infty \frac{\sin t}{t^z} dt$ 。

当 $\operatorname{Re} z \leq 2-\delta$ ($\delta > 0$) 时, 在 $t \in (0, 1)$ 上有 $\left| \frac{\sin t}{t^z} \right| \leq \frac{t}{t^{2-\delta}} = \frac{1}{t^{1-\delta}}$ 。由于 $\int_0^1 \frac{1}{t^{1-\delta}} dt$ 收敛,

所以 $\int_0^1 \frac{\sin t}{t^z} dt$ 在 $\operatorname{Re} z \leq 2-\delta$ 上一致收敛。

当 $\operatorname{Re} z \geq \delta > 0$ 时, $\left| \frac{1}{t^z} \right| \leq \frac{1}{t^\delta}$ 一致单调趋于零, $\left| \int_1^b \sin t dt \right| = |\cos 1 - \cos b| \leq 2$, 由狄里克莱

判别法可知 $\int_1^\infty \frac{\sin t}{t^z} dt$ 一致收敛。所以原积分在 $\delta \leq \operatorname{Re} z \leq 2-\delta$ 上一致收敛。

(3) $\operatorname{Re} z \geq \delta > 0$ 时, $|e^{-zt^2}| \leq e^{-\delta t^2}$, 由于 $\int_0^\infty e^{-\delta t^2} dt$ 收敛, 所以原积分一致收敛。

(4) $\operatorname{Re} z \geq \delta > 0$ 时, $|e^{-zt}| \leq e^{-\delta t}$, 所以 $\int_0^\infty e^{-zt} dt$ 一致收敛, 又由于 $\frac{1}{1+t}$ 单调有界, 由 Abel 判别法可知原积分一致收敛。

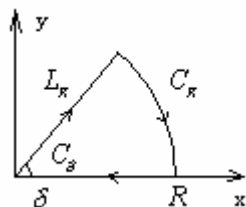
158. 证明: (1) $\Gamma(z) = \int_0^1 \left(\ln \frac{1}{x} \right)^{z-1} dx$, $\operatorname{Re} z > 0$;

(2) $\Gamma(z) = \int_L t^{z-1} e^{-t} dt$, $\operatorname{Re} z > 0$, L 是自原点发出的射线, $0 < |t| < \infty$, $|\arg t| < \frac{\pi}{2}$;

(3) $\Gamma(z) = \sum_{n=0}^\infty \frac{(-1)^n}{n!} \frac{1}{n+z} + \int_1^\infty t^{z-1} e^{-t} dt$, $z \neq 0, -1, -2, \dots$ 。

(1) 作代换 $\ln \frac{1}{x} = t$, 即 $x = e^{-t}$, 则 $\int_0^1 \left(\ln \frac{1}{x} \right)^{z-1} dx = -\int_{\infty}^0 t^{z-1} e^{-t} dt = \Gamma(z)$ 。

(2) 令 $f(t) = t^{z-1} e^{-t}$, 在如下的围道上积分:



$$\left(\int_L + \int_{C_R} + \int_{C_\delta} \right) f(t) dt - \int_\delta^R f(x) dx = 0.$$

由于 $\lim_{t \rightarrow 0} t f(t) = \lim_{t \rightarrow 0} t^z e^{-t} = 0$ ($\operatorname{Re} z > 0$), 所以 $\lim_{\delta \rightarrow 0} \int_{C_\delta} f(t) dt = 0$,

由于 $\lim_{t \rightarrow \infty} t f(t) = \lim_{t \rightarrow \infty} t^z e^{-t} = 0$, 所以 $\lim_{R \rightarrow \infty} \int_{C_R} f(t) dt = 0$ 。

令围道积分 $\delta \rightarrow 0, R \rightarrow \infty$ 既可得 $\int_L e^{-t} t^{z-1} dt = \int_0^\infty e^{-x} x^{z-1} dx = \Gamma(z)$ 。

$$(3) \Gamma(z) = \int_0^1 t^{z-1} e^{-t} dt + \int_1^\infty t^{z-1} e^{-t} dt = \int_0^1 \sum_{n=0}^\infty \frac{(-1)^n t^{n+z-1}}{n!} dt + \int_1^\infty t^{z-1} e^{-t} dt$$

可求得级数 $\sum_{n=0}^\infty \frac{(-1)^n t^{n+z-1}}{n!}$ 的收敛半径为 ∞ , 所以上式中积分与求和可交换顺序, 即

$$\Gamma(z) = \sum_{n=0}^\infty \frac{(-1)^n}{n!} \int_0^1 t^{n+z-1} dt + \int_1^\infty t^{z-1} e^{-t} dt. \operatorname{Re} z > 0 \text{ 时有 } \int_0^1 t^{n+z-1} dt = \frac{1}{n+z}, \text{ 即}$$

$$\Gamma(z) = \sum_{n=0}^\infty \frac{(-1)^n}{n!} \frac{1}{n+z} + \int_1^\infty t^{z-1} e^{-t} dt.$$

任取一有界单连通闭区域 $\bar{G}$ (排除 $z = 0, -1, -2, \dots$), 有

$$\left| \frac{(-1)^n}{n!} \frac{1}{n+z} \right| = \frac{1}{n!} \frac{1}{\sqrt{(n+x)^2 + y^2}} \leq \frac{1}{n! |y|}. \text{ 由于 } \bar{G} \text{ 有界且不含原点, 必存在 } M, \text{ 使在 } \bar{G} \text{ 上}$$

有 $\frac{1}{|y|} \leq M$ 。由于 $\sum \frac{M}{n!}$ 收敛, 所以 $\sum \frac{(-1)^n}{n!} \frac{1}{n+z}$ 在 $\bar{G}$ 上一致收敛, 所以在 G 上解析。

由 $\bar{G}$ 的任意性可知该级数在全平面 (排除 $z = 0, -1, -2, \dots$) 上解析。上式可作为 $\Gamma(z)$ 在全平面上的解析延拓。

159. 将下列连乘积用 Γ 函数表示出来: (1) $(2n)!!$; (2) $(2n-1)!!$;

$$(3) (1+\rho)(2+\rho)\cdots(n+\rho); (4) \left[n(n+1)-\rho(\rho+1) \right] \left[(n-1)n-\rho(\rho+1) \right] \cdots \left[0-\rho(\rho+1) \right].$$

$$(1) (2n)!! = (2n)(2n-2)(2n-4)\cdots 2 = 2^n n(n-1)(n-2)\cdots 1 = 2^n n! = 2^n \Gamma(n+1).$$

$$(2) (2n-1)!! = (2n-1)(2n-3)\cdots 1 = \frac{(2n)(2n-1)(2n-2)(2n-3)\cdots 2 \cdot 1}{(2n)!!}$$

$$= \frac{(2n)!}{(2n)!!} = \frac{\Gamma(2n+1)}{2^n \Gamma(n+1)}.$$

$$(3) \Gamma(\rho+n+1) = (\rho+n)\Gamma(\rho+n) = (\rho+n)(\rho+n-1)\Gamma(\rho+n-1) = \cdots$$

$$= (\rho+n)(\rho+n-1)\cdots(\rho+1)\Gamma(\rho+1)$$

$$\text{所以 } (1+\rho)(2+\rho)\cdots(n+\rho) = \frac{\Gamma(\rho+n+1)}{\Gamma(\rho+1)}.$$

$$(4) n(n+1)-\rho(\rho+1) = -[\rho^2 + \rho - n(n+1)] = (\rho+n+1)(n-\rho),$$

$$\text{原式} = (\rho+n+1)(\rho+n)\cdots(\rho+1) \cdot (n-\rho)(n-1-\rho)\cdots(-\rho) = \frac{\Gamma(\rho+n+2)}{\Gamma(\rho+1)} \frac{\Gamma(n+1-\rho)}{\Gamma(-\rho)}.$$

160. 设 $\psi(z) = \frac{d}{dz} \ln \Gamma(z) = \frac{\Gamma'(z)}{\Gamma(z)}$, 证明: (1) $\psi(z+1) = \frac{1}{z} + \psi(z)$;

$$(2) \psi(z+n) - \psi(z) = \frac{1}{z} + \frac{1}{z+1} + \cdots + \frac{1}{z+n-1}; (3) \psi(1-z) - \psi(z) = \pi \cot(\pi z);$$

$$(4) 2\psi(2z) - \psi(z) - \psi\left(z + \frac{1}{2}\right) = 2\ln 2.$$

(1) 对 $\Gamma(z+1) = z\Gamma(z)$ 两边取对数, 再微商即可证。

(2) 由 $\Gamma(z+n) = (z+n-1)(z+n-2)\cdots(z+1)z\Gamma(z)$ 可证。

(3) 由 $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$ 可得。

(4) 由 $\Gamma(2z) = \frac{1}{\sqrt{\pi}} 2^{2z-1} \Gamma(z) \Gamma\left(z + \frac{1}{2}\right)$ 可得。

161. 证明: $\psi(z)$ 仍以 0 及负整数为其一阶极点, 并求其留数。

由上题第 (2) 小题结论, $\psi(z) = \psi(z+n+1) - \frac{1}{z} - \frac{1}{z+1} - \cdots - \frac{1}{z+n}$ 。 $\psi(z+n+1)$ 在 $z = -n$ 的足够小邻域内是解析的(因为 $\psi(z+n+1) = \frac{d}{dz} \ln \Gamma(z+n+1)$, 而 $\Gamma(z+n+1)$ 是解析的), 所以 $z = -n$ ($n = 0, 1, 2, \cdots$) 为其一阶极点。

$$\begin{aligned} \operatorname{res} \psi(-n) &= \lim_{z \rightarrow -n} (z+n) \psi(z) \\ &= \lim_{z \rightarrow -n} \left[(z+n) \psi(z+n+1) - \frac{(z+n)}{z} - \frac{(z+n)}{z+1} - \cdots - \frac{(z+n)}{z+n-1} - 1 \right] = -1 \end{aligned}$$

162. 定义 $J_\nu(z) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+\nu+1)} \left(\frac{z}{2}\right)^{2k+\nu}$, 试导出 $Y_n(z) = \lim_{\nu \rightarrow n} \frac{J_\nu(z) \cos \nu\pi - J_{-\nu}(z)}{\sin \nu\pi}$,

$n = 0, 1, 2, \cdots$ 的级数表达式。

由于 $z = -n$ ($n = 0, 1, 2, \cdots$) 是 $\Gamma(z)$ 的极点, 所以有 $\frac{1}{\Gamma(-n)} = 0$ 。由此可导出 $J_n(z)$ 的

一个性质: $J_{-n}(z) = (-1)^n J_n(z)$ 。

$$\begin{aligned} J_{-n}(z) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k-n+1)} \left(\frac{z}{2}\right)^{2k-n} = \sum_{k=n}^{\infty} \frac{(-1)^k}{k! \Gamma(k-n+1)} \left(\frac{z}{2}\right)^{2k-n} = \sum_{k=0}^{\infty} \frac{(-1)^{k+n}}{(k+n)! \Gamma(k+1)} \left(\frac{z}{2}\right)^{2k+n} \\ &= (-1)^n \sum_{k=n}^{\infty} \frac{(-1)^k}{k! \Gamma(k+1)} \left(\frac{z}{2}\right)^{2k+n} = (-1)^n J_n(z)。 \end{aligned}$$

当 $\nu \neq n$ 时, $J_\nu(z)$ 是多值函数, 0 和 ∞ 是其枝点, 可把负实轴作为割线, 规定 $|\arg z| < \pi$ 。

用 $u_k(z, \nu)$ 表示 $J_\nu(z)$ 级数表达式的通项, 当 $|z| \leq M, |\nu| \leq L$ (M, L 是任意正数), k 充分

大时, $\left| \frac{u_{k+1}(z, \nu)}{u_k(z, \nu)} \right| = \frac{1}{(k+1)|k+\nu+1|} \left| \frac{z}{2} \right|^2 \leq \frac{1}{(k+1)(k+1-L)} \left(\frac{M}{2}\right)^2 \rightarrow 0$, 所以该级数在

给定范围内对于 z 和 ν 是一致收敛的, 可对 z 和 ν 逐项微分; $J_\nu(z)$ 是 ν 的整函数, 在 z 的单值分枝内是 z 的解析函数。

由于 $z = -n$ 是 $\Gamma(z)$ 和 $\psi(z)$ 的一阶极点, 且 $\operatorname{res} \Gamma(-n) = \frac{(-1)^n}{n!}$, $\operatorname{res} \psi(-n) = -1$,

$$\text{所以可设 } \Gamma(z) = \frac{1}{z+n} \left[\frac{(-1)^n}{n!} + a_1(z+n) + a_2(z+n)^2 + \dots \right],$$

$$\psi(z) = \frac{1}{z+n} \left[-1 + b_1(z+n) + b_2(z+n)^2 + \dots \right], \text{ 由此得 } \frac{\psi(-n)}{\Gamma(-n)} = (-1)^{n+1} n!.$$

$$\frac{\partial J_\nu(z)}{\partial \nu} = - \sum_{k=0}^{\infty} \frac{(-1)^k \Gamma'(k+\nu+1)}{k! \Gamma^2(k+\nu+1)} \left(\frac{z}{2}\right)^{2k+\nu} + \ln\left(\frac{z}{2}\right) \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+\nu+1)} \left(\frac{z}{2}\right)^{2k+\nu}$$

$$= - \sum_{k=0}^{\infty} \frac{(-1)^k \psi(k+\nu+1)}{k! \Gamma(k+\nu+1)} \left(\frac{z}{2}\right)^{2k+\nu} + J_\nu(z) \ln\left(\frac{z}{2}\right)$$

$$\left. \frac{\partial J_\nu(z)}{\partial \nu} \right|_{\nu=n} = - \sum_{k=0}^{\infty} \frac{(-1)^k \psi(k+n+1)}{k! (k+n)!} \left(\frac{z}{2}\right)^{2k+n} + J_n(z) \ln\left(\frac{z}{2}\right)$$

$$\frac{\partial J_{-\nu}(z)}{\partial \nu} = \sum_{k=0}^{\infty} \frac{(-1)^k \psi(k-\nu+1)}{k! \Gamma(k-\nu+1)} \left(\frac{z}{2}\right)^{2k-\nu} - J_{-\nu}(z) \ln\left(\frac{z}{2}\right)$$

$$\left. \frac{\partial J_{-\nu}(z)}{\partial \nu} \right|_{\nu=n} = \sum_{k=0}^{\infty} \frac{(-1)^k \psi(k-n+1)}{k! \Gamma(k-n+1)} \left(\frac{z}{2}\right)^{2k-n} - J_{-n}(z) \ln\left(\frac{z}{2}\right)$$

$$= \sum_{k=0}^{n-1} \frac{(-1)^k \psi(k-n+1)}{k! \Gamma(k-n+1)} \left(\frac{z}{2}\right)^{2k-n} + \sum_{k=n}^{\infty} \frac{(-1)^k \psi(k-n+1)}{k! \Gamma(k-n+1)} \left(\frac{z}{2}\right)^{2k-n} - (-1)^n J_n(z) \ln\left(\frac{z}{2}\right)$$

$$= (-1)^n \left[\sum_{k=0}^{n-1} \frac{(n-k-1)!}{k!} \left(\frac{z}{2}\right)^{2k-n} + \sum_{k=0}^{\infty} \frac{(-1)^k \psi(k+1)}{k! (k+n)!} \left(\frac{z}{2}\right)^{2k+n} - J_n(z) \ln\left(\frac{z}{2}\right) \right]$$

当 $n=0$ 时, 上式右边没有第一项求和。

利用洛比达法则,

$$Y_n(z) = \lim_{\nu \rightarrow n} \frac{\frac{\partial J_\nu(z)}{\partial \nu} \cos \nu \pi - \pi J_\nu(z) \sin \nu \pi - \frac{\partial J_{-\nu}(z)}{\partial \nu}}{\pi \cos \nu \pi} = \frac{1}{\pi} \left[\frac{\partial J_\nu(z)}{\partial \nu} - (-1)^n \frac{\partial J_{-\nu}(z)}{\partial \nu} \right]_{\nu=n}$$

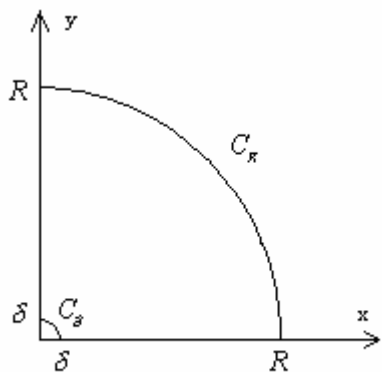
$$= \frac{2}{\pi} J_n(z) \ln\left(\frac{z}{2}\right) - \frac{1}{\pi} \sum_{k=0}^{n-1} \frac{(n-k-1)!}{k!} \left(\frac{z}{2}\right)^{2k-n}$$

$$- \frac{1}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (k+n)!} [\psi(k+n+1) + \psi(k+1)] \left(\frac{z}{2}\right)^{2k+n}$$

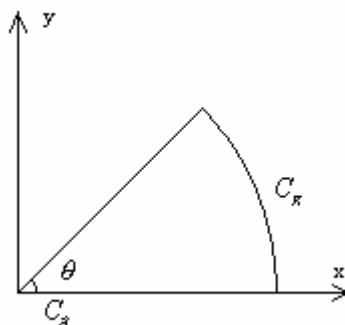
163. 计算下列积分: (1) $\int_0^\infty x^{-\alpha} \sin x dx, \quad 0 < \alpha < 2, \quad \int_0^\infty x^{-\alpha} \cos x dx, \quad 0 < \alpha < 1;$

(2) $\int_0^\infty x^{\alpha-1} e^{-x \cos \theta} \cos(x \sin \theta) dx, \int_0^\infty x^{\alpha-1} e^{-x \cos \theta} \sin(x \sin \theta) dx, \alpha > 0, |\theta| < \frac{\pi}{2}.$

(1) 令 $f(z) = z^{-\alpha} e^{-z}$, 规定 $-\frac{\pi}{2} < \arg z < \frac{3\pi}{2}$, 在下面左图路径上积分:



(1)



(2)

$$\int_\delta^R r^{-\alpha} e^{-r} dr + \int_R^\delta (re^{i\pi/2})^{-\alpha} e^{-ir} d(re^{i\pi/2}) + \left(\int_{C_R} + \int_{C_\delta} \right) f(z) dz = 0$$

当 $0 < \alpha < 1$ 时, $0 < 1 - \alpha < 1$, $\lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} z^{1-\alpha} e^{-z} = 0$, 所以 $\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = 0$ 。

又因为 $\operatorname{Re} z > 0$, 所以 $\lim_{z \rightarrow \infty} z f(z) = \lim_{z \rightarrow \infty} z^{1-\alpha} e^{-z} = 0$, 因此 $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$ 。

令围道积分 $\delta \rightarrow 0, R \rightarrow \infty$ 得 $\int_0^\infty x^{-\alpha} e^{-ix} dx = -ie^{\frac{i\alpha\pi}{2}} \int_0^\infty x^{-\alpha} e^{-x} dx = -ie^{\frac{i\alpha\pi}{2}} \Gamma(1-\alpha)$ 。

取实部和虚部既可得 $0 < \alpha < 1$ 时有 $\int_0^\infty x^{-\alpha} \sin x dx = \Gamma(1-\alpha) \cos \frac{\alpha\pi}{2}$,

$$\int_0^\infty x^{-\alpha} \cos x dx = \Gamma(1-\alpha) \sin \frac{\alpha\pi}{2}.$$

157 题第 (2) 小题已证明 $\int_0^\infty x^{-\alpha} \sin x dx$ 在区间 $\alpha \in [\delta, 2-\delta]$ ($\forall \delta > 0$) 上是一致收敛的,

所以它在 $\alpha \in (\delta, 2-\delta)$ 上是 α 的解析函数, 由 δ 的任意性, 它是 $(0, 2)$ 上的解析函数。由

解析延拓原理, $0 < \alpha < 2$ 时有 $\int_0^\infty x^{-\alpha} \sin x dx = \Gamma(1-\alpha) \cos \frac{\alpha\pi}{2}$ 。

(2) 取 $f(z) = z^{\alpha-1} e^{-z}$ 在上面右图路径上的积分,

$$\int_\delta^R r^{\alpha-1} e^{-r} dr + \int_R^\delta (re^{i\theta})^{\alpha-1} e^{-r(\cos \theta + i \sin \theta)} d(re^{i\theta}) + \left(\int_{C_R} + \int_{C_\delta} \right) f(z) dz = 0$$

由于 $\alpha > 0$, $\lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} z^\alpha e^{-z} = 0$, 所以 $\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = 0$ 。

因为 $\operatorname{Re} z > 0$ ($|\theta| < \frac{\pi}{2}$), 所以 $\lim_{z \rightarrow \infty} z f(z) = \lim_{z \rightarrow \infty} z^\alpha e^{-z} = 0$, 因此 $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$ 。

令围道积分 $\delta \rightarrow 0, R \rightarrow \infty$ 得 $\int_0^\infty x^{\alpha-1} e^{-x \cos \theta} e^{-ix \sin \theta} dx = e^{-i\alpha\theta} \Gamma(\alpha)$

取实部和虚部得 $\int_0^\infty x^{\alpha-1} e^{-x \cos \theta} \cos(x \sin \theta) dx = \Gamma(\alpha) \cos \alpha \theta$,

$$\int_0^\infty x^{\alpha-1} e^{-x \cos \theta} \sin(x \sin \theta) dx = \Gamma(\alpha) \sin \alpha \theta.$$

164. 试用下面的方法导出 $\Gamma(z)$ 的渐近公式: (1) 通过变量代换将 $\Gamma(z+1)$ 的积分表达式

改写成 $\Gamma(z+1) = z^z e^{-z} \int_{-z}^\infty \exp \left[z \ln \left(1 + \frac{s}{z} \right) - s \right] ds$; (2) 将上述积分中被积函数的指数作

展开而只保留最主要的一项, 并将积分下限近似地换成 $-\infty$, 这样就得到 $\Gamma(z+1)$ 在 z 大时

的渐近公式: $\Gamma(z+1) \sim \sqrt{2\pi z} z^z e^{-z}$.

令 $t = s + z$, 则 $\Gamma(z+1) = \int_0^\infty t^z e^{-t} dt = \int_0^\infty \exp(z \ln t - t) dt = \int_{-z}^\infty \exp(z \ln(s+z) - s - z) ds$

$$= e^{z \ln z - z} \int_{-z}^\infty \exp[z \ln(s+z) - z \ln z - s] ds = z^z e^{-z} \int_{-z}^\infty \exp \left[z \ln \left(1 + \frac{s}{z} \right) - s \right] ds.$$

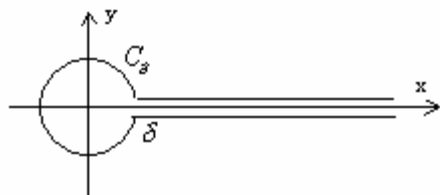
$$\ln \left(1 + \frac{s}{z} \right) = \frac{s}{z} - \frac{s^2}{2z^2} + \frac{s^3}{3z^3} - \dots, \quad z \ln \left(1 + \frac{s}{z} \right) - s = -\frac{s^2}{2z} + \frac{s^3}{3z^2} - \dots.$$

保留第一项 $-\frac{s^2}{2z}$, 并取积分下限为 $-\infty$, 得到 $\Gamma(z+1) \approx z^z e^{-z} \int_{-\infty}^\infty e^{-\frac{s^2}{2z}} ds = \sqrt{2\pi z} z^z e^{-z}$.

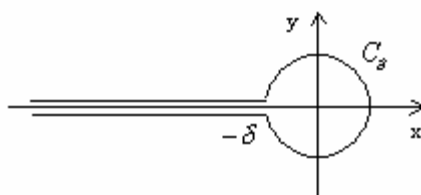
165. 证明 $\Gamma(z)$ 的下列积分表示 (对一切 z 都成立):

(1) $\Gamma(z) = \frac{1}{e^{2i\pi z} - 1} \int_C \zeta^{z-1} e^{-\zeta} d\zeta$, C 如下面左图所示, 规定割线上岸 $\arg \zeta = 0$;

(2) $\frac{1}{\Gamma(z)} = \frac{1}{2\pi i} \int_{C'} \zeta^{-z} e^{\zeta} d\zeta$, C' 如下面右图所示, 规定割线下岸 $\arg \zeta = -\pi$.



(1)



(2)

(1) 记 $f(\zeta) = e^{-\zeta} \zeta^{z-1}$, 围线积分为:

$$\int_C e^{-\zeta} \zeta^{z-1} d\zeta = \int_R^\delta e^{-x} x^{z-1} dx + e^{2i\pi z} \int_\delta^R e^{-x} x^{z-1} dx + \int_{C_\delta} f(\zeta) d\zeta.$$

设 $\operatorname{Re} z > 0$, 则 $\lim_{\zeta \rightarrow 0} \zeta f(\zeta) = \lim_{\zeta \rightarrow 0} e^{-\zeta} \zeta^z = 0$, 所以 $\lim_{\delta \rightarrow 0} \int_{C_\delta} f(\zeta) d\zeta = 0$.

令围道积分 $\delta \rightarrow 0$ 得 $\Gamma(z) = \frac{1}{e^{2i\pi z} - 1} \int_C \zeta^{z-1} e^{-\zeta} d\zeta$. 继续可得

$$\Gamma(z) = \frac{e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} \int_C \zeta^{z-1} e^{-\zeta} d\zeta = \frac{e^{-i\pi z}}{2i \sin \pi z} \int_C \zeta^{z-1} e^{-\zeta} d\zeta. \text{ 由 } \Gamma(z) \Gamma(1-z) = \frac{\pi}{\sin \pi z} \text{ 可得}$$

$$\frac{1}{\Gamma(1-z)} = \frac{e^{-i\pi z}}{2\pi i} \int_C \zeta^{z-1} e^{-\zeta} d\zeta, \text{ 即 } \frac{1}{\Gamma(z)} = -\frac{e^{i\pi z}}{2\pi i} \int_C \zeta^{-z} e^{-\zeta} d\zeta. \text{ 作代换 } \zeta = te^{i\pi}, \text{ 则}$$

$$\frac{1}{\Gamma(z)} = \frac{1}{2\pi i} \int_{C'} t^{-z} e^t dt.$$

166. 从公式 $\Gamma(z) = \lim_{n \rightarrow \infty} \int_0^n \left(1 - \frac{t}{n}\right)^n t^{z-1} dt$ 出发, 证明:

$$(1) \Gamma(z) = \frac{1}{z} \prod_{n=1}^{\infty} \left[\left(1 + \frac{z}{n}\right)^{-1} \left(1 + \frac{1}{n}\right)^z \right]; (2) \Gamma(z) = \frac{1}{z} e^{-\gamma z} \prod_{n=1}^{\infty} \left[\left(1 + \frac{z}{n}\right)^{-1} e^{\frac{z}{n}} \right],$$

其中 $\gamma = \lim_{n \rightarrow \infty} \left(-\ln n + \sum_{k=1}^n \frac{1}{k} \right)$.

(1) 作代换 $t = nx$, 则 $\int_0^n \left(1 - \frac{t}{n}\right)^n t^{z-1} dt = n^z \int_0^1 (1-x)^n x^{z-1} dx = n^z B(n+1, z)$

$$= n^z \frac{\Gamma(n+1)\Gamma(z)}{\Gamma(z+n+1)} = n^z \frac{1 \cdot 2 \cdots (n-1)n}{z(z+1)(z+2) \cdots (z+n-1)(z+n)}$$

$$= n^z \frac{1}{z(1+z) \left(1 + \frac{z}{2}\right) \cdots \left(1 + \frac{z}{n-1}\right) \left(1 + \frac{z}{n}\right)} = n^z \frac{1}{z \left(1 + \frac{z}{n}\right)} \prod_{k=1}^{n-1} \left(1 + \frac{z}{k}\right)^{-1}$$

$$= \left(\frac{2}{1}\right)^z \left(\frac{3}{2}\right)^z \cdots \left(\frac{n}{n-1}\right)^z \frac{1}{z \left(1 + \frac{z}{n}\right)} \prod_{k=1}^{n-1} \left(1 + \frac{z}{k}\right)^{-1}$$

$$= \frac{1}{z \left(1 + \frac{z}{n}\right)} \prod_{k=1}^{n-1} \left[\left(1 + \frac{z}{k}\right)^{-1} \left(1 + \frac{1}{k}\right)^z \right]$$

令 $n \rightarrow \infty$ 即得 $\Gamma(z) = \frac{1}{z} \prod_{n=1}^{\infty} \left[\left(1 + \frac{z}{n}\right)^{-1} \left(1 + \frac{1}{n}\right)^z \right]$ 。

(2) $n^z = e^{z \ln n} = \exp\left(z \ln n - z \sum_{k=1}^n \frac{1}{k}\right) \exp\left(z \sum_{k=1}^n \frac{1}{k}\right) = \exp\left[z\left(\ln n - \sum_{k=1}^n \frac{1}{k}\right)\right] \prod_{k=1}^n e^{\frac{z}{k}}$,

$$\int_0^n \left(1 - \frac{t}{n}\right)^n t^{z-1} dt = n^z \frac{1}{z} \prod_{k=1}^n \left(1 + \frac{z}{k}\right)^{-1} = \frac{1}{z} \exp\left[z\left(\ln n - \sum_{k=1}^n \frac{1}{k}\right)\right] \prod_{k=1}^n \left(1 + \frac{z}{k}\right)^{-1} e^{\frac{z}{k}}$$

令 $n \rightarrow \infty$ 即得 $\Gamma(z) = \frac{1}{z} e^{-\gamma z} \prod_{n=1}^{\infty} \left[\left(1 + \frac{z}{n}\right)^{-1} e^{\frac{z}{n}} \right]$ 。

167. 利用上题结果, 证明: (1) $\Gamma(1)=1$; (2) $\Gamma(z+1)=z\Gamma(z)$;

(3) $\psi(z) = -\gamma - \frac{1}{z} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+z} \right)$; (4) $\Gamma'(1) = -\gamma$;

(5) $\Gamma'\left(\frac{1}{2}\right) = -(\gamma + 2\ln 2)\sqrt{\pi}$ 。

(1) 令上题第 (1) 小题中 $z=1$ 即可得。

$$\begin{aligned} (2) \quad \Gamma(z+1) &= \frac{1}{z+1} \lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{z+1}{k}\right)^{-1} \left(1 + \frac{1}{k}\right)^{z+1} \\ &= \frac{1}{z+1} \lim_{n \rightarrow \infty} \prod_{k=1}^n \left(\frac{k+1+z}{k}\right)^{-1} \left(\frac{k}{k+1}\right)^{-1} \left(1 + \frac{1}{k}\right)^z = \frac{1}{z+1} \lim_{n \rightarrow \infty} \prod_{k=1}^n \left(\frac{k+1+z}{k+1}\right)^{-1} \left(1 + \frac{1}{k}\right)^z \\ &= \frac{1}{z+1} \lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{z}{k+1}\right)^{-1} \left(1 + \frac{1}{k}\right)^z = \frac{1}{z+1} \lim_{n \rightarrow \infty} (1+z) \left(1 + \frac{z}{n+1}\right)^{-1} \prod_{k=1}^n \left(1 + \frac{z}{k}\right)^{-1} \left(1 + \frac{1}{k}\right)^z \\ &= \prod_{k=1}^{\infty} \left(1 + \frac{z}{k}\right)^{-1} \left(1 + \frac{1}{k}\right)^z = z\Gamma(z)。 \end{aligned}$$

(3) $\Gamma(z) = \frac{1}{z} e^{-\gamma z} \lim_{n \rightarrow \infty} \prod_{k=1}^n \left[\left(1 + \frac{z}{k}\right)^{-1} e^{\frac{z}{k}} \right]$,

$$\ln \Gamma(z) = -\ln z - \gamma z + \lim_{n \rightarrow \infty} \ln \prod_{k=1}^n \left[\left(1 + \frac{z}{k}\right)^{-1} e^{\frac{z}{k}} \right] = -\ln z - \gamma z + \lim_{n \rightarrow \infty} \left[-\sum_{k=1}^n \ln \left(1 + \frac{z}{k}\right) + z \sum_{k=1}^n \frac{1}{k} \right]$$

$$\psi(z) = \frac{d}{dz} \ln \Gamma(z) = -\gamma - \frac{1}{z} + \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+z} \right) = -\gamma - \frac{1}{z} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+z} \right)。$$

$$(4) \quad \psi(1) = -\gamma - 1 + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1} \right) = -\gamma - 1 + \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right) = -\gamma,$$

$$\Gamma'(1) = \psi(1)\Gamma(1) = -\gamma.$$

$$(5) \quad \psi\left(\frac{1}{2}\right) = -\gamma - 2 + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1/2} \right) = -\gamma - 2 \left[1 + \sum_{k=1}^{\infty} \left(-\frac{1}{2k} + \frac{1}{2k+1} \right) \right]$$

$$= -\gamma - 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = -\gamma - 2 \ln 2,$$

$$\Gamma'\left(\frac{1}{2}\right) = \psi\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right) = -(\gamma + 2 \ln 2)\sqrt{\pi}.$$

168. 试证 B 函数的下列性质: (1) $B(p, q+1) = \frac{q}{p+q} B(p, q)$;

(2) $B(p, q) = B(p+1, q) + B(p, q+1)$; (3) $pB(p, q+1) = qB(p+1, q)$;

(4) $B(p, q)B(p+q, r) = B(q, r)B(q+r, p)$;

(5) $B(p, q) = \int_0^1 \frac{t^{q-1}}{(1+t)^{p+q}} dt$, $\operatorname{Re} p > 0$, $\operatorname{Re} q > 0$ 。

(1) $B(p, q+1) = \frac{\Gamma(p)\Gamma(q+1)}{\Gamma(p+q+1)} = \frac{q}{p+q} \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} = \frac{q}{p+q} B(p, q)$ 。由 $B(p, q)$ 关于

p, q 的对称性有 $B(p+1, q) = \frac{p}{p+q} B(p, q)$, 由此可得出 (2) (3) 小题结论。

(4) 左边 = $\frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} \frac{\Gamma(p+q)\Gamma(r)}{\Gamma(p+q+r)} = \frac{\Gamma(p)\Gamma(q)\Gamma(r)}{\Gamma(p+q+r)}$,

右边 = $\frac{\Gamma(q)\Gamma(r)}{\Gamma(q+r)} \frac{\Gamma(q+r)\Gamma(p)}{\Gamma(p+q+r)} = \frac{\Gamma(p)\Gamma(q)\Gamma(r)}{\Gamma(p+q+r)}$ 。

(5) 作变量代换 $x = \frac{t}{1+t}$, 则

$$B(p, q) = \int_0^1 (1-x)^{p-1} x^{q-1} dx = \int_0^1 \left(\frac{1}{1+t} \right)^{p-1} \left(\frac{t}{1+t} \right)^{q-1} \frac{dt}{(1+t)^2} = \int_0^1 \frac{t^{q-1}}{(1+t)^{p+q}} dt。$$

169. 计算下列积分: (1) $\int_{-1}^1 (1-x)^p (1+x)^q dx$, $\operatorname{Re} p > -1$, $\operatorname{Re} q > -1$, $\int_{-1}^1 (1-x^2)^n dx$;

(2) $\int_0^{\pi/2} \tan^\alpha \theta d\theta$, $\int_0^{\pi/2} \cot^\alpha \theta d\theta$, $|\alpha| < 1$ 。

(1) 作代换 $1+x=2u$, 即 $(1-x)=2(1-u)$, 则

$$\int_{-1}^1 (1-x)^p (1+x)^q dx = 2^{p+q+1} \int_0^1 (1-u)^p u^q du = 2^{p+q+1} B(p+1, q+1)。$$

作代换 $x^2=u$, 则 $\int_{-1}^1 (1-x^2)^n dx = 2 \int_0^1 (1-x^2)^n dx = \int_0^1 (1-u)^n u^{-1/2} du = B\left(n+1, \frac{1}{2}\right)。$

(2) 作代换 $\tan^2 \theta = x$, 则

$$\int_0^{\pi/2} \tan^\alpha \theta d\theta = \frac{1}{2} \int_0^\infty x^{\frac{\alpha}{2}} \frac{1}{(1+x)\sqrt{x}} dx = \frac{1}{2} \int_0^\infty \frac{x^{\frac{\alpha-1}{2}}}{(1+x)} dx = \frac{1}{2} \int_0^\infty \frac{x^{\frac{1+\alpha}{2}-1}}{(1+x)^{\frac{1-\alpha}{2}+\frac{1+\alpha}{2}}} dx$$

由上题第(5)小题, 上式 $= \frac{1}{2} B\left(\frac{1+\alpha}{2}, \frac{1-\alpha}{2}\right) = \frac{1}{2} B\left(1-\frac{1-\alpha}{2}, \frac{1-\alpha}{2}\right) = \frac{\pi}{2 \sin \pi \left(\frac{1-\alpha}{2}\right)}$

$$= \frac{\pi}{2 \cos \frac{\alpha\pi}{2}}。$$

作代换 $\varphi = \frac{\pi}{2} - \theta$ 可得 $\int_0^{\pi/2} \tan^\alpha \theta d\theta = \int_0^{\pi/2} \cot^\alpha \varphi d\varphi。$

170. 计算积分 $\iiint_V x^{\alpha-1} y^{\beta-1} z^{\gamma-1} dx dy dz$, 其中积分区域 V 为:

(1) 平面 $x=0$, $y=0$, $z=0$ 以及 $x+y+z=1$ 包围的区域;

(2) 平面 $x=0$, $y=0$, $z=0$ 以及曲面 $\left(\frac{x}{a}\right)^p + \left(\frac{y}{b}\right)^q + \left(\frac{z}{c}\right)^r = 1$ 包围的区域。

(1) 原积分 $= \int_0^1 x^{\alpha-1} dx \int_0^{1-x} y^{\beta-1} dy \int_0^{1-x-y} z^{\gamma-1} dz = \int_0^1 x^{\alpha-1} dx \int_0^{1-x} \frac{1}{\gamma} (1-x-y)^\gamma y^{\beta-1} dy$

$$= \frac{1}{\gamma} \int_0^1 (1-x)^{\beta+\gamma} x^{\alpha-1} dx \int_0^1 (1-u)^\gamma u^{\beta-1} du \quad (\text{作代换 } \frac{y}{1-x} = u)$$

$$= \frac{1}{\gamma} B(\beta+\gamma+1, \alpha) B(\gamma+1, \beta) = \frac{1}{\gamma} \frac{\Gamma(\beta+\gamma+1) \Gamma(\alpha)}{\Gamma(\alpha+\beta+\gamma+1)} \frac{\Gamma(\gamma+1) \Gamma(\beta)}{\Gamma(\beta+\gamma+1)}$$

$$= \frac{\Gamma(\alpha)\Gamma(\beta)\Gamma(\gamma)}{\Gamma(\alpha+\beta+\gamma+1)}.$$

$$(2) \text{ 作代换 } \begin{cases} x = a(R \sin \theta \cos \varphi)^{\frac{2}{p}} \\ y = b(R \sin \theta \sin \varphi)^{\frac{2}{q}} \\ z = c(R \cos \theta)^{\frac{2}{r}} \end{cases}, \text{ 则 } J = \frac{\partial(x, y, z)}{\partial(R, \theta, \varphi)} = \begin{vmatrix} \frac{\partial x}{\partial R} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial R} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \varphi} \\ \frac{\partial z}{\partial R} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \varphi} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{2a}{p} R^{\frac{2}{p}-1} \sin^{\frac{2}{p}} \theta \cos^{\frac{2}{p}} \varphi & \frac{2a}{p} R^{\frac{2}{p}} \sin^{\frac{2}{p}-1} \theta \cos \theta \cos^{\frac{2}{p}} \varphi & -\frac{2a}{p} R^{\frac{2}{p}} \sin^{\frac{2}{p}} \theta \cos^{\frac{2}{p}-1} \varphi \sin \varphi \\ \frac{2b}{q} R^{\frac{2}{q}-1} \sin^{\frac{2}{q}} \theta \sin^{\frac{2}{q}} \varphi & \frac{2b}{q} R^{\frac{2}{q}} \sin^{\frac{2}{q}-1} \theta \cos \theta \sin^{\frac{2}{q}} \varphi & \frac{2b}{q} R^{\frac{2}{q}} \sin^{\frac{2}{q}} \theta \sin^{\frac{2}{q}-1} \varphi \cos \varphi \\ \frac{2c}{r} R^{\frac{2}{r}-1} \cos^{\frac{2}{r}} \theta & -\frac{2c}{r} R^{\frac{2}{r}} \cos^{\frac{2}{r}-1} \theta \sin \theta & 0 \end{vmatrix}$$

$$= \frac{8abc}{pqr} R^{\frac{2}{p}+\frac{2}{q}+\frac{2}{r}-1} \sin^{\frac{2}{p}+\frac{2}{q}-1} \theta \cos^{\frac{2}{r}-1} \theta \sin^{\frac{2}{q}} \varphi \cos^{\frac{2}{p}-1} \varphi \begin{vmatrix} \sin \theta \cos \varphi & \cos \theta \cos \varphi & -\sin \varphi \\ \sin \theta \sin \varphi & \cos \theta \sin \varphi & \cos \varphi \\ \cos \theta & -\sin \theta & 0 \end{vmatrix}$$

$$= \frac{8abc}{pqr} R^{\frac{2}{p}+\frac{2}{q}+\frac{2}{r}-1} \sin^{\frac{2}{p}+\frac{2}{q}-1} \theta \cos^{\frac{2}{r}-1} \theta \sin^{\frac{2}{q}} \varphi \cos^{\frac{2}{p}-1} \varphi$$

原积分

$$= \iiint_V a^{\alpha-1} (R \sin \theta \cos \varphi)^{\frac{2}{p}(\alpha-1)} b^{\beta-1} (R \sin \theta \sin \varphi)^{\frac{2}{q}(\beta-1)} c^{\gamma-1} (R \cos \theta)^{\frac{2}{r}(\gamma-1)} |J| dR d\theta d\varphi$$

$$= \frac{8a^{\alpha}b^{\beta}c^{\gamma}}{pqr} \int_0^{\frac{\pi}{2}} \sin^{\frac{2\beta}{q}-1} \varphi \cos^{\frac{2\alpha}{p}-1} \varphi d\varphi \int_0^{\frac{\pi}{2}} \sin^{\frac{2\alpha}{p}+\frac{2\beta}{q}-1} \theta \cos^{\frac{2\gamma}{r}-1} \theta d\theta \int_0^1 R^{\frac{2\alpha}{p}+\frac{2\beta}{q}+\frac{2\gamma}{r}-1} dR$$

$$= \frac{a^{\alpha}b^{\beta}c^{\gamma}}{pqr} \frac{1}{\frac{\alpha}{p}+\frac{\beta}{q}+\frac{\gamma}{r}} B\left(\frac{\alpha}{p}, \frac{\beta}{q}\right) B\left(\frac{\alpha}{p}+\frac{\beta}{q}, \frac{\gamma}{r}\right) = \frac{a^{\alpha}b^{\beta}c^{\gamma}}{pqr} \frac{\Gamma\left(\frac{\alpha}{p}\right)\Gamma\left(\frac{\beta}{q}\right)\Gamma\left(\frac{\gamma}{r}\right)}{\left(\frac{\alpha}{p}+\frac{\beta}{q}+\frac{\gamma}{r}\right)\Gamma\left(\frac{\alpha}{p}+\frac{\beta}{q}+\frac{\gamma}{r}\right)}$$

$$= \frac{a^{\alpha}b^{\beta}c^{\gamma}}{pqr} \frac{\Gamma\left(\frac{\alpha}{p}\right)\Gamma\left(\frac{\beta}{q}\right)\Gamma\left(\frac{\gamma}{r}\right)}{\Gamma\left(\frac{\alpha}{p}+\frac{\beta}{q}+\frac{\gamma}{r}+1\right)}.$$

171. 若 $\alpha > 0$, $0 < r < 1$, 证明: $\frac{1}{2\pi} \int_0^{2\pi} (1+re^{i\theta})^\alpha (1+re^{-i\theta})^\alpha d\theta = \sum_{k=0}^{\infty} \binom{\alpha}{k}^2 r^{2k}$,

$\frac{1}{2\pi} \int_0^{2\pi} (1+re^{i\theta})^\alpha (1-re^{-i\theta})^\alpha d\theta = \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k}^2 r^{2k}$, 因此根据 Abel 第二定理, 有

$$\sum_{k=0}^{\infty} \binom{\alpha}{k}^2 = \frac{2^\alpha}{\pi} \int_0^\pi (1+\cos\theta)^\alpha d\theta = \frac{\Gamma(2\alpha+1)}{\Gamma^2(\alpha+1)},$$

$$\sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k}^2 = \frac{2^\alpha}{\pi} \cos \frac{\alpha\pi}{2} \int_0^\pi \sin^\alpha \theta d\theta = \frac{\Gamma(\alpha+1)}{\Gamma^2(\alpha/2+1)} \cos \frac{\alpha\pi}{2},$$

其中 $\binom{\alpha}{k} = \frac{\Gamma(\alpha+1)}{k!\Gamma(\alpha-k+1)}$ 是普遍的二项式系数。

$$\text{证: } (1+re^{i\theta})^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} r^k e^{ik\theta}, \quad (1+re^{-i\theta})^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} r^k e^{-ik\theta}.$$

$(1+re^{i\theta})^\alpha (1+re^{-i\theta})^\alpha = \sum_{k=0}^{\infty} \sum_{n=0}^k \binom{\alpha}{n} \binom{\alpha}{k-n} r^k e^{i(2n-k)\theta}$ 。在收敛域内可逐项积分:

$$\frac{1}{2\pi} \int_0^{2\pi} (1+re^{i\theta})^\alpha (1+re^{-i\theta})^\alpha d\theta = \sum_{k=0}^{\infty} \sum_{n=0}^k \binom{\alpha}{n} \binom{\alpha}{k-n} r^k \frac{1}{2\pi} \int_0^{2\pi} e^{i(2n-k)\theta} d\theta.$$

由于 $\frac{1}{2\pi} \int_0^{2\pi} e^{i(2n-k)\theta} d\theta = \begin{cases} 1, 2n=k \\ 0, 2n \neq k \end{cases}$, 所以上式右边 k 只取偶数, 令 $k=2m$, 则 n 只取 m ,

$$\frac{1}{2\pi} \int_0^{2\pi} (1+re^{i\theta})^\alpha (1+re^{-i\theta})^\alpha d\theta = \sum_{m=0}^{\infty} \binom{\alpha}{m}^2 r^{2m}.$$

同样可得 $\frac{1}{2\pi} \int_0^{2\pi} (1+re^{i\theta})^\alpha (1-re^{-i\theta})^\alpha d\theta = \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k}^2 r^{2k}$ 。

$$\lim_{k \rightarrow \infty} k \left[\binom{\alpha}{k}^2 / \left(\binom{\alpha}{k+1}^2 - 1 \right) \right] = \lim_{k \rightarrow \infty} k \left[\left(\frac{k+1}{k-\alpha} \right)^2 - 1 \right] = \lim_{k \rightarrow \infty} \frac{2(1+\alpha)k^2 + (1-\alpha^2)k}{(k-\alpha)^2} = 2(1+\alpha) > 1,$$

由极限形式的 Raabe 判别法可知 $\sum_{k=0}^{\infty} \binom{\alpha}{k}^2$ 和 $\sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k}^2$ 收敛。由 Abel 第二定理得

$$\sum_{k=0}^{\infty} \binom{\alpha}{k}^2 = \frac{2^\alpha}{\pi} \int_0^\pi (1+\cos\theta)^\alpha d\theta = \frac{2^{2\alpha}}{\pi} \int_0^\pi \cos^{2\alpha} \frac{\theta}{2} d\theta = \frac{2^{2\alpha}}{\pi} \int_0^\pi 2 \cos^{2\alpha} \varphi d\varphi = \frac{2^{2\alpha}}{\pi} B\left(\alpha + \frac{1}{2}, \frac{1}{2}\right)$$

$$= \frac{2^{2\alpha}}{\sqrt{\pi}} \frac{\Gamma\left(\alpha + \frac{1}{2}\right)}{\Gamma(\alpha + 1)} = \frac{\frac{1}{\sqrt{\pi}} 2^{2\left(\alpha + \frac{1}{2}\right)-1} \Gamma\left(\alpha + \frac{1}{2}\right) \Gamma\left(\alpha + \frac{1}{2} + \frac{1}{2}\right)}{\Gamma^2(\alpha + 1)} = \frac{\Gamma(2\alpha + 1)}{\Gamma^2(\alpha + 1)}。$$

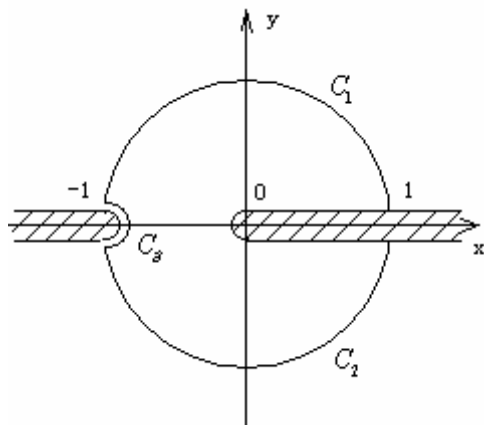
$\sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k}^2 = \frac{1}{2\pi} \int_0^{2\pi} (1+e^{i\theta})^\alpha (1-e^{-i\theta})^\alpha d\theta$ ，可以看到，被积函数是多值的，为确定其

单值分枝，应将其表示为复变积分进行讨论。令 $e^{i\theta} = z$ ，则

$$\frac{1}{2\pi} \int_0^{2\pi} (1+e^{i\theta})^\alpha (1-e^{-i\theta})^\alpha d\theta = \frac{1}{2\pi i} \int_C \frac{(z+1)^\alpha (z-1)^\alpha}{z^{\alpha+1}} dz，$$

可看出， $\pm 1, 0, \infty$ 是被积函数的枝点，上式中 C 是向右绕开-1，在 1 点被截断了的单位圆，

规定 $z=1$ 处的割线上岸 $\arg z = 0$ ， $\arg(z+1) = 0$ ， $\arg(z-1) = \frac{\pi}{2}$ （如下图）。



由于 $\lim_{z \rightarrow -1} (z+1) \frac{(z+1)^\alpha (z-1)^\alpha}{z^{\alpha+1}} = 0$ ，所以 $\lim_{\delta \rightarrow 0} \int_{C_\delta} \frac{(z+1)^\alpha (z-1)^\alpha}{z^{\alpha+1}} dz = 0$ ，即可以不考虑

这段积分。 z 在 C_1 上变化时， $z-1 = \cos \theta - 1 + i \sin \theta = 2 \sin \frac{\theta}{2} e^{i\left(\frac{\pi}{2} + \frac{\theta}{2}\right)}$ ，

$z+1 = 1 + \cos \theta + i \sin \theta = 2 \cos \frac{\theta}{2} e^{i\frac{\theta}{2}}$ ，所以

$$\begin{aligned} \frac{1}{2\pi i} \int_{C_1} \frac{(z+1)^\alpha (z-1)^\alpha}{z^{\alpha+1}} dz &= \frac{1}{2\pi i} \int_0^\pi \frac{\left(2 \cos \frac{\theta}{2} e^{i\frac{\theta}{2}}\right)^\alpha \left[2 \sin \frac{\theta}{2} e^{i\left(\frac{\pi}{2} + \frac{\theta}{2}\right)}\right]^\alpha}{e^{i\theta(\alpha+1)}} d(e^{i\theta}) \\ &= \frac{2^\alpha e^{i\frac{\alpha\pi}{2}}}{2\pi} \int_0^\pi \sin^\alpha \theta d\theta。 \end{aligned}$$

当 z 在 C_2 上变化时, $z-1 = \cos \theta - 1 + i \sin \theta = 2 \sin \frac{\theta}{2} e^{i\left(\frac{\pi}{2} + \frac{\theta}{2}\right)}$,

注意到 $\arg(z+1)$ 在 C_δ 上由 $\pi/2$ 减小到 $-\pi/2$, 在 C_2 上由 $-\pi/2$ 增加到 0 , 所以

$$z+1 = 1 + \cos \theta + i \sin \theta = -2 \cos \frac{\theta}{2} e^{i\left(\frac{\theta}{2} - \pi\right)}, \text{ 因此}$$

$$\begin{aligned} \frac{1}{2\pi i} \int_{C_2} \frac{(z+1)^\alpha (z-1)^\alpha}{z^{\alpha+1}} dz &= \frac{1}{2\pi i} \int_\pi^{2\pi} \frac{\left[-2 \cos \frac{\theta}{2} e^{i\left(\frac{\theta}{2} - \pi\right)}\right]^\alpha \left[2 \sin \frac{\theta}{2} e^{i\left(\frac{\pi}{2} + \frac{\theta}{2}\right)}\right]^\alpha}{e^{i\theta(\alpha+1)}} d(e^{i\theta}) \\ &= \frac{2^\alpha e^{-i\frac{\alpha\pi}{2}}}{2\pi} \int_\pi^{2\pi} (-\sin \theta)^\alpha d\theta = \frac{2^\alpha e^{-i\frac{\alpha\pi}{2}}}{2\pi} \int_0^\pi \sin^\alpha \varphi d\varphi \quad (\theta - \pi = \varphi). \end{aligned}$$

$$\begin{aligned} \text{所以 } \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k}^2 &= \frac{2^\alpha}{2\pi} \left(e^{i\frac{\alpha\pi}{2}} + e^{-i\frac{\alpha\pi}{2}} \right) \int_0^\pi \sin^\alpha \theta d\theta = \frac{2^\alpha}{\pi} \cos \frac{\alpha\pi}{2} \int_0^\pi \sin^\alpha \theta d\theta \\ &= \frac{2^{2\alpha}}{\pi} \cos \frac{\alpha\pi}{2} \int_0^\pi \sin^\alpha \frac{\theta}{2} \cos^\alpha \frac{\theta}{2} d\theta = \frac{2^{2\alpha}}{\pi} \cos \frac{\alpha\pi}{2} \int_0^{\pi/2} 2 \sin^\alpha \varphi \cos^\alpha \varphi d\varphi \\ &= \frac{2^{2\alpha}}{\pi} \cos \frac{\alpha\pi}{2} B\left(\frac{\alpha+1}{2}, \frac{\alpha+1}{2}\right) = \frac{2^{2\alpha}}{\pi} \frac{\Gamma^2\left(\frac{\alpha}{2} + \frac{1}{2}\right)}{\Gamma(\alpha+1)} \cos \frac{\alpha\pi}{2} \\ &= \frac{\left[\frac{1}{\sqrt{\pi}} 2^{2\left(\frac{\alpha}{2} + \frac{1}{2}\right)-1} \Gamma\left(\frac{\alpha}{2} + \frac{1}{2}\right) \Gamma\left(\frac{\alpha}{2} + 1\right)\right]^2}{\Gamma(\alpha+1) \Gamma^2\left(\frac{\alpha}{2} + 1\right)} \cos \frac{\alpha\pi}{2} = \frac{\Gamma^2(\alpha+1)}{\Gamma(\alpha+1) \Gamma^2\left(\frac{\alpha}{2} + 1\right)} \cos \frac{\alpha\pi}{2} \\ &= \frac{\Gamma(\alpha+1)}{\Gamma^2\left(\frac{\alpha}{2} + 1\right)} \cos \frac{\alpha\pi}{2}. \end{aligned}$$

不得不说一句, 这道题..... 太变态了!!!!!!!!!!!!!!!!!!!!

$$172. \text{ 证明: (1) } \frac{\Gamma'(a)}{\Gamma(a)} = \lim_{b \rightarrow 0} [\Gamma(b) - B(a, b)]; \quad (2) \quad \frac{\Gamma'(a)}{\Gamma(a)} + \gamma = \int_0^1 \frac{1-t^{a-1}}{1-t} dt, \text{ 其中}$$

$\gamma = -\Gamma'(1)$ (见第 166 及 167 题)。

(1) 由于 $z=0$ 是 $\Gamma(z)$ 的一阶极点, 留数为 1, 所以有 $\lim_{b \rightarrow 0} b\Gamma(b) = 1$ 。

$$\lim_{b \rightarrow 0} [\Gamma(b) - B(a, b)] = \lim_{b \rightarrow 0} \frac{\Gamma(b) [\Gamma(a+b) - \Gamma(a)]}{\Gamma(a+b)}$$

$$= \lim_{b \rightarrow 0} \left[\frac{1}{\Gamma(a+b)} \cdot b\Gamma(b) \cdot \frac{\Gamma(a+b) - \Gamma(a)}{b} \right] = \frac{\Gamma'(a)}{\Gamma(a)} \circ$$

$$(2) \quad \int_0^1 \frac{1-t^{a-1}}{1-t} dt = \lim_{b \rightarrow 0} \int_0^1 \frac{1-t^{a-1}}{(1-t)^{1-b}} dt = \lim_{b \rightarrow 0} \left[\int_0^1 (1-t)^{b-1} dt - \int_0^1 t^{a-1} (1-t)^{b-1} dt \right]$$

$$= \lim_{b \rightarrow 0} [B(1, b) - B(a, b)],$$

$$\frac{\Gamma'(a)}{\Gamma(a)} - \int_0^1 \frac{1-t^{a-1}}{1-t} dt = \lim_{b \rightarrow 0} \{ [\Gamma(b) - B(a, b)] - [B(1, b) - B(a, b)] \}$$

$$= \lim_{b \rightarrow 0} [\Gamma(b) - B(1, b)] = \frac{\Gamma'(1)}{\Gamma(1)} = -\gamma \circ$$

173. 证明 Laplace 变换的下列性质 (假定有关函数的 Laplace 变换均存在, 其象函数用相应的大写字母表示): (1) $c_1 f_1(t) + c_2 f_2(t) \xrightarrow{LT} c_1 F_1(t) + c_2 F_2(t)$;

$$(2) \int_0^\infty f(t, \tau) d\tau \xrightarrow{LT} \int_0^\infty F(p, \tau) d\tau; (3) f(t - \tau) \xrightarrow{LT} e^{-p\tau} F(p);$$

$$(4) e^{p_0 t} f(t) \xrightarrow{LT} F(p - p_0); (5) f(at) \xrightarrow{LT} \frac{1}{a} F\left(\frac{p}{a}\right), a > 0;$$

$$(6) \int_t^\infty \frac{f(\tau)}{\tau} d\tau \xrightarrow{LT} \frac{1}{p} \int_0^p F(q) dq.$$

$$(1) \int_0^\infty [c_1 f_1(t) + c_2 f_2(t)] e^{-pt} dt = c_1 \int_0^\infty f_1(t) e^{-pt} dt + c_2 \int_0^\infty f_2(t) e^{-pt} dt = c_1 F_1(t) + c_2 F_2(t).$$

$$(2) \int_0^\infty \left[\int_0^\infty f(t, \tau) d\tau \right] e^{-pt} dt = \int_0^\infty \left[\int_0^\infty f(t, \tau) e^{-pt} dt \right] d\tau = \int_0^\infty F(p, \tau) d\tau \quad (\text{假设可交换积分次序}).$$

$$(3) \int_0^\infty f(t - \tau) e^{-pt} dt = \int_\tau^\infty f(t - \tau) e^{-pt} dt = e^{-p\tau} \int_0^\infty f(x) e^{-px} dx = e^{-p\tau} F(p) \quad (\text{这里 } f(t - \tau) = f(t - \tau) \eta(t - \tau)).$$

$$(4) \int_0^\infty [e^{p_0 t} f(t)] e^{-pt} dt = \int_0^\infty f(t) e^{-(p-p_0)t} dt = F(p - p_0).$$

$$(5) \int_0^\infty f(at) e^{-pt} dt = \frac{1}{a} \int_0^\infty f(x) e^{-\frac{p}{a}x} dx = \frac{1}{a} F\left(\frac{p}{a}\right).$$

$$\begin{aligned} (6) \int_0^\infty \left[\int_t^\infty \frac{f(\tau)}{\tau} d\tau \right] e^{-pt} dt &= -\frac{1}{p} \int_0^\infty \left[\int_t^\infty \frac{f(\tau)}{\tau} d\tau \right] \frac{d}{dt}(e^{-pt}) dt \\ &= \frac{1}{p} \int_0^\infty \frac{f(\tau)}{\tau} d\tau - \frac{1}{p} \int_0^\infty \frac{f(t)}{t} e^{-pt} dt = \frac{1}{p} \left[\int_0^\infty F(q) dq - \int_p^\infty F(q) dq \right] \\ &= \frac{1}{p} \int_0^p F(q) dq. \end{aligned}$$

174. 若 $f(t)$ 为周期函数, 周期为 a , 即 $f(t+a) = f(t)$, $t \geq 0$ 。设 $f(t)$ 的拉式变换 $F(p)$

存在, 证明: $F(p) = \frac{1}{1 - e^{-ap}} \int_0^a f(t) e^{-pt} dt$ 。

$$F(p) = \int_0^\infty f(t) e^{-pt} dt = \sum_{n=0}^\infty \int_{na}^{(n+1)a} f(t) e^{-pt} dt = \sum_{n=0}^\infty e^{-nap} \int_0^a f(x) e^{-px} dx,$$

当 $\operatorname{Re} p > 0$ 时, $\sum_{n=0}^{\infty} e^{-nap} = \frac{1}{1-e^{-ap}}$, $F(p) = \frac{1}{1-e^{-ap}} \int_0^a f(t) e^{-pt} dt$ 。

175. 求下列函数的象函数: (1) t^n , $n=0,1,2,\dots$; (2) t^α , $\operatorname{Re} \alpha > -1$; (3) $e^{-\lambda t} \sin \omega t$;

(4) $\frac{1-\cos \omega t}{t^2}$; (5) $\int_t^\infty \frac{\cos \tau}{\tau} d\tau$; (6) $f(t) = \begin{cases} e^t, & 0 < t < 1 \\ 0, & t > 1 \end{cases}$; (7) $|\sin \omega t|$; (8) $t - a \left[\frac{t}{a} \right]$,

$a > 0$ 。

(1) 由 $1 \xrightarrow{LT} \frac{1}{p}$ 和 $(-t)^n f(t) \xrightarrow{LT} F^{(n)}(p)$ 可得 $t^n \xrightarrow{LT} \frac{n!}{p^{n+1}}$ 。

(2) 令 $pt = z$, 则 $\int_0^\infty t^\alpha e^{-pt} dt = \frac{1}{p^{\alpha+1}} \int_L z^\alpha e^{-z} dz$, 其中 L 是从原点出发, 辐角为 $|\arg p|$ 的

射线, 若 $|\arg p| < \frac{\pi}{2}$, 则 $\int_L z^\alpha e^{-z} dz = \Gamma(\alpha+1)$, $t^\alpha \xrightarrow{LT} \frac{\Gamma(\alpha+1)}{p^{\alpha+1}}$ 。

(3) $e^{-\lambda t} \sin \omega t = \frac{1}{2i} [e^{-(\lambda-i\omega)t} - e^{-(\lambda+i\omega)t}] \xrightarrow{LT} \frac{1}{2i} \left(\frac{1}{p+\lambda-i\omega} - \frac{1}{p+\lambda+i\omega} \right) = \frac{\omega}{(p+\lambda)^2 + \omega^2}$ 。

(4) $1 - \cos \omega t \xrightarrow{LT} \frac{1}{p} - \frac{p}{p^2 + \omega^2}$,

$\frac{1 - \cos \omega t}{t} \xrightarrow{LT} \int_p^\infty \left(\frac{1}{q} - \frac{q}{q^2 + \omega^2} \right) dq = \ln \frac{q}{\sqrt{q^2 + \omega^2}} \Big|_p^\infty = -\ln \frac{p}{\sqrt{p^2 + \omega^2}}$,

$\frac{1 - \cos \omega t}{t^2} \xrightarrow{LT} = -\int_p^\infty \ln \frac{q}{\sqrt{q^2 + \omega^2}} dq = -q \ln \frac{q}{\sqrt{q^2 + \omega^2}} \Big|_p^\infty + \int_p^\infty \frac{\omega^2}{q^2 + \omega^2} dq$

$= \frac{p}{2} \ln \frac{p^2}{p^2 + \omega^2} + \omega \arctan \frac{\omega}{p}$ 。

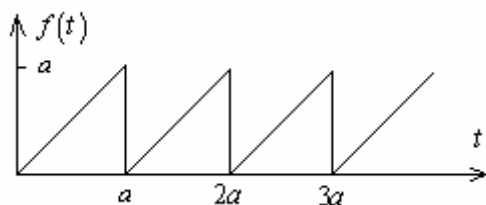
(5) 由 $\cos t \xrightarrow{LT} \frac{p}{p^2 + 1}$ 及上题第 (6) 小题结论, $\int_t^\infty \frac{\cos \tau}{\tau} d\tau \xrightarrow{LT} \frac{1}{p} \int_0^p \frac{q}{q^2 + 1} dq$

$= \frac{1}{2p} \ln(p^2 + 1)$ 。

$$(6) \int_0^{\infty} f(t) e^{-pt} dt = \int_0^1 e^{(1-p)t} dt = \frac{e^{1-p} - 1}{1-p}.$$

$$\begin{aligned} (7) \int_0^{\infty} |\sin \omega t| e^{-pt} dt &= \sum_{n=0}^{\infty} \left[\int_{2n\frac{\pi}{\omega}}^{(2n+1)\frac{\pi}{\omega}} (\sin \omega t) e^{-pt} dt - \int_{(2n+1)\frac{\pi}{\omega}}^{(2n+2)\frac{\pi}{\omega}} (\sin \omega t) e^{-pt} dt \right] \\ &= \int_0^{\pi/\omega} (\sin \omega x) e^{-px} dx \sum_{n=0}^{\infty} \left[e^{-2n\frac{\pi p}{\omega}} + e^{-(2n+1)\frac{\pi p}{\omega}} \right] = \frac{\omega}{p^2 + \omega^2} \frac{\left(1 + e^{-\frac{\pi p}{\omega}}\right)^2}{1 - e^{-\frac{2\pi p}{\omega}}} \quad (\text{设 } \operatorname{Re} p > 0) \\ &= \frac{\omega}{p^2 + \omega^2} \frac{2 \operatorname{ch}^2 \frac{\pi p}{2\omega}}{\operatorname{sh} \frac{\pi p}{\omega}} = \frac{\omega}{p^2 + \omega^2} \frac{2 \operatorname{ch}^2 \frac{\pi p}{2\omega}}{2 \operatorname{sh} \frac{\pi p}{2\omega} \operatorname{ch} \frac{\pi p}{2\omega}} = \frac{\omega}{p^2 + \omega^2} \coth \frac{\pi p}{2\omega}. \end{aligned}$$

(8) 设 $\operatorname{Re} p > 0$,



$$\begin{aligned} \int_0^{\infty} f(t) e^{-pt} dt &= \sum_{n=0}^{\infty} \int_{na}^{(n+1)a} (t-na) e^{-pt} dt = \sum_{n=0}^{\infty} \int_{na}^{(n+1)a} (t-na) e^{-pt} dt \\ &= \int_0^a x e^{-px} dx \sum_{n=0}^{\infty} e^{-nap} = \left[-\frac{1}{p} a e^{-ap} + \frac{1}{p^2} (1 - e^{-ap}) \right] \frac{1}{1 - e^{-ap}} = \frac{1}{p^2} - \frac{a}{p} \frac{e^{-ap}}{1 - e^{-ap}}. \end{aligned}$$

176. 求下列函数的原函数: (1) $\frac{a^3}{p(p+a)^3}$; (2) $\frac{\omega}{p(p^2+\omega^2)}$; (3) $\frac{4p-1}{(p^2+p)(4p^2-1)}$;

(4) $\frac{p^2 - \omega^2}{(p^2 + \omega^2)^2}$; (5) $\frac{e^{-p\tau}}{p^2}$, $\tau > 0$; (6) $\frac{1}{p} \frac{e^{-ap}}{1 - e^{-ap}}$, $a > 0$.

(1) $\frac{a^3}{p(p+a)^3} = \frac{1}{p} - \frac{a^2}{(p+a)^3} - \frac{a}{(p+a)^2} - \frac{1}{p+a}$, 由于 $1 \xrightarrow{LT} \frac{1}{p}$,

$\frac{1}{2} t^2 e^{-at} \xrightarrow{LT} \frac{1}{(p+a)^3}$, $te^{-at} \xrightarrow{LT} \frac{1}{(p+a)^2}$, $e^{-at} \xrightarrow{LT} \frac{1}{p+a}$, 所以原函数

为 $1 - \left(1 + at + \frac{a^2}{2}t^2\right)e^{-at}$, $t > 0$ 。

$$(2) \frac{\omega}{p(p^2 + \omega^2)} = \frac{1}{\omega} \left(\frac{1}{p} - \frac{p}{p^2 + \omega^2} \right), \text{ 由于 } 1 \xrightarrow{LT} \frac{1}{p}, \cos \omega t \xrightarrow{LT} \frac{p}{p^2 + \omega^2}. \text{ 所以}$$

原函数为 $\frac{1}{\omega}(1 - \cos \omega t)$ 。

$$(3) \frac{4p-1}{(p^2+p)(4p^2-1)} = \frac{1}{p} + \frac{5}{3} \frac{1}{p+1} + \frac{1}{3} \frac{1}{p-1/2} - 3 \frac{1}{p+1/2}, \text{ 所以原函数为}$$

$$1 + \frac{5}{3}e^{-t} + \frac{1}{3}e^{t/2} - 3e^{-t/2}.$$

$$(4) \frac{p^2 - \omega^2}{(p^2 + \omega^2)^2} = -\frac{d}{dp} \left(\frac{p}{p^2 + \omega^2} \right), \text{ 由于 } \cos \omega t \xrightarrow{LT} \frac{p}{p^2 + \omega^2},$$

$(-t)f(t) \xrightarrow{LT} \frac{d}{dp}F(p)$, 所以原函数为 $t \cos \omega t$ 。

$$(5) \text{ 由于 } t \xrightarrow{LT} \frac{1}{p^2}, f(t-\tau) \xrightarrow{LT} F(p)e^{-p\tau}, \text{ 所以原函数为 } t-\tau, t > \tau.$$

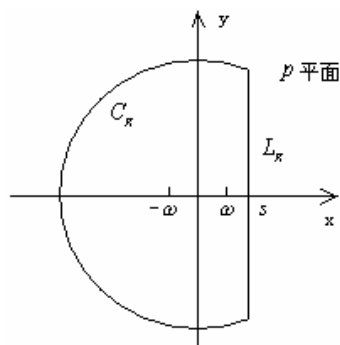
$$(6) \frac{1}{p} \frac{e^{-ap}}{1 - e^{-ap}} = \sum_{n=1}^{\infty} \frac{e^{-nap}}{p} \text{ (设 } \operatorname{Re} p > 0 \text{)}, \text{ 由于 } \eta(t-na) \xrightarrow{LT} \frac{e^{-nap}}{p}, \text{ 所以原函数为}$$

$$\sum_{n=1}^{\infty} \eta(t-na) = \left[\frac{t}{a} \right].$$

177. 用普遍反演公式求下列函数的原函数: (1) $\frac{p}{p^2 - \omega^2}$; (2) $\frac{e^{-p\tau}}{p^4 + 4\omega^4}$;

$$(3) \frac{1}{p} e^{-\alpha\sqrt{p}}, \alpha > 0; (4) \frac{1}{p} \frac{\operatorname{ch}(l-x)\sqrt{p}}{\operatorname{chl}\sqrt{p}}, 0 < x < l.$$

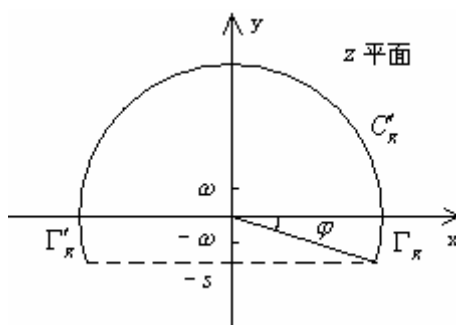
(1) 取如下积分路径:



记 $F(p) = \frac{p}{p^2 - \omega^2}$, 则 $\text{res}[F(p)e^{pt}]_{p=\pm\omega} = \lim_{p \rightarrow \pm\omega} \frac{p}{p \pm \omega} e^{pt} = \frac{1}{2} e^{\pm\omega t}$ 。围道积分为

$$\frac{1}{2\pi i} \left(\int_{L_R} + \int_{C_R} \right) F(p) e^{pt} dp = \text{res}[F(p)e^{pt}]_{p=\omega} + \text{res}[F(p)e^{pt}]_{p=-\omega} = \text{ch } \omega t. (*)$$

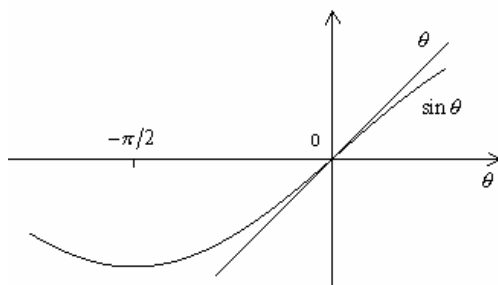
令 $p = iz$, 则 $\int_{C_R} F(p) e^{pt} dp = \int_{C'_R + \Gamma_R + \Gamma'_R} \frac{z}{z^2 + \omega^2} e^{itz} dz$, 其中 $C'_R + \Gamma_R + \Gamma'_R$ 如下图:



可解得 $\varphi = \arctan \frac{s}{\sqrt{R^2 - s^2}}$ 。设 $z \rightarrow \infty$ 时 $h(z)$ 一致 (与辐角无关) 趋于 0 (显然这里的

$\frac{z}{z^2 + \omega^2}$ 满足此条件), 则对任意的 $\varepsilon > 0$, 当 $|z|$ 充分大时有 $|h(z)| < \varepsilon$,

$\int_{\Gamma_R} h(z) e^{itz} dz = iR \int_{-\varphi}^0 h(Re^{i\theta}) e^{-tR \sin \theta} e^{itR \cos \theta} e^{i\theta} d\theta$, $\left| \int_{\Gamma_R} h(z) e^{itz} dz \right| \leq \varepsilon R \int_{-\varphi}^0 e^{-tR \sin \theta} d\theta$ 。当 $\theta \leq 0$ 时有 $\sin \theta \geq \theta$ (如下图), $t > 0$ 时,



$$\left| \int_{\Gamma_R} h(z) e^{itz} dz \right| \leq \varepsilon R \int_{-\varphi}^0 e^{-tR\theta} d\theta = \frac{\varepsilon}{t} (e^{tR\varphi} - 1) \leq \varepsilon R \varphi \leq \varepsilon R \frac{s}{\sqrt{R^2 - s^2}}, \text{ 由于 } \frac{R}{\sqrt{R^2 - s^2}} \rightarrow 1,$$

所以 R 充分大时有 $\frac{R}{\sqrt{R^2 - s^2}} < 1 + \varepsilon$, 所以 $\left| \int_{\Gamma_R} h(z) e^{itz} dz \right| < s\varepsilon(1 + \varepsilon)$, 即

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} h(z) e^{itz} dz = 0, \text{ 同样可得 } \lim_{R \rightarrow \infty} \int_{\Gamma'_R} h(z) e^{itz} dz = 0.$$

由 Jordan 引理可得 $\lim_{R \rightarrow \infty} \int_{C'_R} h(z) e^{itz} dz = 0$, 所以令 (*) 式 $R \rightarrow \infty$ 可得

$$\frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} \frac{p}{p^2 - \omega^2} e^{pt} dp = \text{ch } \omega t, \text{ 即 } \frac{p}{p^2 - \omega^2} \text{ 的原函数为 } \text{ch } \omega t \quad (t > 0).$$

(2) 记 $F(p) = \frac{1}{p^4 + 4\omega^4}$, 则

$$\operatorname{res}[F(p)e^{pt}]_{p=-\sqrt{2}\omega e^{-i\pi/4}} = \lim_{p \rightarrow -\sqrt{2}\omega e^{-i\pi/4}} \frac{e^{pt}}{(p - \sqrt{2}\omega e^{-i\pi/4})(p^2 - 2\omega^2 i)} = \frac{e^{-\omega t} e^{i\omega t}}{16\omega^3 i} (1+i),$$

$$\operatorname{res}[F(p)e^{pt}]_{p=\sqrt{2}\omega e^{-i\pi/4}} = \lim_{p \rightarrow \sqrt{2}\omega e^{-i\pi/4}} \frac{e^{pt}}{(p + \sqrt{2}\omega e^{-i\pi/4})(p^2 - 2\omega^2 i)} = \frac{e^{\omega t} e^{-i\omega t}}{16\omega^3 i} (-1-i),$$

$$\operatorname{res}[F(p)e^{pt}]_{p=-\sqrt{2}\omega e^{i\pi/4}} = \lim_{p \rightarrow -\sqrt{2}\omega e^{i\pi/4}} \frac{e^{pt}}{(p^2 + 2\omega^2 i)(p - \sqrt{2}\omega e^{i\pi/4})} = \frac{e^{-\omega t} e^{-i\omega t}}{16\omega^3 i} (-1+i),$$

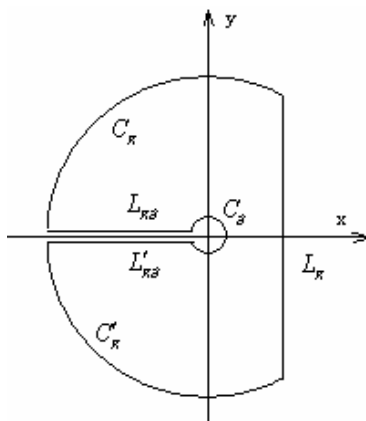
$$\operatorname{res}[F(p)e^{pt}]_{p=\sqrt{2}\omega e^{i\pi/4}} = \lim_{p \rightarrow \sqrt{2}\omega e^{i\pi/4}} \frac{e^{pt}}{(p^2 + 2\omega^2 i)(p + \sqrt{2}\omega e^{i\pi/4})} = \frac{e^{\omega t} e^{i\omega t}}{16\omega^3 i} (1-i).$$

积分路径同上小题 (取 $s > \omega$), 可得

$$\frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} \frac{1}{p^4 + 4\omega^4} e^{pt} dp = \frac{1}{4\omega^3} (\sin \omega t \operatorname{ch} \omega t - \cos \omega t \operatorname{sh} \omega t), \text{ 所以 } \frac{e^{-p\tau}}{p^4 + 4\omega^4} \text{ 的原函数为}$$

$$\frac{1}{4\omega^3} [\sin \omega(t-\tau) \operatorname{ch} \omega(t-\tau) - \cos \omega(t-\tau) \operatorname{sh} \omega(t-\tau)], \quad t > \tau.$$

(3) 记 $F(p) = \frac{1}{p} e^{-\alpha\sqrt{p}}$, 取如下积分路径 (规定 $-\pi \leq \arg p \leq \pi$):



在 C_R 上 $0 < \arg p \leq \pi$, $0 < \arg \sqrt{p} \leq \pi/2$, 所以 $\lim_{p \rightarrow \infty} F(p) = \lim_{p \rightarrow \infty} \frac{1}{p} e^{-\alpha\sqrt{p}} = 0$,

在 C'_R 上 $-\pi \leq \arg p < 0$, $-\pi/2 \leq \arg \sqrt{p} < 0$, 所以 $\lim_{p \rightarrow \infty} F(p) = 0$, 所以

$$\lim_{R \rightarrow \infty} \left(\int_{C_R} + \int_{C'_R} \right) F(p) e^{pt} dp = 0. \text{ 又有 } \lim_{\delta \rightarrow 0} \frac{1}{2\pi i} \int_{C_\delta} F(p) e^{pt} dp = -\lim_{p \rightarrow 0} pF(p) e^{pt} = -1.$$

$$\lim_{\delta \rightarrow 0} \frac{1}{2\pi i} \left(\int_{L_{R\delta}} + \int_{L'_{R\delta}} \right) F(p) e^{pt} dp = \frac{1}{\pi} \int_0^\infty \frac{1}{r} \sin \alpha \sqrt{r} e^{-rt} dr = \frac{2}{\pi} \int_0^\infty \frac{1}{x} \sin \alpha x e^{-x^2 t} dx.$$

$$\text{记 } g(\alpha) = \int_0^\infty \frac{1}{x} \sin \alpha x e^{-x^2 t} dx, \text{ 有 } g'(\alpha) = \int_0^\infty \cos \alpha x e^{-x^2 t} dx = \frac{1}{2} \sqrt{\frac{\pi}{t}} e^{-\frac{\alpha^2}{4t}},$$

由于 $g(0) = 0$, 所以 $g(\alpha) = \frac{1}{2} \sqrt{\frac{\pi}{t}} \int_0^\alpha e^{-\frac{u^2}{4t}} du$ 。令围道积分 $\delta \rightarrow 0, R \rightarrow \infty$ 得

$$\frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} F(p) e^{pt} dp - 1 + \frac{2}{\pi} g(\alpha) = 0。 \text{ 即原函数为}$$

$$1 - \frac{2}{\pi} g(\alpha) = 1 - \frac{2}{\sqrt{\pi}} \int_0^\alpha e^{-x^2} dx = 1 - \operatorname{erf}\left(\frac{\alpha}{\sqrt{t}}\right) = \operatorname{erfc}\left(\frac{\alpha}{\sqrt{t}}\right)。$$

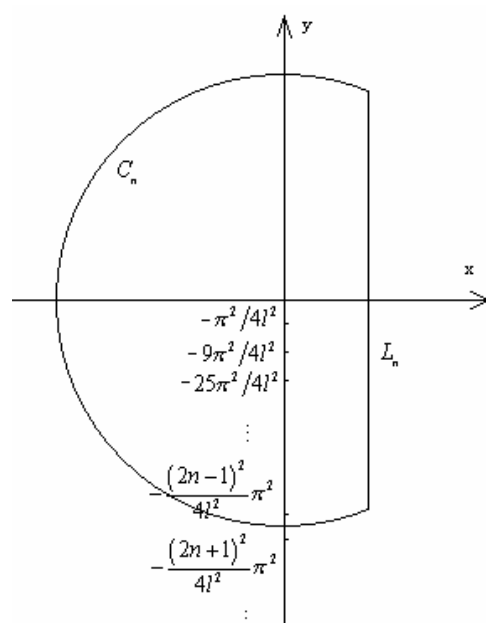
$$(4) \text{ 记 } F(p) = \frac{1}{p} \frac{\operatorname{ch}(l-x)\sqrt{p}}{\operatorname{ch} l \sqrt{p}}, \text{ 它是单值函数 (参考习题 03 第 53 题)}。$$

$$p=0 \text{ 以及 } p = -\frac{(2k+1)^2}{4l^2} \pi^2 \quad (k=0,1,2,\dots) \text{ 是它的一阶极点。}$$

$$\operatorname{res}\left[F(p)e^{pt}\right]_{p=0} = \lim_{p \rightarrow 0} \frac{\operatorname{ch}(l-x)\sqrt{p}}{\operatorname{ch} l \sqrt{p}} e^{pt} = 1,$$

$$\begin{aligned} \operatorname{res}\left[F(p)e^{pt}\right]_{p=-\frac{(2k+1)^2}{4l^2}\pi^2} &= \lim_{p \rightarrow -\frac{(2k+1)^2}{4l^2}\pi^2} \frac{2 \operatorname{ch}(l-x)\sqrt{p}}{l\sqrt{p} \operatorname{sh} l \sqrt{p}} e^{pt} \\ &= -\frac{4}{\pi} \frac{1}{2k+1} \sin \frac{2k+1}{2l} \pi x e^{-\frac{(2k+1)^2}{4l^2}\pi^2 t}。 \end{aligned}$$

取如下积分路径 ($s > 0$), 其中大圆半径为 $n^2 \pi^2 / l^2$ 。



当 $0 < \arg p < \pi$ 时, $0 < \arg \sqrt{p} < \pi/2$, $\operatorname{Re} \sqrt{p} > 0$,

$$\lim_{p \rightarrow \infty} \frac{1}{p} \frac{\operatorname{ch}(l-x)\sqrt{p}}{\operatorname{ch} l\sqrt{p}} = \lim_{p \rightarrow \infty} \frac{1}{p} \frac{e^{(l-x)\sqrt{p}} + e^{-(l-x)\sqrt{p}}}{e^{l\sqrt{p}} + e^{-l\sqrt{p}}} = \lim_{p \rightarrow \infty} \frac{1}{p} \frac{e^{(l-x)\sqrt{p}}}{e^{l\sqrt{p}}} = \lim_{p \rightarrow \infty} \frac{1}{p} e^{-x\sqrt{p}} = 0。$$

当 $\pi < \arg p < 2\pi$ 时, $\pi/2 < \arg \sqrt{p} < \pi$, $\operatorname{Re} \sqrt{p} < 0$,

$$\lim_{p \rightarrow \infty} \frac{1}{p} \frac{\operatorname{ch}(l-x)\sqrt{p}}{\operatorname{ch} l\sqrt{p}} = \lim_{p \rightarrow \infty} \frac{1}{p} \frac{e^{-(l-x)\sqrt{p}}}{e^{-l\sqrt{p}}} = \lim_{p \rightarrow \infty} \frac{1}{p} e^{x\sqrt{p}} = 0。$$

当 $p = -n^2\pi^2/l^2$, 即为圆 C_n 上辐角为 π 的点时, 有

$$\left| \frac{\operatorname{ch}(l-x)\sqrt{p}}{\operatorname{ch} l\sqrt{p}} \right| = \left| \frac{\cos \frac{l-x}{l} n\pi}{\cos n\pi} \right| = \left| \cos \frac{l-x}{l} n\pi \right| \leq 1, \text{ 所以 } \lim_{n \rightarrow \infty} \frac{1}{p} \frac{\operatorname{ch}(l-x)\sqrt{p}}{\operatorname{ch} l\sqrt{p}} = 0。$$

综上, 当圆 C_n 上点 p 无论辐角为何值, 只要模趋于 ∞ , 就有 $F(p) \rightarrow 0$, 即

$$\lim_{n \rightarrow \infty} \int_{C_n} F(p) e^{pt} dp = 0。 \text{ 所以}$$

$$\begin{aligned} \frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} F(p) e^{pt} dp &= \operatorname{res} [F(p) e^{pt}]_{p=0} + \sum_{k=0}^{\infty} \operatorname{res} [F(p) e^{pt}]_{p=-\frac{(2k+1)^2}{4l^2}\pi^2} \\ &= 1 - \frac{4}{\pi} \sum_{k=0}^{\infty} \left[\frac{1}{2k+1} \sin \frac{2k+1}{2l} \pi x e^{-\frac{(2k+1)^2}{4l^2}\pi^2 t} \right] \end{aligned}$$

178. 设 $f(t) \xrightarrow{LT} F(p)$, $f_1(t) \xrightarrow{LT} F_1(p)$, $f_2(t) \xrightarrow{LT} F_2(p)$, 试用 Laplace 反

演公式证明: (1) $f(t-\tau) \xrightarrow{LT} F(p) e^{-p\tau}$;

$$(2) \int_0^t f_1(\tau) f_2(t-\tau) d\tau \xrightarrow{LT} F_1(p) F_2(p)。$$

$$(1) \frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} [F(p) e^{-p\tau}] e^{pt} dp = \frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} F(p) e^{p(t-\tau)} dp = f(t-\tau)。$$

(2) 假设可交换积分次序,

$$\begin{aligned} \frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} [F_1(p) F_2(p)] e^{pt} dp &= \frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} \left[\int_0^\infty f_1(\tau) e^{-p\tau} d\tau \right] F_2(p) e^{pt} dp \\ &= \int_0^\infty f_1(\tau) \left[\frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} F_2(p) e^{p(t-\tau)} dp \right] d\tau = \int_0^\infty f_1(\tau) f_2(t-\tau) d\tau \\ &= \int_0^t f_1(\tau) f_2(t-\tau) d\tau。 \end{aligned}$$

179. 利用 Laplace 变换计算下列积分: (1) $\int_0^\infty \frac{\sin xt}{t} dt$; (2) $\int_0^\infty \frac{\sin xt}{\sqrt{x}} dx$;

(3) $\int_0^\infty \frac{\cos xt}{x^2 + a^2} dx$; (4) $\int_0^\infty \frac{\sin xt}{x(x^2 + 1)} dx$ 。

(1) 由于 $\sin xt \xrightarrow{LT} \frac{x}{p^2 + x^2}$, $\int_0^\infty \frac{f(t)}{t} dt = \int_0^\infty F(p) dp$, 所以

$$\int_0^\infty \frac{\sin xt}{t} dt = \int_0^\infty \frac{x}{p^2 + x^2} dp = \begin{cases} \pi/2, & x > 0 \\ -\pi/2, & x < 0 \end{cases} = \frac{\pi}{2} \operatorname{sgn} x.$$

(2) 记 $f(t, x) = \frac{\sin xt}{\sqrt{x}}$, 则其对 t 的拉式变换为 $F(p, x) = \frac{\sqrt{x}}{p^2 + x^2}$ 。

由 173 题第 (2) 小题结论, $\int_0^\infty \frac{\sin xt}{\sqrt{x}} dx \xrightarrow{LT} \int_0^\infty \frac{\sqrt{z}}{p^2 + z^2} dz$, 注意到 $f(t, x)$ 拉式换式的

收敛域为 $\operatorname{Re} p > 0$, 即 $-\frac{\pi}{2} < \arg p < \frac{\pi}{2}$, 多值函数 $\frac{\sqrt{z}}{p^2 + z^2}$ 在 $0 \leq \arg z \leq 2\pi$ 的单值分枝内的

奇点为 $pe^{i\pi/2}$ 和 $pe^{i3\pi/2}$ 。用多值函数的积分法可求得右边的积分 $= \frac{\pi}{\sqrt{2p}} = \sqrt{\frac{\pi}{2}} \frac{\Gamma(-1/2+1)}{p^{-1/2+1}}$,

即 $\int_0^\infty \frac{\sin xt}{\sqrt{x}} dx \xrightarrow{LT} \sqrt{\frac{\pi}{2}} \frac{\Gamma(-1/2+1)}{p^{-1/2+1}}$ 。由 175 题第 (2) 题结论, $\int_0^\infty \frac{\sin xt}{\sqrt{x}} dx = \sqrt{\frac{\pi}{2t}}$ 。

上面是针对 $t > 0$ 的情况, 当 $t < 0$ 时, $\int_0^\infty \frac{\sin xt}{\sqrt{x}} dx = -\int_0^\infty \frac{\sin x(-t)}{\sqrt{x}} dx = -\sqrt{\frac{\pi}{2(-t)}}$ 。综上,

$$\int_0^\infty \frac{\sin xt}{\sqrt{x}} dx = \sqrt{\frac{\pi}{2|t|}} \operatorname{sgn} t.$$

(3) $t > 0$ 时, $\int_0^\infty \frac{\cos xt}{x^2 + a^2} dx \xrightarrow{LT} p \int_0^\infty \frac{1}{(x^2 + a^2)(x^2 + p^2)} dx$ 。因为 $\frac{\cos xt}{x^2 + a^2}$ 的拉式换式

收敛域为 $\operatorname{Re} p > 0$, 所以 pi 位于上半平面, $-pi$ 位于下半平面, 可求得上式右边积分

$$= \frac{\pi}{2a} \frac{1}{p+a}.$$

所以 $\int_0^\infty \frac{\cos xt}{x^2 + a^2} dx = \frac{\pi}{2a} e^{-at}$ 。

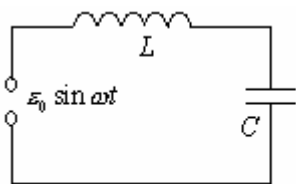
当 $t < 0$ 时, 原积分 $= \int_0^\infty \frac{\cos x(-t)}{x^2 + a^2} dx = \frac{\pi}{2a} e^{at}$ 。综上, $\int_0^\infty \frac{\cos xt}{x^2 + a^2} dx = \frac{\pi}{2a} e^{-a|t|}$ 。

(4) $t > 0$ 时, 原积分 $\xrightarrow{LT} \int_0^\infty \frac{1}{(x^2+1)(x^2+p^2)} dx = \frac{\pi}{2} \left(\frac{1}{p} - \frac{1}{p+1} \right)$, 原积分 $= \frac{\pi}{2} (1 - e^{-t})$ 。

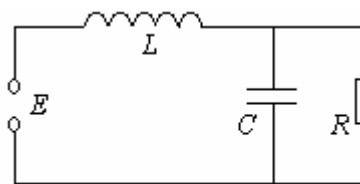
所以原积分 $= \frac{\pi}{2} (1 - e^{-|t|}) \operatorname{sgn} t$ 。

180. 利用 Laplace 变换求解下列微分方程(组)或积分方程: (1) 已知 $i(0)=0$, $q(0)=0$,

求 $i(t)$; (2) 已知 $i(0)=0$, $q(0)=0$, 求 $i(t)$ 。



(1)



(2)

$$(3) \begin{cases} x'' - x + y + z = 0 \\ x + y'' - y + z = 0, \quad x(0)=1, \quad y(0)=z(0)=0, \quad x'(0)=y'(0)=z'(0)=0; \\ x + y + z'' - z = 0 \end{cases}$$

(4) $y(t) = a \sin t - 2 \int_0^t y(\tau) \cos(t-\tau) d\tau$;

(5) $y(t) = a \sin bt + c \int_0^t y(\tau) \sin b(t-\tau) d\tau$, $b > c > 0$;

(6) $f(t) + 2 \int_0^t f(\tau) \cos(t-\tau) d\tau = 9e^{2t}$ 。

(1) $\varepsilon = u_L + u_C$, 代入 $u_C = \frac{q}{C}$, $u_L = L \frac{di}{dt} = L \frac{d^2 q}{dt^2}$ 得 $\frac{d^2}{dt^2} q + \omega_0^2 q = \frac{1}{L} \varepsilon$ (记 $\omega_0^2 = \frac{1}{LC}$),

两边取拉式变换得 $p^2 Q + \omega_0^2 Q = \frac{\varepsilon_0}{L} \frac{\omega}{p^2 + \omega^2}$, 所以 $Q = \frac{\varepsilon_0 \omega}{L} \frac{1}{(p^2 + \omega^2)(p^2 + \omega_0^2)}$ 。

当 $\omega \neq \omega_0$ 时, $Q = \frac{\varepsilon_0 \omega}{L(\omega_0^2 - \omega^2)} \left(\frac{1}{p^2 + \omega^2} - \frac{1}{p^2 + \omega_0^2} \right)$, 所以

$$q(t) = \frac{\varepsilon_0 \omega}{L(\omega_0^2 - \omega^2)} \left(\frac{1}{\omega} \sin \omega t - \frac{1}{\omega_0} \sin \omega_0 t \right)。$$

当 $\omega = \omega_0$ 时, $Q = \frac{\varepsilon_0}{L} \frac{\omega}{(p^2 + \omega^2)^2} = -\frac{\varepsilon_0}{2L} \frac{1}{p} \frac{d}{dp} \left(\frac{\omega}{p^2 + \omega^2} \right)$,

所以 $q(t) = \frac{\varepsilon_0}{2L} \int_0^t \tau \sin \omega \tau d\tau$ 。

$$i(t) = \frac{d}{dt} q(t) = \begin{cases} \frac{\varepsilon_0 \omega}{L(\omega_0^2 - \omega^2)} (\cos \omega t - \cos \omega_0 t), \omega_0 \neq \omega \\ \frac{\varepsilon_0}{2L} t \sin \omega t, \omega_0 = \omega \end{cases}。$$

(2) $C \frac{du_C}{dt} = i - i_R = i - \frac{u_C}{R}$, 两边取拉式变换得 ($u_C(0) = \frac{q_C(0)}{C} = 0$) $U_C = \frac{R}{RCp+1} I$ 。

$$E = u_L + u_C = L \frac{di}{dt} + u_C, \text{ 两边取拉式变换, 代入上式得 } I = \frac{E}{L} \frac{p + \frac{1}{RC}}{p \left(p^2 + \frac{1}{RC} p + \frac{1}{LC} \right)}$$

$$= \frac{E}{L} \frac{p + 2\beta}{p(p^2 + 2\beta p + \gamma^2)}, \text{ 其中 } \beta = \frac{1}{2RC}, \gamma = \frac{1}{\sqrt{LC}}。 \text{ 根据上式分母中一元二次式}$$

$p^2 + 2\beta p + \gamma^2$ 的判别式 $\Delta = 4(\beta^2 - \gamma^2)$ 的不同情况分别讨论:

(i) 当 $\Delta > 0$, 即 $L > 4CR^2$ 时, $I = \frac{E}{L} \frac{p + 2\beta}{p(p + \alpha_1)(p + \alpha_2)}$, 其中 $\alpha_1 = \beta - \sqrt{\beta^2 - \gamma^2}$,

$$\alpha_2 = \beta + \sqrt{\beta^2 - \gamma^2}。 I \text{ 可分解为 } I = \frac{E}{L} \left[\frac{2\beta}{\alpha_1 \alpha_2} \frac{1}{p} + \frac{\alpha_1 - 2\beta}{\alpha_1(\alpha_2 - \alpha_1)} \frac{1}{p + \alpha_1} + \frac{\alpha_2 - 2\beta}{\alpha_2(\alpha_1 - \alpha_2)} \frac{1}{p + \alpha_2} \right]。$$

注意到 $\alpha_1 \alpha_2 = \gamma^2 = \frac{1}{LC}$, 上式中 $\frac{1}{p}$ 的系数 $\frac{2\beta}{\alpha_1 \alpha_2} = \frac{L}{R}$, $\frac{1}{p + \alpha_1}$ 的系数

$$\frac{\alpha_1 - 2\beta}{\alpha_1(\alpha_2 - \alpha_1)} = \frac{1 - \frac{1}{RC\alpha_1}}{\alpha_2 - \alpha_1} = -\frac{1 - \frac{L}{R} \frac{1}{LC\alpha_1}}{\alpha_1 - \alpha_2} = -\frac{1 - \frac{L}{R} \alpha_2}{\alpha_1 - \alpha_2}, \quad \frac{1}{p + \alpha_2} \text{ 的系数 } \frac{\alpha_2 - 2\beta}{\alpha_2(\alpha_1 - \alpha_2)} = \frac{1 - \frac{L}{R} \alpha_1}{\alpha_1 - \alpha_2},$$

$$\text{即 } I = \frac{E}{R} \left(1 - \frac{A}{p + \alpha_1} + \frac{B}{p + \alpha_2} \right), \text{ 其中 } A = \frac{R/L - \alpha_2}{\alpha_1 - \alpha_2}, B = \frac{R/L - \alpha_1}{\alpha_1 - \alpha_2}, \text{ 所以}$$

$$i(t) = \frac{E}{R} (1 - Ae^{-\alpha_1 t} + Be^{-\alpha_2 t})。$$

(ii) 当 $\Delta = 0$, 即 $L = 4CR^2$ 时, $I = \frac{E}{L} \frac{p + 2\beta}{p(p + \beta)^2} = \frac{E}{R} \left[\frac{1}{p} - \frac{1}{p + \beta} - \frac{\beta}{2} \frac{1}{(p + \beta)^2} \right],$

$$i(t) = \frac{E}{R} \left(1 - e^{-\beta t} - \frac{\beta}{2} t e^{-\beta t} \right)。$$

(iii) 当 $\Delta < 0$, 即 $L < 4CR^2$ 时, $I = \frac{E}{L} \frac{p+2\beta}{p(p+\beta-i\omega_0)(p+\beta+i\omega_0)}$,

其中 $\omega_0 = \sqrt{\gamma^2 - \beta^2}$, 上式继续化为

$$\begin{aligned} I &= \frac{E}{L} \left[\frac{L}{R} \frac{1}{p} + \left(-\frac{L}{2R} + \frac{\gamma^2 - 2\beta^2}{2i\omega_0\gamma^2} \right) \frac{1}{p+\beta-i\omega_0} + \left(-\frac{L}{2R} - \frac{\gamma^2 - 2\beta^2}{2i\omega_0\gamma^2} \right) \frac{1}{p+\beta+i\omega_0} \right] \\ &= \frac{E}{R} \left\{ \frac{1}{p} + \left[-\frac{1}{2} + \left(\frac{R}{L} - \beta \right) \frac{1}{2i\omega_0} \right] \frac{1}{p+\beta-i\omega_0} + \left[-\frac{1}{2} - \left(\frac{R}{L} - \beta \right) \frac{1}{2i\omega_0} \right] \frac{1}{p+\beta+i\omega_0} \right\} \\ &= \frac{E}{R} \left[\frac{1}{p} - \frac{p+\beta}{(p+\beta)^2 + \omega_0^2} - \left(\beta - \frac{R}{L} \right) \frac{1}{\omega_0} \frac{\omega_0}{(p+\beta)^2 + \omega_0^2} \right]. \end{aligned}$$

所以 $i(t) = \frac{E}{R} \left\{ 1 - e^{-\beta t} \left[\cos \omega_0 t + \left(\beta - \frac{R}{L} \right) \frac{\sin \omega_0 t}{\omega_0} \right] \right\}.$

(3) 取拉式变换得 $\begin{cases} (p^2-1)X+Y+Z=p \\ X+(p^2-1)Y+Z=0 \\ X+Y+(p^2-1)Z=0 \end{cases}$, 解得 $X = \frac{1}{3} \left(\frac{p}{p^2+1} + \frac{2p}{p^2-2} \right)$,

$Y = Z = \frac{1}{3} \left(\frac{p}{p^2+1} - \frac{p}{p^2-2} \right)$. 所以 $x(t) = \frac{1}{3} (\cos t + 2 \operatorname{ch} \sqrt{2}t)$,

$y(t) = z(t) = \frac{1}{3} (\cos t - \operatorname{ch} \sqrt{2}t)$.

(4) 两边取拉式变换得 $Y = \frac{a}{p^2+1} - 2 \frac{p}{p^2+1} Y$, 所以 $Y = \frac{a}{(p+1)^2}$, 即 $y(t) = ate^{-t}$.

(5) $Y = \frac{ab}{p^2+b^2-bc}$, $y(t) = a \sqrt{\frac{b}{b-c}} \sin \sqrt{b(b-c)}t$.

(6) $F = \frac{5}{p-2} + \frac{4}{p+1} - \frac{6}{(p+1)^2}$, $f(t) = 5e^{2t} + 4e^{-t} - 6te^{-t}$.

181. 求解变系数常微方程初值问题: $\begin{cases} x'' + tx' + x = 0 \\ x(0) = 1, x'(0) = 0 \end{cases}$.

$x' \xrightarrow{LT} pX - 1$, $tx' \xrightarrow{LT} -\frac{d}{dp}(pX - 1) = -X - p \frac{dX}{dp}$, $x'' \xrightarrow{LT} p^2X - p$, 原方程

两边取拉式变换得 $-\frac{dX}{dp} + pX - 1 = 0$ ，再反演得 $x' + tx = 0$ ，解之得 $x(t) = e^{-\frac{1}{2}t^2}$ 。

182. 设有放射性蜕变过程 $A \rightarrow B \rightarrow C \rightarrow \cdots$ ，若其中三种同位素的分子数 $N_1(t)$ ， $N_2(t)$ ，

$$N_3(t) \text{ 遵从方程及初始条件 } \begin{cases} \frac{dN_1}{dt} = -\lambda_1 N_1, N_1(0) = N \\ \frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2, N_2(0) = 0, \text{ 其中 } \lambda_1, \lambda_2, \lambda_3 \text{ 为不相等} \\ \frac{dN_3}{dt} = \lambda_2 N_2 - \lambda_3 N_3, N_3(0) = 0 \end{cases}$$

的常数，试求出 $N_1(t)$ ， $N_2(t)$ ， $N_3(t)$ 。

用上面加波浪线的字母表示相应的拉式变换 $\begin{cases} p\tilde{N}_1 - N = -\lambda_1 \tilde{N}_1 \\ p\tilde{N}_2 = \lambda_1 \tilde{N}_1 - \lambda_2 \tilde{N}_2 \\ p\tilde{N}_3 = \lambda_2 \tilde{N}_2 - \lambda_3 \tilde{N}_3 \end{cases}$ ，

$$\text{解得 } \tilde{N}_1 = \frac{N}{p + \lambda_1}, \quad \tilde{N}_2 = \frac{\lambda_1 N}{\lambda_2 - \lambda_1} \left(\frac{1}{p + \lambda_1} - \frac{1}{p + \lambda_2} \right),$$

$$\tilde{N}_3 = \lambda_1 \lambda_2 N \left[\frac{1}{(\lambda_2 - \lambda_1)(\lambda_3 - \lambda_1)} \frac{1}{p + \lambda_1} + \frac{1}{(\lambda_1 - \lambda_2)(\lambda_3 - \lambda_2)} \frac{1}{p + \lambda_2} + \frac{1}{(\lambda_1 - \lambda_3)(\lambda_2 - \lambda_3)} \frac{1}{p + \lambda_3} \right]$$

$$\text{所以 } N_1(t) = N e^{-\lambda_1 t}, \quad N_2(t) = \frac{\lambda_1 N}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}),$$

$$N_3(t) = \lambda_1 \lambda_2 N \left[\frac{1}{(\lambda_2 - \lambda_1)(\lambda_3 - \lambda_1)} e^{-\lambda_1 t} + \frac{1}{(\lambda_1 - \lambda_2)(\lambda_3 - \lambda_2)} e^{-\lambda_2 t} + \frac{1}{(\lambda_1 - \lambda_3)(\lambda_2 - \lambda_3)} e^{-\lambda_3 t} \right].$$

183. 定义零阶 Bessel 函数 $J_0(t) = \frac{1}{\pi} \int_0^\pi \cos(t \cos \theta) d\theta$ 。(1) 求 $J_0(t)$ 的像函数；

(2) 利用卷积定理证明： $\int_0^t J_0(\tau) J_0(t - \tau) d\tau = \sin t$ 。

(1) 当 $\operatorname{Re} p \geq \delta > 0$ 时 (δ 是任意小的正数)， $\left| \int_0^\pi \cos(t \cos \theta) e^{-pt} dt \right| \leq \int_0^\pi e^{-\delta t} dt$ ，即左边积分一致收敛，因此可交换积分次序：

$$J_0(t) \xrightarrow{LT} \frac{1}{\pi} \int_0^\pi \left[\int_0^\pi \cos(t \cos \theta) d\theta \right] e^{-pt} dt = \frac{1}{\pi} \int_0^\pi \left[\int_0^\pi \cos(t \cos \theta) e^{-pt} dt \right] d\theta。$$

$$\int_0^{\infty} \cos(t \cos \theta) e^{-pt} dt = \frac{1}{2} \left(\int_0^{\infty} e^{it \cos \theta} e^{-pt} dt + \int_0^{\infty} e^{-it \cos \theta} e^{-pt} dt \right) = \frac{1}{2} \left(\frac{1}{p - i \cos \theta} + \frac{1}{p + i \cos \theta} \right),$$

$$\begin{aligned} J_0(t) &\xrightarrow{LT} \frac{1}{2\pi} \left(\int_0^{\pi} \frac{1}{p - i \cos \theta} d\theta + \int_0^{\pi} \frac{1}{p + i \cos \theta} d\theta \right) = \frac{1}{2\pi} \left(\int_0^{\pi} \frac{1}{p - i \cos \theta} d\theta + \int_{\pi}^{2\pi} \frac{1}{p - i \cos \varphi} d\varphi \right) \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{p - i \cos \theta} d\theta. \end{aligned}$$

利用 $e^{i\theta} = z$ 将上面积分化成复变积分计算可得 $J_0(t) \xrightarrow{LT} \frac{1}{\sqrt{p^2 + 1}}$ 。

$$(2) \int_0^t J_0(\tau) J_0(t - \tau) d\tau \xrightarrow{LT} \frac{1}{\sqrt{p^2 + 1}} \cdot \frac{1}{\sqrt{p^2 + 1}} = \frac{1}{p^2 + 1}, \text{ 反演得}$$

$$\int_0^t J_0(\tau) J_0(t - \tau) d\tau = \sin t.$$

184. 试证明 $f(t) = 2te^{t^2} \sin(e^{t^2})$ 的拉氏变换存在。

证：令 $\operatorname{Re} p \geq \delta > 0$ 时 (δ 是任意小的正数)，则 $|f(t)e^{-pt}| \leq 2te^{t^2} \sin(e^{t^2})e^{-\delta t}$ 。

作代换 $e^{t^2} = x$ ，即 $t = \sqrt{\ln x}$ ，则 $\int_0^{\infty} 2te^{t^2} \sin(e^{t^2})e^{-\delta t} dt = \int_1^{\infty} \sin x e^{-\delta \sqrt{\ln x}} dx$ ，

由于 $e^{-\delta \sqrt{\ln x}}$ 单调趋于 0， $\int_1^b \sin x dx$ 有界，所以 $\int_1^{\infty} \sin x e^{-\delta \sqrt{\ln x}} dx$ 收敛，因此

$\int_0^{\infty} f(t)e^{-pt} dt$ 收敛，即 $f(t)$ 的拉氏变换存在。

185. 设 $f(x)$ 的 Fourier 变换和 Fourier 反变换为： $F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{-i\omega x} dx$ ，

$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\omega)e^{i\omega x} d\omega$ 。证明：

$$(1) f(x - x_0) \xrightarrow{FT} e^{-i\omega x_0} F(\omega), F(\omega - \omega_0) \xrightarrow{FT^{-1}} e^{i\omega_0 x} f(x);$$

$$(2) f'(x) \xrightarrow{FT} i\omega F(\omega), F'(\omega) \xrightarrow{FT^{-1}} -ixf(x);$$

$$(3) \int_{-\infty}^x f(t) dt \xrightarrow{FT} \frac{1}{i\omega} F(\omega); (4) f_1(x) * f_2(x) \xrightarrow{FT} F_1(\omega) F_2(\omega),$$

$F_1(\omega) * F_2(\omega) \xrightarrow{FT^{-1}} f_1(x) f_2(x)$, 其中 $f_1(x) * f_2(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f_1(t) f_2(x-t) dt$ 。

$$(1) \quad \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-x_0) e^{-i\omega x} dx = \frac{1}{\sqrt{2\pi}} e^{-i\omega x_0} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt = e^{-i\omega x_0} F(\omega),$$

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\omega - \omega_0) e^{i\omega x} d\omega = \frac{1}{\sqrt{2\pi}} e^{i\omega_0 x} \int_{-\infty}^{\infty} F(u) e^{iu x} du = e^{i\omega_0 x} f(x);$$

$$(2) \quad \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f'(x) e^{-i\omega x} dx = \frac{1}{\sqrt{2\pi}} f(x) e^{-i\omega x} \Big|_{-\infty}^{\infty} + i\omega \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

一般有 $f(\pm\infty) = 0$, 上式右边 $= i\omega F(\omega)$,

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F'(\omega) e^{i\omega x} d\omega = \frac{1}{\sqrt{2\pi}} F(x) e^{i\omega x} \Big|_{-\infty}^{\infty} - ix \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega = -ix f(x)。$$

$$(3) \quad \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\int_{-\infty}^x f(t) dt \right] e^{-i\omega x} dx = -\frac{1}{i\omega} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\int_{-\infty}^x f(t) dt \right] \frac{d}{dx} e^{-i\omega x} dx$$

$$= -\frac{1}{i\omega} \left[\int_{-\infty}^x f(t) dt \right] e^{-i\omega x} \Big|_{-\infty}^{\infty} + \frac{1}{i\omega} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

若 $F(0) = 0$, 即 $\int_{-\infty}^{\infty} f(x) dx = 0$, 则上式 $= \frac{1}{i\omega} F(\omega)$ 。

$$(4) \quad \text{假设可交换积分次序, } \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f_1(t) f_2(x-t) dt \right] e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f_1(t) \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f_2(x-t) e^{-i\omega x} dx \right] dt = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f_1(t) F_2(\omega) e^{-i\omega t} dt$$

$$= F_1(\omega) F_2(\omega),$$

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F_1(u) F_2(\omega-u) du \right] e^{i\omega x} d\omega = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F_1(u) \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F_2(\omega-u) e^{i\omega x} d\omega \right] du$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F_1(u) e^{i\omega u} f_2(x) du = f_1(x) f_2(x)。$$

186. 证明 δ 函数的下列性质: (1) $\delta(x) = \delta(-x)$; (2) $x\delta(x) = 0$;

$$(3) f(x)\delta(x) = f(0)\delta(x); (4) \delta(ax) = \frac{1}{|a|}\delta(x);$$

$$(5) \delta(x^2 - a^2) = \frac{1}{2|a|}[\delta(x-a) + \delta(x+a)];$$

$$(6) \delta(x-a)\delta(x-b) = \delta(a-b)\delta(x-a)。$$

$$(1) \int_{-\infty}^{\infty} \varphi(x)\delta(x)dx = \varphi(0), \int_{-\infty}^{\infty} \varphi(x)\delta(-x)dx = \int_{-\infty}^{\infty} \varphi(-t)\delta(t)dt = \varphi(0),$$

所以 $\delta(x) = \delta(-x)$ 。

$$(2) \int_{-\infty}^{\infty} \varphi(x)x\delta(x)dx = x\varphi(x)|_{x=0} = 0, \text{ 所以 } x\delta(x) = 0。$$

$$(3) \int_{-\infty}^{\infty} \varphi(x)f(x)\delta(x)dx = \varphi(0)f(0), \int_{-\infty}^{\infty} \varphi(x)f(0)\delta(x)dx = \varphi(0)f(0),$$

所以 $f(x)\delta(x) = f(0)\delta(x)$ 。

$$(4) \int_{-\infty}^{\infty} \varphi(x)\delta(ax)dx = \frac{1}{|a|}\varphi(0), \int_{-\infty}^{\infty} \varphi(x)\frac{1}{|a|}\delta(x)dx = \frac{1}{|a|}\varphi(0),$$

所以 $\delta(ax) = \frac{1}{|a|}\delta(x)$ 。

(5) 不妨设 $a > 0$,

$$\begin{aligned} \int_{-\infty}^{\infty} \varphi(x)\delta(x^2 - a^2)dx &= \int_{-a-\varepsilon}^{-a+\varepsilon} \varphi(x)\delta(x^2 - a^2)dx + \int_{a-\varepsilon}^{a+\varepsilon} \varphi(x)\delta(x^2 - a^2)dx \\ &= \int_{-2a\varepsilon+\varepsilon^2}^{2a\varepsilon+\varepsilon^2} \frac{\varphi(-\sqrt{u+a^2})}{2\sqrt{u+a^2}}\delta(u)du + \int_{-2a\varepsilon+\varepsilon^2}^{2a\varepsilon+\varepsilon^2} \frac{\varphi(\sqrt{u+a^2})}{2\sqrt{u+a^2}}\delta(u)du \\ &= \frac{1}{2a}[\varphi(-a) + \varphi(a)] = \int_{-\infty}^{\infty} \varphi(x)\frac{1}{2a}[\delta(x-a) + \delta(x+a)]dx。 \end{aligned}$$

$$\begin{aligned} (6) \int_{-\infty}^{\infty} \varphi(x)\delta(x-a)\delta(x-b)dx &= \int_{-\infty}^{\infty} \varphi(a)\delta(a-b)\delta(x-a)dx \\ &= \int_{-\infty}^{\infty} \varphi(x)\delta(a-b)\delta(x-a)dx, \text{ 即 } \delta(x-a)\delta(x-b) = \delta(a-b)\delta(x-a)。 \end{aligned}$$

187. 若定义 δ 函数为 $\delta(x) = \lim_{n \rightarrow \infty} \phi_n(x)$, 其中 $\lim_{n \rightarrow \infty} \int_{-\infty}^x \phi_n(t)dt = \begin{cases} 0, x < 0 \\ 1, x > 0 \end{cases}$ 。验证

$\lim_{n \rightarrow \infty} \frac{n}{\pi} \frac{1}{1+(nx)^2}$ 和 $\lim_{n \rightarrow \infty} \frac{\sin nx}{\pi x}$ 都是 δ 函数。

$$\int_{-\infty}^x \frac{n}{\pi} \frac{1}{1+(nt)^2} dt = \frac{1}{\pi} \int_{-\infty}^{nx} \frac{1}{1+u^2} du = \frac{1}{\pi} \left(\tan^{-1} nx + \frac{\pi}{2} \right),$$

$$x < 0 \text{ 时 } \lim_{n \rightarrow \infty} \frac{n}{\pi} \frac{1}{1+(nt)^2} = \lim_{n \rightarrow \infty} \frac{1}{\pi} \left(\tan^{-1} nx + \frac{\pi}{2} \right) = \frac{1}{\pi} \left(-\frac{\pi}{2} + \frac{\pi}{2} \right) = 0,$$

$$x > 0 \text{ 时 } \lim_{n \rightarrow \infty} \frac{n}{\pi} \frac{1}{1+(nt)^2} = \lim_{n \rightarrow \infty} \frac{1}{\pi} \left(\tan^{-1} nx + \frac{\pi}{2} \right) = \frac{1}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = 1.$$

$$\text{当 } x < 0 \text{ 时, } \lim_{n \rightarrow \infty} \int_{-\infty}^x \frac{\sin nt}{\pi t} dt = \lim_{n \rightarrow \infty} \int_{-\infty}^{nx} \frac{\sin u}{\pi u} du = 0,$$

$$\text{当 } x > 0 \text{ 时, } \lim_{n \rightarrow \infty} \int_{-\infty}^x \frac{\sin nt}{\pi t} dt = \lim_{n \rightarrow \infty} \int_{-\infty}^{nx} \frac{\sin u}{\pi u} du = \frac{1}{\pi} \cdot \pi = 1.$$

188. 设 $B_n(\omega) = \frac{2\omega_0}{\pi(\omega^2 - \omega_0^2)} \sin\left(2n\pi \frac{\omega}{\omega_0}\right)$, n 是正整数, 验证

$$\lim_{n \rightarrow \infty} B_n(\omega) = \delta(\omega - \omega_0) - \delta(\omega + \omega_0).$$

$$\begin{aligned} \int_{-\infty}^{\infty} \varphi(\omega) B_n(\omega) d\omega &= \int_{-\infty}^{\infty} \varphi(\omega) \frac{1}{\pi} \left(\frac{1}{\omega - \omega_0} - \frac{1}{\omega + \omega_0} \right) \sin\left(2n\pi \frac{\omega}{\omega_0}\right) d\omega \\ &= \int_{-\infty}^{\infty} \varphi(\omega) \frac{1}{\pi(\omega - \omega_0)} \sin\left(2n\pi \frac{\omega}{\omega_0}\right) d\omega - \int_{-\infty}^{\infty} \varphi(\omega) \frac{1}{\pi(\omega + \omega_0)} \sin\left(2n\pi \frac{\omega}{\omega_0}\right) d\omega \\ &= \int_{-\infty}^{\infty} \varphi\left(\omega_0 + \frac{\omega_0}{2\pi} x\right) \frac{\sin nx}{\pi x} dx - \int_{-\infty}^{\infty} \varphi\left(-\omega_0 + \frac{\omega_0}{2\pi} x\right) \frac{\sin nx}{\pi x} dx. \end{aligned}$$

(分别令 $\frac{2\pi}{\omega_0}(\omega - \omega_0) = x$ 和 $\frac{2\pi}{\omega_0}(\omega + \omega_0) = x$) 上题已证明 $\lim_{n \rightarrow \infty} \frac{\sin nx}{\pi x} = \delta(x)$, 所以令上

$$\text{式 } n \rightarrow \infty \text{ 可得 } \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} \varphi(\omega) B_n(\omega) d\omega = \varphi(\omega_0) - \varphi(-\omega_0),$$

$$\text{即 } \lim_{n \rightarrow \infty} B_n(\omega) = \delta(\omega - \omega_0) - \delta(\omega + \omega_0).$$

189. 定义 三维 δ 函数为 $\delta(\mathbf{r} - \mathbf{r}_0) = \delta(x - x_0) \delta(y - y_0) \delta(z - z_0)$, 求证:

$$(1) \text{ 它在球坐标下的表达式为 } \delta(\mathbf{r} - \mathbf{r}_0) = \frac{1}{r^2} \delta(r - r_0) \delta(\cos \theta - \cos \theta_0) \delta(\varphi - \varphi_0);$$

$$(2) \nabla^2 \frac{1}{|\mathbf{r} - \mathbf{r}_0|} = -4\pi \delta(\mathbf{r} - \mathbf{r}_0).$$

$$\begin{aligned}
(1) \quad & \iiint \varphi(\mathbf{r}) \frac{1}{r^2} \delta(r-r_0) \delta(\cos \theta - \cos \theta_0) \delta(\varphi - \varphi_0) dV \\
&= \int_0^{2\pi} \delta(\varphi - \varphi_0) d\varphi \int_0^\pi \delta(\cos \theta - \cos \theta_0) \sin \theta d\theta \int_0^\infty \varphi(r, \theta, \varphi) \delta(r-r_0) dr \\
&= \int_0^{2\pi} \delta(\varphi - \varphi_0) d\varphi \int_0^\pi \varphi(r_0, \theta, \varphi) \delta(\theta - \theta_0) d\theta \\
&= \int_0^{2\pi} \varphi(r_0, \theta_0, \varphi) \delta(\varphi - \varphi_0) d\varphi = \varphi(r_0, \theta_0, \varphi_0).
\end{aligned}$$

(2) 不妨设 $\mathbf{r}_0 = 0$ 。考虑积分 $\iiint_V \nabla^2 \frac{1}{r} dV$ ，可验证，当 $r \neq 0$ 时， $\nabla^2 \frac{1}{r} = 0$ ，所以对于不包含原点的 V ，有 $\iiint_V \nabla^2 \frac{1}{r} dV = 0$ 。若 V 包含原点，考虑以下极限式：

$$\begin{aligned}
\lim_{a \rightarrow 0} \iiint_V \nabla^2 \frac{1}{\sqrt{r^2 + a^2}} dV &= -3 \lim_{a \rightarrow 0} \iiint_V \frac{a^2}{(r^2 + a^2)^{5/2}} r^2 \sin \theta dr d\theta d\varphi \\
&= -12 \lim_{a \rightarrow 0} \int_0^\infty \frac{a^2}{(r^2 + a^2)^{5/2}} r^2 dr = -12 \lim_{a \rightarrow 0} \int_0^{\pi/2} \frac{a^4 \tan^2 \theta}{a^5 (1 + \tan^2 \theta)^{5/2}} a \sec^2 \theta d\theta \\
&= -12 \int_0^{\pi/2} \sin^2 \theta \cos \theta d\theta = -4\pi.
\end{aligned}$$

190. 求解下列微分方程：(1) $y'' = -\delta(x - x_0)$ ， $y(0) = 0$ ， $y'(1) = 0$ ；

(2) $y'' + y = -\delta(x - x_0)$ ， $y(0) = 0$ ， $y(\pi/2) = 0$ 。

(1) 直接积分得 $y' = -\eta(x - x_0) + A$ ，由 $y'(1) = 0$ 定出 $A = 1$ ，即 $y' = 1 - \eta(x - x_0)$ ，

再积分得 $y = x - (x - x_0)\eta(x - x_0) + B$ ，由 $y(0) = 0$ 定出 $B = 0$ ，所以

$$y = x - (x - x_0)\eta(x - x_0) = \begin{cases} x, & 0 \leq x < x_0 \\ x_0, & x_0 < x \leq 1 \end{cases}.$$

(2) $x < x_0$ 时， $y = A \sin x + B \cos x$ ，由 $y(0) = 0$ 得 $y = A \sin x$ 。

$x > x_0$ 时， $y = C \sin x + D \cos x$ 。由 $y(\pi/2) = 0$ 得 $y = D \cos x$ 。

由 $y(x_0^-) = y(x_0^+)$ ， $y'(x_0^+) - y'(x_0^-) = -1$ 可得 $y = \begin{cases} \cos x_0 \sin x, & 0 \leq x < x_0 \\ \sin x_0 \cos x, & x_0 < x \leq \pi/2 \end{cases}.$

191. 求方程 $y'' - x^2 y = 0$ 在 $x = 0$ 邻域内的两个级数解。

$x = 0$ 为方程常点, 设 $y = \sum_{k=0}^{\infty} a_k x^k$, 代入方程得

$$2a_2 + 6a_3x + \sum_{n=2}^{\infty} [(n+1)(n+2)a_{n+2} - a_{n-2}]x^n = 0,$$

所以 $a_2 = 0$, $a_3 = 0$, $(n+1)(n+2)a_{n+2} - a_{n-2} = 0$ ($n \geq 2$)。

$$\begin{aligned} a_{4n} &= \frac{1}{4n(4n-1)} a_{4(n-1)} = \frac{1}{4n(4n-1)} \frac{1}{[4(n-1)][4(n-1)-1]} a_{4(n-2)} = \cdots \\ &= \frac{1}{4n(4n-1)} \frac{1}{[4(n-1)][4(n-1)-1]} \cdots \frac{1}{4 \times (4-1)} a_0 \\ &= \frac{1}{4^n n(n-1) \cdots \times 1} \frac{1}{4^n \left(n - \frac{1}{4}\right) \left[\left(n-1\right) - \frac{1}{4}\right] \cdots \left[1 - \frac{1}{4}\right]} a_0 \\ &= \frac{\Gamma(3/4)}{n! \Gamma(n+3/4)} \left(\frac{1}{2}\right)^{4n} a_0, \quad (n \geq 1) \end{aligned}$$

$$\begin{aligned} a_{4n+1} &= \frac{1}{(4n+1)(4n)} a_{4(n-1)+1} = \frac{1}{(4n+1)(4n)} \frac{1}{[4(n-1)+1][4(n-1)]} a_{4(n-2)+1} = \cdots \\ &= \frac{\Gamma(5/4)}{n! \Gamma(n+5/4)} \left(\frac{1}{2}\right)^{4n} a_1, \quad (n \geq 1) \end{aligned}$$

由于 $a_2 = 0$, $a_3 = 0$, 由递推关系可得 $a_{4n+2} = a_{4n+3} = 0$ ($n \geq 1$)。

所以两个级数解为 $y_1 = \sum_{n=0}^{\infty} \frac{\Gamma(3/4)}{n! \Gamma(n+3/4)} \left(\frac{x}{2}\right)^{4n}$, $y_2 = \sum_{n=0}^{\infty} \frac{\Gamma(5/4)}{n! \Gamma(n+5/4)} \left(\frac{x}{2}\right)^{4n+1}$ 。

192. 在 $x = 0$ 邻域内求解方程 $y'' - xy = 0$ 。

设 $y = \sum_{k=0}^{\infty} a_k x^k$, 代入方程得 $2a_2 + \sum_{n=1}^{\infty} [(n+1)(n+2)a_{n+2} - a_{n-1}]x^n = 0$, 所以 $a_2 = 0$,

$$a_{3n} = \frac{1}{3n(3n-1)} \frac{1}{[3(n-1)][3(n-1)-1]} \cdots \frac{1}{3 \times (3-1)} a_0 = \frac{\Gamma(2/3)}{n! \Gamma(n+2/3)} \frac{a_0}{3^{2n}},$$

$$a_{3n+1} = \frac{1}{(3n+1)(3n)} \frac{1}{[3(n-1)+1][3(n-1)]} \cdots \frac{1}{(3+1) \times 3} a_1 = \frac{\Gamma(4/3)}{n! \Gamma(n+4/3)} \frac{a_1}{3^{2n}},$$

$a_{3n+2} = 0$, (上面 $n \geq 1$) 所以两个级数解为

$$y_1 = \sum_{n=0}^{\infty} \frac{\Gamma(2/3)}{n! \Gamma(n+2/3)} \frac{x^{3n}}{3^{2n}}, \quad y_2 = \sum_{n=0}^{\infty} \frac{\Gamma(4/3)}{n! \Gamma(n+4/3)} \frac{x^{3n+1}}{3^{2n}}.$$

193. 求厄密方程 $u'' - 2xu' + 2\lambda u = 0$ 在 $x=0$ 的解, 并讨论当 λ 取何值时有一解截断为多项式。

设 $u = \sum_{k=0}^{\infty} a_k x^k$, 代入方程得 $\sum_{n=0}^{\infty} [(n+1)(n+2)a_{n+2} + 2(\lambda-n)a_n] x^n = 0$ 。

$$a_{2n} = \frac{2[2(n-1)-\lambda]}{2n(2n-1)} a_{2(n-1)} = \frac{2[2(n-1)-\lambda]}{2n(2n-1)} \frac{2[2(n-2)-\lambda]}{(2n-2)(2n-3)} \cdots \frac{2(-\lambda)}{2 \times 1} a_0$$

$$= \frac{2^{2n} \Gamma(n-\lambda/2)}{(2n)! \Gamma(-\lambda/2)} a_0, \quad (n \geq 1)$$

$$a_{2n+1} = \frac{2^{2n} \Gamma\left(n + \frac{1-\lambda}{2}\right)}{(2n+1)! \Gamma\left(\frac{1-\lambda}{2}\right)} a_1 \quad (n \geq 1),$$

$$\text{所以两个解为: } u_1 = \sum_{n=0}^{\infty} \frac{\Gamma(n-\lambda/2)}{(2n)! \Gamma(-\lambda/2)} (2x)^{2n}, \quad u_2 = \sum_{n=0}^{\infty} \frac{\Gamma\left(n + \frac{1-\lambda}{2}\right)}{(2n+1)! \Gamma\left(\frac{1-\lambda}{2}\right)} (2x)^{2n+1}.$$

$$\text{当 } \lambda = 2m \quad (m=0, 1, 2, \cdots) \text{ 时, } u_1 = \sum_{n=0}^{\infty} \frac{\Gamma(n-m)}{(2n)! \Gamma(-m)} (2x)^{2n}, \text{ 当 } n > m \text{ 时, } \Gamma(n-m)$$

$$\text{是有限值, } \frac{1}{\Gamma(-m)} = 0, \text{ 所以 } u_1 = \sum_{n=0}^m \frac{\Gamma(n-m)}{(2n)! \Gamma(-m)} (2x)^{2n}. \quad n \leq m \text{ 时,}$$

$$\frac{\Gamma(n-m)}{\Gamma(-m)} = (n-1-m)(n-2-m) \cdots (-m) = (-1)^n m(m-1) \cdots (m-n+1) = (-1)^n \frac{m!}{(m-n)!},$$

$$\text{即 } u_1 = \sum_{n=0}^m \frac{(-1)^n m!}{(2n)! (m-n)!} (2x)^{2n}.$$

同样可得, 当 $\lambda = 2m+1$ 时, $u_2 = \sum_{n=0}^m \frac{(-1)^n m!}{(2n+1)!(m-n)!} (2x)^{2n+1}$ 。

194. 求超几何方程 $z(1-z)\frac{d^2u}{dz^2} + [\gamma - (\alpha + \beta + 1)z]\frac{du}{dz} - \alpha\beta u = 0$ 在 $z=0$ 附近的两个独立解, 其中 α, β, γ 为已知常数, 且 γ 不是整数。

$z=0$ 是方程的正则奇点, 设 $u = z^\rho \sum_{k=0}^{\infty} a_k z^k$, 代入方程得:

$$[\rho(\rho-1) + \gamma\rho]a_0 z^{-1} + \sum_{k=0}^{\infty} [(k+\rho+1)(k+\rho+\gamma)a_{k+1} - (k+\rho+\alpha)(k+\rho+\beta)a_k] z^k = 0$$

所以 $\rho(\rho-1) + \gamma\rho = 0$, 解得 $\rho = 0$ 或 $\rho = 1-\gamma$ 。

$$\begin{aligned} a_k &= \frac{(k-1+\rho+\alpha)(k-1+\rho+\beta)}{(k+\rho)(k-1+\rho+\gamma)} a_{k-1} = \dots \\ &= \frac{(k-1+\rho+\alpha)(k-1+\rho+\beta)}{(k+\rho)(k-1+\rho+\gamma)} \frac{(k-2+\rho+\alpha)(k-2+\rho+\beta)}{(k-1+\rho)(k-2+\rho+\gamma)} \dots \frac{(\rho+\alpha)(\rho+\beta)}{(1+\rho)(\rho+\gamma)} a_0 \\ &= \frac{(k-1+\rho+\alpha)(k-2+\rho+\alpha) \dots (\rho+\alpha)}{(k+\rho)(k-1+\rho) \dots (1+\rho)} \frac{(k-1+\rho+\beta)(k-2+\rho+\beta) \dots (\rho+\beta)}{(k-1+\rho+\gamma)(k-2+\rho+\gamma) \dots (\rho+\gamma)} a_0 \\ &= \frac{\Gamma(k+\rho+\alpha)}{\Gamma(\rho+\alpha)} \frac{\Gamma(\rho+1)}{\Gamma(k+\rho+1)} \frac{\Gamma(k+\rho+\beta)}{\Gamma(\rho+\beta)} \frac{\Gamma(\rho+\gamma)}{\Gamma(k+\rho+\gamma)} a_0 \end{aligned}$$

取 $\rho = 0$ 得第一个解 $u_1 = \sum_{n=0}^{\infty} \frac{1}{k!} \frac{\Gamma(k+\alpha)}{\Gamma(\alpha)} \frac{\Gamma(k+\beta)}{\Gamma(\beta)} \frac{\Gamma(\gamma)}{\Gamma(k+\gamma)} z^k = F(\alpha, \beta, \gamma, z)$,

取 $\rho = 1-\gamma$ 得第二个解 $u_2 = z^{1-\gamma} \sum_{n=0}^{\infty} \frac{1}{k!} \frac{\Gamma(k+1+\alpha-\gamma)}{\Gamma(1+\alpha-\gamma)} \frac{\Gamma(k+1+\beta-\gamma)}{\Gamma(1+\beta-\gamma)} \frac{\Gamma(2-\gamma)}{\Gamma(k+2-\gamma)} z^k$
 $= z^{1-\gamma} F(1+\alpha-\gamma, 1+\beta-\gamma, 2-\gamma, z)$ 。

195. 求方程 $xy'' - xy' + y = 0$ 在 $x=0$ 邻域的两个独立解。

$x=0$ 是方程的正则奇点, 设 $y = x^\rho \sum_{k=0}^{\infty} a_k x^k$, 代入方程得:

$$\rho(\rho-1)a_0x^{-1} + \sum_{k=0}^{\infty} [(k+1+\rho)(k+\rho)a_{k+1} - (k-1+\rho)a_k]x^k = 0$$

所以 $\rho=0$ 或 $\rho=1$, $(k+\rho)(k-1+\rho)a_k - (k-2+\rho)a_{k-1} = 0$ ($k \geq 1$)。

$\rho=0$ 时, $0 \cdot a_1 + a_0 = 0$, $2a_2 - 0 \cdot a_1 = 0$, $a_3 = \frac{1}{6}a_2 = 0$, $a_4 = a_5 = \dots = 0$ 即只有 a_1 不

为 0, 所以一个解为 $y_1 = x$ 。当 $\rho=1$ 时, $a_1 = a_2 = \dots = 0$, 解仍为 x 。

设另一解为 $y = gx \ln x + \sum_{k=0}^{\infty} a_k x^k$, 代入方程得

$$g + a_0 + (2a_2 - g)x + \sum_{k=2}^{\infty} [(k+1)ka_{k+1} - (k-1)a_k]x^k = 0$$

所以 $g = -a_0$, $a_2 = \frac{1}{2}g = -\frac{1}{2}a_0$,

$$\begin{aligned} a_k &= \frac{k-2}{k(k-1)}a_{k-1} = \frac{k-2}{k(k-1)} \frac{k-3}{(k-1)(k-2)} \frac{k-4}{(k-2)(k-3)} \dots \frac{2}{4 \times 3} \times \frac{1}{3 \times 2} a_2 \\ &= \frac{1}{k(k-1)} \frac{1}{(k-1)(k-2)} \dots \frac{1}{4} \times \frac{1}{3} a_2 = \frac{2}{(k-1)k!} a_2 = -\frac{1}{(k-1)k!} a_0 \quad (k \geq 3)。 \end{aligned}$$

$gx \ln x + \sum_{k=0}^{\infty} a_k x^k = -a_0 x \ln x + a_0 + a_1 x - a_0 \sum_{k=2}^{\infty} \frac{1}{(k-1)k!} x^k$, 其中 a_1 项对应解 y_1 , 省略之

得到另一解 $y_2 = x \ln x - 1 + \sum_{k=2}^{\infty} \frac{1}{(k-1)k!} x^k$ 。

196. 求方程 $\frac{d^2 u}{dz^2} + \frac{2}{z} \frac{du}{dz} + m^2 u = 0$ 在 $z=0$ 附近的两个独立解。

设 $u = z^\rho \sum_{k=0}^{\infty} a_k z^k$, 代入方程得:

$$\rho(\rho+1)a_0 z^{-1} + (\rho+1)(\rho+2)a_1 + \sum_{k=1}^{\infty} [(k+1+\rho)(k+2+\rho)a_{k+1} + m^2 a_{k-1}] z^k = 0,$$

$$\rho = 0, -1, -2, \quad a_k = \frac{-m^2}{(k+1+\rho)(k+\rho)} a_{k-2}。$$

$$\text{取 } \rho = 0, \quad a_{2k} = \frac{-m^2}{(2k+1)(2k)} \frac{-m^2}{(2k-1)(2k-2)} \cdots \frac{-m^2}{3 \times 2} a_0 = \frac{(-1)^k m^{2k}}{(2k+1)!} a_0,$$

$$\text{可得一个解 } u_1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} (mz)^{2k} = \frac{1}{mz} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} (mz)^{2k+1} = \frac{\sin mz}{mz}.$$

$$\text{取 } \rho = -2, \quad \text{则 } a_{2k+1} = \frac{-m^2}{(2k)(2k-1)} \frac{-m^2}{(2k-2)(2k-3)} \cdots \frac{-m^2}{2 \times 1} a_1 = \frac{(-1)^k m^{2k}}{(2k)!} a_1,$$

$$\text{可得另一解 } u_2 = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} (mz)^{2k-1} = \frac{\cos mz}{mz}.$$

197. 求零阶 Bessel 方程 $\frac{d^2 u}{dz^2} + \frac{1}{z} \frac{du}{dz} + u = 0$ 在 $z = 0$ 邻域内的两个独立解。

$$\text{设 } u = z^\rho \sum_{k=0}^{\infty} a_k z^k \text{ 可得 } \rho^2 a_0 z^{-1} + (\rho-1)^2 a_1 + \sum_{k=1}^{\infty} \left[(k+1+\rho)^2 a_{k+1} + a_{k-1} \right] z^k = 0.$$

$$\text{可得 } \rho = 0 \text{ 或 } 1, \quad a_k = -\frac{1}{(k+\rho)^2} a_{k-2}.$$

$$\text{取 } \rho = 0, \quad a_{2k} = \frac{-1}{(2k)^2} a_{2(k-1)} = \frac{-1}{(2k)^2} \frac{-1}{[2(k-1)]^2} \cdots \frac{-1}{2^2} a_0 = \frac{(-1)^k}{2^{2k} (k!)^2} a_0,$$

$$\text{可得一个解为: } u_1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{z}{2} \right)^{2k} = J_0(z).$$

设另一解为 $u = gJ_0(z) \ln z + \sum_{k=0}^{\infty} a_k z^k$, 代入方程得:

$$g \left[zJ_0''(z) + J_0'(z) + zJ_0(z) \right] \ln z + 2gJ_0'(z) + \sum_{k=0}^{\infty} (k+1)^2 a_{k+1} z^k + \sum_{k=1}^{\infty} a_{k-1} z^k = 0$$

由于 $J_0(z)$ 是方程的解, 即 $zJ_0''(z) + J_0'(z) + zJ_0(z) = 0$, 代入 $J_0(z)$ 表达式, 并把 z 的偶次幂项和奇次幂项分开写成:

$$a_1 + \sum_{k=1}^{\infty} \left[(2k+1)^2 a_{2k+1} - a_{2k-1} \right] z^{2k} + \sum_{k=1}^{\infty} \left[(2k)^2 a_{2k} + a_{2k-2} + 4g \frac{(-1)^k k}{(k!)^2 2^{2k}} \right] z^{2k-1} = 0.$$

所以 $a_1 = 0$, 由 z 的偶次幂项系数的递推公式可得 $a_{2k+1} = 0$.

$$\begin{aligned}
z \text{ 的奇次幂项系数的递推公式为: } a_{2k} &= -\frac{a_{2k-2}}{(2k)^2} - g \frac{(-1)^k}{k(k!)^2 2^{2k}} \\
&= -\frac{1}{(2k)^2} \left[-\frac{a_{2k-4}}{(2k-2)^2} - g \frac{(-1)^{k-1}}{(k-1)[(k-1)!]^2 2^{2k-2}} \right] - g \frac{(-1)^k}{k(k!)^2 2^{2k}} \\
&= \frac{a_{2k-4}}{2^4 k^2 (k-1)^2} - g \frac{(-1)^k}{(k-1)(k!)^2 2^{2k}} - g \frac{(-1)^k}{k(k!)^2 2^{2k}} \\
&= \frac{(-1)^2 a_{2k-4}}{2^4 k^2 (k-1)^2} - g \frac{(-1)^k}{(k!)^2 2^{2k}} \left(\frac{1}{k} + \frac{1}{k-1} \right) \\
&= \cdots = \frac{(-1)^k}{2^{2k} (k!)^2} a_0 - g \frac{(-1)^k}{(k!)^2 2^{2k}} \left(\frac{1}{k} + \frac{1}{k-1} + \cdots + 1 \right).
\end{aligned}$$

所以另一解为 $u_2 = J_0(z) \ln z + \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{1}{k} + \frac{1}{k-1} + \cdots + 1 \right) \left(\frac{z}{2} \right)^{2k}$ (a_0 项对应解 u_1 , 故省略之)。

198. 求 Legendre 方程 $(1-z^2) \frac{d^2 u}{dz^2} - 2z \frac{du}{dz} + \mu(\mu+1)u = 0$ 在 $z=1$ 的有界解。

设 $u = (z-1)^\rho \sum_{k=0}^{\infty} a_k (z-1)^k$, 代入方程得:

$$2\rho^2 a_0 (z-1)^{-1} + \sum_{k=0}^{\infty} \left[2(k+1+\rho)^2 a_{k+1} + (k+1+\rho+\mu)(k+\rho-\mu)a_k \right] (z-1)^k = 0.$$

所以 $\rho = 0$,

$$\begin{aligned}
a_k &= \frac{(k+\mu)(\mu+1-k)}{2k^2} a_{k-1} = \frac{(k+\mu)(\mu+1-k)(k-1+\mu)(\mu+2-k)}{2k^2 \cdot 2(k-1)^2} \cdots \frac{(1+\mu)\mu}{2} a_0 \\
&= \frac{(k+\mu)(k-1+\mu) \cdots (1+\mu) \cdot (\mu+1-k)(\mu+2-k) \cdots \mu}{2^k k^2 (k-1)^2 \cdots 1} a_0 = \frac{\Gamma(k+1+\mu)}{2^k (k!)^2 \Gamma(-k+1+\mu)} a_0
\end{aligned}$$

所以一个解为 $u_1 = \sum_{k=0}^{\infty} \frac{\Gamma(k+1+\mu)}{(k!)^2 \Gamma(-k+1+\mu)} \left(\frac{z-1}{2} \right)^k$ 。

可求出该级数收敛半径为 2, 即收敛域为 $|z-1| < 2$, 即在收敛域内它是有界解。当 $\mu = n$ (整

数) 时, 若 $k \geq n+1$ 则有 $\frac{1}{\Gamma(-k+1+n)} = 0$, 而 $\Gamma(k+1+\mu)$ 为有限值, 所以 u_1 截断为多

$$\text{项式: } u_1 = \sum_{k=0}^n \frac{\Gamma(k+1+n)}{(k!)^2 \Gamma(-k+1+n)} \left(\frac{z-1}{2}\right)^k = \sum_{k=0}^n \frac{(n+k)!}{(k!)^2 (n-k)!} \left(\frac{z-1}{2}\right)^k.$$

199. 求合流超几何方程 $z \frac{d^2 u}{dz^2} + (b-z) \frac{du}{dz} - au = 0$ 在 $z=0$ 附近的两个独立解, 已知其中

的 a, b 为常数, 且 $a > 0$, $1-b \neq$ 整数。

设 $u = z^\rho \sum_{k=0}^{\infty} c_k z^k$, 代入方程得

$$\rho(\rho+b-1)c_0 z^{-1} + \sum_{k=0}^{\infty} [(k+1+\rho)(k+\rho+b)c_{k+1} - (k+\rho+a)c_k] z^k = 0.$$

$$\rho=0 \text{ 或 } 1-b, \quad c_k = \frac{k-1+\rho+a}{(k+\rho)(k-1+\rho+b)} c_{k-1}.$$

$$\text{取 } \rho=0, \text{ 则 } c_k = \frac{k-1+a}{k(k-1+b)} c_{k-1} = \frac{1}{k!} \frac{\Gamma(k+a)}{\Gamma(a)} \frac{\Gamma(b)}{\Gamma(k+b)} c_0,$$

$$\text{所以一个解为 } u_1 = \sum_{k=0}^{\infty} \frac{1}{k!} \frac{\Gamma(k+a)}{\Gamma(a)} \frac{\Gamma(b)}{\Gamma(k+b)} z^k = F(a, b, z).$$

$$\text{取 } \rho=1-b, \text{ 则 } c_k = \frac{1}{k!} \frac{\Gamma(k+1+a-b)}{\Gamma(1+a-b)} \frac{\Gamma(2-b)}{\Gamma(k+2-b)} c_0, \text{ 所以另一解为}$$

$$u_1 = z^{1-b} \sum_{k=0}^{\infty} \frac{1}{k!} \frac{\Gamma(k+1+a-b)}{\Gamma(1+a-b)} \frac{\Gamma(2-b)}{\Gamma(k+2-b)} z^k = z^{1-b} F(1+a-b, 2-b, z).$$

200. 求方程 $\frac{d^2 u}{dz^2} + \frac{1}{z} \frac{du}{dz} - m^2 u = 0$ 在 $z=0$ 附近的两个独立解。

设 $u = z^\rho \sum_{k=0}^{\infty} a_k z^k$, 代入方程得

$$\rho^2 a_0 z^{-1} + (1-\rho)^2 a_1 + \sum_{k=1}^{\infty} [(k+1+\rho)^2 a_{k+1} - m^2 a_{k-1}] z^k = 0.$$

所以 $\rho = 0$ 或 1 , $a_k = \frac{m^2}{(k+\rho)^2} a_{k-2}$ 。

取 $\rho = 0$, 则 $a_{2k} = \frac{m^2}{(2k)^2} a_{2k-2} = \frac{1}{(k!)^2} \left(\frac{m}{2}\right)^{2k} a_0$, 所以一个解为

$$u_1 = \sum_{k=0}^{\infty} \frac{1}{(k!)^2} \left(\frac{mz}{2}\right)^{2k} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{imz}{2}\right)^{2k} = J_0(imz)。$$

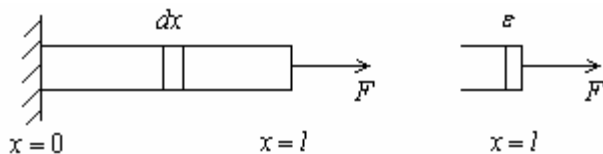
设另一解为 $u = gu_1 \ln z + \sum_{k=0}^{\infty} a_k z^k$, 代入方程得 (参考 197 题)

$$a_1 + \sum_{k=1}^{\infty} \left[(2k+1)^2 a_{2k+1} - m^2 a_{2k-1} \right] z^{2k} \\ + \sum_{k=1}^{\infty} \left[(2k)^2 a_{2k} - m^2 a_{2k-2} + 4g \left(\frac{m}{2}\right)^{2k} \frac{k}{(k!)^2} \right] z^{2k-1} = 0。$$

$$\text{所以 } a_{2k+1} = 0, \quad a_{2k} = \frac{1}{k^2} \left(\frac{m}{2}\right)^2 a_{2k-2} - \frac{g}{k(k!)^2} \left(\frac{m}{2}\right)^{2k} \\ = \frac{1}{k^2} \left(\frac{m}{2}\right)^2 \left[\frac{1}{(k-1)^2} \left(\frac{m}{2}\right)^2 a_{2k-4} - \frac{g}{(k-1)[(k-1)!]^2} \left(\frac{m}{2}\right)^{2k-2} \right] - \frac{g}{k(k!)^2} \left(\frac{m}{2}\right)^{2k} \\ = \frac{1}{k^2 (k-1)^2} \left(\frac{m}{2}\right)^4 a_{2k-4} - \frac{g}{(k!)^2} \left(\frac{m}{2}\right)^{2k} \left(\frac{1}{k} + \frac{1}{k-1}\right) \\ = \cdots = \frac{1}{(k!)^2} \left(\frac{m}{2}\right)^{2k} a_0 - \frac{g}{(k!)^2} \left(\frac{m}{2}\right)^{2k} \left(\frac{1}{k} + \frac{1}{k-1} + \cdots + 1\right)。$$

所以另一解为 $u_2 = J_0(imz) \ln z - \sum_{k=0}^{\infty} \frac{1}{(k!)^2} \left(\frac{1}{k} + \frac{1}{k-1} + \cdots + 1\right) \left(\frac{mz}{2}\right)^{2k}$ 。

201. 一长为 l ，横截面积为 S 的均匀弹性杆，已知一端 ($x=0$) 固定，另一端在杆轴方向上受拉力 F 而平衡。在 $t=0$ 时撤去外力 F 。试推导杆的纵振动所满足的方程，边界条件和初始条件。



假设在垂直杆长方向的任一截面上各点的振动情况相同 $u(x, t)$ 表示杆上 x 处在 t 时刻相对于平衡位置的位移。取杆上长为 dx 的一小段，用 $P(x, t)$ 表示应力，由牛顿第二定律，

$$[P(x+dx, t) - P(x, t)]S = dm \frac{\partial^2 u}{\partial t^2}, \text{ 代入 } dm = \rho S dx \text{ 得 } \frac{\partial P}{\partial x} = \rho \frac{\partial^2 u}{\partial t^2}。 \text{ 由 Hooke 定}$$

$$\text{律 } P = E \frac{\partial u}{\partial x} \text{ 可得 } \frac{\partial^2 u}{\partial x^2} - \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2} = 0, \text{ 其中 } a = \sqrt{\frac{E}{\rho}}。$$

$$\text{取右端长为 } \varepsilon \text{ 的一小段，由牛顿第二定律有 } F(t) - ES \left. \frac{\partial u}{\partial x} \right|_{x=l-\varepsilon} = \rho \varepsilon S \left. \frac{\partial^2 u}{\partial t^2} \right|_{x=l-\alpha \varepsilon}$$

$$(0 < \alpha < 1), \text{ 令 } \varepsilon \rightarrow 0 \text{ 有 } F(t) - ES \left. \frac{\partial u}{\partial x} \right|_{x=l} = 0。 \quad (a)$$

$$\text{当 } t > 0 \text{ 时 } F(t) = 0, \text{ 所以 } \left. \frac{\partial u}{\partial x} \right|_{x=l} = 0。 \text{ 由于左端点固定，故有 } u|_{x=0} = 0。$$

$$\text{令 (a) 式中 } t=0 \text{ 有 } F - ES \left. \frac{\partial u}{\partial x} \right|_{x=l} = 0。 \text{ 因为平衡时应力处处相等，所以该式对于任}$$

$$\text{意 } x \in [0, l] \text{ 都成立，即 } F - ES \left. \frac{\partial u}{\partial x} \right|_{t=0} = 0, \text{ 对 } x \text{ 积分可得 } u|_{t=0} = \frac{F}{ES} x \text{ (注意到}$$

$$u|_{x=0} = 0 \text{)。初始时处于平衡状态，各处速度为 0，即 } \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0。$$

$$\text{综上该定解问题为 } \begin{cases} \frac{\partial^2 u}{\partial x^2} - \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2} = 0 \\ u|_{x=0} = 0, \left. \frac{\partial u}{\partial x} \right|_{x=l} = 0 \\ u|_{t=0} = \frac{F}{ES} x, \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0 \end{cases}。$$

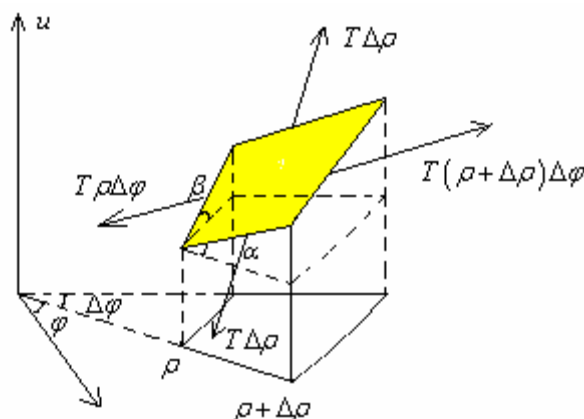
202. 一均匀弹性杆，原处于静止状态。其一端（ $x=0$ ）固定。从 $t=0$ 时刻起，在另一端（ $x=l$ ）单位面积上施加外力 P ，力的方向与杆轴平行。试列出杆的纵振动方程，边界条件和初始条件。

将上题（a）式写成 $P(t)S - ES \frac{\partial u}{\partial x} \Big|_{x=l} = 0$ ，则 $t>0$ 时 $\frac{\partial u}{\partial x} \Big|_{x=l} = \frac{P}{E}$ 。

$t=0$ 时令 $P(t)=0$ 则有 $\frac{\partial u}{\partial x} \Big|_{x=l} = 0$ ，同上题讨论可得 $u \Big|_{t=0} = 0$ ，其他条件与上题同。

$$\text{该定解问题为} \begin{cases} \frac{\partial^2 u}{\partial x^2} - \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2} = 0 \\ u \Big|_{x=0} = 0, \frac{\partial u}{\partial x} \Big|_{x=l} = \frac{P}{E} \\ u \Big|_{t=0} = 0, \frac{\partial u}{\partial t} \Big|_{t=0} = 0 \end{cases}$$

203. 一均匀，各向同性的弹性圆膜，四周固定。试列出膜的横振动方程及边界条件。



设 ρ_m 为面密度，任何方向单位长度张力是 T 。

沿 ρ 方向合张力为 $T(\rho+\Delta\rho)\Delta\varphi \sin \alpha \Big|_{\rho+\Delta\rho} - T\rho\Delta\varphi \sin \alpha \Big|_{\rho}$ ，

φ 方向合张力为 $T\Delta\rho \sin \beta \Big|_{\varphi+\Delta\varphi} - T\Delta\rho \sin \beta \Big|_{\varphi}$ 。

在小振动近似下有 $\sin \alpha \approx \tan \alpha \approx \frac{\Delta u}{\Delta \rho}$ ， $\sin \beta \approx \tan \beta \approx \frac{\Delta u}{\rho \Delta \varphi}$ ，再由牛顿第二定律得

$$T(\rho+\Delta\rho)\Delta\varphi \frac{\Delta u}{\Delta\rho} \Big|_{\rho+\Delta\rho} - T\rho\Delta\varphi \frac{\Delta u}{\Delta\rho} \Big|_{\rho} + T\Delta\rho \frac{\Delta u}{\rho\Delta\varphi} \Big|_{\varphi+\Delta\varphi} - T\Delta\rho \frac{\Delta u}{\rho\Delta\varphi} \Big|_{\varphi}$$

$$= \rho_m \rho \Delta \rho \Delta \varphi \left. \frac{\partial^2 u}{\partial t^2} \right|_{\substack{\rho + \varepsilon_1 \Delta \rho \\ \varphi + \varepsilon_2 \Delta \varphi}} \quad (0 < \varepsilon_1 < 1, \quad 0 < \varepsilon_2 < 1),$$

$$\text{即 } \frac{1}{\rho} \frac{(\rho + \Delta \rho) \left. \frac{\Delta u}{\Delta \rho} \right|_{\rho + \Delta \rho} - \rho \left. \frac{\Delta u}{\Delta \rho} \right|_{\rho}}{\Delta \rho} + \frac{1}{\rho^2} \frac{\left. \frac{\Delta u}{\Delta \varphi} \right|_{\varphi + \Delta \varphi} - \left. \frac{\Delta u}{\Delta \varphi} \right|_{\varphi}}{\Delta \varphi} - \frac{\rho_m}{T} \left. \frac{\partial^2 u}{\partial t^2} \right|_{\substack{\rho + \varepsilon_1 \Delta \rho \\ \varphi + \varepsilon_2 \Delta \varphi}} = 0$$

$$\text{令 } \Delta \rho \rightarrow 0, \Delta \varphi \rightarrow 0 \text{ 得 } \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} - \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2} = 0, \text{ 其中 } a = \sqrt{\frac{T}{\rho_m}}.$$

$$\text{该定解问题为 } \begin{cases} \nabla^2 u - \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2} = 0 \\ u|_{\rho=R} = 0 \end{cases} \quad (\text{极坐标系中 } \nabla^2 = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2}{\partial \varphi^2}).$$

204. 一长为 l 的均匀金属细杆 (可近似看作一维的), 通有恒定电流。设杆的一端 ($x=0$) 温度恒为 0, 另一端 ($x=l$) 恒为 u_0 , 初始时温度分布为 $\frac{u_0}{l}x$ 。试写出杆中温度场所满足的方程, 边界条件与初始条件。

由于热功率为 $I^2 R$, 所以单位时间单位体积产生热量 $\frac{I^2 R}{lS}$ 。所以热传导方程为

$$\rho c \frac{\partial u}{\partial t} - k \frac{\partial^2 u}{\partial x^2} = \frac{I^2 R}{lS}, \text{ 其中 } \rho \text{ 为体密度, } c \text{ 为比热。若用 } \lambda \text{ 表示线密度, 则有 } \rho = \frac{\lambda}{S},$$

$$\text{所以方程为 } \frac{\partial u}{\partial t} - \frac{kS}{\lambda c} \nabla^2 u = \frac{I^2 R}{\lambda cl}.$$

$$\text{该定解问题为 } \begin{cases} \frac{\partial u}{\partial t} - \frac{kS}{\lambda c} \frac{\partial^2 u}{\partial x^2} = \frac{I^2 R}{\lambda cl} \\ u|_{x=0} = 0, u|_{x=l} = u_0, u|_{t=0} = \frac{u_0}{l}x \end{cases}.$$

205. 在铀块中, 除了中子的扩散运动外, 还进行着中子的吸收和增殖过程。设在单位时间内单位体积中, 吸收和增殖的中子数均正比于该时刻该处的中子浓度 $u(\mathbf{r}, t)$, 因而净增中子数可表为 $\alpha u(\mathbf{r}, t)$, α 为比例常数。试导出 $u(\mathbf{r}, t)$ 所满足的方程。

用 $\mathbf{q}$ 表示单位时间流过某单位面积的中子数, 有 $\mathbf{q} = -D \nabla u$ 。取一个六面体

$[x, x + \Delta x] \times [y, y + \Delta y] \times [z, z + \Delta z]$, Δt 时间内沿 x 方向流入该六面体的中子数为

$$(q_x|_x - q_x|_{x+\Delta x}) \Delta y \Delta z \Delta t = D \left(\frac{\partial u}{\partial x} \Big|_{x+\Delta x} - \frac{\partial u}{\partial x} \Big|_x \right) \Delta y \Delta z \Delta t = D \frac{\partial^2 u}{\partial x^2} \Delta x \Delta y \Delta z \Delta t,$$

同样可得沿 y, z 方向流入该六面体的中子数分别为 $D \frac{\partial^2 u}{\partial y^2} \Delta x \Delta y \Delta z \Delta t$ 和 $D \frac{\partial^2 u}{\partial z^2} \Delta x \Delta y \Delta z \Delta t$ 。

六面体内中子数一共增加 $\Delta u \Delta x \Delta y \Delta z \Delta t$, 增加数应等于流入中子数加上净增中子数, 即

$$\Delta u \Delta x \Delta y \Delta z = D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \Delta x \Delta y \Delta z \Delta t + \alpha u \Delta x \Delta y \Delta z \Delta t.$$

两边同除 $\Delta x \Delta y \Delta z \Delta t$, 令 $\Delta t \rightarrow 0$ 得 $\frac{\partial u}{\partial t} = D \nabla^2 u + \alpha u$ 。

206. 设有一均匀杆, 长为 l , 一端固定, 另一端受外力 $F = A \sin \omega t$ 作用, 其方向与杆一致, A 为常数, 列出边界条件。

同 202 题, $\frac{\partial u}{\partial x} \Big|_{x=l} = \frac{F}{ES} = \frac{A}{ES} \sin \omega t$, $\frac{\partial u}{\partial x} \Big|_{x=0} = 0$ 。

207. 有一长为 l 的均匀细杆, 现通过其两端, 在单位时间内, 经单位面积分别供给热量 Q_1 与 Q_2 。试写出边界条件。

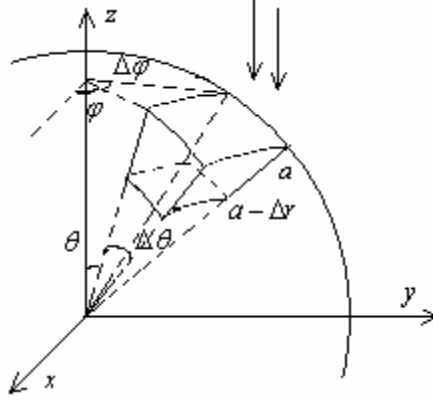
取左端长为 ε 的一小段, 由能量守恒, $Q_1 - q(x, t) \Big|_{x=\varepsilon} = \rho c \varepsilon \frac{\partial u}{\partial t} \Big|_{x=\alpha \varepsilon}$ ($0 < \alpha < 1$)。

代入 $q = -k \frac{\partial u}{\partial x}$ 得 $Q_1 + k \frac{\partial u}{\partial x} \Big|_{x=\varepsilon} = \rho c \varepsilon \frac{\partial u}{\partial t} \Big|_{x=\alpha \varepsilon}$ 。令 $\varepsilon \rightarrow 0$ 得 $\frac{\partial u}{\partial x} \Big|_{x=0} = -\frac{Q_1}{k}$ 。

同样可得 $\frac{\partial u}{\partial x} \Big|_{x=l} = \frac{Q_2}{k}$ 。

208. 有一半半径为 a , 表面涂黑的导体球, 暴晒于日光下, 在垂直于光线的单位面积上, 单位时间内吸收热量 M 。设周围媒质温度为 0 , 球面按牛顿冷却定律散热。试在适当的坐标系中写出边界条件。

牛顿冷却定律: 单位时间流过表面单位面积的热量与表面两边的温度差成正比 (比例系数设为 H)。



取上图的一小块体积元， Δt 时间内外表面（ $r = a$ ）吸收热量为 $M \left(\Delta S_r|_{r=a} \cos \theta \right) \Delta t$ ，

外表面散失热量为 $H u|_{r=a} \Delta S_r|_{r=a} \Delta t$ 。

$r = a - \Delta r$ 面流入热量为 $q_r|_{r=a-\Delta r} \Delta S_r|_{r=a-\Delta r} \Delta t = -k \frac{\partial u}{\partial r} \Big|_{r=a-\Delta r} \Delta S_r|_{r=a-\Delta r} \Delta t$ ，

从 θ 面流入热量 $q_\theta|_\theta \Delta S_\theta|_\theta \Delta t = -\frac{k}{a} \frac{\partial u}{\partial \theta} \Big|_\theta \Delta S_\theta|_\theta \Delta t$ ，

从 $\theta + \Delta \theta$ 面流入热量 $-q_\theta|_{\theta+\Delta \theta} \Delta S_\theta|_{\theta+\Delta \theta} \Delta t = \frac{k}{a} \frac{\partial u}{\partial \theta} \Big|_{\theta+\Delta \theta} \Delta S_\theta|_{\theta+\Delta \theta} \Delta t$ ，

从 φ 面流入热量 $q_\varphi|_\varphi \Delta S_\varphi|_\varphi \Delta t = -\frac{k}{a \sin \theta} \frac{\partial u}{\partial \varphi} \Big|_\varphi \Delta S_\varphi|_\varphi \Delta t$ ，

从 $\varphi + \Delta \varphi$ 面流入热量 $-q_\varphi|_{\varphi+\Delta \varphi} \Delta S_\varphi|_{\varphi+\Delta \varphi} \Delta t = \frac{k}{a \sin \theta} \frac{\partial u}{\partial \varphi} \Big|_{\varphi+\Delta \varphi} \Delta S_\varphi|_{\varphi+\Delta \varphi} \Delta t$ 。

以上各式中 $\Delta S_r|_{r=a} = a^2 \sin \theta \Delta \theta \Delta \varphi$ ， $\Delta S_r|_{r=a-\Delta r} = (a - \Delta r)^2 \sin \theta \Delta \theta \Delta \varphi$ ，

$\Delta S_\theta|_\theta = a \sin \theta \Delta r \Delta \varphi$ ， $\Delta S_\theta|_{\theta+\Delta \theta} = a \sin(\theta + \Delta \theta) \Delta r \Delta \varphi$ ， $\Delta S_\varphi|_\varphi = \Delta S_\varphi|_{\varphi+\Delta \varphi} = a \Delta r \Delta \theta$ 。

该体积元内增加热量为 $\rho c \Delta V \Delta u = \rho c a^2 \sin \theta \Delta r \Delta \theta \Delta \varphi \Delta u$ ，由能量守恒可得

$$\begin{aligned}
 & M \left(\Delta S_r|_{r=a} \cos \theta \right) \Delta t - H u|_{r=a} \Delta S_r|_{r=a} \Delta t - k \frac{\partial u}{\partial r} \Big|_{r=a-\Delta r} \Delta S_r|_{r=a-\Delta r} \Delta t \\
 & - \frac{k}{a} \frac{\partial u}{\partial \theta} \Big|_\theta \Delta S_\theta|_\theta \Delta t + \frac{k}{a} \frac{\partial u}{\partial \theta} \Big|_{\theta+\Delta \theta} \Delta S_\theta|_{\theta+\Delta \theta} \Delta t - \frac{k}{a \sin \theta} \frac{\partial u}{\partial \varphi} \Big|_\varphi \Delta S_\varphi|_\varphi \Delta t \\
 & + \frac{k}{a \sin \theta} \frac{\partial u}{\partial \varphi} \Big|_{\varphi+\Delta \varphi} \Delta S_\varphi|_{\varphi+\Delta \varphi} \Delta t = \rho c a^2 \sin \theta \Delta r \Delta \theta \Delta \varphi \Delta u
 \end{aligned}$$

化简得

$$\begin{aligned} & Ma^2 \sin^2 \theta \cos \theta - H u \Big|_{r=a} a^2 \sin^2 \theta - k \frac{\partial u}{\partial r} \Big|_{r=a-\Delta r} (a - \Delta r)^2 \sin^2 \theta \\ & + k \sin \theta \Delta r \frac{1}{\Delta \theta} \left[\sin(\theta + \Delta \theta) \frac{\partial u}{\partial \theta} \Big|_{\theta+\Delta \theta} - \sin \theta \frac{\partial u}{\partial \theta} \Big|_{\theta} \right] + k \Delta r \frac{1}{\Delta \varphi} \left(\frac{\partial u}{\partial \varphi} \Big|_{\varphi+\Delta \varphi} - \frac{\partial u}{\partial \varphi} \Big|_{\varphi} \right) \\ & = \rho c a^2 \sin^2 \theta \Delta r \frac{\Delta u}{\Delta t} . \end{aligned}$$

$$\text{令 } \Delta r, \Delta \theta, \Delta \varphi, \Delta t \rightarrow 0, \text{ 因为 } \frac{1}{\Delta \theta} \left[\sin(\theta + \Delta \theta) \frac{\partial u}{\partial \theta} \Big|_{\theta+\Delta \theta} - \sin \theta \frac{\partial u}{\partial \theta} \Big|_{\theta} \right] \rightarrow \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right),$$

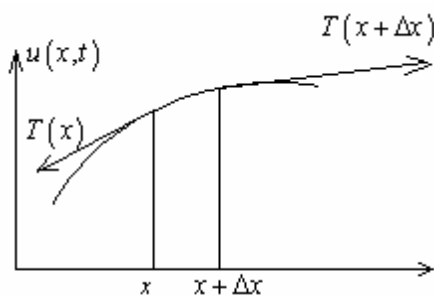
$$\frac{1}{\Delta \varphi} \left(\frac{\partial u}{\partial \varphi} \Big|_{\varphi+\Delta \varphi} - \frac{\partial u}{\partial \varphi} \Big|_{\varphi} \right) \rightarrow \frac{\partial^2 u}{\partial \varphi^2}, \quad \frac{\Delta u}{\Delta t} \rightarrow \frac{\partial u}{\partial t} \text{ 都是有限值, 所以有}$$

$$\left(\frac{\partial u}{\partial r} + \frac{H}{k} u \right) \Big|_{r=a} = \frac{M}{k} \cos \theta .$$

上面的讨论适用于 $0 \leq \theta \leq \pi/2$ 的情况, 即有光线照射到的范围, 对于 $\pi/2 < \theta \leq \pi$, 只需

$$\text{令上式 } M = 0 \text{ 即可, 即 } \left(\frac{\partial u}{\partial r} + \frac{H}{k} u \right) \Big|_{r=a} = \begin{cases} \frac{M}{k} \cos \theta, & 0 \leq \theta \leq \pi/2 \\ 0, & \pi/2 < \theta \leq \pi \end{cases}$$

209. 一完全柔软的均匀细线, 重力可忽略。一端 ($x=0$) 固定在匀速转动的轴上, 角速度为 ω , 另一端 ($x=l$) 自由。由于惯性离心力的作用, 此细线的平衡位置为水平线。试推导细线相对于其平衡位置作横振动的振动方程。



取长为 Δx 的一小段, 水平方向 (纵向) 由牛顿第二定律及向心加速度公式可得

$$T(x) - T(x + \Delta x) = \rho \Delta x \omega^2 x, \text{ 两边同除 } \Delta x \text{ 并令 } \Delta x \rightarrow 0 \text{ 得 } T'(x) = -\rho \omega^2 x .$$

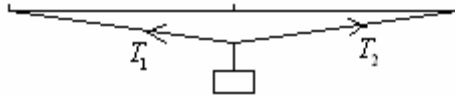
$$\text{将上式积分, 并由 } T(l) = 0 \text{ 可得 } T(x) = \frac{1}{2} \rho \omega^2 (l^2 - x^2) .$$

$$\text{垂直方向 (横向) 可列出牛顿方程 } T(x + \Delta x) \sin \theta \Big|_{x+\Delta x} - T(x) \sin \theta \Big|_x = \rho \Delta x \frac{\partial^2 u}{\partial t^2} \Big|_{x+\alpha \Delta x} .$$

由小振动近似, $\sin \theta \approx \tan \theta \approx \frac{\partial u}{\partial x}$, 代入上式, 两边同除 Δx 并令 $\Delta x \rightarrow 0$ 可得

$$\frac{\partial}{\partial x} \left[T(x) \frac{\partial u}{\partial x} \right] = \rho \frac{\partial^2 u}{\partial t^2}, \text{ 代入 } T(x) \text{ 表达式即可得 } \frac{\partial}{\partial x} \left[(l^2 - x^2) \frac{\partial u}{\partial x} \right] - \frac{2}{\omega^2} \frac{\partial^2 u}{\partial t^2} = 0。$$

210. 一长为 l 的水平均匀弹性弦 (两端固定), 中点处悬一重物, 质量为 M 。试列出弦的横振动方程, 边界条件以及连接条件。设悬线的质量及弹性形变均可忽略。



显然有 $u(x, t) \Big|_{x=\frac{l}{2}-0} = u(x, t) \Big|_{x=\frac{l}{2}+0}$ 。

由于重物没有水平方向的运动, 所以 $T_1 \cos \theta_1 = T_2 \cos \theta_2$ 。由于 $\theta_1 \approx 0, \theta_2 \approx 0$, 所以 $T_1 = T_2$

(记为 T)。垂直方向有 $T_1 \sin \theta_1 + T_2 \sin \theta_2 - Mg = M \frac{\partial^2 u}{\partial t^2} \Big|_{x=\frac{l}{2}}$, 由于 $\sin \theta_1 \approx -\frac{\partial u}{\partial x} \Big|_{x=\frac{l}{2}-0}$,

$$\sin \theta_2 \approx \frac{\partial u}{\partial x} \Big|_{x=\frac{l}{2}+0}, \text{ 所以 } \frac{\partial u}{\partial x} \Big|_{x=\frac{l}{2}+0} - \frac{\partial u}{\partial x} \Big|_{x=\frac{l}{2}-0} = \frac{M}{T} \left(\frac{\partial^2 u}{\partial t^2} \Big|_{x=\frac{l}{2}} + g \right)。$$

$$\text{该定解问题为 } \begin{cases} \frac{\partial^2 u}{\partial x^2} - \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2} = 0 \\ u \Big|_{x=0} = 0, u \Big|_{x=l} = 0 \\ u(x, t) \Big|_{x=\frac{l}{2}-0} = u(x, t) \Big|_{x=\frac{l}{2}+0} \\ \frac{\partial u}{\partial x} \Big|_{x=\frac{l}{2}+0} - \frac{\partial u}{\partial x} \Big|_{x=\frac{l}{2}-0} = \frac{M}{T} \left(\frac{\partial^2 u}{\partial t^2} \Big|_{x=\frac{l}{2}} + g \right) \end{cases}。$$

211. 将下列方程分离变量: (1) $a_1(x) \frac{\partial^2 u}{\partial x^2} + b_1(y) \frac{\partial^2 u}{\partial y^2} + a_2(x) \frac{\partial u}{\partial x} + b_2(y) \frac{\partial u}{\partial y} = 0$;

(2) $\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = 0$; (3) $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) = 0$;

(4) $\frac{\partial}{\partial \alpha} \left(\frac{\sin \alpha}{\operatorname{ch} \beta - \cos \alpha} \frac{\partial u}{\partial \alpha} \right) + \frac{\partial}{\partial \beta} \left(\frac{\sin \alpha}{\operatorname{ch} \beta - \cos \alpha} \frac{\partial u}{\partial \beta} \right) + \frac{1}{\sin \alpha (\operatorname{ch} \beta - \cos \alpha)} \frac{\partial^2 u}{\partial \varphi^2} = 0$ 。

(1) 设 $u(x, y) = X(x)Y(y)$, 代入方程得:

$$a_1(x) X''(x) Y(y) + b_1(y) X(x) Y''(y) + a_2(x) X'(x) Y(y) + b_2(y) X(x) Y'(y) = 0。$$

两边同除 $X(x)Y(y)$ 得 $\frac{a_1(x) X''(x) + a_2(x) X'(x)}{X(x)} = -\frac{b_1(y) Y''(y) + b_2(y) Y'(y)}{Y(y)}$ 。

令两边等于 λ , 则 $\begin{cases} a_1(x) X''(x) + a_2(x) X'(x) - \lambda X(x) = 0 \\ b_1(y) Y''(y) + b_2(y) Y'(y) + \lambda Y(y) = 0 \end{cases}$ 。

(2) 设 $u(\rho, \varphi) = P(\rho)\Phi(\varphi)$, 代入方程得: $\frac{\Phi(\varphi)}{\rho} \frac{d}{d\rho} \left[\rho \frac{dP(\rho)}{d\rho} \right] + \frac{P(\rho)}{\rho^2} \frac{d^2 \Phi(\varphi)}{d\varphi^2} = 0$ 。

两边同乘 $\frac{\rho^2}{P(\rho)\Phi(\varphi)}$ 得 $\frac{\rho}{P(\rho)} \frac{d}{d\rho} \left[\rho \frac{dP(\rho)}{d\rho} \right] = -\frac{1}{\Phi(\varphi)} \frac{d^2 \Phi(\varphi)}{d\varphi^2}$,

令两边等于 λ , 则 $\begin{cases} \frac{1}{\rho} \frac{d}{d\rho} \left[\rho \frac{dP(\rho)}{d\rho} \right] - \frac{\lambda}{\rho^2} P(\rho) = 0 \\ \frac{d^2 \Phi(\varphi)}{d\varphi^2} + \lambda \Phi(\varphi) = 0 \end{cases}$ 。

(3) 设 $u(r, \theta) = R(r)\Theta(\theta)$, 代入方程得

$$\frac{\Theta(\theta)}{r^2} \frac{d}{dr} \left[r^2 \frac{dR(r)}{dr} \right] + \frac{R(r)}{r^2 \sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d\Theta(\theta)}{d\theta} \right] = 0。 两边同乘 $\frac{r^2}{R(r)\Theta(\theta)}$ 得$$

$$\frac{1}{R(r)} \frac{d}{dr} \left[r^2 \frac{dR(r)}{dr} \right] = -\frac{1}{\Theta(\theta) \sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d\Theta(\theta)}{d\theta} \right]。 令两边等于 λ 得$$

$$\begin{cases} \frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{dR(r)}{dr} \right] - \frac{\lambda}{r^2} R(r) = 0 \\ \frac{1}{\sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d\Theta(\theta)}{d\theta} \right] + \lambda \Theta(\theta) = 0 \end{cases}。$$

(4) 令 $u(\alpha, \beta, \varphi) = v(\alpha, \beta, \varphi) \sqrt{\text{ch } \beta - \cos \alpha}$, 则 $\frac{\partial^2 u}{\partial \varphi^2} = \sqrt{\text{ch } \beta - \cos \alpha} \frac{\partial^2 v}{\partial \varphi^2}$,

$$\frac{\partial u}{\partial \alpha} = \sqrt{\text{ch } \beta - \cos \alpha} \frac{\partial v}{\partial \alpha} + \frac{\sin \alpha}{2\sqrt{\text{ch } \beta - \cos \alpha}} v, \quad \frac{\partial u}{\partial \beta} = \sqrt{\text{ch } \beta - \cos \alpha} \frac{\partial v}{\partial \beta} + \frac{\text{sh } \beta}{2\sqrt{\text{ch } \beta - \cos \alpha}} v,$$

代入方程得

$$\begin{aligned} \frac{\partial}{\partial \alpha} \left[\frac{\sin \alpha}{\sqrt{\text{ch } \beta - \cos \alpha}} \frac{\partial v}{\partial \alpha} + \frac{\sin^2 \alpha}{2(\text{ch } \beta - \cos \alpha)^{3/2}} v \right] + \frac{\partial}{\partial \beta} \left[\frac{\sin \alpha}{\sqrt{\text{ch } \beta - \cos \alpha}} \frac{\partial v}{\partial \beta} + \frac{\sin \alpha \text{sh } \beta}{2(\text{ch } \beta - \cos \alpha)^{3/2}} v \right] \\ + \frac{1}{\sin \alpha \sqrt{\text{ch } \beta - \cos \alpha}} \frac{\partial^2 v}{\partial \varphi^2} = 0 \end{aligned}$$

展开得

$$\begin{aligned} \frac{2 \cos \alpha \text{ch } \beta - \cos^2 \alpha - 1}{2(\text{ch } \beta - \cos \alpha)^{3/2}} \frac{\partial v}{\partial \alpha} + \frac{\sin \alpha}{\sqrt{\text{ch } \beta - \cos \alpha}} \frac{\partial^2 v}{\partial \alpha^2} \\ + \frac{4 \sin \alpha \cos \alpha \text{ch } \beta - \sin \alpha \cos^2 \alpha - 3 \sin \alpha}{4(\text{ch } \beta - \cos \alpha)^{5/2}} v + \frac{\sin^2 \alpha}{2(\text{ch } \beta - \cos \alpha)^{3/2}} \frac{\partial v}{\partial \alpha} \\ - \frac{\sin \alpha \text{sh } \beta}{2(\text{ch } \beta - \cos \alpha)^{3/2}} \frac{\partial v}{\partial \beta} + \frac{\sin \alpha}{\sqrt{\text{ch } \beta - \cos \alpha}} \frac{\partial^2 v}{\partial \beta^2} \\ + \frac{3 \sin \alpha - \sin \alpha \text{ch}^2 \beta - 2 \sin \alpha \cos \alpha \text{ch } \beta}{4(\text{ch } \beta - \cos \alpha)^{5/2}} v + \frac{\sin \alpha \text{sh } \beta}{2(\text{ch } \beta - \cos \alpha)^{3/2}} \frac{\partial v}{\partial \beta} \\ + \frac{1}{\sin \alpha \sqrt{\text{ch } \beta - \cos \alpha}} \frac{\partial^2 v}{\partial \varphi^2} = 0 \end{aligned}$$

化简得 $\sin^2 \alpha \frac{\partial^2 v}{\partial \alpha^2} + \sin \alpha \cos \alpha \frac{\partial v}{\partial \alpha} + \sin^2 \alpha \frac{\partial^2 v}{\partial \beta^2} + \frac{\partial^2 v}{\partial \varphi^2} - \frac{1}{4} \sin^2 \alpha v = 0$ 。

设 $v(\alpha, \beta, \varphi) = A(\alpha)B(\beta)\Phi(\varphi)$, 代入上式, 两边同除 $A(\alpha)B(\beta)\Phi(\varphi)$ 得

$$\frac{\sin \alpha}{A(\alpha)} \frac{d}{d\alpha} \left[\sin \alpha \frac{dA(\alpha)}{d\alpha} \right] - \frac{1}{4} \sin^2 \alpha + \frac{\sin^2 \alpha}{B(\beta)} \frac{dB(\beta)}{d\beta} = -\frac{1}{\Phi(\varphi)} \frac{d\Phi(\varphi)}{d\varphi}.$$

令上式两边等于 μ , 则

$$\begin{cases} \frac{1}{\sin \alpha A(\alpha)} \frac{d}{d\alpha} \left[\sin \alpha \frac{dA(\alpha)}{d\alpha} \right] - \frac{\mu}{\sin^2 \alpha} = \frac{1}{4} - \frac{1}{B(\beta)} \frac{dB(\beta)}{d\beta} \\ \frac{d\Phi(\varphi)}{d\varphi} + \mu \Phi(\varphi) = 0 \end{cases}.$$

令上面第一式两边等于 $-\lambda$, 则

$$\begin{cases} \frac{1}{\sin \alpha} \frac{d}{d\alpha} \left[\sin \alpha \frac{dA(\alpha)}{d\alpha} \right] + \left(\lambda - \frac{\mu}{\sin^2 \alpha} \right) A(\alpha) = 0 \\ \frac{dB(\beta)}{d\beta} + \left(\lambda + \frac{1}{4} \right) B(\beta) = 0 \\ \frac{d\Phi(\varphi)}{d\varphi} + \mu \Phi(\varphi) = 0 \end{cases}。$$

212. 求解下列各本征值问题, 证明各题中本征函数的正交性, 并算出归一因子。

$$\begin{aligned} (1) & \begin{cases} X'' + \lambda X = 0 \\ X(0) = 0, X(l) = 0 \end{cases}; \quad (2) \begin{cases} X'' + \lambda X = 0 \\ X(0) = 0, X'(l) = 0 \end{cases}; \quad (3) \begin{cases} X'' + \lambda X = 0 \\ X'(0) = 0, X'(l) = 0 \end{cases}; \\ (4) & \begin{cases} X'' + \lambda X = 0 \\ X(a) = 0, X(b) = 0 \end{cases}; \quad (5) \begin{cases} X'' + \lambda X = 0 \\ X(0) = 0 \\ \alpha X(l) + \beta X'(l) = 0 \end{cases}; \quad (6) \begin{cases} X'' + \lambda X = 0 \\ \alpha_1 X(0) + \beta_1 X'(0) = 0 \\ \alpha_2 X(l) + \beta_2 X'(l) = 0 \end{cases}。 \end{aligned}$$

(1) 方程两边同乘 X , 并对 x 积分得

$$\lambda \int_0^l X^2(x) dx = - \int_0^l X''(x) X(x) dx = -X'(x) X(x) \Big|_0^l + \int_0^l X'^2(x) dx = \int_0^l X'^2(x) dx。$$

$$\text{当 } X(x) \neq 0 \text{ 时, } \lambda = \frac{\int_0^l X'^2(x) dx}{\int_0^l X^2(x) dx} \geq 0。 \text{ 当 } \lambda = 0 \text{ 时, 即 } X'' = 0, \text{ 则解为 } X(x) = ax + b,$$

代入边界条件得 $X(x) = 0$, 所以只有 $\lambda > 0$ 。

$$\lambda > 0 \text{ 时解为 } X(x) = a \sin \sqrt{\lambda} x + b \cos \sqrt{\lambda} x, \text{ 由边界条件可得 } b = 0, \lambda = \left(\frac{n\pi}{l} \right)^2。$$

$$\text{所以对应本征值 } \lambda_n = \left(\frac{n\pi}{l} \right)^2 \text{ 的本征函数为 } X_n(x) = \sin \frac{n\pi}{l} x \quad (n = 1, 2, \dots)。$$

设两对本征值和本征函数为 (λ_1, X_1) , (λ_2, X_2) , $(\lambda_1 \neq \lambda_2)$ 即

$$\begin{cases} X_1'' + \lambda_1 X_1 = 0 \\ X_1(0) = 0, X_1(l) = 0 \end{cases}, \quad \begin{cases} X_2'' + \lambda_2 X_2 = 0 \\ X_2(0) = 0, X_2(l) = 0 \end{cases}。$$

令第一个方程两边同乘 X_2 减去第二个方程两边同乘 X_1 , 并对 x 积分得

$$\begin{aligned} (\lambda_1 - \lambda_2) \int_0^l X_1(x) X_2(x) dx &= \int_0^l X_1(x) X_2''(x) - X_1''(x) X_2(x) dx \\ &= X_1(x) X_2'(x) \Big|_0^l - X_1'(x) X_2(x) \Big|_0^l = 0。 \end{aligned}$$

由于 $\lambda_1 - \lambda_2 \neq 0$ ，所以 $\int_0^l X_1(x) X_2(x) dx = 0$ ，即不同本征值的本征函数正交。

$$\int_0^l X_n^2(x) dx = \int_0^l \sin^2 \frac{n\pi}{l} x dx = \frac{l}{2}, \text{ 所以归一化因子为 } \sqrt{\frac{2}{l}}.$$

$$(2) \quad \lambda_n = \left(\frac{2n+1}{2l} \pi \right)^2 \quad (n=0,1,2,\dots), \quad X_n(x) = \sin \frac{2n+1}{2l} \pi x, \text{ 归一因子 } \sqrt{\frac{2}{l}}.$$

$$(3) \text{ 此时本征值 } \lambda \text{ 可取 } 0. \lambda > 0 \text{ 时可求出 } \lambda_n = \left(\frac{n\pi}{l} \right)^2 \quad (n=1,2,\dots), \quad X_n(x) = \cos \frac{n\pi}{l} x$$

$$(n=1,2,\dots). \lambda = 0 \text{ 时可求出 } X_0(x) = 1, \text{ 因此本征值为 } \lambda_n = \left(\frac{n\pi}{l} \right)^2 \quad (n=0,1,2,\dots),$$

$$\text{本征函数 } X_n(x) = \cos \frac{n\pi}{l} x \quad (n=0,1,2,\dots).$$

$$\int_0^l X_n^2(x) dx = \begin{cases} l, n=0 \\ \frac{l}{2}, n>0 \end{cases}, \text{ 所以归一因子为 } \sqrt{\frac{2}{l(1+\delta_{n,0})}}.$$

$$(4) \text{ 解为 } X(x) = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x, \text{ 代入边界条件得}$$

$$C_1 \sin \sqrt{\lambda} a + C_2 \cos \sqrt{\lambda} a = 0, \quad C_1 \sin \sqrt{\lambda} b + C_2 \cos \sqrt{\lambda} b = 0,$$

$$\text{写成矩阵形式为 } \begin{pmatrix} \sin \sqrt{\lambda} a & \cos \sqrt{\lambda} a \\ \sin \sqrt{\lambda} b & \cos \sqrt{\lambda} b \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = 0,$$

$$\text{要使 } \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} \text{ 有非零解, 应有 } \begin{vmatrix} \sin \sqrt{\lambda} a & \cos \sqrt{\lambda} a \\ \sin \sqrt{\lambda} b & \cos \sqrt{\lambda} b \end{vmatrix} = \sin \sqrt{\lambda} (a-b) = 0,$$

$$\text{即本征值为 } \lambda_n = \left(\frac{n\pi}{b-a} \right)^2 \quad (n=1,2,\dots), \text{ 将 } C_2 = -\frac{\sin \sqrt{\lambda_n} a}{\cos \sqrt{\lambda_n} a} C_1 \text{ 代入 } X(x) \text{ 表达式可得本}$$

$$\text{征函数 } X_n(x) = \sin \frac{n\pi(x-a)}{b-a}. \text{ 归一化因子 } \sqrt{\frac{2}{b-a}}.$$

$$(5) \text{ 解为 } X(x) = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x, \text{ 代入边界条件得 } C_2 = 0,$$

$$\tan \sqrt{\lambda} l + \frac{\beta}{\alpha} \sqrt{\lambda} = 0, \text{ 本征值 } \lambda_n \text{ 是左边方程的第 } n \text{ } (=1,2,\dots) \text{ 个正根, 对应的本征函数}$$

$$\text{为 } X_n(x) = \sin \sqrt{\lambda_n} x.$$

设本征值 λ_1 对应本征函数 X_1 ，本征值 λ_2 ($\neq \lambda_1$) 对应本征函数 X_2 ，则

$$\begin{aligned}
(\lambda_1 - \lambda_2) \int_0^l X_1(x) X_2(x) dx &= \int_0^l [X_1(x) X_2''(x) - X_1''(x) X_2(x)] dx \\
&= X_1(x) X_2'(x) \Big|_0^l - X_1'(x) X_2(x) \Big|_0^l = X_1(l) X_2'(l) - X_1'(l) X_2(l) \\
&= -\frac{\alpha}{\beta} X_1(l) X_2(l) + \frac{\alpha}{\beta} X_1(l) X_2(l) = 0,
\end{aligned}$$

即不同本征值的本征函数是正交的。

$$\int_0^l X_n^2(x) dx = \frac{l}{2} - \frac{1}{2} \int_0^l \cos 2\sqrt{\lambda_n} x dx = \frac{l}{2} - \frac{1}{4\sqrt{\lambda_n}} \sin 2\sqrt{\lambda_n} l = \frac{l}{2} - \frac{1}{2\sqrt{\lambda_n}} \frac{\tan \sqrt{\lambda_n} l}{1 + \tan^2 \sqrt{\lambda_n} l},$$

$$\text{代入 } \tan \sqrt{\lambda_n} l = -\frac{\beta}{\alpha} \sqrt{\lambda_n}, \text{ 则上式} = \frac{l}{2} + \frac{1}{2} \frac{\alpha\beta}{\alpha^2 + \beta^2 \lambda_n}, \text{ 所以归一化因子为 } \sqrt{\frac{1}{\frac{l}{2} + \frac{\alpha\beta}{2(\alpha^2 + \beta^2 \lambda_n)}}}.$$

(6) 解为 $X(x) = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x$, 代入边界条件得

$$\begin{cases} \alpha_1 C_2 + \beta_1 C_1 \sqrt{\lambda} = 0 \\ \alpha_2 C_1 \sin \sqrt{\lambda} l + \alpha_2 C_2 \cos \sqrt{\lambda} l + \beta_2 C_1 \sqrt{\lambda} \cos \sqrt{\lambda} l - \beta_2 C_2 \sqrt{\lambda} \sin \sqrt{\lambda} l = 0 \end{cases}$$

由第一式得 $C_2 = -C_1 \sqrt{\lambda} \frac{\beta_1}{\alpha_1}$, 代入第二式得

$$(\alpha_1 \alpha_2 + \beta_1 \beta_2 \lambda) \tan \sqrt{\lambda} l + (\alpha_1 \beta_2 - \alpha_2 \beta_1) \sqrt{\lambda} = 0. \text{ 本征值 } \lambda_n \text{ 为该方程的第 } n \text{ 个正根。}$$

$$\text{将 } C_2 = -C_1 \sqrt{\lambda_n} \frac{\beta_1}{\alpha_1} \text{ 代入 } X(x) \text{ 表达式为 } \sin \sqrt{\lambda_n} x - \frac{\beta_1}{\alpha_1} \sqrt{\lambda_n} \cos \sqrt{\lambda_n} x,$$

本征函数可表示为 $X_n(x) = \sin(\sqrt{\lambda_n} x - \delta_n)$, 其中 $\tan \delta_n = \frac{\beta_1}{\alpha_1} \sqrt{\lambda_n}$ 。

$$\int_0^l X_n^2(x) dx = \frac{l}{2} - \frac{1}{4\sqrt{\lambda_n}} [\sin 2(\sqrt{\lambda_n} l - \delta_n) + \sin 2\delta_n].$$

由 $\tan \sqrt{\lambda_n} l = -\sqrt{\lambda_n} \frac{\alpha_1 \beta_2 - \alpha_2 \beta_1}{\alpha_1 \alpha_2 + \beta_1 \beta_2 \lambda}$ 可得:

$$\sin 2\sqrt{\lambda_n} l = \frac{2 \tan \sqrt{\lambda_n} l}{1 + \tan^2 \sqrt{\lambda_n} l} = -2\sqrt{\lambda_n} \frac{\alpha_1^2 \alpha_2 \beta_2 - \alpha_1 \alpha_2^2 \beta_1 + \alpha_1 \beta_1 \beta_2^2 \lambda_n - \alpha_2 \beta_1^2 \beta_2 \lambda_n}{(\alpha_1^2 + \beta_1^2 \lambda_n)(\alpha_2^2 + \beta_2^2 \lambda_n)},$$

$$\cos 2\sqrt{\lambda_n} l = \frac{1 - \tan^2 \sqrt{\lambda_n} l}{1 + \tan^2 \sqrt{\lambda_n} l} = \frac{\alpha_1^2 \alpha_2^2 - \alpha_1^2 \beta_2^2 \lambda_n - \alpha_2^2 \beta_1^2 \lambda_n + 4\alpha_1 \alpha_2 \beta_1 \beta_2 \lambda_n + \beta_1^2 \beta_2^2 \lambda_n^2}{(\alpha_1^2 + \beta_1^2 \lambda_n)(\alpha_2^2 + \beta_2^2 \lambda_n)},$$

$$\sin 2\delta_n = \frac{2 \tan \delta_n}{1 + \tan^2 \delta_n} = 2\sqrt{\lambda_n} \frac{\alpha_1 \beta_1}{\alpha_1^2 + \beta_1^2 \lambda_n}, \quad \cos 2\delta_n = \frac{1 - \tan^2 \delta_n}{1 + \tan^2 \delta_n} = \frac{\alpha_1^2 - \beta_1^2 \lambda_n}{\alpha_1^2 + \beta_1^2 \lambda_n},$$

$$\sin 2(\sqrt{\lambda_n} l - \delta_n) = \sin 2\sqrt{\lambda_n} l \cos 2\delta_n - \cos 2\sqrt{\lambda_n} l \sin 2\delta_n = -2\sqrt{\lambda_n} \frac{\alpha_2 \beta_2}{\alpha_2^2 + \beta_2^2 \lambda_n}.$$

$$\text{所以 } \int_0^l X_n^2(x) dx = \frac{l}{2} + \frac{1}{2} \left(\frac{\alpha_2 \beta_2}{\alpha_2^2 + \beta_2^2 \lambda_n} - \frac{\alpha_1 \beta_1}{\alpha_1^2 + \beta_1^2 \lambda_n} \right),$$

$$\text{所以归一因子为 } \sqrt{\frac{1}{\frac{l}{2} + \frac{1}{2} \left(\frac{\alpha_2 \beta_2}{\alpha_2^2 + \beta_2^2 \lambda_n} - \frac{\alpha_1 \beta_1}{\alpha_1^2 + \beta_1^2 \lambda_n} \right)}}.$$

213. 如果我们采用最小二乘法用 $\sum_{n=1}^N a_n \sin \frac{n\pi}{l} x$ 去逼近函数 $f(x)$, $f(x) \approx \sum_{n=1}^N a_n \sin \frac{n\pi}{l} x$,

即要求均方误差 $\varepsilon = \int_0^l \left[f(x) - \sum_{n=1}^N a_n \sin \frac{n\pi}{l} x \right]^2 dx$ 取极小, 试确定展开系数 a_n 。

令 ε 对 $a_1, a_2, \dots, a_N$ 的偏导为零:

$$\frac{\partial \varepsilon}{\partial a_k} = -2 \int_0^l \left[f(x) - \sum_{n=1}^N a_n \sin \frac{n\pi}{l} x \right] \sin \frac{k\pi}{l} x dx = 0 \quad (k=1, 2, \dots, N), \text{ 即}$$

$$\sum_{n=1}^N a_n \int_0^l \sin \frac{n\pi}{l} x \sin \frac{k\pi}{l} x dx = \int_0^l f(x) \sin \frac{k\pi}{l} x dx. \text{ 由 } \left\{ \sin \frac{k\pi}{l} x \right\} \text{ 的正交性可得}$$

$$a_k = \frac{2}{l} \int_0^l f(x) \sin \frac{k\pi}{l} x dx.$$

$$214. \text{ 解第 201 题。} \begin{cases} \frac{\partial^2 u}{\partial x^2} - \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2} = 0 \\ u|_{x=0} = 0, \frac{\partial u}{\partial x} \Big|_{x=l} = 0 \\ u|_{t=0} = \frac{F}{ES} x, \frac{\partial u}{\partial t} \Big|_{t=0} = 0 \end{cases}$$

$$\text{设 } u(x, t) = X(x)T(t), \text{ 代入方程得 } \frac{X''(x)}{X(x)} = \frac{1}{a^2} \frac{T''(t)}{T(t)} = -\lambda.$$

本征值问题为 $\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = 0, X'(l) = 0 \end{cases}$ 。212 题第 (2) 小题已解出该本征值问题：

$$\lambda_n = \left(\frac{2n+1}{2l} \pi \right)^2, \quad X_n(x) = \sin \frac{2n+1}{2l} \pi x。$$

$$\text{解出 } T_n(t) = A_n \sin \frac{2n+1}{2l} a \pi t + B_n \cos \frac{2n+1}{2l} a \pi t，$$

$$u = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} \left(A_n \sin \frac{2n+1}{2l} a \pi t + B_n \cos \frac{2n+1}{2l} a \pi t \right) \sin \frac{2n+1}{2l} \pi x。$$

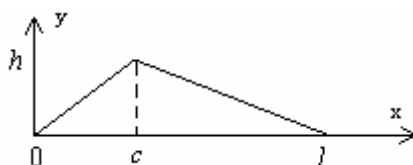
$$\text{由 } u|_{t=0} = \sum_{n=1}^{\infty} B_n \sin \frac{2n+1}{2l} \pi x = \frac{F}{ES} x \text{ 可定出}$$

$$B_n = \frac{2F}{lES} \int_0^l x \sin \frac{2n+1}{l} \pi x dx = \frac{8Fl}{ES\pi^2} \frac{(-1)^n}{(2n+1)^2}。$$

$$\text{由 } \frac{\partial u}{\partial t} \Big|_{t=0} = \sum_{n=1}^{\infty} A_n \frac{2n+1}{2l} a \pi \sin \frac{2n+1}{2l} \pi x = 0 \text{ 可定出 } A_n = 0。$$

$$\text{所以 } u(x, t) = \frac{8Fl}{ES\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)^2} \cos \frac{2n+1}{2l} a \pi t \sin \frac{2n+1}{2l} \pi x。$$

215. 一长为 l ，两端固定的均匀弦，初始时，弦被拉开，待达到平衡后突然放手。试求解此问题。



方程与上题同，边界条件为 $u|_{x=0} = 0$ ， $u|_{x=l} = 0$ 。

$$\text{初始条件为: } u|_{t=0} = \begin{cases} \frac{h}{c} x, & 0 \leq x \leq c \\ \frac{h(l-x)}{l-c}, & c \leq x \leq l \end{cases}, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = 0。$$

本征函数为 $X_n(x) = \sin \frac{n\pi}{l} x$ ($n=1, 2, \dots$)，解出 $T_n(t) = A_n \sin \frac{n\pi}{l} at + B_n \cos \frac{n\pi}{l} at$ ，

$$u = \sum_{n=1}^{\infty} \left(A_n \sin \frac{n\pi}{l} at + B_n \cos \frac{n\pi}{l} at \right) \sin \frac{n\pi}{l} x。$$

由初始条件定出 $B_n = \frac{2hl^2}{c(l-c)\pi^2} \frac{1}{n^2} \sin \frac{n\pi}{l} c$, $A_n = 0$,

所以 $u(x, t) = \frac{2hl^2}{c(l-c)\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{l} c \cos \frac{n\pi}{l} at \sin \frac{n\pi}{l} x$ 。

216. 两端固定的均匀弦, 在硬质平锤的打击下以如下初速度分布振动:

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = \begin{cases} 0, 0 \leq x < c - \delta \\ v_0, c - \delta < x < c + \delta \\ 0, c + \delta < x \leq l \end{cases} \text{。若初位移为 } 0, \text{ 求解弦的横振动。}$$

仍有 $u = \sum_{n=1}^{\infty} \left(A_n \sin \frac{n\pi}{l} at + B_n \cos \frac{n\pi}{l} at \right) \sin \frac{n\pi}{l} x$ 。

$$u|_{t=0} = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi}{l} x = 0, \text{ 所以 } B_n = 0,$$

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = \sum_{n=1}^{\infty} A_n \frac{n\pi}{l} a \sin \frac{n\pi}{l} x, \text{ 所以}$$

$$A_n = \frac{2}{n\pi a} \int_0^l \left. \frac{\partial u}{\partial t} \right|_{t=0} \sin \frac{n\pi}{l} x dx = \frac{4lv_0}{n^2 \pi^2 a} \sin \frac{n\pi}{l} c \sin \frac{n\pi}{l} \delta。$$

所以 $u(x, t) = \frac{4lv_0}{\pi^2 a} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{l} c \sin \frac{n\pi}{l} \delta \sin \frac{n\pi}{l} at \sin \frac{n\pi}{l} x$ 。

217. 两端固定的均匀弦, 其 $x = c$ 点受到尖锤的打击而获得冲量 I 。若初位移为 0, 求解弦的自由横振动。

假设冲量 I 均匀分布于 $c - \delta < x < c + \delta$ 上, 由动量定理, $2\rho\delta v_0 = I$ (ρ 是线密度),

所以 $v_0 = \frac{I}{2\rho\delta}$, 代入上题结果,

$$u_{\delta}(x, t) = \frac{2II}{\pi^2 a \rho \delta} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{l} c \sin \frac{n\pi}{l} \delta \sin \frac{n\pi}{l} at \sin \frac{n\pi}{l} x。$$

$$\left| \frac{1}{n^2} \sin \frac{n\pi}{l} c \sin \frac{n\pi}{l} \delta \sin \frac{n\pi}{l} at \sin \frac{n\pi}{l} x \right| \leq \frac{1}{n^2}, \text{ 所以上面的级数是一致收敛的, 令 } \delta \rightarrow 0,$$

则求极限与求和可交换顺序, 即

$$\begin{aligned}\lim_{\delta \rightarrow 0} u_{\delta}(x, t) &= \frac{2I}{\pi^2 a \rho} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{l} c \left(\lim_{\delta \rightarrow 0} \frac{\sin \frac{n\pi}{l} \delta}{\delta} \right) \sin \frac{n\pi}{l} at \sin \frac{n\pi}{l} x \\ &= \frac{2I}{\pi a \rho} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi}{l} c \sin \frac{n\pi}{l} at \sin \frac{n\pi}{l} x.\end{aligned}$$

218. 一长为 $2l$ 的均匀杆, 两端受力作用而分别压缩了 εl 。在 $t=0$ 时撤去外力, 试解杆的纵振动。

以杆的中点为坐标原点, 由于两端自由, 所以边界条件为 $\left. \frac{\partial u}{\partial x} \right|_{x=-l} = 0$, $\left. \frac{\partial u}{\partial x} \right|_{x=l} = 0$ 。

由于杆均匀, 故初始时刻位移是线性形式, 即 $u|_{t=0} = -\varepsilon x$; 初始时静止, 所以 $\left. \frac{\partial u}{\partial t} \right|_{t=0} = 0$ 。

同 212 题第 (4) 小题作法可得本征函数 $X_n(x) = \cos \frac{n\pi}{2l}(x+l)$ 。

$$T_n(t) = A_n \cos \frac{n\pi}{2l} at + B_n \sin \frac{n\pi}{2l} at,$$

$$u = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi}{2l} at + B_n \sin \frac{n\pi}{2l} at \right) \cos \frac{n\pi}{2l}(x+l)。$$

由 $\left. \frac{\partial u}{\partial t} \right|_{t=0} = 0$ 可得 $B_n = 0$, 由 $u|_{t=0} = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{2l}(x+l) = -\varepsilon x$ 可得

$$A_n = \frac{-\varepsilon}{l} \int_{-l}^l x \cos \frac{n\pi}{2l}(x+l) dx = \frac{4l\varepsilon}{\pi^2 n^2} [1 - (-1)^n], \text{ 所以 } A_{2k} = 0, \quad A_{2k+1} = \frac{8l\varepsilon}{\pi^2 (2k+1)^2},$$

$$\begin{aligned}\text{所以 } u(x, t) &= \frac{8l\varepsilon}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos \frac{(2k+1)\pi}{2l} at \cos \frac{(2k+1)\pi}{2l}(x+l) \\ &= \frac{8l\varepsilon}{\pi^2} \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{(2k+1)^2} \cos \frac{(2k+1)\pi}{2l} at \sin \frac{(2k+1)\pi}{2l} x.\end{aligned}$$

219. 设长为 l 的细杆, $x=0$ 端绝热, 另一端与外界按牛顿冷却定律交换热量, 外界温度为 0。杆身的散热可忽略不计。初始时杆的温度为 u_0 。求杆中温度的分布与变化。

取右端长为 ε 的一小段, 由牛顿冷却定律, Δt 时间内流出该段热量为 $H u|_{x=l} S \Delta t$ (S 为杆

的横截面积), 从内侧流入热量为 $qS\Delta t = -k \frac{\partial u}{\partial x} \Big|_{x=l-\varepsilon} S\Delta t$, 该段内吸收热量为 $\rho c \varepsilon \Delta u$, 由

能量守恒可得 $-H u \Big|_{x=l} S\Delta t - k \frac{\partial u}{\partial x} \Big|_{x=l-\varepsilon} S\Delta t = \rho c \varepsilon \Delta u$, 即 $\frac{\partial u}{\partial x} \Big|_{x=l-\varepsilon} + \frac{H}{k} u \Big|_{x=l} = -\varepsilon \frac{\rho c}{k S} \frac{\Delta u}{\Delta t}$,

令 $\Delta t \rightarrow 0, \varepsilon \rightarrow 0$, 由于 $\frac{\Delta u}{\Delta t} \rightarrow \frac{\partial u}{\partial t}$ 为有限值, 所以 $\frac{\partial u}{\partial x} \Big|_{x=l} + h u \Big|_{x=l} = 0$ ($h = \frac{H}{k}$)。

对于左端, 可认为 $H = 0$, 所以有 $\frac{\partial u}{\partial x} \Big|_{x=0} = 0$ 。

$$\text{即该定解问题为} \begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0 \\ \frac{\partial u}{\partial x} \Big|_{x=0} = 0, \left(\frac{\partial u}{\partial x} + h u \right) \Big|_{x=l} = 0 \\ u \Big|_{t=0} = u_0 \end{cases}$$

令 $u(x, t) = X(x)T(t)$ 可得 $X''(x) + \lambda X(x) = 0$, $T'(t) + \kappa \lambda T(t) = 0$ 。

$$\text{本征值问题为} \begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = 0, X'(l) + h X(l) = 0 \end{cases}。$$

解得本征值 λ_n 是方程 $\sqrt{\lambda} \tan \sqrt{\lambda} l = h$ 的第 n 个正根, 本征函数为 $X_n(x) = \cos \sqrt{\lambda_n} x$ 。

设本征值 λ_1 对应本征函数 X_1 , 本征值 λ_2 对应本征函数 X_2 , 可得

$$\begin{aligned} (\lambda_1 - \lambda_2) \int_0^l X_1(x) X_2(x) dx &= \int_0^l \left[X_1(x) X_2''(x) - X_1''(x) X_2(x) \right] dx \\ &= X_1(x) X_2'(x) \Big|_0^l - X_1'(x) X_2(x) \Big|_0^l = X_1(l) X_2'(l) - X_1'(l) X_2(l) \\ &= -h X_1(l) X_2(l) + h X_1(l) X_2(l) = 0。 \end{aligned}$$

即证明了本征函数的正交性。

解得 $T_n(t) = A_n e^{-\kappa \lambda_n t}$, 则 $u = \sum_{n=1}^{\infty} A_n \cos \sqrt{\lambda_n} x e^{-\kappa \lambda_n t}$ 。

由初始条件 $u_0 = \sum_{n=1}^{\infty} A_n \cos \sqrt{\lambda_n} x$ 及本征函数的正交性有 $A_n = u_0 \frac{\int_0^l \cos \sqrt{\lambda_n} x dx}{\int_0^l \cos^2 \sqrt{\lambda_n} x dx}$ 。

$$\int_0^l \cos \sqrt{\lambda_n} x dx = \frac{1}{\sqrt{\lambda_n}} \sin \sqrt{\lambda_n} l = \frac{(-1)^{n-1}}{\sqrt{\lambda_n}} \frac{\tan \sqrt{\lambda_n} l}{\sqrt{1 + \tan^2 \sqrt{\lambda_n} l}} = (-1)^{n-1} \frac{h}{\sqrt{\lambda_n (\lambda_n + h^2)}},$$

$$\begin{aligned} \int_0^l \cos^2 \sqrt{\lambda_n} x dx &= \frac{l}{2} + \frac{1}{2} \int_0^l \cos 2\sqrt{\lambda_n} x dx = \frac{l}{2} + \frac{1}{4\sqrt{\lambda_n}} \sin 2\sqrt{\lambda_n} l \\ &= \frac{l}{2} + \frac{1}{2\sqrt{\lambda_n}} \frac{\tan \sqrt{\lambda_n} l}{1 + \tan^2 \sqrt{\lambda_n} l} = \frac{1}{2} \left(l + \frac{h}{\lambda_n + h^2} \right). \end{aligned}$$

$$\text{所以 } A_n = (-1)^{n-1} 2hu_0 \frac{1}{\sqrt{\lambda_n (\lambda_n + h^2)}} \frac{1}{\left(l + \frac{h}{\lambda_n + h^2} \right)},$$

$$u(x, t) = 2hu_0 \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{\sqrt{\lambda_n (\lambda_n + h^2)}} \frac{1}{\left(l + \frac{h}{\lambda_n + h^2} \right)} \cos \sqrt{\lambda_n} x e^{-\kappa \lambda_n t}.$$

220. 求解细杆的导热问题, 杆长为 l , 两端 ($x=0$ 及 $x=l$) 均保持为 0 度, 初始温度分布 $u|_{t=0} = b x(l-x)/l^2$ 。

可得本征函数 $X_n(x) = \sin \frac{n\pi}{l} x$, 解出 $T_n(t) = A_n e^{-\kappa \left(\frac{n\pi}{l}\right)^2 t}$, $u = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{l} x e^{-\kappa \left(\frac{n\pi}{l}\right)^2 t}$ 。

代入初始条件, $\sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{l} x = \frac{b}{l^2} x(l-x)$, 所以 $A_n = \frac{2b}{l^3} \int_0^l x(l-x) \sin \frac{n\pi}{l} x dx$

$$= \frac{4b}{\pi^3 n^3} [1 - (-1)^n], \text{ 则 } A_{2k} = 0, \quad A_{2k+1} = \frac{8b}{\pi^3 (2k+1)^3},$$

$$\text{所以 } u(x, t) = \frac{8b}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3} \sin \frac{(2k+1)\pi}{l} x e^{-\kappa \frac{(2k+1)^2 \pi^2}{l^2} t}.$$

$$221. \text{ 求解: } \begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \\ u|_{x=0} = u_0, u|_{x=a} = u_0 y \\ \frac{\partial u}{\partial y} \Big|_{y=0} = 0, \frac{\partial u}{\partial y} \Big|_{y=b} = 0 \end{cases}$$

设 $u(x, y) = X(x)Y(y)$, 则 $\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = \lambda$ 。可得到本征值问题 $\begin{cases} Y''(y) + \lambda Y(y) = 0 \\ Y'(0) = 0, Y'(b) = 0 \end{cases}$ 。

$$\lambda \int_0^b Y^2(y) dy = - \int_0^b Y''(y) Y(y) dy = -Y'(y) Y(y) \Big|_0^b + \int_0^b Y'^2(y) dy = \int_0^b Y'^2(y) dy, \quad \text{当}$$

$$Y(y) \neq 0 \text{ 时有 } \lambda = \frac{\int_0^b Y'^2(y) dy}{\int_0^b Y^2(y) dy} \geq 0。$$

可得本征函数 $Y_n(y) = \cos \frac{n\pi}{b} y$ ($n = 0, 1, 2, \dots$)。

解得 $X_n(x) = Cx + A_0 + A_n \operatorname{ch} \frac{n\pi}{b} x + B_n \operatorname{sh} \frac{n\pi}{b} x$ ($n = 1, 2, \dots$), 其中 $Cx + A_0$ 对应本征值

$$\lambda_0 = 0。 \text{ 则 } u(x, y) = A_0 + Cx + \sum_{n=1}^{\infty} \left(A_n \operatorname{ch} \frac{n\pi}{b} x + B_n \operatorname{sh} \frac{n\pi}{b} x \right) \cos \frac{n\pi}{b} y,$$

由 $u|_{x=0} = u_0$ 可得 $u_0 = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{b} y$, 所以 $A_0 = u_0$, $A_n = 0$ ($n = 1, 2, \dots$)。

由 $u|_{x=a} = u_0 y$ 可得 $u_0 y = u_0 + Ca + \sum_{n=1}^{\infty} B_n \operatorname{sh} \frac{n\pi}{b} a \cos \frac{n\pi}{b} y$,

$$B_n = \frac{2u_0}{b \operatorname{sh} \frac{n\pi}{b} a} \int_0^b (y-1) \cos \frac{n\pi}{b} y dy = \frac{2bu_0}{n^2 \pi^2} \frac{1}{\operatorname{sh} \frac{n\pi}{b} a} [(-1)^n - 1],$$

$$B_{2k} = 0, \quad B_{2k+1} = -\frac{4bu_0}{\pi^2} \frac{1}{(2k+1)^2 \operatorname{sh} \frac{(2k+1)\pi}{b} a}。$$

$$C = \frac{u_0}{ab} \int_0^b (y-1) dy = \frac{b-2}{2a} u_0, \quad \text{所以}$$

$$u(x, y) = u_0 + \frac{b-2}{2a} u_0 x - \frac{4bu_0}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2 \operatorname{sh} \frac{(2k+1)\pi}{b} a} \operatorname{sh} \frac{(2k+1)\pi}{b} x \cos \frac{(2k+1)\pi}{b} y$$

$$222. \text{ 在带状区域 } 0 \leq x \leq a, \quad 0 \leq y < \infty \text{ 中求解 } \begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \\ u|_{x=0} = 0, u|_{x=a} = 0 \\ u|_{y=0} = A \left(1 - \frac{x}{a} \right), \lim_{y \rightarrow \infty} u = 0 \end{cases}。$$

可求得本征函数 $X_n(x) = \sin \frac{n\pi}{a} x$ ($n=1, 2, \dots$)。

由 $\lim_{y \rightarrow \infty} u = 0$ 得 $Y_n(y) = C_n e^{-\frac{n\pi}{a}y}$, 则 $u(x, y) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{a} x e^{-\frac{n\pi}{a}y}$ 。

由 $u|_{y=0} = A \left(1 - \frac{x}{a}\right)$ 得 $C_n = \frac{2A}{a} \int_0^a \left(1 - \frac{x}{a}\right) \sin \frac{n\pi}{a} x dx = \frac{2A}{n\pi}$ 。

$$\begin{aligned} \text{所以 } u(x, y) &= \frac{2A}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi}{a} x e^{-\frac{n\pi}{a}y} = \frac{2A}{\pi} \frac{1}{2i} \left[\sum_{n=1}^{\infty} \frac{1}{n} e^{\frac{n\pi}{a}(-y+ix)} - \sum_{n=1}^{\infty} \frac{1}{n} e^{\frac{n\pi}{a}(-y-ix)} \right] \\ &= \frac{2A}{\pi} \frac{1}{2i} \left\{ -\ln \left[1 - e^{\frac{\pi}{a}(-y+ix)} \right] + \ln \left[1 - e^{\frac{\pi}{a}(-y-ix)} \right] \right\} \quad (\ln(1-z) = -\sum_{n=1}^{\infty} \frac{z^n}{n}, |z| < 1) \\ &= \frac{2A}{\pi} \frac{1}{2i} \ln \frac{1 - e^{\frac{\pi}{a}(-y-ix)}}{1 - e^{\frac{\pi}{a}(-y+ix)}} = \frac{2A}{\pi} \frac{1}{2i} \ln \frac{1 - e^{-\frac{\pi}{a}y} \cos \frac{\pi}{a} x + i e^{-\frac{\pi}{a}y} \sin \frac{\pi}{a} x}{1 - e^{-\frac{\pi}{a}y} \cos \frac{\pi}{a} x - i e^{-\frac{\pi}{a}y} \sin \frac{\pi}{a} x} \\ &= \frac{2A}{\pi} \frac{1}{2i} \ln \frac{1 + i \frac{e^{-\frac{\pi}{a}y} \sin \frac{\pi}{a} x}{1 - e^{-\frac{\pi}{a}y} \cos \frac{\pi}{a} x}}{1 - i \frac{e^{-\frac{\pi}{a}y} \sin \frac{\pi}{a} x}{1 - e^{-\frac{\pi}{a}y} \cos \frac{\pi}{a} x}} = \frac{2A}{\pi} \arctan \frac{e^{-\frac{\pi}{a}y} \sin \frac{\pi}{a} x}{1 - e^{-\frac{\pi}{a}y} \cos \frac{\pi}{a} x}。 \end{aligned}$$

223. 当层状铀块的厚度超过一定值（称为临界厚度）时，中子浓度将随时间增加而增加，以致引起铀块爆炸。这就是原子弹爆炸的基本过程。试估计层状铀块的临界厚度。假定中子浓度满足齐次的第二类边界条件。方程见 205 题。

方程（一维）： $\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + \alpha u$ ，边界条件： $u|_{x=0} = 0$ ， $u|_{x=l} = 0$ 。

分离变量可得本征值 $\lambda_n = D \left(\frac{n\pi}{l} \right)^2 - \alpha$ ($n=1, 2, \dots$)，所以 $T_n = A_n e^{-\lambda_n t} = A_n e^{\left[\alpha - D \left(\frac{n\pi}{l} \right)^2 \right] t}$ 。

当 $\alpha - D \left(\frac{n\pi}{l} \right)^2 > 0$ 时，此函数递增。由此可得临界厚度 $l_c = \pi \sqrt{\frac{D}{\alpha}}$ 。

224. 求解两端固定弦的阻尼振动问题:
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} + 2h \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2} \\ u|_{x=0} = 0, u|_{x=l} = 0 \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t}\bigg|_{t=0} = \psi(x) \end{cases}.$$

分离变量得 $\begin{cases} X''(x) + \lambda X(x) = 0 \\ T''(t) + 2hT'(t) + a^2 \lambda T(t) = 0 \end{cases}$ 。本征值为 $\lambda_n = \left(\frac{n\pi}{l}\right)^2$ ，本征函数为

$$X_n(x) = \sin \frac{n\pi}{l} x \quad (n=1, 2, \dots).$$

$$T_n(t) = e^{-ht} (A_n \cos \omega_n t + B_n \sin \omega_n t), \text{ 其中 } \omega_n = \sqrt{\left(\frac{n\pi a}{l}\right)^2 - h^2} \quad (\text{设 } h < \frac{\pi a}{l}).$$

$$\text{则 } u(x, t) = e^{-ht} \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) \sin \frac{n\pi}{l} x.$$

$$\text{由 } u|_{t=0} = \varphi(x) \text{ 可得 } A_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx,$$

$$\frac{\partial u}{\partial t}\bigg|_{t=0} = \sum_{n=1}^{\infty} (B_n \omega_n - h A_n) \sin \frac{n\pi}{l} x = \psi(x), \text{ 所以 } B_n = \frac{h}{\omega_n} A_n + \frac{2}{l \omega_n} \int_0^l \psi(x) \sin \frac{n\pi}{l} x dx.$$

225. 一个均匀的，各向同性的弹性方形膜， $0 \leq x \leq l$ ， $0 \leq y \leq l$ ，四周夹紧。初始形状为

$Axy(l-x)(l-y)$ ，初速度为 0，求解膜的横振动。

$$\text{定解问题为 } \begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2} = 0 \\ u|_{x=0} = 0, u|_{x=l} = 0 \\ u|_{y=0} = 0, u|_{y=l} = 0 \\ u|_{t=0} = Axy(l-x)(l-y), \frac{\partial u}{\partial t}\bigg|_{t=0} = 0 \end{cases}. \text{ 设 } u(x, y, t) = X(x)w(y, t), \text{ 则}$$

$$\frac{X''}{X} = \frac{1}{w} \left(\frac{1}{a^2} \frac{\partial^2 w}{\partial t^2} - \frac{\partial^2 w}{\partial y^2} \right) = -\lambda. \text{ 即 } \begin{cases} X'' + \lambda X = 0 \\ \left(\frac{1}{a^2} \frac{\partial^2 w}{\partial t^2} - \frac{\partial^2 w}{\partial y^2} \right) + \lambda w = 0 \end{cases}.$$

$$\text{设 } w(x, y) = Y(y)T(t), \text{ 则 } \frac{Y''}{Y} = \frac{1}{a^2} \frac{T''}{T} + \lambda = -\mu,$$

$$\text{所以} \begin{cases} X'' + \lambda X = 0 \\ Y'' + \mu Y = 0 \\ T'' + a^2(\lambda + \mu)T = 0 \end{cases}。$$

$$\text{可得 } \lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin \frac{n\pi}{l} x \quad (n=1, 2, \dots)。$$

$$\mu_m = \left(\frac{m\pi}{l}\right)^2, \quad Y_m(y) = \sin \frac{m\pi}{l} y \quad (m=1, 2, \dots)。$$

$$T_{mn}(t) = A_{mn} \sin \omega_{mn} t + B_{mn} \cos \omega_{mn} t, \quad \text{其中 } \omega_{mn} = \sqrt{n^2 + m^2} \frac{\pi a}{l}。$$

$$u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (A_{mn} \sin \omega_{mn} t + B_{mn} \cos \omega_{mn} t) \sin \frac{n\pi}{l} x \sin \frac{m\pi}{l} y。$$

$$\text{由 } \frac{\partial u}{\partial t} \Big|_{t=0} = 0 \text{ 可得 } A_{mn} = 0。$$

$$\text{由 } u|_{t=0} = Axy(l-x)(l-y) \text{ 得}$$

$$B_{mn} = \frac{4A}{l^2} \int_0^l y(l-y) \sin \frac{m\pi}{l} y dy \int_0^l x(l-x) \sin \frac{n\pi}{l} x dx = \frac{16Al^4}{\pi^6 m^2 n^2} [1 - (-1)^n] [1 - (-1)^m],$$

$$B_{2j, 2i} = 0, \quad B_{2j+1, 2i} = 0, \quad B_{2j, 2i+1} = 0, \quad B_{2j+1, 2i+1} = \frac{64Al^4}{\pi^6 (2i+1)^2 (2j+1)^2}。$$

所以

$$u(x, y, t) = \frac{64Al^4}{\pi^6} \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \frac{1}{(2i+1)^2 (2j+1)^2} \sin \frac{(2i+1)\pi}{l} x \sin \frac{(2j+1)\pi}{l} y \cos \omega_{2j+1, 2i+1} t。$$

226. 一个均匀的, 各向同性的弹性方形膜, $0 \leq x \leq l$, $0 \leq y \leq l$, 四周夹紧。若初始时在

$$\text{中心附近受到敲击, 使得 } \frac{\partial u}{\partial t} \Big|_{t=0} = \begin{cases} v_0, \frac{l}{2} - \delta < x < \frac{l}{2} + \delta, \frac{l}{2} - \delta < y < \frac{l}{2} + \delta \\ 0, \text{others} \end{cases}, \text{ 而初位移为 } 0。$$

求解膜的横振动。

$$\text{同上题有 } u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (A_{mn} \sin \omega_{mn} t + B_{mn} \cos \omega_{mn} t) \sin \frac{n\pi}{l} x \sin \frac{m\pi}{l} y,$$

$$\text{其中 } \omega_{mn} = \sqrt{n^2 + m^2} \frac{\pi a}{l}。$$

由 $u|_{t=0} = 0$ 有 $B_{mn} = 0$ 。

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{mn} \omega_{mn} \sin \frac{n\pi}{l} x \sin \frac{m\pi}{l} y = \begin{cases} v_0, \frac{l}{2} - \delta < x < \frac{l}{2} + \delta, \frac{l}{2} - \delta < y < \frac{l}{2} + \delta \\ 0, \text{others} \end{cases}.$$

$$\begin{aligned} \text{所以 } A_{mn} &= \frac{4v_0}{l^2 \omega_{mn}} \int_{l/2-\delta}^{l/2+\delta} \sin \frac{n\pi}{l} x dx \int_{l/2-\delta}^{l/2+\delta} \sin \frac{m\pi}{l} y dy \\ &= \frac{16v_0}{mn\pi^2 \omega_{mn}} \sin \frac{n\pi}{2} \sin \frac{m\pi}{2} \sin \frac{n\pi\delta}{l} \sin \frac{m\pi\delta}{l}. \end{aligned}$$

$$A_{2j,2i} = A_{2j+1,2i} = A_{2j,2i+1} = 0,$$

$$A_{2j+1,2i+1} = \frac{(-1)^{i+j} 16v_0}{(2i+1)(2j+1)\pi^2 \omega_{2j+1,2i+1}} \sin \frac{(2i+1)\pi\delta}{l} \sin \frac{(2j+1)\pi\delta}{l}.$$

$$\begin{aligned} \text{所以 } u(x, y, t) &= \frac{16v_0}{\pi^2} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{(-1)^{i+j}}{(2i+1)(2j+1)\omega_{2j+1,2i+1}} \sin \frac{(2i+1)\pi\delta}{l} \sin \frac{(2j+1)\pi\delta}{l} \\ &\quad \cdot \sin \frac{(2i+1)\pi}{l} x \sin \frac{(2j+1)\pi}{l} y \sin \omega_{2j+1,2i+1} t \end{aligned}$$

227. 均匀, 各向同性的弹性方膜, $0 \leq x \leq l$, $0 \leq y \leq l$, 四周夹紧。若初始时在中心附近受到敲击, 使中心点得到冲量 I , 而初位移为 0, 试求解膜的横振动。

同 217 题, $v_0 = \frac{I}{4\rho\delta^2}$, 其中 ρ 为面密度。

$$\begin{aligned} u(x, y, t) &= \frac{4I}{\pi^2 \rho} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{(-1)^{i+j}}{(2i+1)(2j+1)\omega_{2j+1,2i+1}} \lim_{\delta \rightarrow 0} \frac{\sin \frac{(2i+1)\pi\delta}{l}}{\delta} \lim_{\delta \rightarrow 0} \frac{\sin \frac{(2j+1)\pi\delta}{l}}{\delta} \\ &\quad \cdot \sin \frac{(2i+1)\pi}{l} x \sin \frac{(2j+1)\pi}{l} y \sin \omega_{2j+1,2i+1} t \\ &= \frac{4I}{\rho l^2} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{(-1)^{i+j}}{\omega_{2j+1,2i+1}} \sin \frac{(2i+1)\pi}{l} x \sin \frac{(2j+1)\pi}{l} y \sin \omega_{2j+1,2i+1} t. \end{aligned}$$

228. 一长为 l 的均匀园杆作微小扭转振动。在振动过程中, 杆的各横截面仍保持为平面而

绕杆轴扭转,轴向上不发生位移。杆的一端固定,另一端连接在圆盘上,则偏转角 θ 所满足

$$\text{的方程和边界条件为} \begin{cases} \frac{\partial^2 \theta}{\partial t^2} - a^2 \frac{\partial^2 \theta}{\partial x^2} = 0 \\ \theta|_{x=0} = 0, \frac{\partial^2 \theta}{\partial t^2} \Big|_{x=l} = -c^2 \frac{\partial \theta}{\partial x} \Big|_{x=l} \end{cases}, \quad a \text{ 和 } c \text{ 均为实常数。}$$

(1) 求相应的本征值 λ_n 及本征函数 $X_n(x)$; (2) 计算积分 $\int_0^l X_n(x) X_m(x) dx$;

(3) 计算积分 $\int_0^l X'_n(x) X'_m(x) dx$ 。

$$(1) \text{ 设 } \theta(x, t) = X(x)T(t), \text{ 则 } \frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda,$$

$$\theta|_{x=0} = X(0)T(t) = 0, \text{ 所以 } X(0) = 0.$$

$$\frac{\partial^2 \theta}{\partial t^2} \Big|_{x=l} = X(l)T''(t) = -c^2 \frac{\partial \theta}{\partial x} \Big|_{x=l} = -c^2 X'(l)T(t), \text{ 所以}$$

$$X'(l) = -\frac{1}{c^2} X(l) \frac{T''(t)}{T(t)} = \frac{a^2}{c^2} \lambda X(l).$$

$$\text{即本征值问题为} \begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = 0, X'(l) = \frac{a^2}{c^2} \lambda X(l) \end{cases}.$$

$$\begin{aligned} \lambda \int_0^l X^2(x) dx &= -\int_0^l X''(x) X(x) dx = -X'(x) X(x) \Big|_0^l + \int_0^l X'^2(x) dx \\ &= -X'(l) X(l) + \int_0^l X'^2(x) dx = -\frac{a^2}{c^2} \lambda X^2(l) + \int_0^l X'^2(x) dx, \end{aligned}$$

$$\text{所以 } \lambda = \frac{\int_0^l X'^2(x) dx}{\int_0^l X^2(x) dx + \frac{a^2}{c^2} X^2(l)} \geq 0 \quad (X(x) \text{ 非恒零}), \text{ 若 } X'(x) = 0, \text{ 即 } X(x) = C \text{ (常}$$

数), 由 $X(0) = 0$ 知 $X(x) = 0$, 所以一定有 $\lambda > 0$ 。

解得本征值 λ_n 为方程 $\sqrt{\lambda} \tan \sqrt{\lambda} l = \frac{c^2}{a^2}$ 的第 n 个正根, 本征函数 $X_n(x) = \sin \sqrt{\lambda_n} x$ 。

(2) 设 λ_n, λ_m 分别对应 $X_n(x), X_m(x)$ ($n \neq m$), 即

$$\lambda_n X_n(x) = -X''_n(x), \quad \lambda_m X_m(x) = -X''_m(x).$$

第一式两边同乘 $X_m(x)$ 减去第二式两边同乘 $X_n(x)$, 再两边积分得

$$\begin{aligned}
 (\lambda_n - \lambda_m) \int_0^l X_n(x) X_m(x) dx &= \int_0^l [X_n(x) X_m''(x) - X_n''(x) X_m(x)] dx \\
 &= X_n(l) X_m'(l) - X_n'(l) X_m(l) = \frac{a^2}{c^2} (\lambda_m - \lambda_n) X_n(l) X_m(l),
 \end{aligned}$$

由于 $\lambda_n \neq \lambda_m$, 所以 $\int_0^l X_n(x) X_m(x) dx = -\frac{a^2}{c^2} X_n(l) X_m(l) = -\frac{a^2}{c^2} \sin \sqrt{\lambda_n} l \sin \sqrt{\lambda_m} l$

当 $n = m$ 时, $\int_0^l X_n(x) X_m(x) dx = \int_0^l \sin^2 \sqrt{\lambda_n} x dx = \frac{l}{2} - \frac{1}{2} \int_0^l \cos 2\sqrt{\lambda_n} x dx$

$$= \frac{l}{2} - \frac{1}{4\sqrt{\lambda_n}} \sin 2\sqrt{\lambda_n} l = \frac{l}{2} - \frac{1}{2\sqrt{\lambda_n}} \frac{\tan \sqrt{\lambda_n} l}{1 + \tan^2 \sqrt{\lambda_n} l} = \frac{l}{2} - \frac{\left(\frac{a}{c}\right)^2}{2 \left[\left(\frac{a}{c}\right)^4 \lambda_n + 1 \right]}.$$

$$\begin{aligned}
 (3) \quad \lambda_n \int_0^l X_n(x) X_m(x) dx &= -\int_0^l X_n''(x) X_m(x) dx \\
 &= -X_n'(l) X_m(l) + \int_0^l X_n'(x) X_m'(x) dx,
 \end{aligned}$$

所以 $\int_0^l X_n'(x) X_m'(x) dx = X_n'(l) X_m(l) + \lambda_n \int_0^l X_n(x) X_m(x) dx$

$$= \lambda_n \left[\frac{a^2}{c^2} X_n(l) X_m(l) + \int_0^l X_n(x) X_m(x) dx \right].$$

$n \neq m$ 时由于 $\int_0^l X_n(x) X_m(x) dx = -\frac{a^2}{c^2} X_n(l) X_m(l)$, 所以 $\int_0^l X_n'(x) X_m'(x) dx = 0$ 。

$n = m$ 时, $\int_0^l X_n'(x) X_m'(x) dx = \lambda_n \int_0^l \cos^2 \sqrt{\lambda_n} x dx = \lambda_n \left[\frac{l}{2} + \frac{1}{2} \int_0^l \cos 2\sqrt{\lambda_n} x dx \right]$

$$= \lambda_n \left\{ \frac{l}{2} + \frac{\left(\frac{a}{c}\right)^2}{2 \left[\left(\frac{a}{c}\right)^4 \lambda_n + 1 \right]} \right\}.$$

229. 求解枢轴的扭转振动问题:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = 0, \frac{\partial^2 u}{\partial t^2} \Big|_{x=l} = -c^2 \frac{\partial u}{\partial x} \Big|_{x=l} \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t} \Big|_{t=0} = \psi(x) \end{cases}$$

由上题, $u(x, t) = \sum_{n=1}^{\infty} X_n(x) T_n(t) = \sum_{n=1}^{\infty} (A_n \sin a\sqrt{\lambda_n}t + B_n \cos a\sqrt{\lambda_n}t) \sin \sqrt{\lambda_n}x$ 。

则 $\frac{\partial u}{\partial x} = \sum_{n=1}^{\infty} (A_n \sin a\sqrt{\lambda_n}t + B_n \cos a\sqrt{\lambda_n}t) X'_n(x)$ (假设可逐项微分)。

初始条件写为 $\frac{\partial u}{\partial x} \Big|_{t=0} = \varphi'(x), \frac{\partial^2 u}{\partial x \partial t} \Big|_{t=0} = \psi'(x)$ 。

根据 $\{X'_n(x)\}$ 的正交性, 可由初始条件定出系数 A_n, B_n 。

$$B_n = \frac{1}{\lambda_n N_n} \int_0^l \varphi'(x) X'_n(x) dx = \frac{1}{\sqrt{\lambda_n} N_n} \int_0^l \varphi'(x) \cos \sqrt{\lambda_n} x dx,$$

$$A_n = \frac{1}{a\sqrt{\lambda_n} \lambda_n N_n} \int_0^l \psi'(x) X'_n(x) dx = \frac{1}{a\lambda_n N_n} \int_0^l \psi'(x) \cos \sqrt{\lambda_n} x dx。$$

$$\text{其中 } N_n = \frac{l}{2} + \frac{\left(\frac{a}{c}\right)^2}{2 \left[\left(\frac{a}{c}\right)^4 \lambda_n + 1 \right]}。$$

$$230. \text{ 求解下列定解问题: } \begin{cases} \frac{\partial^2 u}{\partial t^2} + a^2 \frac{\partial^4 u}{\partial x^4} = 0 \\ u|_{x=0} = 0, u|_{x=l} = 0, \frac{\partial^2 u}{\partial x^2} \Big|_{x=0} = 0, \frac{\partial^2 u}{\partial x^2} \Big|_{x=l} = 0 \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t} \Big|_{t=0} = \psi(x) \end{cases}$$

分离变量得 $\frac{T''(t)}{a^2 T(t)} = -\frac{X^{(4)}(x)}{X(x)} = -\lambda$ 。

$$\text{本征值问题} \begin{cases} X^{(4)}(x) - \lambda X(x) = 0 \\ X(0) = 0, X(l) = 0, X''(0) = 0, X''(l) = 0 \end{cases}.$$

$$\begin{aligned} \lambda \int_0^l X^2(x) dx &= \int_0^l X^{(4)}(x) X(x) dx = X'''(x) X(x) \Big|_0^l - \int_0^l X^{(3)}(x) X'(x) dx \\ &= -X''(x) X'(x) \Big|_0^l + \int_0^l X''^2(x) dx = \int_0^l X''^2(x) dx, \end{aligned}$$

$$\text{所以 } \lambda = \frac{\int_0^l X''^2(x) dx}{\int_0^l X^2(x) dx} > 0.$$

$$X \text{ 的通解为 } X(x) = C_1 \operatorname{sh} \sqrt[4]{\lambda} x + C_2 \operatorname{ch} \sqrt[4]{\lambda} x + C_3 \sin \sqrt[4]{\lambda} x + C_4 \cos \sqrt[4]{\lambda} x,$$

$$\text{由 } X(0) = 0, X''(0) = 0 \text{ 可得 } C_2 = C_4 = 0, \text{ 再由 } X(l) = 0, X''(l) = 0 \text{ 可得}$$

$$\begin{cases} C_1 \operatorname{sh} \sqrt[4]{\lambda} l + C_3 \sin \sqrt[4]{\lambda} l = 0 \\ C_1 \operatorname{sh} \sqrt[4]{\lambda} l - C_3 \sin \sqrt[4]{\lambda} l = 0 \end{cases}, \text{ 两式相加得 } C_1 \operatorname{sh} \sqrt[4]{\lambda} l = 0, \text{ 因为 } \operatorname{sh} \sqrt[4]{\lambda} l \neq 0 \ (\lambda > 0), \text{ 所}$$

$$\text{以 } C_1 = 0. \text{ 两式相减得 } C_3 \sin \sqrt[4]{\lambda} l = 0, \text{ 可得本征值 } \lambda_n = \left(\frac{n\pi}{l} \right)^4 \ (n=1, 2, \dots),$$

$$\text{本征函数 } X_n(x) = \sin \frac{n\pi}{l} x.$$

$$\text{所以 } u(x, t) = \sum_{n=1}^{\infty} \left[A_n \sin \left(\frac{n\pi}{l} \right)^2 at + B_n \cos \left(\frac{n\pi}{l} \right)^2 at \right] \sin \frac{n\pi}{l} x.$$

$$\text{由初始条件可得 } A_n = \frac{2l}{n^2 \pi^2 a} \int_0^l \psi(x) \sin \frac{n\pi}{l} x dx, \quad B_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx.$$

231. 在矩形区域 $0 < x < a, -\frac{b}{2} < y < \frac{b}{2}$ 上求解: (1) $\nabla^2 u = -2$, (2) $\nabla^2 u = -x^2 y$, 而 u 在边界上数值为 0.

(1) 可设方程的一个特解为 $v(x)$, 则 $v''(x) = -2$, 使之满足齐次边界条件可解得

$$v = x(a-x), \text{ 令 } u = v + w, \text{ 则 } \begin{cases} \nabla^2 w = 0 \\ w|_{x=0} = 0, w|_{x=a} = 0 \\ w|_{y=-b/2} = -x(a-x), w|_{y=b/2} = -x(a-x) \end{cases}.$$

$$\text{可得 } w = \sum_{n=1}^{\infty} \left(A_n \operatorname{sh} \frac{n\pi}{a} y + B_n \operatorname{ch} \frac{n\pi}{a} y \right) \sin \frac{n\pi}{a} x,$$

由 y 的边界条件可得

$$\sum_{n=1}^{\infty} \left(-A_n \operatorname{sh} \frac{n\pi b}{2a} + B_n \operatorname{ch} \frac{n\pi b}{2a} \right) \sin \frac{n\pi}{a} x = -x(a-x),$$

$$\sum_{n=1}^{\infty} \left(A_n \operatorname{sh} \frac{n\pi b}{2a} + B_n \operatorname{ch} \frac{n\pi b}{2a} \right) \sin \frac{n\pi}{a} x = -x(a-x)。$$

两式相减可得 $A_n = 0$ ，两式相加可得

$$B_n = \frac{2}{a \operatorname{ch} \frac{n\pi b}{2a}} \int_0^a x(x-a) \sin \frac{n\pi}{a} x dx = \frac{4a^2}{n^3 \pi^3 \operatorname{ch} \frac{n\pi b}{2a}} \left[(-1)^n - 1 \right],$$

$$B_{2k} = 0, \quad B_{2k+1} = -\frac{8a^2}{(2k+1)^3 \pi^3 \operatorname{ch} \frac{(2k+1)\pi b}{2a}}。$$

$$\text{所以 } w(x, t) = -\frac{8a^2}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3 \operatorname{ch} \frac{(2k+1)\pi b}{2a}} \operatorname{ch} \frac{(2k+1)\pi}{a} y \sin \frac{(2k+1)\pi}{a} x,$$

$$u(x, t) = x(a-x) + w(x, t)$$

$$= x(a-x) - \frac{8a^2}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3 \operatorname{ch} \frac{(2k+1)\pi b}{2a}} \operatorname{ch} \frac{(2k+1)\pi}{a} y \sin \frac{(2k+1)\pi}{a} x。$$

(2) 将非齐次项 $-x^2 y$ 按 x 的本征函数展开，即 $-x^2 y = \sum_{n=1}^{\infty} g_n(y) \sin \frac{n\pi}{a} x$ ，

$$\text{可得 } g_n(y) = -\frac{2}{a} \int_0^a x^2 y \sin \frac{n\pi}{a} x dx = \frac{2a^2}{n\pi} \left\{ (-1)^n + \frac{2}{n^2 \pi^2} \left[1 - (-1)^n \right] \right\} y,$$

设 $u(x, y) = \sum_{n=1}^{\infty} Y_n(y) \sin \frac{n\pi}{a} x$ ，代入方程得

$$-\sum_{n=1}^{\infty} \left(\frac{n\pi}{a} \right)^2 Y_n(y) \sin \frac{n\pi}{a} x + \sum_{n=1}^{\infty} Y_n''(y) \sin \frac{n\pi}{a} x = \sum_{n=1}^{\infty} g_n(y) \sin \frac{n\pi}{a} x,$$

$$\text{所以 } \begin{cases} Y_n''(y) - \left(\frac{n\pi}{a} \right)^2 Y_n(y) = g_n(y) \\ Y_n\left(-\frac{b}{2}\right) = 0, Y_n\left(\frac{b}{2}\right) = 0 \end{cases}。$$

$$\text{解得 } Y_n(y) = \frac{2a^4}{n^3\pi^3} \left\{ (-1)^n + \frac{2}{n^2\pi^2} [1 - (-1)^n] \right\} \left(\frac{b}{2} \frac{\operatorname{sh} \frac{n\pi}{a} y}{\operatorname{sh} \frac{n\pi b}{2a}} - y \right)。$$

$$\begin{aligned} \text{所以 } u(x, y) &= \sum_{n=1}^{\infty} Y_n(y) \sin \frac{n\pi}{a} x = \frac{2a^4}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \left(\frac{b}{2} \frac{\operatorname{sh} \frac{n\pi}{a} y}{\operatorname{sh} \frac{n\pi b}{2a}} - y \right) \sin \frac{n\pi}{a} x \\ &\quad + \frac{4a^4}{\pi^5} \sum_{n=1}^{\infty} \frac{1}{n^5} [1 - (-1)^n] \left(\frac{b}{2} \frac{\operatorname{sh} \frac{n\pi}{a} y}{\operatorname{sh} \frac{n\pi b}{2a}} - y \right) \sin \frac{n\pi}{a} x。 \end{aligned}$$

后一项 n 只取奇数, 则

$$\begin{aligned} u(x, y) &= \frac{2a^4}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \left(\frac{b}{2} \frac{\operatorname{sh} \frac{n\pi}{a} y}{\operatorname{sh} \frac{n\pi b}{2a}} - y \right) \sin \frac{n\pi}{a} x \\ &\quad + \frac{8a^4}{\pi^5} \sum_{n=1}^{\infty} \frac{1}{(2n+1)^5} \left[\frac{b}{2} \frac{\operatorname{sh} \frac{(2n+1)\pi}{a} y}{\operatorname{sh} \frac{(2n+1)\pi b}{2a}} - y \right] \sin \frac{(2n+1)\pi}{a} x。 \end{aligned}$$

$$232. \text{ 求解: } \begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = bx(l-x) \\ u|_{x=0} = 0, u|_{x=l} = 0 \\ u|_{t=0} = 0, \frac{\partial u}{\partial t}|_{t=0} = 0 \end{cases}。$$

设方程一个特解为 $v(x)$, 则 $v'' = -\frac{b}{a^2} x(l-x)$, 使之满足齐次边界条件, 解之得

$v = \frac{b}{12a^2} x(x^3 - 2lx^2 + l^3)$ 。设 $u = v + w$, 则 w 满足

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0, w|_{x=l} = 0 \\ w|_{t=0} = -\frac{b}{12a^2} x(x^3 - 2lx^2 + l^3), \frac{\partial w}{\partial t}|_{t=0} = 0 \end{cases}。$$

$$w(x, t) = \sum_{n=1}^{\infty} \left(A_n \sin \frac{n\pi}{l} at + B_n \cos \frac{n\pi}{l} at \right) \sin \frac{n\pi}{l} x,$$

$$A_n = 0, \quad B_n = -\frac{b}{6a^2l} \int_0^l x(x^3 - 2lx^2 + l^3) \sin \frac{n\pi}{l} x dx = \frac{4l^4b}{n^5\pi^5a^2} [(-1)^n - 1],$$

$$\text{所以 } u(x, t) = \frac{b}{12a^2} x(x^3 - 2lx^2 + l^3) - \frac{8l^4b}{\pi^5a^2} \sum_{n=1}^{\infty} \frac{1}{(2k+1)^5} \cos \frac{(2k+1)\pi}{l} at \sin \frac{(2k+1)\pi}{l} x.$$

233. 解第 202 题。

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = 0, \frac{\partial u}{\partial x}|_{x=l} = \frac{P}{E} \\ u|_{t=0} = 0, \frac{\partial u}{\partial t}|_{t=0} = 0 \end{cases}$$

设 $v = \frac{P}{E}x$, 令 $u = v + w$, 则

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0, \frac{\partial w}{\partial x}|_{x=l} = 0 \\ w|_{t=0} = -\frac{P}{E}x, \frac{\partial w}{\partial t}|_{t=0} = 0 \end{cases}.$$

$$\text{可得 } w(x, t) = \sum_{n=1}^{\infty} \left(A_n \sin \frac{2n+1}{2l} a\pi t + B_n \cos \frac{2n+1}{2l} a\pi t \right) \sin \frac{2n+1}{2l} \pi x.$$

$$\text{由 } \frac{\partial w}{\partial t}|_{t=0} = 0 \text{ 可得 } A_n = 0.$$

$$\text{由 } w|_{t=0} = -\frac{P}{E}x \text{ 可得 } B_n = -\frac{2P}{El} \int_0^l x \sin \frac{2n+1}{2l} \pi x dx = -\frac{8Pl(-1)^2}{(2n+1)^2 \pi^2 E}.$$

$$\text{所以 } u(x, t) = \frac{P}{E}x - \frac{8Pl}{\pi^2 E} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)^2} \cos \frac{2n+1}{2l} a\pi t \sin \frac{2n+1}{2l} \pi x.$$

234. 一细长杆, $x=0$ 端固定, $x=l$ 端受周期力 $A \sin \omega t$ 作用。求解此杆的纵振动, 设初位移及初速度均为 0。

见 206 题, 该定解问题为:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = 0, \frac{\partial u}{\partial x} \Big|_{x=l} = \frac{A}{ES} \sin \omega t . \\ u|_{t=0} = 0, \frac{\partial u}{\partial t} \Big|_{t=0} = 0 \end{cases}$$

设 $v(x, t) = f(x) \sin \omega t$ 满足方程和边界条件, 则 $\begin{cases} f''(x) + \frac{\omega^2}{a^2} f(x) = 0 \\ f(0) = 0, f'(l) = \frac{A}{ES} \end{cases}$, 解之得

$$v(x, t) = \frac{Aa}{ES\omega} \frac{\sin \frac{\omega}{a} x}{\cos \frac{\omega}{a} l} \sin \omega t .$$

令 $u = v + w$, 则 w 满足 $\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0, \frac{\partial w}{\partial x} \Big|_{x=l} = 0 \\ w|_{t=0} = 0, \frac{\partial w}{\partial t} \Big|_{t=0} = -\frac{Aa}{ES \cos \frac{\omega}{a} l} \sin \frac{\omega}{a} x \end{cases} .$

可得 $w = \sum_{n=0}^{\infty} \left(A_n \sin \frac{2n+1}{2l} a \pi t + B_n \cos \frac{2n+1}{2l} a \pi t \right) \sin \frac{2n+1}{2l} \pi x ,$

$$B_n = 0 ,$$

$$\begin{aligned} A_n &= -\frac{4A}{\pi ES (2n+1) \cos \frac{\omega}{a} l} \int_0^l \sin \frac{\omega}{a} x \sin \frac{2n+1}{2l} \pi x dx \\ &= \frac{2A}{\pi ES (2n+1) \cos \frac{\omega}{a} l} \int_0^l \left[\cos \left(\frac{\omega}{a} + \frac{2n+1}{2l} \pi \right) x - \cos \left(\frac{\omega}{a} - \frac{2n+1}{2l} \pi \right) x \right] dx \\ &= \frac{2A}{\pi ES (2n+1) \cos \frac{\omega}{a} l} \left[\frac{(-1)^n}{\frac{\omega}{a} + \frac{2n+1}{2l} \pi} \cos \frac{\omega}{a} l - \int_0^l \cos \left(\frac{\omega}{a} - \frac{2n+1}{2l} \pi \right) x dx \right] . \end{aligned}$$

若不存在正整数 m , 使得 $\frac{\omega}{a} = \frac{2m+1}{2l} \pi$, 则

$$A_n = \frac{2A}{\pi ES (2n+1) \cos \frac{\omega}{a} l} \left[\frac{(-1)^n}{\frac{\omega}{a} + \frac{2n+1}{2l} \pi} \cos \frac{\omega}{a} l + \frac{(-1)^n}{\frac{\omega}{a} - \frac{2n+1}{2l} \pi} \cos \frac{\omega}{a} l \right]$$

$$= \frac{4A\omega}{\pi ES a (2n+1)} \frac{(-1)^n}{\left(\frac{\omega}{a}\right)^2 - \left(\frac{2n+1}{2l} \pi\right)^2}。$$

$$\text{所以 } u(x, t) = \frac{Aa}{ES\omega} \frac{\sin \frac{\omega}{a} x}{\cos \frac{\omega}{a} l} \sin \omega t$$

$$+ \frac{4A\omega}{\pi ES a} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \frac{1}{\left(\frac{\omega}{a}\right)^2 - \left(\frac{2n+1}{2l} \pi\right)^2} \sin \frac{2n+1}{2l} a \pi t \sin \frac{2n+1}{2l} \pi x。$$

若存在正整数 m ，使得 $\frac{\omega}{a} = \frac{2m+1}{2l} \pi$ ，则当 $n \neq m$ 时，仍有

$$A_n = \frac{4A\omega}{\pi ES a (2n+1)} \frac{(-1)^n}{\left(\frac{\omega}{a}\right)^2 - \left(\frac{2n+1}{2l} \pi\right)^2}，$$

$$\text{当 } n = m \text{ 时， } A_m = \frac{Aa}{ESl\omega \cos \frac{\omega}{a} l} \left[\frac{(-1)^m a}{2\omega} \cos \frac{\omega}{a} l - l \right]，$$

$$\text{所以 } u(x, t) = \frac{Aa}{ES\omega} \frac{\sin \frac{\omega}{a} x}{\cos \frac{\omega}{a} l} \sin \omega t + A_m \sin \omega t \sin \frac{\omega}{a} x + \sum_{\substack{n=0 \\ n \neq m}}^{\infty} A_n \sin \frac{2n+1}{2l} a \pi t \sin \frac{2n+1}{2l} \pi x$$

$$= \frac{Aa}{ES\omega} \frac{\sin \frac{\omega}{a} x}{\cos \frac{\omega}{a} l} \sin \omega t + \frac{Aa}{ESl\omega \cos \frac{\omega}{a} l} \left[\frac{(-1)^m a}{2\omega} \cos \frac{\omega}{a} l - l \right] \sin \frac{\omega}{a} x \sin \omega t$$

$$+ \frac{4A\omega}{\pi ES a} \sum_{\substack{n=0 \\ n \neq m}}^{\infty} \frac{(-1)^n}{2n+1} \frac{1}{\left(\frac{\omega}{a}\right)^2 - \left(\frac{2n+1}{2l} \pi\right)^2} \sin \frac{2n+1}{2l} a \pi t \sin \frac{2n+1}{2l} \pi x$$

$$= \frac{(-1)^m Aa^2}{2ESl\omega^2} \sin \frac{\omega}{a} x \sin \omega t$$

$$+\frac{4A\omega}{\pi ESa}\sum_{\substack{n=0 \\ n \neq m}}^{\infty} \frac{(-1)^n}{2n+1} \frac{1}{\left(\frac{\omega}{a}\right)^2 - \left(\frac{2n+1}{2l}\pi\right)^2} \sin \frac{2n+1}{2l} a\pi t \sin \frac{2n+1}{2l} \pi x。$$

235. 求下列定解问题之解:
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = \cos \frac{\pi}{l} at, \frac{\partial u}{\partial x}|_{x=l} = 0 \\ u|_{t=0} = \cos \frac{\pi}{l} x, \frac{\partial u}{\partial t}|_{t=0} = \sin \frac{\pi}{2l} x \end{cases}。$$

设 $v(x, t) = f(x) \cos \frac{\pi}{l} at$ 满足方程和边界条件, 解得 $v(x, t) = \cos \frac{\pi}{l} x \cos \frac{\pi}{l} at$ 。

令 $u = v + w$, 则
$$\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0, \frac{\partial w}{\partial x}|_{x=l} = 0 \\ w|_{t=0} = 0, \frac{\partial w}{\partial t}|_{t=0} = \sin \frac{\pi}{2l} x \end{cases}。$$

$$w = \sum_{n=0}^{\infty} \left(A_n \sin \frac{2n+1}{2l} a\pi t + B_n \cos \frac{2n+1}{2l} a\pi t \right) \sin \frac{2n+1}{2l} \pi x,$$

$$B_n = 0, \quad A_n = \frac{4}{(2n+1)a\pi} \int_0^l \sin \frac{1}{2l} \pi x \sin \frac{2n+1}{2l} \pi x dx。$$

$$A_0 = \frac{2l}{a\pi}, \quad A_n = 0 \quad (n=1, 2, \dots)。$$

所以 $u(x, t) = \cos \frac{\pi}{l} x \cos \frac{\pi}{l} at + \frac{2l}{a\pi} \sin \frac{\pi}{2l} at \sin \frac{\pi}{2l} x。$

236. 求解下列定解问题:
$$\begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = Ae^{-\alpha^2 \kappa t}, u|_{x=l} = Be^{-\beta^2 \kappa t} \\ u|_{t=0} = 0 \end{cases}$$

设 $v(x, t) = Af(x)e^{-\alpha^2 \kappa t} + Bg(x)e^{-\beta^2 \kappa t}$ 满足方程和边界条件, 解得

$$v(x, t) = A \frac{\sin \alpha(l-x)}{\sin \alpha l} e^{-\alpha^2 \kappa t} + B \frac{\sin \beta x}{\sin \beta l} e^{-\beta^2 \kappa t}。$$

$$\text{令 } u = v + w, \text{ 则 } \begin{cases} \frac{\partial w}{\partial t} - \kappa \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0, w|_{x=l} = 0 \\ w|_{t=0} = -A \frac{\sin \alpha(l-x)}{\sin \alpha l} - B \frac{\sin \beta x}{\sin \beta l} \end{cases}。$$

$$w = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{l} x e^{-\kappa \left(\frac{n\pi}{l}\right)^2 t},$$

$$\begin{aligned} C_n &= -\frac{2A}{l \sin \alpha l} \int_0^l \sin \alpha(l-x) \sin \frac{n\pi}{l} x dx - \frac{2B}{l \sin \beta l} \int_0^l \sin \beta x \sin \frac{n\pi}{l} x dx \\ &= -\frac{2A}{l} \int_0^l \cos \alpha x \sin \frac{n\pi}{l} x dx + \frac{2A}{l} \cot \alpha l \int_0^l \sin \alpha x \sin \frac{n\pi}{l} x dx - \frac{2B}{l \sin \beta l} \int_0^l \sin \beta x \sin \frac{n\pi}{l} x dx \\ &= -\frac{2An\pi}{(n\pi)^2 - (\alpha l)^2} \left[1 - (-1)^n \cos \alpha l \right] - \frac{2An\pi(-1)^n}{(n\pi)^2 - (\alpha l)^2} \cos \alpha l + \frac{2(-1)^n n\pi B}{(n\pi)^2 - (\beta l)^2} \\ &= -\frac{2An\pi}{(n\pi)^2 - (\alpha l)^2} + \frac{2(-1)^n n\pi B}{(n\pi)^2 - (\beta l)^2}。 \end{aligned}$$

上面假设不存在正整数 n, m , 使 $n\pi = \alpha l$, $m\pi = \beta l$ 。

$$\begin{aligned} u(x, t) &= A \frac{\sin \alpha(l-x)}{\sin \alpha l} e^{-\alpha^2 \kappa t} + B \frac{\sin \beta x}{\sin \beta l} e^{-\beta^2 \kappa t} \\ &\quad - \sum_{n=1}^{\infty} 2n\pi \left[\frac{A}{(n\pi)^2 - (\alpha l)^2} - \frac{(-1)^n B}{(n\pi)^2 - (\beta l)^2} \right] \sin \frac{n\pi}{l} x e^{-\kappa \left(\frac{n\pi}{l}\right)^2 t}。 \end{aligned}$$

$$237. \text{ 求解矩形区域内的第一类边值问题: } \begin{cases} \nabla^2 u = 0 \\ u|_{x=0} = \varphi_1(y), u|_{x=a} = \varphi_2(y) \\ u|_{y=0} = \psi_1(x), u|_{y=b} = \psi_2(x) \end{cases}。$$

设 $u = v + w$, 其中 v, w 分别满足

$$\begin{cases} \nabla^2 v = 0 \\ v|_{x=0} = 0, v|_{x=a} = 0 \\ v|_{y=0} = \psi_1(x), v|_{y=b} = \psi_2(x) \end{cases}, \begin{cases} \nabla^2 w = 0 \\ w|_{x=0} = \varphi_1(y), w|_{x=a} = \varphi_2(y) \\ w|_{y=0} = 0, w|_{y=b} = 0 \end{cases}.$$

$$v = \sum_{n=1}^{\infty} \left(A_n \operatorname{sh} \frac{n\pi}{a} y + B_n \operatorname{ch} \frac{n\pi}{a} y \right) \sin \frac{n\pi}{a} x,$$

$$B_n = \frac{2}{a} \int_0^a \psi_1(x) \sin \frac{n\pi}{a} x dx,$$

$$A_n = -B_n \coth \frac{n\pi b}{a} + \frac{2}{a \operatorname{sh} \frac{n\pi b}{a}} \int_0^a \psi_2(x) \sin \frac{n\pi}{a} x dx.$$

$$w = \sum_{n=1}^{\infty} \left(C_n \operatorname{sh} \frac{n\pi}{b} x + D_n \operatorname{ch} \frac{n\pi}{b} x \right) \sin \frac{n\pi}{b} y,$$

$$D_n = \frac{2}{b} \int_0^b \varphi_1(y) \sin \frac{n\pi}{b} y dy,$$

$$C_n = -D_n \coth \frac{n\pi a}{b} + \frac{2}{b \operatorname{sh} \frac{n\pi a}{b}} \int_0^b \varphi_2(y) \sin \frac{n\pi}{a} y dy.$$

238. 求矩形区域 $0 \leq x \leq a$, $0 \leq y \leq b$ 内满足边界条件 $\begin{cases} u|_{x=0} = Ay(b-y), u|_{x=a} = 0 \\ u|_{y=0} = B \sin \frac{\pi}{a} x, u|_{y=b} = 0 \end{cases}$ 的调和函数。

由上题结论, $B_n = \frac{2B}{a} \int_0^a \sin \frac{\pi}{a} x \sin \frac{n\pi}{a} x dx$, $B_1 = B$, $B_n = 0$ ($n=1, 2, \dots$)。

$$A_n = -B_n \coth \frac{n\pi b}{a}, \quad A_1 = -B \coth \frac{\pi b}{a}, \quad A_n = 0 \quad (n=1, 2, \dots).$$

$$D_n = \frac{2A}{b} \int_0^b y(b-y) \sin \frac{n\pi}{b} y dy = \frac{4Ab^2}{n^3 \pi^3} [1 - (-1)^n], \quad D_{2k} = 0, \quad D_{2k+1} = \frac{8Ab^2}{(2k+1)^3 \pi^3}.$$

$$C_n = -D_n \coth \frac{n\pi a}{b}, \quad C_{2k} = 0, \quad C_{2k+1} = -\frac{8Ab^2}{(2k+1)^3 \pi^3} \coth \frac{(2k+1)\pi a}{b}$$

$$\begin{aligned} \text{所以 } u(x, y) &= B \frac{\operatorname{sh} \frac{\pi}{a} (b-y)}{\operatorname{sh} \frac{\pi b}{a}} \sin \frac{\pi}{a} x \\ &+ \frac{8Ab^2}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3 \operatorname{sh} (2k+1)\pi a/b} \operatorname{sh} \frac{(2k+1)\pi}{b} (a-x) \sin \frac{(2k+1)\pi}{b} y. \end{aligned}$$

239. 求解 204 题
$$\begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = f \\ u|_{x=0} = 0, u|_{x=l} = u_0 \\ u|_{t=0} = \frac{u_0}{l} x \end{cases}$$

可得一个特解 $v(x) = -\frac{f}{2\kappa}x^2 + \left(\frac{u_0}{l} + \frac{fl}{2\kappa}\right)x$ 。令 $u = v + w$ ，则
$$\begin{cases} \frac{\partial w}{\partial t} - \kappa \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0, w|_{x=l} = 0 \\ w|_{t=0} = \frac{f}{2\kappa}(x^2 - lx) \end{cases}。$$

$$w = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{l} x e^{-\kappa \left(\frac{n\pi}{l}\right)^2 t}, \quad A_n = \frac{f}{\kappa l} \int_0^l (x^2 - lx) \sin \frac{n\pi}{l} x dx = \frac{2fl^2}{\kappa n^3 \pi^3} [(-1)^n - 1],$$

$$A_{2k} = 0, \quad A_{2k+1} = -\frac{4fl^2}{\kappa(2k+1)^3 \pi^3}。$$

所以 $u(x, t) = -\frac{f}{2\kappa}x^2 + \left(\frac{u_0}{l} + \frac{fl}{2\kappa}\right)x - \frac{4fl^2}{\kappa\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3} \sin \frac{(2k+1)\pi}{l} x e^{-\kappa \left(\frac{2k+1}{l}\pi\right)^2 t}。$

240. 竖直悬挂的一弹性杆，上端（ $x=0$ ）固定，下端（ $x=l$ ）挂有重物。杆的单位质量上受外力 $f(x)$ 作用（沿杆方向，重力包括在内）。试讨论杆的纵振动，设初始条件为

$$u|_{t=0} = \varphi(x), \quad \frac{\partial u}{\partial t}\bigg|_{t=0} = \psi(x)。提示：x=l 端的边界条件为 $\frac{\partial^2 u}{\partial t^2}\bigg|_{x=l} = -c^2 \frac{\partial u}{\partial x}\bigg|_{x=l} + g。$$$

该定解问题为
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x) \\ u|_{x=0} = 0, \frac{\partial^2 u}{\partial t^2}\bigg|_{x=l} = -c^2 \frac{\partial u}{\partial x}\bigg|_{x=l} + g \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t}\bigg|_{t=0} = \psi(x) \end{cases}$$

令 $u = \frac{g}{c^2}x + v$ ，则
$$\begin{cases} \frac{\partial^2 v}{\partial t^2} - a^2 \frac{\partial^2 v}{\partial x^2} = f(x) \\ v|_{x=0} = 0, \frac{\partial^2 v}{\partial t^2}\bigg|_{x=l} = -c^2 \frac{\partial v}{\partial x}\bigg|_{x=l} \\ v|_{t=0} = \varphi(x) - \frac{g}{c^2}x, \frac{\partial v}{\partial t}\bigg|_{t=0} = \psi(x) \end{cases}。$$

参考 228, 229 题, $v = \sum_{n=1}^{\infty} T_n(t) X_n(x)$, 其中 $X_n(x) = \sin \sqrt{\lambda_n} x$,

λ_n 为方程 $\sqrt{\lambda} \tan \sqrt{\lambda} l = \frac{c^2}{a^2}$ 的第 n 个正根。

令 $f(x) = \sum_{n=1}^{\infty} f_n X_n(x)$, 由 $\{X'_n(x)\}$ 的正交性可得

$$f_n = \frac{1}{\lambda_n N_n} \int_0^l f'(x) X'_n(x) dx = \frac{1}{\sqrt{\lambda_n} N_n} \int_0^l f'(x) \cos \sqrt{\lambda_n} x dx,$$

$$\text{其中 } N_n = \frac{l}{2} + \frac{\left(\frac{a}{c}\right)^2}{2 \left[\left(\frac{a}{c}\right)^4 \lambda_n + 1 \right]}.$$

将 $v = \sum_{n=1}^{\infty} T_n(t) X_n(x)$ 和 $f(x) = \sum_{n=1}^{\infty} f_n X_n(x)$ 代入方程得 $T_n''(t) + a^2 \lambda_n T_n(t) = f_n$. (*)

由初始条件, $\sum_{n=1}^{\infty} T_n(0) X_n(x) = \varphi(x) - \frac{g}{c^2} x$, 则

$$\begin{aligned} T_n(0) &= \frac{1}{\lambda_n N_n} \int_0^l \left[\varphi'(x) - \frac{g}{c^2} \right] X'_n(x) dx \\ &= \frac{1}{\sqrt{\lambda_n} N_n} \int_0^l \varphi'(x) \cos \sqrt{\lambda_n} x dx - \frac{g}{\sqrt{\lambda_n} N_n c^2} \int_0^l \cos \sqrt{\lambda_n} x dx = \varphi_n - \frac{g}{\lambda_n N_n c^2} \sin \sqrt{\lambda_n} l. \end{aligned}$$

$\sum_{n=1}^{\infty} T'_n(0) X_n(x) = \psi(x)$, 则

$$T'_n(0) = \frac{1}{\lambda_n N_n} \int_0^l \psi'(x) X'_n(x) dx = \frac{1}{\sqrt{\lambda_n} N_n} \int_0^l \psi'(x) \cos \sqrt{\lambda_n} x dx = \psi_n.$$

解方程 (*) 得

$$T_n(t) = \frac{\psi_n}{a \sqrt{\lambda_n}} \sin a \sqrt{\lambda_n} t + \left(\varphi_n - \frac{g}{\lambda_n N_n c^2} \sin \sqrt{\lambda_n} l - \frac{f_n}{a^2 \lambda_n} \right) \cos a \sqrt{\lambda_n} t + \frac{f_n}{a^2 \lambda_n}.$$

所以 $u(x, t) = \frac{g}{c^2} x + \sum_{n=1}^{\infty} T_n(t) \sin \sqrt{\lambda_n} x$.

241. 求解 210 题, 设初位移及初速度分别为 $\varphi(x), \psi(x)$ 。

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0, x \neq \frac{l}{2} \\ u|_{x=0} = 0, u|_{x=l} = 0, u|_{x=l/2-0} = u|_{x=l/2+0}, \frac{T}{M} \left(\frac{\partial u}{\partial x} \Big|_{x=l/2+0} - \frac{\partial u}{\partial x} \Big|_{x=l/2-0} \right) = \frac{\partial^2 u}{\partial t^2} \Big|_{x=l/2} + g. \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t} \Big|_{t=0} = \psi(x) \end{cases}$$

令 $u(x, t) = \frac{Mg}{2T} \left(\left| x - \frac{l}{2} \right| - \frac{l}{2} \right) + v(x, t)$, 则 $v(x, t)$ 满足

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} - a^2 \frac{\partial^2 v}{\partial x^2} = 0, x \neq \frac{l}{2} \\ v|_{x=0} = 0, v|_{x=l} = 0, v|_{x=l/2-0} = v|_{x=l/2+0}, \frac{\partial^2 v}{\partial t^2} \Big|_{x=l/2} + c^2 \frac{\partial v}{\partial x} \Big|_{x=l/2-0} = c^2 \frac{\partial v}{\partial x} \Big|_{x=l/2+0} - \frac{\partial^2 v}{\partial t^2} \Big|_{x=l/2}. \\ v|_{t=0} = \varphi(x) - \frac{Mg}{2T} \left(\left| x - \frac{l}{2} \right| - \frac{l}{2} \right), \frac{\partial v}{\partial t} \Big|_{t=0} = \psi(x) \end{cases}$$

其中 $c^2 = \frac{2T}{M}$ 。令 $v = v_1 + v_2$, v_1, v_2 分别满足:

$$\begin{cases} \frac{\partial^2 v_1}{\partial t^2} - a^2 \frac{\partial^2 v_1}{\partial x^2} = 0, x \neq \frac{l}{2} \\ v_1|_{x=0} = 0, v_1|_{x=l} = 0, v_1|_{x=l/2-0} = 0, v_1|_{x=l/2+0} = 0, \\ 2 \frac{\partial^2 v_1}{\partial t^2} \Big|_{x=l/2} = c^2 \left(\frac{\partial v_1}{\partial x} \Big|_{x=l/2+0} - \frac{\partial v_1}{\partial x} \Big|_{x=l/2-0} \right) \\ \frac{\partial^2 v_2}{\partial t^2} - a^2 \frac{\partial^2 v_2}{\partial x^2} = 0, x \neq \frac{l}{2} \\ v_2|_{x=0} = 0, v_2|_{x=l} = 0, \frac{\partial^2 v_2}{\partial t^2} \Big|_{x=l/2} + c^2 \frac{\partial v_2}{\partial x} \Big|_{x=l/2-0} = 0, c^2 \frac{\partial v_2}{\partial x} \Big|_{x=l/2+0} - \frac{\partial^2 v_2}{\partial t^2} \Big|_{x=l/2} = 0. \\ v_2|_{x=l/2-0} = v_2|_{x=l/2+0} \end{cases}$$

可看出 v_1 在 $[0, l/2)$ 和 $(l/2, 0]$ 上的本征函数都是 $\sin \frac{2n\pi}{l} x$, 由条件

$$2 \frac{\partial^2 v_1}{\partial t^2} \Big|_{x=l/2} = c^2 \left(\frac{\partial v_1}{\partial x} \Big|_{x=l/2+0} - \frac{\partial v_1}{\partial x} \Big|_{x=l/2-0} \right) \text{ 可知在 } [0, l] \setminus \{l/2\} \text{ 上有}$$

$$v_1 = \sum_{n=1}^{\infty} \left(A_n \sin \frac{2n\pi}{l} at + B_n \cos \frac{2n\pi}{l} at \right) \sin \frac{2n\pi}{l} x。$$

$$\text{同 228, 229 题可得 } v_2 \text{ 在 } [0, l] \setminus \{l/2\} \text{ 上的本征函数为 } X_n(x) = \begin{cases} \sin \sqrt{\lambda_n} x, 0 \leq x < l/2 \\ \sin \sqrt{\lambda_n} (l-x), l/2 < x \leq l \end{cases},$$

$$\text{其中本征值 } \lambda_n \text{ 为方程 } \sqrt{\lambda} \tan \frac{\sqrt{\lambda} l}{2} = \frac{c^2}{a^2} \text{ 的正根。}$$

$$\text{由条件 } v_2|_{x=l/2-0} = v_2|_{x=l/2+0} \text{ 可得 } v_2 = \sum_{n=1}^{\infty} (C_n \sin \sqrt{\lambda_n} at + D_n \cos \sqrt{\lambda_n} at) X_n(x)$$

$$(x \in [0, l] \setminus \{l/2\})。$$

$$\text{即 } v = \sum_{n=1}^{\infty} \left(A_n \sin \frac{2n\pi}{l} at + B_n \cos \frac{2n\pi}{l} at \right) \sin \frac{2n\pi}{l} x \\ + \sum_{n=1}^{\infty} (C_n \sin \sqrt{\lambda_n} at + D_n \cos \sqrt{\lambda_n} at) X_n(x)。$$

$$\text{将初始条件写成 } \frac{\partial v}{\partial x} \bigg|_{t=0} = \begin{cases} \varphi'(x) + \frac{Mg}{2T}, 0 \leq x < \frac{l}{2} \\ \varphi'(x) - \frac{Mg}{2T}, \frac{l}{2} < x \leq l \end{cases}, \quad \frac{\partial^2 v}{\partial x \partial t} \bigg|_{t=0} = \psi'(x),$$

$$\int_0^l X'_n(x) \cos \frac{2m\pi}{l} x dx = \sqrt{\lambda_n} \int_0^{l/2} \cos \sqrt{\lambda_n} x \cos \frac{2m\pi}{l} x dx - \sqrt{\lambda_n} \int_{l/2}^l \cos \sqrt{\lambda_n} (l-x) \cos \frac{2m\pi}{l} x dx \\ = \sqrt{\lambda_n} \int_0^{l/2} \cos \sqrt{\lambda_n} x \cos \frac{2m\pi}{l} x dx - \sqrt{\lambda_n} \int_0^{l/2} \cos \sqrt{\lambda_n} y \cos \frac{2m\pi}{l} y dy = 0,$$

$$\text{即 } \left\{ \cos \frac{2n\pi}{l} x \right\} \text{ 与 } \{X'_n(x)\} \text{ 在 } [0, l] \text{ 上正交。}$$

$$\int_0^l X_n'^2(x) dx = \int_0^{l/2} X_n'^2(x) dx + \int_{l/2}^l X_n'^2(x) dx = 2 \int_0^{l/2} X_n'^2(x) dx \\ = \lambda_n \left[\frac{l}{2} + \frac{\left(\frac{a}{c}\right)^2}{\left(\frac{a}{c}\right)^4 \lambda_n + 1} \right] = \lambda_n \left(\frac{l}{2} + \frac{2a^2 MT}{M^2 a^4 \lambda_n + 4T^2} \right) = \lambda_n N_n。$$

$$\text{由此可定出系数: } A_n = \frac{1}{2n^2 \pi^2 a} \int_0^l \psi'(x) \cos \frac{2n\pi}{l} x dx,$$

$$B_n = \frac{1}{n\pi} \int_0^l \varphi'(x) \cos \frac{2n\pi}{l} x dx, \quad C_n = \frac{1}{\lambda_n \sqrt{\lambda_n} N_n a} \int_0^l \psi'(x) X'_n(x) dx,$$

$$D_n = \frac{1}{\lambda_n N_n} \left[\int_0^l \varphi'(x) X_n'(x) dx + \frac{Mg}{T} \sin \frac{\sqrt{\lambda_n} l}{2} \right].$$

242. 考虑有界弦的阻尼振动, 如果一端固定, 另一端在外力作用下做周期运动, 经过足够长时间后, 初始条件的影响则因阻尼的作用而衰减殆尽, 因而问题归结为求解无初值问题

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} + h \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2} & \text{的周期解, 试求之。} \\ u|_{x=0} = 0, u|_{x=l} = A \cos \omega t \end{cases}$$

采用复数解。设 $u(x, t) = f(x)e^{i\omega t}$, 原定解问题化为
$$\begin{cases} f''(x) + \left[\left(\frac{\omega}{a} \right)^2 - i \frac{\omega h}{a^2} \right] f(x) = 0 \\ f(0) = 0, f(l) = A \end{cases}.$$

记 $\sqrt{\left(\frac{\omega}{a} \right)^2 - i \frac{\omega h}{a^2}} = \alpha - i\beta$, 可解得 $u(x, t) = A \frac{\sin(\alpha - i\beta)x}{\sin(\alpha - i\beta)l} e^{i\omega t}$ 。

取实部得

$$u(x, t) = \frac{A}{\sin^2 \alpha l + \sin^2 \beta l} \left[(\sin \alpha x \sin \alpha l \operatorname{ch} \beta x \operatorname{ch} \beta l + \cos \alpha x \cos \alpha l \operatorname{sh} \beta x \operatorname{sh} \beta l) \cos \omega t - (\sin \alpha x \cos \alpha l \operatorname{ch} \beta x \operatorname{sh} \beta l - \cos \alpha x \sin \alpha l \operatorname{sh} \beta x \operatorname{ch} \beta l) \sin \omega t \right].$$

243. 热传导问题也存在无初值问题。典型的例子是地表温度的日变化或年变化向地层内传播形成的温度波。把地球设想为均匀半无界空间, 试求无初值问题

$$\begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = A \cos \omega t, |u|_{x \rightarrow \infty} < \infty \end{cases} \quad \text{的周期解。}$$

设 $u = f(x)e^{i\omega t}$, 则
$$\begin{cases} f''(x) - \frac{i\omega}{\kappa} f(x) = 0 \\ f(0) = A, |f(\infty)| < \infty \end{cases}.$$
 解得 $u(x, t) = A e^{-\sqrt{\frac{\omega}{2\kappa}}(1+i)x} e^{i\omega t}$ 。

取实部得 $u(x, t) = A e^{-\sqrt{\frac{\omega}{2\kappa}}x} \cos\left(\sqrt{\frac{\omega}{2\kappa}}x - \omega t\right)$ 。

244. 写出下列正交曲线坐标系中的 Laplace 算子:

(1) 椭圆柱坐标系 (ξ, η, z) : $x = a\xi\eta$, $y = a\sqrt{(\xi^2 - 1)(1 - \eta^2)}$, $z = z$;

(2) 抛物线柱坐标系 (λ, μ, z) : $x = \frac{1}{2}(\lambda - \mu)$, $y = \sqrt{\lambda\mu}$, $z = z$;

(3) 锥面坐标系 (r, λ, μ) : $x = \frac{r}{a}\sqrt{(a^2 - \lambda)(a^2 + \mu)}$, $y = \frac{r}{b}\sqrt{(b^2 + \lambda)(b^2 - \mu)}$,

$z = \frac{r\sqrt{\lambda\mu}}{ab}$, 其中 $a^2 + b^2 = 1$;

(4) 椭球坐标系 (λ, μ, ν) : $x^2 = \frac{(a^2 + \lambda)(a^2 + \mu)(a^2 + \nu)}{(a^2 - b^2)(a^2 - c^2)}$,

$y^2 = \frac{(b^2 + \lambda)(b^2 + \mu)(b^2 + \nu)}{(b^2 - c^2)(b^2 - a^2)}$, $z^2 = \frac{(c^2 + \lambda)(c^2 + \mu)(c^2 + \nu)}{(c^2 - a^2)(c^2 - b^2)}$ 。

(1) $dx = a\eta d\xi + a\xi d\eta$, $dy = a\frac{1-\eta^2}{\sqrt{(\xi^2-1)(1-\eta^2)}}\xi d\xi - a\frac{\xi^2-1}{\sqrt{(\xi^2-1)(1-\eta^2)}}\eta d\eta$,

$ds^2 = dx^2 + dy^2 + dz^2 = a^2 \frac{\xi^2 - \eta^2}{\xi^2 - 1} d\xi^2 + a^2 \frac{\xi^2 - \eta^2}{1 - \eta^2} d\eta^2 + dz^2$ 。

即度规矩阵 $G = \begin{pmatrix} a^2 \frac{\xi^2 - \eta^2}{\xi^2 - 1} & 0 & 0 \\ 0 & a^2 \frac{\xi^2 - \eta^2}{1 - \eta^2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\sqrt{|G|} = a^2 \frac{\xi^2 - \eta^2}{\sqrt{(\xi^2 - 1)(1 - \eta^2)}}$ 。

$d = \frac{\partial}{\partial \xi} d\xi + \frac{\partial}{\partial \eta} d\eta + \frac{\partial}{\partial z} dz$,

$\begin{aligned} *d &= \frac{1}{a^2} \frac{\xi^2 - 1}{\xi^2 - \eta^2} \sqrt{|G|} \frac{\partial}{\partial \xi} d\eta \wedge dz + \frac{1}{a^2} \frac{1 - \eta^2}{\xi^2 - \eta^2} \sqrt{|G|} \frac{\partial}{\partial \eta} dz \wedge d\xi + \sqrt{|G|} \frac{\partial}{\partial z} d\xi \wedge d\eta \\ &= \frac{\xi^2 - 1}{\sqrt{(\xi^2 - 1)(1 - \eta^2)}} \frac{\partial}{\partial \xi} d\eta \wedge dz + \frac{1 - \eta^2}{\sqrt{(\xi^2 - 1)(1 - \eta^2)}} \frac{\partial}{\partial \eta} dz \wedge d\xi + \sqrt{|G|} \frac{\partial}{\partial z} d\xi \wedge d\eta, \end{aligned}$

$d^*d = \frac{\partial}{\partial \xi} \left(\sqrt{\frac{\xi^2 - 1}{1 - \eta^2}} \frac{\partial}{\partial \xi} \right) d\xi \wedge d\eta \wedge dz + \frac{\partial}{\partial \eta} \left(\sqrt{\frac{1 - \eta^2}{\xi^2 - 1}} \frac{\partial}{\partial \eta} \right) d\eta \wedge dz \wedge d\xi$

$$\begin{aligned}
& +a^2 \frac{\partial}{\partial z} \left(\frac{\xi^2 - \eta^2}{\sqrt{(\xi^2 - 1)(1 - \eta^2)}} \frac{\partial}{\partial z} \right) dz \wedge d\xi \wedge d\eta \\
& = \left[\frac{\sqrt{\xi^2 - 1}}{a^2(\xi^2 - \eta^2)} \frac{\partial}{\partial \xi} \left(\sqrt{\xi^2 - 1} \frac{\partial}{\partial \xi} \right) + \frac{\sqrt{1 - \eta^2}}{a^2(\xi^2 - \eta^2)} \frac{\partial}{\partial \eta} \left(\sqrt{1 - \eta^2} \frac{\partial}{\partial \eta} \right) + \frac{\partial^2}{\partial z^2} \right] \sqrt{|G|} d\xi \wedge d\eta \wedge dz \\
\nabla^2 = {}^* d^* d &= \frac{\sqrt{\xi^2 - 1}}{a^2(\xi^2 - \eta^2)} \frac{\partial}{\partial \xi} \left(\sqrt{\xi^2 - 1} \frac{\partial}{\partial \xi} \right) + \frac{\sqrt{1 - \eta^2}}{a^2(\xi^2 - \eta^2)} \frac{\partial}{\partial \eta} \left(\sqrt{1 - \eta^2} \frac{\partial}{\partial \eta} \right) + \frac{\partial^2}{\partial z^2} .
\end{aligned}$$

$$(2) \quad dx = \frac{1}{2} d\lambda - \frac{1}{2} d\mu, \quad dy = \frac{\mu}{2\sqrt{\lambda\mu}} d\lambda + \frac{\lambda}{2\sqrt{\lambda\mu}} d\mu,$$

$$ds^2 = \frac{1}{4} \left(1 + \frac{\mu}{\lambda} \right) d\lambda^2 + \frac{1}{4} \left(1 + \frac{\lambda}{\mu} \right) d\mu^2 + dz^2,$$

$$G = \begin{pmatrix} \frac{1}{4} \left(1 + \frac{\mu}{\lambda} \right) & 0 & 0 \\ 0 & \frac{1}{4} \left(1 + \frac{\lambda}{\mu} \right) & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \sqrt{|G|} = \frac{\lambda + \mu}{4\sqrt{\lambda\mu}}.$$

$$d = \frac{\partial}{\partial \lambda} d\lambda + \frac{\partial}{\partial \mu} d\mu + \frac{\partial}{\partial z} dz,$$

$${}^* d = \sqrt{\frac{\lambda}{\mu}} \frac{\partial}{\partial \lambda} d\mu \wedge dz + \sqrt{\frac{\mu}{\lambda}} \frac{\partial}{\partial \mu} dz \wedge d\lambda + \frac{\lambda + \mu}{4\sqrt{\lambda\mu}} \frac{\partial}{\partial z} d\lambda \wedge d\mu,$$

$$d^* d = \left[\frac{4\sqrt{\lambda}}{\lambda + \mu} \frac{\partial}{\partial \lambda} \left(\sqrt{\lambda} \frac{\partial}{\partial \lambda} \right) + \frac{4\sqrt{\mu}}{\lambda + \mu} \frac{\partial}{\partial \mu} \left(\sqrt{\mu} \frac{\partial}{\partial \mu} \right) + \frac{\partial^2}{\partial z^2} \right] \frac{\lambda + \mu}{4\sqrt{\lambda\mu}} d\lambda d\mu \wedge dz,$$

$$\nabla^2 = \frac{4\sqrt{\lambda}}{\lambda + \mu} \frac{\partial}{\partial \lambda} \left(\sqrt{\lambda} \frac{\partial}{\partial \lambda} \right) + \frac{4\sqrt{\mu}}{\lambda + \mu} \frac{\partial}{\partial \mu} \left(\sqrt{\mu} \frac{\partial}{\partial \mu} \right) + \frac{\partial^2}{\partial z^2}.$$

$$(3) \quad dx = \frac{1}{a} \sqrt{(a^2 - \lambda)(a^2 + \mu)} dr - \frac{r}{2a} \sqrt{\frac{a^2 + \mu}{a^2 - \lambda}} d\lambda + \frac{r}{2a} \sqrt{\frac{a^2 - \lambda}{a^2 + \mu}} d\mu,$$

$$dy = \frac{1}{b} \sqrt{(b^2 + \lambda)(b^2 - \mu)} dr + \frac{r}{2b} \sqrt{\frac{b^2 - \mu}{b^2 + \lambda}} d\lambda - \frac{r}{2b} \sqrt{\frac{b^2 + \lambda}{b^2 - \mu}} d\mu,$$

$$dz = \frac{\sqrt{\lambda\mu}}{ab} dr + \frac{r}{2ab} \sqrt{\frac{\mu}{\lambda}} d\lambda + \frac{r}{2ab} \sqrt{\frac{\lambda}{\mu}} d\mu.$$

$$ds^2 = dr^2 + \frac{r^2(\lambda + \mu)}{4\lambda(a^2 - \lambda)(b^2 + \lambda)} d\lambda^2 + \frac{r^2(\lambda + \mu)}{4\mu(a^2 + \mu)(b^2 - \mu)} d\mu^2,$$

$$G = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{r^2(\lambda + \mu)}{4\lambda(a^2 - \lambda)(b^2 + \lambda)} & 0 \\ 0 & 0 & \frac{r^2(\lambda + \mu)}{4\mu(a^2 + \mu)(b^2 - \mu)} \end{pmatrix},$$

$$\sqrt{|G|} = \frac{r^2(\lambda + \mu)}{4\sqrt{\lambda\mu(a^2 - \lambda)(b^2 + \lambda)(a^2 + \mu)(b^2 - \mu)}}.$$

$$\begin{aligned} {}^*d &= \sqrt{|G|} \frac{\partial}{\partial r} d\lambda \wedge d\mu + \sqrt{\frac{\lambda(a^2 - \lambda)(b^2 + \lambda)}{\mu(a^2 + \mu)(b^2 - \mu)}} \frac{\partial}{\partial \lambda} d\mu \wedge dr \\ &\quad + \sqrt{\frac{\mu(a^2 + \mu)(b^2 - \mu)}{\lambda(a^2 - \lambda)(b^2 + \lambda)}} \frac{\partial}{\partial \mu} dr \wedge d\lambda, \end{aligned}$$

$$\begin{aligned} d^*d &= \left\{ \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{4}{r^2(\lambda + \mu)} \sqrt{\lambda(a^2 - \lambda)(b^2 + \lambda)} \frac{\partial}{\partial \lambda} \left[\sqrt{\lambda(a^2 - \lambda)(b^2 + \lambda)} \frac{\partial}{\partial \lambda} \right] \right. \\ &\quad \left. + \frac{4}{r^2(\lambda + \mu)} \sqrt{\mu(a^2 + \mu)(b^2 - \mu)} \frac{\partial}{\partial \mu} \left[\sqrt{\mu(a^2 + \mu)(b^2 - \mu)} \frac{\partial}{\partial \mu} \right] \right\} \sqrt{|G|} dr \wedge d\lambda \wedge d\mu \end{aligned}$$

$$\begin{aligned} \nabla^2 &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{4}{r^2(\lambda + \mu)} \sqrt{\lambda(a^2 - \lambda)(b^2 + \lambda)} \frac{\partial}{\partial \lambda} \left(\sqrt{\lambda(a^2 - \lambda)(b^2 + \lambda)} \frac{\partial}{\partial \lambda} \right) \\ &\quad + \frac{4}{r^2(\lambda + \mu)} \sqrt{\mu(a^2 + \mu)(b^2 - \mu)} \frac{\partial}{\partial \mu} \left(\sqrt{\mu(a^2 + \mu)(b^2 - \mu)} \frac{\partial}{\partial \mu} \right). \end{aligned}$$

$${}^{(4)}dx = \frac{1}{2} \sqrt{\frac{(a^2 + \lambda)(a^2 + \mu)(a^2 + \nu)}{(a^2 - b^2)(a^2 - c^2)}} \left(\frac{1}{a^2 + \lambda} d\lambda + \frac{1}{a^2 + \mu} d\mu + \frac{1}{a^2 + \nu} d\nu \right),$$

$$dy = \frac{1}{2} \sqrt{\frac{(b^2 + \lambda)(b^2 + \mu)(b^2 + \nu)}{(b^2 - c^2)(b^2 - a^2)}} \left(\frac{1}{b^2 + \lambda} d\lambda + \frac{1}{b^2 + \mu} d\mu + \frac{1}{b^2 + \nu} d\nu \right),$$

$$dz = \frac{1}{2} \sqrt{\frac{(c^2 + \lambda)(c^2 + \mu)(c^2 + \nu)}{(c^2 - a^2)(c^2 - b^2)}} \left(\frac{1}{c^2 + \lambda} d\lambda + \frac{1}{c^2 + \mu} d\mu + \frac{1}{c^2 + \nu} d\nu \right).$$

$$ds^2 = \frac{(\lambda - \mu)(\lambda - \nu)}{4\varphi(\lambda)} d\lambda^2 + \frac{(\mu - \nu)(\mu - \lambda)}{4\varphi(\mu)} d\mu^2 + \frac{(\nu - \lambda)(\nu - \mu)}{4\varphi(\nu)} d\nu^2,$$

其中 $\varphi(x) = (a^2 + x)(b^2 + x)(c^2 + x)$ 。

$$\nabla^2 = \frac{4}{(\lambda - \mu)(\lambda - \nu)(\mu - \nu)} \left[(\mu - \nu) \sqrt{\varphi(\lambda)} \frac{\partial}{\partial \lambda} \left(\sqrt{\varphi(\lambda)} \frac{\partial}{\partial \lambda} \right) \right. \\ \left. + (\lambda - \nu) \sqrt{\varphi(\mu)} \frac{\partial}{\partial \mu} \left(\sqrt{\varphi(\mu)} \frac{\partial}{\partial \mu} \right) + (\lambda - \mu) \sqrt{\varphi(\nu)} \frac{\partial}{\partial \nu} \left(\sqrt{\varphi(\nu)} \frac{\partial}{\partial \nu} \right) \right] .$$

245. 在上述各坐标系中将 Laplace 方程分离变量。

(1) 令 $u(\xi, \eta, z) = v(\xi, \eta) Z(z)$, 代入 $\nabla^2 u = 0$ 得

$$\frac{Z \sqrt{\xi^2 - 1}}{a^2 (\xi^2 - \eta^2)} \frac{\partial}{\partial \xi} \left(\sqrt{\xi^2 - 1} \frac{\partial v}{\partial \xi} \right) + \frac{Z \sqrt{1 - \eta^2}}{a^2 (\xi^2 - \eta^2)} \frac{\partial}{\partial \eta} \left(\sqrt{1 - \eta^2} \frac{\partial v}{\partial \eta} \right) + v Z'' = 0 ,$$

两边同除 vZ 得

$$\frac{\sqrt{\xi^2 - 1}}{a^2 v (\xi^2 - \eta^2)} \frac{\partial}{\partial \xi} \left(\sqrt{\xi^2 - 1} \frac{\partial v}{\partial \xi} \right) + \frac{\sqrt{1 - \eta^2}}{a^2 v (\xi^2 - \eta^2)} \frac{\partial}{\partial \eta} \left(\sqrt{1 - \eta^2} \frac{\partial v}{\partial \eta} \right) = -\frac{Z''}{Z} = \lambda$$

$$\text{所以} \begin{cases} Z'' + \lambda Z = 0 \\ \sqrt{\xi^2 - 1} \frac{\partial}{\partial \xi} \left(\sqrt{\xi^2 - 1} \frac{\partial v}{\partial \xi} \right) + \sqrt{1 - \eta^2} \frac{\partial}{\partial \eta} \left(\sqrt{1 - \eta^2} \frac{\partial v}{\partial \eta} \right) = a^2 \lambda v (\xi^2 - \eta^2) \end{cases} ,$$

令 $v(\xi, \eta) = \Xi(\xi) H(\eta)$ 代入上面第二式得

$$H \sqrt{\xi^2 - 1} \frac{d}{d\xi} \left(\sqrt{\xi^2 - 1} \frac{d\Xi}{d\xi} \right) + \Xi \sqrt{1 - \eta^2} \frac{d}{d\eta} \left(\sqrt{1 - \eta^2} \frac{dH}{d\eta} \right) = a^2 \lambda \Xi H (\xi^2 - 1 + 1 - \eta^2) ,$$

两边同除 ΞH 得

$$\frac{\sqrt{\xi^2 - 1}}{\Xi} \frac{d}{d\xi} \left(\sqrt{\xi^2 - 1} \frac{d\Xi}{d\xi} \right) - a^2 \lambda (\xi^2 - 1) = -\frac{\sqrt{1 - \eta^2}}{H} \frac{d}{d\eta} \left(\sqrt{1 - \eta^2} \frac{dH}{d\eta} \right) + a^2 \lambda (1 - \eta^2) = -\mu ,$$

$$\text{所以} \begin{cases} Z'' + \lambda Z = 0 \\ \sqrt{\xi^2 - 1} \frac{d}{d\xi} \left(\sqrt{\xi^2 - 1} \frac{d\Xi}{d\xi} \right) + [\mu - a^2 \lambda (\xi^2 - 1)] \Xi = 0 \\ \sqrt{1 - \eta^2} \frac{d}{d\eta} \left(\sqrt{1 - \eta^2} \frac{dH}{d\eta} \right) - [\mu + a^2 \lambda (1 - \eta^2)] H = 0 \end{cases} .$$

(2) 令 $u(\lambda, \mu, z) = \Lambda(\lambda) M(\mu) Z(z)$, 代入 $\nabla^2 u = 0$ 得

$$\frac{4MZ\sqrt{\lambda}}{\lambda+\mu} \frac{d}{d\lambda} \left(\sqrt{\lambda} \frac{d\Lambda}{d\lambda} \right) + \frac{4\Lambda Z\sqrt{\mu}}{\lambda+\mu} \frac{d}{d\mu} \left(\sqrt{\mu} \frac{dM}{d\mu} \right) = -\Lambda M Z'',$$

两边同除 $\Lambda M Z$ ，令它等于 σ 得

$$\begin{cases} \frac{\sqrt{\lambda}}{\Lambda} \frac{d}{d\lambda} \left(\sqrt{\lambda} \frac{d\Lambda}{d\lambda} \right) + \frac{\sqrt{\mu}}{M} \frac{d}{d\mu} \left(\sqrt{\mu} \frac{dM}{d\mu} \right) = \sigma \left(\frac{\lambda}{4} + \frac{\mu}{4} \right). \\ Z'' + \sigma Z = 0 \end{cases}$$

$$\text{进一步得到} \begin{cases} \sqrt{\lambda} \frac{d}{d\lambda} \left(\sqrt{\lambda} \frac{d\Lambda}{d\lambda} \right) + \left(\tau - \frac{\sigma}{4} \lambda \right) \Lambda = 0 \\ \sqrt{\mu} \frac{d}{d\mu} \left(\sqrt{\mu} \frac{dM}{d\mu} \right) - \left(\tau + \frac{\sigma}{4} \mu \right) M = 0. \\ Z'' + \sigma Z = 0 \end{cases}$$

$$(3) \quad u(r, \lambda, \mu) = R(r) \Lambda(\lambda) M(\mu),$$

$$\begin{cases} \frac{1}{\Lambda} \sqrt{\lambda(a^2 - \lambda)(b^2 + \lambda)} \frac{d}{d\lambda} \left(\sqrt{\lambda(a^2 - \lambda)(b^2 + \lambda)} \frac{d\Lambda}{d\lambda} \right) \\ + \frac{1}{M} \sqrt{\mu(a^2 + \mu)(b^2 - \mu)} \frac{d}{d\mu} \left(\sqrt{\mu(a^2 + \mu)(b^2 - \mu)} \frac{dM}{d\mu} \right) = \sigma \left(\frac{\lambda}{4} + \frac{\mu}{4} \right) \\ \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \sigma R = 0 \end{cases}$$

$$\begin{cases} \sqrt{\lambda(a^2 - \lambda)(b^2 + \lambda)} \frac{d}{d\lambda} \left(\sqrt{\lambda(a^2 - \lambda)(b^2 + \lambda)} \frac{d\Lambda}{d\lambda} \right) + \left(\tau - \frac{\sigma}{4} \lambda \right) \Lambda = 0 \\ \sqrt{\mu(a^2 + \mu)(b^2 - \mu)} \frac{d}{d\mu} \left(\sqrt{\mu(a^2 + \mu)(b^2 - \mu)} \frac{dM}{d\mu} \right) - \left(\tau + \frac{\sigma}{4} \mu \right) M = 0 \\ \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \sigma R = 0 \end{cases}$$

$$(4) \quad u(\lambda, \mu, \nu) = \Lambda(\lambda) M(\mu) N(\nu), \quad \text{记 } L_\sigma(\Sigma) = \sqrt{\varphi(\sigma)} \frac{d}{d\sigma} \left(\sqrt{\varphi(\sigma)} \frac{d\Sigma}{d\sigma} \right), \quad \text{则}$$

$$\frac{\mu}{(\lambda - \mu)\Lambda} L_\lambda(\Lambda) + \frac{\lambda}{(\lambda - \mu)M} L_\mu(M) - \nu \left[\frac{1}{(\lambda - \mu)M} L_\mu(M) + \frac{1}{(\lambda - \mu)\Lambda} L_\lambda(\Lambda) \right] = -\frac{1}{N} L_\nu(N)$$

$$\text{令 } \frac{\mu}{(\lambda - \mu)\Lambda} L_\lambda(\Lambda) + \frac{\lambda}{(\lambda - \mu)M} L_\mu(M) = \tau, \quad (a)$$

$$\frac{1}{(\lambda - \mu)M} L_{\mu}(M) + \frac{1}{(\lambda - \mu)\Lambda} L_{\lambda}(\Lambda) = \sigma, \quad (b)$$

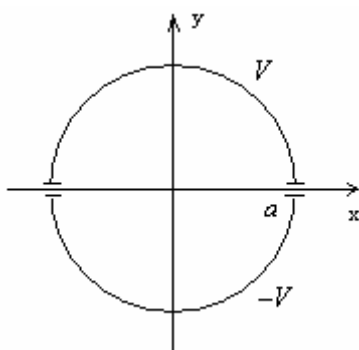
$$\text{则 } \tau - \sigma\nu = -\frac{1}{N} L_{\nu}(N)。$$

$$(a) -\lambda \times (b) \text{ 得 } -\frac{L_{\lambda}(\Lambda)}{\Lambda} = \tau - \sigma\lambda,$$

$$(a) -\mu \times (b) \text{ 得 } \frac{L_{\mu}(M)}{M} = \tau - \sigma\mu。$$

$$\text{即 } \begin{cases} L_{\lambda}(\Lambda) + (\tau - \sigma\lambda)\Lambda = 0 \\ L_{\mu}(M) + (-\tau + \sigma\mu)M = 0。 \\ L_{\nu}(N) + (\tau - \sigma\nu)N = 0 \end{cases}$$

246. 一无穷长空心圆柱导体，分成两半，互相绝缘。一半电势为 V ，另一半为 $-V$ ，求柱内电势分布。



在极坐标系下令 $u(\rho, \varphi) = P(\rho)\Phi(\varphi)$ 对 Laplace 方程分离变量得

$$\begin{cases} \Phi'' + \lambda\Phi = 0 \\ \rho \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) - \lambda P = 0 \end{cases}$$

$$\text{本征值问题 } \begin{cases} \Phi'' + \lambda\Phi = 0 \\ \Phi(-\pi) = \Phi(\pi), \Phi'(-\pi) = \Phi'(\pi) \end{cases},$$

可得 $\lambda = n^2$ ($n = 0, 1, 2, \dots$)，本征函数 $1, \cos n\varphi, \sin n\varphi$ ($n = 1, 2, \dots$)。

做代换 $\rho = e^t$ ，则有 $\rho \frac{d}{d\rho} = \frac{d}{dt}$ ，则 P 的方程化为 $\frac{d^2 P}{dt^2} - \lambda P = 0$ 。

当 $\lambda = 0$ 时，解得 $P_0 = A + Bt = A + B \ln \rho$ ，

当 $\lambda > 0$ 时, 解得 $P_n = C_n e^{nt} + D_n e^{-nt} = C_n \rho^n + D_n \rho^{-n}$ 。

$$\text{所以 } u(\rho, \varphi) = A + B \ln \rho + \sum_{n=1}^{\infty} (C_n \rho^n + D_n \rho^{-n}) \cos n\varphi + \sum_{n=1}^{\infty} (E_n \rho^n + F_n \rho^{-n}) \sin n\varphi。$$

由于 $u(0, \varphi)$ 为有限值, 所以 $B = 0$, $D_n = 0$, $F_n = 0$, 即

$$u(\rho, \varphi) = A + \sum_{n=1}^{\infty} C_n \rho^n \cos n\varphi + \sum_{n=1}^{\infty} E_n \rho^n \sin n\varphi。$$

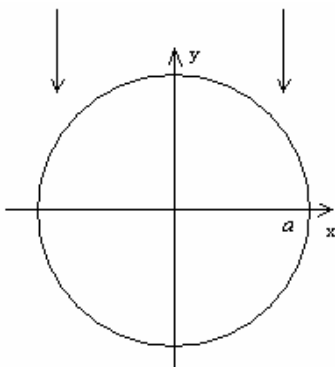
$$\text{边界条件为 } u(a, \varphi) = \begin{cases} -V, & -\pi < \varphi < 0 \\ V, & 0 < \varphi < \pi \end{cases}。$$

将上面右边函数在 $[-\pi, \pi]$ 上展开为 Fourier 级数, 则 $u(a, \varphi) = \frac{4V}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} \sin(2k+1)\varphi$ 。

由此可定出系数 $A = 0$, $C_n = 0$, $E_{2k} = 0$, $E_{2k+1} = \frac{4V}{\pi} \frac{1}{(2k+1)a^{2k+1}}$ 。

$$\text{所以 } u(\rho, \varphi) = \frac{4V}{\pi} \sum_{n=1}^{\infty} \frac{1}{2k+1} \left(\frac{\rho}{a}\right)^{2k+1} \sin(2k+1)\varphi。$$

247. 半径为 a , 表面熏黑的均匀长圆柱, 平放在地上, 受到阳光照射, 其垂直于光线的单位面积上单位时间内吸收热量为 M , 同时, 柱面按牛顿冷却定律向外散热, 外界温度为 0 。试求柱内温度分布。



$$\text{类似于习题 11 第 208 题的讨论, 可得该定解问题为 } \begin{cases} \nabla^2 u = 0 \\ \left(\frac{\partial u}{\partial \rho} + hu \right)_{\rho=a} = \begin{cases} 0, & -\pi < \varphi < 0 \\ \frac{M}{k} \sin \varphi, & 0 < \varphi < \pi \end{cases} \end{cases}。$$

$$\text{同上题可得 } u(\rho, \varphi) = A_0 + \sum_{n=1}^{\infty} A_n \rho^n \cos n\varphi + \sum_{n=1}^{\infty} B_n \rho^n \sin n\varphi。$$

边界条件写成 Fourier 级数: $\left(\frac{\partial u}{\partial \rho} + hu\right)_{\rho=a} = \frac{M}{\pi k} + \frac{M}{2k} \sin \varphi - \frac{2M}{\pi k} \sum_{m=1}^{\infty} \frac{1}{4m^2 - 1} \cos 2m\varphi$ 。

$$\begin{aligned} \text{即 } A_0 h + \sum_{n=1}^{\infty} A_n a^n \left(\frac{n}{a} + h\right) \cos n\varphi + \sum_{n=1}^{\infty} B_n a^n \left(\frac{n}{a} + h\right) \sin n\varphi \\ = \frac{M}{\pi k} + \frac{M}{2k} \sin \varphi - \frac{2M}{\pi k} \sum_{m=1}^{\infty} \frac{1}{4m^2 - 1} \cos 2m\varphi, \end{aligned}$$

$$\text{可得 } A_0 = \frac{M}{\pi h k}, \quad A_{2m+1} = 0 \quad (m=0, 1, 2, \dots), \quad A_{2m} = -\frac{2M}{\pi k} \frac{1}{a^{2m} \left(\frac{2m}{a} + h\right)} \frac{1}{4m^2 - 1}$$

$$(m=1, 2, \dots), \quad B_1 = \frac{M}{2k} \frac{1}{\left(\frac{1}{a} + h\right)a}, \quad B_n = 0 \quad (n=2, 3, \dots).$$

$$\text{所以 } u(\rho, \varphi) = \frac{M}{\pi h k} + \frac{M}{2k \left(\frac{1}{a} + h\right)} \frac{\rho}{a} \sin \varphi - \frac{2M}{\pi k} \sum_{m=1}^{\infty} \frac{1}{\left(\frac{2m}{a} + h\right)(4m^2 - 1)} \left(\frac{\rho}{a}\right)^{2m} \cos 2m\varphi.$$

248. 求环形区域 $a \leq \rho \leq b$ 内满足边界条件 $u|_{\rho=a} = f(\varphi)$, $u|_{\rho=b} = g(\varphi)$ 的调和函数。

$$u(\rho, \varphi) = A + B \ln \rho + \sum_{n=1}^{\infty} (C_n \rho^n + D_n \rho^{-n}) \cos n\varphi + \sum_{n=1}^{\infty} (E_n \rho^n + F_n \rho^{-n}) \sin n\varphi.$$

令 $f(\varphi) = f_0 + \sum_{n=1}^{\infty} f_{sn} \sin n\varphi + f_{cn} \cos n\varphi$, $g(\varphi) = g_0 + \sum_{n=1}^{\infty} g_{sn} \sin n\varphi + g_{cn} \cos n\varphi$, 则

$$\begin{aligned} A + B \ln a + \sum_{n=1}^{\infty} (C_n a^n + D_n a^{-n}) \cos n\varphi + \sum_{n=1}^{\infty} (E_n a^n + F_n a^{-n}) \sin n\varphi \\ = f_0 + \sum_{n=1}^{\infty} f_{sn} \sin n\varphi + f_{cn} \cos n\varphi, \end{aligned}$$

$$\begin{aligned} A + B \ln b + \sum_{n=1}^{\infty} (C_n b^n + D_n b^{-n}) \cos n\varphi + \sum_{n=1}^{\infty} (E_n b^n + F_n b^{-n}) \sin n\varphi \\ = g_0 + \sum_{n=1}^{\infty} g_{sn} \sin n\varphi + g_{cn} \cos n\varphi. \end{aligned}$$

$$\text{比较系数得 } A = \frac{f_0 \ln b - g_0 \ln a}{\ln b - \ln a}, \quad B = \frac{g_0 - f_0}{\ln b - \ln a}, \quad C_n = \frac{g_{cn} a^{-n} - f_{cn} b^{-n}}{\left(\frac{b}{a}\right)^n - \left(\frac{a}{b}\right)^n},$$

$$D_n = \frac{f_{cn} b^n - g_{cn} a^n}{\left(\frac{b}{a}\right)^n - \left(\frac{a}{b}\right)^n}, \quad E_n = \frac{g_{sn} a^{-n} - f_{sn} b^{-n}}{\left(\frac{b}{a}\right)^n - \left(\frac{a}{b}\right)^n}, \quad F_n = \frac{f_{sn} b^n - g_{sn} a^n}{\left(\frac{b}{a}\right)^n - \left(\frac{a}{b}\right)^n}.$$

$$\begin{aligned} \text{所以 } u(\rho, \varphi) = & \frac{g_0 \ln \frac{\rho}{a} + f_0 \ln \frac{b}{\rho}}{\ln b - \ln a} + \sum_{n=1}^{\infty} \left[\frac{\left(\frac{\rho}{a}\right)^n - \left(\frac{a}{\rho}\right)^n}{\left(\frac{b}{a}\right)^n - \left(\frac{a}{b}\right)^n} g_{cn} + \frac{\left(\frac{b}{\rho}\right)^n - \left(\frac{\rho}{b}\right)^n}{\left(\frac{b}{a}\right)^n - \left(\frac{a}{b}\right)^n} f_{cn} \right] \cos n\varphi \\ & + \sum_{n=1}^{\infty} \left[\frac{\left(\frac{\rho}{a}\right)^n - \left(\frac{a}{\rho}\right)^n}{\left(\frac{b}{a}\right)^n - \left(\frac{a}{b}\right)^n} g_{sn} + \frac{\left(\frac{b}{\rho}\right)^n - \left(\frac{\rho}{b}\right)^n}{\left(\frac{b}{a}\right)^n - \left(\frac{a}{b}\right)^n} f_{sn} \right] \sin n\varphi. \end{aligned}$$

249. 求扇形区域 $0 \leq \rho \leq a$, $0 \leq \varphi \leq \alpha$ 内的稳定温度分布。设区域内无热源，在扇形的直边上温度为 0，而在弧形边界上温度为 $f(\varphi)$ 。

可得本征值问题 $\begin{cases} \Phi'' + \lambda \Phi = 0 \\ \Phi(0) = 0, \Phi(\alpha) = 0 \end{cases}$ ，本征值 $\lambda_n = \left(\frac{n\pi}{\alpha}\right)^2$ ，本征函数 $\Phi_n(\varphi) = \sin \frac{n\pi}{\alpha} \varphi$

($n=1, 2, \dots$)， $P(\rho) = A_n \rho^{\frac{n\pi}{\alpha}} + B_n \rho^{-\frac{n\pi}{\alpha}}$ ，所以 $u(\rho, \varphi) = \sum_{n=1}^{\infty} A_n \rho^{\frac{n\pi}{\alpha}} \sin \frac{n\pi}{\alpha} \varphi$ 。

$u(a, \varphi) = \sum_{n=1}^{\infty} A_n a^{\frac{n\pi}{\alpha}} \sin \frac{n\pi}{\alpha} \varphi = f(\varphi)$ ，则 $A_n = \frac{2}{\alpha a^{\frac{n\pi}{\alpha}}} \int_0^{\alpha} f(\varphi) \sin \frac{n\pi}{\alpha} \varphi d\varphi$ ，

$u(\rho, \varphi) = \sum_{n=1}^{\infty} A'_n \left(\frac{\rho}{a}\right)^{\frac{n\pi}{\alpha}} \sin \frac{n\pi}{\alpha} \varphi$ ，其中 $A'_n = \frac{2}{\alpha} \int_0^{\alpha} f(\varphi) \sin \frac{n\pi}{\alpha} \varphi d\varphi$ 。

250. 讨论上题中 $f(\varphi) = A$ (常数) 且 $\alpha = 2\pi$ 的情况，证明沿正实轴：

(1) 当 $\varphi \rightarrow 0$ 及 $\varphi \rightarrow 2\pi$ 时温度分布连续；

(2) 当 $\varphi \rightarrow 0$ 及 $\varphi \rightarrow 2\pi$ 时温度梯度 $\frac{1}{\rho} \frac{\partial u}{\partial \varphi}$ 不连续。

$$A'_n = \frac{A}{\pi} \int_0^{2\pi} \sin \frac{n}{2} \varphi d\varphi = \frac{2A}{\pi n} [1 - (-1)^n], \quad A'_{2k} = 0, \quad A'_{2k+1} = \frac{4A}{\pi n},$$

$$u(\rho, \varphi) = \frac{4A}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} \left(\frac{\rho}{a}\right)^{\frac{2k+1}{2}} \sin \frac{2k+1}{2} \varphi.$$

当 $\varphi \rightarrow 0$ 及 $\varphi \rightarrow 2\pi$ 时都有 $u(\rho, \varphi) \rightarrow 0$, 即温度连续。

$$\frac{1}{\rho} \frac{\partial u}{\partial \varphi} = \frac{2A}{\pi \sqrt{a\rho}} \sum_{k=0}^{\infty} \left(\frac{\rho}{a}\right)^k \cos \frac{2k+1}{2} \varphi,$$

$$\left. \frac{1}{\rho} \frac{\partial u}{\partial \varphi} \right|_{\varphi=0} = \frac{2A}{\pi} \sqrt{\frac{a}{\rho}} \frac{1}{a-\rho}, \quad \left. \frac{1}{\rho} \frac{\partial u}{\partial \varphi} \right|_{\varphi=2\pi} = -\frac{2A}{\pi} \sqrt{\frac{a}{\rho}} \frac{1}{a-\rho}, \quad \text{即温度梯度不连续。}$$

251. 在圆域 $0 \leq \rho \leq a$ 上求解: (1) $\begin{cases} \nabla^2 u = -4 \\ u|_{\rho=a} = 0 \end{cases}$; (2) $\begin{cases} \nabla^2 u = -4\rho \sin \varphi \\ u|_{\rho=a} = 0 \end{cases}$;

(3) $\begin{cases} \nabla^2 u = -4\rho^2 \sin 2\varphi \\ u|_{\rho=a} = 0 \end{cases}$ 。

(1) 设特解只是 ρ 的函数, 可解出一个特解 $-\rho^2$ 。令 $u = -\rho^2 + v$, 则 $\begin{cases} \nabla^2 v = 0 \\ v|_{\rho=a} = a^2 \end{cases}$ 。

$$v = A_0 + \sum_{n=1}^{\infty} A_n \rho^n \cos n\varphi + \sum_{n=1}^{\infty} B_n \rho^n \sin n\varphi, \quad \text{由边界条件得 } v = a^2, \quad \text{所以 } u = a^2 - \rho^2.$$

(2) 设特解具有形式 $\rho^3 f(\varphi)$, 代入方程得 $f''(\varphi) + 9f(\varphi) = -4\sin \varphi$, 设

$$f(\varphi) = A \sin \varphi, \quad \text{可得一个特解 } v = -\frac{1}{2} \rho^3 \sin \varphi.$$

$$\text{令 } u = v + w, \quad \text{则 } \begin{cases} \nabla^2 w = 0 \\ w|_{\rho=a} = \frac{1}{2} a^3 \sin \varphi \end{cases}, \quad w = A_0 + \sum_{n=1}^{\infty} A_n \rho^n \cos n\varphi + \sum_{n=1}^{\infty} B_n \rho^n \sin n\varphi,$$

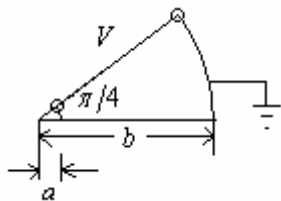
$$\text{由边界条件得 } A_n = 0 \quad (n=0, 1, 2, \dots), \quad B_1 = \frac{1}{2} a^2, \quad B_n = 0 \quad (n=2, 3, \dots),$$

$$\text{即 } w = \frac{1}{2} a^2 \rho \sin \varphi, \quad \text{所以 } u = \frac{1}{2} (a^2 - \rho^2) \rho \sin \varphi.$$

(3) 设特解具有形式 $\rho^4 f(\varphi)$, 则 $f''(\varphi) + 16f(\varphi) = -4\sin 2\varphi$, 可得一个特解

$$v = -\frac{1}{3} \rho^4 \sin 2\varphi, \quad \text{令 } u = v + w, \quad \text{则 } \begin{cases} \nabla^2 w = 0 \\ w|_{\rho=a} = \frac{1}{3} a^4 \sin 2\varphi \end{cases}, \quad \text{可得 } u = \frac{1}{3} (a^2 - \rho^2) \rho^2 \sin 2\varphi.$$

252. 一个由理想导体做成的无穷长波导管，其截面均匀，如下图所示。管内为真空，假定一个平面（即图中一条直边）电势为 V ，其余面上电势为 0 。试求波导管内电势分布。



$$\text{定解问题为 } \begin{cases} \nabla^2 u = 0 \\ u|_{\varphi=0} = 0, u|_{\varphi=\pi/4} = V \\ u|_{\rho=b} = 0, u|_{\rho=a} = 0 \end{cases} \text{。满足 } \varphi \text{ 的边界条件的一个特解为 } \frac{4V}{\pi} \varphi,$$

$$\text{设 } u = \frac{4V}{\pi} \varphi + v, \text{ 则 } \begin{cases} \nabla^2 v = 0 \\ v|_{\varphi=0} = 0, v|_{\varphi=\pi/4} = 0 \\ v|_{\rho=b} = -\frac{4V}{\pi} \varphi, v|_{\rho=a} = -\frac{4V}{\pi} \varphi \end{cases},$$

$$v = \sum_{n=1}^{\infty} (A_n \rho^{4n} + B_n \rho^{-4n}) \sin 4n\varphi, \text{ 由边界条件解得 } A_n = \frac{2V}{\pi} \frac{(-1)^n}{n} \frac{a^{-4n} - b^{-4n}}{\left(\frac{b}{a}\right)^{4n} - \left(\frac{a}{b}\right)^{4n}},$$

$$B_n = \frac{2V}{\pi} \frac{(-1)^n}{n} \frac{b^{4n} - a^{4n}}{\left(\frac{b}{a}\right)^{4n} - \left(\frac{a}{b}\right)^{4n}},$$

$$\text{即 } u = \frac{4V}{\pi} \varphi + \frac{2V}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \frac{\left(\frac{\rho}{a}\right)^{4n} - \left(\frac{a}{\rho}\right)^{4n} + \left(\frac{b}{\rho}\right)^{4n} - \left(\frac{\rho}{b}\right)^{4n}}{\left(\frac{b}{a}\right)^{4n} - \left(\frac{a}{b}\right)^{4n}} \sin 4n\varphi.$$

$$253. \text{ 求解球内定解问题: } \begin{cases} \frac{\partial u}{\partial t} - \frac{\kappa}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = 0 \\ u|_{r=0} \text{ 有界}, u|_{r=1} = A e^{-(p\pi)^2 \kappa t} \\ u|_{t=0} = 0 \end{cases} \text{。提示: } \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} (ru).$$

$$\text{先假定 } p \text{ 不是整数, 设特解具有形式 } A \frac{f(r)}{r} e^{-(p\pi)^2 \kappa t}, \text{ 则 } \begin{cases} f''(r) + (p\pi)^2 f(r) = 0 \\ f(0) = 0, f(1) = 1 \end{cases}, \text{ 可}$$

$$\text{得一个特解 } v = A \frac{\sin p\pi r}{r \sin p\pi} e^{-(p\pi)^2 \kappa t}。 \text{ 令 } u = v + w, \text{ 则 } \begin{cases} \frac{\partial w}{\partial t} - \frac{\kappa}{r} \frac{\partial^2}{\partial r^2} (rw) = 0 \\ w|_{r=0} \text{ 有界}, w|_{r=1} = 0 \\ w|_{t=0} = -A \frac{\sin p\pi r}{r \sin p\pi} \end{cases}。$$

$$\text{分离变量可得本征值问题 } \begin{cases} \frac{d^2}{dr^2} (rR) + \frac{\lambda}{\kappa} (rR) = 0 \\ rR|_{r=0} = 0, rR|_{r=1} = 0 \end{cases}, \text{ 解得本征值 } \lambda_n = \kappa (n\pi)^2, \text{ 本征函数}$$

$$R_n(r) = \frac{\sin n\pi r}{r} \quad (n=1, 2, \dots)。 w = \sum_{n=1}^{\infty} A_n \frac{\sin n\pi r}{r} e^{-\kappa(n\pi)^2 t}, \text{ 由初始条件可得}$$

$$A_n = \frac{2A}{\pi} \frac{(-1)^n n}{n^2 - p^2}, \text{ 所以 } u = A \frac{\sin p\pi r}{r \sin p\pi} e^{-(p\pi)^2 \kappa t} + \frac{2A}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2 - p^2} \frac{\sin n\pi r}{r} e^{-(n\pi)^2 \kappa t}。$$

若 p 是整数 ($=m$), u 写成

$$u = \frac{A}{r} \lim_{p \rightarrow m} \left[\frac{\sin p\pi r}{\sin p\pi} e^{-(p\pi)^2 \kappa t} + \frac{2}{\pi} \frac{(-1)^m m \sin m\pi r}{m^2 - p^2} e^{-(m\pi)^2 \kappa t} \right] \\ + \frac{2A}{\pi} \sum_{\substack{n=1 \\ n \neq m}}^{\infty} \frac{(-1)^n n}{n^2 - m^2} \frac{\sin n\pi r}{r} e^{-(n\pi)^2 \kappa t}$$

上面的极限式

$$= \lim_{p \rightarrow m} \frac{\left[\pi \sin p\pi r (m+p) e^{-(p\pi)^2 \kappa t} - 2(-1)^m m \sin m\pi r \frac{\sin p\pi}{p-m} e^{-(m\pi)^2 \kappa t} \right]}{\pi \frac{\sin p\pi}{p-m} (m+p)}$$

$$= \frac{\frac{d}{dp} \left[\pi \sin p\pi r (m+p) e^{-(p\pi)^2 \kappa t} \right]_{p=m}}{\pi (m+p) \frac{d}{dp} \sin p\pi \Big|_{p=m}}$$

$$= (-1)^m \left(\frac{\sin m\pi r}{2m\pi} + r \cos m\pi r - 2m\pi \kappa t \sin m\pi r \right) e^{-(m\pi)^2 \kappa t},$$

$$\text{所以 } u = (-1)^m A \left(\frac{\sin m\pi r}{2m\pi} + \cos m\pi r - \frac{2m\pi \kappa t \sin m\pi r}{r} \right) e^{-(m\pi)^2 \kappa t}$$

$$+ \frac{2A}{\pi} \sum_{\substack{n=1 \\ n \neq m}}^{\infty} \frac{(-1)^n n}{n^2 - m^2} \frac{\sin n\pi r}{r} e^{-(n\pi)^2 \kappa t}。$$

254. 将下列方程化为 S-L 型方程的标准形式: (1) $x \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + (x + \lambda)y = 0$;

(2) $\frac{d^2 y}{dx^2} + \cot x \frac{dy}{dx} + \lambda y = 0$; (3) $x(1-x) \frac{d^2 y}{dx^2} + (a-bx) \frac{dy}{dx} - \lambda y = 0$;

(4) $x \frac{d^2 y}{dx^2} + (1-x) \frac{dy}{dx} + \lambda y = 0$ 。

(1) 方程两边同乘 x 得: $x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} + x(x + \lambda)y = 0$, 写成:

$$\frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) + (\lambda x + x^2)y = 0。$$

(2) 两边同乘 $\sin x$ 得: $\sin x \frac{d^2 y}{dx^2} + \cos x \frac{dy}{dx} + \lambda y \sin x = 0$, 写成:

$$\frac{d}{dx} \left(\sin x \frac{dy}{dx} \right) + \lambda y \sin x = 0。$$

(3) 两边同乘 $\frac{x^{a-1}}{(1-x)^{a-b+1}}$ 得: $\frac{x^a}{(1-x)^{a-b}} \frac{d^2 y}{dx^2} + \frac{x^{a-1}(a-bx)}{(1-x)^{a-b+1}} \frac{dy}{dx} - \lambda \frac{x^{a-1}}{(1-x)^{a-b+1}} y = 0$,

写成: $\frac{d}{dx} \left[\frac{x^a}{(1-x)^{a-b}} \frac{dy}{dx} \right] - \lambda \frac{x^{a-1}}{(1-x)^{a-b+1}} y = 0。$

(4) 两边同乘 e^{-x} 得: $xe^{-x} \frac{d^2 y}{dx^2} + (1-x)e^{-x} \frac{dy}{dx} + \lambda e^{-x} y = 0$, 写成:

$$\frac{d}{dx} \left(xe^{-x} \frac{dy}{dx} \right) + \lambda e^{-x} y = 0。$$

255. 设有本征值问题
$$\begin{cases} \frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [\lambda \rho(x) - q(x)]y = 0 \\ y'(a) = 0, y'(b) = 0 \end{cases}$$
, 其中 $p(x)$, $\rho(x)$, $q(x)$

在 $a \leq x \leq b$ 上均为连续实函数, 且 $p(x) \geq p_0 > 0$, $\rho(x) \geq \rho_0 > 0$ 。试证明本征函数的正交性。

设本征值 λ_1 对应本征函数 y_1 , 本征值 λ_2 对应本征函数 y_2 ($\lambda_1 \neq \lambda_2$), 即

$$\lambda_1 \rho y_1 = -\frac{d}{dx} \left(p \frac{dy_1}{dx} \right) + q y_1, \quad (\text{a})$$

$$\lambda_2 \rho y_2 = -\frac{d}{dx} \left(p \frac{dy_2}{dx} \right) + q y_2. \quad (\text{b})$$

$$(\text{a}) \times y_2 - (\text{b}) \times y_1 \text{ 得 } (\lambda_1 - \lambda_2) \rho y_1 y_2 = y_1 \frac{d}{dx} \left(p \frac{dy_2}{dx} \right) - y_2 \frac{d}{dx} \left(p \frac{dy_1}{dx} \right),$$

$$\text{两边积分得 } (\lambda_1 - \lambda_2) \int_a^b \rho y_1 y_2 dx = p y_1 \frac{dy_2}{dx} \Big|_a^b - p y_2 \frac{dy_1}{dx} \Big|_a^b = 0$$

由于 $\lambda_1 \neq \lambda_2$, 所以 $\int_a^b \rho y_1 y_2 dx = 0$, 即 y_1, y_2 正交。

$$256. \text{ 假设 S-L 方程的本征值问题 } \begin{cases} \frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [\lambda \rho(x) - q(x)] y = 0 \\ (ay' - by)_{x=0} = 0, (cy' + dy)_{x=l} = 0 \end{cases} \text{ 中,}$$

$p(x) \geq p_0 > 0$, $\rho(x) \geq \rho_0 > 0$, $q(x) \geq 0$, a 与 b 及 c 与 d 均为不同时为 0 的非负常数, 证明本征值 ≥ 0 。

方程两边同乘 y 得 $\lambda \rho y^2 = -y \frac{d}{dx} \left(p \frac{dy}{dx} \right) + q y^2$, 两边积分得

$$\lambda \int_0^l \rho y^2 dx = p [y(0) y'(0) - y(l) y'(l)] + \int_0^l p \left(\frac{dy}{dx} \right)^2 dx + \int_0^l q y^2 dx$$

$$\text{若 } a \neq 0, \text{ 则 } y'(0) = \frac{b}{a} y(0), \quad y(0) y'(0) = \frac{b}{a} y^2(0) \geq 0,$$

$$\text{若 } b \neq 0, \text{ 则 } y(0) = \frac{a}{b} y'(0), \quad y(0) y'(0) = \frac{a}{b} y'^2(0) \geq 0;$$

$$\text{若 } c \neq 0, \text{ 则 } y'(l) = -\frac{d}{c} y(l), \quad -y(l) y'(l) = \frac{d}{c} y^2(l) \geq 0,$$

$$\text{若 } d \neq 0, \text{ 则 } y(l) = \frac{c}{d} y'(l), \quad -y(l) y'(l) = \frac{c}{d} y'^2(l) \geq 0;$$

所以 $\lambda \geq 0$ 。

$$257. \text{ 求解本征值问题: } \begin{cases} \frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \frac{\lambda}{r^2} R = 0 \\ R(a) = 0, R(b) = 0 \end{cases}, \text{ 其中 } b > a > 0.$$

令 $r = e^t$, 则有 $r \frac{d}{dr} = \frac{d}{dt}$ 。方程可写为 $r \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \lambda R = 0$, 即 $\frac{d^2 R}{dt^2} + \lambda R = 0$ 。

由上题结论可知 $\lambda \geq 0$, $\lambda = 0$ 时, 方程通解为 $R = A + Bt = A + B \ln r$, 由边界条件可得 $R = 0$, 所以只有 $\lambda > 0$ 。

通解为 $R = A \sin \sqrt{\lambda} t + B \cos \sqrt{\lambda} t = A \sin(\sqrt{\lambda} \ln r) + B \cos(\sqrt{\lambda} \ln r)$, 由边界条件得

$$\begin{cases} A \sin(\sqrt{\lambda} \ln a) + B \cos(\sqrt{\lambda} \ln a) = 0 \\ A \sin(\sqrt{\lambda} \ln b) + B \cos(\sqrt{\lambda} \ln b) = 0 \end{cases},$$

$$\text{所以} \begin{vmatrix} \sin(\sqrt{\lambda} \ln a) & \cos(\sqrt{\lambda} \ln a) \\ \sin(\sqrt{\lambda} \ln b) & \cos(\sqrt{\lambda} \ln b) \end{vmatrix} = \sin[\sqrt{\lambda}(\ln b - \ln a)] = 0,$$

则本征值 $\lambda_n = \left(\frac{n\pi}{\ln b - \ln a} \right)^2$, 本征函数 $R_n(r) = \sin\left(\frac{\ln r - \ln a}{\ln b - \ln a} n\pi\right)$ ($n = 1, 2, \dots$)。

258. 证明下列奇异的本征值问题是自伴的: (1) $\begin{cases} \frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \lambda y = 0; \\ y(\pm 1) \text{ 有界} \end{cases}$;

$$(2) \begin{cases} \frac{1}{x} \frac{d}{dx} \left(x \frac{dy}{dx} \right) + \lambda y = 0; \\ y(0) \text{ 有界}, y(1) = 0 \end{cases}.$$

(1) 记 $L = -\frac{d}{dx} \left[(1-x^2) \frac{d}{dx} \right]$, 则

$$\begin{aligned} y_1 L y_2 - y_2 L y_1 &= y_2 \frac{d}{dx} \left[(1-x^2) \frac{dy_1}{dx} \right] - y_1 \frac{d}{dx} \left[(1-x^2) \frac{dy_2}{dx} \right] \\ &= y_2 \frac{d(1-x^2)}{dx} \frac{dy_1}{dx} + (1-x^2) y_2 \frac{d^2 y_1}{dx^2} - y_1 \frac{d(1-x^2)}{dx} \frac{dy_2}{dx} - (1-x^2) y_1 \frac{d^2 y_2}{dx^2} \\ &= (1-x^2) \left(y_2 \frac{d^2 y_1}{dx^2} - y_1 \frac{d^2 y_2}{dx^2} \right) + \frac{d(1-x^2)}{dx} \left(y_2 \frac{dy_1}{dx} - y_1 \frac{dy_2}{dx} \right) \\ &= \frac{d}{dx} \left[(1-x^2) \left(y_2 \frac{dy_1}{dx} - y_1 \frac{dy_2}{dx} \right) \right], \end{aligned}$$

两边积分得 $\int_{-1}^1 (y_1 L y_2 - y_2 L y_1) dx = (1-x^2) \left(y_2 \frac{dy_1}{dx} - y_1 \frac{dy_2}{dx} \right) \Big|_{x=-1}^{x=1}$,

由于 $y(\pm 1)$ 有界, 所以 $(1-x^2) \left(y_2 \frac{dy_1}{dx} - y_1 \frac{dy_2}{dx} \right) \Big|_{x=\pm 1} = 0$, $\int_{-1}^1 y_1 L(y_2) dx = \int_{-1}^1 L(y_1) y_2 dx$,

即 L 为自伴算符。

(2) 记 $L = -\frac{d}{dx} \left(x \frac{d}{dx} \right)$, 重复上小题过程有

$$\int_{-1}^1 (y_1 L y_2 - y_2 L y_1) dx = x \left(y_2 \frac{dy_1}{dx} - y_1 \frac{dy_2}{dx} \right) \Big|_{x=0}^{x=1} = -x \left(y_2 \frac{dy_1}{dx} - y_1 \frac{dy_2}{dx} \right) \Big|_{x=0},$$

由于 $y(0)$ 有界, 所以上式等于 0, 即 L 为自伴算符。

259. 设有本征值问题:
$$\begin{cases} \frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [\lambda \rho(x) - q(x)] y = 0 \\ y(b) = a_{11} y(a) + a_{12} y'(a), y'(b) = a_{21} y(a) + a_{22} y'(a) \end{cases},$$

其中 $p(a) = p(b)$ 。证明: 当 $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = 1$ 时, 对应不同本征值的本征函数正交。

设本征值 λ_1 对应本征函数 y_1 , 本征值 λ_2 对应本征函数 y_2 ($\lambda_1 \neq \lambda_2$), 255 题已得:

$$\begin{aligned} (\lambda_1 - \lambda_2) \int_a^b \rho y_1 y_2 dx &= (p y_1 y_2' - p y_2 y_1') \Big|_a^b = p(a) (y_1 y_2' - y_2 y_1') \Big|_a^b \\ &= p(a) \left(\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} - 1 \right) [y_1(a) y_2'(a) - y_1'(a) y_2(a)] \end{aligned}$$

所以当 $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = 1$ 时, y_1 和 y_2 正交。

260. 两条质料不同, 长各为 l_1 与 l_2 的均匀弦连接在一起, 而两端 ($x=0$ 及 $x=l_1+l_2$) 固定。试决定弦的横振动本征频率, 并验证本征函数的正交性。

令 u_1 , u_2 分别表示两段弦的横向位移, 由弦的连续性可得连接条件 $u_1 \Big|_{x=l_1} = u_2 \Big|_{x=l_1}$,

$\frac{\partial u_1}{\partial x} \Big|_{x=l_1} = \frac{\partial u_2}{\partial x} \Big|_{x=l_1}$ 。该问题为:

$$\begin{cases} \frac{\partial^2 u_1}{\partial t^2} - a_1^2 \frac{\partial^2 u_1}{\partial x^2} = 0, 0 < x < l_1 \\ \frac{\partial^2 u_2}{\partial t^2} - a_2^2 \frac{\partial^2 u_2}{\partial x^2} = 0, l_1 < x < l_1 + l_2 \\ u_1|_{x=0} = 0, u_2|_{x=l_1+l_2} = 0 \\ u_1|_{x=l_1} = u_2|_{x=l_1}, \frac{\partial u_1}{\partial x}|_{x=l_1} = \frac{\partial u_2}{\partial x}|_{x=l_1} \end{cases} \quad \text{。令 } u_1 = X_1(x)e^{i\omega t}, \quad u_2 = X_2(x)e^{i\omega t}, \text{ 代入方程及边}$$

$$\text{界条件和连接条件得: } \begin{cases} X_1''(x) + \left(\frac{\omega}{a_1}\right)^2 X_1(x) = 0, 0 < x < l_1 \\ X_2''(x) + \left(\frac{\omega}{a_2}\right)^2 X_2(x) = 0, l_1 < x < l_1 + l_2 \\ X_1(0) = 0, X_2(l_1 + l_2) = 0 \\ X_1(l_1) = X_2(l_1), X_1'(l_1) = X_2'(l_1) \end{cases} \quad \text{。}$$

当 $\omega = 0$ 时只有零解，故 $\omega > 0$ ($\omega < 0$ 是同一解)。

$$\text{由方程及 } X_1(0) = 0, X_2(l_1 + l_2) = 0 \text{ 可得 } X(x) = \begin{cases} X_1(x) = A \sin \frac{\omega}{a_1} x, 0 < x < l_1 \\ X_2(x) = B \sin \frac{\omega}{a_2} (l_1 + l_2 - x), l_1 < x < l_1 + l_2 \end{cases} \quad \text{。}$$

$$\text{代入条件 } X_1(l_1) = X_2(l_1), X_1'(l_1) = X_2'(l_1) \text{ 得 } \begin{cases} A \sin \frac{\omega}{a_1} l_1 - B \sin \frac{\omega}{a_2} l_2 = 0 \\ A \frac{\omega}{a_1} \cos \frac{\omega}{a_1} l_1 + B \frac{\omega}{a_2} \cos \frac{\omega}{a_2} l_2 = 0 \end{cases}, (*)$$

上式中，由于 A, B 都不为 0，所以 $\sin \frac{\omega}{a_1} l_1$ 与 $\sin \frac{\omega}{a_2} l_2$ 同为零或同为非零， $\cos \frac{\omega}{a_1} l_1$ 与

$\cos \frac{\omega}{a_2} l_2$ 同为零或同为非零。

(1) $\cos \frac{\omega}{a_1} l_1 \neq 0, \cos \frac{\omega}{a_2} l_2 \neq 0, \sin \frac{\omega}{a_1} l_1 \neq 0, \sin \frac{\omega}{a_2} l_2 \neq 0$ 。由 (*), 由于 A, B 都不

$$\text{为 0, 所以 } \begin{vmatrix} \sin \frac{\omega}{a_1} l_1 & -\sin \frac{\omega}{a_2} l_2 \\ \frac{\omega}{a_1} \cos \frac{\omega}{a_1} l_1 & \frac{\omega}{a_2} \cos \frac{\omega}{a_2} l_2 \end{vmatrix} = 0, \text{ 即 } a_1 \sin \frac{\omega}{a_1} l_1 \cos \frac{\omega}{a_2} l_2 + a_2 \sin \frac{\omega}{a_2} l_2 \cos \frac{\omega}{a_1} l_1 = 0,$$

两边同除 $\cos \frac{\omega}{a_1} l_1 \cos \frac{\omega}{a_2} l_2$ 得 $a_1 \tan \frac{\omega}{a_1} l_1 + a_2 \tan \frac{\omega}{a_2} l_2 = 0$ ，本征频率 ω_n 即为该方程的第 n

个正根。由 (*) 第一式可取 $A = \frac{1}{\sin \frac{\omega_n}{a_1} l_1}$, $B = \frac{1}{\sin \frac{\omega_n}{a_2} l_2}$,

$$\text{即 } X_n(x) = \begin{cases} X_{1n}(x) = \frac{\sin \frac{\omega_n}{a_1} x}{\sin \frac{\omega_n}{a_1} l_1}, 0 < x < l_1 \\ X_{2n}(x) = \frac{\sin \frac{\omega_n}{a_2} (l_1 + l_2 - x)}{\sin \frac{\omega_n}{a_2} l_2}, l_1 < x < l_1 + l_2 \end{cases} \quad \circ$$

(2) $\sin \frac{\omega}{a_1} l_1 = 0$, $\sin \frac{\omega}{a_2} l_2 = 0$ (此时 $\cos \frac{\omega}{a_1} l_1$ 和 $\cos \frac{\omega}{a_2} l_2$ 为 ± 1), 则存在互质整数 r, s 使

$$\omega = \frac{r\pi a_1}{l_1} = \frac{s\pi a_2}{l_2} \quad (\text{此时参数满足 } \frac{l_1 a_2}{l_2 a_1} = \frac{r}{s}), \text{ 该频率为基频, 本征频率为}$$

$$\omega_n = \frac{nr\pi a_1}{l_1} = \frac{ns\pi a_2}{l_2} \quad (n=1, 2, \dots)。代入 (*) 第二式得 \frac{A}{a_1} (-1)^{nr} = -\frac{B}{a_2} (-1)^{sn}, \text{ 可取}$$

$$A = (-1)^{nr} a_1, \quad B = -(-1)^{sn} a_2,$$

$$\text{即 } X_n(x) = \begin{cases} X_{1n}(x) = (-1)^m a_1 \sin \frac{nr\pi}{l_1} x, 0 < x < l_1 \\ X_{2n}(x) = -(-1)^{sn} a_2 \sin \frac{ns\pi}{l_2} (l_1 + l_2 - x) = -a_2 \sin \frac{ns\pi}{l_2} (l_1 - x), l_1 < x < l_1 + l_2 \end{cases} \quad \circ$$

(3) $\cos \frac{\omega}{a_1} l_1 = 0$, $\cos \frac{\omega}{a_2} l_2 = 0$ (此时 $\sin \frac{\omega}{a_1} l_1$ 和 $\sin \frac{\omega}{a_2} l_2$ 为 ± 1), 则存在互质的

$$2r+1, 2s+1 \text{ 使 } \frac{l_1 a_2}{l_2 a_1} = \frac{2r+1}{2s+1}, \text{ 本征频率 } \omega_n = (2n+1) \frac{2r+1}{2l_1} a_1 \pi = (2n+1) \frac{2s+1}{2l_2} a_2 \pi$$

($n=0, 1, 2, \dots$)。代入 (*) 第一式得 $(-1)^{n+r} A = (-1)^{n+s} B$, 可取 $A = (-1)^r$, $B = (-1)^s$,

$$\text{即 } X_n(x) = \begin{cases} X_{1n}(x) = (-1)^r \sin \frac{(2r+1)(2n+1)}{2l_1} \pi x, 0 < x < l_1 \\ X_{2n}(x) = (-1)^s \sin \frac{(2s+1)(2n+1)}{2l_2} \pi (l_1 + l_2 - x) \\ = (-1)^n \cos \frac{(2s+1)(2n+1)}{2l_2} \pi (l_1 - x), l_1 < x < l_1 + l_2 \end{cases} \quad \circ$$

$$\begin{aligned} \frac{(\omega_n^2 - \omega_m^2)}{a_1^2} \int_0^{l_1} X_{1n} X_{1m} dx &= (X_{1n} X'_{1m} - X'_{1n} X_{1m}) \Big|_0^{l_1} = X_{1n}(l_1) X'_{1m}(l_1) - X'_{1n}(l_1) X_{1m}(l_1), \\ \frac{(\omega_n^2 - \omega_m^2)}{a_2^2} \int_{l_1}^{l_1+l_2} X_{2n} X_{2m} dx &= (X_{2n} X'_{2m} - X'_{2n} X_{2m}) \Big|_{l_1}^{l_1+l_2} \\ &= -X_{2n}(l_1) X'_{2m}(l_1) + X'_{2n}(l_1) X_{2m}(l_1), \end{aligned}$$
$$\begin{aligned} & \text{以上两式相加得} \left(\omega_n^2 - \omega_m^2 \right) \left(\int_0^{l_1} \frac{1}{a_1^2} X_{1n} X_{1m} dx + \int_{l_1}^{l_1+l_2} \frac{1}{a_2^2} X_{2n} X_{2m} dx \right) \\ & = X_{1n}(l_1) X'_{1m}(l_1) - X_{2n}(l_1) X'_{2m}(l_1) - X'_{1n}(l_1) X_{1m}(l_1) + X'_{2n}(l_1) X_{2m}(l_1) = 0。 \end{aligned}$$

$$\text{记 } \rho(x) = \begin{cases} \frac{1}{a_1^2}, 0 < x < l_1 \\ \frac{1}{a_2^2}, l_1 < x < l_1 + l_2 \end{cases}, \text{ 上式即为 } \int_0^{l_1+l_2} \rho X_n X_m dx = 0。$$

令 u_1 , u_2 分别表示两段杆的纵向位移。在连接处取一小段 (如图), 则

$$\frac{P_1(l_1 - \varepsilon)S \leftarrow \boxed{} \boxed{} \boxed{} \rightarrow P_2(l_1 + \varepsilon)S}{l_1 - \varepsilon \quad l_1 \quad l_1 + \varepsilon}$$

$$P_2(l_1 + \varepsilon)S - P_1(l_1 - \varepsilon)S = 2\rho S \varepsilon \left. \frac{\partial^2 u}{\partial t^2} \right|_{x=\xi}, \text{ 其中 } \xi \in (l_1 - \varepsilon, l_1 + \varepsilon)。 \text{ 代入 } P = E \frac{\partial u}{\partial x}, \text{ 并令}$$

$\varepsilon \rightarrow 0$ 得 $E_1 \frac{\partial u_1}{\partial x} \Big|_{x=l_1} = E_2 \frac{\partial u_2}{\partial x} \Big|_{x=l_1}$ 。另外显然有 $u_1|_{x=l_1} = u_2|_{x=l_1}$ 。

$$\text{该问题为} \begin{cases} \frac{\partial^2 u_1}{\partial t^2} - a_1^2 \frac{\partial^2 u_1}{\partial x^2} = 0, 0 < x < l_1 \\ \frac{\partial^2 u_2}{\partial t^2} - a_2^2 \frac{\partial^2 u_2}{\partial x^2} = 0, l_1 < x < l \\ u_1|_{x=0} = 0, \frac{\partial u_2}{\partial x} \Big|_{x=l} = 0 \\ u_1|_{x=l_1} = u_2|_{x=l_1}, E_1 \frac{\partial u_1}{\partial x} \Big|_{x=l_1} = E_2 \frac{\partial u_2}{\partial x} \Big|_{x=l_1} \end{cases} \quad \text{其中 } l = l_1 + l_2。$$

同上题, 令 $u_1 = X_1(x)e^{i\omega t}$, $u_2 = X_2(x)e^{i\omega t}$, 可得

$$\begin{cases} X_1''(x) + \left(\frac{\omega}{a_1}\right)^2 X_1(x) = 0, 0 < x < l_1 \\ X_2''(x) + \left(\frac{\omega}{a_2}\right)^2 X_2(x) = 0, l_1 < x < l \\ X_1(0) = 0, X_2'(l) = 0 \\ X_1(l_1) = X_2(l_1), E_1 X_1'(l_1) = E_2 X_2'(l_1) \end{cases}, \text{ 可得 } X(x) = \begin{cases} X_1(x) = A \sin \frac{\omega}{a_1} x, 0 < x < l_1 \\ X_2(x) = B \cos \frac{\omega}{a_2} (l - x), l_1 < x < l \end{cases}.$$

$$\text{由连接条件得} \begin{cases} A \sin \frac{\omega}{a_1} l_1 = B \cos \frac{\omega}{a_2} l_2 \\ A \frac{E_1}{a_1} \cos \frac{\omega}{a_1} l_1 = B \frac{E_2}{a_2} \sin \frac{\omega}{a_2} l_2 \end{cases}. \quad (*)$$

(1) $\sin \frac{\omega}{a_1} l_1 \neq 0$, $\cos \frac{\omega}{a_2} l_2 \neq 0$, $\sin \frac{\omega}{a_2} l_2 \neq 0$, $\cos \frac{\omega}{a_1} l_1 \neq 0$ 。本征频率 ω_n 是方程

$\frac{E_1}{a_1} \cot \frac{\omega}{a_1} l_1 = \frac{E_2}{a_2} \tan \frac{\omega}{a_2} l_2$ 的第 n 个正根,

$$X_n(x) = \begin{cases} X_{1n}(x) = \sin \frac{\omega_n}{a_1} x / \sin \frac{\omega_n}{a_1} l_1, 0 < x < l_1 \\ X_{2n}(x) = \cos \frac{\omega_n}{a_2} (l - x) / \cos \frac{\omega_n}{a_2} l_2, l_1 < x < l \end{cases}.$$

(2) $\sin \frac{\omega}{a_1} l_1 = 0$, $\cos \frac{\omega}{a_2} l_2 = 0$ 。 $\frac{l_1 a_2}{l_2 a_1} = \frac{2r}{2s+1}$ ($2r$ 与 $2s+1$ 互质),

本征频率 $\omega_n = (2n+1) \frac{r a_1 \pi}{l_1} = (2n+1) \frac{(2s+1) a_2 \pi}{2l_2}$ ($n = 0, 1, 2, \dots$)。

$$X_n(x) = \begin{cases} X_{1n}(x) = \frac{(-1)^r a_1}{E_1} \sin \frac{(2n+1)r}{l_1} \pi x, 0 < x < l_1 \\ X_{2n}(x) = \frac{(-1)^{n+s} a_2}{E_2} \cos \frac{(2n+1)(2s+1)}{2l_2} \pi (l - x) \\ = -\frac{a_2}{E_2} \sin \frac{(2n+1)(2s+1)}{2l_2} \pi (l_1 - x), l_1 < x < l \end{cases}$$

(3) $\sin \frac{\omega}{a_2} l_2 = 0$, $\cos \frac{\omega}{a_1} l_1 = 0$ 。 $\frac{l_1 a_2}{l_2 a_1} = \frac{2s+1}{2r}$ ($2r$ 与 $2s+1$ 互质)。

$$\omega_n = (2n+1) \frac{ra_2\pi}{l_2} = (2n+1) \frac{(2s+1)a_1\pi}{2l_1} \quad (n=0,1,2,\dots)。$$

$$X_n(x) = \begin{cases} X_{1n}(x) = (-1)^{n+s} \sin \frac{(2n+1)(2s+1)}{2l_1} \pi x, 0 < x < l_1 \\ X_{2n}(x) = (-1)^r \cos \frac{(2n+1)r}{l_2} \pi(l-x) \\ = \cos \frac{(2n+1)r}{l_2} \pi(l_1-x), l_1 < x < l \end{cases}。$$

262. 三维空间的本征值问题。设本征值问题为
$$\begin{cases} \nabla^2 u + \lambda u = 0, (x, y, z) \in V \\ \left(\alpha u + \beta \frac{\partial u}{\partial n} \right)_{\Sigma} = 0 \end{cases}$$
，其中 Σ 是 V 的

边界面。若对应本征值 λ_n 的本征函数为 u_n ，试证明： $\iiint_V u_m^* u_n dV = 0$ ， $m \neq n$ ，即对应不

同本征值的本征函数正交。

由 Gauss 公式可得：

$$\begin{aligned} \oiint_{\Sigma} \left(u_n \frac{\partial u_m^*}{\partial n} - u_m^* \frac{\partial u_n}{\partial n} \right) dS &= \oiint_{\Sigma} (u_n \nabla u_m^* - u_m^* \nabla u_n) \cdot d\mathbf{S} = \iiint_V \nabla \cdot (u_n \nabla u_m^* - u_m^* \nabla u_n) dV \\ &= \iiint_V (\nabla u_n \cdot \nabla u_m^* + u_n \nabla^2 u_m^* - \nabla u_m^* \cdot \nabla u_n - u_m^* \nabla^2 u_n) dV = \iiint_V (u_n \nabla^2 u_m^* - u_m^* \nabla^2 u_n) dV。 \end{aligned}$$

$$\text{所以 } (\lambda_n - \lambda_m^*) \iiint_V u_m^* u_n dV = \iiint_V (u_n \nabla^2 u_m^* - u_m^* \nabla^2 u_n) dV = \oiint_{\Sigma} \left(u_n \frac{\partial u_m^*}{\partial n} - u_m^* \frac{\partial u_n}{\partial n} \right) dS。$$

$$\text{若 } \alpha \neq 0, \text{ 则 } u|_{\Sigma} = -\frac{\beta}{\alpha} \frac{\partial u}{\partial n} \Big|_{\Sigma},$$

$$\oiint_{\Sigma} \left(u_n \frac{\partial u_m^*}{\partial n} - u_m^* \frac{\partial u_n}{\partial n} \right) dS = \oiint_{\Sigma} \left(-\frac{\beta}{\alpha} \frac{\partial u_n}{\partial n} \frac{\partial u_m^*}{\partial n} + \frac{\beta}{\alpha} \frac{\partial u_m^*}{\partial n} \frac{\partial u_n}{\partial n} \right) dS = 0,$$

$$\text{若 } \beta \neq 0, \text{ 则 } \frac{\partial u}{\partial n} \Big|_{\Sigma} = -\frac{\alpha}{\beta} u|_{\Sigma},$$

$$\oiint_{\Sigma} \left(u_n \frac{\partial u_m^*}{\partial n} - u_m^* \frac{\partial u_n}{\partial n} \right) dS = \oiint_{\Sigma} \left(-\frac{\alpha}{\beta} u_n u_m^* + \frac{\alpha}{\beta} u_m^* u_n \right) dS = 0。$$

即对应不同本征值的本征函数正交。

263. 若上题中的方程改为 $\nabla \cdot [p(x, y, z) \nabla u] + \lambda \rho(x, y, z) u = 0$, 试证明: 对应不同本征值的本征函数以权重 $\rho(x, y, z)$ 正交。

$$\begin{aligned} \oint_{\Sigma} p \left(u_n \frac{\partial u_m^*}{\partial n} - u_m^* \frac{\partial u_n}{\partial n} \right) dS &= \oint_{\Sigma} (u_n p \nabla u_m^* - u_m^* p \nabla u_n) \cdot d\mathbf{S} = \iiint_V \nabla \cdot (u_n p \nabla u_m^* - u_m^* p \nabla u_n) dV \\ &= \iiint_V [\nabla u_n \cdot p \nabla u_m^* + u_n \nabla \cdot (p \nabla u_m^*) - \nabla u_m^* \cdot p \nabla u_n - u_m^* \nabla \cdot (p \nabla u_n)] dV \\ &= \iiint_V [u_n \nabla \cdot (p \nabla u_m^*) - u_m^* \nabla \cdot (p \nabla u_n)] dV. \end{aligned}$$

$$(\lambda_n - \lambda_m^*) \iiint_V u_m^* u_n dV = \iiint_V [u_n \nabla \cdot (p \nabla u_m^*) - u_m^* \nabla \cdot (p \nabla u_n)] dV = \oint_{\Sigma} p \left(u_n \frac{\partial u_m^*}{\partial n} - u_m^* \frac{\partial u_n}{\partial n} \right) dS$$

同上题可得上式 $= 0$ 。

264. 设本征值问题 $\begin{cases} \nabla^2 \Phi + \lambda \Phi = 0 \\ \Phi|_{\Sigma} = 0 \end{cases}$ 的解 (本征函数) 为 $\{\Phi_k\}$, 对应本征值 $\{\lambda_k\}$,

$k = 1, 2, \dots$ 。试证明: 当 $\lambda = 0$ 不是本征值时, Poisson 方程的第一类边值问题 $\begin{cases} \nabla^2 u = -f \\ u|_{\Sigma} = 0 \end{cases}$ 的

解为 $u = \sum_{k=1}^{\infty} \frac{A_k}{\lambda_k} \Phi_k$, A_k 是 f 按 $\{\Phi_k\}$ 展开的系数。

将 u 按 $\{\Phi_k\}$ 展开, 即 $u = \sum_{k=1}^{\infty} u_k \Phi_k$, 代入方程得 $-\sum_{k=1}^{\infty} u_k \lambda_k \Phi_k = -\sum_{k=1}^{\infty} A_k \Phi_k$, 比较系数得

$$u_k = \frac{A_k}{\lambda_k}, \text{ 即 } u = \sum_{k=1}^{\infty} \frac{A_k}{\lambda_k} \Phi_k.$$

267. 证明: 如果将上题中 Φ 与 u 的边界条件改为齐次第二类或第三类边界条件时, 结论仍然成立。

可看出, 由于是齐次条件, 所以 $\{\Phi_k\}$ 的叠加仍满足 u 的边界条件, 上面的运算仍成立。

266. 在与 264 题相同的条件下, 证明: $\begin{cases} \nabla^2 u + \Lambda u = -f \\ u|_{\Sigma} = 0 \end{cases}$, $\Lambda \neq \lambda_k$ 的解为 $u = \sum_{k=1}^{\infty} \frac{A_k}{\lambda_k - \Lambda} \Phi_k$ 。

将 $u = \sum_{k=1}^{\infty} u_k \Phi_k$ 代入方程得 $-\sum_{k=1}^{\infty} \lambda_k u_k \Phi_k + \sum_{k=1}^{\infty} \Lambda u_k \Phi_k = -\sum_{k=1}^{\infty} A_k \Phi_k$, 所以 $u_k = \frac{A_k}{\lambda_k - \Lambda}$, 即

$$u = \sum_{k=1}^{\infty} \frac{A_k}{\lambda_k - \Lambda} \Phi_k.$$

267. 用 264 题方法求解矩形区域 $0 \leq x \leq a$, $0 \leq y \leq b$ 内 Poisson 方程的定解问题

$$\begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -f(x, y) \\ u|_{x=0} = 0, u|_{x=a} = 0 \\ u|_{y=0} = 0, u|_{y=b} = 0 \end{cases}.$$

$$\text{先解本征值问题} \begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \lambda u = 0 \\ u|_{x=0} = 0, u|_{x=a} = 0 \\ u|_{y=0} = 0, u|_{y=b} = 0 \end{cases} \text{。分离变量得} \begin{cases} X'' + \mu X = 0 \\ X(0) = 0, X(a) = 0 \\ Y'' + (\lambda - \mu)Y = 0 \\ Y(0) = 0, Y(b) = 0 \end{cases},$$

$$\text{解得 } \mu = \left(\frac{n\pi}{a}\right)^2, \quad X(x) = \sin \frac{n\pi}{a} x \quad (n=1, 2, \dots),$$

$$\lambda - \mu = \left(\frac{m\pi}{b}\right)^2, \quad Y(y) = \sin \frac{m\pi}{b} y \quad (m=1, 2, \dots).$$

$$\text{即本征值 } \lambda_{m,n} = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2,$$

$$\text{本征函数为 } \Phi_{m,n}(x, y) = X_n(x)Y_m(y) = \sin \frac{n\pi}{a} x \sin \frac{m\pi}{b} y \quad (n, m=1, 2, \dots).$$

$$\text{令 } f(x, y) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{m,n} \Phi_{m,n}(x, y), \text{ 则 } A_{m,n} = \frac{4}{ab} \int_0^b \int_0^a f(x, y) \sin \frac{n\pi}{a} x \sin \frac{m\pi}{b} y dx dy.$$

$$u(x, y) = \sum_{k=1}^{\infty} \frac{A_{m,n}}{\lambda_{m,n}} \Phi_{m,n}(x, y) = \sum_{k=1}^{\infty} \frac{A_{m,n}}{\left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2} \sin \frac{n\pi}{a} x \sin \frac{m\pi}{b} y.$$

268. 用 264 题方法求解 231 题: (1)
$$\begin{cases} \nabla^2 u = -2 \\ u|_{x=0} = 0, u|_{x=a} = 0 \\ u|_{y=-b/2} = 0, u|_{y=b/2} = 0 \end{cases}; (2) \begin{cases} \nabla^2 u = -x^2 y \\ u|_{x=0} = 0, u|_{x=a} = 0 \\ u|_{y=-b/2} = 0, u|_{y=b/2} = 0 \end{cases}.$$

本征值问题
$$\begin{cases} \nabla^2 u + \lambda u = 0 \\ u|_{x=0} = 0, u|_{x=a} = 0 \\ u|_{y=-b/2} = 0, u|_{y=b/2} = 0 \end{cases}$$
 的解为: 本征值 $\lambda_{m,n} = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2$, 本征函数

$$\Phi_{m,n} = \sin \frac{n\pi}{a} x \sin \frac{m\pi}{b} \left(y + \frac{b}{2}\right).$$

$$A_{m,n} = -\frac{8}{ab} \int_{-b/2}^{b/2} \int_0^a \sin \frac{n\pi}{a} x \sin \frac{m\pi}{b} \left(y + \frac{b}{2}\right) dx dy = -\frac{8}{\pi^2 mn} [1 - (-1)^n] [1 - (-1)^m],$$

$$A_{2i+1, 2j+1} = -\frac{32}{\pi^2 (2i+1)(2j+1)}, \text{ 其余的 } A_{m,n} = 0.$$

$$\begin{aligned} \text{所以 } u(x, y) &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{A_{m,n}}{\lambda_{m,n}} \Phi_{m,n}(x, y) = \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \frac{A_{2i+1, 2j+1}}{\lambda_{2i+1, 2j+1}} \Phi_{2i+1, 2j+1}(x, y) \\ &= -\frac{32}{\pi^4} \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \frac{\sin \frac{(2j+1)\pi}{a} x \sin \frac{(2i+1)\pi}{b} \left(y + \frac{b}{2}\right)}{(2i+1)(2j+1) \left[\frac{(2j+1)^2}{a^2} + \frac{(2i+1)^2}{b^2} \right]}. \end{aligned}$$

$$\begin{aligned} (2) \quad A_{m,n} &= -\frac{4}{ab} \int_{-b/2}^{b/2} \int_0^a x^2 y \sin \frac{n\pi}{a} x \sin \frac{m\pi}{b} \left(y + \frac{b}{2}\right) dx dy \\ &= (-1)^{m+n} \frac{2a^2 b}{\pi mn} [1 + (-1)^m] \left\{ -1 + \frac{2}{\pi^2 n^2} [1 - (-1)^n] \right\}. \end{aligned}$$

$$u(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{A_{m,n}}{\lambda_{m,n}} \Phi_{m,n}(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{A_{m,n}}{\left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2} \sin \frac{n\pi}{a} x \sin \frac{m\pi}{b} \left(y + \frac{b}{2}\right).$$

269. 设有本征值问题
$$\begin{cases} \frac{d^2}{dx^2} \left[p(x) \frac{d^2 y}{dx^2} \right] + \frac{d}{dx} \left[q(x) \frac{dy}{dx} \right] + [\lambda \rho(x) - r(x)] y = 0 \\ y|_{x=a} = 0, p(x) \frac{d^2 y}{dx^2} \Big|_{x=a} = 0, \frac{dy}{dx} \Big|_{x=b} = 0, \frac{d}{dx} \left[p(x) \frac{d^2 y}{dx^2} \right] \Big|_{x=b} = 0 \end{cases},$$

试证明：对应不同本征值的本征函数正交。

$$\begin{aligned}
 (\lambda_1 - \lambda_2^*) \int_a^b \rho y_1 y_2^* dx &= \int_a^b y_1 \frac{d^2}{dx^2} \left(p \frac{d^2 y_2^*}{dx^2} \right) dx - \int_a^b y_2^* \frac{d^2}{dx^2} \left(p \frac{d^2 y_1}{dx^2} \right) dx \\
 &\quad + \int_a^b y_1 \frac{d}{dx} \left(q \frac{dy_2^*}{dx} \right) dx - \int_a^b y_2^* \frac{d}{dx} \left(q \frac{dy_1}{dx} \right) dx \\
 &= y_1 \frac{d}{dx} \left(p \frac{d^2 y_2^*}{dx^2} \right) \Big|_a^b - y_2^* \frac{d}{dx} \left(p \frac{d^2 y_1}{dx^2} \right) \Big|_a^b + y_1 q \frac{dy_2^*}{dx} \Big|_a^b - y_2^* q \frac{dy_1}{dx} \Big|_a^b \\
 &\quad - \int_a^b \frac{dy_1}{dx} \frac{d}{dx} \left(p \frac{d^2 y_2^*}{dx^2} \right) dx + \int_a^b \frac{dy_2^*}{dx} \frac{d}{dx} \left(p \frac{d^2 y_1}{dx^2} \right) dx \\
 &= y_1 \frac{d}{dx} \left(p \frac{d^2 y_2^*}{dx^2} \right) \Big|_a^b - y_2^* \frac{d}{dx} \left(p \frac{d^2 y_1}{dx^2} \right) \Big|_a^b + y_1 q \frac{dy_2^*}{dx} \Big|_a^b - y_2^* q \frac{dy_1}{dx} \Big|_a^b \\
 &\quad - \frac{dy_1}{dx} \left(p \frac{d^2 y_2^*}{dx^2} \right) \Big|_a^b + \frac{dy_2^*}{dx} \left(p \frac{d^2 y_1}{dx^2} \right) \Big|_a^b = 0.
 \end{aligned}$$

270. 设有 4 阶常微分方程 $\frac{d^2}{dx^2} \left[p(x) \frac{d^2 y}{dx^2} \right] + \frac{d}{dx} \left[q(x) \frac{dy}{dx} \right] + [\lambda \rho(x) - r(x)] y = 0$,

$p(x)$, $q(x)$, $\rho(x)$, $r(x)$ 均为已知, λ 为待定系数。若 $y(x)$ 在端点 $x=a$ 及 $x=b$ 均

满足下列边界条件: $y=0$, $\frac{dy}{dx}=0$ 或 $y=0$, $p \frac{d^2 y}{dx^2}=0$ 或 $\frac{dy}{dx}=0$, $\frac{d}{dx} \left(p \frac{d^2 y}{dx^2} \right) = 0$,

试证明: 对应于不同本征值的本征函数在区间 $[a, b]$ 上以权重 $\rho(x)$ 正交。

$$\begin{aligned}
 \text{同上题, } (\lambda_1 - \lambda_2^*) \int_a^b \rho y_1 y_2^* dx &= y_1 \frac{d}{dx} \left(p \frac{d^2 y_2^*}{dx^2} \right) \Big|_a^b - y_2^* \frac{d}{dx} \left(p \frac{d^2 y_1}{dx^2} \right) \Big|_a^b + y_1 q \frac{dy_2^*}{dx} \Big|_a^b - y_2^* q \frac{dy_1}{dx} \Big|_a^b \\
 &\quad - \frac{dy_1}{dx} \left(p \frac{d^2 y_2^*}{dx^2} \right) \Big|_a^b + \frac{dy_2^*}{dx} \left(p \frac{d^2 y_1}{dx^2} \right) \Big|_a^b,
 \end{aligned}$$

可看出, 只要满足题目所给任一组条件, 上式就等于 0。

271. 设有本征值问题 $\begin{cases} X^{(4)} + \lambda X = 0 \\ X(0) = 0, X(l) = 0 \\ X''(0) = 0, X''(l) = 0 \end{cases}$, 证明: 本征值 $\lambda_n = - \frac{\int_0^l |X_n''|^2 dx}{\int_0^l |X_n|^2 dx} < 0$ 。

见习题 12 第 230 题。

272. 试根据 Legendre 方程在 $x=1$ 的有界解 (见习题 10 第 198 题) 求解本征值问题

$$\begin{cases} \frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \mu(\mu+1)y = 0 \\ y(\pm 1) \text{ 有界} \end{cases}.$$

第 198 题已求得 $x=1$ 的有界解为 $y(x) = \sum_{k=0}^{\infty} \frac{1}{(k!)^2} \frac{\Gamma(\mu+k+1)}{\Gamma(\mu-k+1)} \left(\frac{x-1}{2} \right)^k$, 则

$$y(-1) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \frac{\Gamma(\mu+k+1)}{\Gamma(\mu-k+1)}, \text{ 记该级数通项为 } u_k, \text{ 若 } \mu \text{ 不是整数, 当 } k \text{ 充分大时,}$$

$$\frac{u_k}{u_{k+1}} = - \frac{[(k+1)!]^2}{(k!)^2} \frac{\Gamma(\mu+k+1)}{\Gamma(\mu-k+1)} \frac{\Gamma(\mu-k)}{\Gamma(\mu+k+2)} = \frac{(k+1)^2}{(\mu+k+1)(k-\mu)} = 1 + \frac{1}{k} + O\left(\frac{1}{k^2}\right),$$

所以该级数发散。

若 $\mu = n$ ($n=0,1,2,\dots$), 当 $k \geq n+1$ 时有 $\frac{1}{\Gamma(n-k+1)} = 0$, 所以 $y(x)$ 截断为多项式:

$$y_n(x) = \sum_{k=0}^n \frac{1}{(k!)^2} \frac{(n+k)!}{(n-k)!} \left(\frac{x-1}{2} \right)^k, \text{ 这就是该本征值问题的解。}$$

273. 证明: (1) $P_{2k}(0) = (-1)^k \frac{(2k)!}{2^{2k}(k!)^2} = (-1)^{k+1} \frac{2}{B(k+1, -1/2)}, P_{2k+1}(0) = 0;$

(2) $P'_{2k}(0) = 0, P'_{2k+1}(0) = (-1)^k \frac{(2k+1)!}{2^{2k}(k!)^2} = (-1)^k \frac{2}{B(k+1, 1/2)};$

(3) $P'_k(1) = \frac{1}{2}k(k+1), P'_k(-1) = \frac{(-1)^{k-1}}{2}k(k+1), P''_k(1) = \frac{1}{8}(k-1)k(k+1)(k+2).$

(1) $P_n(x) = \sum_{r=0}^{[n/2]} \frac{(-1)^r (2n-2r)!}{2^n r!(n-r)!(n-2r)!} x^{n-2r}, \text{ 所以}$

$$P_{2k}(x) = \sum_{r=0}^k \frac{(-1)^r (4k-2r)!}{2^{2k} r!(2k-r)!(2k-2r)!} x^{2k-2r}, P_{2k+1}(x) = \sum_{r=0}^k \frac{(-1)^r (4k+2-2r)!}{2^{2k+1} r!(2k+1-r)!(2k+1-2r)!} x^{2k+1-2r},$$

显然有 $P_{2k+1}(0) = 0$, 取 $P_{2k}(x)$ 的常数项即为 $P_{2k}(0)$, 即

$$P_{2k}(x) = \frac{(-1)^k (2k)!}{2^{2k} (k!)^2} = \frac{(-1)^k}{2^{2k}} \frac{\Gamma(2k+1)}{\Gamma^2(k+1)} = \frac{(-1)^k}{2^{2k}} \frac{2^{2k} \Gamma(k+1/2) \Gamma(k+1)}{\Gamma^2(k+1)}$$

$$= -(-1)^k \frac{2\Gamma(k+1/2)}{\Gamma(k+1)(-2\sqrt{\pi})} = (-1)^{k+1} \frac{2\Gamma(k+1/2)}{\Gamma(k+1)\Gamma(-1/2)} = (-1)^{k+1} \frac{2}{B(k+1, -1/2)}.$$

$$(2) \quad P'_n(x) = \sum_{r=0}^{[(n-1)/2]} \frac{(-1)^r (2n-2r)!}{2^n r!(n-r)!(n-2r-1)!} x^{n-2r-1}, \text{ 则}$$

$$P'_{2k}(x) = \sum_{r=0}^{k-1} \frac{(-1)^r (4k-2r)!}{2^{2k} r!(2k-r)!(2k-2r-1)!} x^{2k-2r-1}, \quad P'_{2k+1}(x) = \sum_{r=0}^k \frac{(-1)^r (4k+2-2r)!}{2^{2k+1} r!(2k+1-r)!(2k-2r)!} x^{2k-2r},$$

$$\begin{aligned} \text{所以 } P'_{2k}(0) &= 0, \quad P'_{2k+1}(0) = \frac{(-1)^k (2k+2)!}{2^{2k+1} k!(k+1)!} = \frac{(-1)^k (2k+1)!}{2^{2k} (k!)^2} = \frac{(-1)^k}{2^{2k}} \frac{\Gamma(2k+2)}{\Gamma^2(k+1)} \\ &= \frac{(-1)^k}{2^{2k}} \frac{2^{2k+1} \Gamma(k+1) \Gamma(k+3/2)}{\Gamma^2(k+1) \Gamma(1/2)} = (-1)^k \frac{2\Gamma(k+3/2)}{\Gamma(k+1)\Gamma(1/2)} = (-1)^k \frac{2}{B(k+1, 1/2)}. \end{aligned}$$

$$(3) \quad \text{由 } P_k(x) = \sum_{n=0}^k \frac{1}{(n!)^2} \frac{(k+n)!}{(k-n)!} \left(\frac{x-1}{2}\right)^n \text{ 得 } P'_k(x) = \sum_{n=1}^k \frac{n}{2(n!)^2} \frac{(k+n)!}{(k-n)!} \left(\frac{x-1}{2}\right)^{n-1}.$$

$$\text{所以 } P'_k(1) = \frac{(k+1)!}{2(k-1)!} = \frac{1}{2} k(k+1).$$

$$\text{由 Legendre 多项式的微分表示可得 } P'_k(x) = \frac{1}{2^k k!} \frac{d^{k+1}}{dx^{k+1}} (x^2-1)^k, \text{ 所以}$$

$$P'_k(-x) = \frac{1}{2^k k!} \frac{d^{k+1}}{d(-x)^{k+1}} (x^2-1)^k = (-1)^{k+1} P'_k(x), \text{ 所以 } P'_k(-1) = \frac{(-1)^{k-1}}{2} k(k+1).$$

$$\text{又有 } P''_k(x) = \sum_{n=2}^k \frac{n(n-1)}{2^2 (n!)^2} \frac{(k+n)!}{(k-n)!} \left(\frac{x-1}{2}\right)^{n-2},$$

$$\text{所以 } P''_k(1) = \frac{(k+2)!}{8(k-2)!} = \frac{1}{8} (k+2)(k+1)k(k-1).$$

$$274. \text{ 证明: } \int_x^1 P_k(t) P_l(t) dt = \frac{(1-x^2) [P'_k(x) P_l(x) - P'_l(x) P_k(x)]}{k(k+1) - l(l+1)}, \quad k \neq l.$$

$$\begin{aligned} \frac{d}{dt} \frac{(1-t^2) [P'_k(t) P_l(t) - P'_l(t) P_k(t)]}{k(k+1) - l(l+1)} &= \frac{1}{k(k+1) - l(l+1)} \{ -2t [P'_k(t) P_l(t) - P'_l(t) P_k(t)] \\ &\quad + (1-t^2) [P''_k(t) P_l(t) - P''_l(t) P_k(t)] \} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{k(k+1)-l(l+1)} \left\{ P_l(t) \frac{d}{dx} [(1-t^2)P_k'(t)] \right. \\
&\quad \left. - P_k(t) \frac{d}{dx} [(1-t^2)P_l'(t)] \right\} \\
&= \frac{1}{k(k+1)-l(l+1)} [-P_l(t)k(k+1)P_k(t) + P_k(t)l(l+1)P_l(t)] \\
&= -P_l(t)P_k(t).
\end{aligned}$$

两边积分即得证。

275. 计算积分 $\int_{-1}^1 x^k P_l(x) dx$, 并由此导出 $\int_{-1}^1 P_k(x) P_l(x) dx = \frac{2}{2l+1} \delta_{kl}$ 。

$$\int_{-1}^1 x^k P_l(x) dx = \frac{1}{2^l l!} \int_{-1}^1 x^k \frac{d^l}{dx^l} (x^2-1)^l dx = \frac{1}{2^l l!} x^k \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l \Big|_{-1}^1 - \frac{1}{2^l l!} \int_{-1}^1 \frac{dx^k}{dx} \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l dx$$

$$\text{其中 } \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l = \frac{d^{l-1}}{dx^{l-1}} [(x+1)^l (x-1)^l] = \sum_{k=0}^{l-1} \frac{(l-1)!}{k!(l-1-k)!} [(x+1)^l]^{(l-1-k)} [(x-1)^l]^{(k)},$$

可看出上式中 $(x+1)$ 和 $(x-1)$ 的最低次幂都是 1, 所以 $\frac{1}{2^l l!} x^k \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l \Big|_{-1}^1 = 0$, 即

$$\begin{aligned}
\int_{-1}^1 x^k P_l(x) dx &= -\frac{1}{2^l l!} \int_{-1}^1 \frac{dx^k}{dx} \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l dx = (-1)^2 \frac{1}{2^l l!} \int_{-1}^1 \frac{d^2 x^k}{dx^2} \frac{d^{l-2}}{dx^{l-2}} (x^2-1)^l dx \\
&= \cdots = (-1)^l \frac{1}{2^l l!} \int_{-1}^1 (x^2-1)^l \frac{d^l x^k}{dx^l} dx = \frac{1}{2^l l!} \int_{-1}^1 (1-x^2)^l \frac{d^l x^k}{dx^l} dx.
\end{aligned}$$

若 $k < l$, 则 $\frac{d^l x^k}{dx^l} = 0$, 所以 $\int_{-1}^1 x^k P_l(x) dx = 0$ 。

$k \geq l$ 时, 若 $k+l$ 是奇数, 那么 $x^k P_l(x)$ 是奇函数, 一定有 $\int_{-1}^1 x^k P_l(x) dx = 0$, 所以可令

$$k = l + 2n \quad (n = 0, 1, \cdots), \text{ 此时 } \int_{-1}^1 x^{l+2n} P_l(x) dx = \frac{1}{2^l l!} \frac{(l+2n)!}{(2n)!} \int_0^1 2(1-x^2)^l x^{2n} dx,$$

$$\text{作代换 } x^2 = t, \text{ 则 } \int_{-1}^1 x^{l+2n} P_l(x) dx = \frac{1}{2^l l!} \frac{(l+2n)!}{(2n)!} \int_0^1 (1-t)^l t^{n-1/2} dt = \frac{1}{2^l l!} \frac{(l+2n)!}{(2n)!} B\left(l+1, n+\frac{1}{2}\right)$$

$$= \frac{1}{2^l l!} \frac{(l+2n)!}{(2n)!} \frac{\Gamma(l+1)\Gamma(n+1/2)}{\Gamma(l+n+3/2)} = \frac{(l+2n)!}{2^l (2n)!} \frac{\Gamma(n+1/2)}{\Gamma(l+n+3/2)} \quad (*)$$

$$= \frac{(l+2n)!}{2^l (2n)!} \frac{(n-1+1/2)(n-2+1/2)\cdots(1/2)\Gamma(1/2)}{(l+n+1/2)(l+n-1+1/2)\cdots(1/2)\Gamma(1/2)} = \frac{2(l+2n)!(2n-1)!!}{(2n)!(2l+2n+1)!!}$$

$$= \frac{2(l+2n)!(2l+2n)(2l+2n-2)\cdots 2}{(2n)(2n-2)\cdots 2 \cdot (2l+2n+1)!} = \frac{2(l+2n)!2^{l+n}(l+n)!}{2^n n!(2l+2n+1)!} = \frac{2^{l+1}(l+2n)!(l+n)!}{n!(2l+2n+1)!}.$$

综上有 $\int_{-1}^1 x^k P_l(x) dx = \begin{cases} \frac{2^{l+1}(l+2n)!(l+n)!}{n!(2l+2n+1)!}, & k=l+2n \\ 0, & \text{others} \end{cases}$ 其中 $n=0,1,\cdots$.

取 $n=0$ 可得 $\int_{-1}^1 x^l P_l(x) dx = \frac{2^{l+1}(l!)^2}{(2l+1)!}$.

若 $k < l$, 必有 $\int_{-1}^1 P_k(x) P_l(x) dx = 0$.

设 $P_l(x) = c_l x^l + c_{l-2} x^{l-2} + \cdots$, 则由 $P_l(x) = \sum_{r=0}^{[l/2]} \frac{(-1)^r (2l-2r)!}{2^l r! (l-r)! (l-2r)!} x^{l-2r}$ 可知 $c_l = \frac{(2l)!}{2^l (l!)^2}$,

则 $\int_{-1}^1 P_l^2(x) dx = \int_{-1}^1 (c_l x^l + c_{l-2} x^{l-2} + \cdots) P_l(x) dx = c_l \int_{-1}^1 x^l P_l(x) dx = \frac{2}{2l+1}$.

即 $\int_{-1}^1 P_k(x) P_l(x) dx = \frac{2}{2l+1} \delta_{kl}$.

276. 利用 Rodrigues 公式证明: $\int_{-1}^1 (1+x)^k P_l(x) dx = \frac{2^{k+1}(k!)^2}{(k-l)!(k+l+1)!}, k \geq l$.

若 $k < l$ 又如何?

$$\begin{aligned} \int_{-1}^1 (1+x)^k P_l(x) dx &= \frac{1}{2^l l!} \int_{-1}^1 (1+x)^k \frac{d^l}{dx^l} (x^2-1)^l dx \\ &= \frac{1}{2^l l!} (1+x)^k \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l \Big|_{-1}^1 - \frac{1}{2^l l!} \int_{-1}^1 \frac{d(1+x)^k}{dx} \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l dx \\ &= -\frac{1}{2^l l!} \int_{-1}^1 \frac{d(1+x)^k}{dx} \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l dx = \cdots = \frac{1}{2^l l!} \int_{-1}^1 (1-x^2)^l \frac{d^l (1+x)^k}{dx^l} dx \quad (*) \\ &= \frac{k!}{2^l l! (k-l)!} \int_{-1}^1 (1-x^2)^l (1+x)^{k-l} dx = \frac{k!}{2^l l! (k-l)!} \int_{-1}^1 (1-x)^l (1+x)^k dx \end{aligned}$$

作代换 $1+x=2t$, 则上式 $= \frac{2^{k+1} k!}{l! (k-l)!} \int_0^1 (1-t)^l t^k dt = \frac{2^{k+1} k!}{l! (k-l)!} B(l+1, k+1)$

$$= \frac{2^{k+1}k!}{l!(k-l)!} \frac{\Gamma(l+1)\Gamma(k+1)}{\Gamma(l+k+2)} = \frac{2^{k+1}(k!)^2}{(k-l)!(l+k+1)!}。$$

$k < l$ 时由 (*) 式可知积分=0。

277. 试由 Rodrigues 公式出发, 将 Legendre 多项式表示成围道积分, 从而导出 Legendre 多项式的积分表示: $P_l(x) = \frac{1}{\pi} \int_0^\pi \left(x \pm \sqrt{x^2-1} \cos \varphi \right)^l d\varphi。$

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2-1)^l = \frac{1}{2^l} \frac{1}{2\pi i} \oint_C \frac{(z^2-1)^l}{(z-x)^{l+1}} dz, \text{ 这里用了解析函数的高阶微商公式,}$$

其中 C 为任意包围 x 的围道, 这里取为以 x 为圆心, $\sqrt{x^2-1}$ 为半径的圆, 即 $z-x = \sqrt{x^2-1}e^{i\varphi}$,

$$\begin{aligned} \text{则 } P_l(x) &= \frac{1}{2^l} \frac{1}{2\pi} \int_{-\pi}^\pi \frac{\left(x + \sqrt{x^2-1}e^{i\varphi} + 1 \right)^l \left(x + \sqrt{x^2-1}e^{i\varphi} - 1 \right)^l}{\left(\sqrt{x^2-1} \right)^{l+1} e^{i(l+1)\varphi}} \sqrt{x^2-1} e^{i\varphi} d\varphi \\ &= \frac{1}{2\pi} \int_{-\pi}^\pi \left\{ \frac{\left(x+1+\sqrt{x^2-1}e^{i\varphi} \right) \left[(x-1)e^{-i\varphi} + \sqrt{x^2-1} \right]}{2\left(\sqrt{x^2-1} \right)} \right\}^l d\varphi \\ &= \frac{1}{2\pi} \int_{-\pi}^\pi \left(x + \sqrt{x^2-1} \cos \varphi \right)^l d\varphi = \frac{1}{\pi} \int_0^\pi \left(x + \sqrt{x^2-1} \cos \varphi \right)^l d\varphi \end{aligned}$$

作代换 $\varphi = \pi - \theta$ 即可得 $P_l(x) = \frac{1}{\pi} \int_0^\pi \left(x - \sqrt{x^2-1} \cos \theta \right)^l d\theta。$

278. 证明: $P_l(x)$ 的零点均为实数, 且全都在区间 $(-1,1)$ 内。

± 1 是 $(x^2-1)^l$ 的零点, 根据 Rolle 定理, $(-1,1)$ 内有 $\frac{d}{dx}(x^2-1)^l$ 的一个零点 (记为 ξ_1)。

又因为 ± 1 也是 $\frac{d}{dx}(x^2-1)^l$ 的零点, 所以 $(-1, \xi_1)$ 和 $(\xi_1, 1)$ 上有 $\frac{d^2}{dx^2}(x^2-1)^l$ 的零点, 而 ± 1 又

是 $\frac{d^2}{dx^2}(x^2-1)^l$ 的零点.....由此下去 $(-1,1)$ 内有 $\frac{d^l}{dx^l}(x^2-1)^l$ 的 l 个零点, 而 ± 1 不是

$\frac{d^l}{dx^l}(x^2-1)^l$ 的零点。 $P_l(x)$ 是 l 次多项式, 只能有 l 个零点。

279. 设 (x, y, z) 是空间一点坐标, θ 是矢径 $\mathbf{r}$ 与 z 轴夹角, $r = |\mathbf{r}|$, 证明:

$$P_l(\cos \theta) = \frac{(-1)^l r^{l+1}}{l!} \frac{\partial^l}{\partial z^l} \left(\frac{1}{r} \right).$$

用数学归纳法。 $l=0$ 时显然成立，设 l 成立，即 $\frac{\partial^l}{\partial z^l} \left(\frac{1}{r} \right) = \frac{(-1)^l l!}{r^{l+1}} P_l(\cos \theta)$ 。

由直角坐标与球坐标关系可得

$$\frac{\partial}{\partial z} = \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} = \cos \theta \frac{\partial}{\partial r} + \frac{\sin^2 \theta}{r} \frac{\partial}{\partial (\cos \theta)} = x \frac{\partial}{\partial r} + \frac{1-x^2}{r} \frac{\partial}{\partial x}, \text{ 其中 } x = \cos \theta.$$

$$\begin{aligned} \text{所以 } \frac{\partial^{l+1}}{\partial z^{l+1}} \left(\frac{1}{r} \right) &= \frac{\partial}{\partial z} \left[\frac{(-1)^l l!}{r^{l+1}} P_l(\cos \theta) \right] = \left(x \frac{\partial}{\partial r} + \frac{1-x^2}{r} \frac{\partial}{\partial x} \right) \left[\frac{(-1)^l l!}{r^{l+1}} P_l(x) \right] \\ &= \frac{(-1)^l l!}{r^{l+2}} \left[-(l+1)x P_l(x) + (1-x^2) P'_l(x) \right]. \end{aligned}$$

$$\text{所以 } \frac{(-1)^{l+1} r^{l+2}}{(l+1)!} \frac{\partial^{l+1}}{\partial z^{l+1}} \left(\frac{1}{r} \right) = -\frac{1}{l+1} \left[-(l+1)x P_l(x) + (1-x^2) P'_l(x) \right] = P_{l+1}(x).$$

280. 从 Legendre 多项式的生成函数出发证明：(1) $P_l(-x) = (-1)^l P_l(x)$ ；

$$(2) P_l\left(-\frac{1}{2}\right) = \sum_{k=0}^{2l} P_k\left(-\frac{1}{2}\right) P_{2l-k}\left(\frac{1}{2}\right); (3) P_l(\cos 2\theta) = \sum_{k=0}^{2l} (-1)^k P_k(\cos \theta) P_{2l-k}(\cos \theta);$$

$$(4) \int_{-1}^1 P_k(x) P_l(x) dx = \frac{2}{2l+1} \delta_{kl}.$$

$$(1) \sum_{l=0}^{\infty} P_l(x) t^l = \frac{1}{\sqrt{1-2xt+t^2}} = \frac{1}{\sqrt{1-2(-x)(-t)+(-t)^2}} = \sum_{l=0}^{\infty} (-1)^l P_l(-x) t^l,$$

所以 $P_l(-x) = (-1)^l P_l(x)$ 。

(2) 令 (3) 中 $\theta = \pi/3$ 即得。

$$\begin{aligned} (3) \sum_{l=0}^{\infty} P_l(\cos 2\theta) t^l &= \frac{1}{\sqrt{1-2t \cos 2\theta + t^2}} = \frac{1}{\sqrt{1-2t(2\cos^2 \theta - 1) + t^2}} = \frac{1}{\sqrt{(1+t)^2 - 4t \cos^2 \theta}} \\ &= \frac{1}{\sqrt{1+2\sqrt{t} \cos \theta + t}} \cdot \frac{1}{\sqrt{1-2\sqrt{t} \cos \theta + t}} = \sum_{k=0}^{\infty} P_k(\cos \theta) (-\sqrt{t})^k \cdot \sum_{l=0}^{\infty} P_l(\cos \theta) (\sqrt{t})^l \\ &= \sum_{l=0}^{\infty} \sum_{k=0}^l (-1)^k P_k(\cos \theta) P_{l-k}(\cos \theta) t^{l/2} = \sum_{l=0}^{\infty} \sum_{k=0}^{2l} (-1)^k P_k(\cos \theta) P_{2l-k}(\cos \theta) t^l. \end{aligned}$$

所以 $P_l(\cos 2\theta) = \sum_{k=0}^{2l} (-1)^k P_k(\cos \theta) P_{2l-k}(\cos \theta)$ 。

$$(4) \quad \frac{1}{1-2xt+t^2} = \sum_{l=0}^{\infty} P_l(x)t^l \cdot \sum_{k=0}^{\infty} P_k(x)t^k = \sum_{l=0}^{\infty} \left[\sum_{k=0}^{\infty} P_k(x)P_l(x)t^k \right] t^l。$$

$$\text{两边积分得 } \int_{-1}^1 \frac{1}{1-2xt+t^2} dx = \frac{1}{t} [\ln(1+t) - \ln(1-t)] = \frac{1}{t} \left[\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} t^k + \sum_{k=1}^{\infty} \frac{1}{k} t^k \right]$$

$$= \sum_{l=0}^{\infty} \frac{2}{2l+1} t^{2l} = \sum_{l=0}^{\infty} \left[\sum_{k=0}^{\infty} \int_{-1}^1 P_k(x)P_l(x) dx t^k \right] t^l。$$

$$\text{所以 } \sum_{k=0}^{\infty} \int_{-1}^1 P_k(x)P_l(x) dx t^k = \frac{2}{2l+1} t^l, \text{ 即 } \int_{-1}^1 P_k(x)P_l(x) dx = \frac{2}{2l+1} \delta_{kl}。$$

$$281. \text{ 证明: } P_l(\cos \theta) = \frac{1}{2^{2l}} \sum_{k=0}^l \frac{(2l-2k)!(2k)!}{(k!)^2 [(l-k)!]^2} \cos(l-2k)\theta。$$

$$\begin{aligned} \sum_{l=0}^{\infty} P_l(\cos \theta) t^l &= \frac{1}{\sqrt{1-2t \cos \theta + t^2}} = \frac{1}{\sqrt{1-t(e^{i\theta} + e^{-i\theta}) + t^2}} = \frac{1}{\sqrt{1-te^{i\theta}}} \frac{1}{\sqrt{1-te^{-i\theta}}} \\ &= \sum_{n=0}^{\infty} \binom{-1/2}{n} (-te^{i\theta})^n \cdot \sum_{k=0}^{\infty} \binom{-1/2}{k} (-te^{-i\theta})^k = \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} \frac{(2n)!(2k)!}{2^{2(n+k)} (n!)^2 (k!)^2} t^{n+k} e^{i(n-k)\theta}。 \end{aligned}$$

令上式右边 $n+k=l$ ，则

$$\begin{aligned} \sum_{l=0}^{\infty} P_l(\cos \theta) t^l &= \sum_{l=0}^{\infty} \sum_{k=0}^l \frac{(2l-2k)!(2k)!}{2^{2l} (k!)^2 [(l-k)!]^2} e^{i(l-2k)\theta} t^l \\ &= \sum_{l=0}^{\infty} \sum_{k=0}^l \frac{(2l-2k)!(2k)!}{2^{2l} (k!)^2 [(l-k)!]^2} \cos(l-2k)\theta t^l + i \sum_{l=0}^{\infty} \sum_{k=0}^l \frac{(2l-2k)!(2k)!}{2^{2l} (k!)^2 [(l-k)!]^2} \sin(l-2k)\theta t^l。 \end{aligned}$$

对于上式右边第二项，当 l 为偶数 ($=2m$) 时，

$$\begin{aligned} \sum_{k=0}^l \frac{(2l-2k)!(2k)!}{(k!)^2 [(l-k)!]^2} \sin(l-2k)\theta &= \sum_{k=0}^{m-1} \frac{(4m-2k)!(2k)!}{(k!)^2 [(2m-k)!]^2} \sin(2m-2k)\theta \\ &\quad + \sum_{k=m+1}^{2m} \frac{(4m-2k)!(2k)!}{(k!)^2 [(2m-k)!]^2} \sin(2m-2k)\theta \end{aligned}$$

$$\text{令上面右边第二式 } n=2m-k \text{ 可得 } \sum_{k=0}^l \frac{(2l-2k)!(2k)!}{(k!)^2 [(l-k)!]^2} \sin(l-2k)\theta = 0，$$

当 l 为奇数时同样有 $\sum_{k=0}^l \frac{(2l-2k)!(2k)!}{(k!)^2 [(l-k)!]^2} \sin(l-2k)\theta = 0$ 。

所以 $\sum_{l=0}^{\infty} P_l(\cos \theta) t^l = \sum_{l=0}^{\infty} \sum_{k=0}^l \frac{(2l-2k)!(2k)!}{2^{2l} (k!)^2 [(l-k)!]^2} \cos(l-2k)\theta t^l$, 比较系数即得

$$P_l(\cos \theta) = \frac{1}{2^{2l}} \sum_{k=0}^l \frac{(2l-2k)!(2k)!}{(k!)^2 [(l-k)!]^2} \cos(l-2k)\theta。$$

282. 如果 $|x|$ 足够小, 且 $f(x) = \sum_{k=0}^{\infty} \frac{a_k}{k!} x^k$, 证明: $f(x)$ 可按 Legendre 多项式展开:

$$f(x) = \sum_{l=0}^{\infty} c_l P_l(x), \text{ 其中 } c_l = (2l+1) \sum_{n=0}^{\infty} \frac{\Gamma(3/2) a_{l+2n}}{2^{l+2n} n! \Gamma(l+n+3/2)}。$$

由 Legendre 展开系数公式, $c_l = \frac{2l+1}{2} \sum_{k=0}^{\infty} \frac{a_k}{k!} \int_{-1}^1 x^k P_l(x) dx$ 。由 275 题可知, 只有 $k = l+2n$

时, 该式右边的积分才不为 0, 即 $c_l = \frac{2l+1}{2} \sum_{n=0}^{\infty} \frac{a_{l+2n}}{(l+2n)!} \int_{-1}^1 x^{l+2n} P_l(x) dx$ 。

由 275 题 (*) 式, $\int_{-1}^1 x^{l+2n} P_l(x) dx = \frac{(l+2n)!}{2^l (2n)!} \frac{\Gamma(n+1/2)}{\Gamma(l+n+3/2)}$

$$= \frac{(l+2n)!}{2^l (2n)! \Gamma(l+n+3/2)} \left(n-1+\frac{1}{2}\right) \left(n-2+\frac{1}{2}\right) \cdots \left(1+\frac{1}{2}\right) \Gamma\left(\frac{3}{2}\right)$$

$$= \frac{(l+2n)!}{2^{l+n-1} (2n)! \Gamma(l+n+3/2)} (2n-1)(2n-3) \cdots 3 \Gamma\left(\frac{3}{2}\right) = \frac{(l+2n)! \Gamma(3/2)}{2^{l+2n-1} n! \Gamma(l+n+3/2)}。$$

所以 $c_l = \frac{2l+1}{2} \sum_{n=0}^{\infty} \frac{a_{l+2n}}{(l+2n)!} \frac{(l+2n)! \Gamma(3/2)}{2^{l+2n-1} n! \Gamma(l+n+3/2)} = (2l+1) \sum_{n=0}^{\infty} \frac{\Gamma(3/2) a_{l+2n}}{2^{l+2n} n! \Gamma(l+n+3/2)}。$

283. 计算下列积分: (1) $\int_{-1}^1 P'_k(x) P_l(x) dx$; (2) $\int_{-1}^1 \frac{P_l(x)}{(1-2xt+t^2)^{3/2}} dx$ 。

(1) $P'_k(x)$ 是 $k-1$ 次多项式, 所以当 $k \leq l$ 时 $\int_{-1}^1 P'_k(x) P_l(x) dx = 0$ 。 $k > l$ 时,

$$\int_{-1}^1 P'_k(x) P_l(x) dx = P_k(x) P_l(x) \Big|_{-1}^1 - \int_{-1}^1 P_k(x) P'_l(x) dx = P_k(x) P_l(x) \Big|_{-1}^1 = 1 - (-1)^{k+l}。$$

所以当 $k = l + 2n + 1$ ($n = 0, 1, 2, \dots$) 时, 积分值为 2, 其余都为 0。

$$(2) \quad t = 0 \text{ 时, 原积分} = \int_{-1}^1 P_l(x) dx = \int_{-1}^1 P_0(x) P_l(x) dx = 2\delta_{l0}。$$

$$\begin{aligned} t \neq 0 \text{ 时, 原积分} &= \frac{1}{t} \int_{-1}^1 P_l(x) d \frac{1}{\sqrt{1-2xt+t^2}} = \frac{1}{t} \frac{P_l(x)}{\sqrt{1-2xt+t^2}} \Big|_{-1}^1 - \frac{1}{t} \int_{-1}^1 \frac{P_l'(x)}{\sqrt{1-2xt+t^2}} dx \\ &= \frac{1}{t} \left[\frac{1}{\sqrt{(1-t)^2}} - \frac{(-1)^l}{\sqrt{(1+t)^2}} \right] - \frac{1}{t} \int_{-1}^1 \frac{P_l'(x)}{\sqrt{1-2xt+t^2}} dx。 \end{aligned}$$

$$\text{当 } |t| < 1 \text{ 时, } \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{k=0}^{\infty} P_k(x) t^k, \text{ 所以}$$

$$\text{原积分} = \frac{1}{t} \left[\frac{1}{1-t} - \frac{(-1)^l}{1+t} \right] - \frac{1}{t} \sum_{k=0}^{\infty} \left[\int_{-1}^1 P_l'(x) P_k(x) dx \right] t^k = \frac{1}{t} \left[\frac{1}{1-t} - \frac{(-1)^l}{1+t} \right] - \frac{2}{t} \sum_{n=0}^{\left[\frac{l-1}{2}\right]} t^{l-2n-1}。$$

这里用到了上小题结论。对上式分别代入 l 为奇数和偶数可得原积分 $= \frac{2t^l}{1-t^2}$ 。

$$\text{当 } |t| > 1 \text{ 时, } \frac{1}{\sqrt{1-2xt+t^2}} = \frac{1}{|t| \sqrt{1-2x\frac{1}{t} + \frac{1}{t^2}}} = \frac{1}{|t|} \sum_{k=0}^{\infty} P_k(x) t^{-k},$$

$$\begin{aligned} t > 1 \text{ 时, 原积分} &= \frac{1}{t} \left[\frac{1}{t-1} - \frac{(-1)^l}{t+1} \right] - \frac{1}{t^2} \sum_{k=0}^{\infty} \left[\int_{-1}^1 P_l'(x) P_k(x) dx \right] t^{-k} \\ &= \frac{1}{t} \left[\frac{1}{t-1} - \frac{(-1)^l}{t+1} \right] - \frac{2}{t^2} \sum_{n=0}^{\left[\frac{l-1}{2}\right]} t^{2n-l+1} = \frac{2}{t^{l+1}(t^2-1)}。 \end{aligned}$$

$$\begin{aligned} t < -1 \text{ 时, 原积分} &= \frac{1}{t} \left[\frac{1}{1-t} + \frac{(-1)^l}{1+t} \right] + \frac{1}{t^2} \sum_{k=0}^{\infty} \left[\int_{-1}^1 P_l'(x) P_k(x) dx \right] t^{-k} \\ &= \frac{1}{t} \left[\frac{1}{1-t} + \frac{(-1)^l}{1+t} \right] + \frac{2}{t^2} \sum_{n=0}^{\left[\frac{l-1}{2}\right]} t^{2n-l+1} = \frac{2}{t^{l+1}(1-t^2)}。 \end{aligned}$$

284. 证明: 对于足够小的 $|t|$, 下式成立: $\frac{1-t^2}{(1-2xt+t^2)^{3/2}} = \sum_{l=0}^{\infty} (2l+1) P_l(x) t^l$ 。

$$\begin{aligned}
\frac{1-t^2}{(1-2xt+t^2)^{3/2}} &= \left(\frac{1}{t}-t\right) \frac{d}{dx} \frac{1}{\sqrt{1-2xt+t^2}} = \left(\frac{1}{t}-t\right) \sum_{l=0}^{\infty} P'_l(x) t^l \\
&= \sum_{l=0}^{\infty} P'_l(x) t^{l-1} - \sum_{l=0}^{\infty} P'_l(x) t^{l+1} = 1 + \sum_{l=1}^{\infty} P'_{l+1}(x) t^l - \sum_{l=1}^{\infty} P'_{l-1}(x) t^l \\
&= 1 + \sum_{l=1}^{\infty} [P'_{l+1}(x) - P'_{l-1}(x)] t^l = \sum_{l=0}^{\infty} (2l+1) P_l(x) t^l.
\end{aligned}$$

285. 利用 $\frac{1}{\sqrt{1-2xt+t^2}} = \frac{1}{1-xt} \frac{1}{\sqrt{1-t^2(x^2-1)/(1-xt)^2}}$ 证明:

$$P_l(x) = \sum_{k=0}^{[l/2]} \frac{l!}{2^{2k} (k!)^2 (l-2k)!} (x^2-1)^k x^{l-2k}.$$

$$\begin{aligned}
\sum_{l=0}^{\infty} P_l(x) t^l &= \frac{1}{\sqrt{1-2xt+t^2}} = \frac{1}{1-xt} \frac{1}{\sqrt{1-\frac{t^2(x^2-1)}{(1-xt)^2}}} = \frac{1}{1-xt} \sum_{k=0}^{\infty} \binom{-\frac{1}{2}}{k} \left[\frac{-t^2(x^2-1)}{(1-xt)^2} \right]^k \\
&= \sum_{k=0}^{\infty} \frac{(2k)!}{2^{2k} (k!)^2} t^{2k} (x^2-1)^k (1-xt)^{-2k-1} = \sum_{k=0}^{\infty} \frac{(2k)! t^{2k} (x^2-1)^k}{2^{2k} (k!)^2} \sum_{n=0}^{\infty} \binom{-2k-1}{n} (-xt)^n \\
&= \sum_{k=0}^{\infty} \sum_{n=0}^{\infty} \frac{(2k+n)!}{2^{2k} (k!)^2 n!} (x^2-1)^k x^n t^{2k+n}.
\end{aligned}$$

$$\text{令 } 2k+n=l, \text{ 则上式} = \sum_{l=0}^{\infty} \sum_{k=0}^{[l/2]} \frac{l!}{2^{2k} (k!)^2 (l-2k)!} (x^2-1)^k x^{l-2k} t^l,$$

$$\text{所以 } P_l(x) = \sum_{k=0}^{[l/2]} \frac{l!}{2^{2k} (k!)^2 (l-2k)!} (x^2-1)^k x^{l-2k}.$$

286. 利用上题结果证明: $e^{xt} J_0(t\sqrt{1-x^2}) = \sum_{l=0}^{\infty} \frac{P_l(x)}{l!} t^l$, 其中 $J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2}\right)^{2k}$.

$$\sum_{l=0}^{\infty} \frac{P_l(x)}{l!} t^l = \sum_{l=0}^{\infty} \sum_{k=0}^{[l/2]} \frac{1}{2^{2k} (k!)^2 (l-2k)!} (x^2-1)^k x^{l-2k} t^l, \text{ 令 } 2k+n=l, \text{ 则}$$

$$\begin{aligned}\text{上式} &= \sum_{k=0}^{\infty} \sum_{n=0}^{\infty} \frac{1}{2^{2k} (k!)^2 n!} (x^2 - 1)^k x^n t^{2k+n} = \sum_{n=0}^{\infty} \frac{(xt)^n}{n!} \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{t\sqrt{1-x^2}}{2} \right)^{2k} \\ &= e^{xt} J_0 \left(t\sqrt{1-x^2} \right).\end{aligned}$$

287. 在上题中分别令 $x = \cos \alpha$, $t = r \sin \beta$ 及 $x = \cos \beta$, $t = r \sin \alpha$, 从而推出:

$$P_l(\cos \alpha) = \left(\frac{\sin \alpha}{\sin \beta} \right)^l \sum_{k=0}^l \frac{l!}{k!(l-k)!} \left[\frac{\sin(\beta - \alpha)}{\sin \alpha} \right]^{l-k} P_k(\cos \beta).$$

上题中分别令 $x = \cos \alpha$, $t = r \sin \beta$ 及 $x = \cos \beta$, $t = r \sin \alpha$ 可得:

$$e^{r \cos \alpha \sin \beta} J_0(r \sin \alpha \sin \beta) = \sum_{l=0}^{\infty} \frac{P_l(\cos \alpha) \sin^l \beta}{l!} r^l,$$

$$e^{r \sin \alpha \cos \beta} J_0(r \sin \alpha \sin \beta) = \sum_{l=0}^{\infty} \frac{P_l(\cos \beta) \sin^l \alpha}{l!} r^l.$$

由以上两式消去 $J_0(r \sin \alpha \sin \beta)$ 可得

$$\begin{aligned}\sum_{l=0}^{\infty} \frac{P_l(\cos \alpha) \sin^l \beta}{l!} r^l &= e^{r \sin(\beta - \alpha)} \sum_{l=0}^{\infty} \frac{P_l(\cos \beta) \sin^l \alpha}{l!} r^l \\ &= \sum_{k=0}^{\infty} \frac{\sin^k(\beta - \alpha)}{k!} r^k \cdot \sum_{l=0}^{\infty} \frac{P_l(\cos \beta) \sin^l \alpha}{l!} r^l = \sum_{l=0}^{\infty} \sum_{k=0}^l \frac{P_k(\cos \beta) \sin^k \alpha}{k!} r^k \frac{\sin^{l-k}(\beta - \alpha)}{(l-k)!} r^{l-k} \\ &= \sum_{l=0}^{\infty} \sum_{k=0}^l \frac{1}{k!(l-k)!} \sin^k \alpha \sin^{l-k}(\beta - \alpha) P_k(\cos \beta) r^l.\end{aligned}$$

$$\text{两边比较系数得 } \frac{P_l(\cos \alpha) \sin^l \beta}{l!} = \sum_{k=0}^l \frac{1}{k!(l-k)!} \sin^k \alpha \sin^{l-k}(\beta - \alpha) P_k(\cos \beta),$$

$$\text{即 } P_l(\cos \alpha) = \left(\frac{\sin \alpha}{\sin \beta} \right)^l \sum_{k=0}^l \frac{l!}{k!(l-k)!} \left[\frac{\sin(\beta - \alpha)}{\sin \alpha} \right]^{l-k} P_k(\cos \beta).$$

288. 计算下列积分: (1) $\int_{-1}^1 (1-x^2) P'_k(x) P'_l(x) dx$; (2) $\int_{-1}^1 P'_k(x) P'_l(x) dx$.

$$(1) \text{ 原积分} = (1-x^2) P_k(x) P'_l(x) \Big|_{-1}^1 - \int_{-1}^1 P_k(x) \frac{d}{dx} [(1-x^2) P'_l(x)] dx$$

$$= l(l+1) \int_{-1}^1 P_k(x) P_l(x) dx = \frac{2l(l+1)}{2l+1} \delta_{kl}.$$

(2) 不妨设 $k \geq l$, 则原积分

$$= P_k(x) P_l'(x) \Big|_{-1}^1 - \int_{-1}^1 P_k(x) P_l''(x) dx = \frac{1}{2} l(l+1) [1 - (-1)^{l+k+1}].$$

当 $k+l$ 为偶数时, 积分值为 $l(l+1)$, $k+l$ 为奇数时, 积分值为 0.

289. 计算 下列积分: (1) $\int_0^1 P_k(x) P_l(x) dx$; (2) $\int_{-1}^1 x P_k(x) P_{k+1}(x) dx$;

(3) $\int_{-1}^1 x^2 P_k(x) P_{k+2}(x) dx$; (4) $\int_{-1}^1 [x P_k(x)]^2 dx$.

(1) 当 $k+l$ 为偶数时, $\int_0^1 P_k(x) P_l(x) dx = \frac{1}{2} \int_{-1}^1 P_k(x) P_l(x) dx = \frac{1}{2l+1} \delta_{kl}$. (**)

当 $k+l$ 为奇数时, 设 $k=2n$, $l=2m+1$, 令 274 题中 $x=0$ 得

$$\begin{aligned} \int_0^1 P_k(t) P_l(t) dt &= \frac{P'_{2n}(0) P_{2m+1}(0) - P'_{2m+1}(0) P_{2n}(0)}{2n(2n+1) - (2m+1)(2m+2)} \\ &= \frac{(-1)^{m+n}}{(2m+1)(2m+2) - 2n(2n+1)} \frac{(2n)!(2m+1)!}{2^{2(m+n)} (m!)^2 (n!)^2}. \quad (*) \end{aligned}$$

(2) 由递推关系, $x P_k(x) = \frac{k+1}{2k+1} P_{k+1}(x) + \frac{k}{2k+1} P_{k-1}(x)$,

$$\text{原积分} = \frac{k+1}{2k+1} \int_{-1}^1 P_{k+1}^2(x) dx + \frac{k}{2k+1} \int_{-1}^1 P_{k-1}(x) P_{k+1}(x) dx = \frac{2(k+1)}{(2k+1)(2k+3)}.$$

(3) $x P_{k+2}(x) = \frac{k+3}{2k+5} P_{k+3}(x) + \frac{k+2}{2k+5} P_{k+1}(x)$, 原积分

$$\begin{aligned} &= \int_{-1}^1 \left[\frac{k+1}{2k+1} P_{k+1}(x) + \frac{k}{2k+1} P_{k-1}(x) \right] \left[\frac{k+3}{2k+5} P_{k+3}(x) + \frac{k+2}{2k+5} P_{k+1}(x) \right] dx \\ &= \frac{(k+1)(k+2)}{(2k+1)(2k+5)} \int_{-1}^1 P_{k+1}^2(x) dx = \frac{2(k+1)(k+2)}{(2k+1)(2k+3)(2k+5)}. \end{aligned}$$

(4) 原积分

$$\begin{aligned} &= \int_{-1}^1 \left[\frac{k+1}{2k+1} P_{k+1}(x) + \frac{k}{2k+1} P_{k-1}(x) \right]^2 dx = \left(\frac{k+1}{2k+1} \right)^2 \int_{-1}^1 P_{k+1}^2(x) dx + \left(\frac{k}{2k+1} \right)^2 \int_{-1}^1 P_{k-1}^2(x) dx \\ &= \left(\frac{k+1}{2k+1} \right)^2 \frac{2}{2k+3} + \left(\frac{k}{2k+1} \right)^2 \frac{2}{2k-1} = \frac{2(2k^2+2k-1)}{(2k+3)(2k+1)(2k-1)}. \end{aligned}$$

290. 将下列函数按 Legendre 多项式展开: (1) $f(x) = x^2$; (2) $f(x) = |x|$;

$$(3) f(x) = \begin{cases} 0, & -1 \leq x < 0 \\ x, & 0 \leq x \leq 1 \end{cases}; (4) f(x) = \sqrt{1-2xt+t^2}.$$

$$(1) f(x) = a_2 P_2(x) + a_0 P_0(x), \quad a_2 = \frac{5}{2} \int_{-1}^1 x^2 P_2(x) dx = \frac{2}{3}, \quad a_0 = \frac{1}{2} \int_{-1}^1 x^2 dx = \frac{1}{3},$$

$$\text{即 } f(x) = \frac{2}{3} P_2(x) + \frac{1}{3} P_0(x).$$

$$(2) f(x) = \sum_{k=0}^{\infty} a_{2k} P_{2k}(x).$$

$$a_{2k} = \frac{4k+1}{2} \int_{-1}^1 |x| P_{2k}(x) dx = (4k+1) \int_0^1 x P_{2k}(x) dx = (4k+1) \int_0^1 P_1(x) P_{2k}(x) dx,$$

$$\text{由上题 (*) 式得 } a_{2k} = \frac{(-1)^{k+1} (2k)! (4k+1)}{2^{2k+1} (k!)^2 (k+1)(2k-1)}.$$

$$\text{即 } f(x) = \sum_{k=0}^{\infty} \frac{(-1)^{k+1} (2k)! (4k+1)}{2^{2k+1} (k!)^2 (k+1)(2k-1)} P_{2k}(x).$$

$$(3) f(x) = \frac{1}{2}x + \frac{1}{2}|x| = \frac{1}{2}P_1(x) + \frac{1}{2}|x| = \frac{1}{2}P_1(x) + \sum_{k=0}^{\infty} \frac{(-1)^{k+1} (2k)! (4k+1)}{2^{2(k+1)} (k!)^2 (k+1)(2k-1)} P_{2k}(x).$$

$$(4) \text{ 记 } F(x, u) = \sqrt{1-2xu+u^2}, \text{ 则}$$

$$\begin{aligned} \frac{\partial F(x, u)}{\partial u} &= \frac{u-x}{\sqrt{1-2xu+u^2}} = \sum_{l=0}^{\infty} P_l(x) u^{l+1} - \sum_{l=0}^{\infty} x P_l(x) u^l \\ &= \sum_{l=0}^{\infty} P_l(x) u^{l+1} - P_1(x) - \sum_{l=1}^{\infty} \frac{l+1}{2l+1} P_{l+1}(x) u^l - \sum_{l=1}^{\infty} \frac{l}{2l+1} P_{l-1}(x) u^l \\ &= \sum_{l=0}^{\infty} P_l(x) u^{l+1} - P_1(x) - \sum_{l=2}^{\infty} \frac{l}{2l-1} P_l(x) u^{l-1} - \sum_{l=0}^{\infty} \frac{l+1}{2l+3} P_l(x) u^{l+1} \\ &= \frac{2}{3} u P_0(x) + \sum_{l=1}^{\infty} \left(\frac{l+2}{2l+3} u^{l+1} - \frac{l}{2l-1} u^{l-1} \right) P_l(x). \end{aligned}$$

两边对 u 从 0 积到 t 得

$$f(x) = \left(\frac{1}{3} t^2 + 1 \right) P_0(x) + \sum_{l=1}^{\infty} \left(\frac{t^{l+2}}{2l+3} - \frac{t^l}{2l-1} \right) P_l(x) = \sum_{l=0}^{\infty} \left(\frac{t^{l+2}}{2l+3} - \frac{t^l}{2l-1} \right) P_l(x).$$

291. 定义 $Q_l(x) = \frac{1}{2} \int_{-1}^1 \frac{P_l(t)}{x-t} dt$, (1) 证明 $Q_l(x)$ 是 Legendre 方程的解;

(2) 求出 $Q_0(x)$, $Q_1(x)$ 和 $Q_2(x)$ 的表达式。

$$(1) \quad \frac{dQ_l(x)}{dx} = -\frac{1}{2} \int_{-1}^1 \frac{P_l(t)}{(x-t)^2} dt,$$

$$\frac{d}{dx} \left[(1-x^2) \frac{dQ_l(x)}{dx} \right] = -\frac{1}{2} \int_{-1}^1 P_l(t) \frac{d}{dx} \frac{1-x^2}{(x-t)^2} dt = -\int_{-1}^1 P_l(t) \frac{xt-1}{(x-t)^3} dt.$$

$$\begin{aligned} l(l+1)Q_l(x) &= \frac{1}{2} \int_{-1}^1 \frac{l(l+1)P_l(t)}{x-t} dt = -\frac{1}{2} \int_{-1}^1 \frac{1}{x-t} \frac{d}{dt} [(1-t^2)P'_l(t)] dt \\ &= -\frac{1}{x-t} (1-t^2)P'_l(t) \Big|_{-1}^1 + \frac{1}{2} \int_{-1}^1 P'_l(t) \frac{1-t^2}{(x-t)^2} dt \\ &= \frac{1}{2} P_l(t) \frac{1-t^2}{(x-t)^2} \Big|_{-1}^1 + \int_{-1}^1 P_l(t) \frac{xt-1}{(x-t)^3} dt = \int_{-1}^1 P_l(t) \frac{xt-1}{(x-t)^3} dt. \end{aligned}$$

$$\text{所以 } \frac{d}{dx} \left[(1-x^2) \frac{dQ_l(x)}{dx} \right] + l(l+1)Q_l(x) = 0.$$

(2) 若 $x \in (-1, 1)$, 则 $\int_{-1}^1 \frac{P_l(t)}{x-t} dt$ 是瑕积分, 这时取其主值。

$$Q_0(x) = \frac{1}{2} \int_{-1}^1 \frac{1}{x-t} dt = -\frac{1}{2} \lim_{\varepsilon \rightarrow 0^+} \left[\ln|x-t| \Big|_{-1}^{x-\varepsilon} + \ln|x-t| \Big|_{x+\varepsilon}^1 \right] = \frac{1}{2} \ln \frac{x+1}{x-1}.$$

$$Q_1(x) = \frac{1}{2} \int_{-1}^1 \frac{t}{x-t} dt = -\frac{1}{2} \int_{-1}^1 \left(1 - \frac{x}{x-t} \right) dt = -1 + xQ_0(x) = \frac{x}{2} \ln \frac{x+1}{x-1} - 1.$$

$$\begin{aligned} Q_2(x) &= \frac{1}{2} \int_{-1}^1 \frac{P_2(t)}{x-t} dt = \frac{3}{2} \cdot \frac{1}{2} \int_{-1}^1 \frac{tP_1(t)}{x-t} dt - \frac{1}{2} \cdot \frac{1}{2} \int_{-1}^1 \frac{P_0(t)}{x-t} dt \\ &= \frac{3x}{2} Q_1(x) - \frac{1}{2} Q_0(x) = \frac{1}{4} (3x^2 - 1) \ln \frac{x+1}{x-1} - \frac{3}{2} x. \end{aligned}$$

$x > 1$ 或 $x < -1$ 时也可得上面结果。

292. 求解下列本征值问题: (1) $\begin{cases} \frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \lambda y = 0 \\ y(0) = 0, y(1) \text{ 有界} \end{cases}$; (2) $\begin{cases} \frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \lambda y = 0 \\ y'(0) = 0, y(1) \text{ 有界} \end{cases}$ 。

令 $\lambda = \nu(\nu+1)$, 求出方程在 $x=0$ 的两个独立级数解:

$$y_0(x) = \sum_{n=0}^{\infty} \frac{(2n-1+\nu)(2n-3+\nu)\cdots(1+\nu)}{(2n)!} [2(n-1)-\nu][2(n-2)-\nu]\cdots(-\nu)x^{2n},$$

$$y_1(x) = \sum_{n=0}^{\infty} \frac{(2n+\nu)(2n-2+\nu)\cdots(2+\nu)}{(2n+1)!} [2(n-1)+1-\nu][2(n-2)+1-\nu]\cdots(1-\nu)x^{2n+1}.$$

$y_0(x)$ 为偶函数, $y_1(x)$ 为奇函数。一般地, $y_0(x)$ 和 $y_1(x)$ 在 $x = \pm 1$ 是发散的。

(1) 为使 $y(0) = 0$, 只能取 $y_1(x)$ 。当 $\nu = 2l+1$ ($l = 0, 1, \cdots$) 时, 因子

$$[2(n-1)+1-\nu][2(n-2)+1-\nu]\cdots(1-\nu) = [2(n-1)-2l][2(n-2)-2l]\cdots(-2l)$$

当 $n > l$ 时为 0, 即 $y_1(x)$ 截断为多项式:

$$\begin{aligned} y_1(x) &= \sum_{n=0}^l \frac{(2l+2n+1)(2l+2n-1)\cdots(2l+3)}{(2n+1)!} (-1)^n (2l)(2l-2)\cdots[2l-2(n-1)]x^{2n+1} \\ &= \sum_{n=0}^l \frac{(2l+2n+2)!}{2^{n+1}(2n+1)!(l+n+1)(l+n)\cdots(l+1)(2l+1)!} \frac{(-1)^n 2^n l!}{(l-n)!} x^{2n+1} \\ &= (-1)^l \frac{2^{2l} (l!)^2}{(2l+1)!} \sum_{n=0}^l \frac{(-1)^{l+n} (2l+2n+2)!}{2^{2l+1} (l-n)!(l+n+1)!(2n+1)!} x^{2n+1}, \end{aligned}$$

令 $n = l - k$, 则

$$y_1(x) = (-1)^l \frac{2^{2l} (l!)^2}{(2l+1)!} \sum_{k=0}^l \frac{(-1)^k (4l-2k+2)!}{2^{2l+1} k!(2l-k+1)!(2l-2k+1)!} x^{2l+1-2k} = (-1)^l \frac{2^{2l} (l!)^2}{(2l+1)!} P_{2l+1}(x)$$

所以本征值 $\lambda_l = (2l+1)(2l+2)$, 本征函数 $y_l(x) = P_{2l+1}(x)$ ($l = 0, 1, \cdots$)。

(2) 为使 $y'(0) = 0$, 只能取 $y_0(x)$ 。当 $\nu = 2l$ ($l = 0, 1, \cdots$) 时, $y_0(x)$ 截断为多项式:

$$\begin{aligned} y_0(x) &= (-1)^l \frac{2^{2l} (l!)^2}{(2l)!} \sum_{n=0}^l \frac{(-1)^{l+n} (2l+2n)!}{2^{2l} (l-n)!(l+n)!(2n)!} x^{2n} \\ &= (-1)^l \frac{2^{2l} (l!)^2}{(2l)!} \sum_{k=0}^l \frac{(-1)^k (4l-2k)!}{2^{2l} k!(2l-k)!(2l-2k)!} x^{2l-2k} = (-1)^l \frac{2^{2l} (l!)^2}{(2l)!} P_{2l}(x). \end{aligned}$$

所以本征值 $\lambda_l = 2l(2l+1)$, 本征函数 $y_l(x) = P_{2l}(x)$ ($l = 0, 1, \cdots$)。

293. 求解球内定解问题:
$$\begin{cases} \nabla^2 u = 0, r < a \\ u|_{r=a} = \begin{cases} u_0, 0 \leq \theta \leq \alpha \\ 0, \alpha < \theta \leq \pi \end{cases} \end{cases}$$

将 $r = a$ 的边界条件展开成 Legendre 级数: $u|_{r=a} = \begin{cases} u_0, 0 \leq \theta \leq \alpha \\ 0, \alpha < \theta \leq \pi \end{cases} = \sum_{l=0}^{\infty} c_l P_l(\cos \theta)$, 则

$$c_0 = \frac{1}{2} \int_0^\alpha u_0 \sin \theta d\theta = \frac{u_0}{2} (1 - \cos \alpha).$$

$$\begin{aligned} l > 0 \text{ 时, } c_l &= \frac{2l+1}{2} u_0 \int_0^\alpha P_l(\cos \theta) \sin \theta d\theta = \frac{2l+1}{2} u_0 \int_{\cos \alpha}^1 P_l(x) dx \\ &= \frac{1}{2} u_0 \left[\int_{\cos \alpha}^1 P'_{l+1}(x) dx - \int_{\cos \alpha}^1 P'_{l-1}(x) dx \right] = \frac{1}{2} u_0 [P_{l-1}(\cos \alpha) - P_{l+1}(\cos \alpha)]. \end{aligned}$$

可看出问题与 φ 无关。分离变量可得本征函数 $\Theta_l(\theta) = P_l(\cos \theta)$ ($l = 0, 1, \dots$)。

$$u(r, \theta) = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-l-1}) P_l(\cos \theta), \text{ 由于 } u(0, \theta) \text{ 有界, 所以 } B_l = 0, \text{ 即}$$

$$u(r, \theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta). \text{ 所以 } u(a, \theta) = \sum_{l=0}^{\infty} A_l a^l P_l(\cos \theta) = \sum_{l=0}^{\infty} c_l P_l(\cos \theta),$$

比较系数得 $A_0 = c_0 = \frac{u_0}{2} (1 - \cos \alpha)$, $A_l = \frac{c_l}{a^l} = \frac{u_0}{2a^l} [P_{l-1}(\cos \alpha) - P_{l+1}(\cos \alpha)]$, 所以

$$u(r, \theta) = \frac{u_0}{2} (1 - \cos \alpha) + \frac{u_0}{2} \sum_{l=1}^{\infty} [P_{l-1}(\cos \alpha) - P_{l+1}(\cos \alpha)] \left(\frac{r}{a}\right)^l P_l(\cos \theta).$$

294. 求解习题 11 第 208 题, 假定温度已达稳定。
$$\begin{cases} \nabla^2 u = 0, r < a \\ \left(\frac{\partial u}{\partial r} + \frac{H}{k} u \right) \Big|_{r=a} = \begin{cases} \frac{M}{k} \cos \theta, 0 \leq \theta \leq \pi/2 \\ 0, \pi/2 < \theta \leq \pi \end{cases} \end{cases}$$

$$\text{令 } \left(\frac{\partial u}{\partial r} + \frac{H}{k} u \right) \Big|_{r=a} = \sum_{l=0}^{\infty} c_l P_l(\cos \theta), \text{ 则}$$

$$c_l = \frac{2l+1}{2} \int_0^{\pi/2} \frac{M}{k} \cos \theta P_l(\cos \theta) \sin \theta d\theta = \frac{2l+1}{2} \frac{M}{k} \int_0^1 P_1(x) P_l(x) dx.$$

由 289 题 (**) 式, $c_1 = \frac{M}{2k}$, 由 289 题 (*) 式,

$$c_{2n} = \frac{4n+1}{2} \frac{M}{k} \int_0^1 P_1(x) P_{2n}(x) dx = \frac{M}{2k} \frac{(-1)^{n+1} (4n+1)(2n)!}{2^{2n+1} (2n-1)(n+1)(n!)^2}. \quad (n = 0, 1, \dots)$$

$$u(r, \theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta),$$

$$\left(\frac{\partial u}{\partial r} + \frac{H}{k} u \right) \Big|_{r=a} = \frac{H}{k} A_0 + \sum_{l=1}^{\infty} A_l \left(l a^{l-1} + \frac{H}{k} a^l \right) P_l(\cos \theta) = \sum_{l=0}^{\infty} c_l P_l(\cos \theta),$$

$$\text{对比系数得 } A_1 = \frac{M}{2} \frac{1}{Ha + k},$$

$$A_{2n} = \frac{kc_{2n}}{H + \frac{2nk}{a}} \frac{1}{a^{2n}} = \frac{M}{2} \frac{1}{H + \frac{2nk}{a}} \frac{(-1)^{n+1} (4n+1)(2n)!}{2^{2n+1} (2n-1)(n+1)(n!)^2} \frac{1}{a^{2n}} \quad (n=0, 1, \dots).$$

$$u(r, \theta) = \frac{M}{2 \left(H + \frac{k}{a} \right)} \frac{r}{a} P_1(\cos \theta) + \frac{M}{2} \sum_{n=0}^{\infty} \frac{1}{H + \frac{2nk}{a}} \frac{(-1)^{n+1} (4n+1)(2n)!}{2^{2n+1} (2n-1)(n+1)(n!)^2} \left(\frac{r}{a} \right)^{2n} P_{2n}(\cos \theta).$$

295. 求解下列定解问题:
$$\begin{cases} \nabla^2 u = 0, a < r < b \\ u|_{r=a} = u_0, u|_{r=b} = u_0 \cos^2 \theta \end{cases}$$

$$u(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta),$$

$$u|_{r=a} = \sum_{l=0}^{\infty} \left(A_l a^l + \frac{B_l}{a^{l+1}} \right) P_l(\cos \theta) = u_0 P_0(\cos \theta),$$

$$u|_{r=b} = \sum_{l=0}^{\infty} \left(A_l b^l + \frac{B_l}{b^{l+1}} \right) P_l(\cos \theta) = u_0 \cos^2 \theta = \frac{1}{3} u_0 P_0(\cos \theta) + \frac{2}{3} u_0 P_2(\cos \theta).$$

$$\text{比较系数得 } A_0 = \frac{b-3a}{3(b-a)} u_0, \quad B_0 = \frac{2ab}{3(b-a)} u_0, \quad A_2 = \frac{2b^3}{3(b^5-a^5)} u_0, \quad B_2 = \frac{2a^5 b^3}{3(a^5-b^5)} u_0,$$

其他 $A_l = 0, \quad B_l = 0$ 。所以

$$u(r, \theta) = \frac{b-3a}{3(b-a)} u_0 + \frac{2b}{3(b-a)} \frac{a}{r} u_0 + \frac{2b^3 a^2 u_0}{3(b^5-a^5)} \left[\left(\frac{r}{a} \right)^2 - \left(\frac{a}{r} \right)^3 \right] P_2(\cos \theta).$$

296. 解习题 11 第 209 题:
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - \frac{\omega^2}{2} \frac{\partial}{\partial x} \left[(l^2 - x^2) \frac{\partial u}{\partial x} \right] = 0 \\ u|_{x=0} = 0, u|_{x=l} \text{ 有界} \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t} \Big|_{t=0} = \psi(x) \end{cases}.$$

分离变量得
$$\begin{cases} \frac{d}{dx} \left[(l^2 - x^2) \frac{dX(x)}{dx} \right] + \lambda X(x) = 0 \\ T''(t) + \frac{\omega^2}{2} \lambda T(t) = 0 \end{cases}.$$
 可得本征值问题
$$\begin{cases} \frac{d}{dx} \left[(l^2 - x^2) \frac{dX}{dx} \right] + \lambda X = 0 \\ X(0) = 0, X(l) \text{ 有界} \end{cases},$$

令 $u = \frac{x}{l}$, 用 y 表示 X , 则
$$\begin{cases} \frac{d}{du} \left[(1 - u^2) \frac{dy}{du} \right] + \lambda y = 0 \\ y(0) = 0, y(1) \text{ 有界} \end{cases}.$$

第 292 题第 (1) 小题已得出 $\lambda_k = (2k+1)(2k+2)$, $y_k(u) = P_{2k+1}(u)$ ($k=0,1,\dots$),

所以 $X_k(x) = P_{2k+1}\left(\frac{x}{l}\right)$.

可解出 $T_k(t) = A_k \sin \omega_k t + B_k \cos \omega_k t$, 其中 $\omega_k = \sqrt{(k+1)(2k+1)}\omega$.

所以 $u(x,t) = \sum_{k=0}^{\infty} (A_k \sin \omega_k t + B_k \cos \omega_k t) P_{2k+1}\left(\frac{x}{l}\right)$.

由 289 题 (**) 式可得 $\int_0^1 P_{2k+1}(x) P_{2n+1}(x) dx = \frac{1}{4k+3} \delta_{kn}$,

所以有 $\int_0^l P_{2k+1}\left(\frac{x}{l}\right) P_{2n+1}\left(\frac{x}{l}\right) dx = \frac{l}{4k+3} \delta_{kn}$.

由初始条件以及上式的正交性可定出 $A_k = \frac{4k+3}{l\omega_k} \int_0^l \psi(x) P_{2k+1}\left(\frac{x}{l}\right) dx$,

$B_k = \frac{4k+3}{l} \int_0^l \varphi(x) P_{2k+1}\left(\frac{x}{l}\right) dx$.

297. 设有一半径为 a 的导体半球, 球面温度为 1°C , 底面温度为 0°C , 求半球内的稳定温度

分布。
$$\begin{cases} \nabla^2 u = 0, r < a \\ u|_{\theta=\pi/2} = 0, u|_{r=a} = 1 \end{cases}$$

$$\text{分离变量可得本征值问题} \begin{cases} \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \lambda \Theta = 0, & \text{则} \\ \Theta(0) \text{有界}, \Theta(\pi/2) = 0 \end{cases}$$

本征值 $\lambda_l = (2l+1)(2l+2)$, 本征函数 $\Theta_l(\theta) = P_{2l+1}(\cos \theta)$ ($l=0,1,\dots$)。

$$\text{所以 } u(r, \theta) = \sum_{l=0}^{\infty} A_l r^{2l+1} P_{2l+1}(\cos \theta)。$$

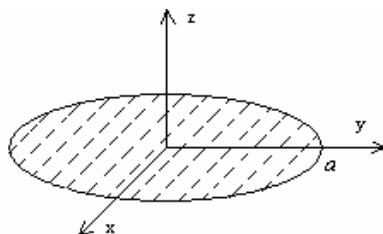
$$\text{由 289 题 (**) 式可得 } \int_0^{\pi/2} P_{2k+1}(\cos \theta) P_{2l+1}(\cos \theta) \sin \theta d\theta = \frac{1}{4l+3} \delta_{kl}。$$

$$\text{由 289 题 (*) 式可得 } \int_0^{\pi/2} P_0(\cos \theta) P_{2l+1}(\cos \theta) \sin \theta d\theta = \frac{(-1)^l (2l)!}{2^{2l+1} (l+1) (l!)^2}。$$

$$\text{由初始条件定出 } A_l = \frac{(-1)^l (2l)! (4l+3)}{2^{2l+1} (l+1) (l!)^2} \frac{1}{a^{2l+1}}, \text{ 所以}$$

$$u(r, \theta) = \sum_{l=0}^{\infty} \frac{(-1)^l (2l)! (4l+3)}{2^{2l+1} (l+1) (l!)^2} \left(\frac{r}{a} \right)^{2l+1} P_{2l+1}(\cos \theta)。$$

298. 一个均匀圆盘, 总质量 M , 半径 a , 求空间引力势。



$$\begin{cases} \nabla^2 u = \frac{4GM}{a^2} \frac{1}{r} \delta\left(\theta - \frac{\pi}{2}\right) \eta(a-r) \\ u|_{\theta=0} \text{有界}, u|_{\theta=\pi} \text{有界} \\ u|_{r=0} \text{有界}, u|_{r \rightarrow \infty} = 0 \end{cases}。$$

将 $\delta(\theta - \pi/2)$ 展开成 Legendre 级数: $\delta(\theta - \pi/2) = \sum_{l=0}^{\infty} c_l P_l(\cos \theta)$, 则

$$c_l = \frac{2l+1}{2} \int_0^{\pi} \delta\left(\theta - \frac{\pi}{2}\right) P_l(\cos \theta) \sin \theta d\theta = \frac{2l+1}{2} P_l(0),$$

$$\text{即 } \delta\left(\theta - \frac{\pi}{2}\right) = \sum_{l=0}^{\infty} \frac{2l+1}{2} P_l(0) P_l(\cos \theta)。$$

用 u_1 表示 $r < a$ 的 u , u_2 表示 $r > a$ 的 u , 则

$$\begin{cases} \nabla^2 u_1 = \frac{4GM}{a^2} \frac{1}{r} \delta\left(\theta - \frac{\pi}{2}\right) \\ u_1|_{\theta=0} \text{ 有界}, u_1|_{\theta=\pi} \text{ 有界}, u_1|_{r=0} \text{ 有界} \end{cases}, \begin{cases} \nabla^2 u_2 = 0 \\ u_2|_{\theta=0} \text{ 有界}, u_2|_{\theta=\pi} \text{ 有界}, u_2|_{r \rightarrow \infty} = 0 \end{cases},$$

$$\text{连接条件 } u_1|_{r=a-0} = u_2|_{r=a+0}, \quad \frac{\partial u_1}{\partial r}\bigg|_{r=a-0} = \frac{\partial u_2}{\partial r}\bigg|_{r=a+0}.$$

$$u_2 \text{ 解为 } u_2(r, \theta) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta).$$

$$\text{将 } u_1 \text{ 展开为 } u_1(r, \theta) = \sum_{l=0}^{\infty} R_l(r) P_l(\cos \theta), \text{ 代入 } u_1 \text{ 方程得}$$

$$\sum_{l=0}^{\infty} \left[\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR_l}{dr} \right) - \frac{l(l+1)}{r^2} R_l \right] P_l(\cos \theta) = \frac{4GM}{a^2 r} \sum_{l=0}^{\infty} \frac{2l+1}{2} P_l(0) P_l(\cos \theta),$$

$$\text{即 } \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR_l}{dr} \right) - \frac{l(l+1)}{r^2} R_l = \frac{4GM}{a^2 r} \frac{2l+1}{2} P_l(0).$$

$$\text{可解得 } R_l(r) = A_l r^l - \frac{2GM}{a^2} \frac{2l+1}{(l+2)(l-1)} P_l(0) r, \text{ 这里已去掉了无界项 } \frac{1}{r^{l+1}}.$$

$$\text{所以 } u_1(r, \theta) = \sum_{l=0}^{\infty} \left[A_l r^l - \frac{2GM}{a^2} \frac{2l+1}{(l+2)(l-1)} P_l(0) r \right] P_l(\cos \theta).$$

$$\text{由连接条件可得 } A_l a^l - \frac{2GM}{a} \frac{2l+1}{(l+2)(l-1)} P_l(0) = \frac{B_l}{a^{l+1}},$$

$$l A_l a^{l-1} - \frac{2GM}{a^2} \frac{2l+1}{(l+2)(l-1)} P_l(0) = -\frac{(l+1)B_l}{a^{l+2}} \quad (l = 0, 1, \dots).$$

$$\text{解得 } A_l = \frac{2GM}{a(l-1)} P_l(0) \frac{1}{a^l}, \quad B_l = -\frac{2GM}{a(l+2)} P_l(0) a^{l+1}.$$

$$\text{再由 } P_{2k}(0) = (-1)^k \frac{(2k)!}{2^{2k} (k!)^2}, \quad P_{2k+1}(0) = 0 \text{ 可得}$$

$$u_1(r, \theta) = \frac{2GM}{a} \sum_{k=0}^{\infty} \left[\frac{1}{2k-1} \left(\frac{r}{a} \right)^{2k} - \frac{4k+1}{(2k+2)(2k-1)} \frac{r}{a} \right] \frac{(-1)^k (2k)!}{2^{2k} (k!)^2} P_{2k}(\cos \theta),$$

$$u_2(r, \theta) = -\frac{GM}{a} \sum_{k=0}^{\infty} \frac{(-1)^k (2k)!}{2^{2k} (k+1)(k!)^2} \left(\frac{a}{r}\right)^{2k+1} P_{2k}(\cos \theta)。$$

299. 有一半径为 b 的接地导体球壳，球壳内放一圆环，环半径为 a ，环心与球心重合，环上均匀带电，总电荷为 Q 。求球内电势。

$$\text{以球心为圆点，垂直于环面的轴为 } z \text{ 轴。} \begin{cases} \nabla^2 u = -\frac{Q}{2\pi\epsilon_0 a^2} \delta(r-a) \delta\left(\theta - \frac{\pi}{2}\right) \\ u|_{\theta=0} \text{ 有界}, u|_{\theta=\pi} \text{ 有界} \\ u|_{r=0} \text{ 有界}, u|_{r=b} = 0 \end{cases}。$$

用 u_1 表示 $r < a$ 的 u ， u_2 表示 $a < r < b$ 的 u ，则

$$\begin{cases} \nabla^2 u_1 = 0 \\ u_1|_{\theta=0} \text{ 有界}, u_1|_{\theta=\pi} \text{ 有界}, u_1|_{r=0} \text{ 有界} \end{cases}, \begin{cases} \nabla^2 u_2 = 0 \\ u_2|_{\theta=0} \text{ 有界}, u_2|_{\theta=\pi} \text{ 有界}, u_2|_{r=b} = 0 \end{cases}，$$

u 在 $r = a$ 处是关于 r 连续的，即 $u_1|_{r=a-0} = u_2|_{r=a+0}$ ，否则原方程右边会出现 $\delta'(r-a)$ 项。

$$\text{原方程写为: } \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) = -\frac{r^2 Q}{2\pi\epsilon_0 a^2} \delta(r-a) \delta\left(\theta - \frac{\pi}{2}\right)，$$

两边对 r 在 $[a-\varepsilon, a+\varepsilon]$ 上积分，取极限得

$$\frac{\partial u_2}{\partial r} \Big|_{r=a+0} - \frac{\partial u_1}{\partial r} \Big|_{r=a-0} = -\frac{Q}{2\pi\epsilon_0 a^2} \delta\left(\theta - \frac{\pi}{2}\right) = -\frac{Q}{2\pi\epsilon_0 a^2} \sum_{l=0}^{\infty} \frac{2l+1}{2} P_l(0) P_l(\cos \theta)。$$

$$\text{综上，连接条件为} \begin{cases} u_1|_{r=a-0} = u_2|_{r=a+0} \\ \frac{\partial u_2}{\partial r} \Big|_{r=a+0} - \frac{\partial u_1}{\partial r} \Big|_{r=a-0} = -\frac{Q}{2\pi\epsilon_0 a^2} \sum_{l=0}^{\infty} \frac{2l+1}{2} P_l(0) P_l(\cos \theta) \end{cases}。$$

$$\text{可得 } u_1(r, \theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta), \quad u_2(r, \theta) = \sum_{l=0}^{\infty} \left(B_l r^l + \frac{C_l}{r^{l+1}} \right) P_l(\cos \theta)。$$

$$\text{由边界条件 } u_2|_{r=b} = 0 \text{ 得 } C_l = -B_l b^{2l+1}, \text{ 即 } u_2(r, \theta) = \sum_{l=0}^{\infty} \left(r^l - \frac{b^{2l+1}}{r^{l+1}} \right) B_l P_l(\cos \theta)。$$

$$\text{由连接条件得 } A_l a^l = \left(a^l - \frac{b^{2l+1}}{a^{l+1}} \right) B_l, \quad \left[l a^{l-1} + \frac{(l+1)b^{2l+1}}{a^{l+2}} \right] B_l - l A_l a^{l-1} = -\frac{Q(2l+1)}{4\pi\epsilon_0 a^2} P_l(0)。$$

$$\text{解得 } A_l = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a^{l+1}} - \frac{a^l}{b^{2l+1}} \right) P_l(0), \quad B_l = -\frac{Q}{4\pi\epsilon_0} \frac{a^l}{b^{2l+1}} P_l(0)。$$

$$\begin{aligned} \text{所以 } u_1(r, \theta) &= \frac{Q}{4\pi\epsilon_0 a} \sum_{l=0}^{\infty} \left[\left(\frac{r}{a} \right)^l - \left(\frac{a}{b} \right)^{l+1} \left(\frac{r}{b} \right)^l \right] P_l(0) P_l(\cos \theta) \\ &= \frac{Q}{4\pi\epsilon_0 a} \sum_{k=0}^{\infty} \frac{(-1)^k (2k)!}{2^{2k} (k!)^2} \left[\left(\frac{r}{a} \right)^{2k} - \left(\frac{a}{b} \right)^{2k+1} \left(\frac{r}{b} \right)^{2k} \right] P_{2k}(\cos \theta)。 \end{aligned}$$

$$\begin{aligned} u_2(r, \theta) &= \frac{Q}{4\pi\epsilon_0 r} \sum_{l=0}^{\infty} \left[\left(\frac{a}{r} \right)^l - \left(\frac{a}{b} \right)^l \left(\frac{r}{b} \right)^{l+1} \right] P_l(0) P_l(\cos \theta) \\ &= \frac{Q}{4\pi\epsilon_0 r} \sum_{k=0}^{\infty} \frac{(-1)^k (2k)!}{2^{2k} (k!)^2} \left[\left(\frac{a}{r} \right)^{2k} - \left(\frac{a}{b} \right)^{2k} \left(\frac{r}{b} \right)^{2k+1} \right] P_{2k}(\cos \theta)。 \end{aligned}$$

300. 将下列函数按球谐函数 $Y_l^m(\theta, \varphi)$ 展开: (1) $\sin^2 \theta \cos^2 \varphi$; (2) $(1+3\cos \theta) \sin \theta \cos \varphi$ 。

$$\begin{aligned} (1) \quad \sin^2 \theta \cos^2 \varphi &= \frac{1}{2}(1-x^2)(1+\cos 2\varphi) = \frac{1}{2}(1-x^2) + \frac{1}{4}(1-x^2)(e^{2i\varphi} + e^{-2i\varphi}) \\ &= \frac{1}{3}P_0^0(x) - \frac{1}{3}P_2^0(x) + \frac{1}{12}P_2^2(x)(e^{2i\varphi} + e^{-2i\varphi}) \\ &= \frac{2\sqrt{\pi}}{3}Y_0^0(\theta, \varphi) - \frac{2}{3}\sqrt{\frac{\pi}{5}}Y_2^0(\theta, \varphi) + \sqrt{\frac{2\pi}{15}}Y_2^2(\theta, \varphi) + \sqrt{\frac{2\pi}{15}}Y_2^{-2}(\theta, \varphi)。 \end{aligned}$$

$$\begin{aligned} (2) \quad (1+3\cos \theta) \sin \theta \cos \varphi &= \frac{1}{2}(1+3x)(1-x^2)^{\frac{1}{2}}(e^{i\varphi} + e^{-i\varphi}) \\ &= -\frac{1}{2}P_1^1(x)(e^{i\varphi} + e^{-i\varphi}) - \frac{1}{2}P_2^1(x)(e^{i\varphi} + e^{-i\varphi}) \\ &= -\sqrt{\frac{2\pi}{3}}Y_1^1(\theta, \varphi) - \sqrt{\frac{2\pi}{3}}Y_1^{-1}(\theta, \varphi) - \sqrt{\frac{6\pi}{5}}Y_2^1(\theta, \varphi) - \sqrt{\frac{6\pi}{5}}Y_2^{-1}(\theta, \varphi)。 \end{aligned}$$

301. 在半径为 a 的 (1) 球内区域, (2) 球外区域, 求解:
$$\begin{cases} \nabla^2 u = 0 \\ \left. \frac{\partial u}{\partial r} \right|_{r=a} = f(\theta, \varphi) \end{cases}$$

$$(1) \quad u(r, \theta, \varphi) = A_{0,0} + \sum_{l=1}^{\infty} \sum_{m=-l}^l A_{l,m} r^l Y_l^m(\theta, \varphi), \quad \text{这里去掉了无界项 } \frac{1}{r^{l+1}}。$$

$$\left. \frac{\partial u}{\partial r} \right|_{r=a} = \sum_{l=1}^{\infty} \sum_{m=-l}^l l A_{l,m} a^{l-1} Y_l^m(\theta, \varphi) = f(\theta, \varphi),$$

$$\text{求得 } A_{l,m} = \frac{1}{l a^{l-1}} \int_0^{2\pi} \int_0^{\pi} f(\theta, \varphi) Y_l^{m*}(\theta, \varphi) \sin \theta d\theta d\varphi \quad (l=1, 2, \dots).$$

由于所给边界条件只是导数值，所以 $A_{0,0}$ 不定，即零电位点可任意选取。

$$(2) \quad u(r, \theta, \varphi) = A_{0,0} + \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{B_{l,m}}{r^{l+1}} Y_l^m(\theta, \varphi), \text{ 这里去掉了有界项 } r^l \quad (l=1, 2, \dots).$$

$$B_{l,m} = -\frac{a^{l+2}}{l+1} \int_0^{2\pi} \int_0^{\pi} f(\theta, \varphi) Y_l^{m*}(\theta, \varphi) \sin \theta d\theta d\varphi.$$

302. 一半径为 a 的均匀导体球，表面温度为 (1) $u|_{r=a} = P_1^1(\cos \theta) \cos \varphi$,

(2) $u|_{r=a} = P_1^1(\cos \theta) \sin \theta \cos \varphi$ ，求出球内的稳定温度分布。

$$(1) \quad u = \sum_{l=0}^{\infty} \sum_{m=0}^l r^l P_l^m(\cos \theta) (A_{l,m} \cos m\varphi + B_{l,m} \sin m\varphi).$$

由边界条件可看出 $u(r, \theta, \varphi) = \frac{r}{a} P_1^1(\cos \theta) \cos \varphi = -\frac{r}{a} \sin \theta \cos \varphi$ 。

$$(2) \quad u|_{r=a} = -\frac{1}{3} P_2^1(\cos \theta) \cos \varphi,$$

可看出 $u(r, \theta, \varphi) = -\frac{1}{3} \left(\frac{r}{a}\right)^2 P_2^1(\cos \theta) \cos \varphi = \left(\frac{r}{a}\right)^2 \sin \theta \cos \theta \cos \varphi$ 。

303. 求解球内问题:
$$\begin{cases} \nabla^2 u = A + Br^2 \sin 2\theta \cos \varphi \\ u|_{r=a} = 0 \end{cases}.$$

令 $u(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=0}^l P_l^m(\cos \theta) [R_{l,m}(r) \sin m\varphi + S_{l,m}(r) \cos m\varphi]$ ，代入方程得

$$\begin{aligned} & \sum_{l=0}^{\infty} \sum_{m=0}^l \frac{1}{r^2} \left[\frac{d}{dr} \left(r^2 \frac{dR_{l,m}}{dr} \right) - l(l+1) R_{l,m} \right] P_l^m(\cos \theta) \sin m\varphi \\ & + \sum_{l=0}^{\infty} \sum_{m=0}^l \frac{1}{r^2} \left[\frac{d}{dr} \left(r^2 \frac{dS_{l,m}}{dr} \right) - l(l+1) S_{l,m} \right] P_l^m(\cos \theta) \cos m\varphi \end{aligned}$$

$$= A + Br^2 \sin 2\theta \cos \varphi = AP_0^0(\cos \theta) - \frac{2}{3} Br^2 P_2^1(\cos \theta) \cos \varphi。$$

$$\text{所以 } \frac{d}{dr} \left(r^2 \frac{dR_{l,m}}{dr} \right) - l(l+1) R_{l,m} = 0, \quad \frac{d}{dr} \left(r^2 \frac{dS_{0,0}}{dr} \right) = Ar^2, \quad \frac{d}{dr} \left(r^2 \frac{dS_{2,1}}{dr} \right) - 6S_{2,1} = -\frac{2}{3} Br^4,$$

$$\text{其他 } \frac{d}{dr} \left(r^2 \frac{dS_{l,m}}{dr} \right) - l(l+1) S_{l,m} = 0。$$

由边界条件 $R_{l,m}(a) = 0$, $S_{l,m}(a) = 0$ 以及自然条件 $R_{l,m}(0)$, $S_{l,m}(0)$ 有界, 解以上各常微分方程可得

$$R_{l,m}(r) = 0, \quad S_{0,0}(r) = \frac{A}{6}(r^2 - a^2), \quad S_{2,1}(r) = \frac{1}{21} Br^2(a^2 - r^2), \quad \text{其他 } S_{l,m}(r) = 0。$$

$$\begin{aligned} \text{所以 } u(r, \theta, \varphi) &= \frac{A}{6}(r^2 - a^2) + \frac{B}{21} r^2(a^2 - r^2) P_2^1(\cos \theta) \cos \varphi \\ &= \frac{A}{6}(r^2 - a^2) + \frac{B}{14} r^2(r^2 - a^2) \sin 2\theta \cos \varphi。 \end{aligned}$$

附录:

$$\text{关于 } P_l^m(x) \text{ 的定义, 这里采用 } P_l^m(x) = (-1)^m (1-x^2)^{\frac{m}{2}} \frac{d^m P_l(x)}{dx^m},$$

$$\text{原习题集答案采用 } P_l^m(x) = (1-x^2)^{\frac{m}{2}} \frac{d^m P_l(x)}{dx^m}。$$

304. 计算 Wronski 行列式 $W(J_\nu, J_{-\nu})$ 及 $W(J_\nu, Y_\nu)$, 其中 $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ 。

由 Bessel 方程, $\frac{d}{dx} \left[x \frac{dJ_\nu(x)}{dx} \right] + x \left(1 - \frac{\nu^2}{x^2} \right) J_\nu(x) = 0$,

$$\frac{d}{dx} \left[x \frac{dJ_{-\nu}(x)}{dx} \right] + x \left(1 - \frac{\nu^2}{x^2} \right) J_{-\nu}(x) = 0。$$

第一式两边乘 $J_{-\nu}(x)$ 减去第二式两边乘 $J_\nu(x)$ 得

$$J_{-\nu}(x) \frac{d}{dx} \left[x \frac{dJ_\nu(x)}{dx} \right] - J_\nu(x) \frac{d}{dx} \left[x \frac{dJ_{-\nu}(x)}{dx} \right] = 0。$$

继续化为 $J_{-\nu}(x) J_\nu'(x) - J_\nu(x) J_{-\nu}'(x) + x [J_{-\nu}(x) J_\nu''(x) - J_\nu(x) J_{-\nu}''(x)] = 0$,

$$\text{即 } \frac{d}{dx} \{ x [J_{-\nu}(x) J_\nu'(x) - J_\nu(x) J_{-\nu}'(x)] \} = 0,$$

所以 $x [J_{-\nu}(x) J_\nu'(x) - J_\nu(x) J_{-\nu}'(x)] = C$ (常数)。

$$\text{由 } J_\nu(x) = \left(\frac{x}{2} \right)^\nu \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(\nu + k + 1)} \left(\frac{x}{2} \right)^{2k}, \quad J_{-\nu}(x) = \left(\frac{x}{2} \right)^{-\nu} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(-\nu + k + 1)} \left(\frac{x}{2} \right)^{2k},$$

$$J_\nu'(x) = \left(\frac{x}{2} \right)^\nu \sum_{k=0}^{\infty} \frac{(2k + \nu)(-1)^k}{2^{2k} k! \Gamma(\nu + k + 1)} x^{2k-1}, \quad J_{-\nu}'(x) = \left(\frac{x}{2} \right)^{-\nu} \sum_{k=0}^{\infty} \frac{(2k - \nu)(-1)^k}{2^{2k} k! \Gamma(-\nu + k + 1)} x^{2k-1}$$

可确定常数 C 。

$$C = \frac{1}{\Gamma(-\nu+1)} \frac{\nu}{\Gamma(\nu+1)} - \frac{1}{\Gamma(\nu+1)} \frac{-\nu}{\Gamma(-\nu+1)} = \frac{2\nu}{\Gamma(\nu+1)\Gamma(-\nu+1)} = \frac{2}{\Gamma(\nu)\Gamma(1-\nu)} = \frac{2\sin\pi\nu}{\pi},$$

$$\text{所以 } W(J_\nu, J_{-\nu}) = J_\nu(x) J_{-\nu}'(x) - J_{-\nu}(x) J_\nu'(x) = -\frac{C}{x} = -\frac{2\sin\pi\nu}{\pi x},$$

$$W(J_\nu, Y_\nu) = \begin{vmatrix} J_\nu & Y_\nu \\ J_\nu' & Y_\nu' \end{vmatrix} = \begin{vmatrix} J_\nu & \cot\pi\nu J_\nu - \frac{1}{\sin\pi\nu} J_{-\nu} \\ J_\nu' & \cot\pi\nu J_\nu' - \frac{1}{\sin\pi\nu} J_{-\nu}' \end{vmatrix} = \cot\pi\nu \begin{vmatrix} J_\nu & J_\nu \\ J_\nu' & J_\nu' \end{vmatrix} - \frac{1}{\sin\pi\nu} \begin{vmatrix} J_\nu & J_{-\nu} \\ J_\nu' & J_{-\nu}' \end{vmatrix} = \frac{2}{\pi x}。$$

305. 利用上题结果计算下列积分: (1) $\int \frac{dx}{xJ_\nu^2(x)}$; (2) $\int \frac{dx}{xY_\nu^2(x)}$; (3) $\int \frac{dx}{xJ_\nu(x)Y_\nu(x)}$;

$$(4) \int \frac{dx}{x[J_\nu^2(x) + Y_\nu^2(x)]}。$$

(1) 将 $J_\nu(x)J'_{-\nu}(x) - J_{-\nu}(x)J'_\nu(x) = -\frac{2\sin\pi\nu}{\pi x}$ 两边同乘 $-\frac{\pi}{2\sin\pi\nu} \frac{1}{J_\nu^2(x)}$ 得

$$\frac{1}{xJ_\nu^2(x)} = -\frac{\pi}{2\sin\pi\nu} \frac{J_\nu(x)J'_{-\nu}(x) - J_{-\nu}(x)J'_\nu(x)}{J_\nu^2(x)} = -\frac{\pi}{2\sin\pi\nu} \frac{d}{dx} \frac{J_{-\nu}(x)}{J_\nu(x)},$$

$$\begin{aligned} \text{所以 } \int \frac{dx}{xJ_\nu^2(x)} &= -\frac{\pi}{2\sin\pi\nu} \int d \frac{J_{-\nu}(x)}{J_\nu(x)} = -\frac{\pi}{2\sin\pi\nu} \frac{J_{-\nu}(x)}{J_\nu(x)} + C \\ &= \frac{\pi}{2} \left[\cot\pi\nu - \frac{1}{\sin\pi\nu} \frac{J_{-\nu}(x)}{J_\nu(x)} \right] + C' = \frac{\pi}{2} \frac{\cos\pi\nu J_\nu(x) - J_{-\nu}(x)}{\sin\pi\nu J_\nu(x)} + C' \\ &= \frac{\pi}{2} \frac{Y_\nu(x)}{J_\nu(x)} + C'. \end{aligned}$$

(2) 由 $W(J_\nu, Y_\nu) = J_\nu(x)Y'_\nu(x) - J'_\nu(x)Y_\nu(x) = \frac{2}{\pi x}$ 可得 $\frac{1}{xY_\nu^2(x)} = -\frac{\pi}{2} \frac{d}{dx} \frac{J_\nu(x)}{Y_\nu(x)}$,

$$\text{所以 } \int \frac{dx}{xY_\nu^2(x)} = -\frac{\pi}{2} \frac{J_\nu(x)}{Y_\nu(x)} + C.$$

(3) 将 $J_\nu Y'_\nu - J'_\nu Y_\nu = \frac{2}{\pi x}$ 两边同乘 $\frac{\pi}{2} \frac{1}{J_\nu Y_\nu}$ 得 $\frac{1}{xJ_\nu(x)Y_\nu(x)} = \frac{\pi}{2} \left[\frac{Y'_\nu(x)}{Y_\nu(x)} - \frac{J'_\nu(x)}{J_\nu(x)} \right]$,

$$\text{所以 } \int \frac{dx}{xJ_\nu(x)Y_\nu(x)} = \frac{\pi}{2} \int \left[\frac{Y'_\nu(x)}{Y_\nu(x)} - \frac{J'_\nu(x)}{J_\nu(x)} \right] dx = \frac{\pi}{2} \ln \frac{Y_\nu(x)}{J_\nu(x)} + C.$$

(4) 将 $J_\nu Y'_\nu - J'_\nu Y_\nu = \frac{2}{\pi x}$ 两边同乘 $\frac{\pi}{2} \frac{1}{J_\nu^2 + Y_\nu^2}$ 得

$$\frac{1}{x[J_\nu^2(x) + Y_\nu^2(x)]} = \frac{\pi}{2} \frac{1}{1 + [Y_\nu(x)/J_\nu(x)]^2} \frac{d}{dx} \frac{Y_\nu(x)}{J_\nu(x)}, \text{ 所以}$$

$$\int \frac{dx}{x[J_\nu^2(x) + Y_\nu^2(x)]} = \frac{\pi}{2} \arctan \frac{Y_\nu(x)}{J_\nu(x)} + C.$$

306. 有很多方程经过适当的自变量或因变量变换可化为 Bessel 方程而得到他的解。例如,

$$\text{方程 } u'' + \frac{1-2\alpha}{z} u' + \left[\left(\beta \gamma z^{\gamma-1} \right)^2 + \frac{\alpha^2 - \gamma^2 \nu^2}{z^2} \right] u = 0 \text{ 的通解为 } c_1 z^\alpha J_\nu(\beta z^\gamma) + c_2 z^\alpha Y_\nu(\beta z^\gamma).$$

试验证此结果。

令 $x = \beta z^\gamma$, $u = z^\alpha y$, 则

$$\frac{du}{dz} = \alpha z^{\alpha-1} y + z^\alpha \frac{dy}{dz} = \alpha z^{\alpha-1} y + \beta \gamma z^{\alpha+\gamma-1} \frac{dy}{dx} = \alpha z^{\alpha-1} y + \gamma x z^{\alpha-1} \frac{dy}{dx},$$

$$\frac{d^2 u}{dz^2} = \alpha(\alpha-1) z^{\alpha-2} y + \alpha z^{\alpha-1} \frac{dy}{dz} + \beta \gamma (\alpha + \gamma - 1) z^{\alpha+\gamma-2} \frac{dy}{dx} + \beta \gamma z^{\alpha+\gamma-1} \frac{d}{dz} \frac{dy}{dx}$$

$$= \alpha(\alpha-1) z^{\alpha-2} y + \alpha \beta \gamma z^{\alpha+\gamma-2} \frac{dy}{dx} + \beta \gamma (\alpha + \gamma - 1) z^{\alpha+\gamma-2} \frac{dy}{dx} + \beta^2 \gamma^2 z^{\alpha+2\gamma-2} \frac{d^2 y}{dx^2}$$

$$= \alpha(\alpha-1) z^{\alpha-2} y + \beta \gamma (2\alpha + \gamma - 1) z^{\alpha+\gamma-2} \frac{dy}{dx} + \beta^2 \gamma^2 z^{\alpha+2\gamma-2} \frac{d^2 y}{dx^2}$$

$$= \alpha(\alpha-1) z^{\alpha-2} y + \gamma (2\alpha + \gamma - 1) x z^{\alpha-2} \frac{dy}{dx} + \gamma^2 x^2 z^{\alpha-2} \frac{d^2 y}{dx^2}.$$

代入方程，化简得 $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - \nu^2) y = 0$,

这是 ν 阶 Bessel 方程，通解为 $y = c_1 J_\nu(x) + c_2 Y_\nu(x)$,

所以 $u = z^\alpha y = c_1 z^\alpha J_\nu(x) + c_2 z^\alpha Y_\nu(x) = c_1 z^\alpha J_\nu(\beta z^\gamma) + c_2 z^\alpha Y_\nu(\beta z^\gamma)$ 。

307. 利用上题结果，解下列常微分方程：(1) $u'' + az^b u = 0$ ；(2) $z^2 u'' - 2zu' + 4(z^4 - 1)u = 0$ ；

(3) $zu'' - 3u' + zu = 0$ ；(4) $zu'' - u' + 4z^3 u = 0$ ；(5) $z^2 u'' + zu' - (z^2 + 1/4)u = 0$ ；

(6) $zu'' - u' - zu = 0$ ；(7) $u'' - z^2 u = 0$ ；(8) 一单摆在平衡位置附近作微小振动，若摆长以等速率 b 增长，而初始时摆长为 a ，则其动力学方程为 $(a + bt)\ddot{\theta} + 2b\dot{\theta} + g\theta = 0$ 。

设 $t = 0$ 时单摆静止于 $\theta(0) = \theta_0$ 处，试求 $\theta(t)$ 。

(1) 可看出，令上题中 $\alpha = \frac{1}{2}$, $\beta = \frac{2\sqrt{a}}{b+2}$, $\gamma = \frac{b}{2} + 1$, $\nu = \frac{1}{b+2}$ 即可得该方程，因此

$$u = c_1 \sqrt{z} J_{\frac{1}{b+2}} \left(\frac{2\sqrt{a}}{b+2} z^{\frac{b}{2}+1} \right) + c_2 \sqrt{z} Y_{\frac{1}{b+2}} \left(\frac{2\sqrt{a}}{b+2} z^{\frac{b}{2}+1} \right).$$

(2) $\alpha = \frac{3}{2}$, $\beta = 1$, $\gamma = 2$, $\nu = \frac{5}{4}$, $u = c_1 z^{3/2} J_{5/4}(z^2) + c_2 z^{3/2} Y_{5/4}(z^2)$ 。

(3) $\alpha = 2$, $\beta = 1$, $\gamma = 1$, $\nu = 2$, $u = c_1 z^2 J_2(z) + c_2 z^2 Y_2(z)$ 。

(4) $\alpha = 1$, $\beta = 1$, $\gamma = 2$, $\nu = 1/2$, $u = c_1 z J_{1/2}(z^2) + c_2 z Y_{1/2}(z^2)$ 。

$$(5) \alpha=0, \beta=i, \gamma=1, \nu=1/2, u=c_1 J_{1/2}(iz)+c_2 Y_{1/2}(iz)=c_1' I_{1/2}(z)+c_2' K_{1/2}(z)。$$

$$(6) \alpha=1, \beta=i, \gamma=1, \nu=1, u=c_1 z I_1(z)+c_2 z K_1(z)。$$

$$(7) \alpha=1/2, \beta=i/2, \gamma=2, \nu=1/2, u=c_1 \sqrt{z} I_{1/2}\left(\frac{1}{2} z^2\right)+c_2 \sqrt{z} K_{1/2}\left(\frac{1}{2} z^2\right)。$$

$$(8) \text{ 令 } a+bt=x, \text{ 则方程化为 } x \frac{d^2 \theta}{dx^2}+2 \frac{d \theta}{dx}+\frac{g}{b^2} \theta=0, \text{ 令上题 } \alpha=-\frac{1}{2}, \beta=\frac{2}{b} \sqrt{g},$$

$\gamma=1/2, \nu=1$ 即为该方程, 所以

$$\theta=\frac{1}{\sqrt{x}}\left[c_1 J_1\left(\frac{2}{b} \sqrt{g x}\right)+c_2 Y_1\left(\frac{2}{b} \sqrt{g x}\right)\right]=\frac{1}{\sqrt{a+b t}}\left[c_1 J_1\left(\frac{2}{b} \sqrt{g(a+b t)}\right)+c_2 Y_1\left(\frac{2}{b} \sqrt{g(a+b t)}\right)\right]。$$

代入初始条件 $\theta(0)=\theta_0, \dot{\theta}(0)=0$ 可得

$$c_1=\frac{b \theta_0}{2 \sqrt{g}} \frac{Y_1\left(\frac{2}{b} \sqrt{g a}\right)-\frac{2}{b} \sqrt{g a} Y_1'\left(\frac{2}{b} \sqrt{g a}\right)}{J_1'\left(\frac{2}{b} \sqrt{g a}\right) Y_1\left(\frac{2}{b} \sqrt{g a}\right)-J_1\left(\frac{2}{b} \sqrt{g a}\right) Y_1'\left(\frac{2}{b} \sqrt{g a}\right)}, \text{ 根据 304 题求出的 Wronski}$$

行列式可知 $J_1'\left(\frac{2}{b} \sqrt{g a}\right) Y_1\left(\frac{2}{b} \sqrt{g a}\right)-J_1\left(\frac{2}{b} \sqrt{g a}\right) Y_1'\left(\frac{2}{b} \sqrt{g a}\right)=-\frac{b}{\pi \sqrt{g a}}$, 所以

$$c_1=\frac{\pi \sqrt{a}}{2} \theta_0\left[-Y_1\left(\frac{2}{b} \sqrt{g a}\right)+\frac{2}{b} \sqrt{g a} Y_1'\left(\frac{2}{b} \sqrt{g a}\right)\right], \text{ 还可得}$$

$$c_2=\frac{\pi \sqrt{a}}{2} \theta_0\left[J_1\left(\frac{2}{b} \sqrt{g a}\right)-\frac{2}{b} \sqrt{g a} J_1'\left(\frac{2}{b} \sqrt{g a}\right)\right]。$$

$$\text{308. 证明: (1) } \cos x=J_0(x)+2 \sum_{n=1}^{\infty}(-1)^n J_{2 n}(x), \sin x=2 \sum_{n=0}^{\infty}(-1)^n J_{2 n+1}(x)。$$

$$(2) J_0^2(x)+2 \sum_{n=1}^{\infty} J_n^2(x)=1; (3) x=2 \sum_{n=0}^{\infty}(2 n+1) J_{2 n+1}(x); (4) x^2=2 \sum_{n=1}^{\infty}(2 n)^2 J_{2 n}(x)。$$

$$(1) \text{ 令 } \exp \left[\frac{x}{2}\left(t-\frac{1}{t}\right)\right]=\sum_{n=-\infty}^{\infty} J_n(x) t^n \text{ 中 } t=e^{i \theta}, \text{ 则有 } e^{i x \sin \theta}=\sum_{n=-\infty}^{\infty} J_n(x) e^{i n \theta}, \quad (\text{a})$$

这是 $e^{i x \sin \theta}$ 的 Fourier 级数表示。比较两边实部和虚部有

$$\begin{aligned}
\cos(x \sin \theta) &= \sum_{n=-\infty}^{\infty} J_n(x) \cos n\theta = J_0(x) + \sum_{n=1}^{\infty} J_n(x) \cos n\theta + \sum_{n=-1}^{-\infty} J_n(x) \cos n\theta \\
&= J_0(x) + \sum_{n=1}^{\infty} J_n(x) \cos n\theta + \sum_{n=1}^{\infty} J_{-n}(x) \cos n\theta = J_0(x) + \sum_{n=1}^{\infty} J_n(x) [1 + (-1)^n] \cos n\theta \\
&= J_0(x) + 2 \sum_{n=1}^{\infty} J_{2n}(x) \cos 2n\theta. \tag{b}
\end{aligned}$$

令 $\theta = \pi/2$ 既可得 $\cos x = J_0(x) + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(x)$ 。

$$\begin{aligned}
\sin(x \sin \theta) &= \sum_{n=-\infty}^{\infty} J_n(x) \sin n\theta = \sum_{n=1}^{\infty} J_n(x) \sin n\theta + \sum_{n=-1}^{-\infty} J_n(x) \sin n\theta \\
&= \sum_{n=1}^{\infty} J_n(x) \sin n\theta - \sum_{n=1}^{\infty} J_{-n}(x) \sin n\theta = \sum_{n=1}^{\infty} J_n(x) [1 - (-1)^n] \sin n\theta \\
&= 2 \sum_{n=0}^{\infty} J_{2n+1}(x) \sin(2n+1)\theta. \tag{c}
\end{aligned}$$

令 $\theta = \pi/2$ 既可得 $\sin x = 2 \sum_{n=0}^{\infty} (-1)^n J_{2n+1}(x)$ 。

(2) 由复 Fourier 级数的 Parseval 等式, 对于 (a) 式有

$$J_0^2(x) + 2 \sum_{n=1}^{\infty} J_n^2(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |e^{ix \sin \theta}|^2 d\theta = 1.$$

(3) 将 (a) 式两边对 θ 求导得 $ix \cos \theta e^{ix \sin \theta} = \sum_{n=-\infty}^{\infty} in J_n(x) e^{in\theta}$, \tag{d}

$$\begin{aligned}
\text{令 } \theta = 0 \text{ 得 } x &= \sum_{n=-\infty}^{\infty} n J_n(x) = \sum_{n=1}^{\infty} n J_n(x) + \sum_{n=-1}^{-\infty} n J_n(x) = \sum_{n=1}^{\infty} n J_n(x) - \sum_{n=1}^{\infty} n J_{-n}(x) \\
&= \sum_{n=1}^{\infty} n [1 - (-1)^n] J_n(x) = 2 \sum_{n=0}^{\infty} (2n+1) J_{2n+1}(x).
\end{aligned}$$

(4) 将 (d) 式两边对 θ 求导得 $-x \sin \theta e^{ix \sin \theta} + ix^2 \cos^2 \theta e^{ix \sin \theta} = i \sum_{n=-\infty}^{\infty} n^2 J_n(x) e^{in\theta}$,

令 $\theta = 0$ 得 $x^2 = \sum_{n=-\infty}^{\infty} n^2 J_n(x) = \sum_{n=1}^{\infty} n^2 [1 + (-1)^n] J_n(x) = 2 \sum_{n=1}^{\infty} (2n)^2 J_{2n}(x)$ 。

309. 将函数 $\cos(x \sin \theta)$ 和 $\sin(x \sin \theta)$ 展为 Fourier 级数。(见上题 (b) (c) 式)

310. 将函数 $\cos(z \cos \varphi)$ 展开为 z 的幂级数, 逐项积分, 证明:

$$\begin{aligned} J_\nu(z) &= \frac{(z/2)^\nu}{\sqrt{\pi}\Gamma(\nu+1/2)} \int_0^\pi \cos(z \cos \varphi) \sin^{2\nu} \varphi d\varphi \\ &= \frac{(z/2)^\nu}{\sqrt{\pi}\Gamma(\nu+1/2)} \int_{-1}^1 \cos(z\xi) (1-\xi^2)^{\nu-\frac{1}{2}} d\xi. \end{aligned}$$

其中 $\operatorname{Re} \nu > -\frac{1}{2}$ 。这个结果可以用来把“李萨如图形”展开成 Fourier 级数。作为一个例子,

试将 $y = \sqrt{\pi^2 - x^2}$ 在 $[-\pi, \pi]$ 上展成 Fourier 级数。

$$\cos(z \cos \varphi) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (z \cos \varphi)^{2n},$$

$$\begin{aligned} \int_0^\pi \cos(z \cos \varphi) \sin^{2\nu} \varphi d\varphi &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n} \int_0^\pi \cos^{2n} \varphi \sin^{2\nu} \varphi d\varphi \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n} \int_0^{\pi/2} 2 \cos^{2n} \varphi \sin^{2\nu} \varphi d\varphi = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n} B\left(n+\frac{1}{2}, \nu+\frac{1}{2}\right) \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n} \frac{\Gamma\left(n+\frac{1}{2}\right)\Gamma\left(\nu+\frac{1}{2}\right)}{\Gamma(n+\nu+1)} = \sqrt{\pi}\Gamma\left(\nu+\frac{1}{2}\right) \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\Gamma(n+\nu+1)} \left(\frac{z}{2}\right)^{2n} \\ &= \frac{\sqrt{\pi}\Gamma(\nu+1/2)}{(z/2)^\nu} J_\nu(z), \end{aligned}$$

即 $J_\nu(z) = \frac{(z/2)^\nu}{\sqrt{\pi}\Gamma(\nu+1/2)} \int_0^\pi \cos(z \cos \varphi) \sin^{2\nu} \varphi d\varphi$, 再令 $\cos \varphi = \xi$ 即可得

$$J_\nu(z) = \frac{(z/2)^\nu}{\sqrt{\pi}\Gamma(\nu+1/2)} \int_{-1}^1 \cos(z\xi) (1-\xi^2)^{\nu-\frac{1}{2}} d\xi.$$

$\sqrt{\pi^2 - x^2}$ 是偶函数, 可令 $\sqrt{\pi^2 - x^2} = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$, 则 $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \sqrt{\pi^2 - x^2} \cos nx dx$,

作代换 $x = \pi\xi$, 则 $a_n = \pi \int_{-1}^1 \sqrt{1-\xi^2} \cos(n\pi\xi) d\xi = \pi \frac{\sqrt{\pi}\Gamma(3/2)J_1(n\pi)}{n\pi/2} = \frac{\pi J_1(n\pi)}{n}$,

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} \sqrt{\pi^2 - x^2} dx = \frac{\pi^2}{2}, \text{ 所以 } \sqrt{\pi^2 - x^2} = \frac{\pi^2}{4} + \sum_{n=1}^{\infty} \frac{\pi J_1(n\pi)}{n} \cos nx.$$

311. 证明: $\int x J_{\nu}^2(x) dx = \frac{1}{2} x^2 [J_{\nu}^2(x) + J_{\nu+1}^2(x)] - \nu x J_{\nu}(x) J_{\nu+1}(x) + C$, 其中 $\operatorname{Re} \nu \geq 0$, C 为积分常数。如果把式中的 Bessel 函数换成其他的柱函数, 公式还成立吗?

Bessel 方程 $\frac{1}{x} \frac{d}{dx} [x J_{\nu}'(x)] + \left(1 - \frac{\nu^2}{x^2}\right) J_{\nu}(x) = 0$ 两边同乘 $x^2 J_{\nu}'(x)$ 得

$$x J_{\nu}'(x) \frac{d}{dx} [x J_{\nu}'(x)] + (x^2 - \nu^2) J_{\nu}(x) J_{\nu}'(x) = 0, \text{ 两边积分得}$$

$$\frac{1}{2} x^2 J_{\nu}'^2(x) + \int x^2 J_{\nu}(x) J_{\nu}'(x) dx - \frac{1}{2} \nu^2 J_{\nu}^2(x) + C = 0. \quad (*)$$

$$\begin{aligned} \text{其中 } \int x^2 J_{\nu}(x) J_{\nu}'(x) dx &= x^2 J_{\nu}^2(x) - \int J_{\nu}(x) \frac{d}{dx} [x^2 J_{\nu}(x)] dx \\ &= x^2 J_{\nu}^2(x) - 2 \int x J_{\nu}^2(x) dx - \int x^2 J_{\nu}(x) J_{\nu}'(x) dx, \end{aligned}$$

所以 $\int x^2 J_{\nu}(x) J_{\nu}'(x) dx = \frac{1}{2} x^2 J_{\nu}^2(x) - \int x J_{\nu}^2(x) dx$, 该式代入 (*) 式得

$$\int x J_{\nu}^2(x) dx = \frac{1}{2} x^2 J_{\nu}'^2(x) + \frac{1}{2} x^2 J_{\nu}^2(x) - \frac{1}{2} \nu^2 J_{\nu}^2(x) + C \quad (**)$$

将递推公式 $\frac{d}{dx} [x^{-\nu} J_{\nu}(x)] = -x^{-\nu} J_{\nu+1}(x)$ 展开为 $-\nu x^{-\nu-1} J_{\nu}(x) + x^{-\nu} J_{\nu}'(x) = -x^{-\nu} J_{\nu+1}(x)$,

两边同乘 $x^{\nu+1}$ 可得 $x J_{\nu}'(x) = \nu J_{\nu}(x) - x J_{\nu+1}(x)$, 该式代入 (**) 式即得

$$\int x J_{\nu}^2(x) dx = \frac{1}{2} x^2 [J_{\nu}^2(x) + J_{\nu+1}^2(x)] - \nu x J_{\nu}(x) J_{\nu+1}(x) + C.$$

上面计算只用到了 Bessel 方程和柱函数的递推公式, 所以对其他柱函数也适用。

312. 设 μ_i 是 $J_n(x)$ 的正零点, 试证: $\int_0^1 J_n(\mu_i x) J_n(\alpha x) x dx = -\frac{\mu_i J_n(\alpha) J_n'(\mu_i)}{\mu_i^2 - \alpha^2}$ 。然后,

令 $\alpha \rightarrow \mu_i$, 由此计算出积分 $\int_0^1 J_n^2(\mu_i x) x dx$ 。

$$\frac{d}{dx} \left[x \frac{dJ_n(\mu_i x)}{dx} \right] + \left(\mu_i^2 x - \frac{n^2}{x} \right) J_n(\mu_i x) = 0, \quad \frac{d}{dx} \left[x \frac{dJ_n(\alpha x)}{dx} \right] + \left(\alpha^2 x - \frac{n^2}{x} \right) J_n(\alpha x) = 0.$$

第一式两边乘 $J_n(\alpha x)$ 减去第二式两边乘 $J_n(\mu_i x)$, 积分得

$$(\mu_i^2 - \alpha^2) \int_0^1 J_n(\mu_i x) J_n(\alpha x) x dx = \int_0^1 \left\{ J_n(\mu_i x) \frac{d}{dx} \left[x \frac{dJ_n(\alpha x)}{dx} \right] - J_n(\alpha x) \frac{d}{dx} \left[x \frac{dJ_n(\mu_i x)}{dx} \right] \right\} dx$$

$$\begin{aligned}
&= x J_n(\mu_i x) \frac{dJ_n(\alpha x)}{dx} \Big|_0^1 - x J_n(\alpha x) \frac{dJ_n(\mu_i x)}{dx} \Big|_0^1 \\
&= \alpha x J_n(\mu_i x) J_n'(\alpha x) \Big|_0^1 - \mu_i x J_n(\alpha x) J_n'(\mu_i x) \Big|_0^1 \\
&= -\mu_i J_n(\alpha) J_n'(\mu_i).
\end{aligned}$$

所以 $\int_0^1 J_n(\mu_i x) J_n(\alpha x) x dx = -\frac{\mu_i J_n(\alpha) J_n'(\mu_i)}{\mu_i^2 - \alpha^2} \quad (\alpha \neq \mu_i).$

$$\lim_{\alpha \rightarrow \mu_i} \frac{\mu_i J_n(\alpha) J_n'(\mu_i)}{\alpha^2 - \mu_i^2} = \lim_{\alpha \rightarrow \mu_i} \frac{\mu_i J_n'(\alpha) J_n'(\mu_i)}{2\alpha} = \frac{1}{2} [J_n'(\mu_i)]^2,$$

即 $\int_0^1 J_n^2(\mu_i x) x dx = \frac{1}{2} J_n'^2(\mu_i).$ (*)

若令 $x = \frac{\rho}{a}$, 则有 $\int_0^a J_n^2\left(\frac{\mu_i}{a} \rho\right) \rho d\rho = \frac{a^2}{2} J_n'^2(\mu_i).$

313. 设 μ_i 是 $J_n'(x)$ 的正零点, 重复上题步骤, 计算积分 $\int_0^1 J_n^2(\mu_i x) x dx$.

$$\begin{aligned}
(\mu_i^2 - \alpha^2) \int_0^1 J_n(\mu_i x) J_n(\alpha x) x dx &= \alpha x J_n(\mu_i x) J_n'(\alpha x) \Big|_0^1 - \mu_i x J_n(\alpha x) J_n'(\mu_i x) \Big|_0^1 \\
&= \alpha J_n(\mu_i) J_n'(\alpha),
\end{aligned}$$

$$\int_0^1 J_n(\mu_i x) J_n(\alpha x) x dx = \frac{\alpha J_n'(\alpha) J_n(\mu_i)}{\mu_i^2 - \alpha^2},$$

$$\begin{aligned}
\int_0^1 J_n^2(\mu_i x) x dx &= \lim_{\alpha \rightarrow \mu_i} \frac{\alpha J_n'(\alpha) J_n(\mu_i)}{\mu_i^2 - \alpha^2} = -\lim_{\alpha \rightarrow \mu_i} \frac{J_n'(\alpha) J_n(\mu_i) + \alpha J_n''(\alpha) J_n(\mu_i)}{2\alpha} \\
&= -\frac{1}{2} J_n''(\mu_i) J_n(\mu_i),
\end{aligned}$$

取 Bessel 方程 $J_n''(x) + \frac{1}{x} J_n'(x) + \left(1 - \frac{n^2}{x^2}\right) J_n(x) = 0$ 中 $x = \mu_i$ 可得

$$J_n''(\mu_i) = -\left(1 - \frac{n^2}{\mu_i^2}\right) J_n(\mu_i), \text{ 所以 } \int_0^1 J_n^2(\mu_i x) x dx = \frac{1}{2} \left(1 - \frac{n^2}{\mu_i^2}\right) J_n^2(\mu_i).$$

若令 $x = \frac{\rho}{a}$, 则有 $\int_0^a J_n^2\left(\frac{\mu_i}{a} \rho\right) \rho d\rho = \frac{a^2}{2} \left(1 - \frac{n^2}{\mu_i^2}\right) J_n^2(\mu_i).$

314. 若 $\operatorname{Re} \nu > -1$, 证明: $\frac{1}{2} \int_0^x J_\nu(t) dt = \sum_{n=0}^{\infty} J_{\nu+2n+1}(x)$ 。

由递推公式 $2J'_\nu(x) = J_{\nu-1}(x) - J_{\nu+1}(x)$ 可得

$$\begin{aligned} 2 \sum_{n=0}^N J'_{\nu+2n+1}(x) &= 2J'_{\nu+1}(x) + 2J'_{\nu+3}(x) + \cdots + 2J'_{\nu+2N+1}(x) \\ &= J_\nu(x) - J_{\nu+2}(x) + J_{\nu+2}(x) - J_{\nu+4}(x) - \cdots + J_{\nu+2N}(x) - J_{\nu+2N+2}(x) \\ &= J_\nu(x) - J_{\nu+2N+2}(x)。 \end{aligned}$$

写出 $J_{\nu+2N+2}(x)$ 的级数表达式 $J_{\nu+2N+2}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(\nu+2N+3+n)} \left(\frac{x}{2}\right)^{2n+\nu+2N+2}$, 由

$\Gamma(z)$ 当 $z \rightarrow \infty$ 时的渐进公式可知 $\lim_{N \rightarrow \infty} J_{\nu+2N+2}(x) = 0$, 即 $\frac{1}{2} J'_\nu(x) = \sum_{n=0}^{\infty} J'_{\nu+2n+1}(x)$ 。

两边积分, 由于 $\operatorname{Re} \nu > -1$, 故有 $J_{\nu+2n+1}(0) = 0$, 所以 $\frac{1}{2} \int_0^x J_\nu(t) dt = \sum_{n=0}^{\infty} J_{\nu+2n+1}(x)$ 。

315. 计算下列积分: (1) $\int_0^x t^{-n} J_{n+1}(t) dt$; (2) $\int_0^a x^3 J_0(x) dx$; (3) $\int_0^\infty e^{-ax} J_0(\sqrt{bx}) dx$, $a > 0, b \geq 0$; (4) $\int_0^t J_0(\sqrt{x(t-x)}) dx$ 。

$$(1) \int_0^x t^{-n} J_{n+1}(t) dt = - \int_0^x \frac{d}{dt} [t^{-n} J_n(t)] dt = -x^{-n} J_n(x) + \frac{J_n(t)}{t^n} \Big|_{t=0}。$$

由 $J_n(x)$ 当 $x \rightarrow 0$ 时的渐进公式 $J_n(x) \sim \frac{1}{n!} \left(\frac{x}{2}\right)^n$ 可知 $\frac{J_n(t)}{t^n} \Big|_{t=0} = \frac{1}{2^n n!}$, 所以

$$\int_0^x t^{-n} J_{n+1}(t) dt = -x^{-n} J_n(x) + \frac{1}{2^n n!}。$$

$$\begin{aligned} (2) \int_0^a x^3 J_0(x) dx &= \int_0^a x^2 x J_0(x) dx = \int_0^a x^2 \frac{d}{dx} [x J_1(x)] dx = x^3 J_1(x) \Big|_0^a - 2 \int_0^a x^2 J_1(x) dx \\ &= a^3 J_1(a) - 2x^2 J_2(x) \Big|_0^a = a^3 J_1(a) - 2a^2 J_2(a)。 \end{aligned}$$

$$(3) \int_0^\infty e^{-ax} J_0(\sqrt{bx}) dx = \int_0^\infty e^{-ax} \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{\sqrt{bx}}{2}\right)^{2n} dx = \int_0^\infty \sum_{n=0}^{\infty} \frac{1}{(n!)^2} \left(-\frac{bx}{4}\right)^n e^{-ax} dx。$$

在 x 的任一有界区域 $0 \leq x \leq M$ 有 $\left| \frac{1}{(n!)^2} \left(-\frac{bx}{4} \right)^n e^{-ax} \right| \leq \frac{1}{(n!)^2} \left(\frac{bM}{4} \right)^n$,

级数 $\sum_{n=0}^{\infty} \frac{1}{(n!)^2} \left(\frac{bM}{4} \right)^n$ 显然收敛, 所以有限积分可与求和交换次序:

$$\int_0^M \sum_{n=0}^{\infty} \frac{1}{(n!)^2} \left(-\frac{bx}{4} \right)^n e^{-ax} dx = \sum_{n=0}^{\infty} \frac{1}{(n!)^2} \left(-\frac{b}{4} \right)^n \int_0^M x^n e^{-ax} dx, \quad (*)$$

因为 $\left| \frac{1}{(n!)^2} \left(-\frac{b}{4} \right)^n \int_0^M x^n e^{-ax} dx \right| \leq \frac{1}{(n!)^2} \left(\frac{b}{4} \right)^n \int_0^M x^n e^{-ax} dx \leq \frac{1}{(n!)^2} \left(\frac{b}{4} \right)^n \int_0^{\infty} x^n e^{-ax} dx$, 右边

积分是一个由 a 决定, 与 M 无关的 Γ 函数值 (有限), 所以 $\sum_{n=0}^{\infty} \frac{1}{(n!)^2} \left(-\frac{b}{4} \right)^n \int_0^M x^n e^{-ax} dx$ 对

于 M 是一致收敛的。(*) 式两边令 $M \rightarrow \infty$, 则求极限与求和可交换顺序, 即

$$\begin{aligned} \int_0^{\infty} \sum_{n=0}^{\infty} \frac{1}{(n!)^2} \left(-\frac{bx}{4} \right)^n e^{-ax} dx &= \sum_{n=0}^{\infty} \frac{1}{(n!)^2} \left(-\frac{b}{4} \right)^n \int_0^{\infty} x^n e^{-ax} dx \\ &= \frac{1}{a} \sum_{n=0}^{\infty} \frac{1}{(n!)^2} \left(-\frac{b}{4a} \right)^n \int_0^{\infty} t^n e^{-t} dt = \frac{1}{a} \sum_{n=0}^{\infty} \frac{1}{(n!)^2} \left(-\frac{b}{4a} \right)^n \Gamma(n+1) \\ &= \frac{1}{a} \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{b}{4a} \right)^n = \frac{1}{a} e^{-\frac{b}{4a}}. \end{aligned}$$

下面关于求和与积分交换顺序合法性讨论省略。

$$\begin{aligned} (4) \quad \int_0^t J_0(\sqrt{x(t-x)}) dx &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2 2^{2n}} \int_0^t x^n (t-x)^n dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2 2^{2n}} t^{2n+1} \int_0^1 u^n (1-u)^n du \\ &= 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{t}{2} \right)^{2n+1} B(n+1, n+1) = 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \left(\frac{t}{2} \right)^{2n+1} = 2 \sin \frac{t}{2}. \end{aligned}$$

$$316. \text{ 证明: } \int_0^t \left[\sqrt{x(t-x)} \right]^n J_n \left[\sqrt{x(t-x)} \right] dx = \frac{\sqrt{\pi}}{2^n} t^{n+\frac{1}{2}} J_{n+1/2} \left(\frac{t}{2} \right).$$

$$\begin{aligned} \int_0^t \left[\sqrt{x(t-x)} \right]^n J_n \left[\sqrt{x(t-x)} \right] dx &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(n+k+1) 2^{2k+n}} \int_0^t x^{k+n} (t-x)^{k+n} dx \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (n+k)! 2^{2k+n}} t^{2k+2n+1} B(k+n+1, k+n+1) = \sqrt{2} t^{n+\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k (n+k)!}{k! (2k+2n+1)!} \left(\frac{t}{2} \right)^{2k+n+\frac{1}{2}} \end{aligned}$$

$$\begin{aligned}
&= \sqrt{2} t^{n+\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k (n+k)!}{k! (2k+2n+1)(2k+2n)(2k+2n-1) \cdots 3 \times 2 \times 1} \left(\frac{t}{2}\right)^{2k+n+\frac{1}{2}} \\
&= \sqrt{2} t^{n+\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k (n+k)!}{k! 2^{2k+2n+1} (n+k)! \left(k+n+\frac{1}{2}\right) \left(k+n-\frac{1}{2}\right) \cdots \frac{3}{2} \times \frac{1}{2}} \left(\frac{t}{2}\right)^{2k+n+\frac{1}{2}} \\
&= \frac{\sqrt{\pi}}{2^n} t^{n+\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma\left(n+\frac{1}{2}+k+1\right)} \left(\frac{t}{4}\right)^{2k+n+\frac{1}{2}} = \frac{\sqrt{\pi}}{2^n} t^{n+\frac{1}{2}} J_{n+1/2} \left(\frac{t}{2}\right).
\end{aligned}$$

317. 证明: $\int_0^1 (1-x)^{c-1} x^{\frac{n}{2}} J_n(\alpha\sqrt{x}) dx = \left(\frac{2}{\alpha}\right)^c \Gamma(c) J_{n+c}(\alpha)$ 。

$$\begin{aligned}
\int_0^1 (1-x)^{c-1} x^{\frac{n}{2}} J_n(\alpha\sqrt{x}) dx &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (n+k)!} \left(\frac{\alpha}{2}\right)^{2k+n} \int_0^1 x^{k+n} (1-x)^{c-1} dx \\
&= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (n+k)!} \left(\frac{\alpha}{2}\right)^{2k+n} B(k+n+1, c) = \Gamma(c) \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(n+c+k+1)} \left(\frac{\alpha}{2}\right)^{2k+n} \\
&= \left(\frac{2}{\alpha}\right)^c \Gamma(c) J_{n+c}(\alpha).
\end{aligned}$$

318. 设 $\nu > -1$, $a > 0$, $b > 0$, 证明:

$$(1) \int_0^{\infty} e^{-ax} J_{\nu}(bx) x^{\nu+1} dx = \frac{2a(2b)^{\nu} \Gamma(\nu+3/2)}{\sqrt{\pi}(a^2+b^2)^{\nu+3/2}};$$

$$(2) \int_0^{\infty} e^{-a^2 x^2} J_{\nu}(bx) x^{\nu+1} dx = \frac{b^{\nu}}{(2a^2)^{\nu+1}} e^{-\frac{b^2}{4a^2}}.$$

$$\begin{aligned}
(1) \int_0^{\infty} e^{-ax} J_{\nu}(bx) x^{\nu+1} dx &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(\nu+n+1)} \left(\frac{b}{2}\right)^{2n+\nu} \int_0^{\infty} x^{2n+2\nu+1} e^{-ax} dx \\
&= \frac{1}{a^{2\nu+2}} \left(\frac{b}{2}\right)^{\nu} \sum_{n=0}^{\infty} \frac{(-1)^n}{n! 2^{2n} \Gamma(\nu+n+1)} \left(\frac{b}{a}\right)^{2n} \Gamma(2n+2\nu+2) \\
&= \frac{1}{a^{2\nu+2}} \left(\frac{b}{2}\right)^{\nu} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1+2\nu)(2n+2\nu)(2n-1+2\nu) \cdots (2\nu) \Gamma(2\nu)}{n! 2^{2n} \Gamma(\nu+n+1)} \left(\frac{b}{a}\right)^{2n}
\end{aligned}$$

$$\begin{aligned}
&= \frac{\Gamma(2\nu)}{a^{2\nu+2}} \left(\frac{b}{2}\right)^\nu \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1+2\nu)(2n-1+2\nu)\cdots(1+2\nu)(2n+2\nu)(2n-2+2\nu)\cdots(2\nu)}{n! 2^{2n} \Gamma(\nu+n+1)} \left(\frac{b}{a}\right)^{2n} \\
&= \frac{\Gamma(2\nu)}{a^{2\nu+2}} \left(\frac{b}{2}\right)^\nu \sum_{n=0}^{\infty} \frac{(-1)^n}{n! 2^{2n} \Gamma(\nu+n+1)} \frac{2^{n+1} \Gamma(n+\nu+3/2)}{\Gamma(\nu+1/2)} \frac{2^{n+1} \Gamma(n+\nu+1)}{\Gamma(\nu)} \left(\frac{b}{a}\right)^{2n} \\
&= \frac{2^2 \Gamma(2\nu)}{a^{2\nu+2} \Gamma(\nu) \Gamma(\nu+1/2)} \left(\frac{b}{2}\right)^\nu \sum_{n=0}^{\infty} \frac{(-1)^n \Gamma(n+\nu+3/2)}{n!} \left(\frac{b}{a}\right)^{2n} \\
&= \frac{2(2b)^\nu}{\sqrt{\pi} a^{2\nu+2}} \sum_{n=0}^{\infty} \frac{(-1)^n (n-1+\nu+3/2)(n-2+\nu+3/2)\cdots(\nu+3/2) \Gamma(\nu+3/2)}{n!} \left(\frac{b}{a}\right)^{2n} \\
&= \frac{2(2b)^\nu \Gamma(\nu+3/2)}{\sqrt{\pi} a^{2\nu+2}} \sum_{n=0}^{\infty} \frac{(-\nu-3/2)(-\nu-3/2-1)\cdots(-\nu-3/2-n+1)}{n!} \left(\frac{b}{a}\right)^{2n} \\
&= \frac{2(2b)^\nu \Gamma(\nu+3/2)}{\sqrt{\pi} a^{2\nu+2}} \sum_{n=0}^{\infty} \binom{-\nu-\frac{3}{2}}{n} \left(\frac{b}{a}\right)^{2n} = \frac{2a(2b)^\nu \Gamma(\nu+3/2)}{\sqrt{\pi} a^{2\nu+3}} \left(1 + \frac{b^2}{a^2}\right)^{-\nu-\frac{3}{2}} \\
&= \frac{2a(2b)^\nu \Gamma(\nu+3/2)}{\sqrt{\pi} (a^2 + b^2)^{\nu+3/2}}.
\end{aligned}$$

$$(2) \int_0^\infty e^{-a^2 x^2} J_\nu(bx) x^{\nu+1} dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(\nu+n+1)} \left(\frac{b}{2}\right)^{2n+\nu} \int_0^\infty x^{2n+2\nu+1} e^{-a^2 x^2} dx$$

$$\begin{aligned}
&= \frac{a^2 x^2 = t}{2a^{2\nu+2}} \left(\frac{b}{2}\right)^\nu \sum_{n=0}^{\infty} \frac{1}{n! \Gamma(\nu+n+1)} \left(-\frac{b^2}{4a^2}\right)^n \int_0^\infty t^{n+\nu} e^{-t} dt \\
&= \frac{b^\nu}{2^{\nu+1} a^{2\nu+2}} \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{b^2}{4a^2}\right)^n = \frac{b^\nu}{(2a^2)^{\nu+1}} e^{-\frac{b^2}{4a^2}}.
\end{aligned}$$

319. 证明: (1) $\int_0^\infty \frac{\sin p}{p} J_0(rp) dp = \begin{cases} \pi/2, 0 \leq r \leq 1 \\ \sin^{-1} \frac{1}{r}, r > 1 \end{cases};$

$$(2) \int_0^\infty \sin p J_0(rp) dp = \begin{cases} \frac{1}{\sqrt{1-r^2}}, 0 < r < 1 \\ \infty, r = 1 \\ 0, r > 1 \end{cases}; \quad (3) \int_0^\infty J_0(rp) J_1(ap) dp = \begin{cases} \frac{1}{a}, r < a \\ \frac{1}{2a}, r = a \\ 0, r > a \end{cases}.$$

$$\begin{aligned}
(1) \quad \int_0^\infty \frac{\sin p}{p} J_0(rp) dp &= \int_0^\infty \frac{\sin p}{p} \frac{1}{\pi} \int_0^\pi \cos(rp \sin \theta) d\theta dp = \int_0^\pi d\theta \frac{1}{\pi} \int_0^\infty \frac{\sin p}{p} \cos(rp \sin \theta) dp \\
&= \int_0^\pi d\theta \frac{1}{2\pi} \int_0^\infty \frac{\sin p}{p} (e^{irp \sin \theta} + e^{-irp \sin \theta}) dp = \int_0^\pi d\theta \frac{1}{2\pi} \int_{-\infty}^\infty \frac{\sin p}{p} e^{irp \sin \theta} dp.
\end{aligned}$$

上面的无穷积分正是 $\frac{\sin p}{p}$ 的 Fourier 反演, 易知 $\text{rect}(x) \xrightarrow{FT} \frac{2}{p} \sin \frac{p}{2}$, 其中

$$\text{rect}(x) = \begin{cases} 1, & |x| < \frac{1}{2} \\ 0, & \text{others} \end{cases}, \text{ 所以上面积分} = \frac{1}{2} \int_0^\pi \text{rect}\left(\frac{r}{2} \sin \theta\right) d\theta.$$

当 $0 < r < 1$ 时, $\left|\frac{r}{2} \sin \theta\right| < \frac{1}{2}$, $\text{rect}\left(\frac{r}{2} \sin \theta\right) = 1$, 所以原积分 $= \frac{\pi}{2}$,

当 $r > 1$ 时, $0 < \theta < \sin^{-1} \frac{1}{r}$, $\pi - \sin^{-1} \frac{1}{r} < \theta < \pi$ 时, $\text{rect}\left(\frac{r}{2} \sin \theta\right) = 1$, 其余 $\text{rect}\left(\frac{r}{2} \sin \theta\right) = 0$,

$$\text{所以原积分} = \frac{1}{2} \left(\int_0^{\sin^{-1} \frac{1}{r}} d\theta + \int_{\pi - \sin^{-1} \frac{1}{r}}^\pi d\theta \right) = \sin^{-1} \frac{1}{r}.$$

关于上面交换积分次序的合法性讨论书上有。

(2) 书上已求出 $\int_0^\infty J_0(rt) e^{-pt} dt = \frac{1}{\sqrt{p^2 + r^2}}$ ($\text{Re } p > 0$), 右边函数以 $\pm ir$ 为枝点, 以连

接两枝点的虚轴线段为割线, 规定割线右岸 $\arg(p - ir) = -\frac{\pi}{2}$, $\arg(p + ir) = \frac{\pi}{2}$, 则令

$$p \rightarrow i \text{ 得 } \int_0^\infty J_0(rt) e^{-it} dt = \begin{cases} -i \frac{1}{\sqrt{1-r^2}}, & 0 < r < 1 \\ \infty, & r = 1 \\ \frac{1}{\sqrt{r^2-1}}, & r > 1 \end{cases}, \text{ 取虚部即可。}$$

$$\begin{aligned}
(3) \quad \int_0^\infty J_0(rp) J_1(ap) dp &= \frac{1}{\pi} \int_0^\infty J_1(ap) dp \int_0^\pi \cos(rp \sin \theta) d\theta \\
&= \frac{1}{\pi} \int_0^\pi d\theta \int_0^\infty J_1(ap) \cos(rp \sin \theta) dp = -\frac{1}{a\pi} \int_0^\pi d\theta \int_0^\infty \cos(rp \sin \theta) dJ_0(ap).
\end{aligned}$$

其中 $\int_0^\infty \cos(rp \sin \theta) dJ_0(ap) = J_0(ap) \cos(rp \sin \theta) \Big|_{p=0}^{p \rightarrow \infty} + r \sin \theta \int_0^\infty \sin(rp \sin \theta) J_0(ap) dp$

$$= -1 + \int_0^\infty \sin x J_0\left(\frac{a}{r \sin \theta} x\right) dx,$$

$$\text{所以 } \int_0^\infty J_0(rp)J_1(ap)dp = \frac{1}{a\pi} \int_0^\pi \left[1 - \int_0^\infty \sin x J_0\left(\frac{a}{r \sin \theta} x\right) dx \right] d\theta.$$

$$\text{由上小题结论, 当 } r < a \text{ 时, } \frac{a}{r \sin \theta} > 1, \quad \int_0^\infty \sin x J_0\left(\frac{a}{r \sin \theta} x\right) dx = 0,$$

$$\text{所以 } \int_0^\infty J_0(rp)J_1(ap)dp = \frac{1}{a\pi} \int_0^\pi d\theta = \frac{1}{a}.$$

$$\text{当 } r > a \text{ 时, } \int_0^\infty \sin x J_0\left(\frac{a}{r \sin \theta} x\right) dx = \begin{cases} \frac{r \sin \theta}{\sqrt{r^2 \sin^2 \theta - a^2}}, \sin \theta > \frac{a}{r} \\ \infty, \sin \theta = \frac{a}{r} \\ 0, \sin \theta < \frac{a}{r} \end{cases}, \quad \text{记 } \theta_0 = \sin^{-1} \frac{a}{r},$$

$$\begin{aligned} \text{则 } \int_0^\infty J_0(rp)J_1(ap)dp &= \frac{1}{a\pi} \left[\int_0^{\theta_0} d\theta + \int_{\theta_0}^{\pi-\theta_0} \left(1 - \frac{r \sin \theta}{\sqrt{r^2 \sin^2 \theta - a^2}} \right) d\theta + \int_{\pi-\theta_0}^\pi d\theta \right] \\ &= \frac{2}{a\pi} \left[\theta_0 + \int_{\theta_0}^{\pi/2} \left(1 - \frac{r \sin \theta}{\sqrt{r^2 \sin^2 \theta - a^2}} \right) d\theta \right] = \frac{2}{a\pi} \left(\frac{\pi}{2} - \int_{\theta_0}^{\pi/2} \frac{r \sin \theta}{\sqrt{r^2 \sin^2 \theta - a^2}} d\theta \right). \end{aligned}$$

$$\text{其中 } - \int_{\theta_0}^{\pi/2} \frac{r \sin \theta}{\sqrt{r^2 \sin^2 \theta - a^2}} d\theta = \int_{\theta_0}^{\pi/2} \frac{r}{\sqrt{r^2 - a^2 - r^2 \cos^2 \theta}} d \cos \theta$$

$$\begin{aligned} &= \int_{x=\cos \theta}^0 \frac{r}{\sqrt{1-\frac{a^2}{r^2}} \sqrt{r^2 - a^2 - r^2 x^2}} dx \stackrel{\frac{x}{\sqrt{1-\frac{a^2}{r^2}}} = y}{=} \int_1^0 \frac{1}{\sqrt{1-y^2}} dy = \sin^{-1} y \Big|_1^0 = -\frac{\pi}{2}, \end{aligned}$$

$$\text{所以 } \int_0^\infty J_0(rp)J_1(ap)dp = 0.$$

$$\text{当 } r = a \text{ 时, } \int_0^\infty J_0(rp)J_1(ap)dp = -\frac{1}{a} \int_0^\infty J_0(ap)dJ_0(ap) = \frac{1}{2a} J_0^2(ap) \Big|_{p \rightarrow \infty}^{p=0} = \frac{1}{2a}.$$

320. 根据 Neumann 函数 $Y_\nu(z)$ 的定义 $Y_\nu(z) = \frac{\cos \nu \pi J_\nu(z) - J_{-\nu}(z)}{\sin \nu \pi}$, 证明:

$$(1) \quad Y_{-\nu}(z) = \sin \nu \pi J_\nu(z) + \cos \nu \pi Y_\nu(z), \quad Y_\nu(ze^{im\pi}) = e^{-im\nu\pi} Y_\nu(z) + 2i \sin m\nu\pi \cot \nu \pi J_\nu(z),$$

$$Y_{-\nu}(ze^{im\pi}) = e^{-im\nu\pi} Y_{-\nu}(z) + 2i \sin m\nu\pi \csc \nu \pi J_\nu(z);$$

$$(2) \quad Y_\nu(z) \text{ 的递推关系与 } J_\nu(z) \text{ 相同, 即 } \frac{d}{dz} [z^\nu Y_\nu(z)] = z^\nu Y_{\nu-1}(z),$$

$$\frac{d}{dz} \left[z^{-\nu} Y_{\nu}(z) \right] = -z^{-\nu} Y_{\nu+1}(z)。$$

$$\begin{aligned} (1) \quad \cos \nu \pi Y_{\nu}(z) &= \frac{\cos^2 \nu \pi J_{\nu}(z) - \cos \nu \pi J_{-\nu}(z)}{\sin \nu \pi} = \frac{J_{\nu}(z) - \sin^2 \nu \pi J_{\nu}(z) - \cos \nu \pi J_{-\nu}(z)}{\sin \nu \pi} \\ &= -\frac{\cos \nu \pi J_{-\nu}(z) - J_{\nu}(z)}{\sin \nu \pi} - \sin \nu \pi J_{\nu}(z) = Y_{-\nu}(z) - \sin \nu \pi J_{\nu}(z)。 \end{aligned}$$

由 $J_{\nu}(z)$ 级数表达式有 $J_{\nu}(ze^{im\pi}) = e^{im\nu\pi} J_{\nu}(z)$ ，所以

$$Y_{\nu}(ze^{im\pi}) = \cot \nu \pi e^{im\nu\pi} J_{\nu}(z) - \csc \nu \pi e^{-im\nu\pi} J_{-\nu}(z)，又有$$

$$e^{-im\nu\pi} Y_{\nu}(z) = \cot \nu \pi e^{-im\nu\pi} J_{\nu}(z) - \csc \nu \pi e^{-im\nu\pi} J_{-\nu}(z)，两式相减得$$

$$Y_{\nu}(ze^{im\pi}) - e^{-im\nu\pi} Y_{\nu}(z) = \cot \nu \pi (e^{im\nu\pi} - e^{-im\nu\pi}) J_{\nu}(z) = 2i \sin m\nu\pi \cot \nu \pi J_{\nu}(z)。$$

$$Y_{-\nu}(ze^{im\pi}) = -\cot \nu \pi e^{-im\nu\pi} J_{-\nu}(z) + \csc \nu \pi e^{im\nu\pi} J_{\nu}(z)，$$

$$e^{-im\nu\pi} Y_{-\nu}(z) = -\cot \nu \pi e^{-im\nu\pi} J_{-\nu}(z) + \csc \nu \pi e^{-im\nu\pi} J_{\nu}(z)，两式相减得$$

$$Y_{-\nu}(ze^{im\pi}) - e^{-im\nu\pi} Y_{-\nu}(z) = 2i \sin m\nu\pi \csc \nu \pi J_{\nu}(z)。$$

$$\begin{aligned} (2) \quad \frac{d}{dz} \left[z^{\nu} Y_{\nu}(z) \right] &= \cot \nu \pi \frac{d}{dz} \left[z^{\nu} J_{\nu}(z) \right] - \csc \nu \pi \frac{d}{dz} \left[z^{\nu} J_{-\nu}(z) \right] \\ &= \cot \nu \pi z^{\nu} J_{\nu-1}(z) + \csc \nu \pi z^{\nu} J_{-\nu+1}(z) \\ &= z^{\nu} \left[\cot(\nu-1) \pi J_{\nu-1}(z) - \csc(\nu-1) \pi J_{-\nu+1}(z) \right] = z^{\nu} Y_{\nu-1}(z)。 \end{aligned}$$

$$\begin{aligned} \frac{d}{dz} \left[z^{-\nu} Y_{\nu}(z) \right] &= \cot \nu \pi \frac{d}{dz} \left[z^{-\nu} J_{\nu}(z) \right] - \csc \nu \pi \frac{d}{dz} \left[z^{-\nu} J_{-\nu}(z) \right] \\ &= -\cot \nu \pi z^{-\nu} J_{\nu+1}(z) - \csc \nu \pi z^{-\nu} J_{-\nu+1}(z) \\ &= -z^{-\nu} \left[\cot(\nu+1) \pi J_{\nu+1}(z) - \csc(\nu+1) \pi J_{-\nu+1}(z) \right] = -z^{-\nu} Y_{\nu+1}(z)。 \end{aligned}$$

321. 设有一柱体半径为 a ，高为 h 。与外界绝热，初始温度为 $u_0 \left(1 - \frac{\rho^2}{a^2} \right)$ ，求此柱体内温度分布与变化。又当时间足够长时该柱体温度应达到稳定，试求此稳定温度。

$$\text{初始温度分布与 } \varphi, z \text{ 无关, 所以该问题与 } \varphi, z \text{ 无关, } \begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) = 0 \\ u|_{\rho=0} \text{ 有界, } \frac{\partial u}{\partial \rho} \Big|_{\rho=a} = 0 \\ u|_{t=0} = u_0 \left(1 - \frac{\rho^2}{a^2} \right) \end{cases} .$$

$$\text{分离变量得本征值问题 } \begin{cases} \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) + \lambda^2 P = 0 \\ P(0) \text{ 有界, } P'(a) = 0 \end{cases} \text{ 及 } T' + \lambda^2 \kappa T = 0 .$$

解本征值问题得 $\lambda_0 = 0$, $\lambda_i = \frac{\mu'_i}{a}$ (μ'_i 是 $J'_0(x)$, 即 $J_1(x)$ 的第 i 个正零点, $i = 1, 2, \dots$),

$$P_0(\rho) = A_0, \quad P_i(\rho) = J_0\left(\frac{\mu'_i}{a} \rho\right), \quad \text{以及 } T(t) = A_i \exp\left[-\kappa \left(\frac{\mu'_i}{a}\right)^2 t\right].$$

$$\text{所以 } u(\rho, t) = A_0 + \sum_{i=1}^{\infty} A_i J_0\left(\frac{\mu'_i}{a} \rho\right) \exp\left[-\kappa \left(\frac{\mu'_i}{a}\right)^2 t\right],$$

$$u|_{t=0} = A_0 + \sum_{i=1}^{\infty} A_i J_0\left(\frac{\mu'_i}{a} \rho\right) = u_0 \left(1 - \frac{\rho^2}{a^2}\right), \quad \text{由 313 题求出的归一化因子可定出}$$

$$A_0 = \frac{2u_0}{a^2} \int_0^a \left(1 - \frac{\rho^2}{a^2}\right) \rho d\rho = \frac{u_0}{2},$$

$$\begin{aligned} A_i &= \frac{2u_0}{a^2 J_0^2(\mu'_i)} \int_0^a \left(1 - \frac{\rho^2}{a^2}\right) \rho J_0\left(\frac{\mu'_i}{a} \rho\right) d\rho = \frac{2u_0}{a^2 J_0^2(\mu'_i)} \left[\int_0^a \rho J_0\left(\frac{\mu'_i}{a} \rho\right) d\rho - \frac{1}{a^2} \int_0^a \rho^3 J_0\left(\frac{\mu'_i}{a} \rho\right) d\rho \right] \\ &= \frac{2u_0}{a \mu'_i J_0^2(\mu'_i)} \int_0^a \frac{d}{d\rho} \left[\rho J_1\left(\frac{\mu'_i}{a} \rho\right) \right] d\rho - \frac{2u_0}{a^3 \mu'_i J_0^2(\mu'_i)} \int_0^a \rho^2 \frac{d}{d\rho} \left[\rho J_1\left(\frac{\mu'_i}{a} \rho\right) \right] d\rho \\ &= \frac{2u_0}{a \mu'_i J_0^2(\mu'_i)} \rho J_1\left(\frac{\mu'_i}{a} \rho\right) \Big|_0^a - \frac{2u_0}{a^3 \mu'_i J_0^2(\mu'_i)} \left\{ \rho^3 J_1\left(\frac{\mu'_i}{a} \rho\right) \Big|_0^a - 2 \int_0^a \rho^2 J_1\left(\frac{\mu'_i}{a} \rho\right) d\rho \right\} \\ &= \frac{4u_0}{a^2 \mu'^2 J_0^2(\mu'_i)} \int_0^a \frac{d}{d\rho} \left[\rho^2 J_2\left(\frac{\mu'_i}{a} \rho\right) \right] d\rho = \frac{4u_0 J_2(\mu'_i)}{\mu'^2 J_0^2(\mu'_i)}, \end{aligned}$$

递推公式 $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_{\nu}(x)$ 中令 $\nu = 1, x = \mu'_i$ 可得 $J_2(\mu'_i) = -J_0(\mu'_i)$, 所以

$$A_i = -\frac{4u_0}{\mu'^2 J_0(\mu'_i)}, \quad \text{即 } u(\rho, t) = \frac{u_0}{2} - 4u_0 \sum_{i=1}^{\infty} \frac{1}{\mu'^2 J_0(\mu'_i)} J_0\left(\frac{\mu'_i}{a} \rho\right) \exp\left[-\kappa \left(\frac{\mu'_i}{a}\right)^2 t\right],$$

$$t \rightarrow \infty \text{ 时 } u \rightarrow \frac{u_0}{2}。$$

322. 半径为 R 的圆形膜, 边缘固定, 初始形状是旋转抛物面 $u|_{t=0} = H\left(1 - \frac{\rho^2}{R^2}\right)$, 初速恒

$$\text{为 } 0, \text{ 求解膜的自由横振动: } \begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) = 0 \\ u|_{\rho=0} \text{ 有界}, u|_{\rho=R} = 0 \\ u|_{t=0} = H\left(1 - \frac{\rho^2}{R^2}\right), \frac{\partial u}{\partial t} \Big|_{t=0} = 0 \end{cases}。$$

$$u(\rho, t) = \sum_{n=1}^{\infty} J_0\left(\frac{\mu_n}{R} \rho\right) \left[A_n \sin \frac{a\mu_n}{R} t + B_n \cos \frac{a\mu_n}{R} t \right] \quad (\mu_n \text{ 是 } J_0(x) \text{ 的正零点}), \text{ 由初始条}$$

$$\text{件得 } A_n = 0, \quad B_n = \frac{2H}{R^2 J_1^2(\mu_n)} \int_0^R \left(1 - \frac{\rho^2}{R^2}\right) J_0\left(\frac{\mu_n}{R} \rho\right) \rho d\rho = \frac{4HJ_2(\mu_n)}{\mu_n^2 J_1^2(\mu_n)}, \text{ 由递推关系可}$$

得

$$J_2(\mu_n) = \frac{2}{\mu_n} J_1(\mu_n), \text{ 所以 } B_n = \frac{8H}{\mu_n^3 J_1(\mu_n)},$$

$$\text{即 } u(\rho, t) = 8H \sum_{n=1}^{\infty} \frac{1}{\mu_n^3 J_1(\mu_n)} J_0\left(\frac{\mu_n}{R} \rho\right) \cos \frac{a\mu_n}{R} t。$$

$$323. \text{ 求解下列定解问题: } \begin{cases} \frac{\partial u}{\partial t} - \kappa \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} \right] = 0 \\ u|_{\rho=0} \text{ 有界}, u|_{\rho=a} = 0, u|_{t=0} = u_0 \sin 2\varphi \end{cases}。$$

$$u(\rho, \varphi, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} J_m\left(\frac{\mu_n^{(m)}}{a} \rho\right) (A_{mn} \sin m\varphi + B_{mn} \cos m\varphi) \exp \left[- \left(\frac{\mu_n^{(m)}}{a} \right)^2 \kappa t \right] \quad (\mu_n^{(m)} \text{ 是}$$

$$J_m(x) \text{ 的正零点})。 \text{ 由于 } u|_{t=0} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} J_m\left(\frac{\mu_n^{(m)}}{a} \rho\right) (A_{mn} \sin m\varphi + B_{mn} \cos m\varphi) = u_0 \sin 2\varphi,$$

所以 $B_{mn} = 0$,

$$A_{2n} = \frac{2u_0}{a^2 J_2'^2(\mu_n^{(2)})} \int_0^a J_2\left(\frac{\mu_n^{(2)}}{a} \rho\right) \rho d\rho = \frac{2u_0}{a^2 J_2'^2(\mu_n^{(2)})} \int_0^a \rho^2 \rho^{-1} J_2\left(\frac{\mu_n^{(2)}}{a} \rho\right) d\rho$$

$$\begin{aligned}
&= -\frac{2u_0}{a\mu_n^{(2)}J_2'^2(\mu_n^{(2)})}\int_0^a \rho^2 \frac{d}{d\rho} \left[\rho^{-1} J_1 \left(\frac{\mu_n^{(2)}}{a} \rho \right) \right] d\rho \\
&= -\frac{2u_0}{\mu_n^{(2)}J_2'^2(\mu_n^{(2)})} J_1(\mu_n^{(2)}) + \frac{4u_0}{a\mu_n^{(2)}J_2'^2(\mu_n^{(2)})} \int_0^a J_1 \left(\frac{\mu_n^{(2)}}{a} \rho \right) d\rho \\
&= -\frac{2u_0}{\mu_n^{(2)}J_2'^2(\mu_n^{(2)})} J_1(\mu_n^{(2)}) + \frac{4u_0}{(\mu_n^{(2)})^2 J_2'^2(\mu_n^{(2)})} \left[1 - J_0(\mu_n^{(2)}) \right] \\
&= \frac{2u_0}{\mu_n^{(2)}J_2'^2(\mu_n^{(2)})} \left[\frac{2}{\mu_n^{(2)}} - J_1(\mu_n^{(2)}) - \frac{2}{\mu_n^{(2)}} J_0(\mu_n^{(2)}) \right].
\end{aligned}$$

递推公式 $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)$ 中令 $\nu=1$, $x=\mu_n^{(2)}$ 可得 $J_1(\mu_n^{(2)}) = \frac{\mu_n^{(2)}}{2} J_0(\mu_n^{(2)})$,

$\frac{d}{dx} [x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x)$ 展开成 $\nu J_\nu(x) + x J_\nu'(x) = x J_{\nu-1}(x)$, 令 $\nu=2$, $x=\mu_n^{(2)}$ 可得

$$J_2'(\mu_n^{(2)}) = J_1(\mu_n^{(2)}) = \frac{\mu_n^{(2)}}{2} J_0(\mu_n^{(2)}), \text{ 所以}$$

$$\begin{aligned}
A_{2n} &= \frac{2u_0}{\mu_n^{(2)} \left[\frac{\mu_n^{(2)}}{2} J_0(\mu_n^{(2)}) \right]^2} \left[\frac{2}{\mu_n^{(2)}} - \frac{\mu_n^{(2)}}{2} J_0(\mu_n^{(2)}) - \frac{2}{\mu_n^{(2)}} J_0(\mu_n^{(2)}) \right] \\
&= 4u_0 \frac{4 - \left[(\mu_n^{(2)})^2 + 4 \right] J_0(\mu_n^{(2)})}{(\mu_n^{(2)})^4 J_0^2(\mu_n^{(2)})},
\end{aligned}$$

其余 $A_{mn} = 0$ 。

$$\text{即 } u(\rho, \varphi, t) = 4u_0 \sum_{n=1}^{\infty} \frac{4 - \left[(\mu_n^{(2)})^2 + 4 \right] J_0(\mu_n^{(2)})}{(\mu_n^{(2)})^4 J_0^2(\mu_n^{(2)})} J_2 \left(\frac{\mu_n^{(2)}}{a} \rho \right) \sin 2\varphi \exp \left[- \left(\frac{\mu_n^{(2)}}{a} \right)^2 \kappa t \right].$$

324. 一长为 π , 半径为 1 的圆柱形导体, 柱体侧面和其上下底的温度均保持为 0, 初始时柱体内温度分布为 $f(\rho) \sin nz$, 求柱体内温度变化与分布。

$$\begin{cases} \frac{\partial u}{\partial t} - \kappa \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{\partial^2 u}{\partial z^2} \right] = 0 \\ u|_{\rho=0} \text{ 有界}, u|_{\rho=a} = 0, u|_{z=0} = 0, u|_{z=\pi} = 0, u|_{t=0} = f(\rho) \sin nz \end{cases}$$

$$\text{分离变量得本征值问题} \begin{cases} \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) + k^2 P = 0 \\ P(0) \text{有界}, P(a) = 0 \end{cases}, \begin{cases} \frac{d^2 Z}{dz^2} + m^2 Z = 0 \\ Z(0) = 0, Z(\pi) = 0 \end{cases} \quad \text{以及}$$

$$\frac{dT}{dt} + (k^2 + m^2) \kappa T = 0.$$

$$u(\rho, z, t) = \sum_{i=1}^{\infty} \sum_{m=1}^{\infty} A_{im} J_0(\mu_i \rho) \sin mz \exp[-(\mu_i^2 + m^2) \kappa t] \quad (\mu_i \text{ 是 } J_0(x) \text{ 的正零点}).$$

$$\text{由初始条件可得 } A_{in} = \frac{2}{J_1^2(\mu_i)} \int_0^1 f(\rho) J_0(\mu_i \rho) \rho d\rho, \text{ 其余 } A_{im} = 0, \text{ 即}$$

$$u(\rho, z, t) = \sum_{i=1}^{\infty} A_{in} J_0(\mu_i \rho) \sin nz \exp[-(\mu_i^2 + n^2) \kappa t].$$

325. 一空心圆柱，内半径为 a ，外半径为 b ，维持内外柱面温度为 0，又设柱体高为 h ，

$$\text{上下底绝热，初温为常数 } u_0, \text{ 求柱体内温度变化与分布: } \begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) = 0 \\ u|_{\rho=a} = 0, u|_{\rho=b} = 0, u|_{t=0} = u_0 \end{cases}.$$

$$\text{分离变量得本征值问题} \begin{cases} \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) + k^2 P = 0 \\ P(a) = 0, P(b) = 0 \end{cases} \quad \text{以及} \quad \frac{dT}{dt} + k^2 \kappa T = 0.$$

$$P(\rho) \text{ 的通解为 } P(\rho) = AJ_0(k\rho) + BY_0(k\rho), \text{ 由边界条件得 } \begin{vmatrix} J_0(ka) & Y_0(ka) \\ J_0(kb) & Y_0(kb) \end{vmatrix} = 0,$$

$$\text{用 } k_i \text{ 表示方程 } J_0(k_i a) Y_0(k_i b) - J_0(k_i b) Y_0(k_i a) = 0 \quad (\text{所以 } Y_0(k_i a) = \frac{J_0(k_i a)}{J_0(k_i b)} Y_0(k_i b),$$

$$Y_0(k_i b) = \frac{J_0(k_i b)}{J_0(k_i a)} Y_0(k_i a)) \text{ 的正根, 即为本征值, 本征函数为}$$

$$P_i(\rho) = Y_0(k_i a) J_0(k_i \rho) - J_0(k_i a) Y_0(k_i \rho), \text{ 可算出:}$$

$$\begin{aligned} \int_a^b P_i(\rho) \rho d\rho &= Y_0(k_i a) \int_a^b \rho J_0(k_i \rho) d\rho - J_0(k_i a) \int_a^b \rho Y_0(k_i \rho) d\rho \\ &= \frac{Y_0(k_i a)}{k_i} \rho J_1(k_i \rho) \Big|_{\rho=a}^{\rho=b} - \frac{J_0(k_i a)}{k_i} \rho Y_1(k_i \rho) \Big|_{\rho=a}^{\rho=b} \end{aligned}$$

$$\begin{aligned}
&= \frac{b}{k_i} \begin{vmatrix} J_0(k_i a) & J'_0(k_i b) \\ Y_0(k_i a) & Y'_0(k_i b) \end{vmatrix} - \frac{a}{k_i} \begin{vmatrix} J_0(k_i a) & J'_0(k_i a) \\ Y_0(k_i a) & Y'_0(k_i a) \end{vmatrix} \\
&= \frac{b}{k_i} \frac{J_0(k_i a)}{J_0(k_i b)} \begin{vmatrix} J_0(k_i b) & J'_0(k_i b) \\ Y_0(k_i b) & Y'_0(k_i b) \end{vmatrix} - \frac{a}{k_i} \begin{vmatrix} J_0(k_i a) & J'_0(k_i a) \\ Y_0(k_i a) & Y'_0(k_i a) \end{vmatrix} \\
&= \frac{b}{k_i} \frac{J_0(k_i a)}{J_0(k_i b)} \frac{2}{\pi k_i b} - \frac{a}{k_i} \frac{2}{\pi k_i a} = \frac{2}{\pi k_i^2} \left[\frac{J_0(k_i a)}{J_0(k_i b)} - 1 \right],
\end{aligned}$$

下面求 $\int_a^b P_i^2(\rho) \rho d\rho$:

令 $P(\rho) = Y_0(ka)J_0(k\rho) - J_0(ka)Y_0(k\rho)$ (显然 $P(a) = 0$), 则

$\frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) + k^2 P = 0$, 又有 $\frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dP_i}{d\rho} \right) + k_i^2 P_i = 0$, 第一式两边乘 $\rho P_i(\rho)$ 减去第

二式两边乘 $\rho P(\rho)$, 两边积分得

$$\begin{aligned}
\int_a^b P P_i \rho d\rho &= \frac{\int_a^b \left[P \frac{d}{d\rho} \left(\rho \frac{dP_i}{d\rho} \right) - P_i \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) \right] d\rho}{k^2 - k_i^2} = \frac{(\rho P P_i)'_a^b - (\rho P_i P)'_a^b}{k^2 - k_i^2} \\
&= \frac{b P(b) P_i'(b)}{k^2 - k_i^2} = \frac{b k_i P(b)}{k^2 - k_i^2} [Y_0(k_i a) J'_0(k_i b) - J_0(k_i a) Y'_0(k_i b)] \\
&= -\frac{b k_i P(b)}{k^2 - k_i^2} \frac{J_0(k_i a)}{J_0(k_i b)} \begin{vmatrix} J_0(k_i b) & J'_0(k_i b) \\ Y_0(k_i b) & Y'_0(k_i b) \end{vmatrix} = -\frac{b k_i P(b)}{k^2 - k_i^2} \frac{J_0(k_i a)}{J_0(k_i b)} \frac{2}{\pi k_i b} \\
&= -\frac{2}{\pi} \frac{J_0(k_i a)}{J_0(k_i b)} \frac{Y_0(ka) J_0(kb) - J_0(ka) Y_0(kb)}{k^2 - k_i^2}.
\end{aligned}$$

所以 $\int_a^b P_i^2(\rho) \rho d\rho = \lim_{k \rightarrow k_i} \int_a^b P P_i \rho d\rho = -\lim_{k \rightarrow k_i} \frac{2}{\pi} \frac{J_0(k_i a)}{J_0(k_i b)} \frac{Y_0(ka) J_0(kb) - J_0(ka) Y_0(kb)}{k^2 - k_i^2}$

$$\begin{aligned}
&= -\frac{2}{\pi} \frac{J_0(k_i a)}{J_0(k_i b)} \lim_{k \rightarrow k_i} \frac{a \begin{vmatrix} J_0(kb) & J'_0(ka) \\ Y_0(kb) & Y'_0(ka) \end{vmatrix} - b \begin{vmatrix} J_0(ka) & J'_0(kb) \\ Y_0(ka) & Y'_0(kb) \end{vmatrix}}{2k} \\
&= -\frac{2}{\pi} \frac{J_0(k_i a)}{J_0(k_i b)} \frac{a \frac{J_0(k_i b)}{J_0(k_i a)} \begin{vmatrix} J_0(k_i a) & J'_0(k_i a) \\ Y_0(k_i a) & Y'_0(k_i a) \end{vmatrix} - b \frac{J_0(k_i a)}{J_0(k_i b)} \begin{vmatrix} J_0(k_i b) & J'_0(k_i b) \\ Y_0(k_i b) & Y'_0(k_i b) \end{vmatrix}}{2k_i}
\end{aligned}$$

$$= \frac{2}{\pi^2 k_i^2} \left\{ \left[\frac{J_0(k_i a)}{J_0(k_i b)} \right]^2 - 1 \right\}.$$

$$u(\rho, t) = \sum_{i=1}^{\infty} A_i P_i(\rho) \exp(-k_i^2 \kappa t) = \sum_{i=1}^{\infty} A_i [Y_0(k_i a) J_0(k_i \rho) - J_0(k_i a) Y_0(k_i \rho)] \exp(-k_i^2 \kappa t).$$

由初始条件, $\sum_{i=1}^{\infty} A_i P_i(\rho) = u_0$, 所以

$$A_i = u_0 \frac{\int_a^b P_i(\rho) \rho d\rho}{\int_a^b P_i^2(\rho) \rho d\rho} = u_0 \frac{\frac{2}{\pi k_i^2} \left[\frac{J_0(k_i a)}{J_0(k_i b)} - 1 \right]}{\frac{2}{\pi^2 k_i^2} \left\{ \left[\frac{J_0(k_i a)}{J_0(k_i b)} \right]^2 - 1 \right\}} = \pi u_0 \frac{J_0(k_i b)}{J_0(k_i a) + J_0(k_i b)}.$$

$$\text{即 } u(\rho, t) = \pi u_0 \sum_{i=1}^{\infty} \frac{J_0(k_i b)}{J_0(k_i a) + J_0(k_i b)} [Y_0(k_i a) J_0(k_i \rho) - J_0(k_i a) Y_0(k_i \rho)] \exp(-k_i^2 \kappa t).$$

326. 半径为 R 的圆形膜, 边缘固定, 在单位质量上受周期力 (1) $f(\rho, t) = A \sin \omega t$, (2)

$f(\rho, t) = A \left(1 - \frac{\rho^2}{R^2} \right) \sin \omega t$ 的作用, 求解膜的强迫振动, 设初位移和初速度均为 0。

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) = f(\rho, t) \\ u|_{\rho=0} = \text{有界}, u|_{\rho=R} = 0, u|_{t=0} = 0, \frac{\partial u}{\partial t} \Big|_{t=0} = 0 \end{cases}.$$

(1) 将 $u(\rho, t)$ 和 $f(\rho, t)$ 按本征函数 $J_0\left(\frac{\mu_n}{R} \rho\right)$ (μ_n 是 $J_0(x)$ 的正零点) 展开:

$$u(\rho, t) = \sum_{n=1}^{\infty} T_n(t) J_0\left(\frac{\mu_n}{R} \rho\right), \quad f(\rho, t) = \sum_{n=1}^{\infty} f_n(t) J_0\left(\frac{\mu_n}{R} \rho\right),$$

$$f_n(t) = \frac{2A \sin \omega t}{R^2 J_1^2(\mu_n)} \int_0^R J_0\left(\frac{\mu_n}{R} \rho\right) \rho d\rho = \frac{2A}{\mu_n J_1(\mu_n)} \sin \omega t.$$

代入方程得

$$\sum_{n=1}^{\infty} T_n''(t) J_0\left(\frac{\mu_n}{R} \rho\right) + \sum_{n=1}^{\infty} \left(\frac{a \mu_n}{R} \right)^2 T_n(t) J_0\left(\frac{\mu_n}{R} \rho\right) = \sum_{n=1}^{\infty} \frac{2A}{\mu_n J_1(\mu_n)} \sin \omega t J_0\left(\frac{\mu_n}{R} \rho\right),$$

$$\text{所以} \begin{cases} T_n''(t) + \left(\frac{a\mu_n}{R}\right)^2 T_n(t) = \frac{2A}{\mu_n J_1(\mu_n)} \sin \omega t, \\ T_n(0) = 0, T_n'(0) = 0 \end{cases},$$

若不存在 m 使得 $\omega = \frac{a\mu_m}{R}$,

$$\text{可解得 } T_n(t) = \frac{2A}{\mu_n J_1(\mu_n) \left[\left(\frac{a\mu_n}{R}\right)^2 - \omega^2 \right]} \left(\sin \omega t - \frac{\omega R}{a\mu_n} \sin \frac{a\mu_n}{R} t \right).$$

若存在 m 使得 $\omega = \frac{a\mu_m}{R}$, 当 $n \neq m$ 时, $T_n(t)$ 解仍如上式, 而

$$T_m(t) = \frac{A}{\omega \mu_m J_1(\mu_m)} \left(\frac{1}{\omega} \sin \omega t - t \cos \omega t \right).$$

$$\text{第一种情况 } u(\rho, t) = 2A \sum_{n=1}^{\infty} \frac{J_0\left(\frac{\mu_n}{R} \rho\right)}{\mu_n J_1(\mu_n) \left[\left(\frac{a\mu_n}{R}\right)^2 - \omega^2 \right]} \left(\sin \omega t - \frac{\omega R}{a\mu_n} \sin \frac{a\mu_n}{R} t \right).$$

$$\begin{aligned} \text{第二种情况 } u(\rho, t) &= 2A \sum_{\substack{n=1 \\ n \neq m}}^{\infty} \frac{J_0\left(\frac{\mu_n}{R} \rho\right)}{\mu_n J_1(\mu_n) \left[\left(\frac{a\mu_n}{R}\right)^2 - \omega^2 \right]} \left(\sin \omega t - \frac{\omega R}{a\mu_n} \sin \frac{a\mu_n}{R} t \right) \\ &\quad + \frac{A}{\omega \mu_m J_1(\mu_m)} \left(\frac{1}{\omega} \sin \omega t - t \cos \omega t \right) J_0\left(\frac{\mu_m}{R} \rho\right). \end{aligned}$$

$$(2) \quad u(\rho, t) = \sum_{n=1}^{\infty} T_n(t) J_0\left(\frac{\mu_n}{R} \rho\right), \quad f(\rho, t) = \sum_{n=1}^{\infty} f_n(t) J_0\left(\frac{\mu_n}{R} \rho\right),$$

$$f_n(t) = \frac{2A \sin \omega t}{R^2 J_1^2(\mu_n)} \int_0^R \left(1 - \frac{\rho^2}{R^2} \right) J_0\left(\frac{\mu_n}{R} \rho\right) \rho d\rho = \frac{8A}{\mu_n^3 J_1(\mu_n)} \sin \omega t.$$

$$\text{代入方程得} \begin{cases} T_n''(t) + \left(\frac{a\mu_n}{R}\right)^2 T_n(t) = \frac{8A}{\mu_n^3 J_1(\mu_n)} \sin \omega t \\ T_n(0) = 0, T_n'(0) = 0 \end{cases}$$

$$\text{第一种情况, } T_n(t) = \frac{8A}{\left(\frac{a\mu_n}{R}\right)^2 - \omega^2} \frac{1}{\mu_n^4 J_1(\mu_n)} \left(\mu_n \sin \omega t - \frac{\omega R}{a} \sin \frac{a\mu_n}{R} t \right),$$

$$u(\rho, t) = 8A \sum_{n=1}^{\infty} \frac{1}{\left(\frac{a\mu_n}{R}\right)^2 - \omega^2} \frac{J_0\left(\frac{\mu_n}{R}\rho\right)}{\mu_n^4 J_1(\mu_n)} \left(\mu_n \sin \omega t - \frac{\omega R}{a} \sin \frac{a\mu_n}{R} t \right).$$

第二种情况, $T_m(t) = \frac{4A}{\omega^2 \mu_m^3 J_1(\mu_m)} (\sin \omega t - \omega t \cos \omega t),$

$$u(\rho, t) = 8A \sum_{\substack{n=1 \\ n \neq m}}^{\infty} \frac{1}{\left(\frac{a\mu_n}{R}\right)^2 - \omega^2} \frac{J_0\left(\frac{\mu_n}{R}\rho\right)}{\mu_n^4 J_1(\mu_n)} \left(\mu_n \sin \omega t - \frac{\omega R}{a} \sin \frac{a\mu_n}{R} t \right) \\ + \frac{4A}{\omega^2 \mu_m^3 J_1(\mu_m)} (\sin \omega t - \omega t \cos \omega t) J_0\left(\frac{\mu_m}{R}\rho\right).$$

327. 求长圆柱形铀块的临界半径 (见习题 11 第 205 题和习题 12 第 223 题)。

$$\begin{cases} \frac{\partial u}{\partial t} = D \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \alpha u \\ u|_{\rho=0} \text{ 有界}, u|_{\rho=a} = 0 \end{cases}.$$

分离变量可得 $u(\rho, t) = \sum_{n=1}^{\infty} A_n J_0\left(\frac{\mu_n}{a}\rho\right) \exp\left\{-\left[D\left(\frac{\mu_n}{a}\right)^2 - \alpha\right]t\right\}$ (μ_n 是 $J_0(x)$ 的正零点)。

可看出, 只要 $a < \sqrt{\frac{D}{\alpha}} \mu_1$, 上面级数第一项趋于无穷, $\sqrt{\frac{D}{\alpha}} \mu_1$ 即为临界厚度。

328. 一完全柔软的均匀线, 密度为 ρ , 上端 ($x=l$) 固定在匀速转动的轴上, 下端 ($x=0$)

自由, 此线相对于平衡位置作横振动, 横振动方程及定解条件为

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - g \frac{\partial}{\partial x} \left(x \frac{\partial u}{\partial x} \right) - \omega^2 u = 0 \\ u|_{x=0} \text{ 有界}, u|_{x=l} = 0 \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t} \Big|_{x=0} = \psi(x) \end{cases}, \text{ 求 } u(x, t).$$

令 $u(x, t) = X(x)T(t)$ 分离变量得
$$\begin{cases} xX'' + X' + k^2 X = 0 \\ T'' + (gk^2 - \omega^2)T = 0 \end{cases}.$$

令 306 题中 $\alpha = 0, \beta = 2k, \gamma = \frac{1}{2}, \nu = 0$ 可得 $X(x) = AJ_0(2k\sqrt{x}) + BY_0(2k\sqrt{x})$, 由边界

条件得到本征值 $k_n = \frac{\mu_n}{2\sqrt{l}}$ (μ_n 是 $J_0(x)$ 的正零点), 本征函数 $X_n(x) = J_0\left(\mu_n\sqrt{\frac{x}{l}}\right)$ 。

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) J_0\left(\mu_n\sqrt{\frac{x}{l}}\right), \text{ 其中 } \omega_n = \sqrt{\frac{g}{4l}} \mu_n^2 - \omega^2。$$

令 312 题 (*) 式 $n = 0$: $\int_0^1 J_0^2(\mu_i x) x dx = \frac{1}{2} J_1^2(\mu_i)$, 作代换 $x = \sqrt{\frac{t}{l}}$, 则有

$$\int_0^l J_0^2\left(\mu_i\sqrt{\frac{t}{l}}\right) dt = l J_1^2(\mu_i)。由初始条件可定出 $A_n = \frac{1}{l J_1^2(\mu_n)} \int_0^l \varphi(x) J_0\left(\mu_n\sqrt{\frac{x}{l}}\right) dx,$$$

$$B_n = \frac{1}{l \omega_n J_1^2(\mu_n)} \int_0^l \psi(x) J_0\left(\mu_n\sqrt{\frac{x}{l}}\right) dx。$$

329. 一完全柔软的均匀线, 上端 ($x = l$) 固定, 下端 ($x = 0$) 自由, 线的密度 $\rho = ax^m$

$$(m > -1), \text{ 则横振动方程及定解条件为 } \begin{cases} \frac{\partial^2 u}{\partial t^2} - \frac{gx}{m+1} \frac{\partial^2 u}{\partial x^2} - g \frac{\partial u}{\partial x} = 0 \\ u|_{x=0} \text{ 有界}, u|_{x=l} = 0 \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t}|_{x=0} = \psi(x) \end{cases}, \text{ 求 } u(x, t)。$$

$$\text{分离变量得 } \begin{cases} xX'' + (m+1)X' + k^2X = 0 \\ T'' + \frac{g}{m+1}k^2T = 0 \end{cases}, \text{ 令 306 题中 } \alpha = -\frac{m}{2}, \beta = 2k, \gamma = \frac{1}{2}, \nu = m \text{ 可得}$$

本征值 $k_n = \frac{\mu_n}{2\sqrt{l}}$, 本征函数 $X_n(x) = x^{-\frac{m}{2}} J_m\left(\mu_n\sqrt{\frac{x}{l}}\right)$ (μ_n 是 $J_m(x)$ 的正零点)。

$$u(x, t) = x^{-\frac{m}{2}} \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) J_m\left(\mu_n\sqrt{\frac{x}{l}}\right), \text{ 其中 } \omega_n = \frac{\mu_n}{2} \sqrt{\frac{g}{l(1+m)}}。$$

$$\text{由 312 题 (*) 式可得 } \int_0^1 J_m^2\left(\mu_n\sqrt{\frac{x}{l}}\right) dx = l J_m'^2(\mu_n),$$

$$\text{由递推关系 } \frac{d}{dx} [x^{-m} J_m(x)] = -x^{-m} J_{m+1}(x) \text{ 可得 } J_m'(\mu_n) = -J_{m+1}(\mu_n),$$

$$\text{所以 } \int_0^1 J_m^2 \left(\mu_n \sqrt{\frac{x}{l}} \right) dx = l J_{m+1}^2 (\mu_n)。$$

$$\text{由初始条件可得 } A_n = \frac{1}{l J_{m+1}^2 (\mu_n)} \int_0^l x^{\frac{m}{2}} \varphi(x) J_0 \left(\mu_n \sqrt{\frac{x}{l}} \right) dx，$$

$$B_n = \frac{1}{l \omega_n J_{m+1}^2 (\mu_n)} \int_0^l x^{\frac{m}{2}} \psi(x) J_0 \left(\mu_n \sqrt{\frac{x}{l}} \right) dx。$$

$$330. \text{ 证明 (1) } H_{-\nu}^{(1)}(z) = e^{i\nu\pi} H_{\nu}^{(1)}(z), \quad H_{-\nu}^{(2)}(z) = e^{-i\nu\pi} H_{\nu}^{(2)}(z);$$

$$(2) \quad H_{\nu}^{(1)}(ze^{im\pi}) = \frac{\sin(1-m)\nu\pi}{\sin\nu\pi} H_{\nu}^{(1)}(z) - e^{-i\nu\pi} \frac{\sin m\nu\pi}{\sin\nu\pi} H_{\nu}^{(2)}(z),$$

$$H_{\nu}^{(2)}(ze^{im\pi}) = \frac{\sin(1+m)\nu\pi}{\sin\nu\pi} H_{\nu}^{(2)}(z) + e^{i\nu\pi} \frac{\sin m\nu\pi}{\sin\nu\pi} H_{\nu}^{(1)}(z)。$$

(1) 利用 320 题结论。

$$H_{-\nu}^{(1)}(z) = J_{-\nu}(z) + iY_{-\nu}(z) = J_{-\nu}(z) + i\sin\nu\pi J_{\nu}(z) + i\cos\nu\pi Y_{\nu}(z),$$

$$e^{i\nu\pi} H_{\nu}^{(1)}(z) = e^{i\nu\pi} J_{\nu}(z) + ie^{i\nu\pi} Y_{\nu}(z)$$

$$= \cos\nu\pi J_{\nu}(z) + i\sin\nu\pi J_{\nu}(z) + i\cos\nu\pi Y_{\nu}(z) - \sin\nu\pi Y_{\nu}(z),$$

$$\text{两式相减得 } H_{-\nu}^{(1)}(z) - e^{i\nu\pi} H_{\nu}^{(1)}(z) = J_{-\nu}(z) - \cos\nu\pi J_{\nu}(z) + \sin\nu\pi Y_{\nu}(z) = 0。$$

$$\text{同样可得 } H_{-\nu}^{(2)}(z) = e^{-i\nu\pi} H_{\nu}^{(2)}(z)。$$

$$(2) \quad \frac{\sin(1-m)\nu\pi}{\sin\nu\pi} H_{\nu}^{(1)}(z) - e^{-i\nu\pi} \frac{\sin m\nu\pi}{\sin\nu\pi} H_{\nu}^{(2)}(z)$$

$$= (\cos m\nu\pi - \cot\nu\pi \sin m\nu\pi) [J_{\nu}(z) + iY_{\nu}(z)]$$

$$- (\cot\nu\pi \sin m\nu\pi - i\sin m\nu\pi) [J_{\nu}(z) - iY_{\nu}(z)]$$

$$= e^{im\nu\pi} J_{\nu}(z) + i [e^{-im\nu\pi} Y_{\nu}(z) + 2i \cot\nu\pi \sin m\nu\pi J_{\nu}(z)]$$

$$= J_{\nu}(ze^{im\pi}) + iY_{\nu}(ze^{im\pi}) = H_{\nu}^{(1)}(ze^{im\pi})。$$

$$\text{同样可得 } H_{\nu}^{(2)}(ze^{im\pi}) = \frac{\sin(1+m)\nu\pi}{\sin\nu\pi} H_{\nu}^{(2)}(z) + e^{i\nu\pi} \frac{\sin m\nu\pi}{\sin\nu\pi} H_{\nu}^{(1)}(z)。$$

331. 若 n 为一正整数, 证明: $J_{n+1/2}(x) = i^{-n} \sqrt{\frac{x}{2\pi}} \int_{-1}^1 e^{i\mu x} P_n(\mu) d\mu$, 并推出

$$i^n \sqrt{2\pi} \int_{-\infty}^{\infty} e^{-itx} J_{n+1/2}(x) x^{-\frac{1}{2}} dx = \begin{cases} 2\pi P_n(t), & |t| < 1 \\ 0, & |t| > 1 \end{cases}.$$

$$\text{证: } \int_{-1}^1 e^{i\mu x} P_n(\mu) d\mu = \int_{-1}^1 \sum_{k=0}^{\infty} \frac{(i\mu x)^k}{k!} P_n(\mu) d\mu = \sum_{k=0}^{\infty} \frac{i^k x^k}{k!} \int_{-1}^1 \mu^k P_n(\mu) d\mu,$$

由习题 15 第 275 题结果,

$$\begin{aligned} \text{上式} &= 2^{n+1} i^n x^n \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m} (n+m)!}{m! (2n+2m+1)!} = 2^{n+1} i^n x^n \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m} \Gamma\left(\frac{1}{2}\right)}{2^{2n+2m+1} m! \Gamma\left(n+m+\frac{3}{2}\right)} \\ &= \sqrt{\frac{2\pi}{x}} i^n \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma\left(n+\frac{1}{2}+m+1\right)} \left(\frac{x}{2}\right)^{2m+n+\frac{1}{2}} = \sqrt{\frac{2\pi}{x}} i^n J_{n+1/2}(x). \end{aligned}$$

$$\begin{aligned} i^n \sqrt{2\pi} \int_{-\infty}^{\infty} e^{-itx} J_{n+1/2}(x) x^{-\frac{1}{2}} dx &= \int_{-\infty}^{\infty} e^{-itx} dx \int_{-1}^1 e^{i\mu x} P_n(\mu) d\mu \\ &= \int_{-1}^1 P_n(\mu) d\mu \int_{-\infty}^{\infty} e^{i(\mu-t)x} dx = 2\pi \int_{-1}^1 P_n(\mu) \delta(\mu-t) d\mu, \end{aligned}$$

当 $|t| < 1$ 时, 上式 $= 2\pi P_n(t)$, 当 $|t| > 1$ 时上式 $= 0$ 。

332. 一导体球, 半径为 a , 初温为常温 u_0 , 球面温度为 0, 求球内温度变化和分布。

$$\begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = 0 \\ u|_{r=0} \text{ 有界}, u|_{r=a} = 0, u|_{t=0} = u_0 \end{cases}$$

$$\text{分离变量得本征值问题} \begin{cases} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + k^2 R = 0 \\ R(0) \text{ 有界}, R(a) = 0 \end{cases} \text{ 及 } T' + k^2 \kappa T = 0.$$

R 的方程是零阶球 Bessel 方程, 通解为 $R(r) = A j_0(kr) + B n_0(kr)$, 由边界条件可得本征

值 $k_n = \frac{\mu_n}{a}$ (μ_n 是 $j_0(x)$ 的正零点, $n=1, 2, \dots$), 本征函数 $R_n(r) = j_0\left(\frac{\mu_n}{a} r\right)$ 。

由于 $j_0(x) = \frac{\sin x}{x}$, 所以 $\mu_n = n\pi$, $k_n = \frac{n\pi}{a}$, $R_n(r) = \frac{a}{n\pi r} \sin \frac{n\pi}{a} r$ 。

$$\text{所以 } u(r, t) = \frac{a}{\pi r} \sum_{n=1}^{\infty} A_n \frac{1}{n} \sin \frac{n\pi}{a} r \exp \left[- \left(\frac{n\pi}{a} \right)^2 \kappa t \right],$$

$$\text{由初始条件, } \sum_{n=1}^{\infty} A_n \frac{1}{n} \sin \frac{n\pi}{a} r = \frac{\pi u_0}{a} r, \text{ 所以 } A_n = \frac{2n\pi u_0}{a^2} \int_0^a r \sin \frac{n\pi}{a} r dr = 2u_0 (-1)^{n+1},$$

$$\text{即 } u(r, t) = \frac{2u_0 a}{\pi r} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{n\pi}{a} r \exp \left[- \left(\frac{n\pi}{a} \right)^2 \kappa t \right].$$

333. 确定球形铀块的临界半径（见习题 11 第 205 题和习题 12 第 223 题）。

$$\begin{cases} \frac{\partial u}{\partial t} = D \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \alpha u \\ u|_{r=0} \text{ 有界}, u|_{r=a} = 0 \end{cases}$$

$$\text{分离变量可得 } u(r, t) = \sum_{n=1}^{\infty} A_n j_0 \left(\frac{n\pi}{a} r \right) \exp \left\{ - \left[D \left(\frac{n\pi}{a} \right)^2 - \alpha \right] t \right\}, \text{ 可得临界厚度为 } \pi \sqrt{\frac{D}{\alpha}}.$$

334. 定义: $K_\nu(z) = \frac{\pi}{2 \sin \nu \pi} [I_{-\nu}(z) - I_\nu(z)]$, 试证明:

$$K_\nu(z) = \begin{cases} \frac{\pi}{2} i e^{i\nu\pi/2} H_\nu^{(1)}(ze^{i\pi/2}), & -\pi < \arg z \leq \frac{\pi}{2} \\ -\frac{\pi}{2} i e^{-i\nu\pi/2} H_\nu^{(2)}(ze^{-i\pi/2}), & -\frac{\pi}{2} < \arg z < \pi \end{cases}.$$

$$-\pi < \arg z \leq \frac{\pi}{2} \text{ 时, } -\frac{\pi}{2} < \arg ze^{i\pi/2} \leq \pi,$$

$$\begin{aligned} e^{i\nu\pi/2} H_\nu^{(1)}(ze^{i\pi/2}) &= e^{i\nu\pi/2} [J_\nu(ze^{i\pi/2}) + iY_\nu(ze^{i\pi/2})] \\ &= e^{i\nu\pi/2} [J_\nu(ze^{i\pi/2}) + i \cot \nu\pi J_\nu(ze^{i\pi/2}) - i \csc \nu\pi J_{-\nu}(ze^{i\pi/2})] \\ &= e^{i\nu\pi/2} [(1 + i \cot \nu\pi) e^{i\nu\pi/2} I_\nu(z) - i \csc \nu\pi e^{-i\nu\pi/2} I_{-\nu}(z)] \\ &= (1 + i \cot \nu\pi) e^{i\nu\pi} I_\nu(z) - i \csc \nu\pi I_{-\nu}(z) \\ &= i \csc \nu\pi [I_\nu(z) - I_{-\nu}(z)] = -\frac{2}{\pi} i K_\nu(z). \end{aligned}$$

$$\text{同样可得 } -\frac{\pi}{2} < \arg z < \pi \text{ 时, } K_\nu(z) = -\frac{\pi}{2} i e^{-i\nu\pi/2} H_\nu^{(2)}(ze^{-i\pi/2}).$$

335. 证明: (1) $\int_0^\infty e^{-\frac{1}{2}ax} \sin bx I_0\left(\frac{1}{2}ax\right) dx = \frac{1}{\sqrt{2b}} \frac{1}{\sqrt{a^2+b^2}} \sqrt{b+\sqrt{a^2+b^2}},$

$\int_0^\infty e^{-\frac{1}{2}ax} \cos bx I_0\left(\frac{1}{2}ax\right) dx = \frac{1}{\sqrt{2b}} \frac{1}{\sqrt{a^2+b^2}} \frac{a}{\sqrt{b+\sqrt{a^2+b^2}}},$ 其中 $a>0, b>0$;

(2) $\int_0^\infty J_0(ax) K_0(bx) x dx = \frac{1}{\sqrt{a^2+b^2}}, \quad a>0, \operatorname{Re} b>0.$

(1) $I_0(x) = J_0(ix) = \frac{1}{2\pi} \int_{-\pi}^\pi e^{-x \cos \theta} d\theta$ (见附录),

$$\begin{aligned} \int_0^\infty e^{-\left(\frac{1}{2}a+ib\right)x} I_0\left(\frac{1}{2}ax\right) dx &= \int_0^\infty e^{-\left(\frac{1}{2}a+ib\right)x} \frac{1}{2\pi} \int_{-\pi}^\pi e^{-\frac{1}{2}ax \cos \theta} d\theta dx \\ &= \frac{1}{2\pi} \int_{-\pi}^\pi d\theta \int_0^\infty e^{-x\left[\frac{1}{2}a(1+\cos \theta)+ib\right]} dx = \frac{1}{a\pi} \int_{-\pi}^\pi \frac{d\theta}{1+\cos \theta + i\frac{2b}{a}} \\ &= \frac{2}{a\pi} \int_0^\pi \frac{d\theta}{\cos \theta + 1 + i\frac{2b}{a}} \stackrel{x=\tan \frac{\theta}{2}}{=} \frac{4}{a\pi} \int_0^\infty \frac{1}{\frac{1-x^2}{1+x^2} + 1 + i\frac{2b}{a}} \frac{dx}{1+x^2} \\ &= \frac{2}{b\pi i} \int_0^\infty \frac{dx}{x^2 + 1 - i\frac{a}{b}}, \end{aligned} \quad (*)$$

设 $-1+i\frac{a}{b}$ 的平方根为 $\xi+i\eta$, 即 $(\xi+i\eta)^2 = -1+i\frac{a}{b}$, 则 $\xi^2-\eta^2 = -1, 2\xi\eta = \frac{a}{b}$, 解得

$\xi_1+i\eta_1 = \frac{a}{\sqrt{2b}} \frac{1}{\sqrt{b+\sqrt{a^2+b^2}}} + i\frac{1}{\sqrt{2b}} \sqrt{b+\sqrt{a^2+b^2}},$ 另一根为 $-\xi_1-i\eta_1$,

$$\begin{aligned} (*) \text{ 式} &= \frac{2}{b\pi i} \pi i \operatorname{res} \left[\frac{1}{(z-\xi_1-i\eta_1)(z+\xi_1+i\eta_1)} \right]_{z=\xi_1+i\eta_1} = \frac{1}{b} \frac{1}{\xi_1+i\eta_1} = \frac{1}{b} \frac{1}{\xi_1^2+\eta_1^2} (\xi_1-i\eta_1) \\ &= \frac{1}{\sqrt{a^2+b^2}} \left(\frac{a}{\sqrt{2b}} \frac{1}{\sqrt{b+\sqrt{a^2+b^2}}} - i\frac{1}{\sqrt{2b}} \sqrt{b+\sqrt{a^2+b^2}} \right). \end{aligned}$$

取实部和虚部即可得欲证两式。

(2) 由 337 题 $K_0(x)$ 的积分表示,

$$\int_0^\infty J_0(ax) K_0(bx) x dx = \int_0^\infty J_0(ax) x dx \int_0^\infty e^{-bx \operatorname{ch} t} dt = \int_0^\infty dt \int_0^\infty e^{-bx \operatorname{ch} t} J_0(ax) x dx,$$

由 318 (1) 题, $\int_0^\infty e^{-bx \operatorname{ch} t} J_0(ax) x dx = \frac{b \operatorname{ch} t}{(b^2 \operatorname{ch}^2 t + a^2)^{3/2}},$

$$\begin{aligned}
\text{所以 } \int_0^\infty J_0(ax) K_0(bx) x dx &= b \int_0^\infty \frac{\operatorname{ch} t}{(b^2 \operatorname{ch}^2 t + a^2)^{3/2}} dt = b \int_0^\infty \frac{d \operatorname{sh} t}{(a^2 + b^2 + b^2 \operatorname{sh}^2 t)^{3/2}} \\
&= \frac{b}{(a^2 + b^2)^{3/2}} \int_0^\infty \frac{dx}{\left(1 + \frac{b^2}{a^2 + b^2} x^2\right)^{3/2}} = \frac{1}{a^2 + b^2} \int_0^\infty \frac{dy}{(1 + y^2)^{3/2}} \\
&\stackrel{y=\tan \theta}{=} \frac{1}{a^2 + b^2} \int_0^{\pi/2} \cos \theta d\theta = \frac{1}{a^2 + b^2}。
\end{aligned}$$

336. 高为 h ，半径为 a 的圆柱体，上下底保持温度为 0，而柱面温度为 $u_0 \sin \frac{2\pi}{h} z$ ，求柱

$$\text{体内的稳定温度分布。} \begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{\partial^2 u}{\partial z^2} = 0 \\ u|_{z=0} = 0, u|_{z=h} = 0 \\ u|_{\rho=0} = \text{有界}, u|_{\rho=a} = u_0 \sin \frac{2\pi}{h} z \end{cases}$$

$$\text{分离变量得本征值问题} \begin{cases} Z'' + k^2 Z = 0 \\ Z(0) = 0, Z(h) = 0 \end{cases} \text{ 及 } \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) - k^2 P = 0。$$

$$\text{解得本征值 } k_n = \frac{n\pi}{h}, \text{ 本征函数 } Z_n(z) = \sin \frac{n\pi}{h} z \quad (n=1, 2, \dots), \quad P_n(\rho) = I_0\left(\frac{n\pi}{h} \rho\right)。$$

$$u(\rho, z) = \sum_{n=1}^{\infty} A_n I_0\left(\frac{n\pi}{h} \rho\right) \sin \frac{n\pi}{h} z。 \quad u|_{\rho=a} = \sum_{n=1}^{\infty} A_n I_0\left(\frac{n\pi a}{h}\right) \sin \frac{n\pi}{h} z = u_0 \sin \frac{2\pi}{h} z,$$

$$\text{所以 } A_2 = \frac{u_0}{I_0\left(\frac{2\pi a}{h}\right)}, \text{ 其余 } A_n = 0, \text{ 即 } u(\rho, z) = \frac{u_0}{I_0\left(\frac{2\pi a}{h}\right)} I_0\left(\frac{2\pi}{h} \rho\right) \sin \frac{2\pi}{h} z。$$

337. 证明: $K_0(x) = \int_0^\infty e^{-x \operatorname{ch} t} dt$ ($x > 0$) 满足零阶虚宗量 Bessel 方程，由此证明当 x 很

大时， $K_0(x)$ 的渐近形式为 $\frac{A}{\sqrt{x}} e^{-x}$ ，定出常数 A 。

$$\begin{aligned}
x K_0''(x) + K_0'(x) - x K_0(x) &= x \int_0^\infty (\operatorname{ch}^2 t - 1) e^{-x \operatorname{ch} t} dt - \int_0^\infty \operatorname{ch} t e^{-x \operatorname{ch} t} dt \\
&= x \int_0^\infty \operatorname{sh}^2 t e^{-x \operatorname{ch} t} dt - \int_0^\infty e^{-x \operatorname{ch} t} d \operatorname{sh} t = x \int_0^\infty \operatorname{sh}^2 t e^{-x \operatorname{ch} t} dt - \operatorname{sh} t e^{-x \operatorname{ch} t} \Big|_{t=0}^{t \rightarrow \infty} - x \int_0^\infty \operatorname{sh}^2 t e^{-x \operatorname{ch} t} dt = 0
\end{aligned}$$

即 $K_0(x) = \int_0^\infty e^{-x \operatorname{ch} t} dt$ 满足零阶虚宗量 Bessel 方程。

作代换 $u = \operatorname{ch} t$ ，即 $t = \ln(u + \sqrt{u^2 - 1})$ ，

$$\text{则 } K_0(x) = \int_0^\infty e^{-x \operatorname{ch} t} dt = \int_1^\infty \frac{e^{-xu}}{\sqrt{u^2 - 1}} du \stackrel{x(u-1)=y}{=} \frac{e^{-x}}{\sqrt{x}} \int_0^\infty \frac{e^{-y}}{\sqrt{2y + \frac{y^2}{x}}} dy,$$

$$\text{令积分式中 } x \rightarrow \infty \text{ 有 } K_0(x) \rightarrow \frac{e^{-x}}{\sqrt{x}} \int_0^\infty \frac{e^{-y}}{\sqrt{2y}} dy \stackrel{y=t^2}{=} \sqrt{2} \frac{e^{-x}}{\sqrt{x}} \int_0^\infty e^{-t^2} dt = \sqrt{\frac{\pi}{2}} \frac{e^{-x}}{\sqrt{x}}.$$

338. 假定零阶虚宗量 Bessel 方程 $y'' + \frac{1}{x}y' - y = 0$ 的形式解为 $y \sim e^{\lambda x} x^{-\rho} \sum_{k=0}^\infty a_k x^{-k}$ ， $a_0 \neq 0$ ，

$$\text{试求出此方程的两个形式解 } y_1 \sim c_1 \frac{e^x}{\sqrt{x}} \left[1 + \frac{1^2}{1 \cdot (8x)} + \frac{1^2 \cdot 3^2}{2! (8x)^2} + \cdots \right],$$

$$y_2 \sim c_2 \frac{e^{-x}}{\sqrt{x}} \left[1 - \frac{1^2}{1 \cdot (8x)} + \frac{1^2 \cdot 3^2}{2! (8x)^2} - \cdots \right]. \text{ 如果取 } c_1 = \frac{1}{\sqrt{2\pi}}, \quad c_2 = \sqrt{\frac{\pi}{2}}, \text{ 这正好就是}$$

$I_0(x)$ 和 $K_0(x)$ 在 $x \rightarrow \infty$ 时的渐进展开。

将 $y = e^{\lambda x} x^{-\rho} \sum_{k=0}^\infty a_k x^{-k}$ 代入方程可得

$$(\lambda^2 - 1) \sum_{k=0}^\infty a_k x^{-k} - \lambda \sum_{k=1}^\infty (2k + 2\rho - 3) a_{k-1} x^{-k} + \sum_{k=2}^\infty (k + \rho - 2)^2 a_{k-2} x^{-k} = 0.$$

由常数项为零可得 $\lambda^2 - 1 = 0$ ，由 x^{-1} 项系数为零可得 $\rho = \frac{1}{2}$ ，

取 $\lambda = 1$ 可得递推关系

$$a_k = \frac{\left(k - \frac{1}{2}\right)^2}{2k} a_{k-1} = \frac{\left(k - \frac{1}{2}\right)^2}{2k} \frac{\left(k - 1 - \frac{1}{2}\right)^2}{2k - 2} \cdots \frac{\left(\frac{1}{2}\right)^2}{2} a_0 = \frac{(2k-1)^2 (2k-3)^2 \cdots 1}{2^{3k} k!} a_0,$$

$$\text{即 } y_1 \sim a_0 \frac{e^x}{\sqrt{x}} \left[1 + \sum_{k=1}^\infty \frac{(2k-1)^2 (2k-3)^2 \cdots 1}{2^{3k} k!} x^{-k} \right].$$

$$\text{取 } \lambda = -1 \text{ 可得 } a_k = -\frac{\left(k - \frac{1}{2}\right)^2}{2k} a_{k-1} = (-1)^k \frac{(2k-1)^2 (2k-3)^2 \cdots 1}{2^{3k} k!} a_0$$

$$\text{即 } y_1 \sim a_0 \frac{e^{-x}}{\sqrt{x}} \left[1 + \sum_{k=1}^{\infty} (-1)^k \frac{(2k-1)^2 (2k-3)^2 \cdots 1}{2^{3k} k!} x^{-k} \right]。$$

附录:

$$\text{令 } \exp \left[\frac{x}{2} \left(t - \frac{1}{t} \right) \right] = \sum_{n=-\infty}^{\infty} J_n(x) t^n \text{ 中 } t = ie^{i\theta} \text{ 得 } e^{ix \cos \theta} = \sum_{n=-\infty}^{\infty} i^n J_n(x) e^{in\theta}, \text{ 所以}$$

$$J_n(x) = \frac{1}{2\pi i^n} \int_{-\pi}^{\pi} e^{i(x \cos \theta - n\theta)} d\theta,$$

$$\text{将 } x \text{ 换成 } ix \text{ 得 } I_n(x) = \frac{(-1)^n}{2\pi} \int_{-\pi}^{\pi} e^{-x \cos \theta - in\theta} d\theta。$$

$$\text{书上已得 } J_n(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(x \sin \theta - n\theta)} d\theta, \text{ 所以 } I_n(x) = \frac{1}{2\pi i^n} \int_{-\pi}^{\pi} e^{-x \sin \theta - in\theta} d\theta。$$

339. 圆内 Laplace 方程第一边值问题的 Green 函数 $\begin{cases} \nabla^2 G(\mathbf{r}, \mathbf{r}') = -\frac{1}{\varepsilon_0} \delta(\mathbf{r} - \mathbf{r}') \\ G(\mathbf{r}, \mathbf{r}')|_{r=a} = 0 \end{cases}$ 是

$$G(\mathbf{r}, \mathbf{r}') = -\frac{1}{2\pi\varepsilon_0} \ln R + \frac{1}{2\pi\varepsilon_0} \ln R_1 - \frac{1}{2\pi\varepsilon_0} \ln \frac{a}{r'}. \text{ 其中 } R = |\mathbf{r} - \mathbf{r}'| = \sqrt{r^2 + r'^2 - 2rr' \cos \theta},$$

$R_1 = |\mathbf{r} - \mathbf{r}_1| = \sqrt{r^2 + r_1^2 - 2rr_1 \cos \theta}$, θ 是 $\mathbf{r}$ 与 $\mathbf{r}'$ 的夹角, $\mathbf{r}_1 = \left(\frac{a}{r'}\right)^2 \mathbf{r}'$, a 是圆半径。试证

明圆内定解问题 $\begin{cases} \nabla^2 u(\mathbf{r}) = 0 \\ u(\mathbf{r})|_{r=a} = f(\varphi) \end{cases}$ 的解可表为 $u(r, \varphi) = \frac{1}{2\pi} \int_0^{2\pi} \frac{(a^2 - r^2) f(\varphi') d\varphi'}{a^2 + r^2 - 2ar \cos(\varphi - \varphi')}$ 。

$$\frac{\partial G}{\partial r'} = -\frac{1}{2\pi\varepsilon_0} \frac{r' - r \cos \theta}{R^2} + \frac{1}{2\pi\varepsilon_0} \frac{-\frac{a^4}{r'^3} + \frac{ra^2}{r'^2} \cos \theta}{R_1^2} + \frac{1}{2\pi\varepsilon_0 r'},$$

$$\left. \frac{\partial G}{\partial r'} \right|_{r'=a} = -\frac{1}{\pi\varepsilon_0} \frac{a - r \cos \theta}{r^2 + a^2 - 2ar \cos \theta} + \frac{1}{2\pi\varepsilon_0 a} = \frac{1}{2\pi\varepsilon_0 a} \frac{r^2 - a^2}{r^2 + a^2 - 2ar \cos \theta}.$$

将 $\nabla^2 u(\mathbf{r}) = 0$ 两边乘 $G(\mathbf{r}, \mathbf{r}')$ 得 $G(\mathbf{r}, \mathbf{r}') \nabla^2 u(\mathbf{r}) = 0$, (a)

将 $\nabla^2 G(\mathbf{r}, \mathbf{r}') = -\frac{1}{\varepsilon_0} \delta(\mathbf{r} - \mathbf{r}')$ 两边乘 $u(\mathbf{r})$ 得 $u(\mathbf{r}) \nabla^2 G(\mathbf{r}, \mathbf{r}') = -\frac{1}{\varepsilon_0} u(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}')$ 。 (b)

(a) - (b), 两边在圆域内积分得 $\iint_S [G(\mathbf{r}, \mathbf{r}') \nabla^2 u(\mathbf{r}) - u(\mathbf{r}) \nabla^2 G(\mathbf{r}, \mathbf{r}')] dS = \frac{1}{\varepsilon_0} u(\mathbf{r}')$,

由 Green 公式得 $u(\mathbf{r}') = \varepsilon_0 \int_C \left[G(\mathbf{r}, \mathbf{r}') \frac{\partial u(\mathbf{r})}{\partial r} - u(\mathbf{r}) \frac{\partial G(\mathbf{r}, \mathbf{r}')}{\partial r} \right] dl$, 其中 C 是圆周 $r = a$ 。

$$\begin{aligned} \text{交换 } \mathbf{r}, \mathbf{r}' \text{ 有 } u(r, \varphi) &= -\varepsilon_0 \int_0^{2\pi} u(\mathbf{r}') \frac{\partial G(\mathbf{r}', \mathbf{r})}{\partial r'} \bigg|_{r'=a} a d\varphi' = \frac{1}{2\pi} \int_0^{2\pi} \frac{(a^2 - r^2) f(\varphi') d\varphi'}{r^2 + a^2 - 2ar \cos \theta} \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{(a^2 - r^2) f(\varphi') d\varphi'}{a^2 + r^2 - 2ar \cos(\varphi - \varphi')}. \end{aligned}$$

340. (1) 用电像法求出球内 Laplace 方程第一边值问题的 Green 函数

$$\begin{cases} \nabla^2 G(\mathbf{r}, \mathbf{r}') = -\frac{1}{\varepsilon_0} \delta(\mathbf{r} - \mathbf{r}') \\ G(\mathbf{r}, \mathbf{r}')|_{r=a} = 0 \end{cases};$$

(2) 求出边界面 (球面 $r = a$) 上各点的感应电荷密度 $\sigma(\theta, \varphi)$;

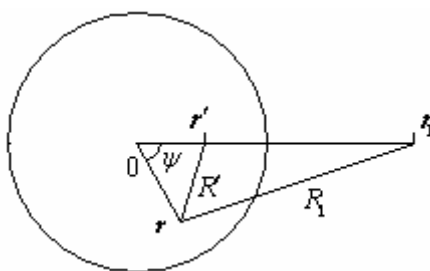
(3) 证明像电荷和感应电荷在球内完全等效;

(4) 证明球内 Laplace 方程第一边值问题 $\begin{cases} \nabla^2 u = 0 \\ u|_{r=a} = f(\theta, \varphi) \end{cases}$ 的解是

$$u(r, \theta, \varphi) = \frac{a}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{(a^2 - r^2) f(\theta', \varphi')}{(r^2 + a^2 - 2ar \cos \psi)^{3/2}} \sin \theta' d\theta' d\varphi', \text{ 其中 } \psi \text{ 是 } \mathbf{r}(r, \theta, \varphi) \text{ 与}$$

$\mathbf{r}'(r', \theta', \varphi')$ 的夹角, $\cos \psi = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\varphi - \varphi')$ 。

(1) 如下图, $\mathbf{r}'$ 处有电荷 $q' = 1$, 由对称性, 像电荷应放置在 $\mathbf{r}'$ 的延长线 $\mathbf{r}_1$ 处, 带电量 q_1 ,



$$\begin{aligned} \text{则 } \mathbf{r} \text{ 处电势为 } G(\mathbf{r}, \mathbf{r}') &= \frac{1}{4\pi\epsilon_0} \left(\frac{q'}{R'} + \frac{q_1}{R_1} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} + \frac{q_1}{|\mathbf{r} - \mathbf{r}_1|} \right) \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{r^2 + r'^2 - 2rr' \cos \psi}} + \frac{q_1}{\sqrt{r^2 + r_1^2 - 2rr_1 \cos \psi}} \right). \end{aligned}$$

$$\begin{aligned} G|_{r=a} &= \frac{1}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{a^2 + r'^2 - 2ar' \cos \psi}} + \frac{q_1}{\sqrt{a^2 + r_1^2 - 2ar_1 \cos \psi}} \right) \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{1}{a \sqrt{1 + \left(\frac{r'}{a}\right)^2 - 2\frac{r'}{a} \cos \psi}} + \frac{q_1}{r_1 \sqrt{1 + \left(\frac{a}{r_1}\right)^2 - 2\frac{a}{r_1} \cos \psi}} \right) \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{1}{a} \sum_{l=0}^{\infty} P_l(\cos \psi) \left(\frac{r'}{a}\right)^l + \frac{q_1}{r_1} \sum_{l=0}^{\infty} P_l(\cos \psi) \left(\frac{a}{r_1}\right)^l \right] = 0, \end{aligned}$$

所以有 $\frac{1}{a} \left(\frac{r'}{a}\right)^l + \frac{q_1}{r_1} \left(\frac{a}{r_1}\right)^l = 0$, 即 $q_1 = -\frac{r_1}{a} \left(\frac{r'r_1}{a^2}\right)^l$ 。可看出 $\left(\frac{r'r_1}{a^2}\right)^l$ 的值与 l 无关, 那么只有

$$\frac{r'r_1}{a^2} = 1, \text{ 即 } r_1 = \frac{a^2}{r'}, \quad q_1 = -\frac{a}{r'}, \text{ 所以}$$

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{r^2 + r'^2 - 2rr' \cos \psi}} - \frac{a}{r'} \frac{1}{\sqrt{r^2 + \frac{a^4}{r'^2} - 2\frac{a^2}{r'} r \cos \psi}} \right).$$

$$\text{其中 } \cos \psi = \frac{\mathbf{r} \cdot \mathbf{r}'}{rr'}$$

$$= \frac{(r \sin \theta \cos \varphi \mathbf{i} + r \sin \theta \sin \varphi \mathbf{j} + r \cos \theta \mathbf{k}) \cdot (r' \sin \theta' \cos \varphi' \mathbf{i} + r' \sin \theta' \sin \varphi' \mathbf{j} + r' \cos \theta' \mathbf{k})}{rr'}$$

$$= \sin \theta \cos \varphi \sin \theta' \cos \varphi' + \sin \theta \sin \varphi \sin \theta' \sin \varphi' + \cos \theta \cos \theta'$$

$$= \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\varphi - \varphi').$$

(2) 由电场边界条件, $\hat{\mathbf{r}} \cdot (\mathbf{D}|_{r=a^+} - \mathbf{D}|_{r=a^-}) = \sigma$ 。由于 $G(\mathbf{r}, \mathbf{r}')|_{r=a} = 0$, 可将边界球面看

作接地的导体球, 即 $\mathbf{D}|_{r=a^+} = 0$, 又由于 $\hat{\mathbf{r}} \cdot \mathbf{D} = D_r = \epsilon_0 E_r = -\epsilon_0 \frac{\partial G}{\partial r}$, 所以 $\sigma = \epsilon_0 \frac{\partial G}{\partial r} \Big|_{r=a^-}$

$$= \frac{1}{4\pi} \left[\frac{-a + r' \cos \psi}{(a^2 + r'^2 - 2ar' \cos \psi)^{3/2}} + \frac{a(ar'^2 - a^2 r' \cos \psi)}{(a^2 r'^2 + a^4 - 2a^3 r' \cos \psi)^{3/2}} \right] = -\frac{1}{4\pi a^2} \frac{1 - \left(\frac{r'}{a}\right)^2}{\left[1 + \left(\frac{r'}{a}\right)^2 - 2\frac{r'}{a} \cos \psi\right]^{3/2}}.$$

取 $\mathbf{r}'$ 方向为 z 方向, 即 $\theta' = 0$, 则 $\cos \psi = \cos \theta$, 所以

$$\sigma(\theta, \varphi) = -\frac{1}{4\pi a^2} \frac{1 - \left(\frac{r'}{a}\right)^2}{\left[1 - 2\frac{r'}{a} \cos \theta + \left(\frac{r'}{a}\right)^2\right]^{3/2}} = -\frac{1}{4\pi a^2} \frac{1 - \left(\frac{r'}{a}\right)^2}{\frac{r'}{a}} \sum_{l=0}^{\infty} P_l(\cos \theta) \left(\frac{r'}{a}\right)^l.$$

(3) 感应电荷在球内产生的电场为 (γ 为球面上点 (a, θ', φ') 与球内点 (r, θ, φ) 的夹角,

$\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\varphi - \varphi')$.):

$$\begin{aligned} G_\sigma(r, \theta, \varphi) &= \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi \frac{\sigma(\theta', \varphi')}{\sqrt{a^2 + r^2 - 2ar \cos \gamma}} a^2 \sin \theta' d\theta' d\varphi' \\ &= \frac{a}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi \frac{\sigma(\theta', \varphi')}{\sqrt{1 - 2\frac{r}{a} \cos \gamma + \left(\frac{r}{a}\right)^2}} \sin \theta' d\theta' d\varphi' \\ &= \frac{a}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi \sigma(\theta', \varphi') \sum_{l=0}^{\infty} P_l(\cos \gamma) \left(\frac{r}{a}\right)^l \sin \theta' d\theta' d\varphi' \end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{4\pi\epsilon_0} \frac{1}{4\pi a} \frac{1 - \left(\frac{r'}{a}\right)^2}{\frac{r'}{a}} \int_0^{2\pi} \int_0^\pi \sum_{l=0}^{\infty} P'_l(\cos \theta') \left(\frac{r'}{a}\right)^l \cdot \sum_{l=0}^{\infty} P_l(\cos \gamma) \left(\frac{r}{a}\right)^l \sin \theta' d\theta' d\varphi' \\
&= -\frac{1}{4\pi\epsilon_0} \frac{1}{4\pi a} \frac{1 - \left(\frac{r'}{a}\right)^2}{\frac{r'}{a}} \sum_{l=0}^{\infty} \left(\frac{r}{a}\right)^l \sum_{k=0}^{\infty} \left(\frac{r'}{a}\right)^k \int_0^{2\pi} \int_0^\pi P'_k(\cos \theta') P_l(\cos \gamma) \sin \theta' d\theta' d\varphi'.
\end{aligned}$$

有 Legendre 多项式的加法公式:

$$P_l(\cos \gamma) = P_l(\cos \theta) P_l(\cos \theta') + 2 \sum_{m=1}^l \frac{(l-m)!}{(l+m)!} P_l^m(\cos \theta) P_l^m(\cos \theta') e^{im(\varphi-\varphi')}.$$

(见王竹溪, 郭敦仁《特殊函数概论》5.14 节) 所以,

$$\begin{aligned}
&\int_0^{2\pi} \int_0^\pi P'_k(\cos \theta') P_l(\cos \gamma) \sin \theta' d\theta' d\varphi' = 2\pi P_l(\cos \theta) \int_0^\pi P'_k(\cos \theta') P_l(\cos \theta') \sin \theta' d\theta' \\
&\quad + 2 \sum_{m=1}^l \frac{(l-m)!}{(l+m)!} P_l^m(\cos \theta) \int_0^\pi P'_k(\cos \theta') P_l^m(\cos \theta') \sin \theta' d\theta' \int_0^{2\pi} e^{im(\varphi-\varphi')} d\varphi' \\
&= 2\pi P_l(\cos \theta) \int_{-1}^1 P'_k(x) P_l(x) dx.
\end{aligned}$$

$$\text{由习题 15 第 283 题结论, } \int_{-1}^1 P'_k(x) P_l(x) dx = \begin{cases} 2, k = l + 2n + 1 \\ 0, \text{others} \end{cases} \quad (n = 0, 1, \dots),$$

即 $k = l + 2n + 1$ 时, $\int_0^{2\pi} \int_0^\pi P'_k(\cos \theta') P_l(\cos \gamma) \sin \theta' d\theta' d\varphi' = 4\pi P_l(\cos \theta)$, 其他情况该积分都为 0, 所以

$$\begin{aligned}
G_\sigma &= -\frac{1}{4\pi\epsilon_0} \frac{1 - \left(\frac{r'}{a}\right)^2}{\frac{r'}{a}} \frac{1}{a} \sum_{l=0}^{\infty} P_l(\cos \theta) \left(\frac{rr'}{a^2}\right)^l \sum_{n=0}^{\infty} \left(\frac{r'}{a}\right)^{2n+1} = -\frac{1}{4\pi\epsilon_0} \frac{1}{a \sqrt{1 + \frac{r^2 r'^2}{a^4} - 2 \frac{rr'}{a^2} \cos \theta}} \\
&= -\frac{1}{4\pi\epsilon_0} \frac{a}{r'} \frac{1}{\sqrt{r^2 + \frac{a^4}{r'^2} - 2r \frac{a^2}{r'} \cos \theta}} = -\frac{1}{4\pi\epsilon_0} \frac{a}{r'} \frac{1}{|\mathbf{r} - \mathbf{r}_1|}
\end{aligned}$$

即与像电荷产生电势相等。

$$(4) \quad \frac{\partial G(\mathbf{r}', \mathbf{r})}{\partial r'} = -\frac{1}{4\pi\epsilon_0} \left[\frac{r' - r \cos \psi}{(r^2 + r'^2 - 2rr' \cos \psi)^{3/2}} - \frac{a \left(\frac{r}{r'}\right)^2 - \left(\frac{a}{r'}\right)^3 r \cos \psi}{\left(r^2 + \frac{a^4}{r'^2} - 2 \frac{a^2}{r'} r \cos \psi\right)^{3/2}} \right],$$

$$\left. \frac{\partial G(\mathbf{r}', \mathbf{r})}{\partial r'} \right|_{r'=a} = \frac{1}{4\pi\epsilon_0 a} \frac{r^2 - a^2}{(a^2 + r^2 - 2ar \cos \psi)^{3/2}}.$$

由 $\nabla^2 u(\mathbf{r}) = 0$ 和 $\nabla^2 G(\mathbf{r}, \mathbf{r}') = -\frac{1}{\epsilon_0} \delta(\mathbf{r} - \mathbf{r}')$ 可得

$$\begin{aligned} u(\mathbf{r}) &= \epsilon_0 \iiint_V [G(\mathbf{r}', \mathbf{r}) \nabla^2 u(\mathbf{r}') - u(\mathbf{r}') \nabla^2 G(\mathbf{r}', \mathbf{r})] dV' \\ &= \epsilon_0 \oint_S \left[G(\mathbf{r}', \mathbf{r}) \frac{\partial u(\mathbf{r}')}{\partial n'} - u(\mathbf{r}') \frac{\partial G(\mathbf{r}', \mathbf{r})}{\partial n'} \right] dS \\ &= -\epsilon_0 \int_0^{2\pi} \int_0^\pi f(\theta', \varphi') \left. \frac{\partial G(\mathbf{r}', \mathbf{r})}{\partial r'} \right|_{r'=a} a^2 \sin \theta' d\theta' d\varphi' \\ &= \frac{a}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{(a^2 - r^2) f(\theta', \varphi')}{(r^2 + a^2 - 2ar \cos \psi)^{3/2}} \sin \theta' d\theta' d\varphi'. \end{aligned}$$

341. 证明互易定理: 若 Φ 是由 V 中体电荷密度 ρ 及面 Σ (V 的边界面) 上面点荷密度 σ 产

生的静电势, Φ' 是由体电荷密度 ρ' 及面点荷密度 σ' 产生的静电势, 则

$$\iiint_V \rho \Phi' dV + \oint_\Sigma \sigma \Phi' dS = \iiint_V \rho' \Phi dV + \oint_\Sigma \sigma' \Phi dS.$$

在 V 内有 $\nabla^2 \Phi = -\frac{1}{\epsilon_0} \rho$, $\nabla^2 \Phi' = -\frac{1}{\epsilon_0} \rho'$. 第一式两边乘 Φ' 减去第二式两边乘 Φ 得

$$\begin{aligned} \frac{1}{\epsilon_0} \iiint_V \rho' \Phi dV - \frac{1}{\epsilon_0} \iiint_V \rho \Phi' dV &= \iiint_V (\Phi' \nabla^2 \Phi - \Phi \nabla^2 \Phi') dV \\ &= \oint_\Sigma \left(\Phi' \frac{\partial \Phi}{\partial n} - \Phi \frac{\partial \Phi'}{\partial n} \right) dS = -\oint_\Sigma (\Phi' E_n - \Phi E'_n) dS, \end{aligned}$$

上面 E_n 表示电场法向分量, 由电场边界条件 $E_n = -\frac{\sigma}{\epsilon_0}$, $E'_n = -\frac{\sigma'}{\epsilon_0}$ (V 外电场为 0) 得

$$\frac{1}{\epsilon_0} \iiint_V \rho' \Phi dV - \frac{1}{\epsilon_0} \iiint_V \rho \Phi' dV = \frac{1}{\epsilon_0} \oint_\Sigma (\sigma \Phi' - \sigma' \Phi) dS, \text{ 即}$$

$$\iiint_V \rho \Phi' dV + \oint_\Sigma \sigma \Phi' dS = \iiint_V \rho' \Phi dV + \oint_\Sigma \sigma' \Phi dS.$$

342. 用 Fourier 变换法求三维无界空间 Helmholtz 方程的 Green 函数

$$(\nabla^2 + k^2)G(\mathbf{r}, \mathbf{r}') = -\frac{1}{\varepsilon_0} \delta(\mathbf{r} - \mathbf{r}').$$

记 $\mathbf{k}' = k'_x \mathbf{i} + k'_y \mathbf{j} + k'_z \mathbf{k}$, $k'^2 = k_x'^2 + k_y'^2 + k_z'^2$, 三维 Fourier 变换和反变换为:

$$\begin{aligned}\tilde{G}(\mathbf{k}', \mathbf{r}') &= \iiint G(\mathbf{r}, \mathbf{r}') \exp(-i\mathbf{k}' \cdot \mathbf{r}) d\mathbf{r} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} G(\mathbf{r}, \mathbf{r}') \exp[-i(k'_x x + k'_y y + k'_z z)] dx dy dz; \\ G(\mathbf{r}, \mathbf{r}') &= \frac{1}{(2\pi)^3} \iiint \tilde{G}(\mathbf{k}', \mathbf{r}') \exp(i\mathbf{k}' \cdot \mathbf{r}) d\mathbf{k}' \\ &= \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \tilde{G}(\mathbf{k}', \mathbf{r}') \exp[i(k'_x x + k'_y y + k'_z z)] dk'_x dk'_y dk'_z.\end{aligned}$$

方程 $(\nabla^2 + k^2)G(\mathbf{r}, \mathbf{r}') = -\frac{1}{\varepsilon_0} \delta(\mathbf{r} - \mathbf{r}')$ 两边对 $\mathbf{r}$ 作三维 Fourier 变换得

$$(-k'^2 + k^2) \tilde{G}(\mathbf{k}', \mathbf{r}') = -\frac{1}{\varepsilon_0} \exp(-i\mathbf{k}' \cdot \mathbf{r}'), \text{ 所以 } \tilde{G}(\mathbf{k}', \mathbf{r}') = \frac{1}{\varepsilon_0} \frac{\exp(-i\mathbf{k}' \cdot \mathbf{r}')}{k'^2 - k^2},$$

$$\text{取 Fourier 反变换得 } G(\mathbf{r}, \mathbf{r}') = \frac{1}{(2\pi)^3 \varepsilon_0} \iiint \frac{\exp[i\mathbf{k}' \cdot (\mathbf{r} - \mathbf{r}')] }{k'^2 - k^2} d\mathbf{k}'$$

可看出 $G(\mathbf{r}, \mathbf{r}')$ 只是 $\mathbf{r} - \mathbf{r}'$ 的函数, 可令 $\mathbf{R} = \mathbf{r} - \mathbf{r}'$, 在 $\mathbf{k}'$ 空间中以 $\mathbf{R}$ 方向为 z 方向建立球

坐标系 (k', θ, φ) 计算这个积分:

$$\begin{aligned}G(\mathbf{r}, \mathbf{r}') &= \frac{1}{(2\pi)^3 \varepsilon_0} \int_0^{2\pi} \int_0^\pi \int_0^\infty \frac{\exp(ik'R \cos \theta)}{k'^2 - k^2} k'^2 \sin \theta dk' d\theta d\varphi \\ &= \frac{1}{(2\pi)^2 \varepsilon_0} \int_0^\infty \frac{k'^2}{k'^2 - k^2} dk' \int_0^\pi e^{ik'R \cos \theta} \sin \theta d\theta \\ &= \frac{1}{(2\pi)^2 iR \varepsilon_0} \int_0^\infty \frac{k'}{k'^2 - k^2} (e^{ik'R} - e^{-ik'R}) dk' = \frac{1}{(2\pi)^2 iR \varepsilon_0} \text{v.p.} \int_{-\infty}^\infty \frac{k' e^{ik'R}}{k'^2 - k^2} dk' \\ &= \frac{1}{(2\pi)^2 iR \varepsilon_0} \lim_{\eta \rightarrow 0^+} \text{v.p.} \int_{-\infty}^\infty \frac{k'}{k'^2 - (k + i\eta)^2} e^{ik'R} dk' .\end{aligned}$$

用留数定理可算出

$$\text{v.p.} \int_{-\infty}^{\infty} \frac{k'}{k'^2 - (k + i\eta)^2} e^{ik'R} dk' = 2\pi i \operatorname{res} \left[\frac{ze^{izR}}{z^2 - (k + i\eta)^2} \right]_{z=k+i\eta} = \pi i e^{-R\eta} e^{ikR},$$

$$\text{所以 } G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi\epsilon_0} \frac{e^{ikR}}{R}.$$

343. 一无穷长弦, $t=t'$ 时其 $x=x'$ 处受到瞬时的打击, 冲量为 I 。试求解弦的横振动, 设初位移和初速度都为 0。

$$\begin{cases} \frac{\partial^2 G(x, t; x', t')}{\partial t^2} - a^2 \frac{\partial^2 G(x, t; x', t')}{\partial x^2} = \frac{I}{\rho} \delta(x - x') \delta(t - t') \\ G(x, t; x', t')|_{x \rightarrow \pm\infty} \text{ 有界}, G(x, t; x', t')|_{t=0} = 0, \frac{\partial G(x, t; x', t')}{\partial t} \Big|_{t=0} = 0 \end{cases},$$

方程两边对变量 t 作 Laplace 变换得

$$\begin{cases} \frac{\partial^2 \tilde{G}(x, p; x', t')}{\partial x^2} - \left(\frac{p}{a}\right)^2 \tilde{G}(x, p; x', t') = -\frac{I}{\rho a^2} e^{-pt'} \delta(x - x') \\ G(x, p; x', t')|_{x \rightarrow \pm\infty} \text{ 有界} \end{cases}, \text{ 可限定 } \operatorname{Re} p > 0.$$

方程两边对 x 在区间 $[x' - \varepsilon, x' + \varepsilon]$ 上积分, 并令 $\varepsilon \rightarrow 0^+$ 得

$$\frac{\partial \tilde{G}}{\partial x} \Big|_{x=x'+0} - \frac{\partial \tilde{G}}{\partial x} \Big|_{x=x'-0} = -\frac{I}{\rho a^2} e^{-pt'}, \text{ 另外还有连续条件 } \tilde{G} \Big|_{x=x'+0} = \tilde{G} \Big|_{x=x'-0} \text{ (若 } \tilde{G} \text{ 在 } x=x' \text{ 点}$$

不连续, 方程右边必出现 $\delta'(x - x')$ 项)。

在 $x \neq x'$ 点, 方程为齐次方程, 因此, 上面的问题即为带连接边界条件的定解问题:

$$\begin{cases} \frac{\partial^2 \tilde{G}_1}{\partial x^2} - \left(\frac{p}{a}\right)^2 \tilde{G}_1 = 0, x < x' \\ \tilde{G}_1|_{x \rightarrow -\infty} = 0 \end{cases}, \begin{cases} \frac{\partial^2 \tilde{G}_2}{\partial x^2} - \left(\frac{p}{a}\right)^2 \tilde{G}_2 = 0, x > x' \\ \frac{\partial \tilde{G}_2}{\partial x} \Big|_{x \rightarrow \infty} = 0 \end{cases}, \begin{cases} \frac{\partial \tilde{G}_2}{\partial x} \Big|_{x=x'^+} - \frac{\partial \tilde{G}_1}{\partial x} \Big|_{x=x'^-} = -\frac{I}{\rho a^2} e^{-pt'} \\ \tilde{G}_2 \Big|_{x=x'^+} = \tilde{G}_1 \Big|_{x=x'^-} \end{cases}.$$

$$\text{可解出 } \tilde{G}(x, p; x', t') = \begin{cases} A e^{\frac{p}{a}x}, x < x' \\ B e^{-\frac{p}{a}x}, x > x' \end{cases}. \text{ 由连接边界条件定出 } A = \frac{I}{2\rho a} \frac{e^{-p\left(t' + \frac{x'}{a}\right)}}{p},$$

$$B_2 = \frac{I}{2\rho a} \frac{e^{-p\left(t' - \frac{x'}{a}\right)}}{p}, \text{ 即 } \tilde{G}(x, p; x', t') = \begin{cases} \frac{I}{2\rho a} \frac{e^{-p\left(t' - \frac{x-x'}{a}\right)}}{p}, x < x' \\ \frac{I}{2\rho a} \frac{e^{-p\left(t' + \frac{x-x'}{a}\right)}}{p}, x > x' \end{cases}. \text{ 取反变换得}$$

$$G(x, t; x', t') = \begin{cases} \frac{I}{2\rho a} \eta\left(t - t' + \frac{x - x'}{a}\right), & x < x' \\ \frac{I}{2\rho a} \eta\left(t - t' - \frac{x - x'}{a}\right), & x > x' \end{cases} = \frac{I}{2\rho a} \eta\left(t - t' - \frac{|x - x'|}{a}\right).$$

344. 两端固定的弦, 长为 l , $t = t'$ 时用细锤敲击弦上 $x = x'$ 点, 使得该点获得冲量 I 。求解弦的横振动, 设初位移和初速度都为 0。

$$\begin{cases} \frac{\partial^2 G(x, t; x', t')}{\partial t^2} - a^2 \frac{\partial^2 G(x, t; x', t')}{\partial x^2} = \frac{I}{\rho} \delta(x - x') \delta(t - t') \\ G(x, t; x', t')|_{x=0} = 0, G(x, t; x', t')|_{x=l} = 0 \\ G(x, t; x', t')|_{t < t'} = 0, \frac{\partial G(x, t; x', t')}{\partial t} \Big|_{t < t'} = 0 \end{cases}.$$

可将 $G(x, t; x', t')$ 用本征函数 $\sin \frac{n\pi}{l} x$ 展开: $G(x, t; x', t') = \sum_{n=1}^{\infty} g_n(t) \sin \frac{n\pi}{l} x$,

将 $\delta(x - x')$ 也按 $\sin \frac{n\pi}{l} x$ 展开: $\delta(x - x') = \sum_{n=1}^{\infty} c_n \sin \frac{n\pi}{l} x$, 可求出

$$c_n = \frac{2}{l} \int_0^l \delta(x - x') \sin \frac{n\pi}{l} x dx = \frac{2}{l} \sin \frac{n\pi}{l} x', \text{ 即 } \delta(x - x') = \frac{2}{l} \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x' \sin \frac{n\pi}{l} x.$$

代入方程得:

$$\sum_{n=1}^{\infty} g_n''(t) \sin \frac{n\pi}{l} x + \left(\frac{n\pi a}{l}\right)^2 \sum_{n=1}^{\infty} g_n(t) \sin \frac{n\pi}{l} x = \frac{2I}{\rho l} \delta(t - t') \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x' \sin \frac{n\pi}{l} x,$$

$$\text{再由初始条件可得} \begin{cases} g_n''(t) + \left(\frac{n\pi a}{l}\right)^2 g_n(t) = \frac{2I}{\rho l} \sin \frac{n\pi}{l} x' \delta(t - t') \\ g_n|_{t < t'} = 0, g_n'|_{t < t'} = 0 \end{cases}.$$

对方程两边在 $t = t'$ 附近积分可得 $g_n'(t'^+) = g_n'(t'^-) + \frac{2I}{\rho l} \sin \frac{n\pi}{l} x' = \frac{2I}{\rho l} \sin \frac{n\pi}{l} x'$, 再由

$g_n(t)$ 在 $t = t'$ 的连续性可得 $g_n(t'^+) = g_n(t'^-) = 0$, 上面问题即为

$$\begin{cases} g_n''(t) + \left(\frac{n\pi a}{l}\right)^2 g_n(t) = 0, t > t' \\ g_n(t') = 0, g_n'(t') = \frac{2I}{\rho l} \sin \frac{n\pi}{l} x' \end{cases}, \text{ 解得 } g_n(t) = \frac{2I}{n\pi a \rho} \sin \frac{n\pi}{l} x' \sin \frac{n\pi a}{l} (t - t').$$

$$\text{即 } G(x, t; x', t') = \frac{2I}{\pi a \rho} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi}{l} x' \sin \frac{n\pi}{l} x \sin \frac{n\pi a}{l} (t-t') \eta(t-t').$$

345. 求解点热源在无穷长细杆上产生的温度分布与变化:

$$\begin{cases} \frac{\partial G(x, t; x', t')}{\partial t} - \kappa \frac{\partial^2 G(x, t; x', t')}{\partial x^2} = \delta(x-x') \delta(t-t') \\ G(x, t; x', t')|_{t=0} = 0 \end{cases}.$$

对变量 t 作 Laplace 变换得 $\frac{\partial^2 \tilde{G}(x, p; x', t')}{\partial x^2} - \frac{p}{\kappa} \tilde{G}(x, p; x', t') = -\frac{e^{-pt'}}{\kappa} \delta(x-x')$, 可限定

$$-\pi/2 < \operatorname{Re} p < \pi/2. \text{ 由自然条件 } \tilde{G}(x, t; x', t')|_{x \rightarrow \pm\infty} \text{ 有界解得 } \tilde{G} = \begin{cases} A e^{\sqrt{\frac{p}{\kappa}} x}, & x < x' \\ B e^{-\sqrt{\frac{p}{\kappa}} x}, & x > x' \end{cases}.$$

由连接条件 $\tilde{G}|_{x=x'^-} = \tilde{G}|_{x=x'^+}$, $\frac{\partial \tilde{G}}{\partial x}|_{x=x'^+} - \frac{\partial \tilde{G}}{\partial x}|_{x=x'^-} = -\frac{e^{-pt'}}{\kappa}$ 可得 $\tilde{G} = \frac{1}{2\sqrt{\kappa p}} e^{-pt'} e^{-\sqrt{\frac{p}{\kappa}}|x-x'|}$.

$$\text{反演得 } G(x, t; x', t') = \frac{1}{2\sqrt{\kappa\pi(t-t')}} \exp\left[-\frac{(x-x')^2}{4\kappa(t-t')}\right] \eta(t-t').$$

346. 试证明一维热传导方程 Green 函数 $G(x, t; x', t')$ 的互易性: $G(x, t; x', t') = G(x', -t'; x, t)$,

$$\text{其中 } G(x, t; x', t') \text{ 满足 } \begin{cases} \frac{\partial G(x, t; x', t')}{\partial t} - \kappa \frac{\partial^2 G(x, t; x', t')}{\partial x^2} = \delta(x-x') \delta(t-t') \\ G|_{x=0} = 0, G|_{x=l} = 0, G|_{t=0} = 0 \end{cases}.$$

$$\text{又可写成 } \begin{cases} \frac{\partial G(x, t; x', t')}{\partial t} - \kappa \frac{\partial^2 G(x, t; x', t')}{\partial x^2} = \delta(x-x') \delta(t-t') \\ G|_{x=0} = 0, G|_{x=l} = 0, G|_{t < t'} = 0 \end{cases}.$$

$$\text{对于 } G(x, -t; x'', -t'') \text{ 有 } \begin{cases} -\frac{\partial G(x, -t; x'', -t'')}{\partial t} - \kappa \frac{\partial^2 G(x, -t; x'', -t'')}{\partial x^2} = \delta(x-x'') \delta(t-t'') \\ G|_{x=0} = 0, G|_{x=l} = 0, G(x, -t; x'', -t'')|_{t > t''} = 0 \end{cases},$$

将 $G(x, t; x', t')$ 的方程两边乘 $G(x, -t; x'', -t'')$ 减去 $G(x, -t; x'', -t'')$ 的方程两边乘 $G(x, t; x', t')$

$$\text{可得: } G(x', -t'; x'', -t'') \delta(x-x') \delta(t-t') - G(x'', t''; x', t') \delta(x-x'') \delta(t-t'')$$

$$\begin{aligned}
&= G(x, -t; x'', -t'') \frac{\partial G(x, t; x', t')}{\partial t} + G(x, t; x', t') \frac{\partial G(x, -t; x'', -t'')}{\partial t} \\
&\quad + \kappa G(x, t; x', t') \frac{\partial^2 G(x, -t; x'', t'')}{\partial x^2} - \kappa G(x, -t; x'', -t'') \frac{\partial^2 G(x, t; x', t')}{\partial x^2} \\
&= \frac{\partial}{\partial t} \left[G(x, -t; x'', -t'') G(x, t; x', t') \right] \\
&\quad + \kappa \frac{\partial}{\partial x} \left[G(x, t; x', t') \frac{\partial G(x, -t; x'', t'')}{\partial x} - G(x, -t; x'', -t'') \frac{\partial G(x, t; x', t')}{\partial x} \right].
\end{aligned}$$

两边对 x 从 0 积到 l 得:

$$\begin{aligned}
&G(x', -t'; x'', -t'') \delta(t - t') - G(x'', t''; x', t') \delta(t - t'') \\
&= \int_0^l \frac{\partial}{\partial t} \left[G(x, -t; x'', -t'') G(x, t; x', t') \right] dx \\
&\quad + \kappa \left[G(x, t; x', t') \frac{\partial G(x, -t; x'', t'')}{\partial x} - G(x, -t; x'', -t'') \frac{\partial G(x, t; x', t')}{\partial x} \right]_{x=0}^{x=l} \\
&= \frac{\partial}{\partial t} \int_0^l \left[G(x, -t; x'', -t'') G(x, t; x', t') \right] dx,
\end{aligned}$$

再对 t 从 0 积到 ∞ , 由于 $G(x, -t; x'', -t'')|_{t > t''} = 0$, 所以

$$G(x', -t'; x'', -t'') - G(x'', t''; x', t') = \int_0^l \left[G(x, -t; x'', -t'') G(x, t; x', t') \right]_{t=0}^{t \rightarrow \infty} dx = 0,$$

即 $G(x', -t'; x'', -t'') = G(x'', t''; x', t')$, 也就是 $G(x, t; x', t') = G(x', -t'; x, t)$ 。

347. 用 Green 函数法解无界弦的横振动问题:
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t}|_{t=0} = \psi(x) \end{cases}.$$

令 343 题中 $\frac{I}{\rho} = 1$, 即
$$\frac{\partial^2 G(x, t; x', t')}{\partial t^2} - a^2 \frac{\partial^2 G(x, t; x', t')}{\partial x^2} = \delta(x - x') \delta(t - t'),$$

将 x 与 x' 互换, t 与 t' 互换得
$$\frac{\partial^2 G(x', t'; x, t)}{\partial t'^2} - a^2 \frac{\partial^2 G(x', t'; x, t)}{\partial x'^2} = \delta(x - x') \delta(t - t'),$$

由 $G(x, t; x', t')$ 的互易性得
$$\frac{\partial^2 G(x, -t; x', -t')}{\partial t'^2} - a^2 \frac{\partial^2 G(x, -t; x', -t')}{\partial x'^2} = \delta(x - x') \delta(t - t'),$$

将 $-t$ 换成 t , $-t'$ 换成 t' 得 $\frac{\partial^2 G(x, t; x', t')}{\partial t'^2} - a^2 \frac{\partial^2 G(x, t; x', t')}{\partial x'^2} = \delta(x - x') \delta(t - t')$ 。(a)

又有 $\frac{\partial^2 u(x', t')}{\partial t'^2} - a^2 \frac{\partial^2 u(x', t')}{\partial x'^2} = 0$ 。(b)

(a) $\times u(x', t') -$ (b) $\times G(x, t; x', t')$ 得:

$$\begin{aligned} u(x, t) \delta(x - x') \delta(t - t') &= \frac{\partial}{\partial t'} \left[u(x', t') \frac{\partial G(x, t; x', t')}{\partial t'} - G(x, t; x', t') \frac{\partial u(x', t')}{\partial t'} \right] \\ &\quad + a^2 \frac{\partial}{\partial x'} \left[G(x, t; x', t') \frac{\partial u(x', t')}{\partial x'} - u(x', t') \frac{\partial G(x, t; x', t')}{\partial x'} \right]. \end{aligned}$$

两边对 x' 在 $(-\infty, \infty)$ 上积分得

$$\begin{aligned} u(x, t) \delta(t - t') &= \int_{-\infty}^{\infty} \frac{\partial}{\partial t'} \left[u(x', t') \frac{\partial G(x, t; x', t')}{\partial t'} - G(x, t; x', t') \frac{\partial u(x', t')}{\partial t'} \right] dx' \\ &\quad + a^2 \left[G(x, t; x', t') \frac{\partial u(x', t')}{\partial x'} - u(x', t') \frac{\partial G(x, t; x', t')}{\partial x'} \right]_{x' \rightarrow -\infty}^{x' \rightarrow \infty}, \end{aligned}$$

第 343 题已求得 $G = \frac{1}{2a} \eta \left(t - t' - \frac{|x - x'|}{a} \right)$, 所以 $G|_{x' \rightarrow \pm\infty} = \frac{1}{2a} \eta(-\infty) = 0$,

$$\frac{\partial G}{\partial x'} = \operatorname{sgn}(x - x') \frac{1}{2a^2} \delta \left(t - t' - \frac{|x - x'|}{a} \right), \quad \frac{\partial G}{\partial x'} \Big|_{x' \rightarrow \pm\infty} = \pm \frac{1}{2a^2} \delta(-\infty) = 0,$$

$$\text{所以 } u(x, t) \delta(t - t') = \int_{-\infty}^{\infty} \frac{\partial}{\partial t'} \left[u(x', t') \frac{\partial G(x, t; x', t')}{\partial t'} - G(x, t; x', t') \frac{\partial u(x', t')}{\partial t'} \right] dx'.$$

两边对 t' 在 $[0, \infty)$ 上积分得

$$u(x, t) = \int_{-\infty}^{\infty} \left[u(x', t') \frac{\partial G(x, t; x', t')}{\partial t'} - G(x, t; x', t') \frac{\partial u(x', t')}{\partial t'} \right]_{t'=0}^{t' \rightarrow \infty} dt'.$$

由于 $G|_{t' \rightarrow \infty} = \frac{1}{2a} \eta(-\infty) = 0$, $G|_{t'=0} = \frac{1}{2a} \eta \left(t - \frac{|x - x'|}{a} \right)$, $\frac{\partial G}{\partial t'} = -\frac{1}{2a} \delta \left(t - t' - \frac{|x - x'|}{a} \right)$,

$$\frac{\partial G}{\partial t'} \Big|_{t' \rightarrow \infty} = -\frac{1}{2a} \delta(-\infty) = 0, \quad \frac{\partial G}{\partial t'} \Big|_{t'=0} = -\frac{1}{2a} \delta \left(t - \frac{|x - x'|}{a} \right),$$

$$\begin{aligned}
\text{所以 } u(x, t) &= - \int_{-\infty}^{\infty} \left[u(x', t') \frac{\partial G(x, t; x', t')}{\partial t'} - G(x, t; x', t') \frac{\partial u(x', t')}{\partial t'} \right]_{t'=0} dx' \\
&= \frac{1}{2a} \int_{-\infty}^{\infty} \left[\varphi(x') \delta\left(t - \frac{|x-x'|}{a}\right) + \eta\left(t - \frac{|x-x'|}{a}\right) \psi(x') \right] dx' \\
&= \frac{1}{2} \int_{-\infty}^x \varphi(x') \delta(x' - x + at) dx' + \frac{1}{2a} \int_{x-at}^x \psi(x') dx' \\
&\quad + \frac{1}{2} \int_x^{\infty} \varphi(x') \delta(x' - x - at) dx' + \frac{1}{2a} \int_x^{x+at} \psi(x') dx' \\
&= \frac{1}{2} [\varphi(x-at) + \varphi(x+at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(x') dx'.
\end{aligned}$$

348. 用 Green 函数法解 215 题:
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = 0, u|_{x=l} = 0, u|_{t=0} = \begin{cases} \frac{h}{c}x, 0 \leq x < c \\ \frac{h(l-x)}{l-c}, c \leq x < l \end{cases}, \frac{\partial u}{\partial t}\bigg|_{t=0} = 0 \end{cases}$$

第 344 题已求得 $G = \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi}{l} x' \sin \frac{n\pi}{l} x \sin \frac{n\pi a}{l} (t-t') \eta(t-t')$ (令 $\frac{l}{\rho} = 1$)。

重复上题步骤可得 $u(x, t) = \int_0^l \left(G \frac{\partial u}{\partial t'} - u \frac{\partial G}{\partial t'} \right)_{t'=0} dx' = - \int_0^l u(x', 0) \frac{\partial G}{\partial t'} \bigg|_{t'=0} dx'$ 。

$\frac{\partial G}{\partial t'} \bigg|_{t'=0} = -\frac{2}{l} \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x' \sin \frac{n\pi}{l} x \cos \frac{n\pi a}{l} t$, 所以

$$\begin{aligned}
u(x, t) &= \frac{2}{l} \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x \cos \frac{n\pi a}{l} t \int_0^l u(x', 0) \sin \frac{n\pi}{l} x' dx' \\
&= \frac{2}{l} \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x \cos \frac{n\pi a}{l} t \left[\frac{h}{c} \int_0^c x' \sin \frac{n\pi}{l} x' dx' + \frac{h}{l-c} \int_c^l (l-x') \sin \frac{n\pi}{l} x' dx' \right] \\
&= \frac{2hl^2}{c(l-c)\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi c}{l} \sin \frac{n\pi}{l} x \cos \frac{n\pi a}{l} t.
\end{aligned}$$

349. 用 Green 法解无界弦的热传导问题:
$$\begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{t=0} = \varphi(x) \end{cases}$$

第 345 题已求得 $G(x, t; x', t') = \frac{1}{2\sqrt{\kappa\pi(t-t')}} \exp\left[-\frac{(x-x')^2}{4\kappa(t-t')}\right] \eta(t-t')$

重复 247 题步骤可得 $u(x, t) = \int_{-\infty}^{\infty} \varphi(x') G|_{t'=0} dx' = \frac{1}{2\sqrt{\kappa\pi t}} \int_{-\infty}^{\infty} \varphi(x') \exp\left[-\frac{(x-x')^2}{4\kappa t}\right] dx'.$

350. 用 342 题方法求三维无界空间波动方程和热传导方程的 Green 函数。

波动方程: $\left(\frac{\partial^2}{\partial t^2} - a^2 \nabla^2\right) G(\mathbf{r}, t; \mathbf{r}', t') = \delta(\mathbf{r} - \mathbf{r}') \delta(t - t').$

对变量 $\mathbf{r}$ 和 t 作 Fourier 变换得

$$(-\omega^2 + a^2 k^2) \tilde{G}(\mathbf{k}, \omega; \mathbf{r}', t') = \exp(-i\mathbf{k} \cdot \mathbf{r}') \exp(-i\omega t'),$$

$$\text{即 } \tilde{G}(\mathbf{k}, \omega; \mathbf{r}', t') = \frac{\exp(-i\mathbf{k} \cdot \mathbf{r}') \exp(-i\omega t')}{a^2 k^2 - \omega^2},$$

反演得 $G(\mathbf{r}, t; \mathbf{r}', t') = \frac{1}{(2\pi)^4 a^2} \int_{-\infty}^{\infty} e^{i\omega(t-t')} d\omega \iiint \frac{\exp[i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')] dk}{k^2 - \left(\frac{\omega}{a}\right)^2}$, 同 342 题, 记

$\mathbf{R} = \mathbf{r} - \mathbf{r}'$, $\tau = t - t'$, 以 $\mathbf{R}$ 方向为 z 方向在 $\mathbf{k}$ 空间建立球坐标系, 有

$$\begin{aligned} G(\mathbf{r}, t; \mathbf{r}', t') &= \frac{1}{(2\pi)^3 a^2} \int_{-\infty}^{\infty} e^{i\omega\tau} d\omega \int_0^{\infty} \frac{k^2 dk}{k^2 - \left(\frac{\omega}{a}\right)^2} \int_0^{\pi} e^{ikR \cos \theta} \sin \theta d\theta \\ &= \frac{1}{(2\pi)^3 a^2 iR} \int_{-\infty}^{\infty} e^{i\omega\tau} d\omega \int_0^{\infty} \frac{k(e^{ikR} - e^{-ikR}) dk}{k^2 - \left(\frac{\omega}{a}\right)^2} = \frac{1}{(2\pi)^3 a^2 iR} \int_{-\infty}^{\infty} e^{i\omega\tau} d\omega \text{v.p.} \int_{-\infty}^{\infty} \frac{ke^{ikR} dk}{k^2 - \left(\frac{\omega}{a}\right)^2} \\ &= \frac{1}{(2\pi)^3 a^2 iR} \int_{-\infty}^{\infty} e^{i\omega\tau} d\omega \lim_{\eta \rightarrow 0^+} \text{v.p.} \int_{-\infty}^{\infty} \frac{ke^{ikR} dk}{k^2 - \left(\frac{\omega}{a} - i\eta\right)^2}. \end{aligned}$$

由留数定理, $\text{v.p.} \int_{-\infty}^{\infty} \frac{ke^{ikR} dk}{k^2 - \left(\frac{\omega}{a} - i\eta\right)^2} = 2\pi i \text{res} \left[\frac{ze^{iRz}}{z^2 - \left(\frac{\omega}{a} - i\eta\right)^2} \right]_{z = -\frac{\omega}{a} + i\eta} = \pi i e^{-R\eta} e^{-i\frac{\omega}{a}R},$

所以 $G(\mathbf{r}, t; \mathbf{r}', t') = \frac{1}{4\pi a^2 R} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega\left(\tau - \frac{R}{a}\right)} d\omega = \frac{1}{4\pi a^2 R} \delta\left(\tau - \frac{R}{a}\right)$

$$= \frac{1}{4\pi a^2 |\mathbf{r} - \mathbf{r}'|} \delta\left(t - t' - \frac{|\mathbf{r} - \mathbf{r}'|}{a}\right)。$$

$$\text{热传导方程: } \left(\frac{\partial}{\partial t} - \kappa \nabla^2\right) G(\mathbf{r}, t; \mathbf{r}', t') = \delta(\mathbf{r} - \mathbf{r}') \delta(t - t')。$$

对变量 $\mathbf{r}$ 作 Fourier 变换, 对变量 t 作 Laplace 变换得

$$(p + \kappa k^2) \tilde{G}(\mathbf{k}, p; \mathbf{r}', t') = \exp(-i\mathbf{k} \cdot \mathbf{r}') \exp(-pt'), \quad \text{限定 } |\arg p| < \frac{\pi}{2}。$$

$$\text{即 } \tilde{G}(\mathbf{k}, p; \mathbf{r}', t') = \frac{1}{\kappa} \frac{\exp(-i\mathbf{k} \cdot \mathbf{r}') \exp(-pt')}{k^2 + \frac{p}{\kappa}}, \quad \text{作 Fourier 反演得}$$

$$\tilde{G}(\mathbf{r}, p; \mathbf{r}', t') = \frac{e^{-pt'}}{\kappa} \frac{1}{(2\pi)^3} \iiint \frac{\exp(i\mathbf{k} \cdot \mathbf{R})}{k^2 + \frac{p}{\kappa}} d\mathbf{k} = \frac{e^{-pt'}}{4\pi^2 i \kappa R} \text{v.p.} \int_{-\infty}^{\infty} \frac{ke^{ikR}}{k^2 + \frac{p}{\kappa}} dk。$$

由留数定理, 由于 $|\arg p| < \frac{\pi}{2}$, 所以 $\frac{3\pi}{4} < \arg i\sqrt{\frac{p}{\kappa}} < \frac{\pi}{4}$, $-\frac{3\pi}{4} < \arg\left(-i\sqrt{\frac{p}{\kappa}}\right) < -\frac{\pi}{4}$, 即

$i\sqrt{\frac{p}{\kappa}}$ 位于上半平面, $-i\sqrt{\frac{p}{\kappa}}$ 位于下半平面,

$$\text{所以 } \text{v.p.} \int_{-\infty}^{\infty} \frac{ke^{ikR}}{k^2 + \frac{p}{\kappa}} dk = 2\pi i \operatorname{res} \left(\frac{ze^{izR}}{z^2 + \frac{p}{\kappa}} \right)_{z=\sqrt{\frac{p}{\kappa}}i} = \pi i e^{-R\sqrt{\frac{p}{\kappa}}},$$

$$\text{因此 } \tilde{G}(\mathbf{r}, p; \mathbf{r}', t') = \frac{e^{-pt'}}{4\pi\kappa R} e^{-R\sqrt{\frac{p}{\kappa}}}。$$

$$\text{书中已求得 } \frac{2}{\sqrt{\pi}} \int_{\frac{\alpha}{2\sqrt{t}}}^{\infty} e^{-x^2} dx \xrightarrow{LT} \frac{1}{p} e^{-\alpha\sqrt{p}},$$

$$\text{所以 } e^{-\alpha\sqrt{p}} \xrightarrow{LT^{-1}} \frac{d}{dt} \frac{2}{\sqrt{\pi}} \int_{\frac{\alpha}{2\sqrt{t}}}^{\infty} e^{-x^2} dx = \frac{\alpha}{2\sqrt{\pi} t^{3/2}} e^{-\frac{\alpha^2}{4t}},$$

再利用 Laplace 变换性质 $f(at) \xrightarrow{LT} \frac{1}{a} F\left(\frac{p}{a}\right)$ 可得

$$G(\mathbf{r}, t; \mathbf{r}', t') = \frac{1}{8[\pi\kappa(t-t')]^{3/2}} e^{-\frac{R^2}{4\kappa(t-t')}} \eta(t-t')。$$

351. 用 Laplace 变换求解半无界问题:
$$\begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0, x > 0, t > 0 \\ u|_{x=0} = u_0, u|_{t=0} = 0 \end{cases}.$$

对变量 t 作 Laplace 变换得
$$\begin{cases} \frac{\partial^2 U(x, p)}{\partial x^2} - \frac{p}{\kappa} U(x, p) = 0, x > 0 \\ U(x, p)|_{x=0} = \frac{u_0}{p} \end{cases},$$

解得 $U(x, p) = \frac{u_0}{p} e^{-\sqrt{\frac{p}{\kappa}} x}$ 。由于 $\operatorname{erfc}\left(\frac{\alpha}{2\sqrt{t}}\right) \xrightarrow{LT} \frac{1}{p} e^{-\alpha\sqrt{p}}$, 令 $\alpha = \frac{x}{\sqrt{\kappa}}$ 即可得

$$u(x, t) = u_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{\kappa t}}\right).$$

352. 一高为 d , 底面积为 S 的圆柱体, 侧面绝热, 单位时间内通过下底供给热量 H , 而上底保持温度为 0。设柱体初温为 0, 证明在 t 时刻单位时间内通过上底流出的热量为

$$Q = H \left[1 - \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} e^{-\left(\frac{2n+1}{2d}\pi\right)^2 \kappa t} \right], \text{ 其中 } \kappa \text{ 为扩散率。}$$

以底面圆心为坐标原点, 圆柱轴线为 z 轴建立柱坐标系, 温度分布与 ρ, φ 无关, 底面热流

密度 $\frac{H}{S} = -k \frac{\partial u}{\partial z} \Big|_{z=0}$, 所以
$$\begin{cases} \frac{\partial u(z, t)}{\partial t} - \kappa \frac{\partial^2 u(z, t)}{\partial z^2} = 0 \\ \frac{\partial u(z, t)}{\partial z} \Big|_{z=0} = -\frac{H}{kS}, u(z, t)|_{z=d} = 0, u(z, t)|_{t=0} = 0 \end{cases}.$$

对变量 t 作 Laplace 变换得
$$\begin{cases} \frac{\partial^2 U(z, p)}{\partial z^2} - \frac{p}{\kappa} U(z, p) = 0 \\ \frac{\partial U(z, p)}{\partial z} \Big|_{z=0} = -\frac{H}{kS} \frac{1}{p}, U(z, p)|_{z=d} = 0 \end{cases},$$

解得 $U(z, p) = \frac{H}{kS} \frac{\operatorname{sh} \sqrt{\frac{p}{\kappa}} (d-z)}{\operatorname{ch} \sqrt{\frac{p}{\kappa}} d} \frac{1}{p} \sqrt{\frac{\kappa}{p}}$ 没, 则 $-k \frac{\partial U(z, p)}{\partial z} \Big|_{z=d} = \frac{H}{S} \frac{1}{p} \frac{1}{\operatorname{ch} \sqrt{\frac{p}{\kappa}} d}$ 。

令习题 09 第 177 (4) 题中 $x = l = \frac{d}{\sqrt{\kappa}}$ 可得上式反演

$$-k \frac{\partial u}{\partial z} \Big|_{z=d} = \frac{H}{S} \left[1 - \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} e^{-\left(\frac{2n+1}{2d}\pi\right)^2 \kappa t} \right], \text{ 该式就是上底热流密度 } \frac{Q}{S}, \text{ 即}$$

$$Q = H \left[1 - \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} e^{-\left(\frac{2n+1}{2d}\pi\right)^2 \kappa t} \right].$$

353. 设有两条半无界杆, 温度分别为 0 和 u_0 。在 $t=0$ 时将两杆端点相接, 求 $t>0$ 时杆中

$$\text{温度分布: } \begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{t=0} = u_0 \eta(x) \end{cases}.$$

$$\text{对变量 } t \text{ 作 Laplace 变换得 } \frac{\partial^2 U(x, p)}{\partial x^2} - \frac{p}{\kappa} U(x, p) = -\frac{u_0}{\kappa} \eta(x), \quad (*)$$

$$\text{即 } \begin{cases} \frac{\partial^2 U(x, p)}{\partial x^2} - \frac{p}{\kappa} U(x, p) = 0, x < 0 \\ U|_{x \rightarrow -\infty} \text{ 有界} \end{cases}, \begin{cases} \frac{\partial^2 U(x, p)}{\partial x^2} - \frac{p}{\kappa} U(x, p) = -\frac{u_0}{\kappa}, x > 0 \\ U|_{x \rightarrow \infty} \text{ 有界} \end{cases},$$

$$\text{解为 } U(x, p) = \begin{cases} Ae^{\sqrt{\frac{p}{\kappa}}x}, x < 0 \\ Be^{-\sqrt{\frac{p}{\kappa}}x} + \frac{u_0}{p}, x > 0 \end{cases}.$$

由方程 (*) 可看出 $U(x, p)$ 和 $\frac{\partial U(x, p)}{\partial x}$ 在 $x=0$ 连续, 否则方程右边会出现冲激或冲激偶。

$$\text{由此可得 } U(x, p) = \begin{cases} \frac{u_0}{2} \frac{1}{p} e^{\sqrt{\frac{p}{\kappa}}x}, x < 0 \\ \frac{u_0}{p} - \frac{u_0}{2} \frac{1}{p} e^{-\sqrt{\frac{p}{\kappa}}x}, x > 0 \end{cases}.$$

$$\text{反演得 } u(x, t) = \begin{cases} \frac{u_0}{2} \operatorname{erfc}\left(-\frac{x}{2\sqrt{\kappa t}}\right), x < 0 \\ u_0 - \frac{u_0}{2} \operatorname{erfc}\left(\frac{x}{2\sqrt{\kappa t}}\right), x > 0 \end{cases}.$$

$$354. \text{ 用 Laplace 变换解习题 12 第 236 题: } \begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = Ae^{-\alpha^2 \kappa t}, u|_{x=l} = Be^{-\beta^2 \kappa t}, u|_{t=0} = 0 \end{cases}.$$

对变量 t 作 Laplace 变换得
$$\begin{cases} \frac{\partial^2 U(x, p)}{\partial x^2} - \frac{p}{\kappa} U(x, p) = 0 \\ U|_{x=0} = \frac{A}{p + \alpha^2 \kappa}, U|_{x=l} = \frac{B}{p + \beta^2 \kappa} \end{cases}。$$

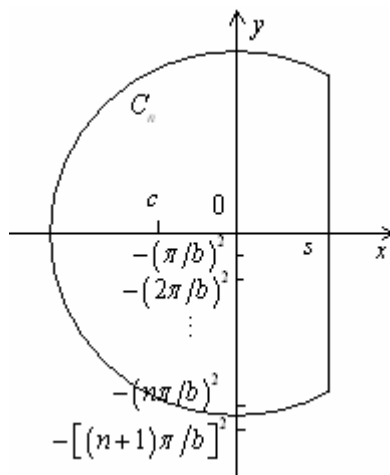
解得
$$U(x, p) = \frac{A}{p + \alpha^2 \kappa} \frac{\operatorname{sh} \sqrt{\frac{p}{\kappa}}(l-x)}{\operatorname{sh} \sqrt{\frac{p}{\kappa}} l} + \frac{B}{p + \beta^2 \kappa} \frac{\operatorname{sh} \sqrt{\frac{p}{\kappa}} x}{\operatorname{sh} \sqrt{\frac{p}{\kappa}} l}。 \quad (*)$$

下面求 $F(p) = \frac{1}{p-c} \frac{\operatorname{sh} a \sqrt{p}}{\operatorname{sh} b \sqrt{p}}$ ($b > a > 0, c \neq 0$) 的反演:

$F(p)$ 的一阶极点: $p=c, p=-\left(\frac{k\pi}{b}\right)^2, k=1, 2, \dots。$

$\operatorname{res}[F(p)e^{pt}]_{p=c} = \frac{\operatorname{sh} a \sqrt{c}}{\operatorname{sh} b \sqrt{c}} e^{ct}, \operatorname{res}[F(p)e^{pt}]_{p=-\left(\frac{k\pi}{b}\right)^2} = \frac{(-1)^k 2k\pi}{k^2 \pi^2 + cb^2} \sin \frac{k\pi a}{b} e^{-\left(\frac{k\pi}{b}\right)^2 t}。$

取如下积分路径: 其中 C_n 半径为 $-\left(\frac{n+1/2}{b}\pi\right)^2$, 该取法的讨论类似习题 09 第 177 (4) 题。



令 $n \rightarrow \infty$ 可得
$$\frac{1}{p-c} \frac{\operatorname{sh} a \sqrt{p}}{\operatorname{sh} b \sqrt{p}} \xrightarrow{LT^{-1}} \frac{\operatorname{sh} a \sqrt{c}}{\operatorname{sh} b \sqrt{c}} e^{ct} + \sum_{k=1}^{\infty} \frac{(-1)^k 2k\pi}{k^2 \pi^2 + cb^2} \sin \frac{k\pi a}{b} \exp \left[-\left(\frac{k\pi}{b}\right)^2 t \right]。$$

所以 (*) 式的反演为
$$u(x, t) = A \frac{\sin \alpha(l-x)}{\sin \alpha l} e^{-\alpha^2 \kappa t} + B \frac{\sin \beta x}{\sin \beta l} e^{-\beta^2 \kappa t}$$

$$+ \sum_{n=1}^{\infty} 2n\pi \left[\frac{A}{\alpha^2 l^2 - n^2 \pi^2} - \frac{(-1)^n B}{\beta^2 l^2 - n^2 \pi^2} \right] \sin \frac{n\pi x}{l} \exp \left[-\left(\frac{n\pi}{l}\right)^2 \kappa t \right]。$$

355. 用 Fourier 变换法解无界弦的振动问题:
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{t=0} = u_0 \exp\left[-\left(\frac{x}{a}\right)^2\right], \frac{\partial u}{\partial t}\bigg|_{t=0} = 0 \end{cases}。$$

对变量 x 作 Fourier 变换得
$$\begin{cases} \frac{\partial^2 U(k, t)}{\partial t^2} + k^2 c^2 U(k, t) = 0 \\ U|_{t=0} = u_0 FT\left\{\exp\left[-\left(\frac{x}{a}\right)^2\right]\right\}, \frac{\partial U}{\partial t}\bigg|_{t=0} = 0 \end{cases}。$$

解得
$$U(k, t) = u_0 FT\left\{\exp\left[-\left(\frac{x}{a}\right)^2\right]\right\} \cos kct = \frac{u_0}{2} FT\left\{\exp\left[-\left(\frac{x}{a}\right)^2\right]\right\} (e^{ikct} + e^{-ikct}),$$

反演得
$$u(x, t) = \frac{u_0}{2} \left\{ \exp\left[-\left(\frac{x+ct}{a}\right)^2\right] + \exp\left[-\left(\frac{x-ct}{a}\right)^2\right] \right\}。$$

356. 用 Fourier 变换和 Laplace 变换解无界弦的横振动问题:
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t}\bigg|_{t=0} = \psi(x) \end{cases}。$$

对变量 x 作 Fourier 变换得
$$\begin{cases} \frac{\partial^2 \tilde{u}(k, t)}{\partial t^2} + k^2 c^2 \tilde{u}(k, t) = 0 \\ \tilde{u}|_{t=0} = \tilde{\varphi}(k), \frac{\partial \tilde{u}}{\partial t}\bigg|_{t=0} = \tilde{\psi}(k) \end{cases},$$
 再对变量 t 作 Laplace 变换得

$$U(k, p) = \frac{p\tilde{\varphi}(k) + \tilde{\psi}(k)}{p^2 + k^2 c^2} = \frac{1}{2} \frac{\tilde{\varphi}(k) - \frac{\tilde{\psi}(k)}{ikc}}{p + ikc} + \frac{1}{2} \frac{\tilde{\varphi}(k) + \frac{\tilde{\psi}(k)}{ikc}}{p - ikc},$$

作 Laplace 反演得
$$\begin{aligned} \tilde{u}(k, t) &= \frac{1}{2} \left[\tilde{\varphi}(k) - \frac{\tilde{\psi}(k)}{ikc} \right] e^{-ikct} + \frac{1}{2} \left[\tilde{\varphi}(k) + \frac{\tilde{\psi}(k)}{ikc} \right] e^{ikct} \\ &= \frac{1}{2} \tilde{\varphi}(k) (e^{-ikct} + e^{ikct}) + \frac{1}{2} \frac{\tilde{\psi}(k)}{ikc} (e^{ikct} - e^{-ikct}). \end{aligned}$$

再作 Fourier 反演得

$$\begin{aligned} u(x, t) &= \frac{1}{2} [\varphi(x-ct) + \varphi(x+ct)] + \frac{1}{2c} \int_{-\infty}^x [\psi(\xi+ct) - \psi(\xi-ct)] d\xi \\ &= \frac{1}{2} [\varphi(x-ct) + \varphi(x+ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(\xi) d\xi。 \end{aligned}$$

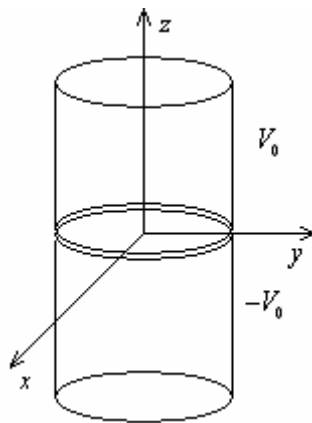
357. 一半无界弦 $x \geq 0$ ，原处于平衡状态。设在 $t > 0$ 时 $x = 0$ 端作微小振动 $A \sin \omega t$ 。试

求弦上各点运动：
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = A \sin \omega t \eta(t), u|_{t=0} = 0, \frac{\partial u}{\partial t} \Big|_{t=0} = 0 \end{cases}。$$

对变量 t 作 Laplace 变换得
$$\begin{cases} \frac{\partial^2 U(x, p)}{\partial x^2} - \left(\frac{p}{c}\right)^2 U(x, p) = 0 \\ U(x, p)|_{x=0} = \frac{A\omega}{p^2 + \omega^2}, U(x, p)|_{x \rightarrow \infty} \text{ 有界} \end{cases}。$$

解得 $U(x, p) = \frac{A\omega}{p^2 + \omega^2} e^{-\frac{p}{c}x}$ ，反演得 $u(x, t) = A \sin \omega \left(t - \frac{x}{c}\right) \eta\left(t - \frac{x}{c}\right)。$

358. 电子光学中常遇到一种简单的静电透镜—等径双筒镜，它的两极是由两个无限接近的等径同轴长圆筒组成，其电势分别为 $-V_0$ 与 V_0 （见下图）。求筒内静电势。



先令边界条件为 $u|_{\rho=a} = V_0 e^{-k|z|} \operatorname{sgn} z$ ，该问题显然与 φ 无关，

即
$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left[\rho \frac{\partial u(\rho, z)}{\partial \rho} \right] + \frac{\partial^2 u(\rho, z)}{\partial z^2} = 0 \\ u|_{\rho=0} \text{ 有界}, u|_{\rho=a} = V_0 e^{-k|z|} \operatorname{sgn} z \end{cases}。$$
 对变量 z 作 Fourier 变换得

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left[\rho \frac{\partial U(\rho, \omega)}{\partial \rho} \right] - \omega^2 U(\rho, \omega) = 0 \\ U|_{\rho=0} \text{ 有界}, U|_{\rho=a} = -2iV_0 \frac{\omega}{k^2 + \omega^2} \end{cases}。$$
 这是零阶虚宗量 Bessel 方程，有界解为

$U(\rho, \omega) = A I_0(\omega \rho)$ ，由边界条件得 $U(\rho, \omega) = -2iV_0 \frac{\omega}{k^2 + \omega^2} \frac{I_0(\omega \rho)}{I_0(\omega a)}。$

$$\begin{aligned} \text{反演得 } u(\rho, z) &= -\frac{V_0}{\pi} \int_{-\infty}^{\infty} \frac{\omega}{k^2 + \omega^2} \frac{I_0(\omega\rho)}{I_0(\omega a)} i e^{i\omega z} d\omega \\ &= -\frac{V_0}{\pi} \int_{-\infty}^{\infty} \frac{\omega}{k^2 + \omega^2} \frac{I_0(\omega\rho)}{I_0(\omega a)} (i \cos \omega z - \sin \omega z) d\omega. \end{aligned}$$

上面的积分理解为主值, 由于 $\frac{\omega}{k^2 + \omega^2} \frac{I_0(\omega\rho)}{I_0(\omega a)} \cos \omega z$ 是 ω 的奇函数, 所以这部分积分为 0,

$$\text{即 } u(\rho, z) = \frac{2V_0}{\pi} \int_0^{\infty} \frac{\omega}{k^2 + \omega^2} \frac{I_0(\omega\rho)}{I_0(\omega a)} \sin \omega z d\omega,$$

$$\text{令 } k \rightarrow 0 \text{ 得 } u(\rho, z) = \frac{2V_0}{\pi} \int_0^{\infty} \frac{I_0(\omega\rho)}{I_0(\omega a)} \frac{\sin \omega z}{\omega} d\omega.$$

359. 设有一半径为 1 的带电圆盘, 圆盘上电势为 V_0 , 求空间电势。

以圆盘圆心为原点, 垂直于盘面的轴为 z 轴建立柱坐标系。该问题与 φ 无关, 且显然关于

$z=0$ 面是对称的, 即 u 是 z 的偶函数, 必有 $\left. \frac{\partial u}{\partial z} \right|_{z=0} = 0$ ($\rho > 1$), 可以只考虑半空间 $z > 0$ 。

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left[\rho \frac{\partial u(\rho, z)}{\partial \rho} \right] + \frac{\partial^2 u(\rho, z)}{\partial z^2} = 0 \\ u|_{z=0, 0 < \rho < 1} = V_0, \left. \frac{\partial u}{\partial z} \right|_{z=0, \rho > 1} = 0, u|_{z \rightarrow \infty} = 0. \\ u|_{\rho=0} \text{ 有界}, u|_{\rho \rightarrow \infty} = 0 \end{cases}$$

对变量 ρ 作 Hankel 变换: $U(k, z) = \int_0^{\infty} u(\rho, z) J_0(k\rho) \rho d\rho$, 则

$$\begin{aligned} \int_0^{\infty} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left[\rho \frac{\partial u(\rho, z)}{\partial \rho} \right] J_0(k\rho) \rho d\rho &= \rho \frac{\partial u(\rho, z)}{\partial \rho} J_0(k\rho) \Big|_{\rho=0}^{\rho \rightarrow \infty} - \int_0^{\infty} \rho \frac{\partial u(\rho, z)}{\partial \rho} \frac{dJ_0(k\rho)}{d\rho} d\rho \\ &= - \int_0^{\infty} \frac{\partial u(\rho, z)}{\partial \rho} \rho \frac{dJ_0(k\rho)}{d\rho} d\rho = -u(\rho, z) \rho \frac{dJ_0(k\rho)}{d\rho} \Big|_{\rho=0}^{\rho \rightarrow \infty} + \int_0^{\infty} u(\rho, z) \frac{d}{d\rho} \left[\rho \frac{dJ_0(k\rho)}{d\rho} \right] d\rho \\ &= -k^2 \int_0^{\infty} u(\rho, z) J_0(k\rho) \rho d\rho = -k^2 U(k, z). \end{aligned}$$

即原方程变换为 $\frac{\partial^2 U(k, z)}{\partial z^2} - k^2 U(k, z) = 0$, 由有界条件可得 $U(k, z) = A(k) e^{-kz}$ 。

反演得 $u(\rho, z) = \int_0^\infty A(k) e^{-kz} J_0(k\rho) k dk$, 代入 z 的边界条件得到积分方程:

$$\int_0^\infty A(k) J_0(k\rho) k dk = V_0 \quad (0 < \rho < 1),$$

$$\int_0^\infty A(k) J_0(k\rho) k^2 dk = 0 \quad (\rho > 1).$$

由习题 16 第 319 题结果可知 $A(k) = \frac{2V_0}{\pi} \frac{\sin k}{k^2}$, 即

$$u(\rho, z) = \frac{2V_0}{\pi} \int_0^\infty e^{-kz} J_0(k\rho) \frac{\sin k}{k} dk.$$

360. 由柱面坐标 (ρ, z) 可以定义扁球面坐标 (μ, ξ) : $z = \mu\xi$, $\rho = \sqrt{(1-\mu^2)(1+\xi^2)}$ 。

$$(1) \text{ 证明: 在此坐标系下, 上题化为 } \begin{cases} \frac{\partial}{\partial \mu} \left[(1-\mu^2) \frac{\partial u}{\partial \mu} \right] + \frac{\partial}{\partial \xi} \left[(1+\xi^2) \frac{\partial u}{\partial \xi} \right] = 0 \\ u|_{\xi=0} = V_0, u|_{\xi \rightarrow \infty} \rightarrow 0 \\ \frac{\partial u}{\partial \mu} \Big|_{\mu=0} = 0, u|_{\mu=1} \text{ 有界} \end{cases}.$$

(2) 求出 $u(\mu, \xi)$ 。

$$(1) \text{ 该坐标系下的 Laplace 算符为 } \nabla^2 = \frac{1}{\mu^2 + \xi^2} \left\{ \frac{\partial}{\partial \mu} \left[(1-\mu^2) \frac{\partial}{\partial \mu} \right] + \frac{\partial}{\partial \xi} \left[(1+\xi^2) \frac{\partial}{\partial \xi} \right] \right\}.$$

对于 $z > 0$ 半空间, (μ, ξ) 取值范围: $0 < \mu < 1$, $\xi > 0$ 。

$z = 0, 0 < \rho < 1$ 对应 $\xi = 0$, 所以 $u|_{z=0, 0 < \rho < 1} = u|_{\xi=0} = V_0$ 。

$$\text{由坐标关系可得 } \frac{\partial}{\partial z} = \frac{\xi \frac{1-\mu^2}{1+\xi^2} \frac{\partial}{\partial \mu} + \mu \frac{\partial}{\partial \xi}}{\xi^2 \frac{1-\mu^2}{1+\xi^2} + \mu^2}, \quad z = 0, \rho > 1 \text{ 对应 } \mu = 0,$$

$$\text{所以 } \frac{\partial u}{\partial z} \Big|_{z=0, \rho > 1} = \frac{\xi \frac{1-\mu^2}{1+\xi^2} \frac{\partial u}{\partial \mu} + \mu \frac{\partial u}{\partial \xi}}{\xi^2 \frac{1-\mu^2}{1+\xi^2} + \mu^2} \Big|_{\mu=0} = \frac{1}{\xi} \frac{\partial u}{\partial \mu} \Big|_{\mu=0} = 0.$$

$z \rightarrow \infty$ 和 $\rho \rightarrow \infty$ 对应 $\xi \rightarrow \infty$, 所以 $u|_{\rho \rightarrow \infty} = u|_{z \rightarrow \infty} = u|_{\xi \rightarrow \infty} \rightarrow 0$ 。

$\rho = 0$ 对应 $\mu = 1$, 所以 $u|_{\rho=0} = u|_{\mu=1}$ 有界。

$$(2) \text{ 令 } u(\mu, \xi) = M(\mu)\Xi(\xi), \text{ 分离变量得 } \begin{cases} \frac{d}{d\mu} \left[(1-\mu^2) \frac{dM}{d\mu} \right] + \lambda M = 0 \\ \frac{d}{d\xi} \left[(1+\xi^2) \frac{d\Xi}{d\xi} \right] - \lambda \Xi = 0 \end{cases}.$$

可得本征值问题 $\begin{cases} \frac{\partial}{\partial \mu} \left[(1-\mu^2) \frac{\partial M}{\partial \mu} \right] + \lambda M = 0 \\ M'(0) = 0, M(1) \text{ 有界} \end{cases}$, 习题 15 第 292 (2) 题已得该问题的解为

$$\lambda_l = 2l(2l+1), \quad M_l(\mu) = P_{2l}(\mu).$$

$$\Xi(\xi) \text{ 方程为 } \frac{d}{d\xi} \left[(1+\xi^2) \frac{d\Xi(\xi)}{d\xi} \right] - 2l(2l+1)\Xi(\xi) = 0. \quad (*)$$

将 ξ 换成 $i\xi$ 得 $\frac{d}{d\xi} \left[(1-\xi^2) \frac{d\Xi(i\xi)}{d\xi} \right] + 2l(2l+1)\Xi(i\xi) = 0$, 所以通解为

$$\Xi_l(\xi) = A_l P_{2l}(i\xi) + B_l Q_{2l}(i\xi). \text{ 由于 } \Xi(\xi) \Big|_{\xi \rightarrow \infty} = 0, \text{ 而当 } l \neq 0 \text{ 时 } P_{2l}(\infty) \text{ 和 } Q_{2l}(\infty) \text{ 都无}$$

界, 所以只有 $l=0$ 。此时解方程 (*) 可得 $\Xi_0(\xi) = A \arctan \xi + B$,

$$\text{所以 } u(\mu, \xi) = M_0(\mu)\Xi_0(\xi) = A \arctan \xi + B,$$

由边界条件 $u|_{\xi=0} = V_0, u|_{\xi \rightarrow \infty} \rightarrow 0$ 得 $A = -\frac{2}{\pi}V_0, B = V_0$,

$$\text{所以 } u(\mu, \xi) = V_0 \left(1 - \frac{2}{\pi} \arctan \xi \right) = \frac{2V_0}{\pi} \left(\frac{\pi}{2} - \arctan \xi \right) = \frac{2V_0}{\pi} \operatorname{arccot} \xi.$$

361. $\zeta = z^2$ 把 z 平面上的下列区域变为 ζ 平面上的什么区域? (1) 上半平面; (2) 上半圆 $|z| < 1, \operatorname{Im} z > 0$; (3) 圆 $|z| < 1$; (4) 双纽线内部 $|z|^2 < \cos(2\arg z)$ 。

(1) $0 < \arg z < \pi$, $0 < \arg \zeta = 2\arg z < 2\pi$, 所以是沿正实轴割开的全平面。

(2) $0 < \arg z < \pi$, $0 < \arg \zeta = 2\arg z < 2\pi$, $|\zeta| = |z|^2 < 1$, 所以是沿正实轴割开的单位圆内部。

(3) $|\zeta| = |z|^2 < 1$, 所以是单位圆内部。

(4) 设 $z = \rho e^{i\varphi}$, $\zeta = re^{i\theta}$, 则由 $\zeta = z^2$ 可得 $r = \rho^2, \theta = 2\varphi$ 。由于 $\rho^2 < \cos 2\varphi$, 所以

$$r < \cos \theta, \text{ 即为圆内部 } \left| \zeta - \frac{1}{2} \right| < \frac{1}{2}。$$

362. 若在分式线性变换 $\zeta = \lambda \frac{z - \mu}{z - \nu}$ 下, z_1, z_2, z_3 各点分别变为 $\zeta_1, \zeta_2, \zeta_3$ 各点, 试证:

$$\frac{\zeta - \zeta_1}{\zeta - \zeta_2} \bigg/ \frac{\zeta_3 - \zeta_1}{\zeta_3 - \zeta_2} = \frac{z - z_1}{z - z_2} \bigg/ \frac{z_3 - z_1}{z_3 - z_2}。$$

变换写成 $\zeta = \lambda + \frac{\lambda(\nu - \mu)}{z - \nu}$, 记 $(z, z_1, z_2, z_3) = \frac{z - z_1}{z - z_2} \bigg/ \frac{z_3 - z_1}{z_3 - z_2}$, 只需证明该值在平移, 反

演, 旋转, 伸缩下不变即可。平移, 旋转, 伸缩变换易证, 而 $\left(\frac{1}{z}, \frac{1}{z_1}, \frac{1}{z_2}, \frac{1}{z_3} \right) = \frac{\frac{1}{z} - \frac{1}{z_1}}{\frac{1}{z} - \frac{1}{z_2}} \frac{\frac{1}{z_3} - \frac{1}{z_1}}{\frac{1}{z_3} - \frac{1}{z_2}}$

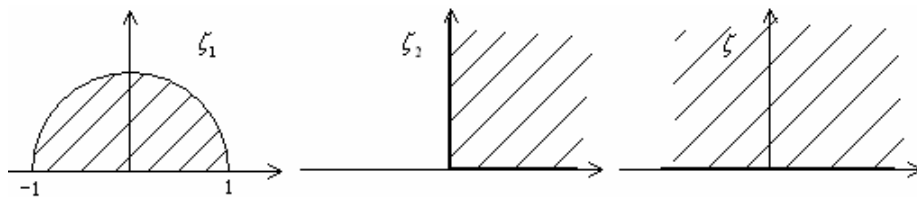
$$= \frac{z_1 z_2 - z z_2}{z_1 z_2 - z z_1} \frac{z_1 z_2 - z_1 z_3}{z_1 z_2 - z_3 z_2} = \frac{z_1 - z}{z_2 - z} \frac{z_2 - z_3}{z_1 - z_3} = (z, z_1, z_2, z_3), \text{ 即该值在反演变换下也不变, 所}$$

以该值在分式线性变换不变。

363. 求一变换, 把上半平面变为单位圆, 并把实轴上的点 $-1, 0, 1$ 分别变为单位圆上的 $1, i, -1$ 。

设 $\zeta = \lambda \frac{z - \mu}{z - \nu}$, 将 (z, ζ) 分别代入 $(-1, 1), (0, i), (1, -1)$ 得到三个方程解出 $\zeta = -i \frac{z - i}{z + i}$ 。

364. z 平面上的单位圆 $|z| < 1$ ，沿正实轴割开，试把此区域变为 ζ 平面上的上半平面。



变换 $\zeta_1 = \sqrt{z}$ 将单位圆内变换为上半单位圆内。

令 $\zeta_2 = -\frac{\zeta_1 + 1}{\zeta_1 - 1}$ ，可看出该变换将 -1 变换为 0 ， 1 变换为 ∞ ，所以 ζ_1 平面实轴上从 -1 到 1

的线段和上半圆弧都变换为 ζ_2 平面上经过原点和 ∞ 的直线，由于 0 变换为 1 ，所以 -1 到 1 的线段变换为正实轴，线段与上半圆弧在 -1 处的夹角为 $\pi/2$ ，由保角性，上半圆弧变换为正虚轴，即变换为第一象限。

令 $\zeta = \zeta_2^2$ ，则变换为上半平面，综上该变换为 $\zeta = \left(\frac{\sqrt{z} + 1}{\sqrt{z} - 1} \right)^2$ 。

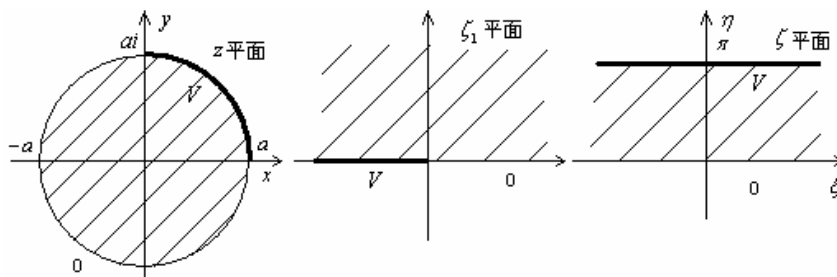
365. 求一变换，把第一象限变为单位圆内。

先由变换 $\zeta_1 = z^2$ 将它变换为上半平面。令 $\zeta = e^{i\zeta_1} = e^{-y} e^{ix}$ ，其中 $\zeta_1 = x + iy$ ，由于 $y > 0$ ，所以 $|\zeta| = e^{-y} < 1$ ，即变为单位圆内。综上 $\zeta = \exp(iz^2)$ 。

366. 一半径为 a 的无穷长导体柱面，第一象限的电势为 V ，第三象限电势 $-V$ ，第二，四象限接地，求导体内电势。

将 u 分成两部分： $u = u_1 + u_2$ ，其中 u_1 在第一象限为 V ，其余象限为 0 ， u_2 在第三象限为 $-V$ ，

其余象限为 0 。先求 u_1 ：



$\zeta_1 = \lambda \frac{z - a}{z - ai}$ ，使 z 平面上的 a 和 ai 分别对应 ζ_1 平面上的 0 和 ∞ ，即 z 平面上圆映射为 ζ_1

平面上通过原点的直线，该直线原点一侧对应 z 平面上第一象限圆弧，原点另一侧对应二，三，四象限的圆弧。由于 $z = -a$ 时 $\zeta_1 = \frac{2\lambda}{1+i}$ ，令 $\lambda = 1+i$ 可使 $-a$ 映射为 2，即二，三，四象限的圆弧映射为正实轴，那么第一象限的圆弧即映射为负实轴。

$\zeta = \ln \zeta_1$ ($\zeta = \xi + i\eta$) 将 ζ_1 平面正实轴映射为 $\eta = 0$ ，负实轴映射为 $\eta = \pi$ ， ζ_1 平面的上

半平面变为 ζ 平面上 $\eta = 0$ 和 $\eta = \pi$ 之间的带状区域，综上 $\zeta = \ln \left[(1+i) \frac{z-a}{z-ai} \right]$ 。

ζ 平面上 u_1 显然与 ξ 无关，所以 $\frac{d^2 \zeta}{d\eta^2} = 0$ ，通解为 $u_1 = A\eta + B$ ，由边界条件定出 $u_1 = \frac{V}{\pi} \eta$ 。

由 $\zeta = \ln \left[(1+i) \frac{z-a}{z-ai} \right]$ 可知 $\eta = \arg \left[(1+i) \frac{z-a}{z-ai} \right] = \frac{\pi}{4} + \arg(\rho e^{i\varphi} - a) - \arg(\rho e^{i\varphi} - ai)$

$$= \frac{\pi}{4} + \tan^{-1} \frac{\rho \sin \varphi}{\rho \cos \varphi - a} - \tan^{-1} \frac{\rho \sin \varphi - a}{\rho \cos \varphi},$$

$$\text{即 } u_1(\rho, \varphi) = \frac{V}{\pi} \left(\frac{\pi}{4} + \tan^{-1} \frac{\rho \sin \varphi}{\rho \cos \varphi - a} - \tan^{-1} \frac{\rho \sin \varphi - a}{\rho \cos \varphi} \right).$$

$$\text{显然 } u_2(\rho, \varphi) = -u_1(\rho, \varphi - \pi) = -\frac{V}{\pi} \left(\frac{\pi}{4} + \tan^{-1} \frac{\rho \sin \varphi}{\rho \cos \varphi + a} - \tan^{-1} \frac{\rho \sin \varphi + a}{\rho \cos \varphi} \right)$$

$$\text{所以 } u = u_1 + u_2 = \frac{V}{\pi} \left(\tan^{-1} \frac{\rho \sin \varphi}{\rho \cos \varphi - a} - \tan^{-1} \frac{\rho \sin \varphi}{\rho \cos \varphi + a} \right.$$

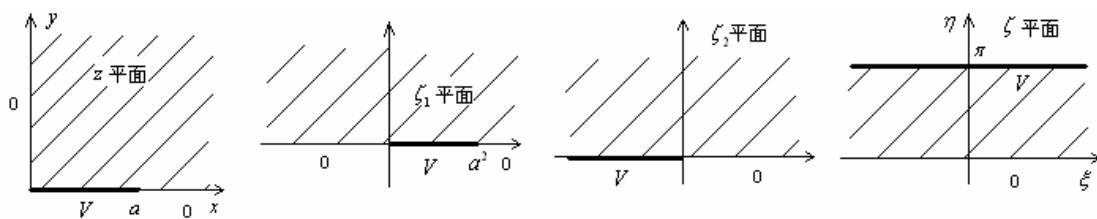
$$\left. + \tan^{-1} \frac{\rho \sin \varphi + a}{\rho \cos \varphi} - \tan^{-1} \frac{\rho \sin \varphi - a}{\rho \cos \varphi} \right)$$

$$= \frac{V}{\pi} \left(\tan^{-1} \frac{\frac{\rho \sin \varphi}{\rho \cos \varphi - a} - \frac{\rho \sin \varphi}{\rho \cos \varphi + a}}{1 + \frac{\rho \sin \varphi}{\rho \cos \varphi - a} \frac{\rho \sin \varphi}{\rho \cos \varphi + a}} + \tan^{-1} \frac{\frac{\rho \sin \varphi + a}{\rho \cos \varphi} - \frac{\rho \sin \varphi - a}{\rho \cos \varphi}}{1 + \frac{\rho \sin \varphi + a}{\rho \cos \varphi} \frac{\rho \sin \varphi - a}{\rho \cos \varphi}} \right)$$

$$= \frac{V}{\pi} \left(\tan^{-1} \frac{2a\rho \sin \varphi}{\rho^2 - a^2} + \tan^{-1} \frac{2a\rho \cos \varphi}{\rho^2 - a^2} \right).$$

367. 实轴上 $(0, a)$ 段电势为 V ，实轴上 $x > a$ 段及正虚轴电势均为 0，求第一象限电势分布。

$\zeta_1 = z^2$ 将第一象限变为上半平面， a 点变为 a^2 。



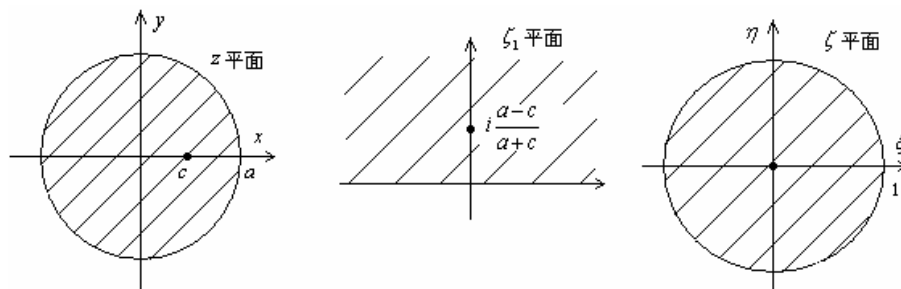
$\zeta_2 = \frac{\zeta_1 - a^2}{\zeta_1}$ 将 a^2 变为 0 , 0 变为 ∞ , 所以 ζ_1 平面的实轴变为 ζ_2 平面过原点的直线, $a^2 + 1$

变为 $\frac{1}{a^2 + 1} > 0$, 所以 ζ_1 平面实轴上 a^2 右边部分和负实轴变为 ζ_2 平面的正实轴, 电势为 V 的部分变为负实轴。

$\zeta = \ln \zeta_2$ 将负实轴变为 $\eta = \pi$, 正实轴变为 $\eta = 0$, 上半平面变为带状区域 $\eta = 0$ 和 $\eta = \pi$ 之

间。综上 $\zeta = \ln \frac{z^2 - a^2}{z^2}$, 所以 $u = \frac{V}{\pi} \eta = \frac{V}{\pi} \arg \frac{z^2 - a^2}{z^2}$ 。

368. 在接地的无穷长金属柱内, 有一条平行于柱轴的均匀带电丝, 线电荷密度为 σ 。设圆柱的半径为 a , 带电丝与柱轴距离为 c 。求柱内电势。



$\zeta_1 = -i \frac{z-a}{z+a}$ 将圆内变为上半平面, c 点变为 $i \frac{a-c}{a+c}$,

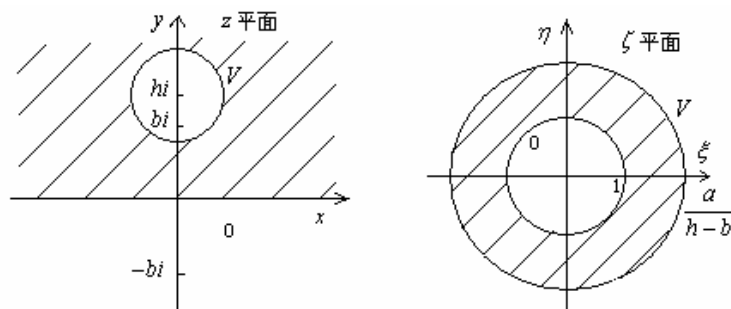
$\zeta = \frac{\zeta_1 - i \frac{a-c}{a+c}}{\zeta_1 + i \frac{a-c}{a+c}}$ 将上半平面变成单位圆内, $i \frac{a-c}{a+c}$ 变为原点。 $\zeta = \frac{a(z-c)}{cz-a^2}$ 。

$$u = \frac{\sigma}{2\pi\epsilon_0} \ln \frac{1}{|\zeta|} = \frac{\sigma}{2\pi\epsilon_0} \ln \left| \frac{cz-a^2}{a(z-c)} \right|。$$

369. 地面上平挂着一无穷长导体圆柱, 圆柱的半径为 a , 柱轴距地面 h ($h > a$)。设圆柱面的电势为 V , 求圆柱外电势。

设 $\pm bi$ ($b > 0$) 是圆柱的反演点对 (当然也是地面的反演点对), 则可求出 $b = \sqrt{h^2 - a^2}$ 。

$\zeta = \frac{z+bi}{z-bi}$ 将 $-bi$ 变为 0, bi 变为无穷大, 由于反演点对的不变性, 地面和圆柱都映射为以原点为圆心的圆。



由于 $\left| \frac{x+bi}{x-bi} \right| = 1$, 所以地面映射成的圆半径为 1, 又 $\frac{(h+a)i+bi}{(h+a)i-bi} = \frac{a}{h-b}$, 即 $(h+a)i$ 映射

为 $\frac{a}{h-b}$, 所以圆柱映射成的圆半径为 $\frac{a}{h-b}$ 。 $u = V \frac{\ln|\zeta|}{\ln \frac{a}{h-b}} = V \frac{\ln \left| \frac{z+bi}{z-bi} \right|}{\ln \frac{a}{h-b}}$ 。

370. 证明: 儒可夫斯基变换 $z = \frac{a}{2} \left(\zeta + \frac{1}{\zeta} \right)$ 把 z 平面上以 $z = \pm a$ 为焦点的共焦椭圆变为 ζ 平面上的同心圆。

记 $z = x + iy$, $\zeta = \rho e^{i\varphi}$, 则
$$\begin{cases} x = \frac{a}{2} \left(\rho + \frac{1}{\rho} \right) \cos \varphi \\ y = \frac{a}{2} \left(\rho - \frac{1}{\rho} \right) \sin \varphi \end{cases}, \text{ 对于 } \zeta \text{ 平面上的圆 } \rho = \rho_0, \text{ 对应}$$

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, \text{ 其中半长轴 } A = \frac{a}{2} \left(\rho_0 + \frac{1}{\rho_0} \right), \text{ 半短轴 } B = \frac{a}{2} \left(\rho_0 - \frac{1}{\rho_0} \right),$$

焦距 $C = \sqrt{A^2 - B^2} = a$, 即 ζ 平面上的同心圆映射为 z 平面上以 $z = \pm a$ 为焦点的共焦椭圆, 反之亦然。

371. 讨论下列方程的类型, 并将他们化为典则形式:

$$(1) \frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} - 3 \frac{\partial^2 u}{\partial y^2} + 2 \frac{\partial u}{\partial x} + 6 \frac{\partial u}{\partial y} = 0;$$

$$(2) \frac{\partial^2 u}{\partial x^2} + 4 \frac{\partial^2 u}{\partial x \partial y} + 5 \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0;$$

$$(3) \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial y^2} + \frac{1}{2} \frac{\partial u}{\partial y} = 0;$$

$$(4) (1+x^2) \frac{\partial^2 u}{\partial x^2} + (1+y^2) \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0;$$

$$(5) \tan^2 x \frac{\partial^2 u}{\partial x^2} - 2y \tan x \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + y \frac{\partial u}{\partial y} = 0.$$

(1) $a=1, b=1, c=-3, d=2, e=6, f=g=0$, $b^2-ac=4>0$, 所以是双曲型方程。

特征线方程为 $\frac{dy}{dx}=3$ 和 $\frac{dy}{dx}=-1$, 特征线为 $y-3x=C_1$ 和 $y+x=C_2$, 所以可令

$\xi=y-3x$, $\eta=y+x$, 则变换后方程系数 $A=C=0, B=-8, D=0, E=8, F=G=0$,

即原方程化为 $-16 \frac{\partial^2 u}{\partial \xi \partial \eta} + 8 \frac{\partial u}{\partial \eta} = 0$, 即 $\frac{\partial^2 u}{\partial \xi \partial \eta} - \frac{1}{2} \frac{\partial u}{\partial \eta} = 0$ 。

令 $\alpha = \frac{\xi+\eta}{2} = y-x, \beta = \frac{\eta-\xi}{2} = 2x$, 则 $\frac{\partial}{\partial \xi} = \frac{1}{2} \left(\frac{\partial}{\partial \alpha} - \frac{\partial}{\partial \beta} \right)$, $\frac{\partial}{\partial \eta} = \frac{1}{2} \left(\frac{\partial}{\partial \alpha} + \frac{\partial}{\partial \beta} \right)$,

所以, $\frac{\partial^2 u}{\partial \xi \partial \eta} = \frac{1}{4} \left(\frac{\partial}{\partial \alpha} - \frac{\partial}{\partial \beta} \right) \left(\frac{\partial u}{\partial \alpha} + \frac{\partial u}{\partial \beta} \right) = \frac{1}{4} \left(\frac{\partial^2 u}{\partial \alpha^2} - \frac{\partial^2 u}{\partial \beta^2} \right)$, 方程 $\frac{\partial^2 u}{\partial \xi \partial \eta} - \frac{1}{2} \frac{\partial u}{\partial \eta} = 0$ 化为

$$\frac{\partial^2 u}{\partial \alpha^2} - \frac{\partial^2 u}{\partial \beta^2} - \frac{\partial u}{\partial \alpha} - \frac{\partial u}{\partial \beta} = 0.$$

(2) $a=1, b=2, c=5, d=1, e=2, f=g=0$, $b^2-ac=-1<0$, 所以是椭圆型方程。

特征线方程 $\frac{dy}{dx} = 2 \pm i$, 特征线为 $y-(2+i)x=C_1$ 和 $y-(2-i)x=C_2$, 令

$\xi=y-(2+i)x$, $\eta=y-(2-i)x$, 则原方程化为 $4 \frac{\partial^2 u}{\partial \xi \partial \eta} - i \frac{\partial u}{\partial \xi} + i \frac{\partial u}{\partial \eta} = 0$,

令 $\rho = \frac{\xi+\eta}{2} = y-2x, \sigma = \frac{\eta-\xi}{2i} = x$, 则方程化为 $\frac{\partial^2 u}{\partial \rho^2} + \frac{\partial^2 u}{\partial \sigma^2} + \frac{\partial u}{\partial \sigma} = 0$ 。

$$(3) a=1, b=0, c=y, d=0, e=\frac{1}{2}, f=g=0, b^2-ac=-y,$$

当 $y > 0$ 时, 方程是椭圆型, $y < 0$ 时方程是双曲型。

$$y < 0 \text{ 时, 特征线方程 } \frac{dy}{dx} = \pm\sqrt{-y}, \text{ 特征线 } 2\sqrt{-y} + x = C_1 \text{ 和 } 2\sqrt{-y} - x = C_2,$$

$$\text{令 } \xi = 2\sqrt{-y} + x, \eta = 2\sqrt{-y} - x, \text{ 则方程化为 } \frac{\partial^2 u}{\partial \xi \partial \eta} = 0,$$

$$\text{令 } \rho = \frac{\xi + \eta}{2} = 2\sqrt{-y}, \sigma = \frac{\xi - \eta}{2} = x, \text{ 方程化为 } \frac{\partial^2 u}{\partial \rho^2} - \frac{\partial^2 u}{\partial \sigma^2} = 0。$$

$$y > 0 \text{ 时, 特征线方程 } \frac{dy}{dx} = \pm i\sqrt{y}, \text{ 特征线 } 2\sqrt{y} + ix = C_1 \text{ 和 } 2\sqrt{y} - ix = C_2,$$

$$\text{令 } \xi = 2\sqrt{y} + ix, \eta = 2\sqrt{y} - ix, \text{ 方程化为 } \frac{\partial^2 u}{\partial \xi \partial \eta} = 0,$$

$$\text{令 } \rho = \frac{\xi + \eta}{2} = 2\sqrt{y}, \sigma = \frac{\xi - \eta}{2i} = x, \text{ 方程化为 } \frac{\partial^2 u}{\partial \rho^2} + \frac{\partial^2 u}{\partial \sigma^2} = 0。$$

$$(4) a=1+x^2, b=0, c=1+y^2, d=x, e=y, f=g=0, b^2-ac=-(1+x^2)(1+y^2) < 0,$$

$$\text{所以是椭圆型方程。特征线方程 } \frac{dy}{dx} = \pm i\sqrt{\frac{1+y^2}{1+x^2}}, \text{ 特征线 } \operatorname{sh}^{-1} y - i \operatorname{sh}^{-1} x = C_1 \text{ 和}$$

$$\operatorname{sh}^{-1} y + i \operatorname{sh}^{-1} x = C_2, \text{ 令 } \xi = \operatorname{sh}^{-1} y - i \operatorname{sh}^{-1} x, \eta = \operatorname{sh}^{-1} y + i \operatorname{sh}^{-1} x, \text{ 方程化为 } \frac{\partial^2 u}{\partial \xi \partial \eta} = 0。$$

$$\text{令 } \rho = \frac{\xi + \eta}{2} = \operatorname{sh}^{-1} y, \sigma = \frac{\eta - \xi}{2i} = \operatorname{sh}^{-1} x, \text{ 方程化为 } \frac{\partial^2 u}{\partial \rho^2} + \frac{\partial^2 u}{\partial \sigma^2} = 0。$$

$$(5) a=\tan^2 x, b=-y \tan x, c=y^2, d=0, e=y, f=g=0, b^2-ac=0, \text{ 所以方程是抛}$$

$$\text{物型。特征线方程 } \frac{dy}{dx} = -y \cot x, \text{ 特征线 } y \sin x = C。 \text{ 可令 } \xi = y \sin x, \eta = y \cos x, \text{ 变换}$$

$$\text{后方程系数 } A=B=0, C=\frac{y^2}{\cos^2 x}, D=-\frac{y \sin x}{\cos^2 x}, E=\frac{y}{\cos x}, F=G=0, \text{ 即方程化为}$$

$$\frac{y^2}{\cos^2 x} \frac{\partial^2 u}{\partial \eta^2} - \frac{y \sin x}{\cos^2 x} \frac{\partial u}{\partial \xi} + \frac{y}{\cos x} \frac{\partial u}{\partial \eta} = 0, \text{ 即 } (\xi^2 + \eta^2) \frac{\partial^2 u}{\partial \eta^2} - \xi \frac{\partial u}{\partial \xi} + \eta \frac{\partial u}{\partial \eta} = 0。$$

372. 有些方程经过适当的因变量变换后可以消去一阶偏导数项。

(1) 证明：在因变量变换 $u = \exp[-(ax+by)]v$ 下，方程 $\nabla^2 u + 2a \frac{\partial u}{\partial x} + 2b \frac{\partial u}{\partial y} = 0$ 化为

Helmholtz 方程 $\nabla^2 v - (a^2 + b^2)v = 0$ ，其中 a, b 为常数；

(2) 寻求适当的变换，使方程 $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} + 2a \frac{\partial u}{\partial x} + 2b \frac{\partial u}{\partial y} = 0$ 在变换后不再含有一阶偏导数项；

(3) 设有方程 $a \frac{\partial^2 u}{\partial x^2} + 2b \frac{\partial^2 u}{\partial x \partial y} + c \frac{\partial^2 u}{\partial y^2} + d \frac{\partial u}{\partial x} + e \frac{\partial u}{\partial y} + fu = \frac{\partial u}{\partial t}$ ，其中 a, b, c, d, e, f 为常

数。证明：只有在 $b^2 - ac \neq 0$ 时可作变换 $u = \exp(\alpha x + \beta y + \gamma t)v$ 使 $v(x, y, t)$ 满足方程

$$a \frac{\partial^2 v}{\partial x^2} + 2b \frac{\partial^2 v}{\partial x \partial y} + c \frac{\partial^2 v}{\partial y^2} = \frac{\partial v}{\partial t}。$$

$$(1) \quad \frac{\partial u}{\partial x} = -ae^{-(ax+by)}v + e^{-(ax+by)} \frac{\partial v}{\partial x}, \quad \frac{\partial u}{\partial y} = -be^{-(ax+by)}v + e^{-(ax+by)} \frac{\partial v}{\partial y},$$

$$\frac{\partial^2 u}{\partial x^2} = a^2 e^{-(ax+by)}v - 2ae^{-(ax+by)} \frac{\partial v}{\partial x} + e^{-(ax+by)} \frac{\partial^2 v}{\partial x^2},$$

$$\frac{\partial^2 u}{\partial y^2} = b^2 e^{-(ax+by)}v - 2be^{-(ax+by)} \frac{\partial v}{\partial y} + e^{-(ax+by)} \frac{\partial^2 v}{\partial y^2}, \text{ 代入原方程得}$$

$$\begin{aligned} & (a^2 + b^2)e^{-(ax+by)}v - 2(a+b)e^{-(ax+by)} \frac{\partial v}{\partial x} + e^{-(ax+by)} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \\ & - 2(a^2 + b^2)e^{-(ax+by)}v + 2(a+b)e^{-(ax+by)} \frac{\partial v}{\partial x} = 0 \end{aligned}$$

$$\text{即 } \nabla^2 v - (a^2 + b^2)v = 0。$$

(2) 令 $u = v \exp(\alpha x + \beta y)$ ，则方程化为

$$\frac{\partial^2 v}{\partial x^2} - \frac{\partial^2 v}{\partial y^2} + 2(\alpha + a) \frac{\partial v}{\partial x} + 2(b - \beta) \frac{\partial v}{\partial y} + (\alpha^2 + 2a\alpha - \beta^2 + 2b\beta)v = 0，$$

$$\text{只需令 } \alpha = -a, \beta = b，\text{ 上面方程为 } \frac{\partial^2 v}{\partial x^2} - \frac{\partial^2 v}{\partial y^2} + (b^2 - a^2)v = 0。$$

(3) 将 $u = \exp(\alpha x + \beta y + \gamma t)v$ 代入原方程得

$$a \frac{\partial^2 v}{\partial x^2} + 2b \frac{\partial^2 v}{\partial x \partial y} + c \frac{\partial^2 v}{\partial y^2} + (2a\alpha + 2b\beta + d) \frac{\partial v}{\partial x} + (2b\alpha + 2c\beta + e) \frac{\partial v}{\partial y} + (a\alpha^2 + 2b\alpha\beta + c\beta^2 + d\alpha + e\beta - \gamma)v = \frac{\partial v}{\partial t}.$$

所以 $2a\alpha + 2b\beta + d = 0$, $2b\alpha + 2c\beta + e = 0$, $a\alpha^2 + 2b\alpha\beta + c\beta^2 + d\alpha + e\beta - \gamma = 0$,

前两式写成矩阵式 $\begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} -\frac{d}{2} \\ -\frac{e}{2} \end{pmatrix}$, 要使其有解, 必须有 $\begin{vmatrix} a & b \\ b & c \end{vmatrix} = b^2 - ac \neq 0$.

373. 求下列各偏微分方程通解: (1) $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial^2 u}{\partial x \partial y} - 3\frac{\partial^2 u}{\partial y^2} = 0$; (2) $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial x \partial y} = 0$;

(3) $(a^2 - b^2)\frac{\partial^2 u}{\partial x^2} + 2b\frac{\partial^2 u}{\partial x \partial t} - \frac{\partial^2 u}{\partial t^2} = 0$, a, b 为常数, $a \neq 0$; (4) $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial^2 u}{\partial x \partial y} + 2\frac{\partial^2 u}{\partial y^2} = 0$.

(1) 取试探解 $u = f(\alpha x + y)$, 则 $\frac{\partial^2 u}{\partial x^2} = \alpha^2 f''(\alpha x + y)$, $\frac{\partial^2 u}{\partial x \partial y} = \alpha f''(\alpha x + y)$,

$\frac{\partial^2 u}{\partial y^2} = f''(\alpha x + y)$, 代入方程得附加方程 $\alpha^2 - 2\alpha - 3 = 0$, 解得 $\alpha_1 = -1$, $\alpha_2 = 3$,

所以通解为 $u = f_1(x - y) + f_2(3x + y)$ 。

(2) 附加方程 $\alpha^2 - \alpha = 0$, 解得 $\alpha_1 = 0$, $\alpha_2 = 1$, 通解 $u = f_1(y) + f_2(x + y)$ 。

(3) 附加方程 $(a^2 - b^2)\alpha^2 + 2b\alpha - 1 = 0$, 解得 $\alpha_1 = \frac{1}{a+b}$, $\alpha_2 = -\frac{1}{a-b}$, 通解

$u = f_1[x + (a+b)t] + f_2[x - (a-b)t]$ 。

(4) 附加方程 $\alpha^2 - 2\alpha + 2 = 0$, $\alpha = 1 \pm i$, $u = f_1[(1+i)x + y] + f_2[(1-i)x + y]$ 。

374. 求偏微分方程 $x^2 \frac{\partial^2 u}{\partial x^2} - 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$ 的通解。

令 $x = e^t$, $y = e^s$, 即 $t = \ln x$, $s = \ln y$, 则 $x \frac{\partial}{\partial x} = \frac{\partial}{\partial t}$, $x^2 \frac{\partial^2}{\partial x^2} = \frac{\partial^2}{\partial t^2} - \frac{\partial}{\partial t}$,

$$y \frac{\partial}{\partial y} = \frac{\partial}{\partial s}, \quad y^2 \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial s^2} - \frac{\partial}{\partial s}, \quad xy \frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial t \partial s}, \quad \text{原方程化为 } \frac{\partial^2 u}{\partial t^2} - 2 \frac{\partial^2 u}{\partial t \partial s} + \frac{\partial^2 u}{\partial s^2} = 0,$$

取试探解为 $u = f(\alpha t + s)$, 可得附加方程 $(\alpha - 1)^2 = 0$, 可得重根 $\alpha = 1$, 通解为

$$u = f_1(t + s) + t f_2(t + s) = f_1(\ln xy) + \ln x f_2(\ln xy).$$

375. 证明方程 $\frac{\partial}{\partial x} \left[\left(1 - \frac{x}{h} \right)^2 \frac{\partial u}{\partial x} \right] - \frac{1}{a^2} \left(1 - \frac{x}{h} \right)^2 \frac{\partial^2 u}{\partial t^2} = 0$ 的通解为 $u = \frac{f(x+at) + g(x-at)}{h-x}$,

由此写出此方程在初始条件 $u|_{t=0} = \varphi(x)$, $\frac{\partial u}{\partial t}|_{t=0} = \psi(x)$ 下的解。

$$\frac{\partial u}{\partial x} = \frac{f(x+at) + g(x-at)}{(h-x)^2} + \frac{f'(x+at) + g'(x-at)}{h-x},$$

$$(h-x)^2 \frac{\partial u}{\partial x} = f(x+at) + g(x-at) + (h-x)[f'(x+at) + g'(x-at)],$$

$$\frac{\partial}{\partial x} \left[(h-x)^2 \frac{\partial u}{\partial x} \right] = (h-x)[f''(x+at) + g''(x-at)],$$

$$\frac{1}{a^2} (h-x)^2 \frac{\partial^2 u}{\partial t^2} = (h-x)[f''(x+at) + g''(x-at)], \quad \text{所以 } u = \frac{f(x+at) + g(x-at)}{h-x} \text{ 是原}$$

方程的解, 它有两个任意函数, 所以是原二阶方程的通解。

$$\text{将通解代入初始条件得, } f(x) + g(x) = (h-x)\varphi(x), \quad (\text{a})$$

$$f'(x) - g'(x) = \frac{1}{a}(h-x)\psi(x). \quad (\text{b})$$

$$(\text{b}) \text{ 式两边积分得 } f(x) - g(x) = \frac{1}{a} \int_0^x (h-\xi)\psi(\xi)d\xi + C, \quad (\text{c})$$

$$(\text{a}) + (\text{c}) \text{ 得 } f(x) = \frac{1}{2}(h-x)\varphi(x) + \frac{1}{2a} \int_0^x (h-\xi)\psi(\xi)d\xi + \frac{C}{2},$$

$$\text{所以 } f(x+at) = \frac{1}{2}(h-x-at)\varphi(x+at) + \frac{1}{2a} \int_0^{x+at} (h-\xi)\psi(\xi)d\xi + \frac{C}{2}.$$

$$(\text{a}) - (\text{c}) \text{ 得 } g(x) = \frac{1}{2}(h-x)\varphi(x) - \frac{1}{2a} \int_0^x (h-\xi)\psi(\xi)d\xi - \frac{C}{2},$$

$$\text{所以 } g(x-at) = \frac{1}{2}(h-x+at)\varphi(x-at) - \frac{1}{2a} \int_0^{x-at} (h-\xi)\psi(\xi)d\xi - \frac{C}{2},$$

$$\text{所以解为 } u = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2} \frac{at}{h-x} [\varphi(x-at) - \varphi(x+at)]$$

$$+ \frac{1}{2a(h-x)} \int_{x-at}^{x+at} (h-\xi) \psi(\xi) d\xi.$$

376. 求解弦振动方程的 Goursat 问题:
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x-at=0} = \varphi(x), u|_{x+at=0} = \psi(x) \\ \varphi(0) = \psi(0) \end{cases}$$

方程通解为 $u = f(x+at) + g(x-at)$, 代入条件 $u|_{x-at=0} = \varphi(x), u|_{x+at=0} = \psi(x)$ 得

$$f(2x) + g(0) = \varphi(x), \quad f(0) + g(2x) = \psi(x),$$

$$\text{所以 } f(x+at) = \varphi\left(\frac{x+at}{2}\right) - g(0), \quad g(x-at) = \psi\left(\frac{x-at}{2}\right) - f(0),$$

$$u = \varphi\left(\frac{x+at}{2}\right) + \psi\left(\frac{x-at}{2}\right) - f(0) - g(0), \quad \text{由于 } u(0,0) = f(0) + g(0) = \varphi(0),$$

$$\text{所以 } u = \varphi\left(\frac{x+at}{2}\right) + \psi\left(\frac{x-at}{2}\right) - \varphi(0).$$

377. 在波动方程 $\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0$ 中用 iy 代替 at 就能得到 Laplace 方程的“初值”问题

$$\begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \\ u|_{y=0} = \varphi(x), \frac{\partial u}{\partial y}|_{y=0} = \psi(x) \end{cases} \quad \text{。其形式解为 } u = \frac{1}{2} [\varphi(x+iy) + \varphi(x-iy)] + \frac{1}{2i} \int_{x-iy}^{x+iy} \psi(\xi) d\xi.$$

(1) 令 $\varphi(x) = x, \psi(x) = e^{-x}$, 可得 $u = x + e^{-x} \sin y$, 验证这个表达式处处满足 Laplace

方程, 也满足 $y=0$ 时的“初始”条件;

(2) 如果 $\varphi(x) = \frac{1}{1+x^2}, \psi(x) = 0$, 形式解为 $u = \frac{1+x^2-y^2}{(1+x^2-y^2)^2 + 4x^2y^2}$, 证明: 这个函

数在点 $(0, \pm 1)$ 不连续, 因此, 至少在这些点上不满足 Laplace 方程。这说明在一般情况下

Laplace 方程的“初值”问题无解。

$$(1) \frac{\partial u}{\partial x} = 1 - e^{-x} \sin y, \quad \frac{\partial^2 u}{\partial x^2} = e^{-x} \sin y, \quad \frac{\partial u}{\partial y} = e^{-x} \cos y, \quad \frac{\partial^2 u}{\partial y^2} = -e^{-x} \sin y, \text{ 所以}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \quad u|_{y=0} = u = x = \varphi(x), \quad \left. \frac{\partial u}{\partial y} \right|_{y=0} = e^{-x} \cos y|_{y=0} = e^{-x} = \psi(x).$$

$$(2) \lim_{x \rightarrow 0} u = \frac{1}{1-y^2}, \quad \lim_{y \rightarrow \pm 1} \lim_{x \rightarrow 0} u = \lim_{y \rightarrow \pm 1} \frac{1}{1-y^2} \rightarrow \infty.$$

378. 如果 $u(x, y, z)$ 满足 Laplace 方程, 证明: $v = (ax + by + cz)u$ 满足 $\nabla^4 v = 0$, 其中 a, b, c 为任意常数。

记 $f = ax + by + cz$, $\mathbf{a} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ 则 $\nabla f = \mathbf{a}$ 。

$$\nabla v = \nabla(fu) = u\nabla f + f\nabla u = u\mathbf{a} + f\nabla u,$$

$$\nabla^2 v = \nabla \cdot (u\mathbf{a}) + \nabla \cdot (f\nabla u) = \mathbf{a} \cdot \nabla u + u\nabla \cdot \mathbf{a} + \mathbf{a} \cdot \nabla u + f\nabla^2 u = 2\mathbf{a} \cdot \nabla u,$$

$$\nabla \nabla^2 v = 2\nabla(\mathbf{a} \cdot \nabla u) = 2[\mathbf{a} \times (\nabla \times \nabla u) + (\mathbf{a} \cdot \nabla)\nabla u + \nabla u \times (\nabla \times \mathbf{a}) + (\nabla u \cdot \nabla)\mathbf{a}] = 2(\mathbf{a} \cdot \nabla)\nabla u$$

$$= 2\left(a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y} + c \frac{\partial}{\partial z}\right)\nabla u,$$

$$\nabla^4 v = 2\nabla \cdot \left[\left(a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y} + c \frac{\partial}{\partial z}\right)\nabla u\right], \text{ 由于 } a, b, c \text{ 为常数, 可交换微分次序, 因此}$$

$$\text{上式} = 2\left(a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y} + c \frac{\partial}{\partial z}\right)\nabla^2 u = 0$$

379. 如果 $u(x, y, z)$ 是直角坐标系下 Laplace 方程的解, 证明 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial u}{\partial z}$ 也是解。

显然可交换微分次序, $\nabla^2 \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \nabla^2 u = 0$, 同样的, $\frac{\partial u}{\partial y}$, $\frac{\partial u}{\partial z}$ 也满足 Laplace 方程。

380. 如果 $u(\rho, \varphi, z)$ 是柱坐标下 Laplace 方程的解, 证明 $\frac{\partial u}{\partial \rho}$, $\frac{\partial u}{\partial \varphi}$ 也是解, 但 $\frac{\partial u}{\partial \rho}$ 一般不

是解。

$$\nabla^2 = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2}{\partial \varphi^2} + \frac{\partial^2}{\partial z^2}, \text{ 显然 } \frac{\partial}{\partial \varphi} \text{ 和 } \frac{\partial}{\partial z} \text{ 可与 } \nabla^2 \text{ 交换次序, 而 } \frac{\partial}{\partial \rho} \text{ 不能与 } \nabla^2 \text{ 交}$$

换次序, 同上题可知 $\frac{\partial u}{\partial \varphi}$, $\frac{\partial u}{\partial z}$ 也是解, 但 $\frac{\partial u}{\partial \rho}$ 一般不是解。

$$\text{三维无界空间波动方程初值问题} \begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ u|_{t=0} = \varphi(x, y, z), \frac{\partial u}{\partial t} \Big|_{t=0} = \psi(x, y, z) \end{cases} \quad \text{的解 (Poisson 公式)}$$

$$\text{为: } u(x, y, z, t) = \frac{1}{4\pi a} \left[\frac{\partial}{\partial t} \oiint_{S_{at}^M} \frac{\varphi(\xi, \eta, \zeta)}{at} dS + \oiint_{S_{at}^M} \frac{\psi(\xi, \eta, \zeta)}{at} dS \right], \text{ 其中 } S_{at}^M \text{ 是以点}$$

$M(x, y, z)$ 为球心, at 为半径的球面。

$$381. \text{ 直接验证 } u(x, y, z, t) = \frac{1}{4\pi a} \oiint_{S_{at}^M} \frac{\psi(\xi, \eta, \zeta)}{at} dS \text{ 是三维波动方程初值问题}$$

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ u|_{t=0} = 0, \frac{\partial u}{\partial t} \Big|_{t=0} = \psi(x, y, z) \end{cases} \text{ 的解, 其中 } S_{at}^M \text{ 是以点 } M(x, y, z) \text{ 为球心, } at \text{ 为半径的球面。}$$

u 的表达式写成:

$$u(x, y, z, t) = \frac{t}{4\pi} \iint \psi(x + at \sin \theta \cos \varphi, y + at \sin \theta \sin \varphi, z + at \cos \theta) d\Omega$$

($d\Omega = \sin \theta d\theta d\varphi$)。显然 $u(x, y, z, 0) = 0$,

$$\begin{aligned} \frac{\partial u}{\partial t} &= \frac{1}{4\pi} \iint \psi(x + at \sin \theta \cos \varphi, y + at \sin \theta \sin \varphi, z + at \cos \theta) d\Omega \\ &\quad + \frac{at}{4\pi} \iint (\psi_\xi \sin \theta \cos \varphi + \psi_\eta \sin \theta \sin \varphi + \psi_\zeta \cos \theta) d\Omega, \end{aligned}$$

$$\frac{\partial u}{\partial t} \Big|_{t=0} = \frac{1}{4\pi} \iint \psi(x, y, z) \sin \theta d\theta d\varphi = \frac{1}{4\pi} \psi(x, y, z) \int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta = \psi(x, y, z),$$

即 $u(x, y, z, t)$ 满足初始条件。

$$\begin{aligned} \text{令 } v(x, y, z, r) &= \frac{1}{4\pi r^2} \iint_{S_r^M} \psi(\xi, \eta, \zeta) dS \\ &= \frac{1}{4\pi} \iint \psi(x + r \sin \theta \cos \varphi, y + r \sin \theta \sin \varphi, z + r \cos \theta) d\Omega, \\ \text{则 } \frac{\partial v}{\partial r} &= \frac{1}{4\pi} \iint (\psi_\xi \sin \theta \cos \varphi + \psi_\eta \sin \theta \sin \varphi + \psi_\zeta \cos \theta) d\Omega \\ &= \frac{1}{4\pi r^2} \iint (\psi_\xi \sin \theta \cos \varphi + \psi_\eta \sin \theta \sin \varphi + \psi_\zeta \cos \theta) r^2 d\Omega \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4\pi r^2} \iint_{S_r^M} (\psi_\xi \sin \theta \cos \varphi + \psi_\eta \sin \theta \sin \varphi + \psi_\zeta \cos \theta) dS \\
&= \frac{1}{4\pi r^2} \iint_{S_r^M} (\psi_\xi \mathbf{i} + \psi_\eta \mathbf{j} + \psi_\zeta \mathbf{k}) \cdot \mathbf{n} dS = \frac{1}{4\pi r^2} \iiint_{B_r^M} (\psi_{\xi\xi} + \psi_{\eta\eta} + \psi_{\zeta\zeta}) dV \\
&= \frac{1}{4\pi r^2} \int_0^r r_1^2 dr_1 \iint (\psi_{\xi\xi} + \psi_{\eta\eta} + \psi_{\zeta\zeta}) d\Omega,
\end{aligned}$$

$$\text{所以 } \frac{\partial}{\partial r} \left(r^2 \frac{\partial v}{\partial r} \right) = \frac{r^2}{4\pi} \iint (\psi_{\xi\xi} + \psi_{\eta\eta} + \psi_{\zeta\zeta}) d\Omega,$$

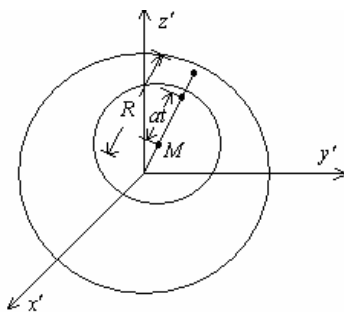
$$\begin{aligned}
\text{而 } \nabla^2 v &= \frac{1}{4\pi} \iint \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi(x + r \sin \theta \cos \varphi, y + r \sin \theta \sin \varphi, z + r \cos \theta) d\Omega \\
&= \frac{1}{4\pi} \iint (\psi_{\xi\xi} + \psi_{\eta\eta} + \psi_{\zeta\zeta}) d\Omega,
\end{aligned}$$

$$\text{所以 } \frac{\partial}{\partial r} \left(r^2 \frac{\partial v}{\partial r} \right) = r^2 \nabla^2 v, \text{ 可写成 } \frac{\partial^2}{\partial r^2} (rv) = \nabla^2 (rv), \text{ 令 } r = at \text{ 即可得 } \frac{\partial^2 u}{\partial t^2} = a^2 \nabla^2 u.$$

$$382. \text{ 求解 } \begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ u|_{t=0} = \begin{cases} u_0, x^2 + y^2 + z^2 < R^2 \\ 0, x^2 + y^2 + z^2 > R^2 \end{cases}, \frac{\partial u}{\partial t} \Big|_{t=0} = 0 \end{cases}.$$

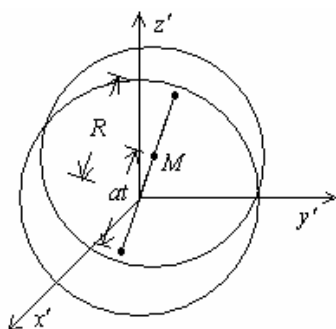
(1) 先讨论 $M(x, y, z)$ 位于球面 $x'^2 + y'^2 + z'^2 = R^2$ 内, 即 $r = \sqrt{x^2 + y^2 + z^2} < R$ 。

(i) $0 \leq t < \frac{R-r}{a}$, 即球面 S_{at}^M 完全位于球面 $x'^2 + y'^2 + z'^2 = R^2$ 内, 如下图:



$$u(M, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \oiint_{S_{at}^M} \frac{u(x', y', z', 0)}{at} dS' = \frac{u_0}{4\pi a} \frac{\partial}{\partial t} \left[\frac{1}{at} 4\pi (at)^2 \right] = u_0.$$

(ii) $\frac{R-r}{a} < t < \frac{R+r}{a}$, 即球面 S_{at}^M 与球面 $x'^2 + y'^2 + z'^2 = R^2$ 有交集, 如下图:

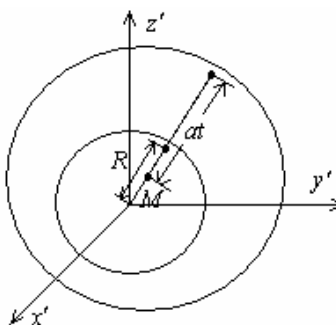


球面 S_{at}^M 在球面 $x'^2 + y'^2 + z'^2 = R^2$ 内的部分是高为 $h = \frac{R^2 - (r - at)^2}{2r}$ 的球冠，面积为

$$S = 2\pi a h = \pi a t \frac{R^2 - (r - at)^2}{r},$$

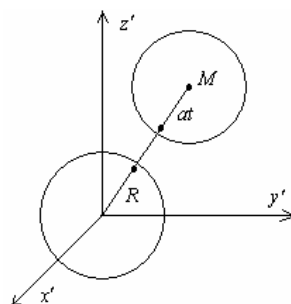
$$u(M, t) = \frac{u_0}{4\pi a} \frac{d}{dt} \left[\pi \frac{R^2 - (r - at)^2}{r} \right] = \frac{u_0}{4\pi a} \frac{2\pi a (r - at)}{r} = u_0 \frac{r - at}{2r}.$$

(iii) $t > \frac{R+r}{a}$ ，即球面 S_{at}^M 完全位于球面 $x'^2 + y'^2 + z'^2 = R^2$ 外，如下图，显然 $u(M, t) = 0$ 。

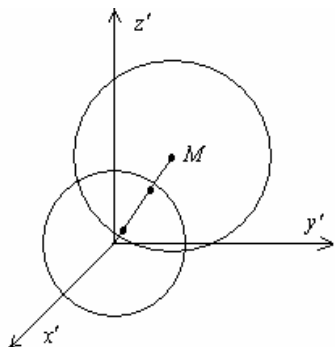


(2) $M(x, y, z)$ 位于球面 $x'^2 + y'^2 + z'^2 = R^2$ 外，即 $r = \sqrt{x^2 + y^2 + z^2} > R$ 。

(i) $0 \leq t < \frac{r-R}{a}$ ，即球面 S_{at}^M 与球面 $x'^2 + y'^2 + z'^2 = R^2$ 分离(如下图)，显然 $u(M, t) = 0$ 。

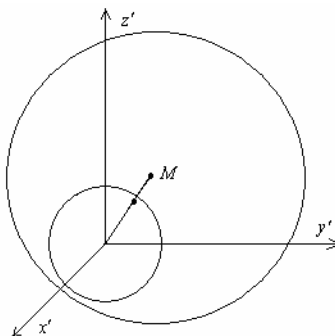


(ii) $\frac{r-R}{a} < t < \frac{r+R}{a}$ ，即球面 S_{at}^M 与球面 $x'^2 + y'^2 + z'^2 = R^2$ 有交集，如下图：



同第一种情况，有 $u(M, t) = u_0 \frac{r - at}{2r}$ 。

(iii) $t > \frac{r+R}{a}$ ，即球面 $x'^2 + y'^2 + z'^2 = R^2$ 在球面 S_{at}^M 内，如下图，显然 $u(M, t) = 0$ 。



$$383. (1) \begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) u \\ u|_{t=0} = \varphi(x), \frac{\partial u}{\partial t} \Big|_{t=0} = \psi(x) \end{cases}; (2) \begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) u \\ u|_{t=0} = \varphi(r), \frac{\partial u}{\partial t} \Big|_{t=0} = \psi(r) \end{cases}, \text{ 其中}$$

$$r = \sqrt{x^2 + y^2 + z^2}.$$

(1) 将坐标架置换一下，初始条件变为 $u|_{t=0} = \varphi(z), \frac{\partial u}{\partial t} \Big|_{t=0} = \psi(z)$ (这样使下面计算方便)。

$$\begin{aligned} u(x, y, z, t) &= \frac{1}{4\pi a} \left[\frac{\partial}{\partial t} \oint\!\!\!\oint_{S_{at}^M} \frac{\varphi(\zeta)}{at} dS + \oint\!\!\!\oint_{S_{at}^M} \frac{\psi(\zeta)}{at} dS \right] \\ &= \frac{1}{4\pi a} \left\{ \frac{\partial}{\partial t} \left[\int_0^{2\pi} d\varphi \int_0^\pi \varphi(z + at \cos \theta) at \sin \theta d\theta \right] + \int_0^{2\pi} d\varphi \int_0^\pi \psi(z + at \cos \theta) at \sin \theta d\theta \right\} \end{aligned}$$

$$\begin{aligned}
& \stackrel{z+at \cos \theta = \xi}{=} \frac{1}{2a} \left[\frac{\partial}{\partial t} \int_{z-at}^{z+at} \varphi(\xi) d\xi + \int_{z-at}^{z+at} \psi(\xi) d\xi \right] \\
& = \frac{1}{2} [\varphi(z-at) + \varphi(z+at)] + \frac{1}{2a} \int_{z-at}^{z+at} \psi(\xi) d\xi.
\end{aligned}$$

(2) 可看出, 解在空间上只与 r 有关, 所以方程可写成 $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right)$,

令 $v = ru$, 由于 $u|_{r=0}$ 有界, 所以

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial r^2}, r > 0 \\ v|_{r=0} = 0, v|_{t=0} = r\varphi(r), \frac{\partial v}{\partial t}|_{t=0} = r\psi(r) \end{cases}. \text{通解为 } v = f(r+at) + g(r-at), \text{ 可定出}$$

$$f(r) = \frac{1}{2} r\varphi(r) + \frac{1}{2a} \int_0^{r+at} \xi\psi(\xi) d\xi + \frac{C}{2}, \quad g(r) = \frac{1}{2} r\varphi(r) - \frac{1}{2a} \int_0^{r-at} \xi\psi(\xi) d\xi - \frac{C}{2}.$$

当 $r > at$ 时直接可得

$$\begin{aligned}
v &= \frac{1}{2} [(r+at)\varphi(r+at) + (r-at)\varphi(r-at)] + \frac{1}{2a} \int_{r-at}^{r+at} \xi\psi(\xi) d\xi, \\
u &= \frac{1}{2r} [(r+at)\varphi(r+at) + (r-at)\varphi(r-at)] + \frac{1}{2ar} \int_{r-at}^{r+at} \xi\psi(\xi) d\xi.
\end{aligned}$$

当 $r < at$ 时, 由 $v|_{r=0} = 0$ 可得 $f(at) + g(-at) = 0$, 所以

$$\begin{aligned}
g(r-at) &= -f(at-r) = -\frac{1}{2} (at-r)\varphi(at-r) - \frac{1}{2a} \int_0^{at-r} \xi\psi(\xi) d\xi - \frac{C}{2}, \\
v &= \frac{1}{2} [(r+at)\varphi(r+at) - (at-r)\varphi(at-r)] + \frac{1}{2a} \int_{at-r}^{at+r} \xi\psi(\xi) d\xi. \\
v &= \frac{1}{2r} [(r+at)\varphi(r+at) - (at-r)\varphi(at-r)] + \frac{1}{2ar} \int_{at-r}^{at+r} \xi\psi(\xi) d\xi.
\end{aligned}$$

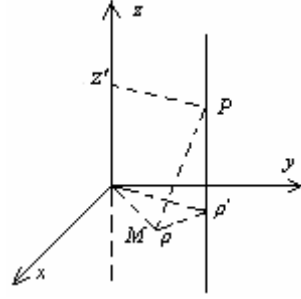
384. 由三维无界空间波动方程初值问题的 Green 函数 $G(\mathbf{r}, t; \mathbf{r}', t') = \frac{1}{4\pi a^2} \frac{\delta\left(t-t' - \frac{|\mathbf{r}-\mathbf{r}'|}{a}\right)}{|\mathbf{r}-\mathbf{r}'|}$

$$\left(\begin{cases} \frac{\partial^2 G(\mathbf{r}, t; \mathbf{r}', t')}{\partial t^2} - a^2 \nabla^2 G(\mathbf{r}, t; \mathbf{r}', t') = \delta(\mathbf{r}-\mathbf{r}') \delta(t-t') \\ G|_{t < t'} = 0, \frac{\partial G}{\partial t} \Big|_{t < t'} = 0 \end{cases} \right) \text{ 求出二维无界空间波动方}$$

程初值问题的 Green 函数 $\left(\begin{cases} \frac{\partial^2 G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')}{\partial t^2} - a^2 \nabla^2 G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t') = \delta(\boldsymbol{\rho}-\boldsymbol{\rho}') \delta(t-t') \\ G|_{t < t'} = 0, \frac{\partial G}{\partial t} \Big|_{t < t'} = 0 \end{cases} \right),$

从而求出二维无界空间波动方程初值问题的 Poisson 公式。

$G(\rho, t; \rho', t')$ 就是三维空间中平行于 z 轴的线源产生的场，如下图，



线源上点 P 到 xy 面上的场点 M 的距离为 $|\mathbf{r} - \mathbf{r}'| = \sqrt{z'^2 + |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}$,

$$G(\rho, t; \rho', t') = \int_{-\infty}^{\infty} G(\mathbf{r}, t; \mathbf{r}', t') dz' = \frac{1}{4\pi a} \int_{-\infty}^{\infty} \frac{\delta\left[\sqrt{z'^2 + |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2} - a(t - t')\right]}{\sqrt{z'^2 + |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}} dz',$$

当 $|\boldsymbol{\rho} - \boldsymbol{\rho}'| < a(t - t')$ 时，

$$\begin{aligned} \delta\left[\sqrt{z'^2 + |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2} - a(t - t')\right] &= \frac{a(t - t')}{\sqrt{a^2(t - t')^2 - |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}} \left\{ \delta\left[z' - \sqrt{a^2(t - t')^2 - |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}\right] \right. \\ &\quad \left. + \delta\left[z' + \sqrt{a^2(t - t')^2 - |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}\right] \right\}, \quad (\text{见附录}) \end{aligned}$$

$$\begin{aligned} \text{所以 } G(\rho, t; \rho', t') &= \frac{1}{4\pi a \sqrt{a^2(t - t')^2 - |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}} \int_{-\infty}^{\infty} \left\{ \delta\left[z' - \sqrt{a^2(t - t')^2 - |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}\right] \right. \\ &\quad \left. + \delta\left[z' + \sqrt{a^2(t - t')^2 - |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}\right] \right\} dz' \\ &= \frac{1}{2\pi a^2 \sqrt{(t - t')^2 - \frac{|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}{a^2}}}. \end{aligned}$$

当 $|\boldsymbol{\rho} - \boldsymbol{\rho}'| > a(t - t')$ 时， $\delta\left[\sqrt{z'^2 + |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2} - a(t - t')\right] = 0$ ，所以 $G(\rho, t; \rho', t') = 0$ 。

$$\text{综上, } G(\rho, t; \rho', t') = \begin{cases} \frac{1}{2\pi a^2 \sqrt{(t - t')^2 - \frac{|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}{a^2}}}, & |\boldsymbol{\rho} - \boldsymbol{\rho}'| < a(t - t') \\ 0, & |\boldsymbol{\rho} - \boldsymbol{\rho}'| > a(t - t') \end{cases}.$$

二维无界空间波动方程初值问题:
$$\begin{cases} \frac{\partial^2 u(\boldsymbol{\rho}, t)}{\partial t^2} - a^2 \nabla^2 u(\boldsymbol{\rho}, t) = 0 \\ u|_{t=0} = \varphi(\boldsymbol{\rho}), \frac{\partial u}{\partial t}\bigg|_{t=0} = \psi(\boldsymbol{\rho}) \end{cases}.$$

将 u 和 G 方程写成 $\frac{\partial^2 u(\boldsymbol{\rho}', t')}{\partial t'^2} - a^2 \nabla'^2 u(\boldsymbol{\rho}', t') = 0,$ (a)

由 Green 函数的互易性有 $\frac{\partial^2 G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')}{\partial t'^2} - a^2 \nabla'^2 G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t') = \delta(\boldsymbol{\rho} - \boldsymbol{\rho}') \delta(t - t').$ (b)

(b) $\times u(\boldsymbol{\rho}', t') -$ (a) $\times G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')$, 对 $x'y'$ 全平面及 $t' \in [0, t + \varepsilon]$ ($\varepsilon > 0$) 积分得

$$\begin{aligned} u(\boldsymbol{\rho}, t) &= \iint dx' dy' \int_0^{t+\varepsilon} \left[u(\boldsymbol{\rho}', t') \frac{\partial^2 G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')}{\partial t'^2} - G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t') \frac{\partial^2 u(\boldsymbol{\rho}', t')}{\partial t'^2} \right] dt' \\ &\quad + a^2 \int_0^{t+\varepsilon} dt' \iint [G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t') \nabla'^2 u(\boldsymbol{\rho}', t') - u(\boldsymbol{\rho}', t') \nabla'^2 G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')] dx' dy' \\ &= \iint dx' dy' \int_0^{t+\varepsilon} \frac{\partial}{\partial t'} \left[u(\boldsymbol{\rho}', t') \frac{\partial G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')}{\partial t'} - G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t') \frac{\partial u(\boldsymbol{\rho}', t')}{\partial t'} \right] dt' \\ &\quad + a^2 \int_0^{t+\varepsilon} dt' \oint \left[G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t') \frac{\partial u(\boldsymbol{\rho}', t')}{\partial n'} - u(\boldsymbol{\rho}', t') \frac{\partial G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')}{\partial n'} \right] dl' \end{aligned}$$

由于 $G \sim \frac{1}{r'}$, $\frac{\partial G}{\partial n'} \sim \frac{1}{r'^2}$, $u \sim \frac{1}{r'}$, $\frac{\partial u}{\partial n'} \sim \frac{1}{r'^2}$, $dl' \sim r'$, 所以后一项积分趋于 0, 所以

$$u(\boldsymbol{\rho}, t) = \iint \left[u(\boldsymbol{\rho}', t') \frac{\partial G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')}{\partial t'} - G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t') \frac{\partial u(\boldsymbol{\rho}', t')}{\partial t'} \right]_{t'=0}^{t'=t+\varepsilon} dx' dy',$$

由于 $G|_{t < t'} = 0, \frac{\partial G}{\partial t}\bigg|_{t < t'} = 0$, 所以 $\left[u(\boldsymbol{\rho}', t') \frac{\partial G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')}{\partial t'} - G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t') \frac{\partial u(\boldsymbol{\rho}', t')}{\partial t'} \right]_{t'=t+\varepsilon} = 0,$

$$\begin{aligned} u(\boldsymbol{\rho}, t) &= - \iint \left[u(\boldsymbol{\rho}', t') \frac{\partial G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t')}{\partial t'} - G(\boldsymbol{\rho}, t; \boldsymbol{\rho}', t') \frac{\partial u(\boldsymbol{\rho}', t')}{\partial t'} \right]_{t'=0} dx' dy' \\ &= \frac{1}{2\pi a^2} \left[\frac{\partial}{\partial t} \iint_{|\boldsymbol{\rho}-\boldsymbol{\rho}'| < a(t-t')} \frac{\varphi(\boldsymbol{\rho}')}{\sqrt{t^2 - \frac{|\boldsymbol{\rho}-\boldsymbol{\rho}'|^2}{a^2}}} dx' dy' + \iint_{|\boldsymbol{\rho}-\boldsymbol{\rho}'| < a(t-t')} \frac{\psi(\boldsymbol{\rho}')}{\sqrt{t^2 - \frac{|\boldsymbol{\rho}-\boldsymbol{\rho}'|^2}{a^2}}} dx' dy' \right]. \end{aligned}$$

385. 稳定问题的平均值定理。设在空间区域 V 内部有 $\nabla^2 u = 0$, 证明: 任意一点 $M(x, y, z)$

处的 u 值等于以该点为球心的任意球面上 u 的平均值。

记 $\bar{u}(M, r) = \frac{1}{4\pi r^2} \iint_{S_r^M} u(\xi, \eta, \zeta) dS$, 同 381 题可得

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial \bar{u}}{\partial r} \right) = \frac{r^2}{4\pi} \iint (u_{\xi\xi} + u_{\eta\eta} + u_{\zeta\zeta}) d\Omega = 0, \text{ 所以 } \bar{u}(M, r) = \frac{A(M)}{r} + B(M),$$

由 $\lim_{r \rightarrow 0} \bar{u}(M, r) = u(M)$ 可定出 $A(M) = 0, B(M) = u(M)$, 即 $\bar{u}(M, r) = u(M)$ 。

386. 用 Riemann 方法求解:
$$\begin{cases} x^2 \frac{\partial^2 u}{\partial x^2} - y^2 \frac{\partial^2 u}{\partial y^2} = 0 \\ u|_{y=1} = \varphi(x), \frac{\partial u}{\partial y} \Big|_{y=1} = \psi(x) \end{cases}.$$

关于 Riemann 方法见郭敦仁《数学物理方法》22.3 节, 或 H.M.Lieberstein《Theory of Partial Differential Equation》Chapter 7。

设 $x > 0, y > 1$ 。特征线方程: $\frac{dy}{dx} = \pm \frac{y}{x}$, 特征线 $xy = C_1, \frac{y}{x} = C_2$, 令 $\xi = xy, \eta = \frac{y}{x}$

($\xi > 0, \eta > 0$), 即 $x = \sqrt{\frac{\xi}{\eta}}, y = \sqrt{\xi\eta}$, 则方程化为 $\frac{\partial^2 u}{\partial \xi \partial \eta} - \frac{1}{2\xi} \frac{\partial u}{\partial \eta} = 0$, $y=1$ 变换为

$$\xi\eta = 1, \text{ 即 } u|_{y=1} = u|_{\xi\eta=1} = \varphi\left(\sqrt{\frac{\xi}{\eta}}\right)_{\eta=\frac{1}{\xi}} = \varphi(\xi), \text{ 所以 } \frac{\partial u}{\partial \eta} \Big|_{\xi=\frac{1}{\eta}} = \frac{\partial u|_{\xi=\frac{1}{\eta}}}{\partial \eta} = \frac{\partial \varphi(\xi)}{\partial \eta} = 0,$$

$$\frac{\partial u}{\partial y} \Big|_{y=1} = \left(x \frac{\partial u}{\partial \xi} + \frac{1}{x} \frac{\partial u}{\partial \eta} \right)_{\xi\eta=1} = \sqrt{\frac{\xi}{\eta}} \frac{\partial u}{\partial \xi} \Big|_{\xi\eta=1} = \xi \frac{\partial u}{\partial \xi} \Big|_{\xi\eta=1} = \psi\left(\sqrt{\frac{\xi}{\eta}}\right)_{\eta=\frac{1}{\xi}} = \psi(\xi),$$

该问题写成
$$\begin{cases} \frac{\partial^2 u}{\partial \xi \partial \eta} - \frac{1}{2\xi} \frac{\partial u}{\partial \eta} = 0 \\ u|_{\xi\eta=1} = \varphi(\xi), \frac{\partial u}{\partial \xi} \Big|_{\xi\eta=1} = \frac{\psi(\xi)}{\xi} \end{cases}.$$

Riemann 函数 $R(\xi, \eta; \xi_0, \eta_0)$ 满足
$$\begin{cases} \frac{\partial^2 R}{\partial \xi \partial \eta} + \frac{1}{2\xi} \frac{\partial R}{\partial \eta} = 0 \\ R(\xi, \eta_0; \xi_0, \eta_0) = \exp\left(-\int_{\xi_0}^{\xi} \frac{1}{2\lambda} d\lambda\right) = \sqrt{\frac{\xi_0}{\xi}}, \\ R(\xi_0, \eta; \xi_0, \eta_0) = 1 \end{cases}$$

R 的方程写成 $\frac{\partial R_\eta}{\partial \xi} + \frac{1}{2\xi} R_\eta = 0$, 可得 $R_\eta = A(\eta) \exp\left(-\int \frac{1}{2\xi} d\xi\right) = \frac{A(\eta)}{\sqrt{\xi}}$,

积分得 $R = \frac{B(\eta)}{\sqrt{\xi}} + C$, 由于 $R(\xi_0, \eta; \xi_0, \eta_0) = \frac{B(\eta)}{\sqrt{\xi_0}} + C = 1$, 所以 $B(\eta) = (1-C)\sqrt{\xi_0}$,

代入 $R(\xi, \eta_0; \xi_0, \eta_0) = \frac{B(\eta_0)}{\sqrt{\xi}} + C = \sqrt{\frac{\xi_0}{\xi}}$ 得 $C = 0$, 所以 $B(\eta) = \sqrt{\xi_0}$, $R = \sqrt{\frac{\xi_0}{\xi}}$ 。

$$\begin{aligned} u(\xi_0, \eta_0) &= u(A)R(A; \xi_0, \eta_0) + \int_{\widehat{AB}} \left[R(u_\xi + bu) d\xi + u(R_\eta - aR) d\eta \right] \\ &= \varphi\left(\frac{1}{\eta_0}\right) \sqrt{\xi_0 \eta_0} + \int_{\frac{1}{\eta_0}}^{\xi_0} \sqrt{\frac{\xi_0}{\xi}} \left[\frac{\psi(\xi)}{\xi} - \frac{1}{2\xi} \varphi(\xi) \right] d\xi \end{aligned}$$

将 (ξ_0, η_0) 代换回 (x_0, y_0) , 再换成 (x, y) 即可得

$$u(x, y) = y\varphi\left(\frac{x}{y}\right) + \sqrt{xy} \int_{\frac{x}{y}}^{xy} \xi^{-3/2} \left[\psi(\xi) - \frac{1}{2} \varphi(\xi) \right] d\xi。$$

$$\begin{aligned} \text{若用公式 } u(\xi_0, \eta_0) &= \frac{1}{2} u(A) R(A) + \frac{1}{2} u(B) R(B) + \int_{\widehat{AB}} \left\{ \left[\frac{1}{2} R u_\xi + \left(bR - \frac{1}{2} R_\xi \right) u \right] d\xi \right. \\ &\quad \left. - \left[\frac{1}{2} R u_\eta + \left(aR - \frac{1}{2} R_\eta \right) u \right] d\eta \right\} \end{aligned}$$

$$\text{则有 } u(x, y) = \frac{1}{2} y\varphi\left(\frac{x}{y}\right) + \frac{1}{2} \varphi(xy) + \frac{1}{2} \sqrt{xy} \int_{\frac{x}{y}}^{xy} \xi^{-3/2} \left[\psi(\xi) - \frac{1}{2} \varphi(\xi) \right] d\xi。$$

387~390 略。

附录:

$$\delta[f(x)] = \sum_{k=1}^n \frac{1}{|f'(x_k)|} \delta(x - x_k), \text{ 其中 } x_k \ (k=1, 2, \dots, n) \text{ 是 } f(x) \text{ 的全部零点。}$$

391. 试根据变分原理导出完全柔软的均匀弦的横振动方程。

取弦上足够短的一段 dx , 该段弦的动能为 $\frac{1}{2}\rho dx\left(\frac{\partial u}{\partial t}\right)^2$, 势能为 $\frac{1}{2}Tdx\left(\frac{\partial u}{\partial x}\right)^2$, 弦的

$$\text{Hamilton 作用量 } S = \int_{t_0}^{t_1} \int_{x_0}^{x_1} F(u_t, u_x) dx dt = \int_{t_0}^{t_1} \int_{x_0}^{x_1} \frac{1}{2} \left[\rho \left(\frac{\partial u}{\partial t} \right)^2 + T \left(\frac{\partial u}{\partial x} \right)^2 \right] dx dt.$$

该泛函的 Euler-Lagrange 方程为 $\frac{\partial}{\partial t} \frac{\partial F}{\partial u_t} + \frac{\partial}{\partial x} \frac{\partial F}{\partial u_x} = \rho \frac{\partial^2 u}{\partial t^2} + T \frac{\partial^2 u}{\partial x^2} = 0$ 。

392. 设 $y = y(x)$, $F(y, y')$ 不显含 x , 证明: $\begin{cases} J[y] = \int_a^b F(y, y') dx \\ y(a) = A, y(b) = B \end{cases}$ 取极值的必要条件

是 $y' \frac{\partial F}{\partial y'} - F = C$ (常数)。

$$\begin{aligned} \delta J[y] &= \int_a^b \left(\frac{\partial F}{\partial y} \delta y + \frac{\partial F}{\partial y'} \delta y' \right) dx = \frac{\partial F}{\partial y'} \delta y \Big|_{x=a}^{x=b} + \int_a^b \left(\frac{\partial F}{\partial y} \delta y - \frac{d}{dx} \frac{\partial F}{\partial y'} \delta y \right) dx \\ &= \int_a^b \left(\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} \right) \delta y dx = 0, \end{aligned}$$

所以 $\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = 0$ 。

$$\text{由于 } \frac{d}{dx} \left(y' \frac{\partial F}{\partial y'} - F \right) = y'' \frac{\partial F}{\partial y'} + y' \frac{d}{dx} \frac{\partial F}{\partial y'} - y' \frac{\partial F}{\partial y} - y'' \frac{\partial F}{\partial y'} = y' \left(\frac{d}{dx} \frac{\partial F}{\partial y'} - \frac{\partial F}{\partial y} \right) = 0,$$

所以 $y' \frac{\partial F}{\partial y'} - F = C$ 。

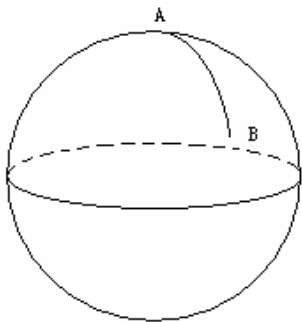
393. 求泛函 $\begin{cases} J[y] = \int_0^1 \sqrt{1+y'^2} dx \\ y(0) = 0, y(1) = 1 \end{cases}$ 的极值曲线。

Euler-Lagrange 方程为 $\frac{d}{dx} \frac{y'}{\sqrt{1+y'^2}} = 0$, 所以 $\sqrt{1+y'^2} = Cy'$, 可得 $y' = C_1$, 积分得

$y(x) = C_1 x + C_2$, 由边界条件得 $y = x$ 。

394. 如下图所示, 写出单位球面上从 A 点到 B 点的“短程线”所满足的微分方程, 并求出短程线。证明此短程线在过 A, B 两点的大圆上。基于对称性的考虑, 不妨取 A 点坐标为

$(\theta_0, \varphi_0) = (0, 0)$, B 点坐标为 (θ_1, φ_1) 。(单位球面上弧元为 $ds = \sqrt{d\theta^2 + \sin^2 \theta d\varphi^2}$)

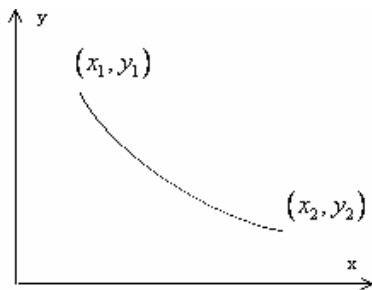


A, B 间弧长为 $s = \int_0^{\theta_1} \sqrt{1 + \sin^2 \theta \varphi'^2} d\theta$ ($\varphi' = d\varphi/d\theta$),

Euler-Lagrange 方程为 $\frac{d}{d\theta} \frac{\sin^2 \theta \varphi'}{\sqrt{1 + \sin^2 \theta \varphi'^2}} = 0$, 即 $\frac{\sin^2 \theta \varphi'}{\sqrt{1 + \sin^2 \theta \varphi'^2}} = C$, 代入 A 点坐标可得

$C = 0$, 所以 $\frac{d\varphi}{d\theta} = 0$, 即 $\varphi = C_1$, 代入 B 点坐标得 $\varphi = \varphi_1$, 这正是在大圆上。

395. 一质点在重力作用下沿光滑曲线由点 (x_1, y_1) 运动至点 (x_2, y_2) (见下图)。试求“捷线”(即质点沿此曲线运动费时最少)所满足的微分方程。



$v = \frac{ds}{dt} = \sqrt{v_0^2 + 2g(y_1 - y)}$, 所以 $t = \int_{(x_1, y_1)}^{(x_2, y_2)} \frac{ds}{\sqrt{v_0^2 + 2g(y_1 - y)}} = \int_{x_1}^{x_2} \sqrt{\frac{1 + y'^2}{v_0^2 + 2g(y_1 - y)}} dx$ 。

记 $F(y, y') = \sqrt{\frac{1 + y'^2}{v_0^2 + 2g(y_1 - y)}}$, 由 392 题结论, $y' \frac{\partial F}{\partial y'} - F = C$, 即

$$\frac{y'^2}{\sqrt{1 + y'^2}} \frac{1}{\sqrt{v_0^2 + 2g(y_1 - y)}} - \sqrt{1 + y'^2} \frac{1}{\sqrt{v_0^2 + 2g(y_1 - y)}} = C, \text{ 还可写成}$$

$$\frac{1}{\sqrt{1 + y'^2}} = -C \sqrt{v_0^2 + 2g(y_1 - y)}.$$

396. 若 $\bar{y}(x)$ 使泛函 $\begin{cases} J[y] = \int_a^b F(x, y, y') dx \\ y(a) = A, y(b) = B \end{cases}$ 在限制条件 $J_1[y] = \int_a^b G(x, y, y') dx = C$ 下

取极值, 且相应的 Lagrange 乘子 $\lambda \neq 0$, 试证明 $\bar{y}(x)$ 也使泛函 $\begin{cases} J_1[y] = \int_a^b G(x, y, y') dx \\ y(a) = A, y(b) = B \end{cases}$ 在

限制条件 $J[y] = \int_a^b F(x, y, y') dx = D$ 下取极值。

第一个泛函极值问题引入 Lagrange 乘子 λ , 则 $\bar{y}(x)$ 满足 $\int_a^b (F - \lambda G) dx$ 的 Euler-Lagrange

方程: $\frac{\partial F}{\partial y} - \lambda \frac{\partial G}{\partial y} - \bar{y}' \frac{\partial F}{\partial y'} + \lambda \bar{y}' \frac{\partial G}{\partial y'} = 0$, 由于 $\lambda \neq 0$, 方程两边乘 $\frac{1}{\lambda}$ 得

$\frac{\partial G}{\partial y} - \frac{1}{\lambda} \frac{\partial F}{\partial y} - \bar{y}' \frac{\partial G}{\partial y'} + \frac{1}{\lambda} \bar{y}' \frac{\partial F}{\partial y'} = 0$, 这正是 $\int_a^b \left(G - \frac{1}{\lambda} F \right) dx$ 的 Euler-Lagrange 方程, 即

$\bar{y}(x)$ 是第二个泛函极值问题的解。

397. 过二已知点 (x_1, y_1) , (x_2, y_2) 作一曲线, 使此曲线绕 x 轴旋转所得曲面面积最小, 求曲线作满足的微分方程。

旋转面面积为 $S = \int_{(x_1, y_1)}^{(x_2, y_2)} 2\pi y ds = 2\pi \int_{x_1}^{x_2} y \sqrt{1 + y'^2} dx$, 由 392 题结论, Euler-Lagrange 方程

为 $y\sqrt{1 + y'^2} - \frac{yy'^2}{\sqrt{1 + y'^2}} = C$, 即 $\frac{y}{\sqrt{1 + y'^2}} = C$ 。

398. 试写出本征值问题 $\begin{cases} \nabla^2 u + \lambda u = 0 \\ \left(\alpha u + \beta \frac{\partial u}{\partial n} \right)_{\Sigma} = 0 \end{cases}$ 所对应的泛函极值问题。设 $\beta \neq 0$ 。

由于 $\nabla \cdot (\delta u \nabla u) = \nabla \delta u \cdot \nabla u + \delta u \nabla^2 u$, 所以 $\delta u \nabla^2 u = \nabla \cdot (\delta u \nabla u) - \nabla \delta u \cdot \nabla u$,

$$\iiint_V (\nabla^2 u + \lambda u) \delta u dV = \iiint_V [\nabla \cdot (\delta u \nabla u) - \nabla \delta u \cdot \nabla u + \lambda \delta u] dV$$

$$= \oint_{\Sigma} \delta u \frac{\partial u}{\partial n} dS - \iiint_V (\nabla \delta u \cdot \nabla u - \lambda \delta u) dV = - \oint_{\Sigma} \frac{\alpha}{\beta} u \delta u dS - \frac{1}{2} \delta \iiint_V (\nabla u \cdot \nabla u - \lambda u^2) dV$$

$$= -\frac{1}{2} \delta \left[\oint_{\Sigma} \frac{\alpha}{\beta} u^2 dS + \iiint_V (\nabla u \cdot \nabla u - \lambda u^2) dV \right] = 0$$

$$\text{即对应泛函} \begin{cases} \oint_{\Sigma} \frac{\alpha}{\beta} u^2 dS + \iiint_V \nabla u \cdot \nabla u dV \\ \left(\alpha u + \beta \frac{\partial u}{\partial n} \right)_{\Sigma} = 0 \end{cases} \quad \text{在条件} \iiint_V u^2 dV = C \text{ 下的极值。}$$

399. 设有一长为 1 的弦, 由同种质料组成, 线密度 $\rho(x) = 1+x$ ($0 \leq x \leq 1$), 则振动方程

$$\text{为} (1+x) \frac{\partial^2 u}{\partial t^2} = T \frac{\partial^2 u}{\partial x^2}, \text{ 试用 Ritz 方法求出两端固定时的最低固有频率。}$$

$$\text{令} u(x, t) = y(x) e^{i\omega t}, \text{ 代入方程得} y'' + \frac{\omega^2}{T} (1+x) y = 0.$$

$$\text{对应泛函} \int_0^1 \left[y'^2 - \frac{\omega^2}{T} (1+x) y^2 \right] dx \text{ 的极值。取一组基函数展开 } y(x): y = \sum_{k=1}^n c_k \varphi_k(x),$$

$$\text{泛函化为} \sum_{k=1}^n \sum_{l=1}^n c_k c_l \int_0^1 \left[\varphi'_k(x) \varphi'_l(x) - \frac{\omega^2}{T} (1+x) \varphi_k(x) \varphi_l(x) \right] dx = \sum_{k=1}^n \sum_{l=1}^n c_k c_l f_{kl}.$$

要使它取极值, 只需使它对 c_k ($k=1, 2, \dots, n$) 的偏导数为 0, 即

$$\sum_{l=1, l \neq k}^n c_l f_{kl} + 2c_k f_{kk} = 0, \quad k=1, 2, \dots, n, \text{ 写成矩阵式} \begin{pmatrix} 2f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & 2f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1} & f_{n2} & \cdots & 2f_{nn} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = 0,$$

解之即可。

$$400. \text{ 用 Ritz 方法求出} \begin{cases} y'' + \lambda y = 0 \\ y(-1) = 0, y(1) = 0 \end{cases} \text{ 的最低两个本征值的近似值, 取试探函数为:}$$

$$(1) y = c_1(1-x^2) + c_2x(1-x^2)^2; (2) y = c_1(1-x^2) + c_2x^2(1-x^2)^2.$$

该边值问题对应泛函 $\int_{-1}^1 y'^2 dx$ 在约束条件 $\int_{-1}^1 y^2 dx = C$ 下的极值问题。后面步骤略。